Support Vector Machine

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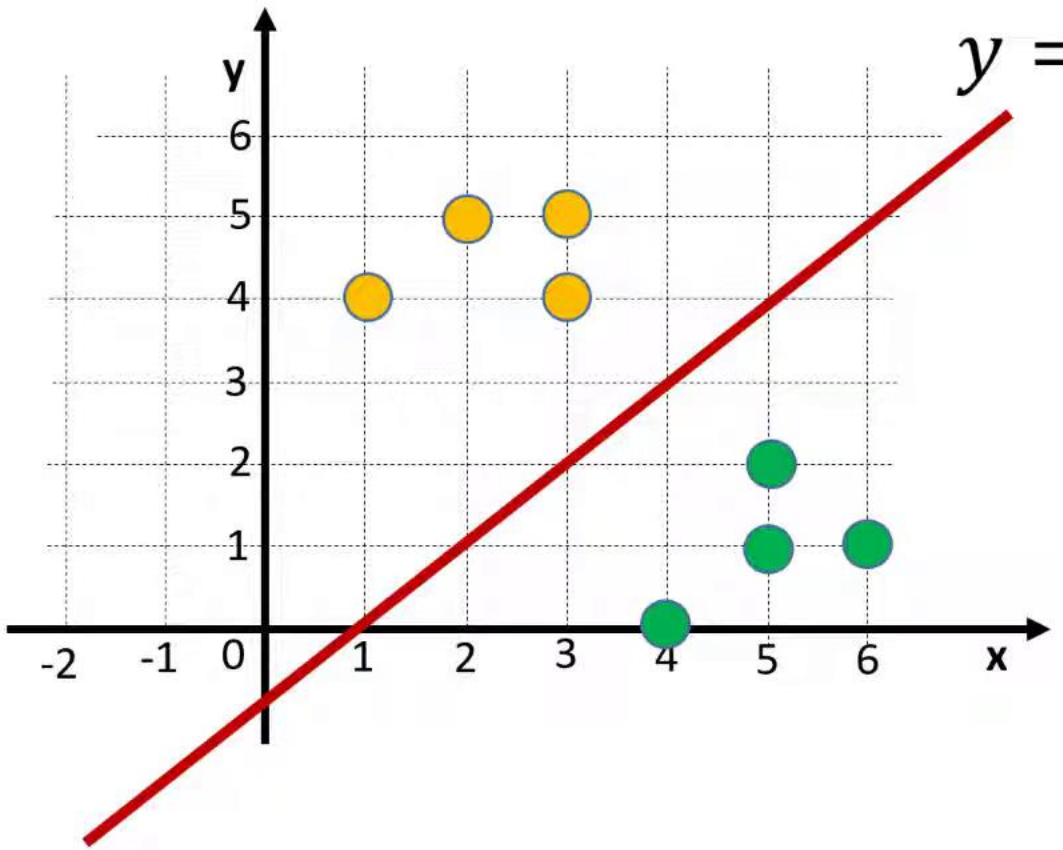
SVM

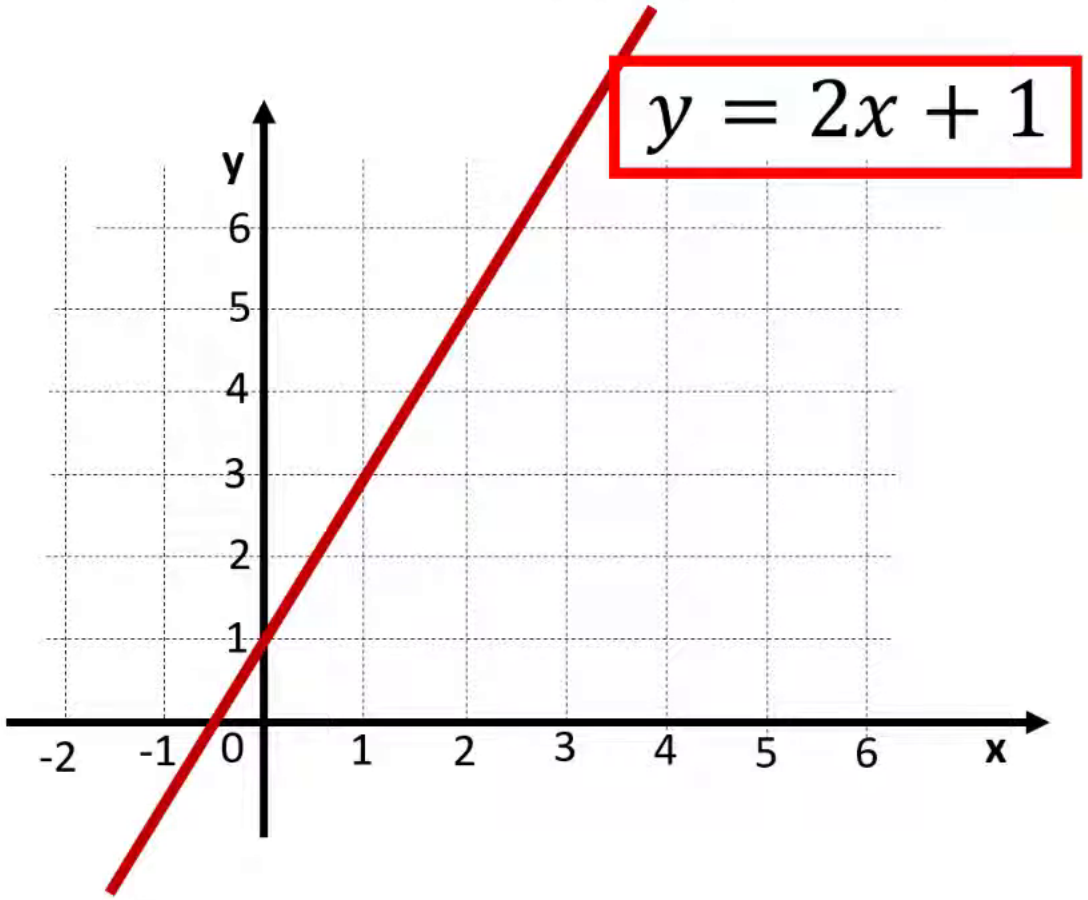
$$Ax + By + C = 0$$



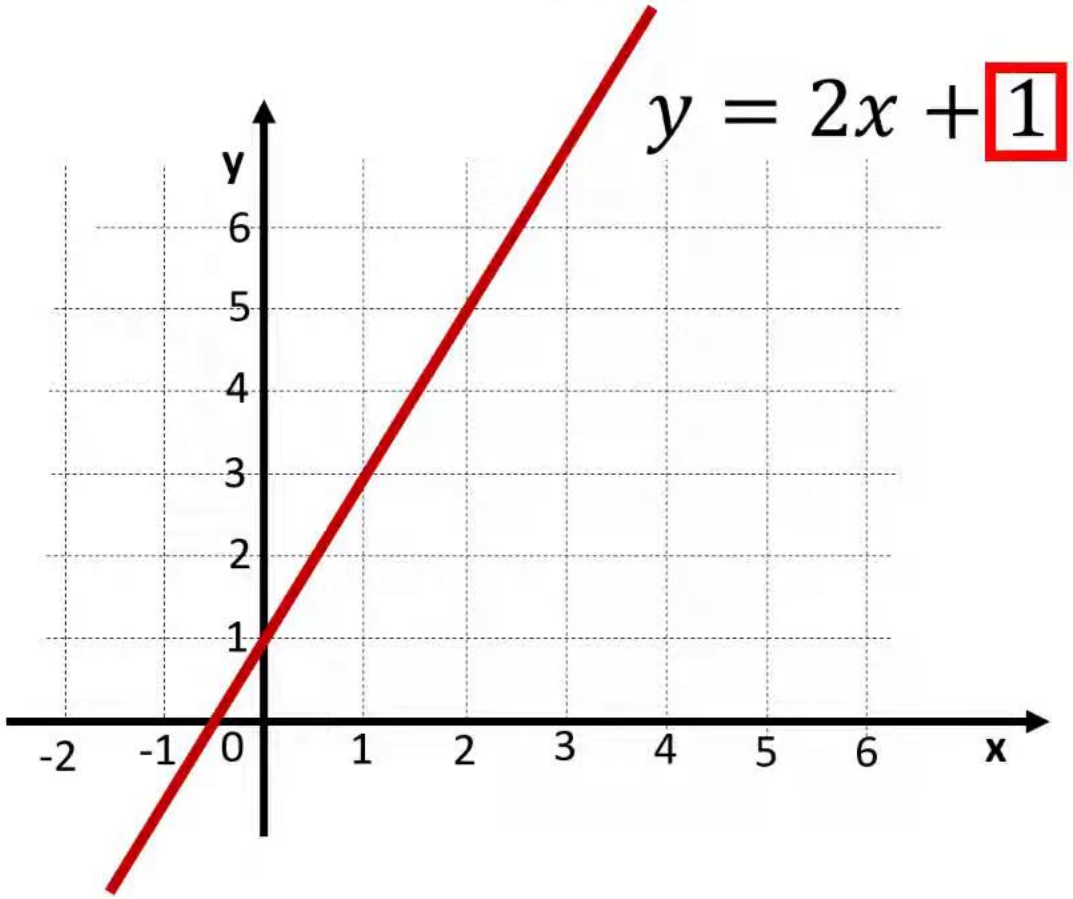
$$y = x - 1$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

The equation of a straight line

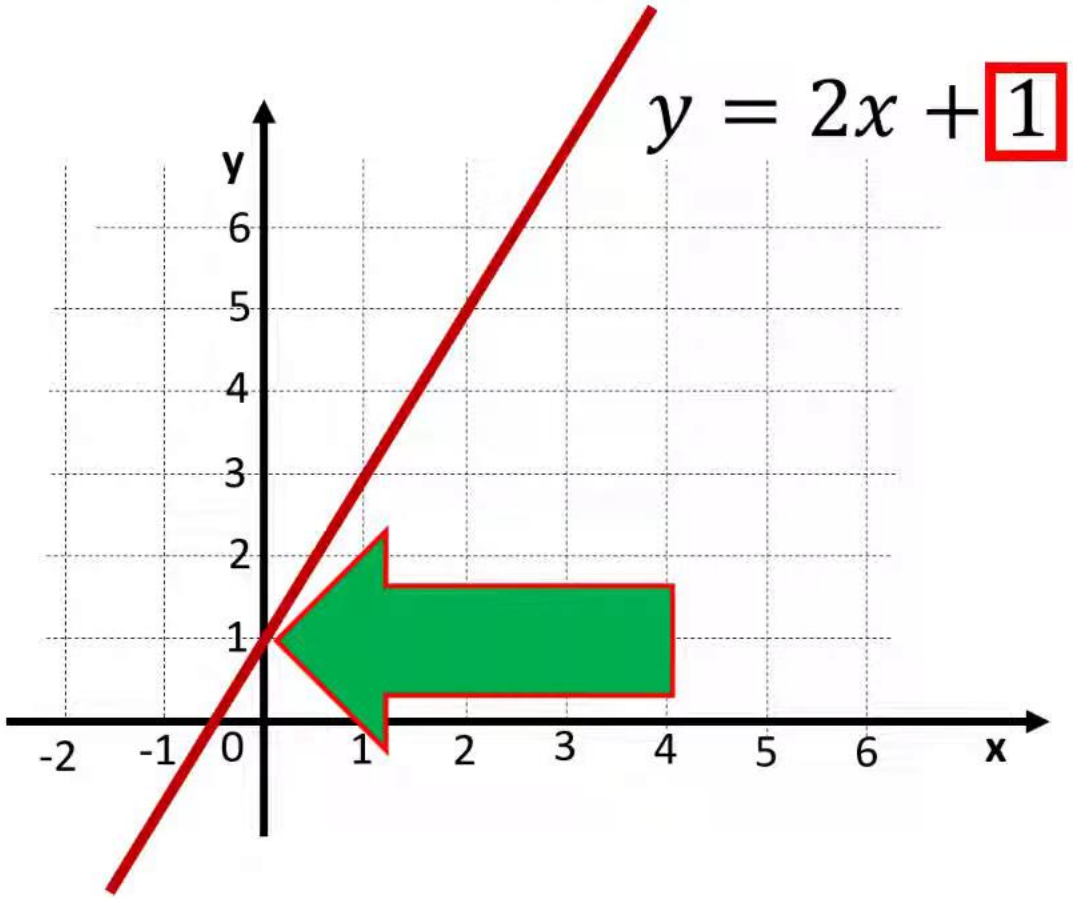


The equation of a straight line



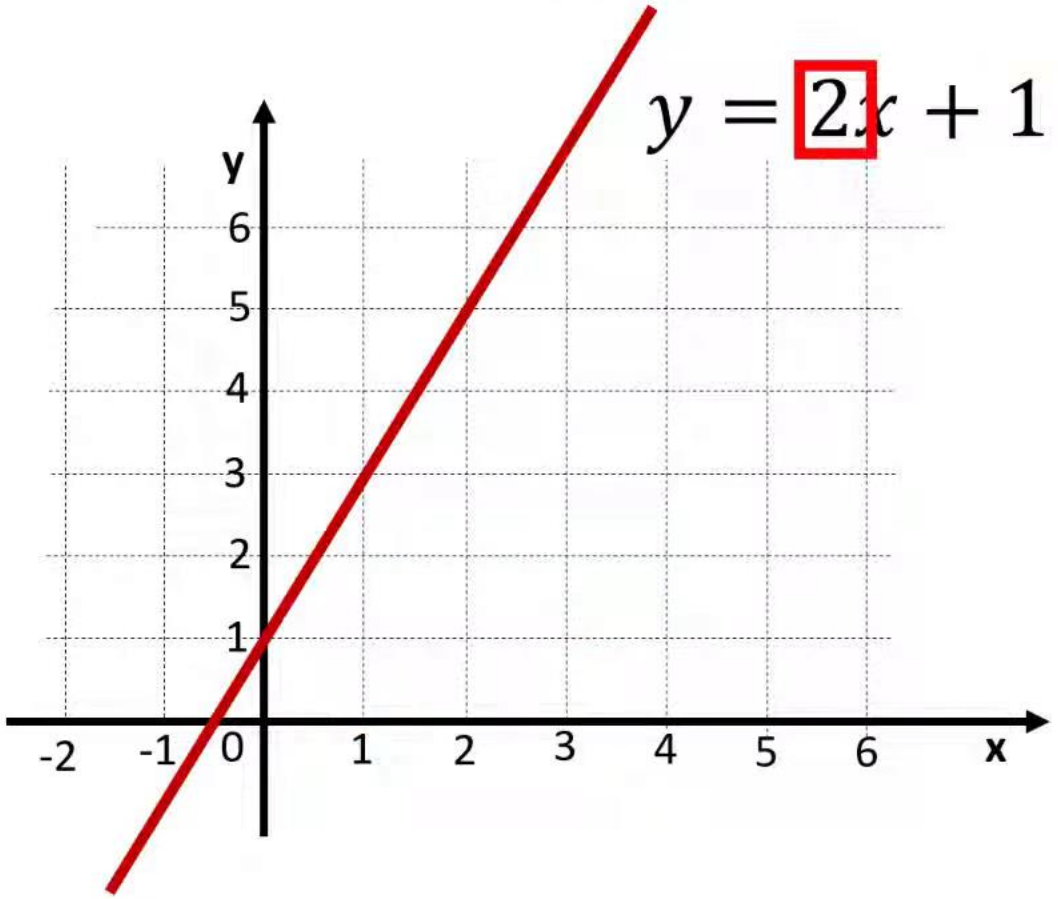
where this constant term represents,

The equation of a straight line



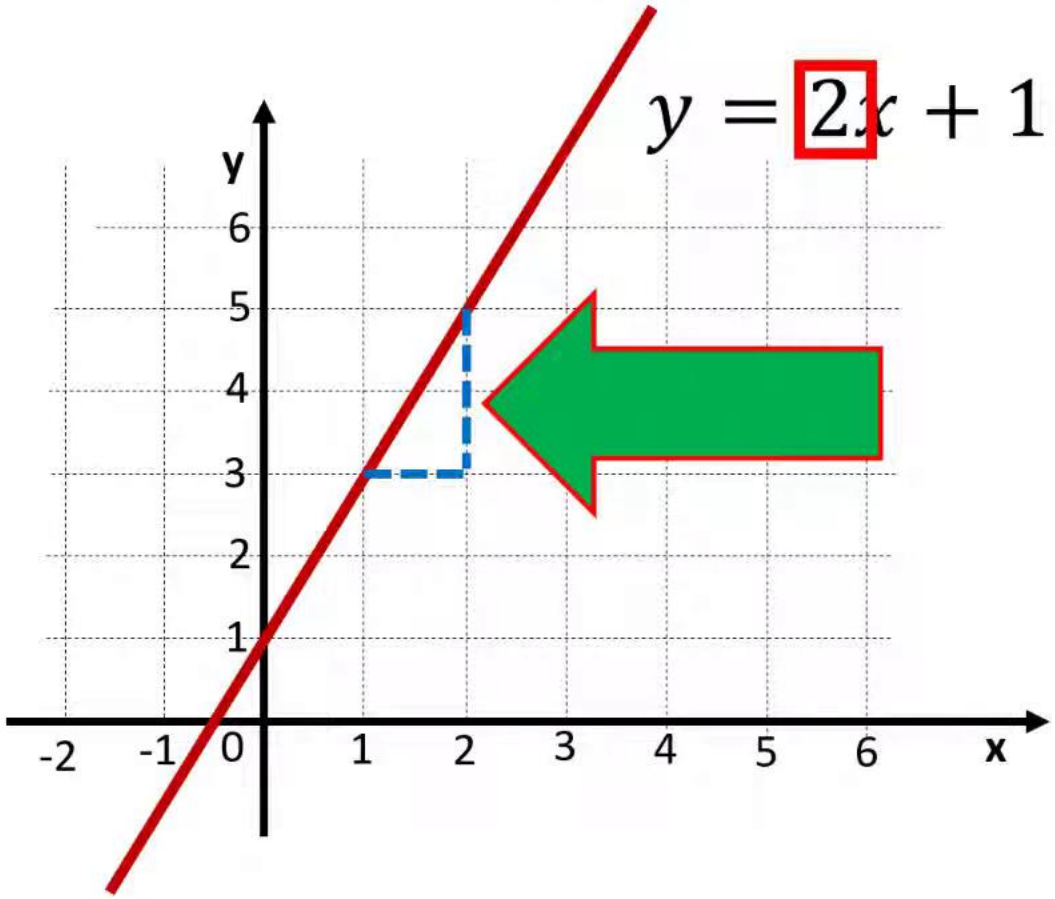
that the line intercepts the y-axis at one,

The equation of a straight line



and this number represents the slope of the line,

The equation of a straight line



which tells us that if x increases by one unit, y increases by two.

SVM – the math

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{array}{l} \geq 1 \text{ if } Y_i = +1 \\ \leq -1 \text{ if } Y_i = -1 \end{array}$$

To really understand support vector machines, you need to understand this kind of math. At the end of this lecture, I hope that you get the main ideas behind its math. Once you know this, it will be a lot easier to read more advanced stuff about SVM.

The equation of a straight line

$$y = mx + b$$

We will therefore see how to go from the simple equation of a straight line,

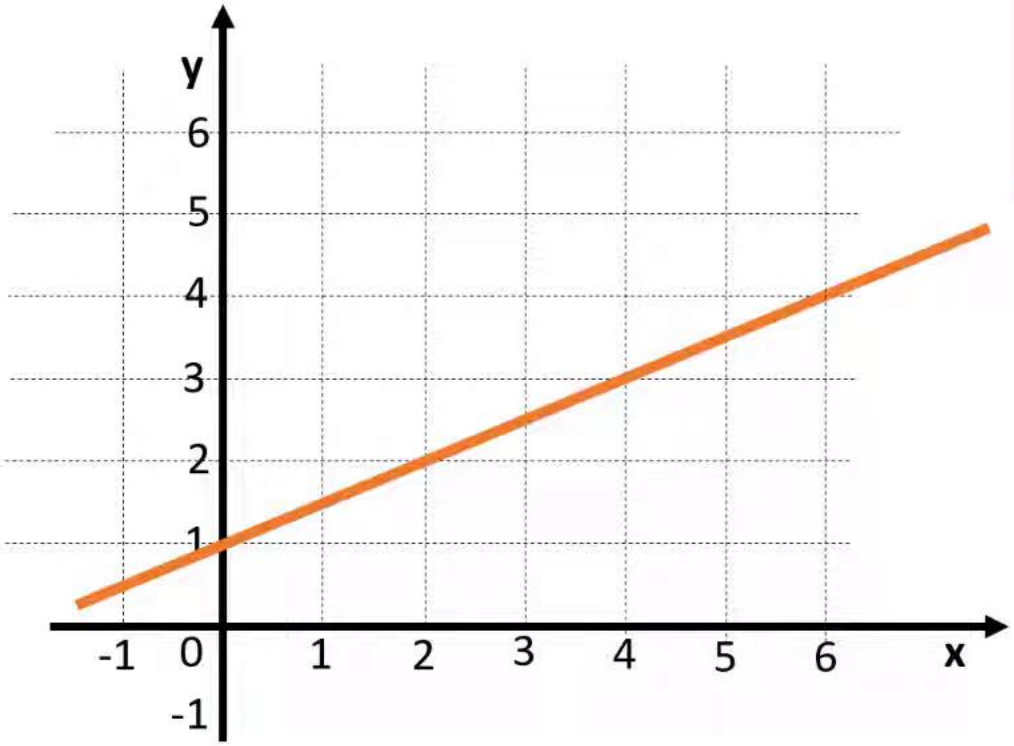
The equation of a straight line

$$y = mx + b$$

$$w^T x + b = 0$$

to this type of equation that is used in SVM, and then see how such an equation can be used for classification.

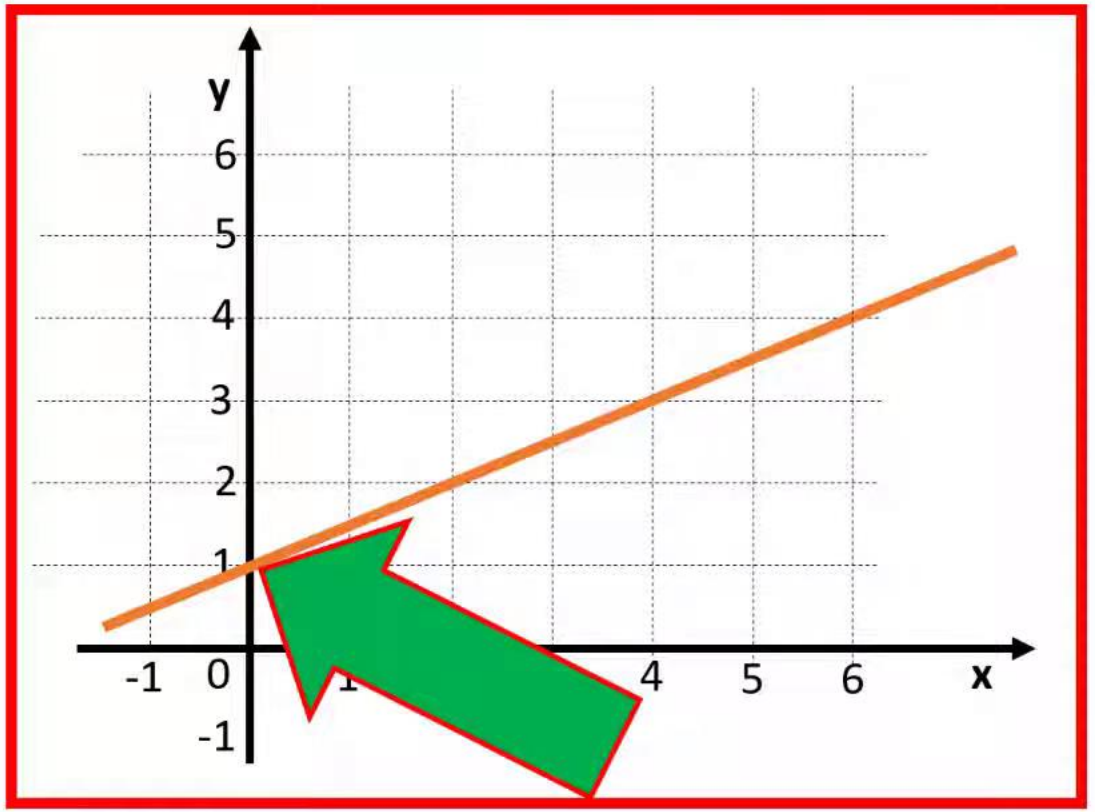
The equation of a straight line



$$y = mx + b$$

Usually, we express the equation of a straight line like this, which is called the slope-intercept form because the equation contains information about the slope and the intercept of the line.

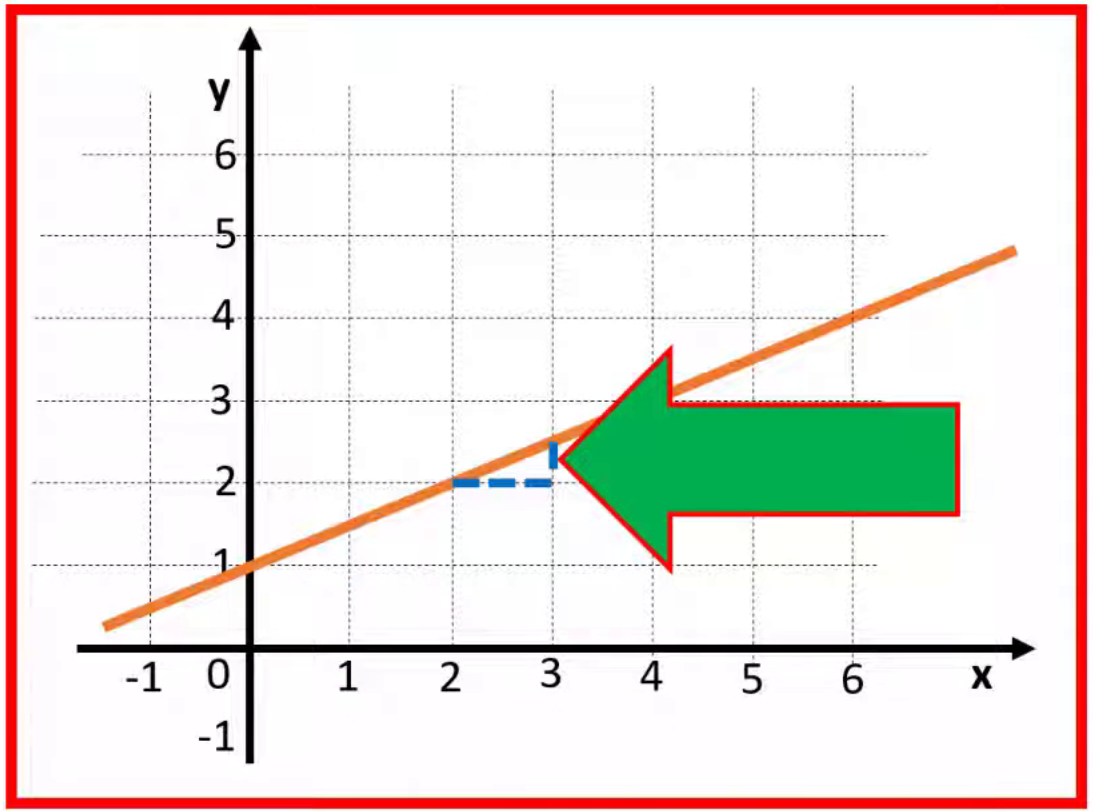
The equation of a straight line



$$y = mx + b$$

Since this line intercepts the y-axis at one,

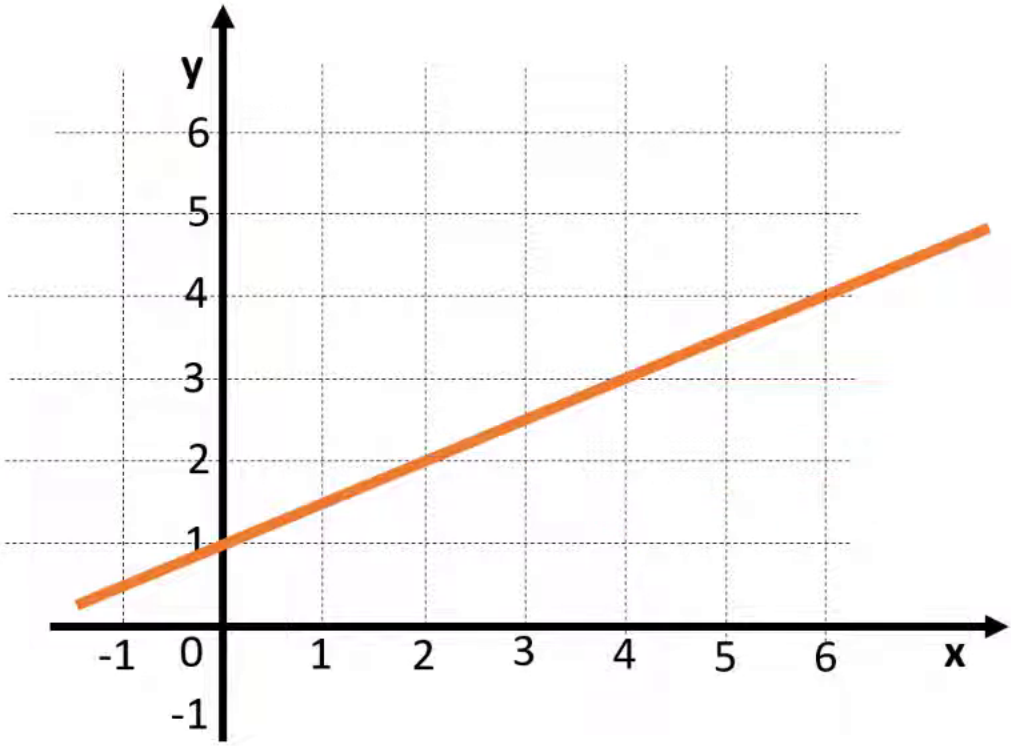
The equation of a straight line



$$y = mx + b$$

and has a slope of 0.5,

The equation of a straight line

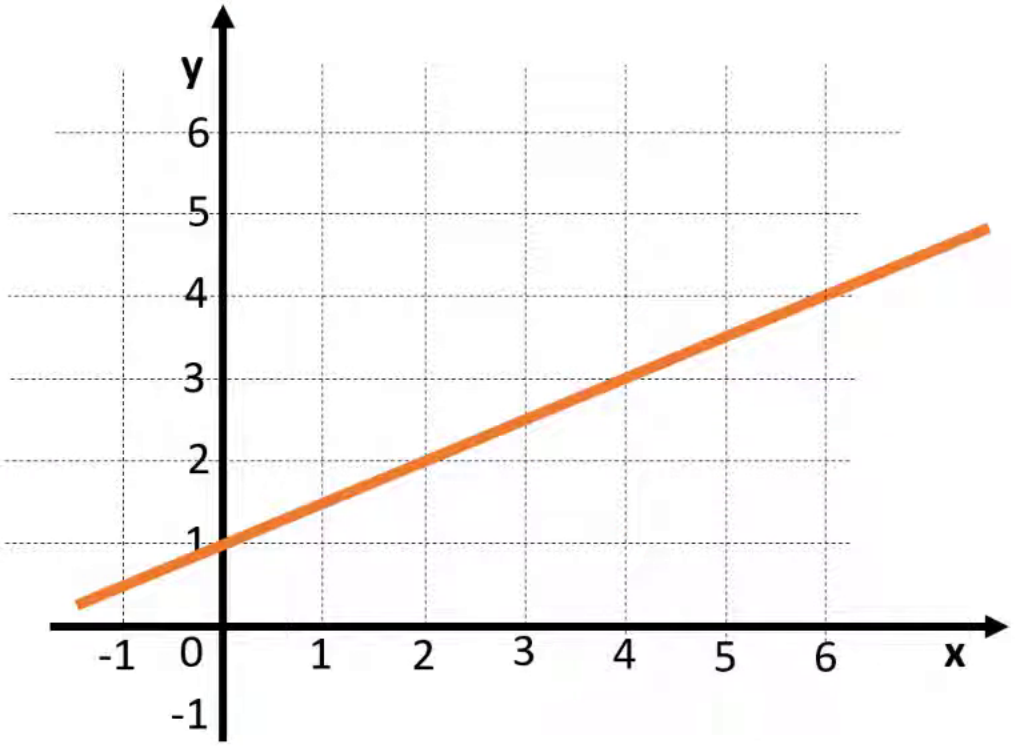


$$y = mx + b$$

$$y = 0.5x + 1$$

m is here equal to 0.5,

The equation of a straight line

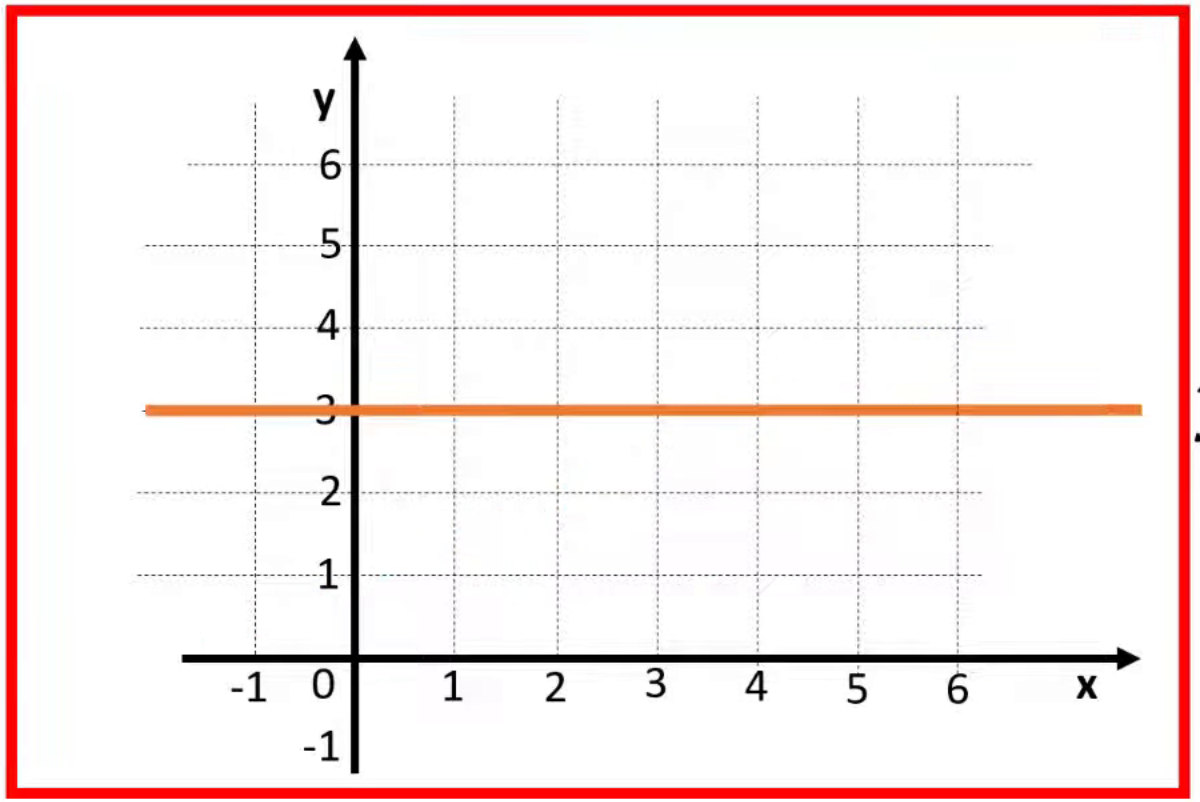


$$y = mx + b$$

$$y = 0.5x + 1$$

and b is equal to one.

The equation of a straight line

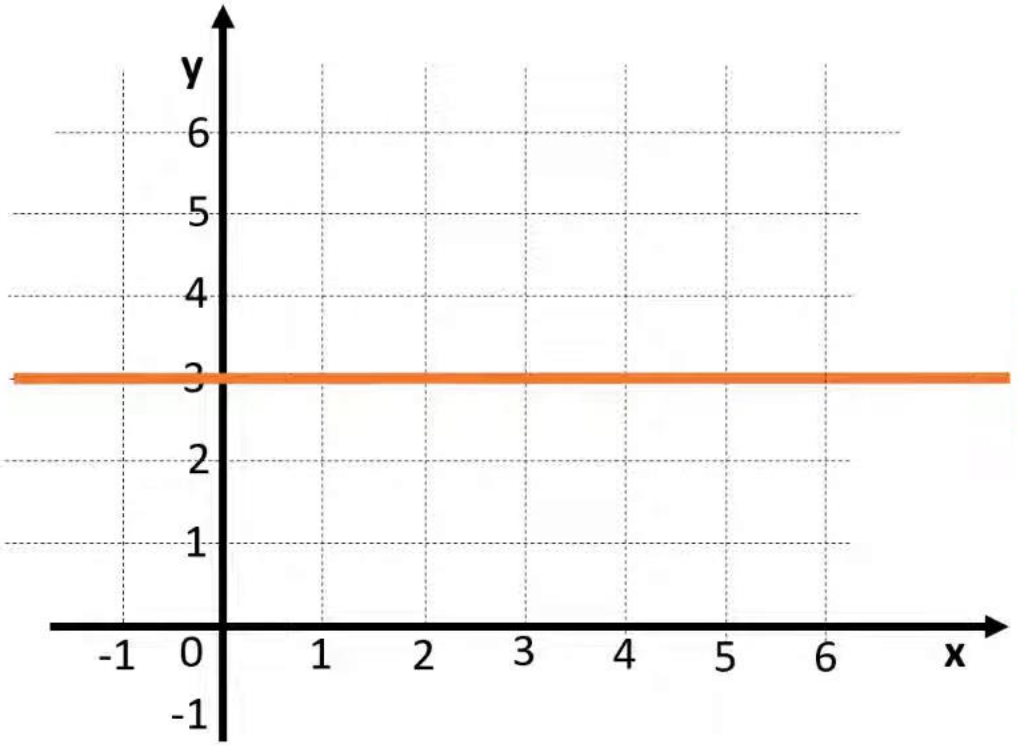


$$y = mx + b$$

$$y = 3$$

For example, the following horizontal line intercepts the y-axis at three and has a slope of zero.

The equation of a straight line

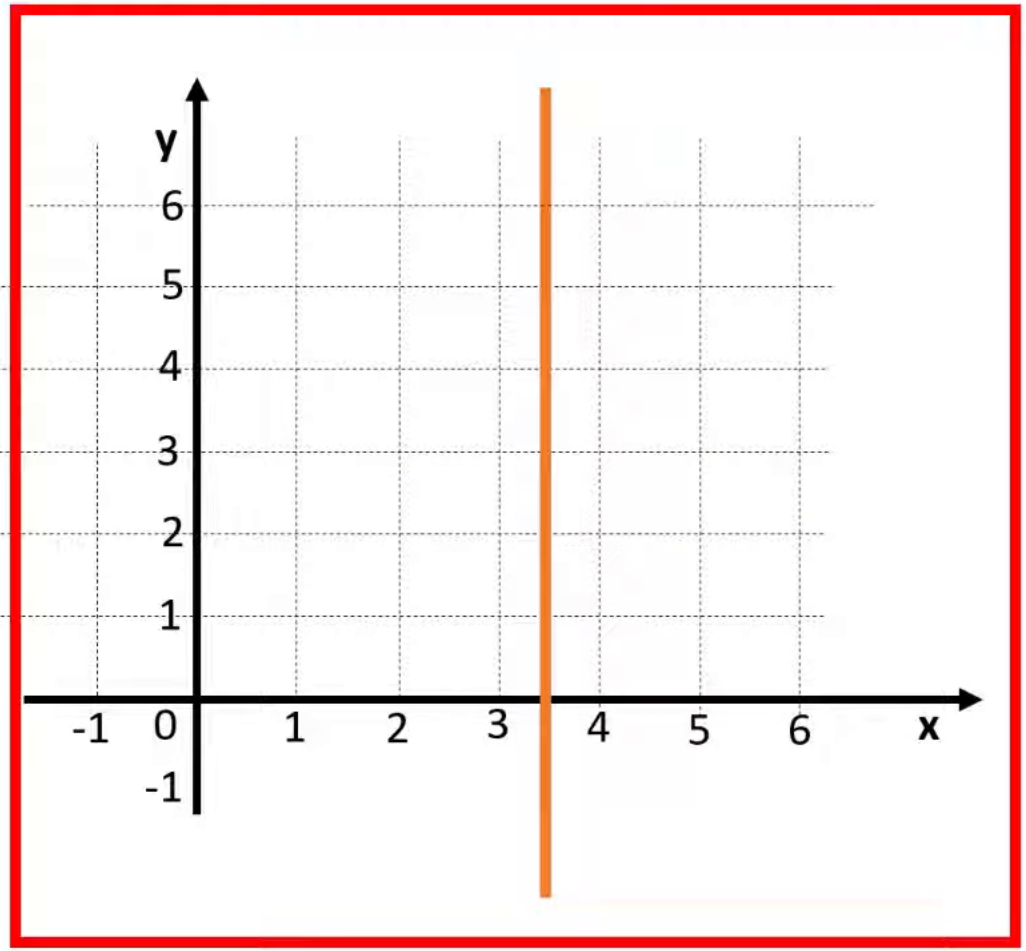


$$y = mx + b$$

$$y = 3$$

Since m is equal to zero, the equation is y equal to 3.

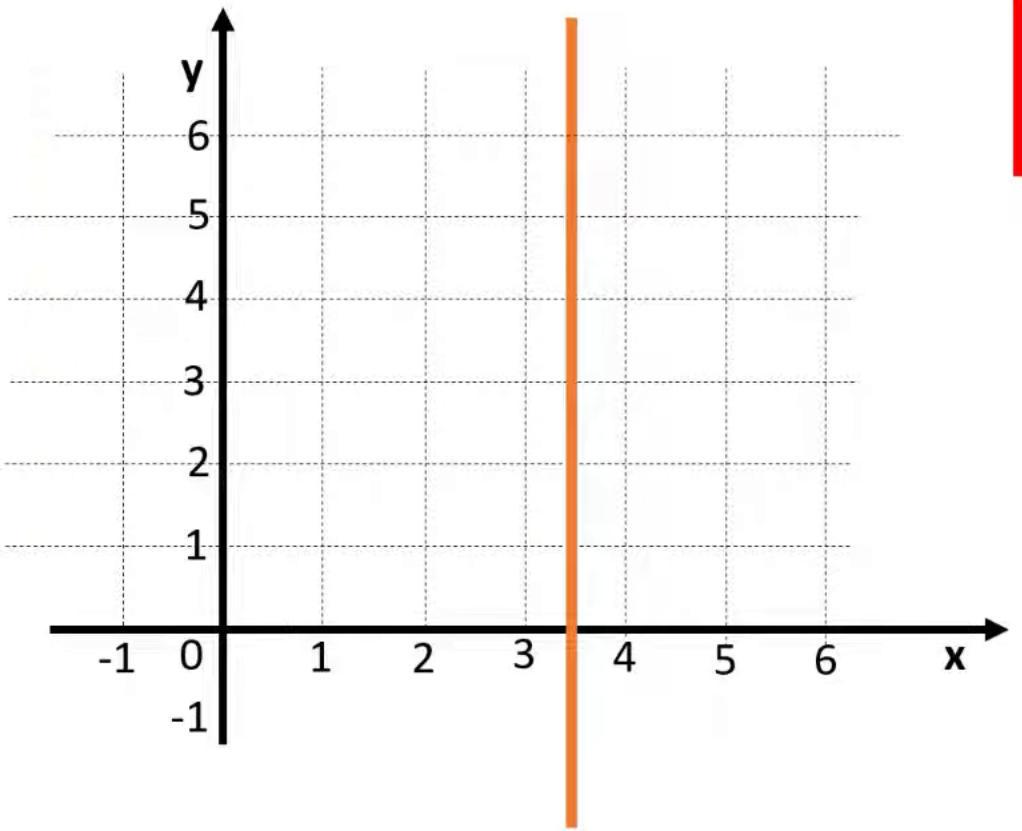
The equation of a straight line



$$y = mx + b$$

However, what is the equation of the following vertical line?

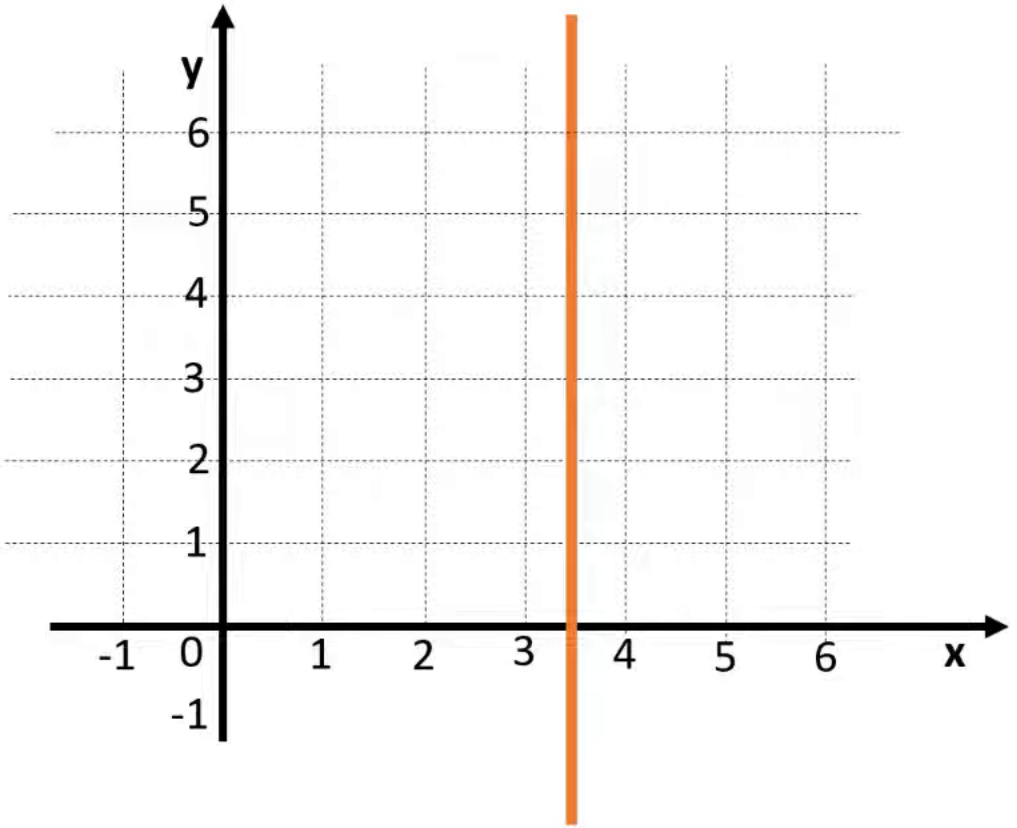
The equation of a straight line



$$y = mx + b$$

It turns out that this form of the equation cannot be used to describe a vertical line,

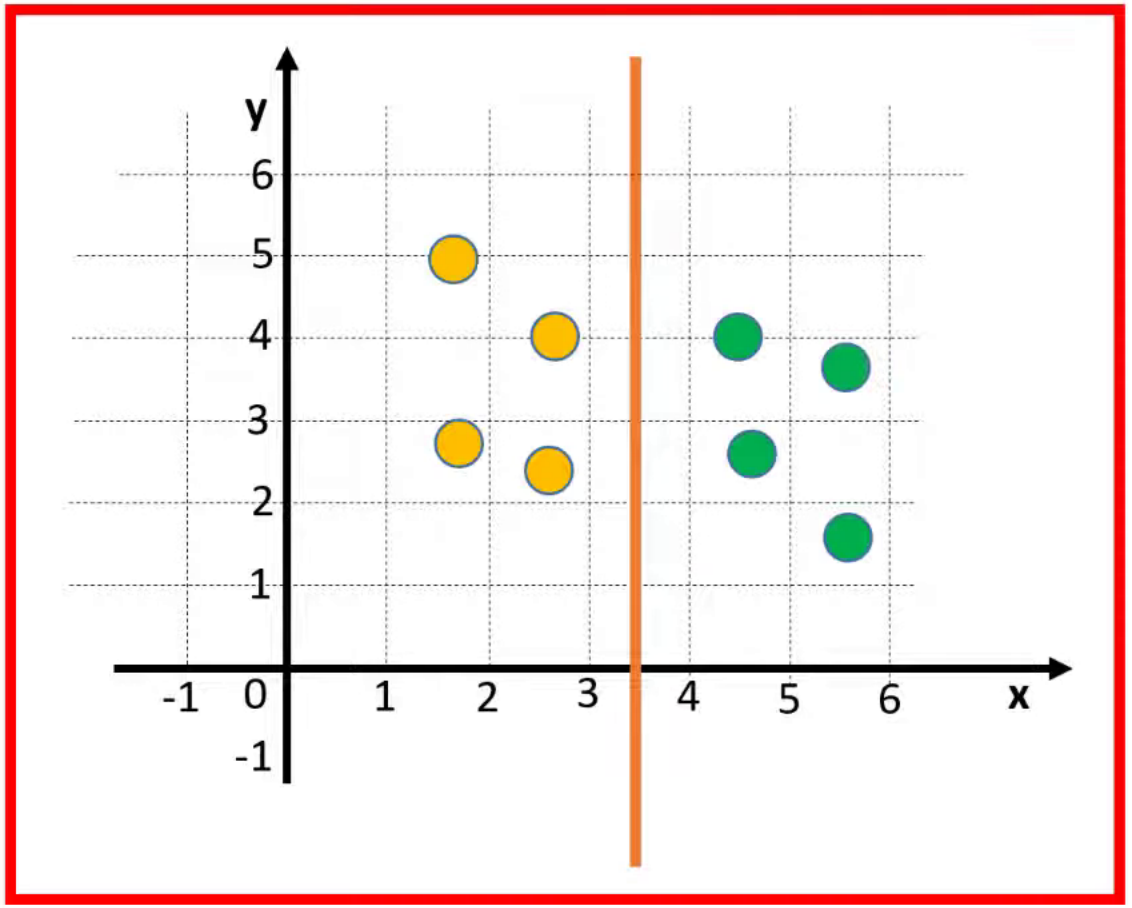
The equation of a straight line



$$y = mx + b$$

because the slope would then be infinitely large.

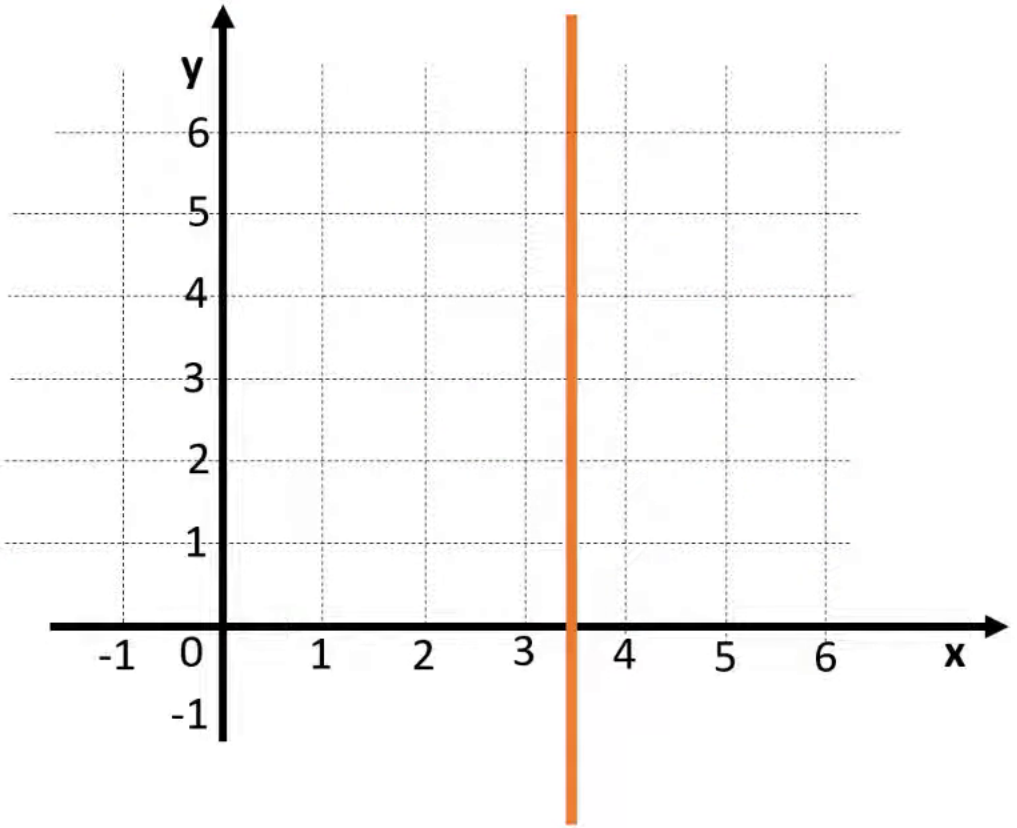
The equation of a straight line



$$y = mx + b$$

Since vertical lines can be used to separate data points of two groups, the slope-intercept form to describe the line is not appropriate to use in SVM.

The equation of a straight line

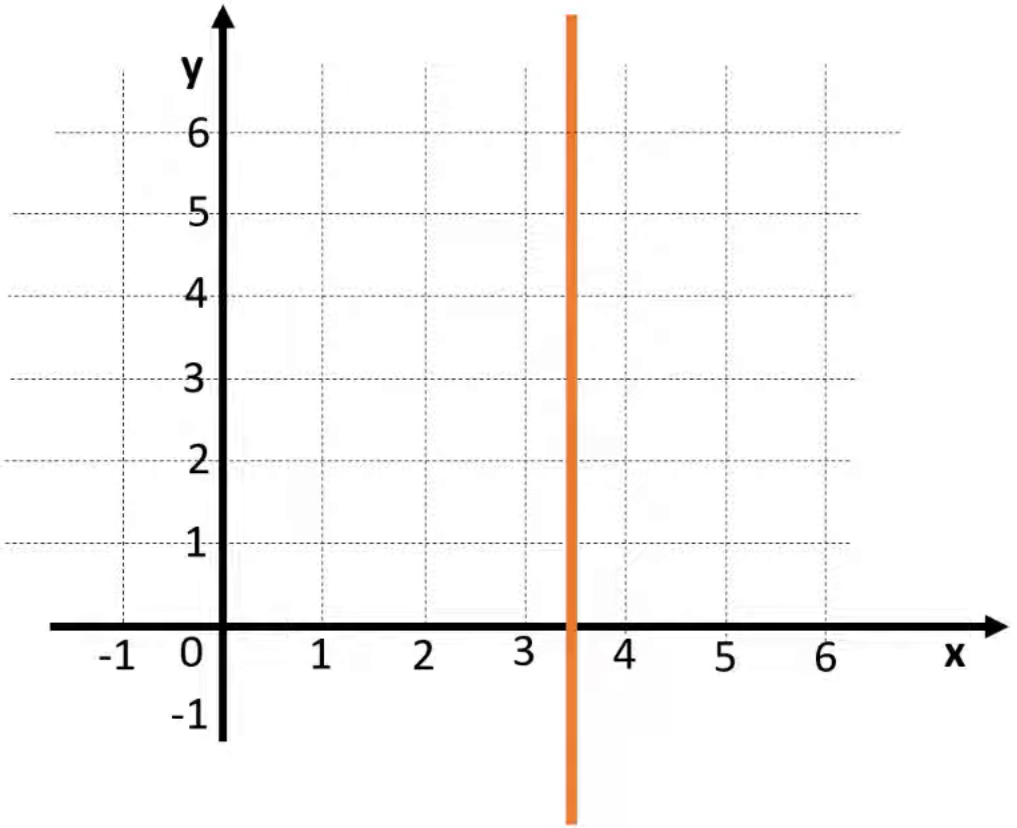


$$y = mx + b$$

$$Ax + By + C = 0$$

This is one reason why we need to use the so-called “general form” of the equation for a straight line.

The equation of a straight line

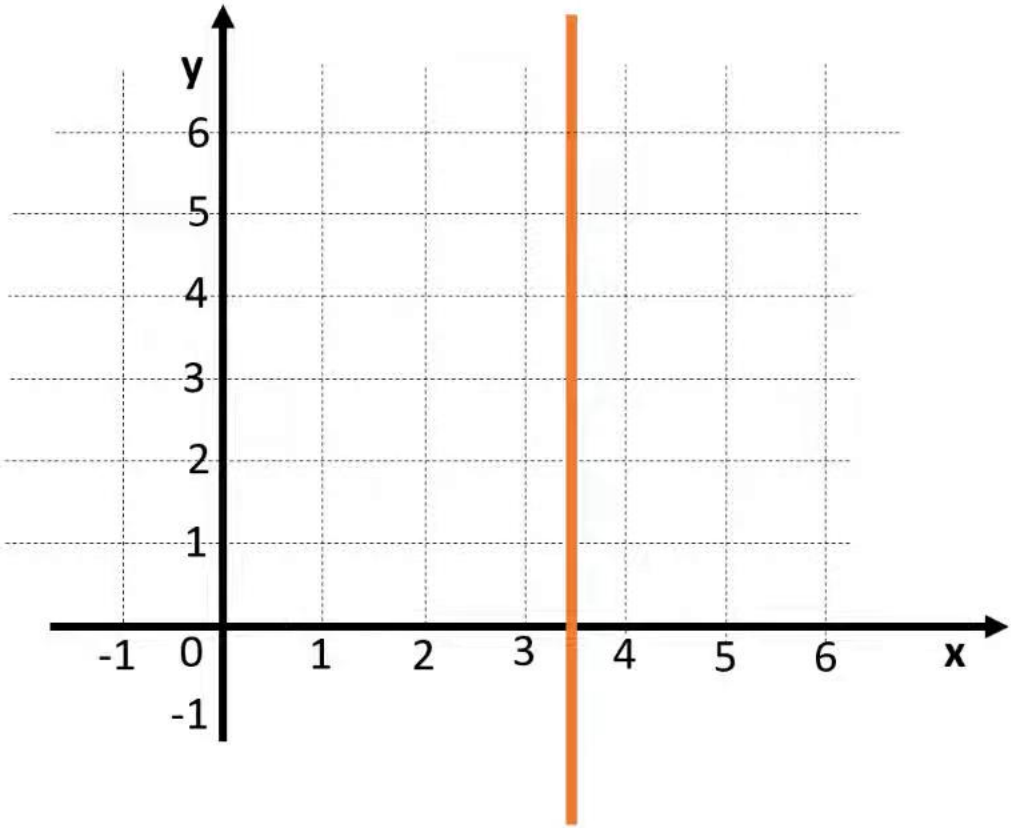


$$y = mx + b$$

$$Ax + By + C = 0$$

If we move these two terms to the right-hand side,

The equation of a straight line

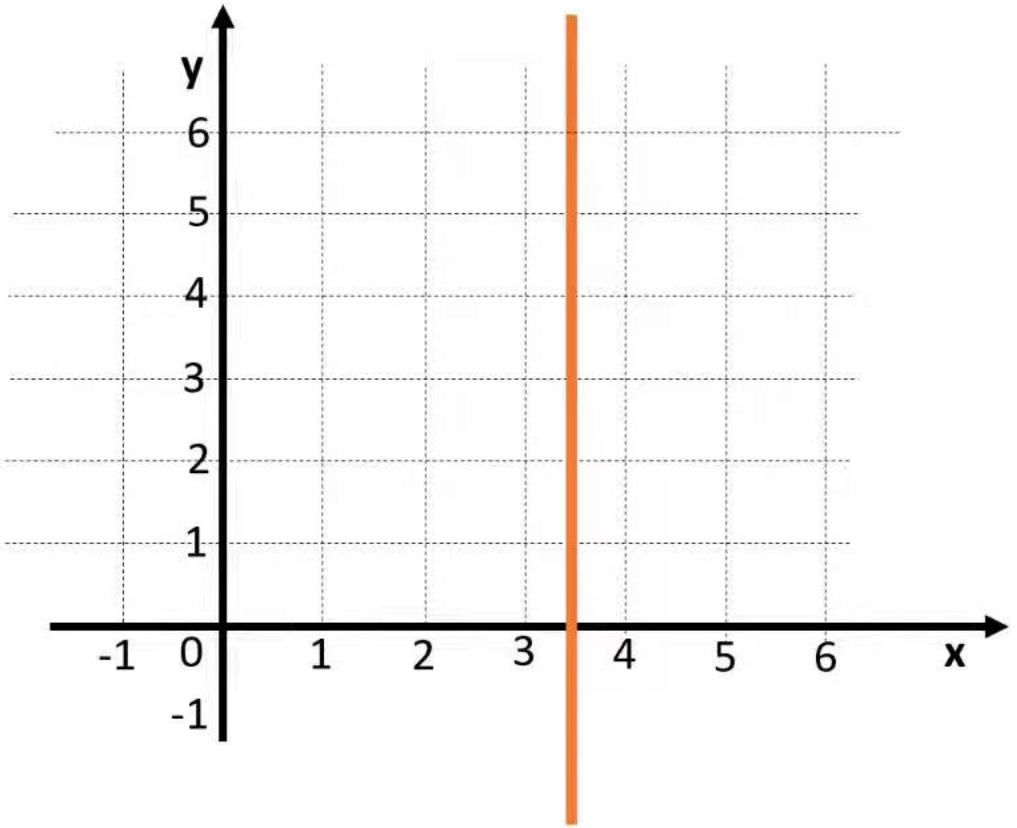


$$y = mx + b$$

$$Ax + By + C = 0$$

and divide by B,

The equation of a straight line



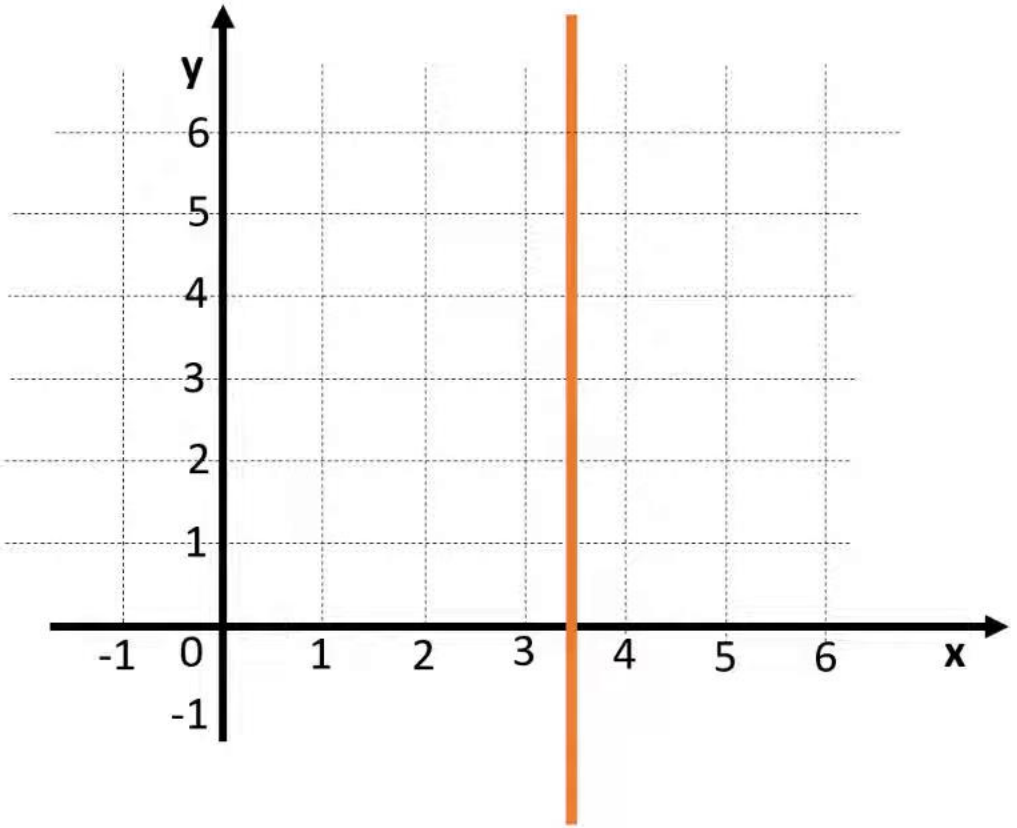
$$y = mx + b$$

$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

we will get the following equation,

The equation of a straight line



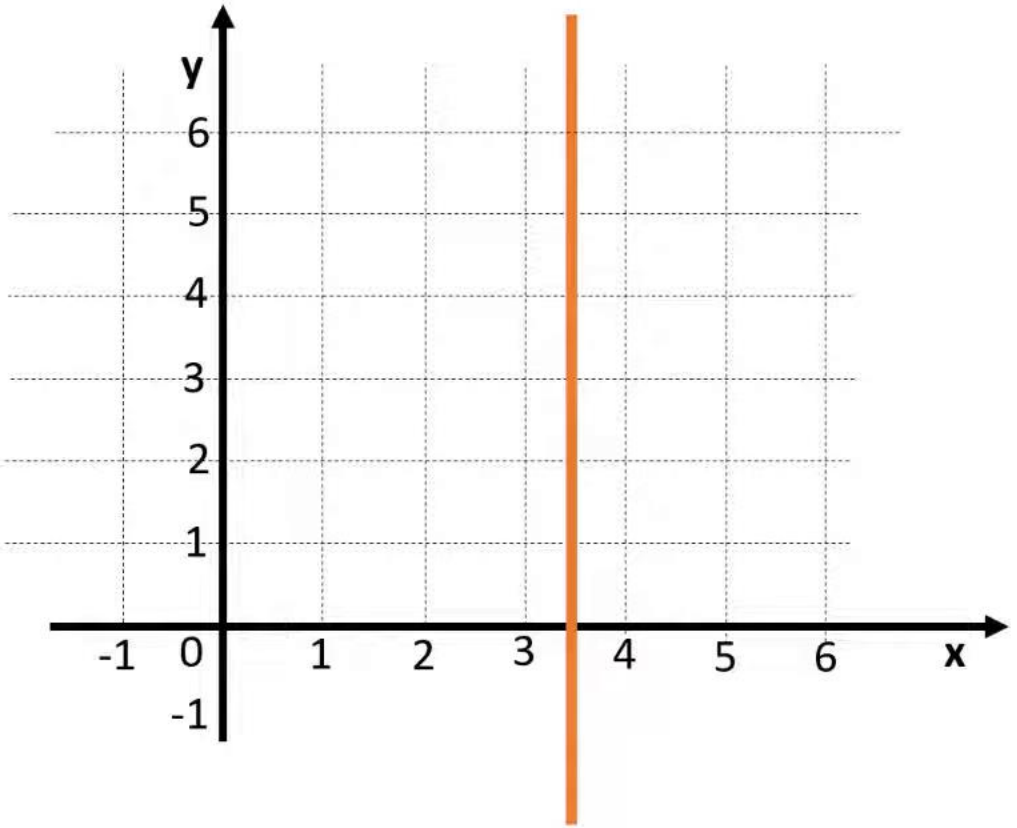
$$y = mx + b$$

$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

where this is the slope of the line,

The equation of a straight line



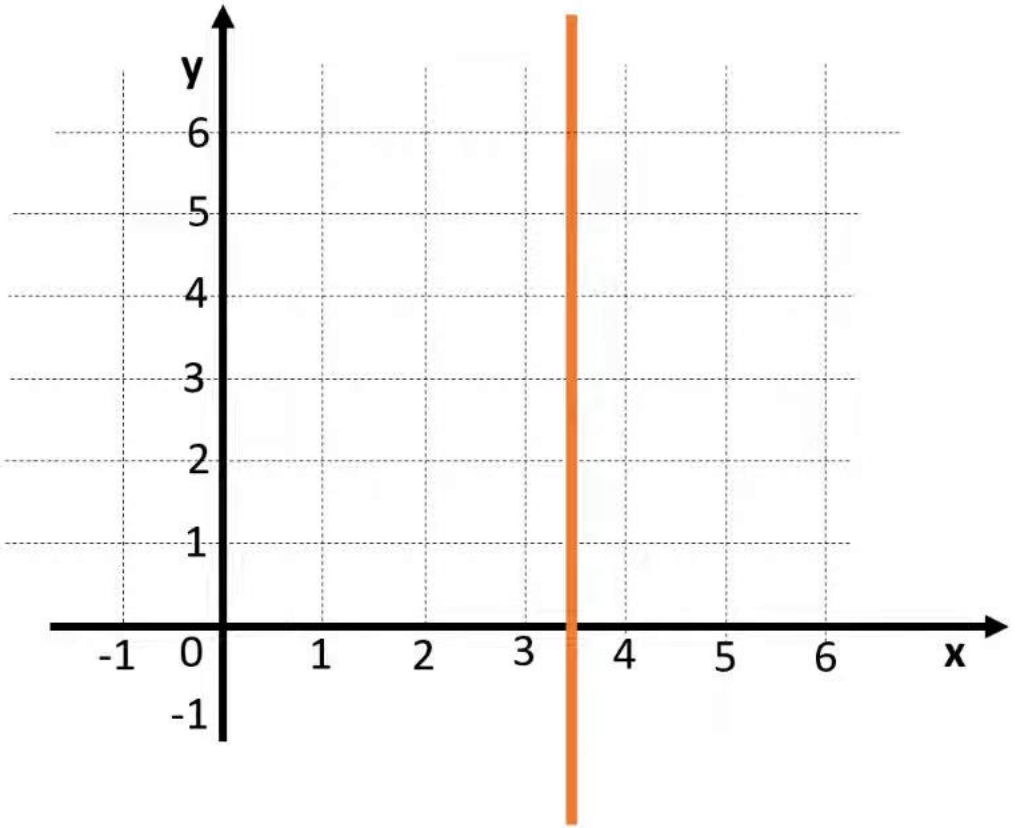
$$y = mx + b$$

$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

and this is the intercept.

The equation of a straight line



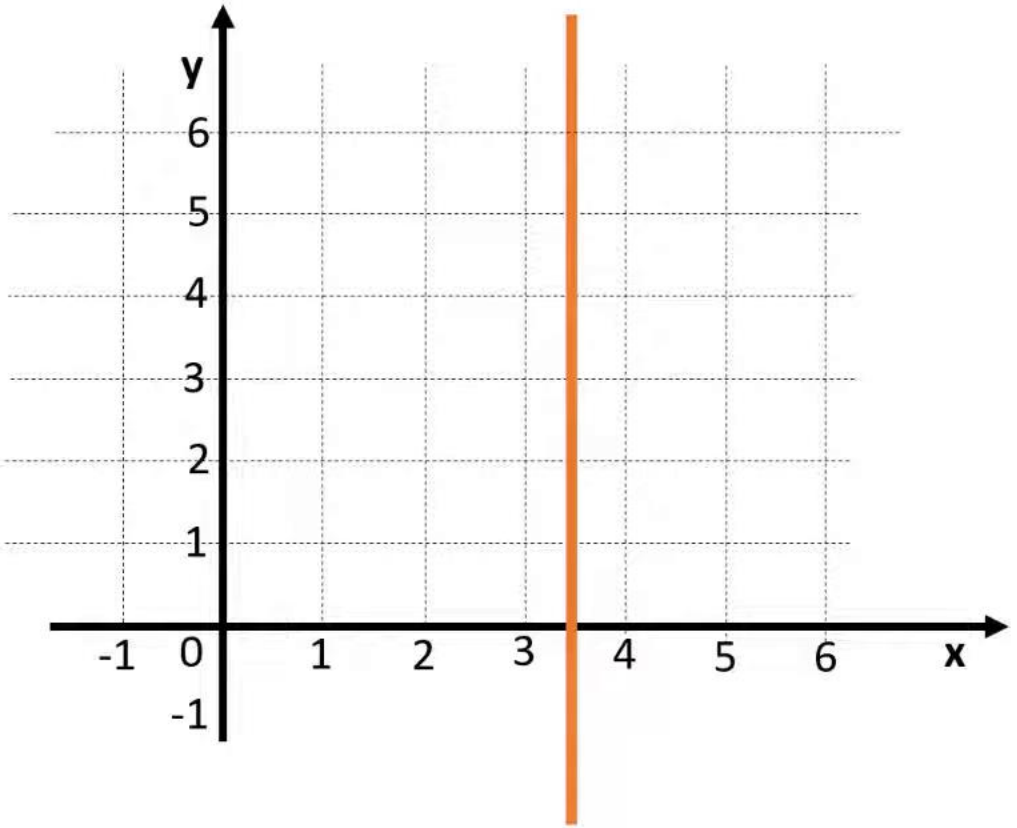
$$y = mx + b$$

$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

We can see that the coefficient A affects the slope,

The equation of a straight line



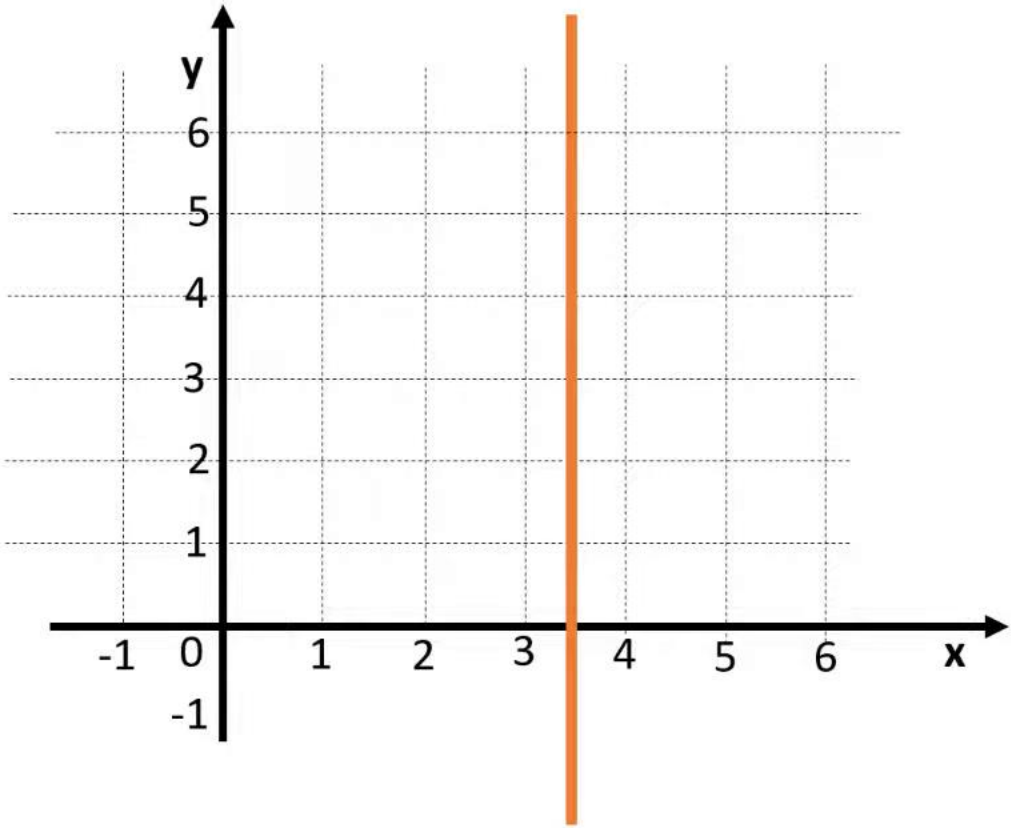
$$y = mx + b$$

$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

and the constant C affects the intercept,

The equation of a straight line



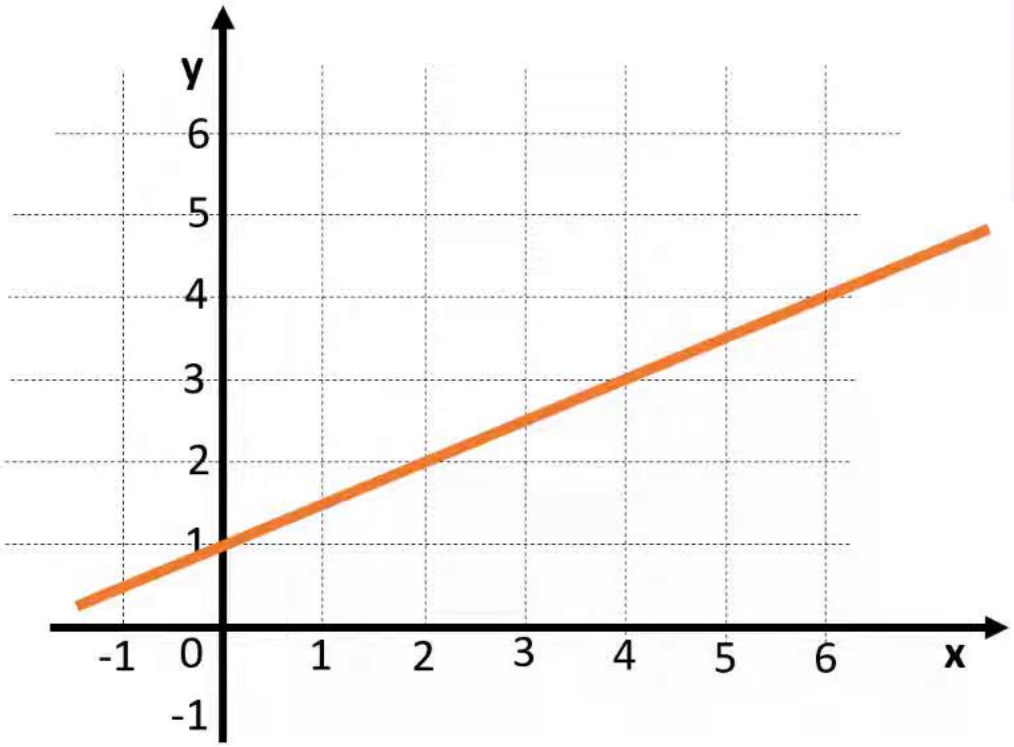
$$y = mx + b$$

$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

whereas the coefficient B affects both the intercept and the slope.

The equation of a straight line



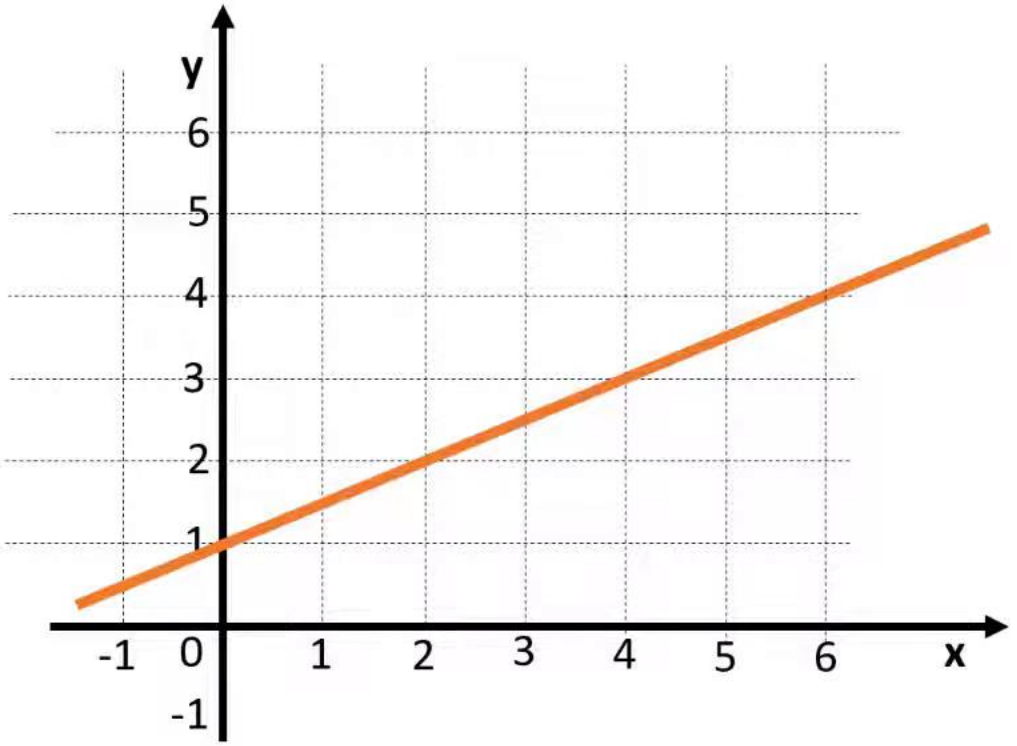
$$y = 0.5x + 1$$

$$y = mx + b$$

$$Ax + By + C = 0$$

Suppose that we have the following line with a slope of 0.5 and an intercept of one.

The equation of a straight line



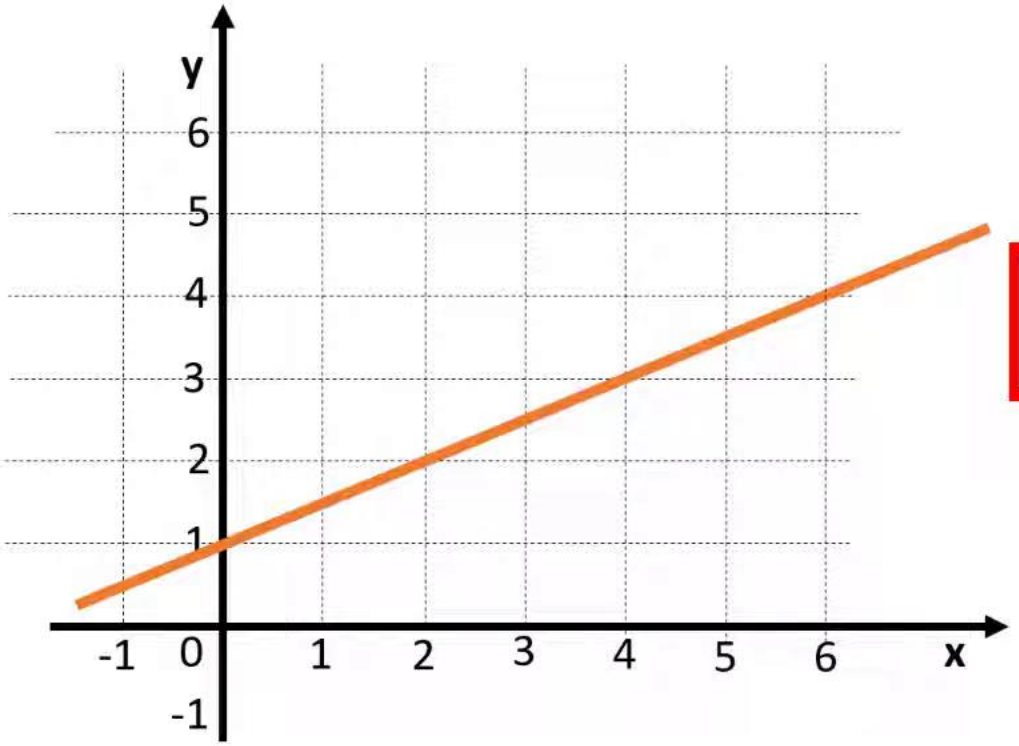
$$y = 0.5x + 1$$

$$y = mx + b$$

$$Ax + By + C = 0$$

We will now convert this equation to the general form.

The equation of a straight line



$$y = 0.5x + 1$$

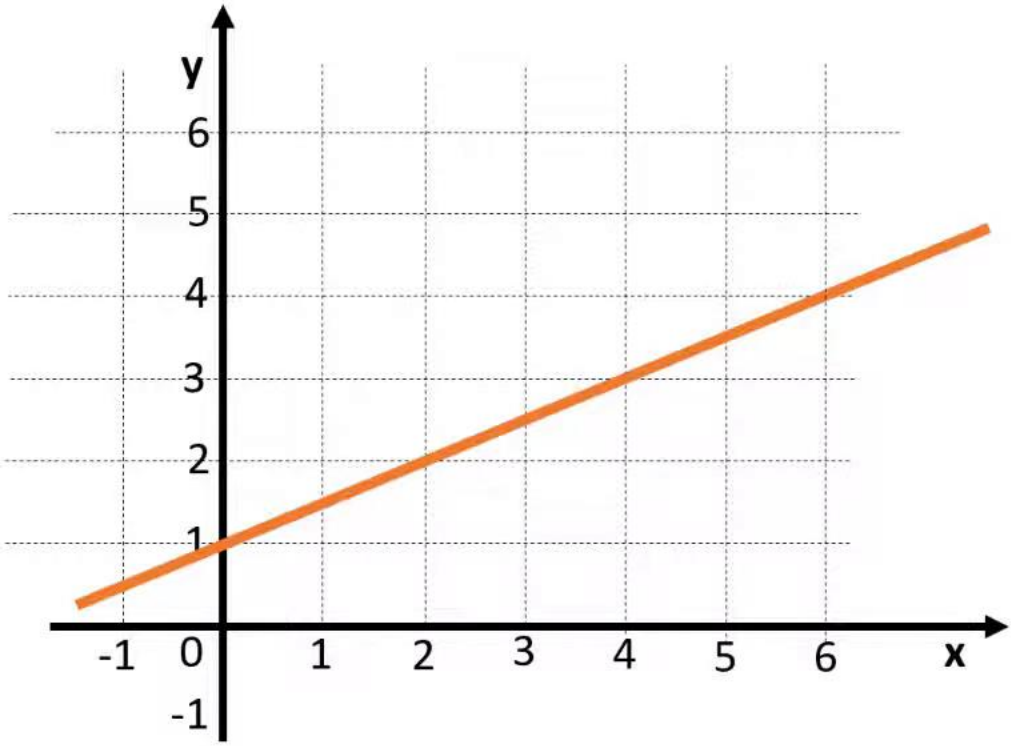
$$y - 0.5x - 1 = 0$$

$$y = mx + b$$

$$Ax + By + C = 0$$

We therefore move all terms to the left-hand side like this.

The equation of a straight line



$$y = 0.5x + 1$$

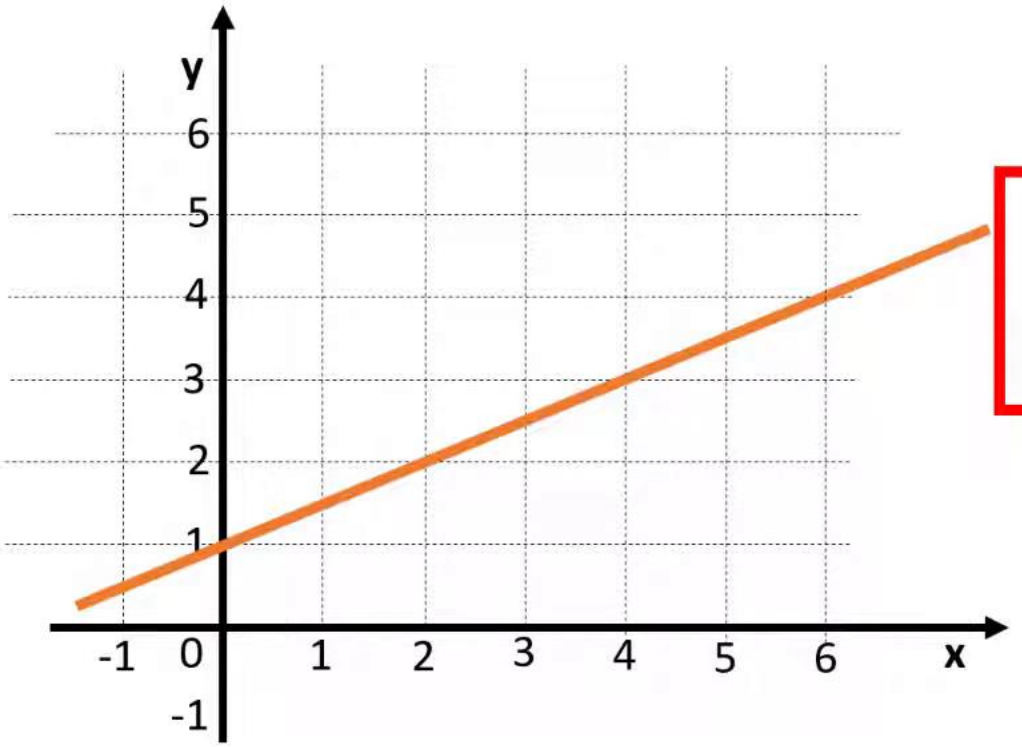
$$y - 0.5x - 1 = 0$$

$$y = mx + b$$

$$Ax + By + C = 0$$

To be consistent with this form,

The equation of a straight line



$$y = 0.5x + 1$$

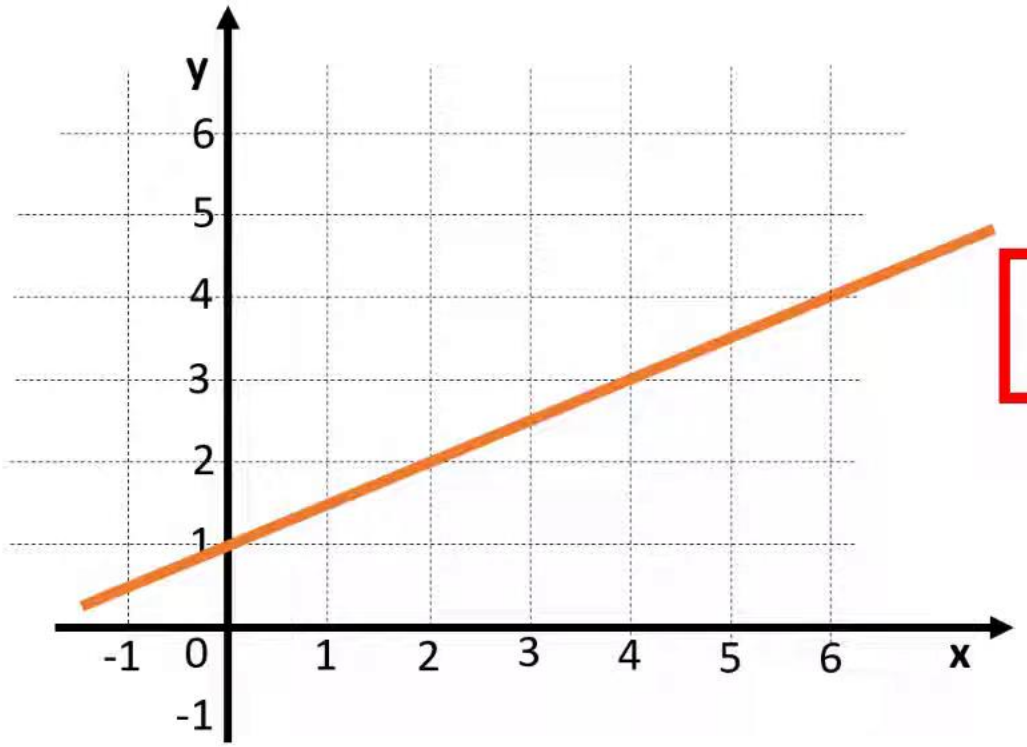
$$y - 0.5x - 1 = 0$$

$$y = mx + b$$

$$Ax + By + C = 0$$

we swap places of these two terms,

The equation of a straight line



$$y = 0.5x + 1$$

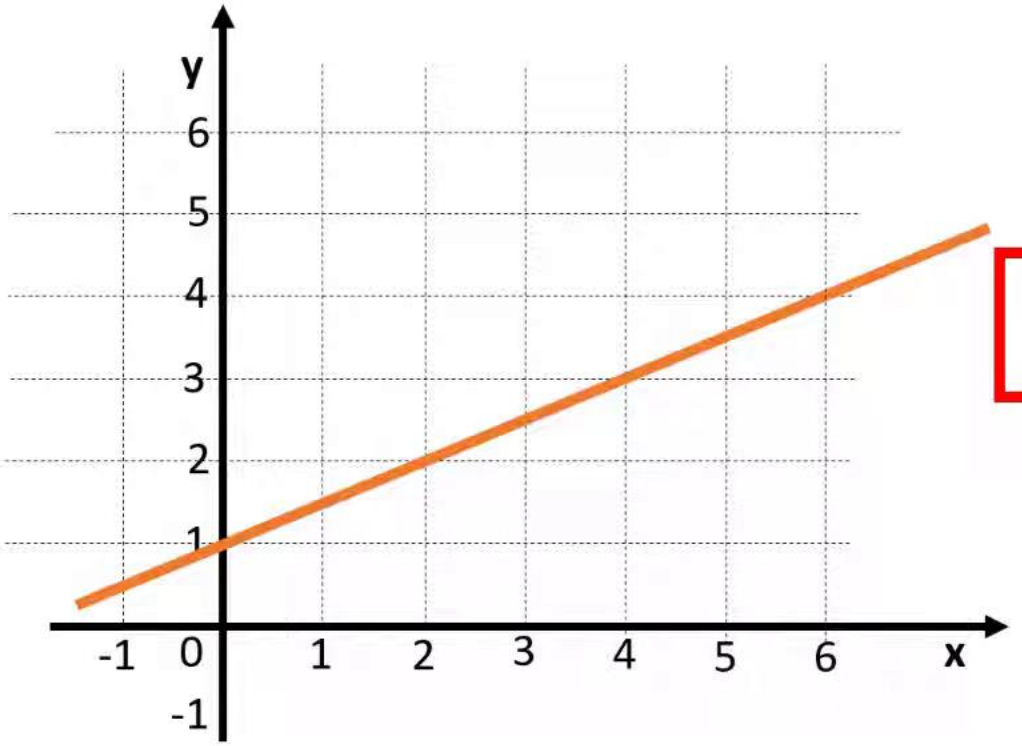
$$-0.5x + y - 1 = 0$$

$$y = mx + b$$

$$Ax + By + C = 0$$

so that we have the following equation that also describes the line.

The equation of a straight line



$$y = 0.5x + 1$$

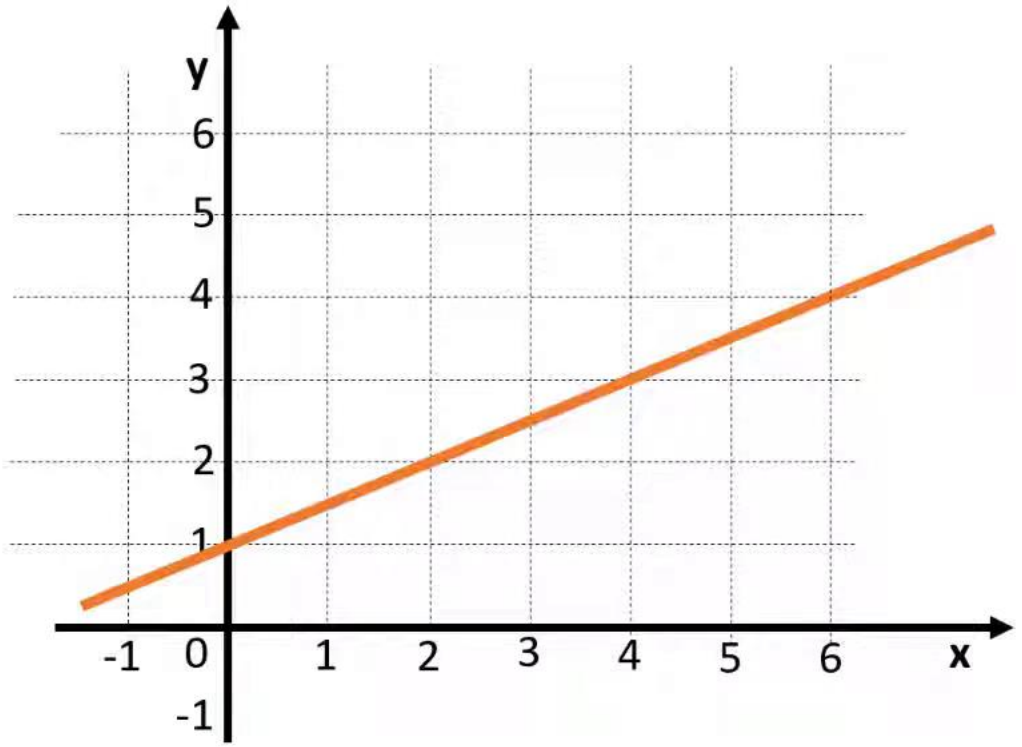
$$-0.5x + y - 1 = 0$$

$$y = mx + b$$

$$Ax + By + C = 0$$

so that we have the following equation that also describes the line.

The equation of a straight line



$$y = 0.5x + 1$$

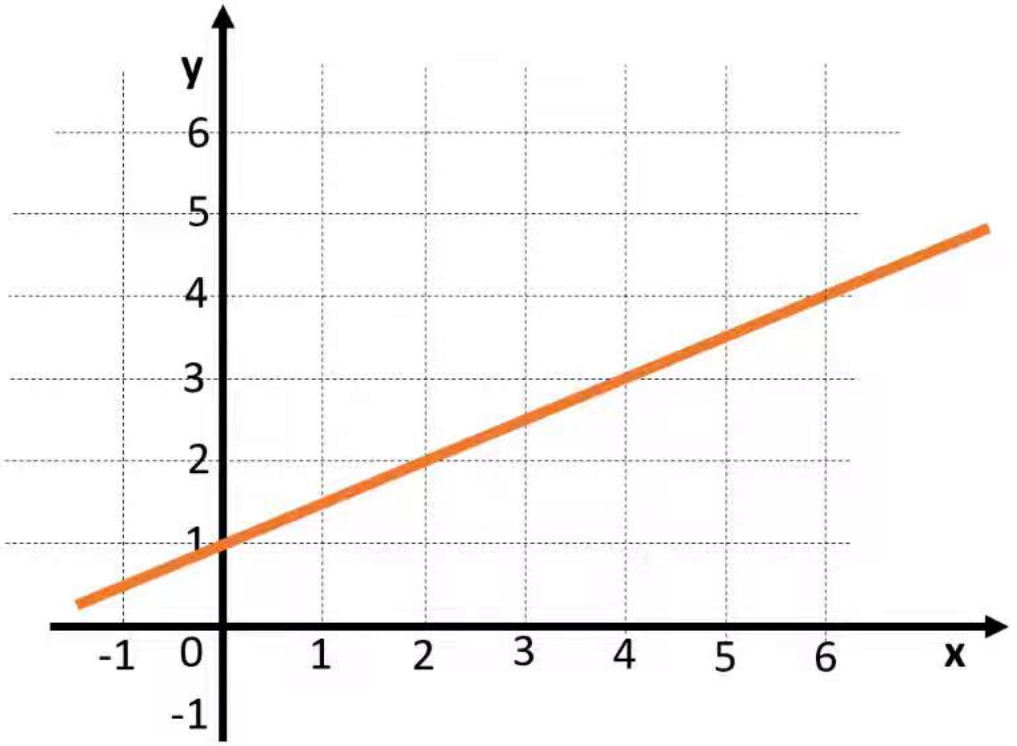
$$-0.5x + y - 1 = 0$$

$y = mx + b$

$Ax + By + C = 0$

The coefficients are usually expressed as integers in this form.

The equation of a straight line



$$y = mx + b$$

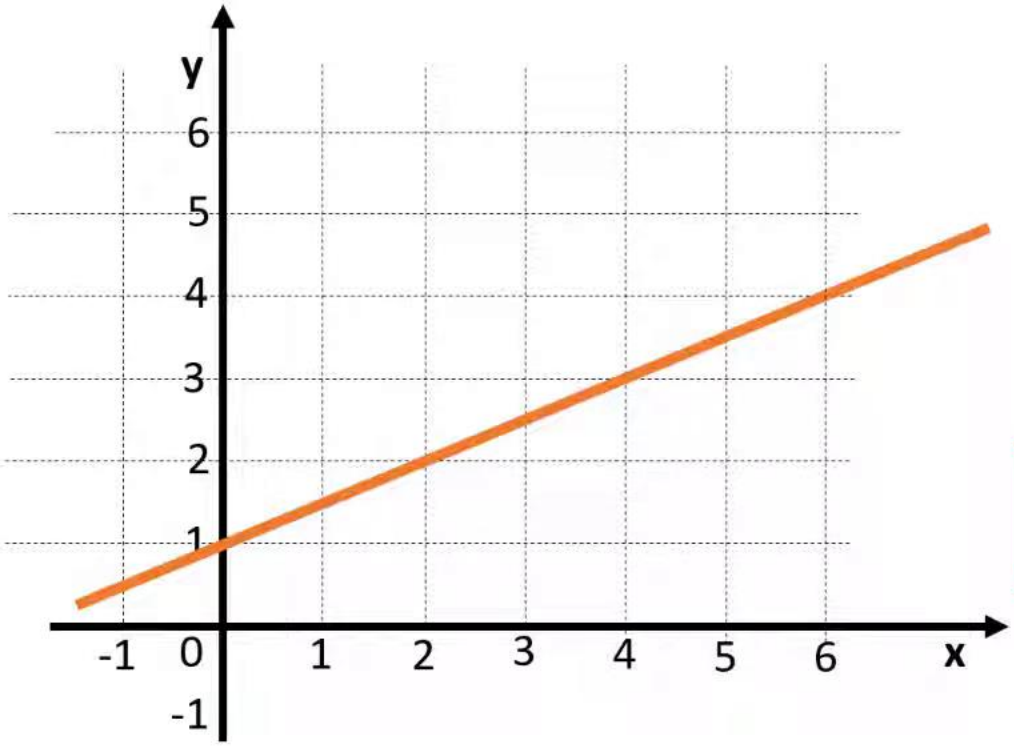
$$Ax + By + C = 0$$

$$y = 0.5x + 1$$

$$-0.5x + y - 1 = 0 \quad \times 4$$

If we, for example, multiply all terms by 4,

The equation of a straight line



$$y = 0.5x + 1$$

$$-0.5x + y - 1 = 0$$

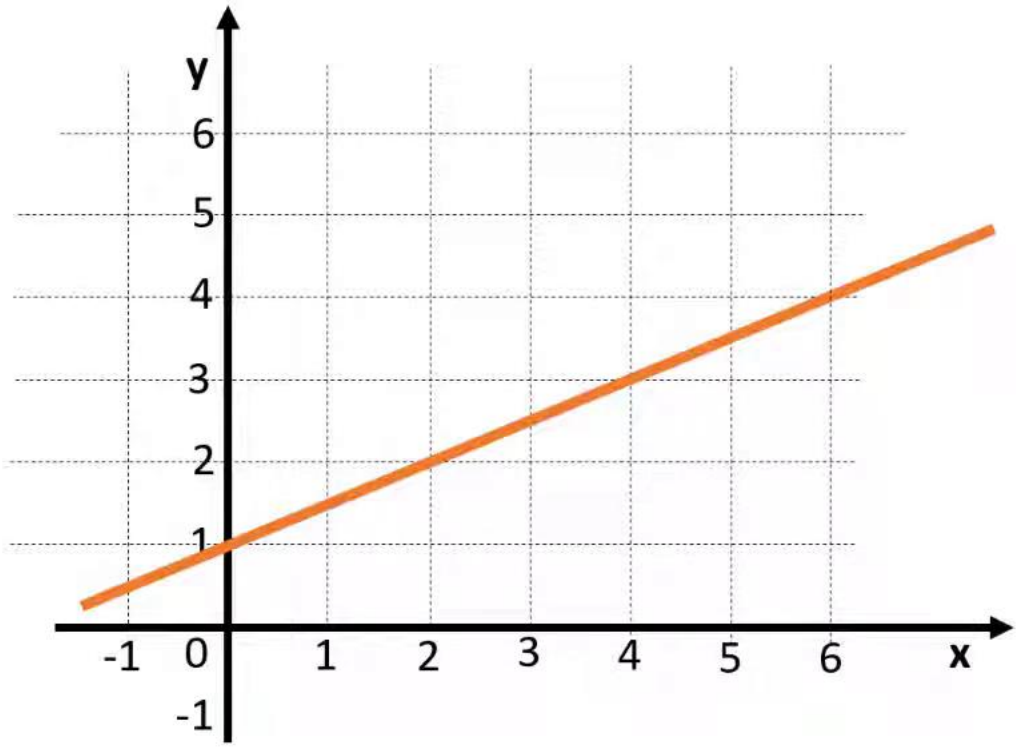
$$-2x + 4y - 4 = 0$$

$$y = mx + b$$

$$Ax + By + C = 0$$

we will get the following equation that also describes the same line,

The equation of a straight line



$$y = 0.5x + 1$$

$$-0.5x + y - 1 = 0$$

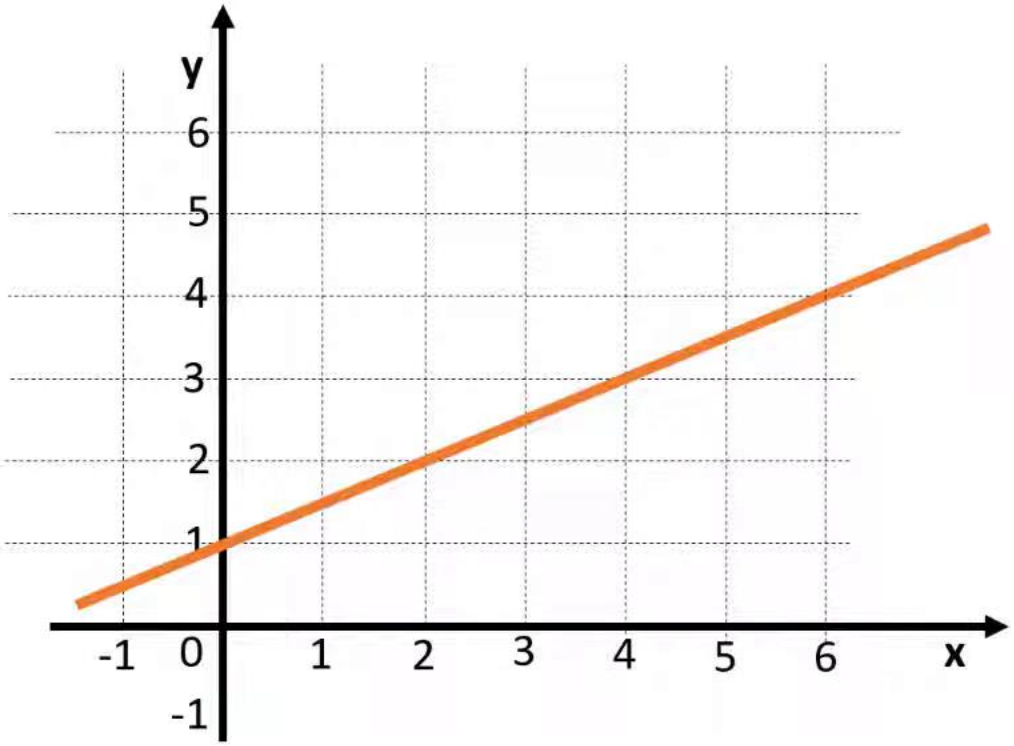
$$-2x + 4y - 4 = 0$$

$$y = mx + b$$

$$Ax + By + C = 0$$

because if we solve this equation for y,

The equation of a straight line



$$y = 0.5x + 1$$

$$y = mx + b$$

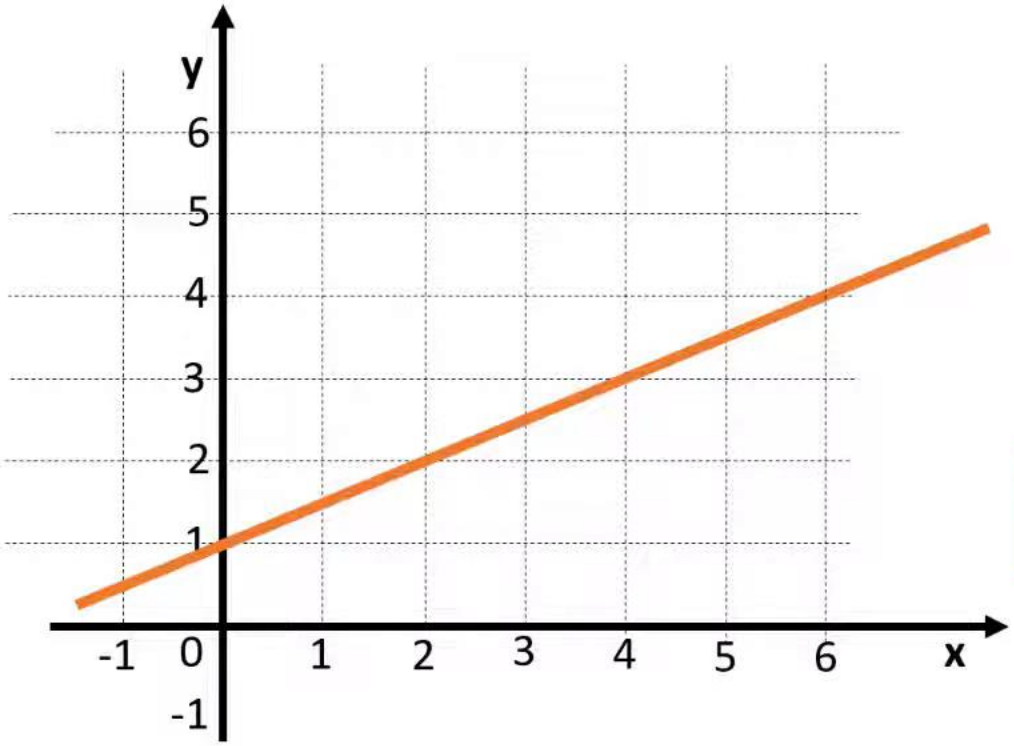
$$Ax + By + C = 0$$

$$-0.5x + y - 1 = 0$$

$$-2x + 4y - 4 = 0$$

we will end up with the following equation.

The equation of a straight line



$$y = mx + b$$

$$Ax + By + C = 0$$

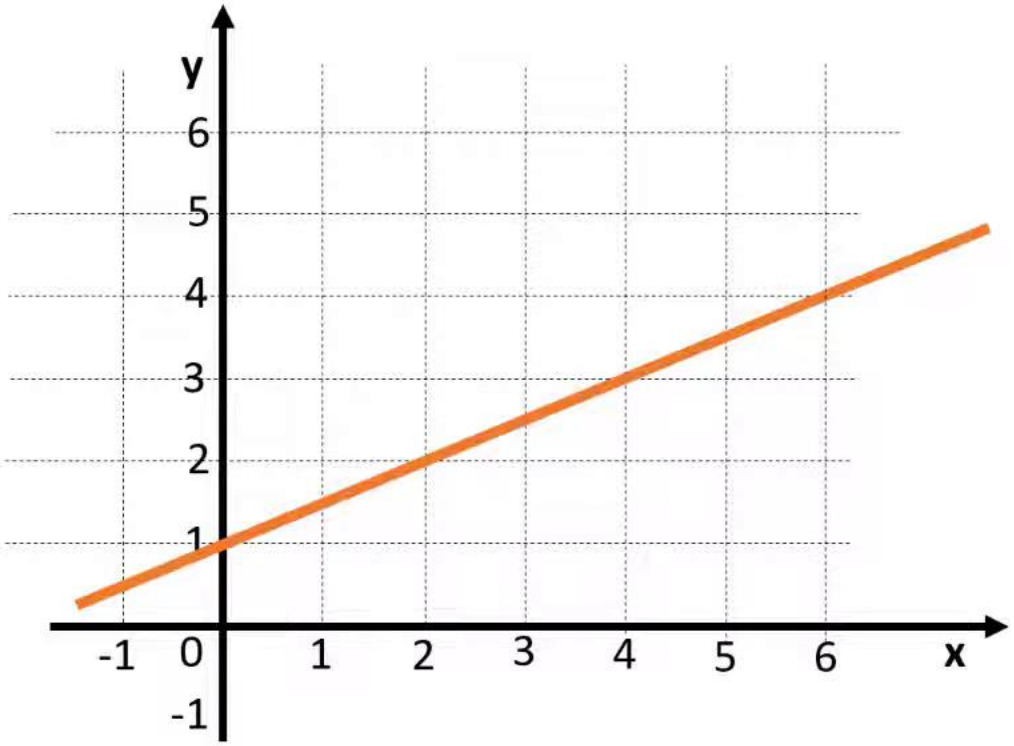
$$y = 0.5x + 1$$

$$-0.5x + y - 1 = 0$$

$$-2x + 4y - 4 = 0$$

In the general form, one should set the coefficient A to be a positive number,

The equation of a straight line



$$y = mx + b$$
$$Ax + By + C = 0$$

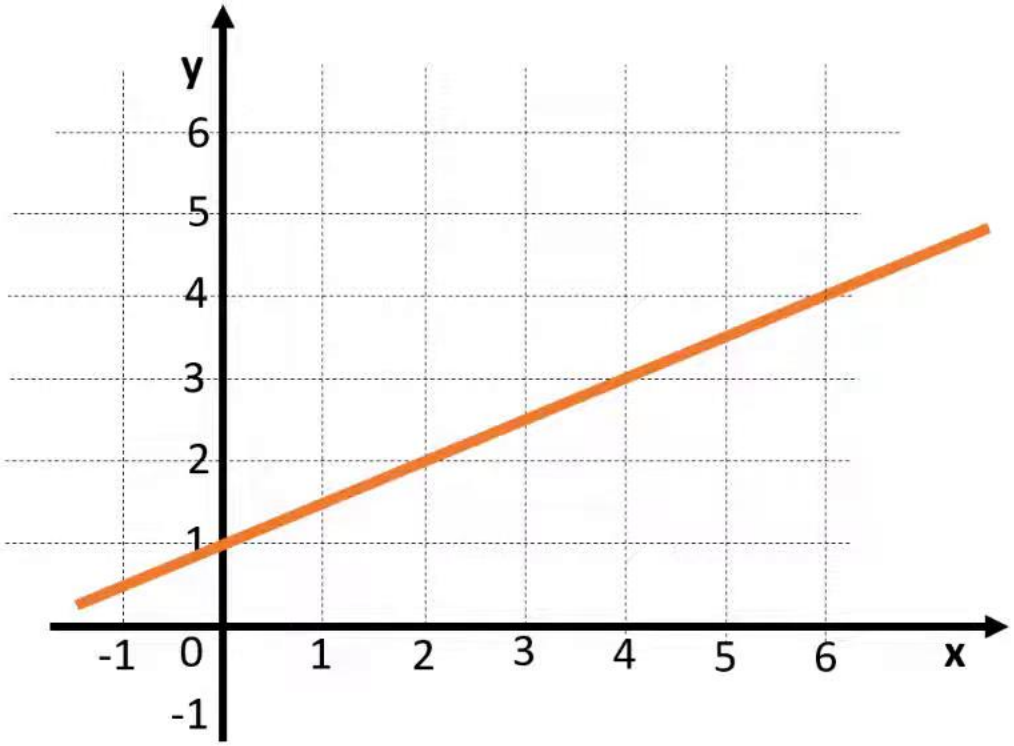
$$y = 0.5x + 1$$

$$-0.5x + y - 1 = 0$$

$$-2x + 4y - 4 = 0 \quad \times (-1)$$

which can be done by multiplying all terms with negative one,

The equation of a straight line



$$y = mx + b$$
$$Ax + By + C = 0$$

$$y = 0.5x + 1$$

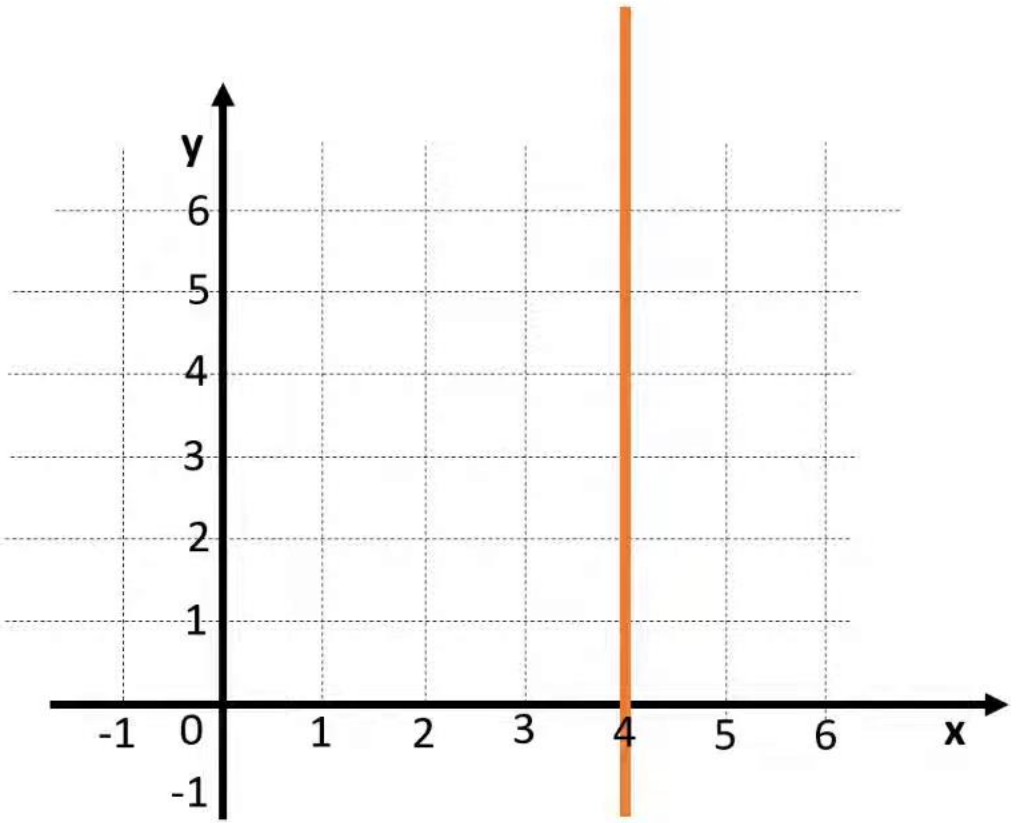
$$-0.5x + y - 1 = 0$$

$$-2x + 4y - 4 = 0 \quad \times(-1)$$

$$\boxed{2x - 4y + 4 = 0}$$

so that we get the following equation that also describes the same line.

The equation of a straight line

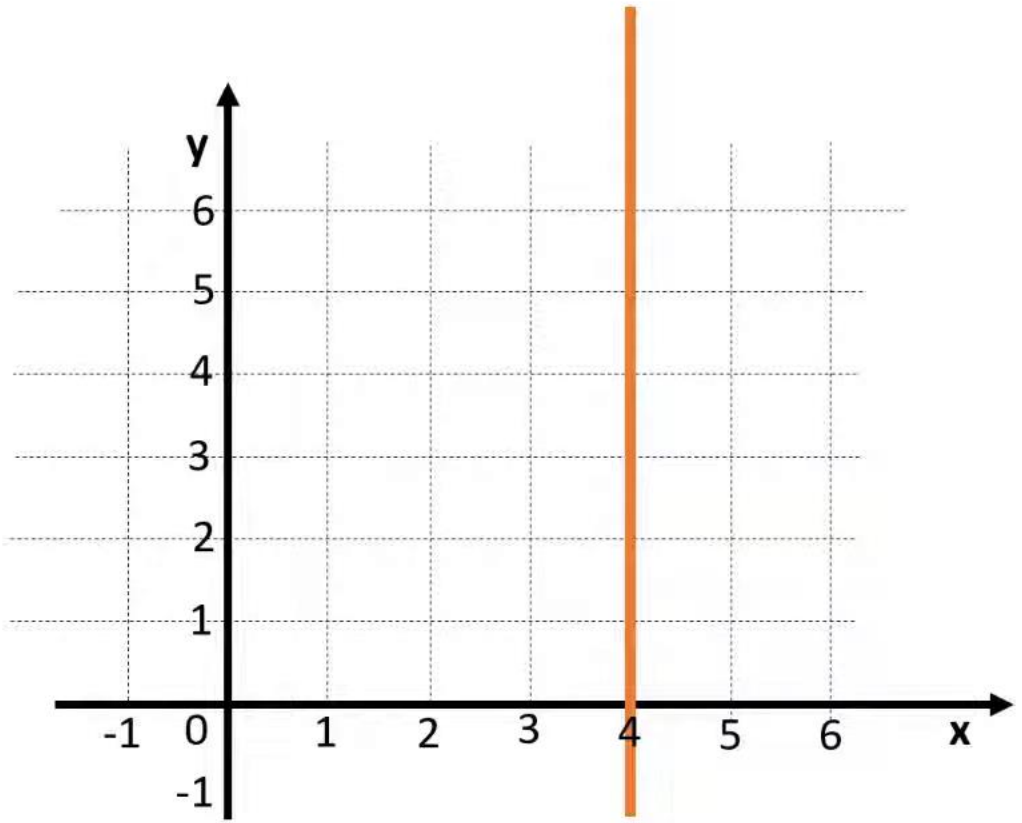


$$y = mx + b$$

$$Ax + By + C = 0$$

By using this form of the equation, we can now describe this vertical line,

The equation of a straight line

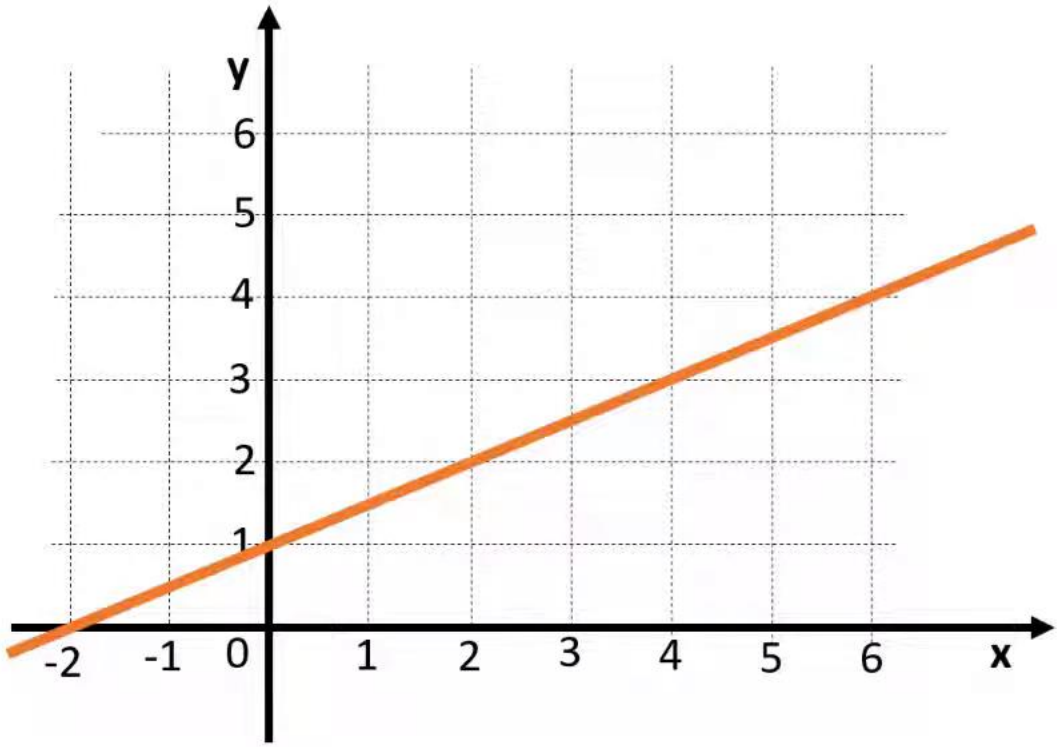


$$y = mx + b$$
$$Ax + By + C = 0$$

$$x - 4 = 0$$

like this because x is equal to 4 along this line.

The equation of a straight line



$$y = 0.5x + 1$$

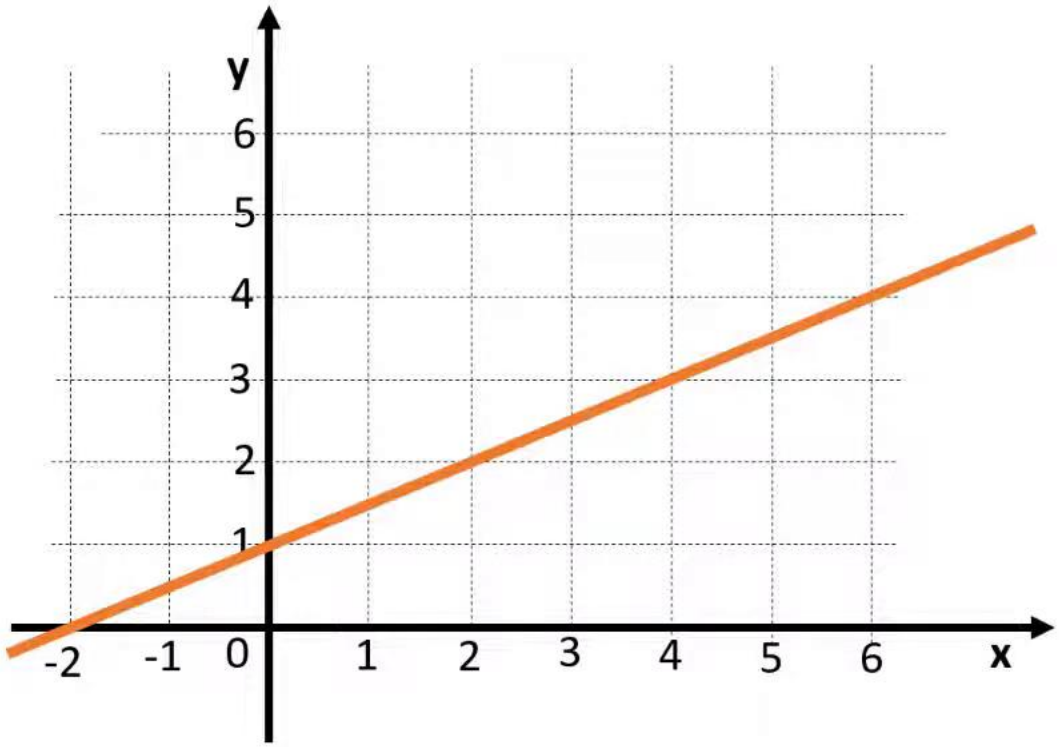
$$-2x + 4y - 4 = 0$$

$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

Remember that we can describe this line with these two different forms of equations,

The equation of a straight line



$$y = 0.5x + 1$$

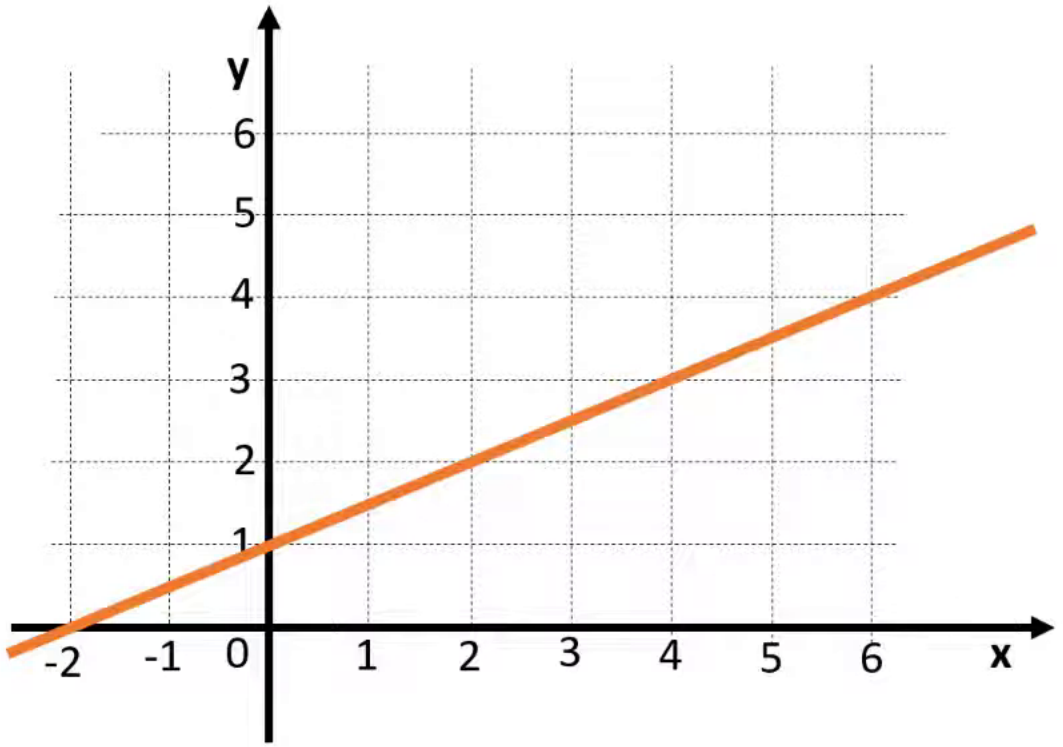
$$-2x + 4y - 4 = 0$$

$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

where we from hereon will focus on the second type of equation because that is the form used in SVM.

The equation of a straight line



$$y = 0.5x + 1$$

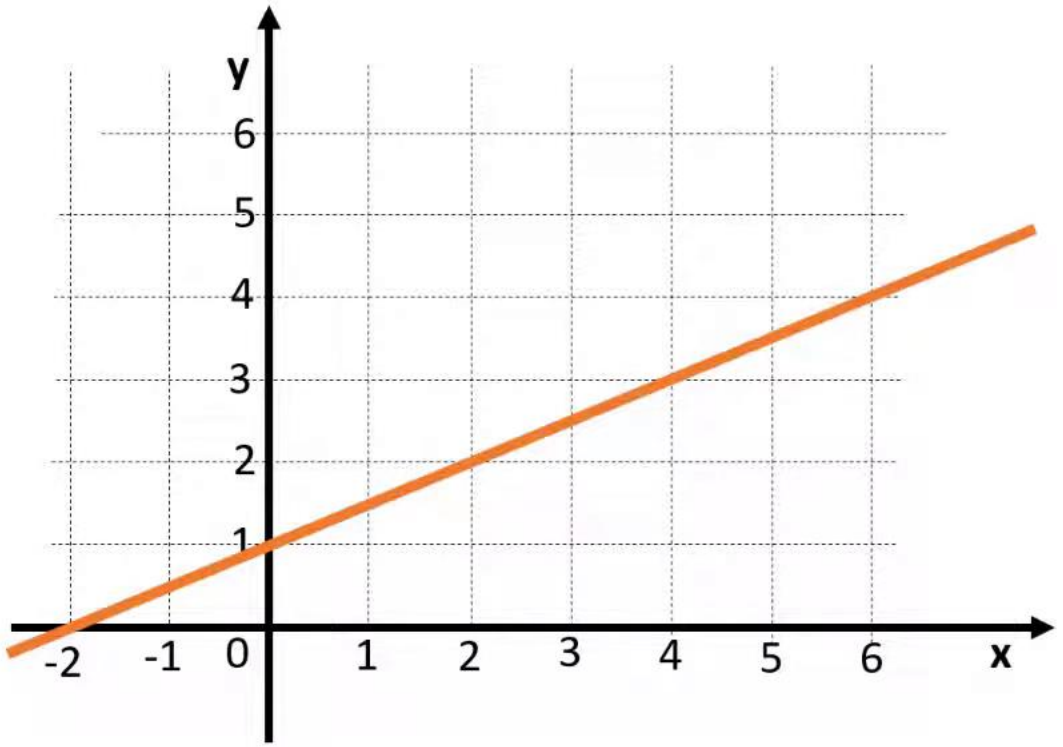
$$-2x + 4y - 4 = 0$$

$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

where we from hereon will focus on the second type of equation because that is the form used in SVM.

The equation of a straight line



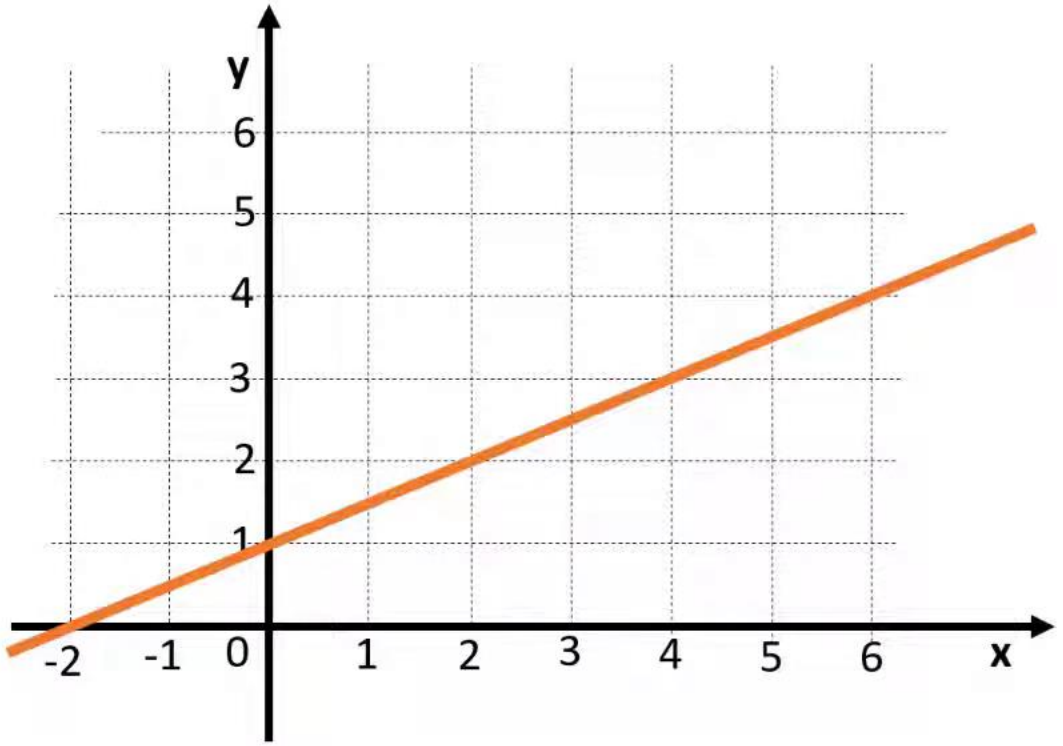
$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

The coefficient A is equal to negative 2,

The equation of a straight line



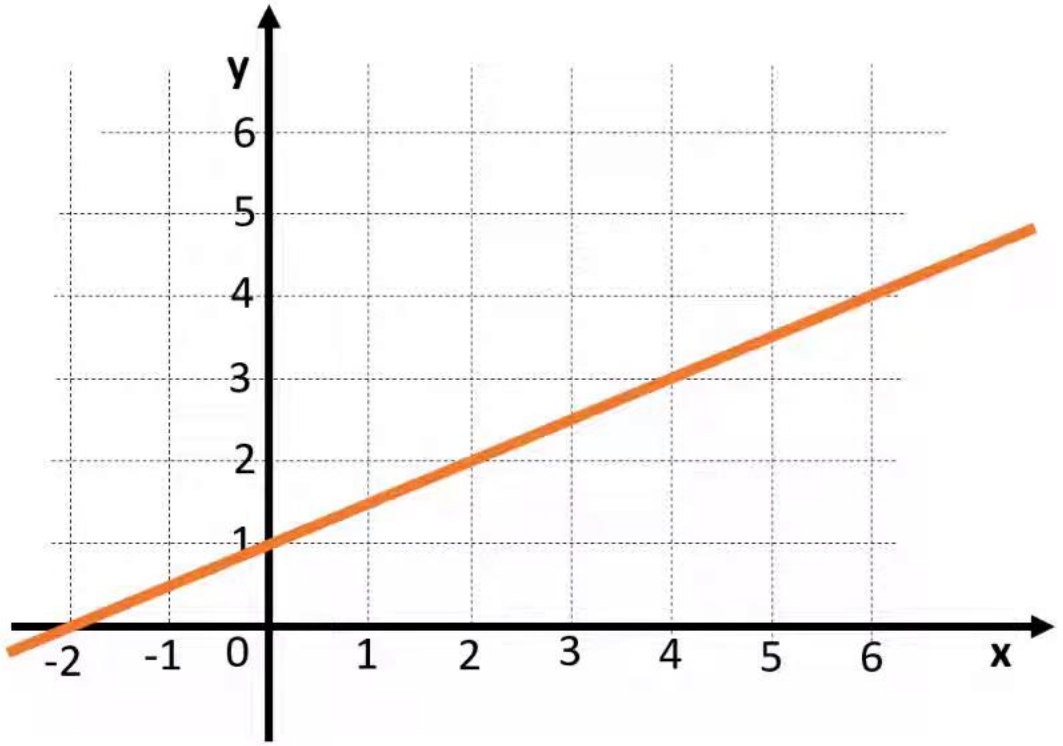
$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

and coefficient B is equal to 4,

The equation of a straight line



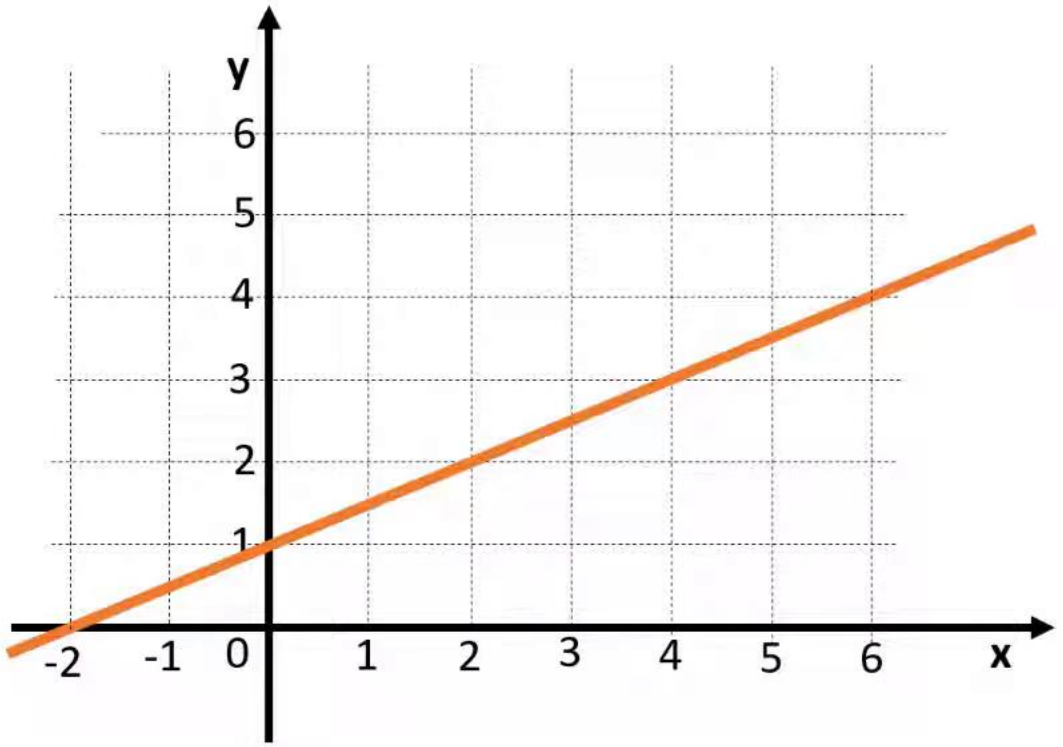
$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

whereas C is equal to negative 4.

The equation of a straight line



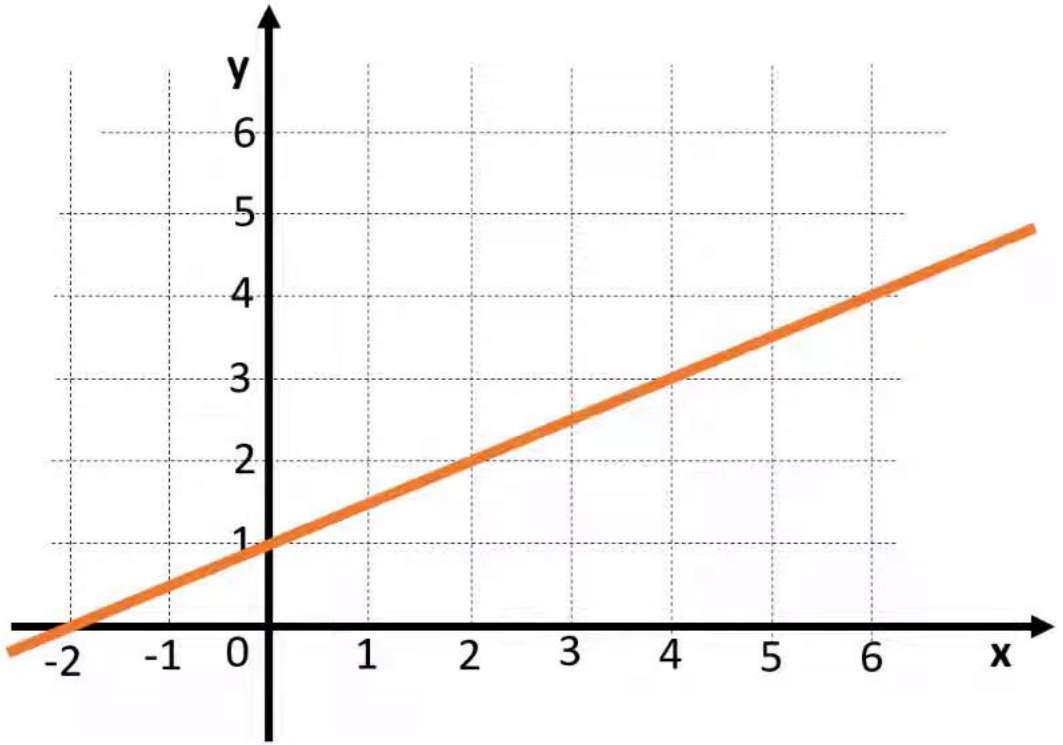
$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$-2x + 4y - 4 = 0$$

Let's plug in these values in this equation,

The equation of a straight line



$$-2x + 4y - 4 = 0$$

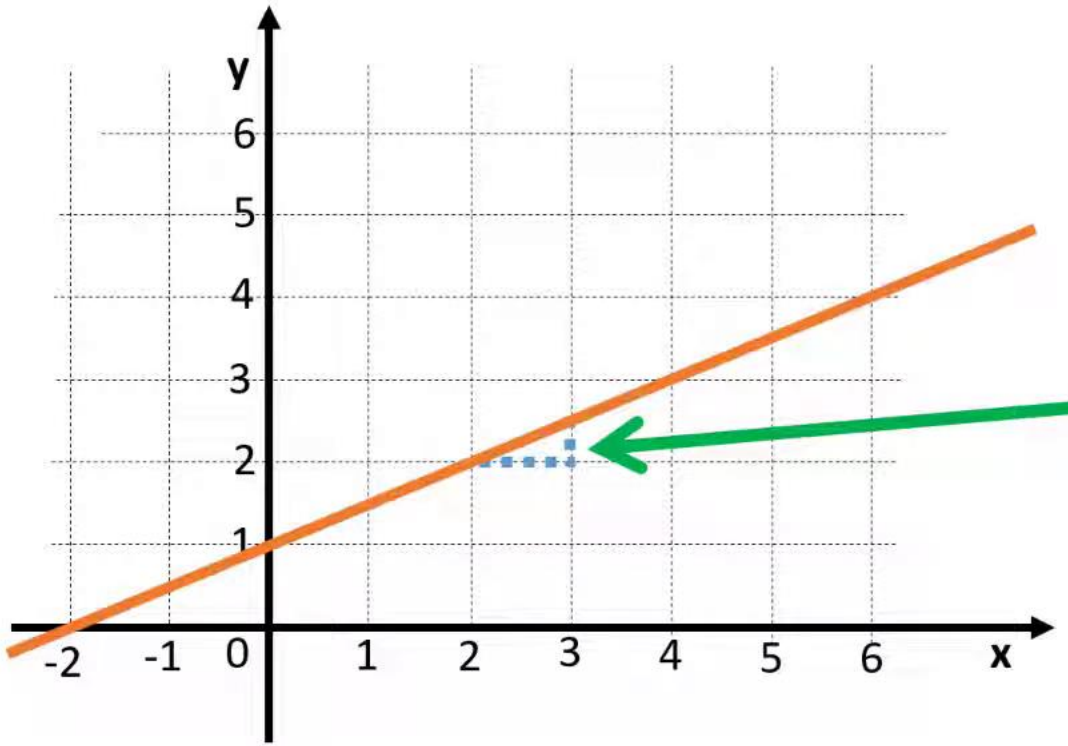
$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$y = -\frac{(-2)}{4}x - \frac{(-4)}{4}$$

so we get the following equation.

The equation of a straight line



$$Ax + By + C = 0$$

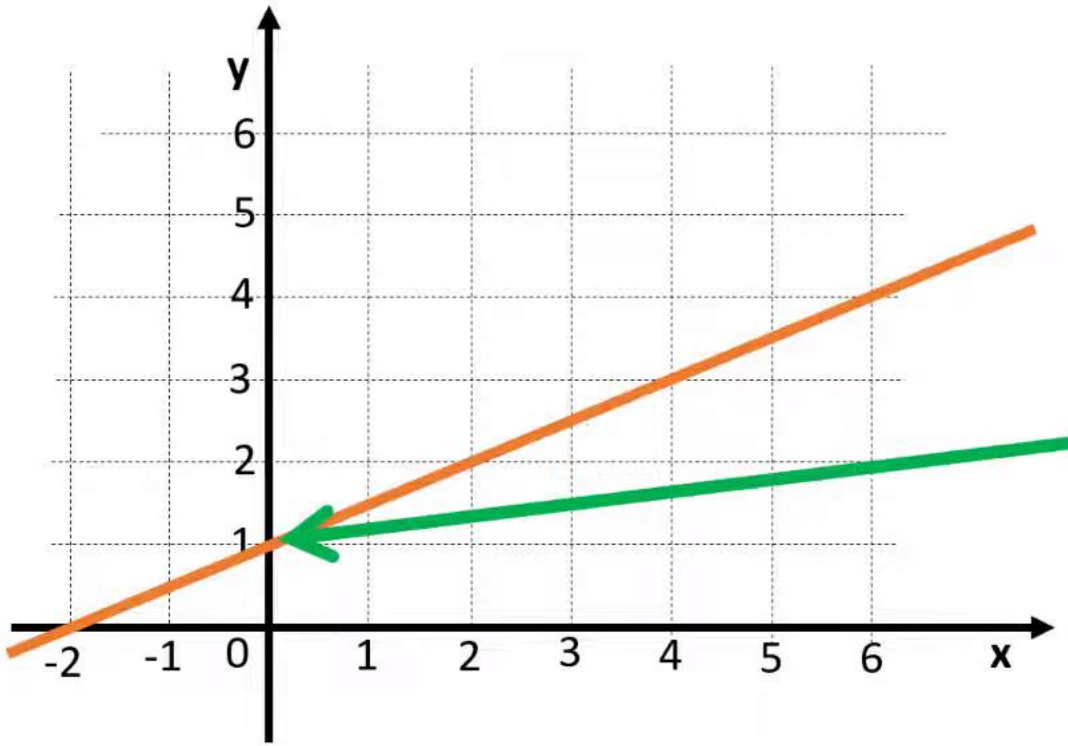
$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$-2x + 4y - 4 = 0$$

$$y = -\frac{(-2)}{4}x - \frac{(-4)}{4}$$

We see that the slope is 0.5 because if we increase one unit in x, y is increased by 0.5,

The equation of a straight line



$$Ax + By + C = 0$$

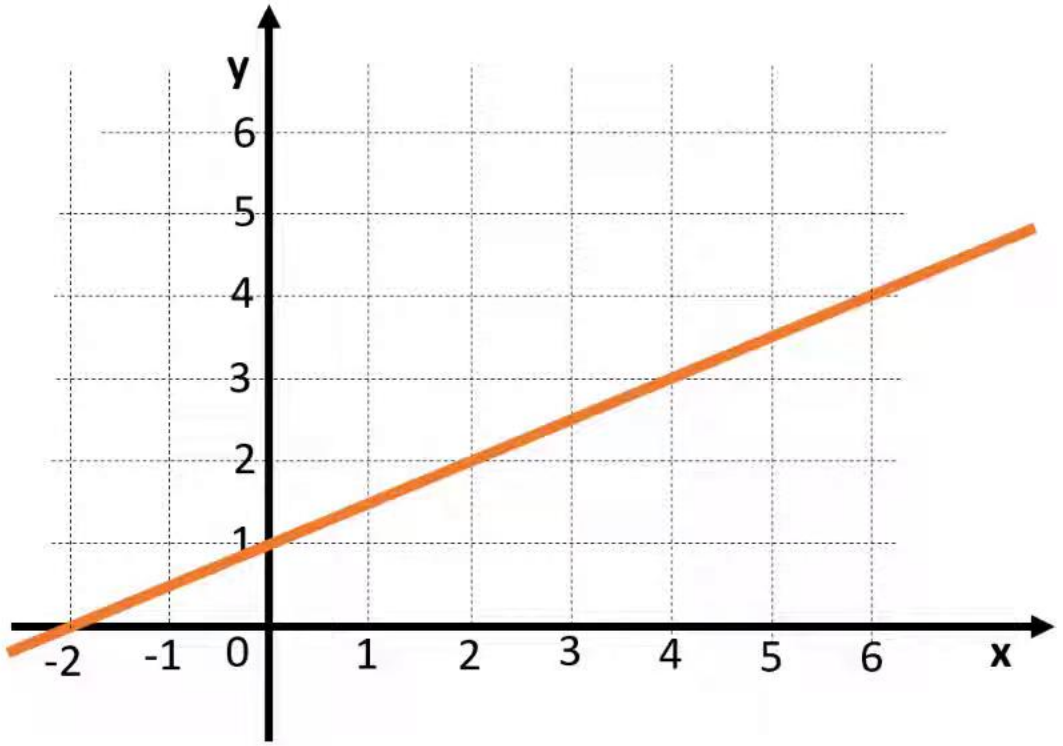
$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$-2x + 4y - 4 = 0$$

$$y = -\frac{(-2)}{4}x - \frac{(-4)}{4}$$

and we see that the intercept is equal to one.

The equation of a straight line



$$Ax + By + C = 0$$

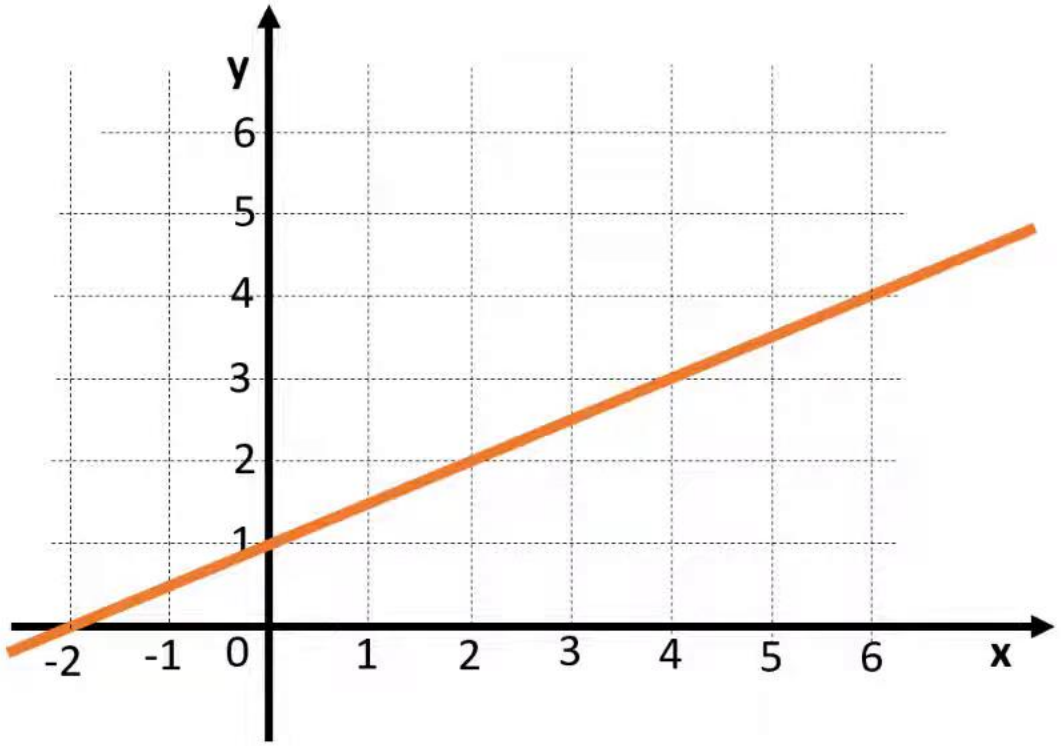
$$-2x + 4y - 4 = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$y = -\frac{(-2)}{4}x - \frac{(-4)}{4}$$

Linear support vector machines find the optimal values of the coefficients A and B, so that the line separates the data points as good as possible. We therefore first need to understand what happens with the line if we change A, B and C.

The equation of a straight line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-x + 4y - 4 = 0$$

$$y = 0.25x + 1$$

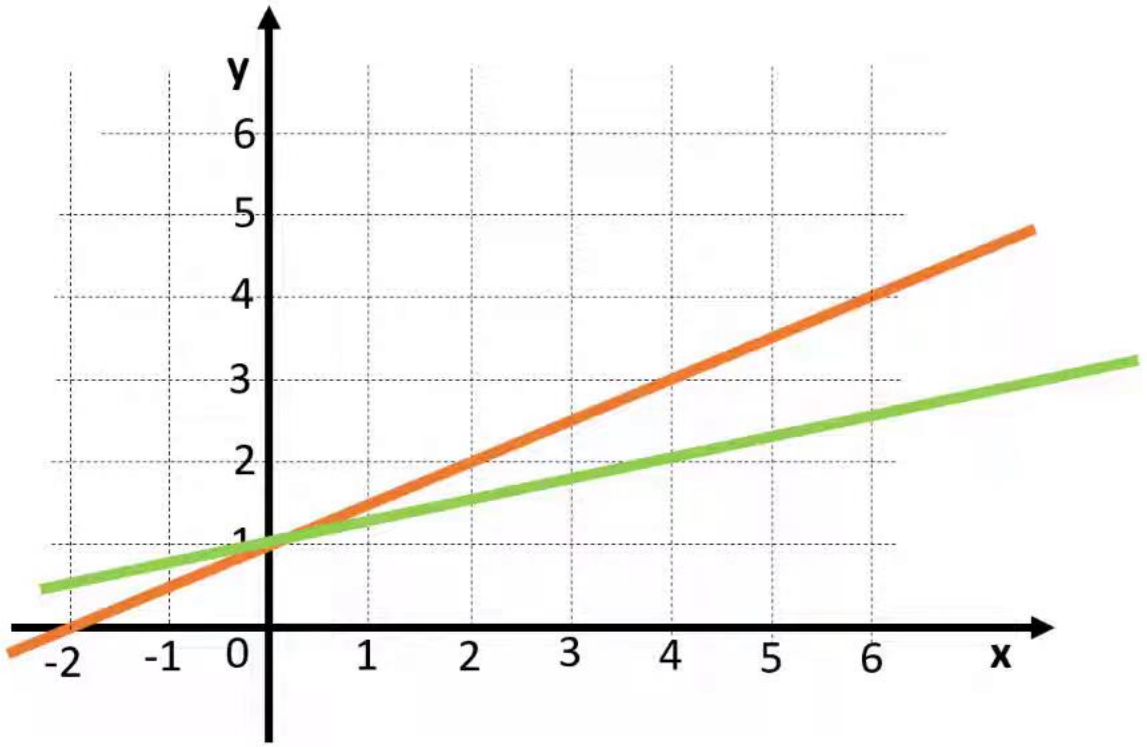
$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$y = -\frac{(-1)}{4}x - \frac{(-4)}{4}$$

For example, if we increase A from negative two to negative one,

The equation of a straight line



$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

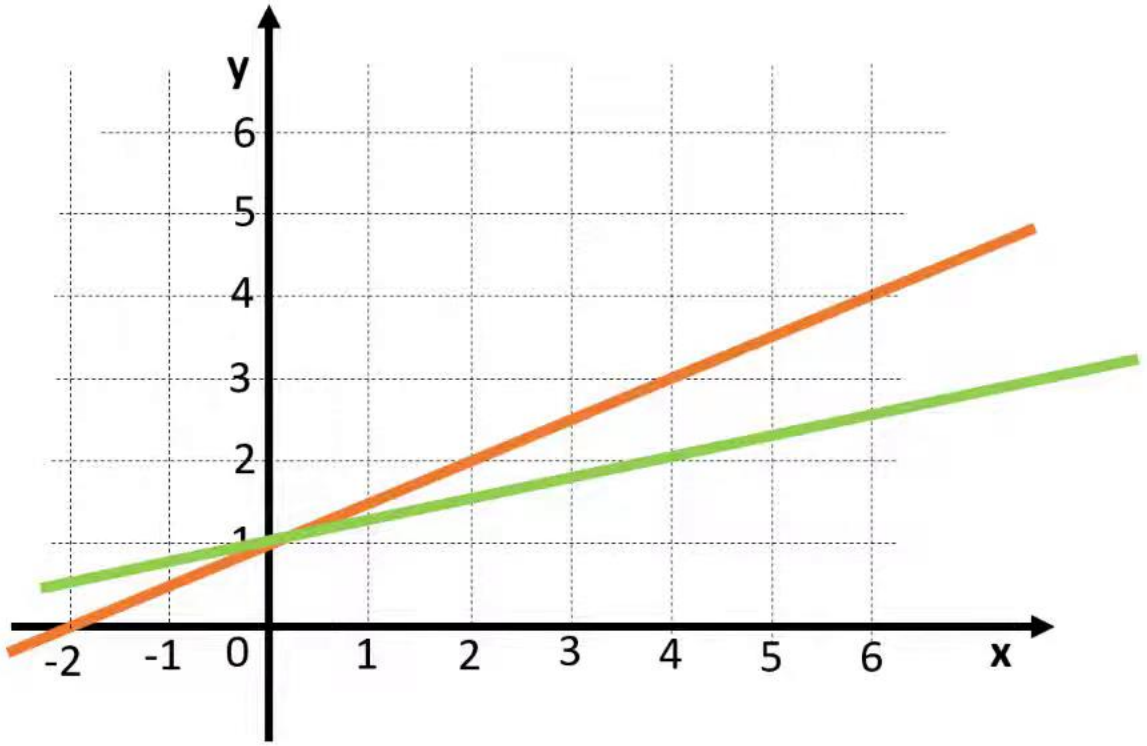
$$-x + 4y - 4 = 0$$

$$y = 0.25x + 1$$

$$y = -\frac{(-1)}{4}x - \frac{(-4)}{4}$$

the slope will be reduced from 0.5 to 0.25.

The equation of a straight line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-4x + 4y - 4 = 0$$

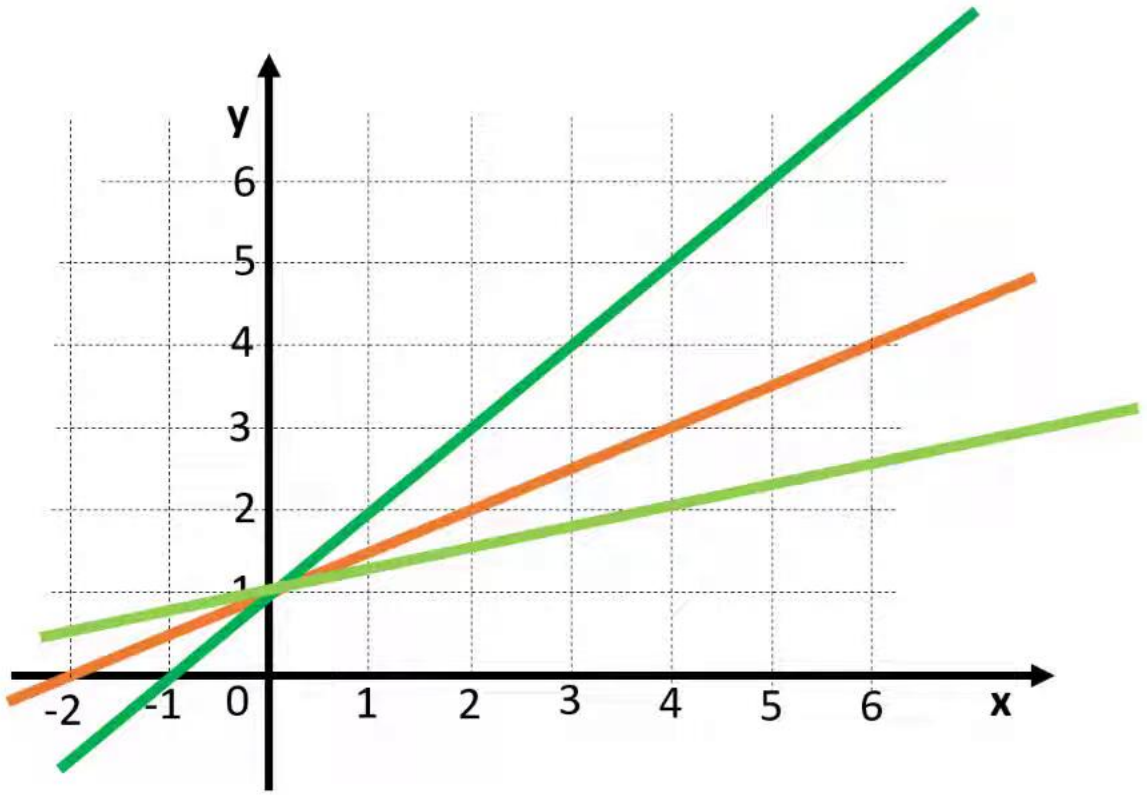
$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$y = -\frac{(-4)}{4}x - \frac{(-4)}{4}$$

And if we instead reduce A from negative two to negative four,

The equation of a straight line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-4x + 4y - 4 = 0$$

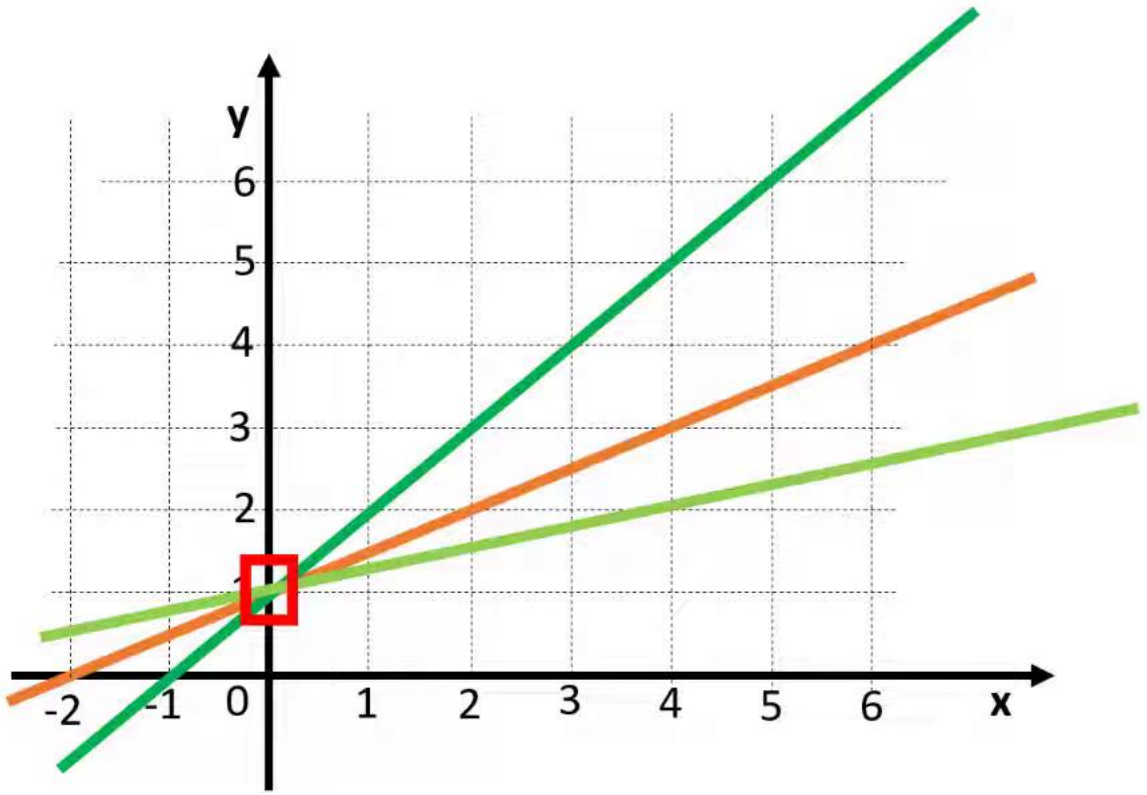
$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$y = -\frac{(-4)}{4}x - \frac{(-4)}{4}$$

the slope will be increased to one.

The equation of a straight line



$$Ax + By + C = 0$$

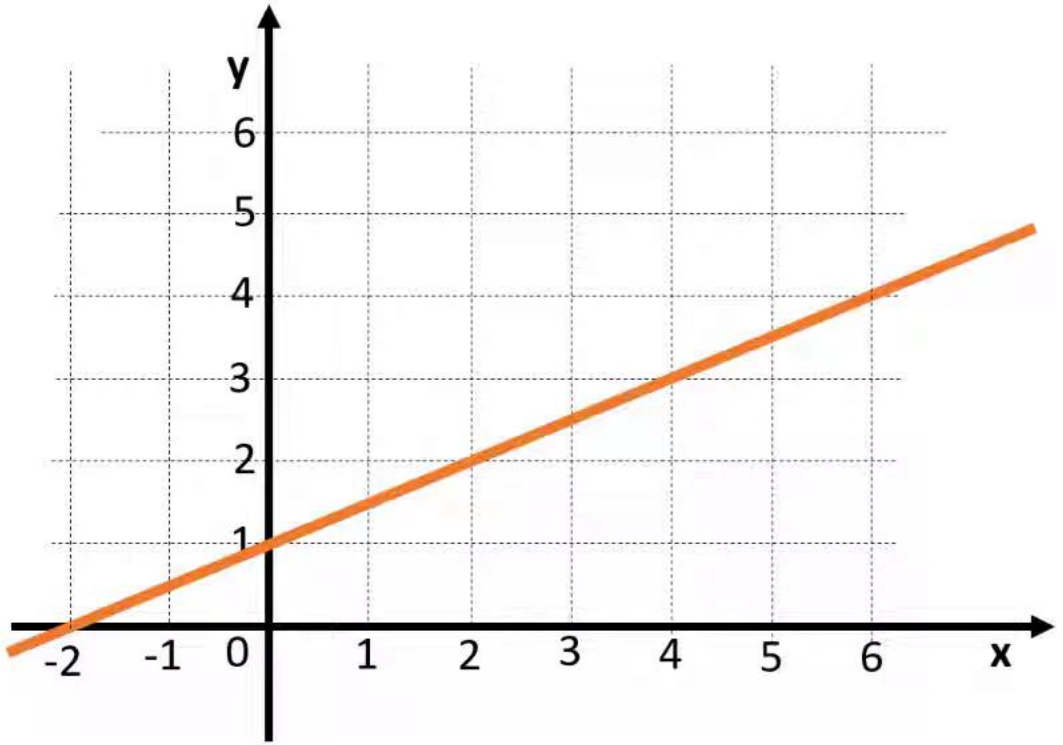
$$-2x + 4y - 4 = 0$$

$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

Note that, when we change A, the line is rotating around this point.

The equation of a straight line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-2x + 4y - 8 = 0$$

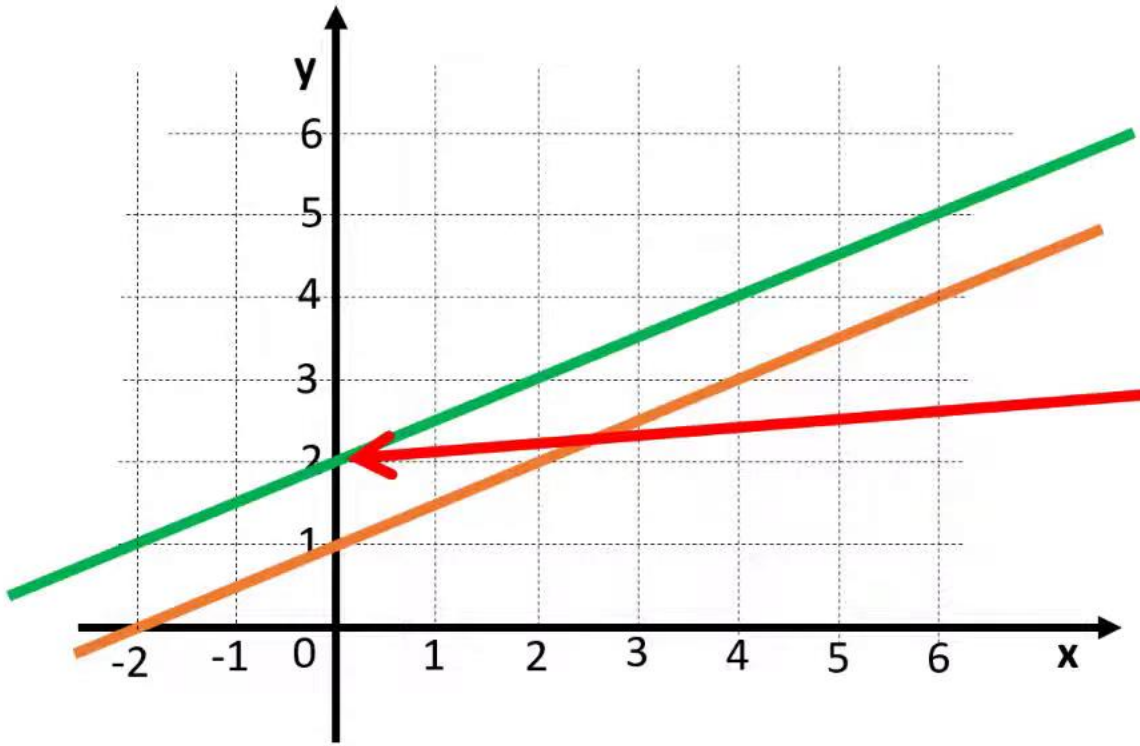
$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$y = -\frac{(-2)}{4}x - \frac{(-8)}{4}$$

If we decrease C from negative four to negative eight,

The equation of a straight line



$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$Ax + By + C = 0$$

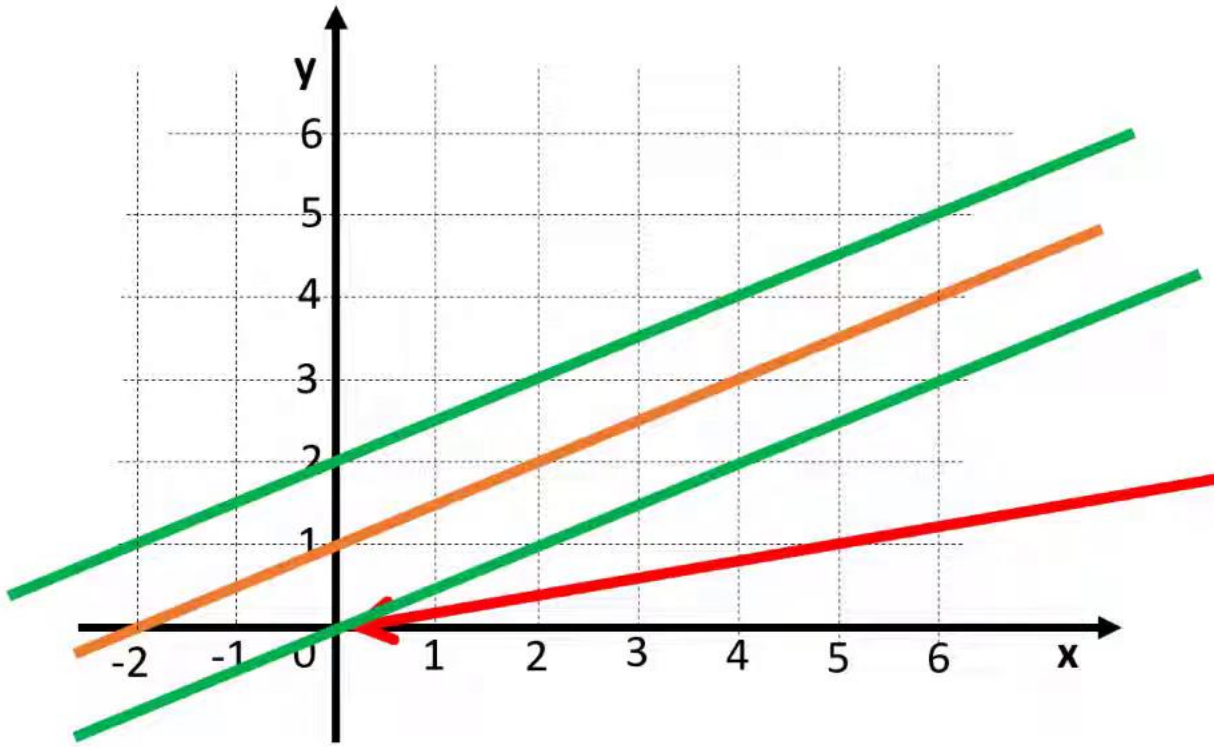
$$-2x + 4y - 4 = 0$$

$$-2x + 4y - 8 = 0$$

$$y = -\frac{(-2)}{4}x - \frac{(-8)}{4}$$

the intercept will be increased to two,

The equation of a straight line



$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$Ax + By + C = 0$$

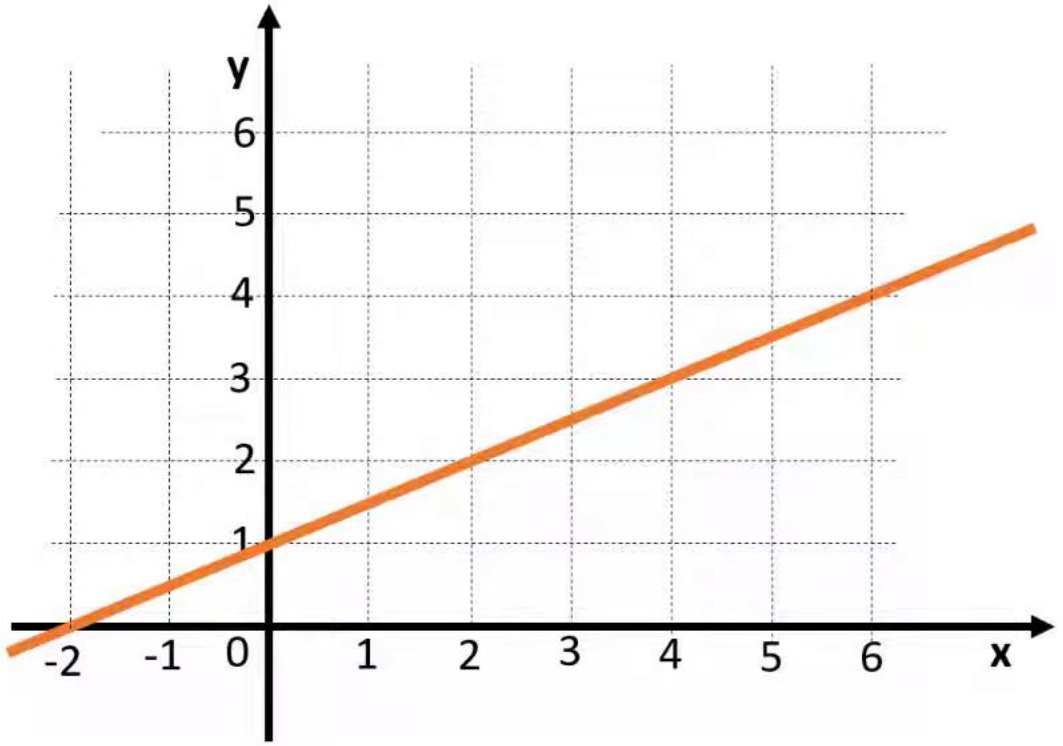
$$-2x + 4y - 4 = 0$$

$$-2x + 4y + 0 = 0$$

$$y = -\frac{(-2)}{4}x - \frac{0}{4}$$

and if we set C to zero, the intercept will be reduced to zero.

The equation of a straight line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-2x + 8y - 4 = 0$$

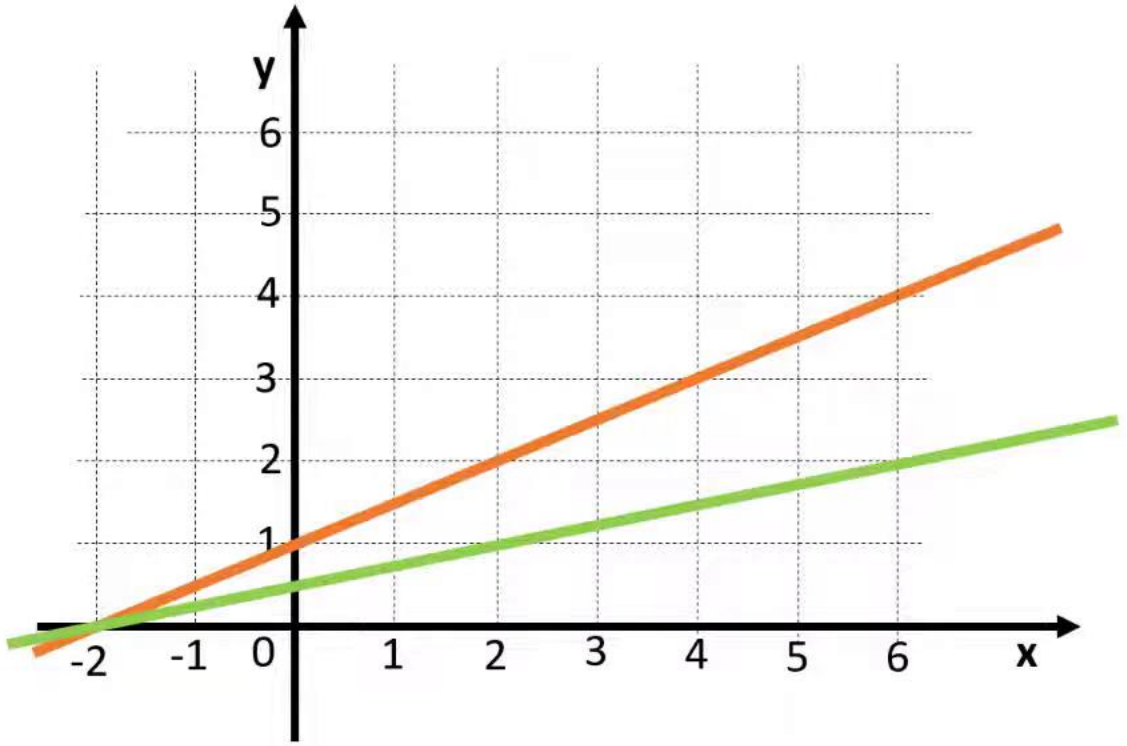
$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$y = -\frac{(-2)}{8}x - \frac{(-4)}{8}$$

If we instead increase B from four to eight,

The equation of a straight line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-2x + 8y - 4 = 0$$

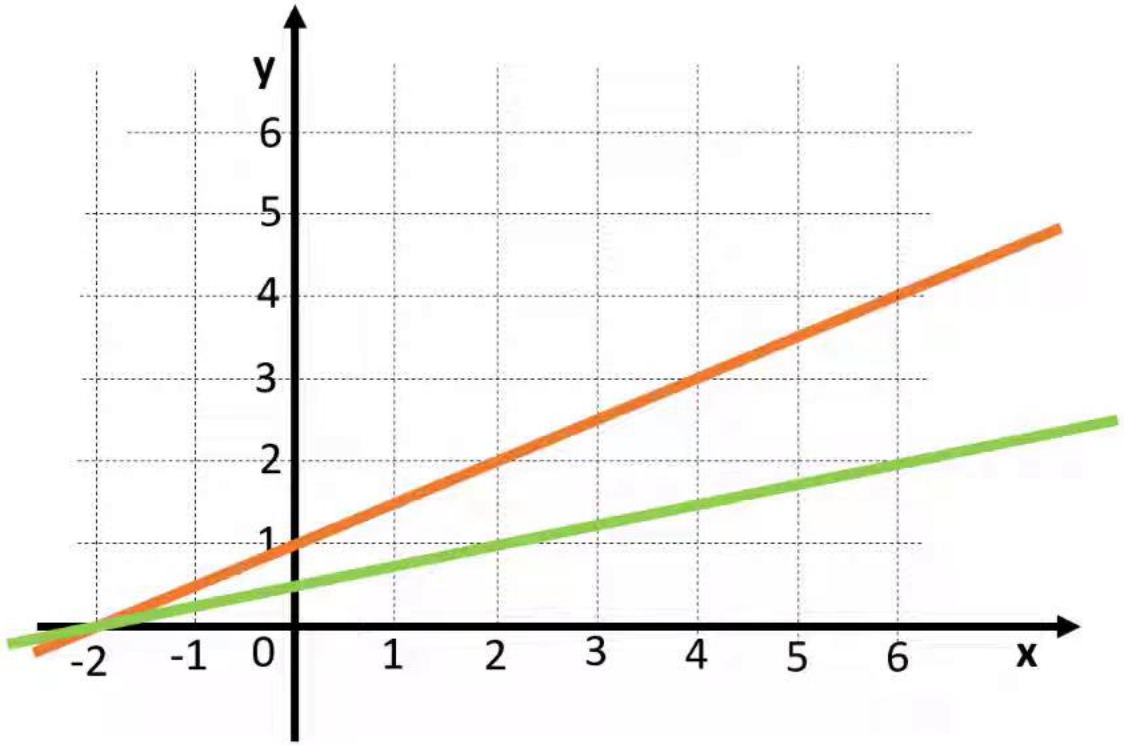
$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$y = -\frac{(-2)}{8}x - \frac{(-4)}{8}$$

we see that the slope is reduced from 0.5 to 0.25,

The equation of a straight line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-2x + 2y - 4 = 0$$

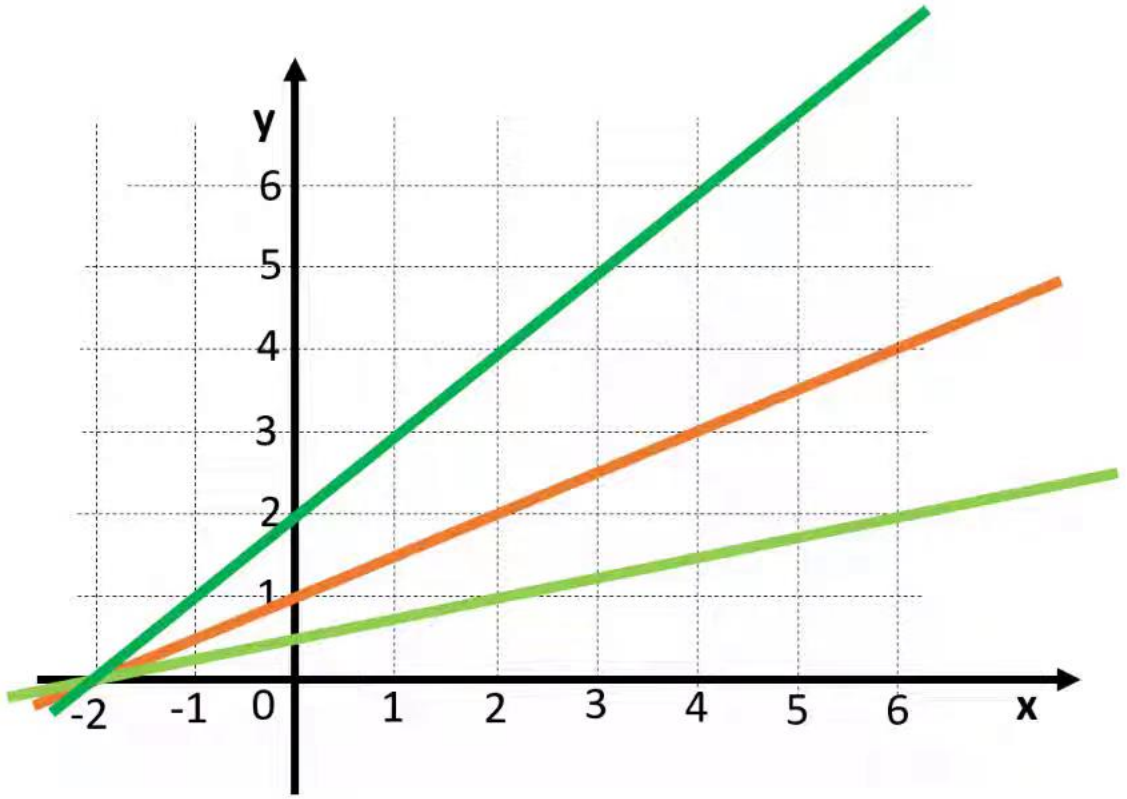
$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$y = -\frac{(-2)}{2}x - \frac{(-4)}{2}$$

If we now reduce B from four to two,

The equation of a straight line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-2x + 2y - 4 = 0$$

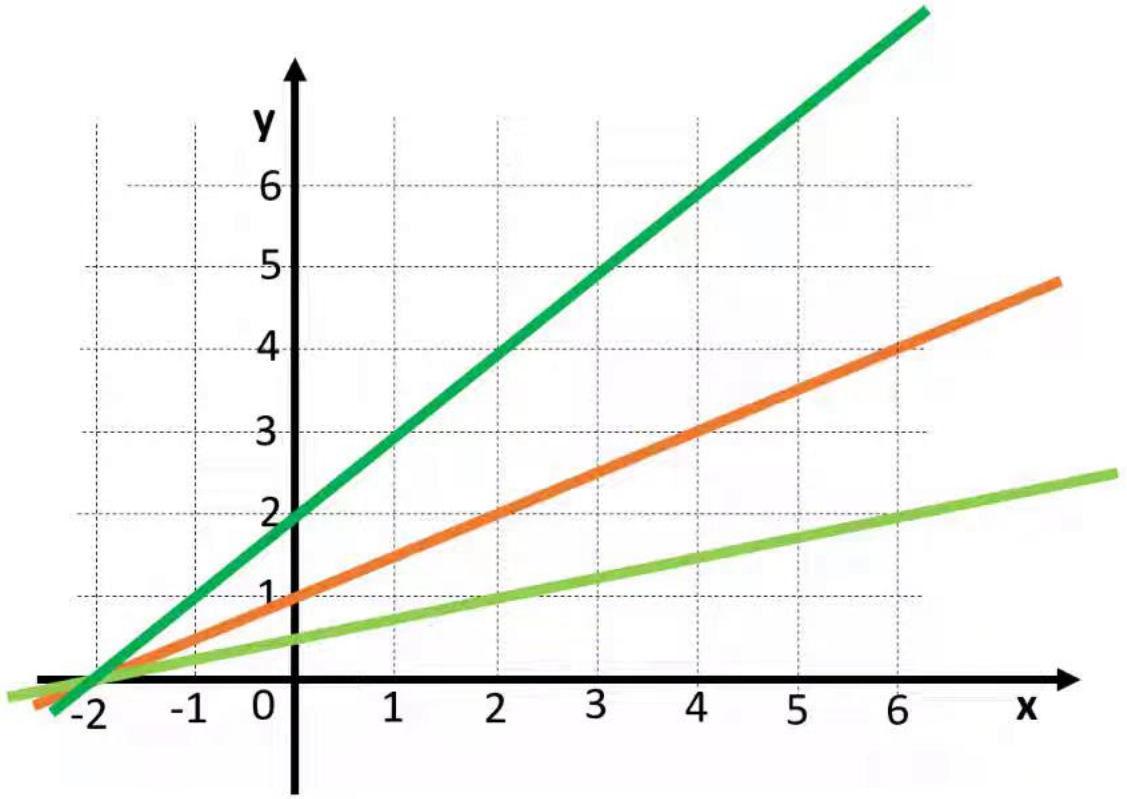
$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$y = -\frac{(-2)}{2}x - \frac{(-4)}{2}$$

we see that the slope has increased from 0.5 to one,

The equation of a straight line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-2x + 2y - 4 = 0$$

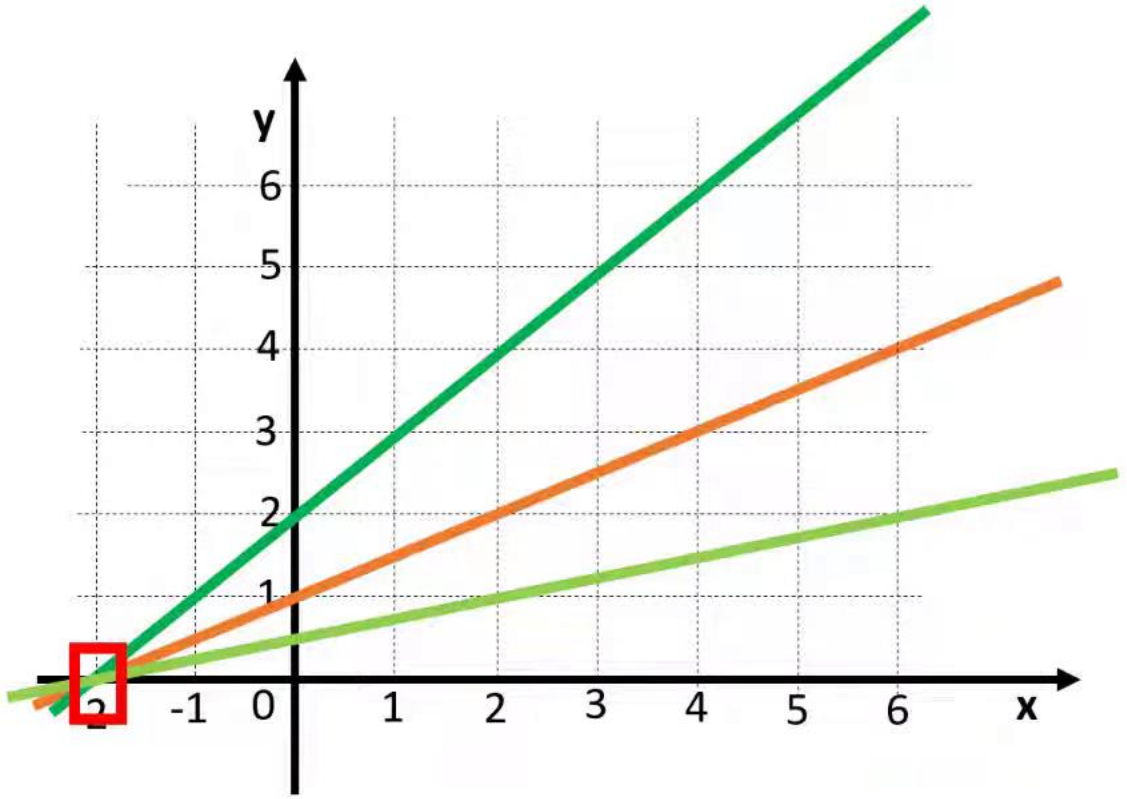
$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$y = -\frac{(-2)}{2}x - \frac{(-4)}{2}$$

and that the intercept has increased from one to two.

The equation of a straight line



$$Ax + By + C = 0$$

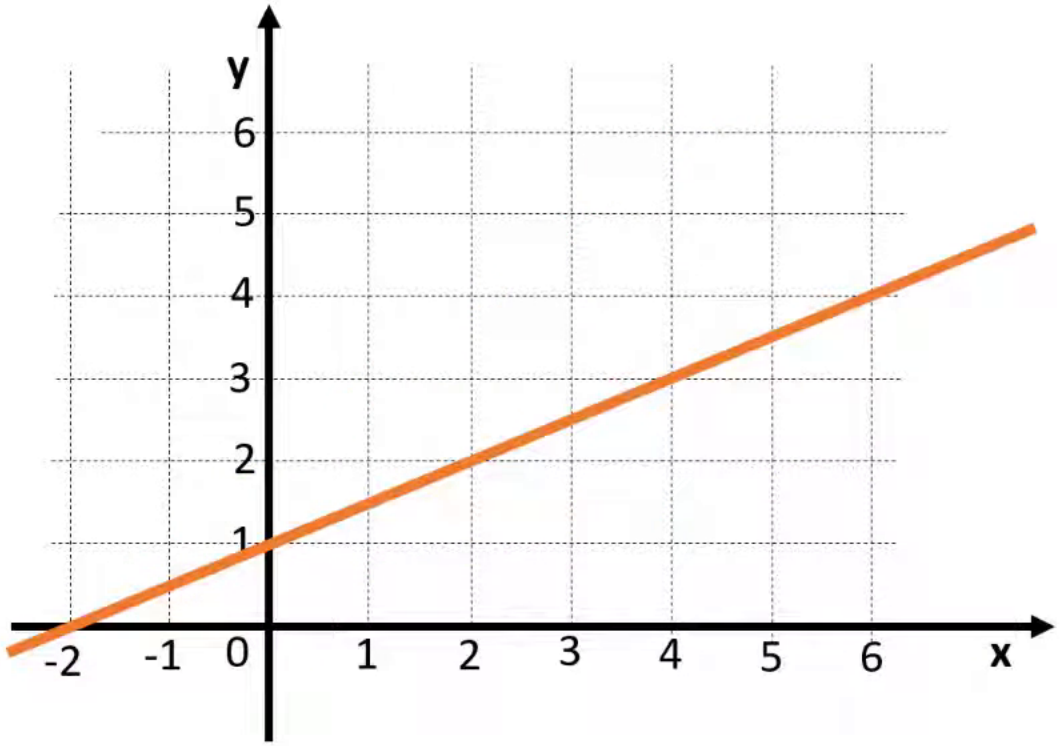
$$Ax + By + C = 0$$

$$y = -\frac{A}{B}x - \frac{C}{B}$$

$$y = -\frac{(-2)}{2}x - \frac{(-4)}{2}$$

Note how the line is rotating around this point.

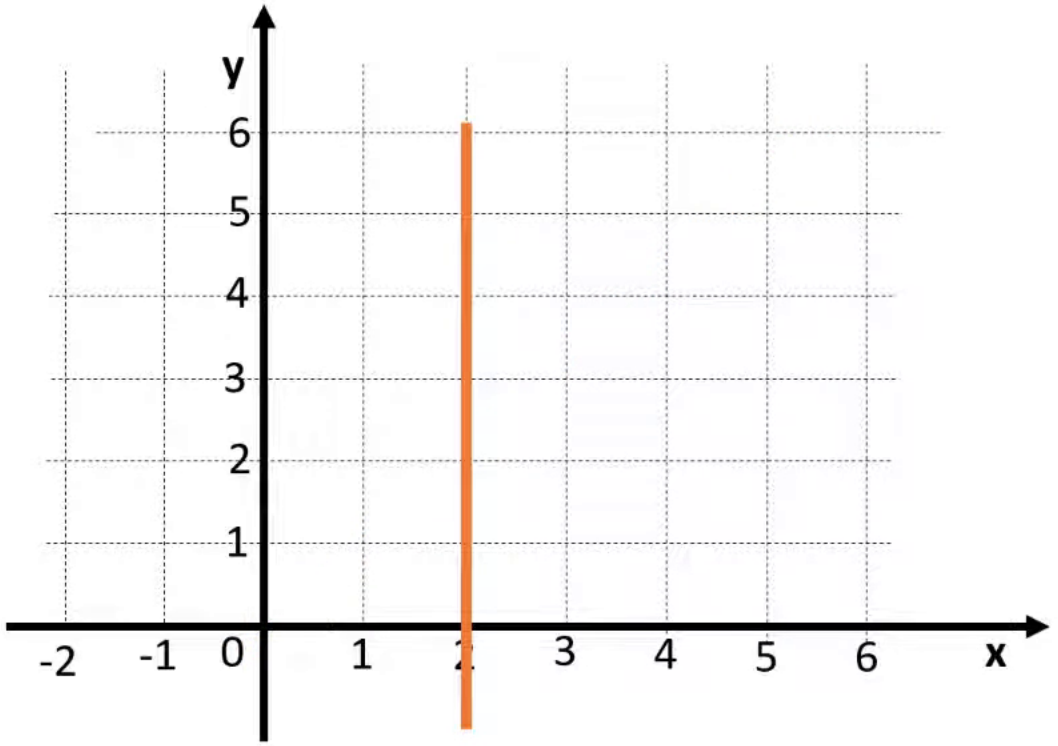
The equation of a straight line



$$Ax + By + C = 0$$

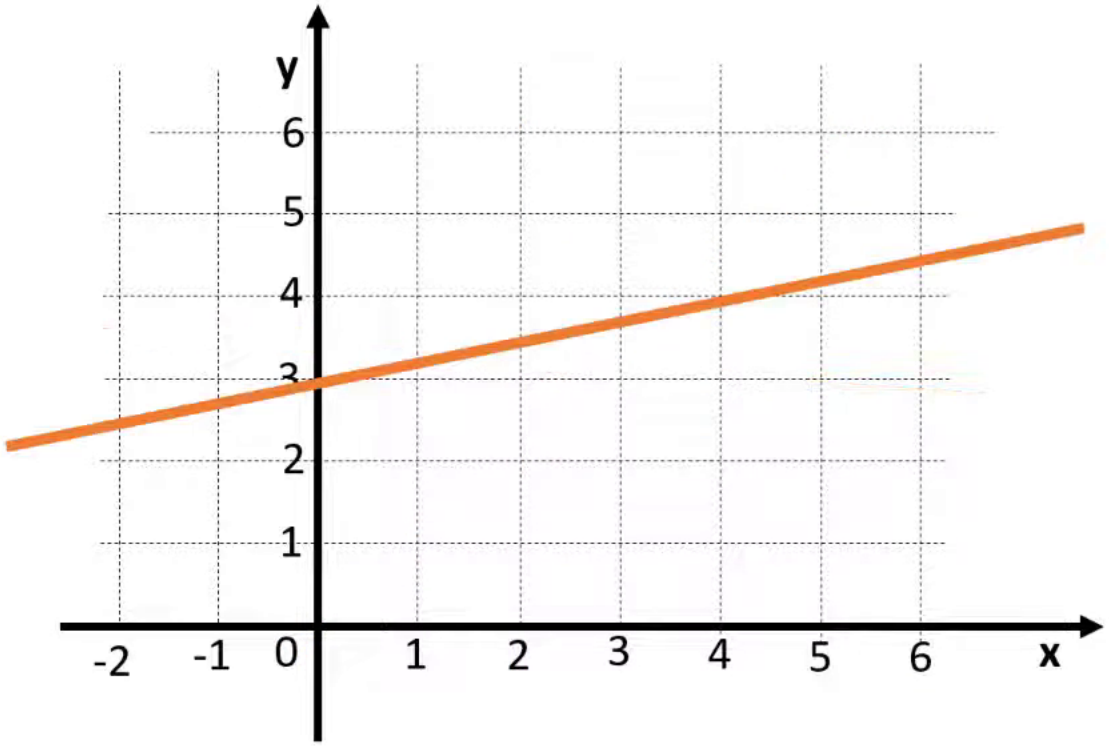
By changing the values of A and B,

The equation of a straight line



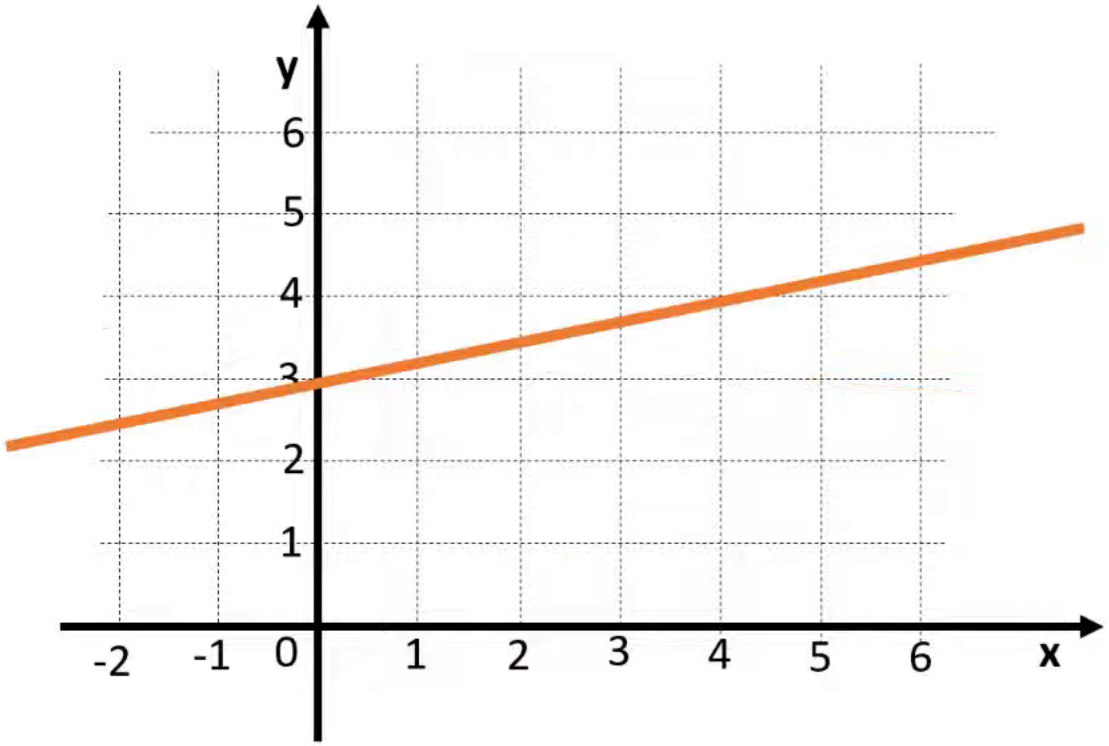
$$Ax + By + C = 0$$

The equation of a straight line



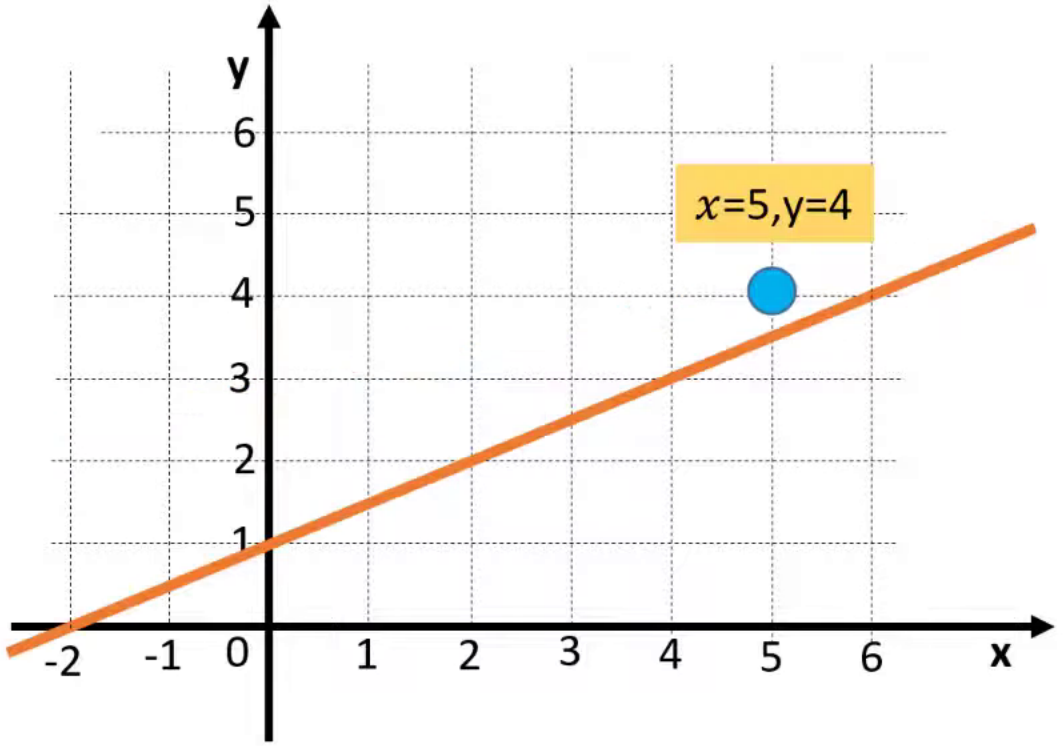
$$Ax + By + C = 0$$

The equation of a straight line



$$Ax + By + C = 0$$

The equation of a straight line

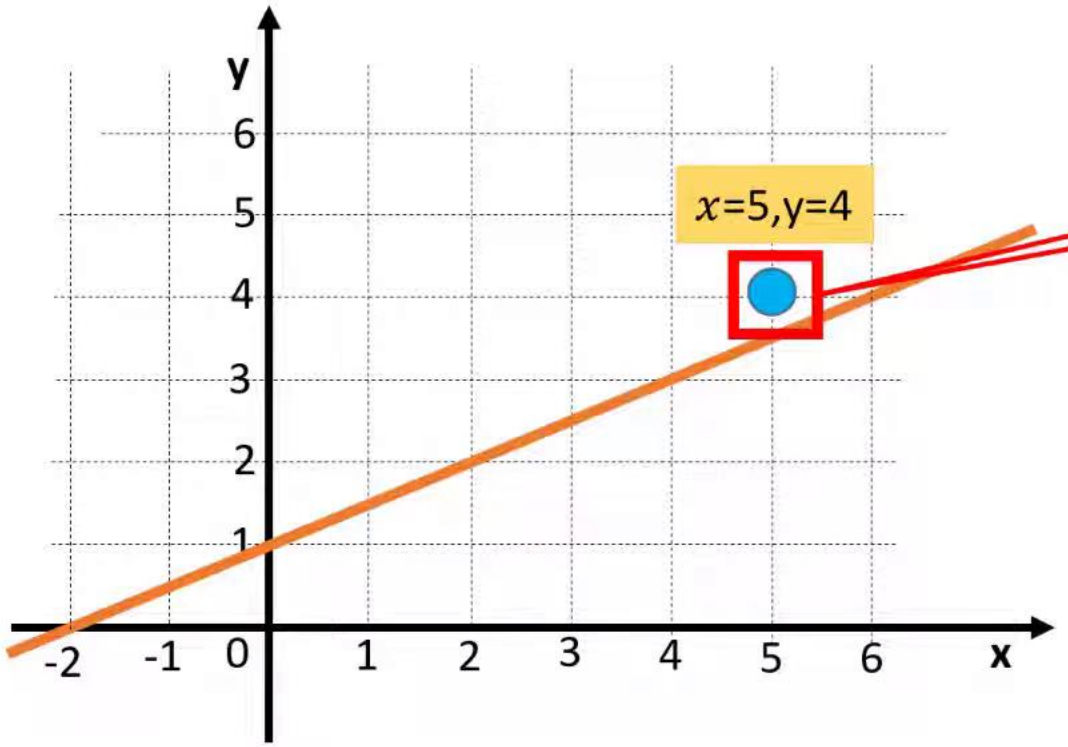


$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

Another nice feature about this form of the equation is that it can tell us which side of the line a data point is located on.

The equation of a straight line

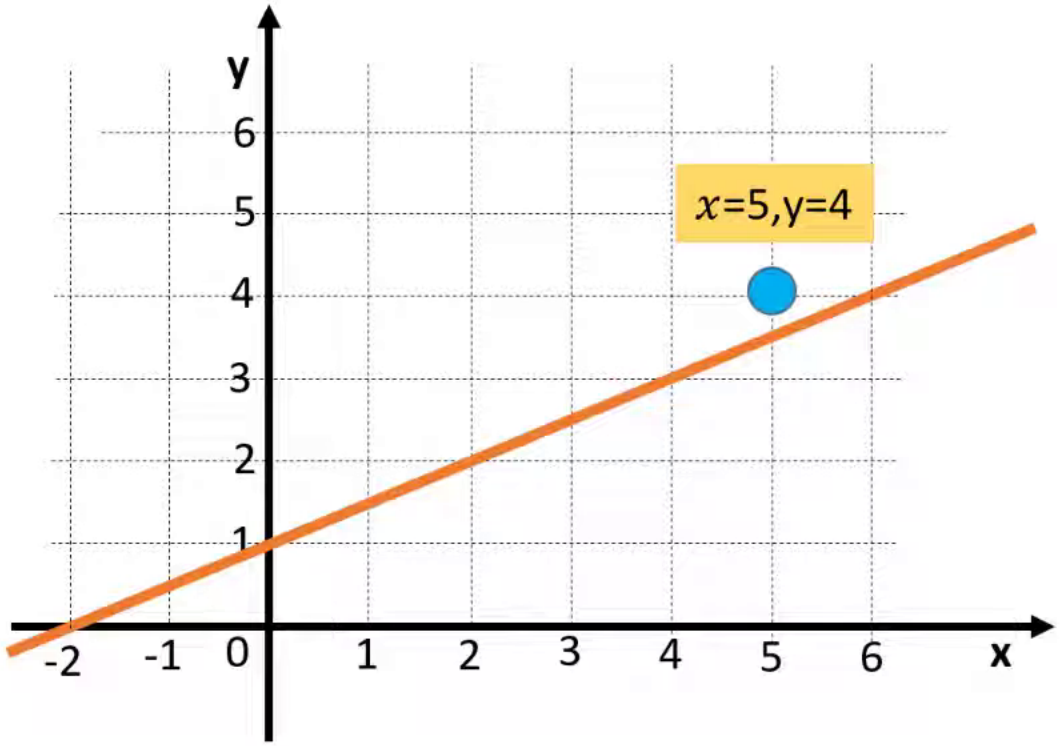


$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

If we plug in the x and y coordinates of this point in the equation,

The equation of a straight line



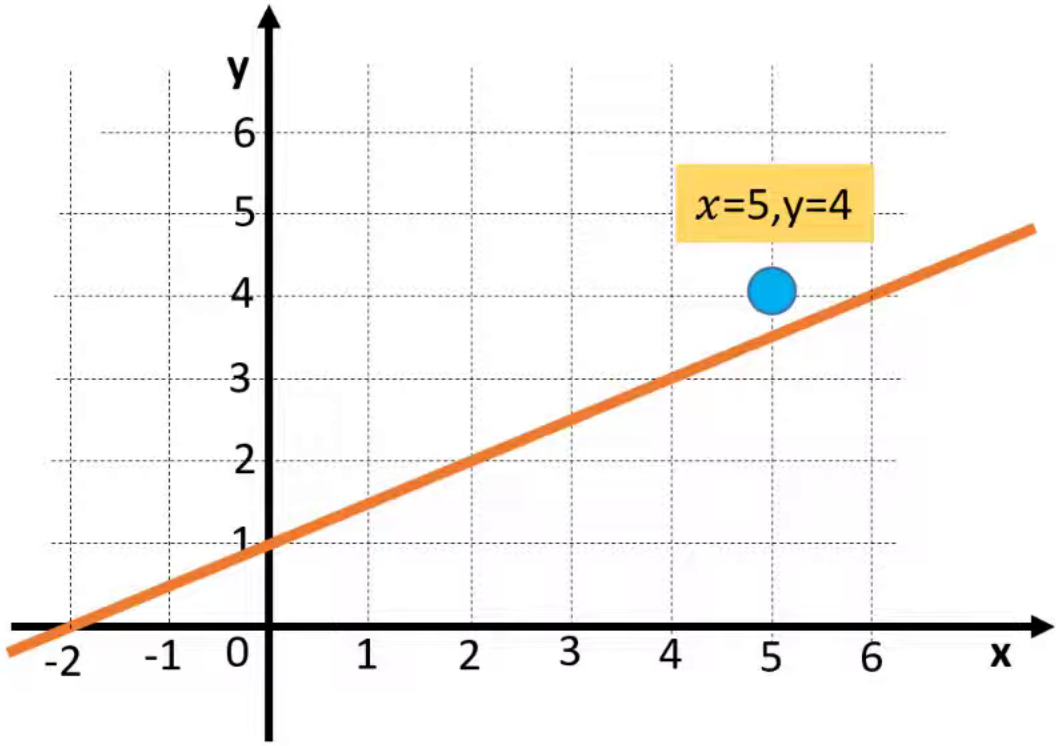
$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-2 \cdot 5 + 4 \cdot 4 - 4 = 2$$

we see that it results in a value that is greater than zero, which in this case means that the data point is above the line.

The equation of a straight line



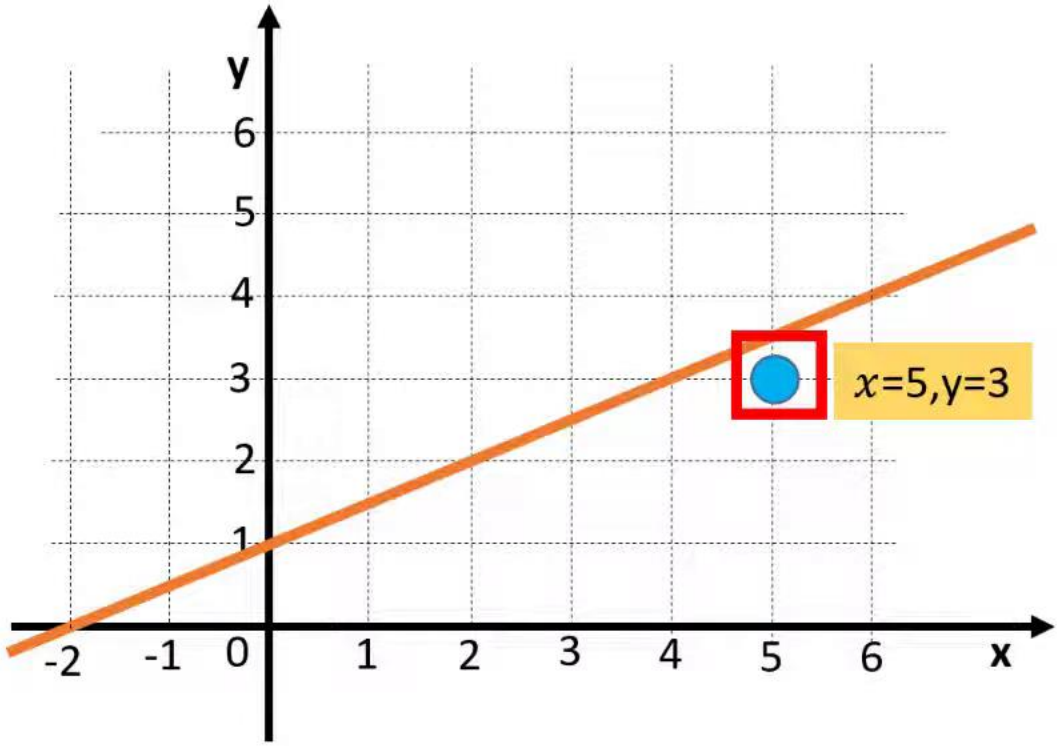
$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-2 \cdot 5 + 4 \cdot 4 - 4 = 2$$

we see that it results in a value that is greater than zero, which in this case means that the data point is above the line.

The equation of a straight line



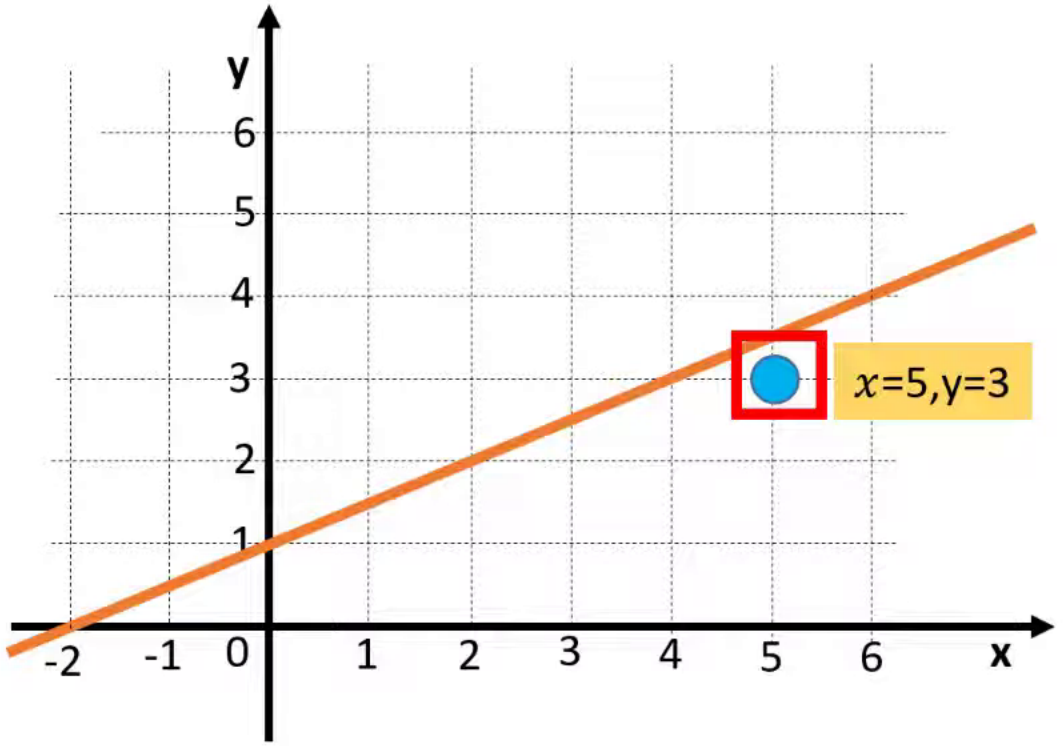
$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-2 \cdot 5 + 4 \cdot 3 - 4 = -2$$

In comparison, this data point results in a negative value, which tells us that the data point is below the line. This is handy when we later will classify data points based on the equation of the line.

The equation of a straight line



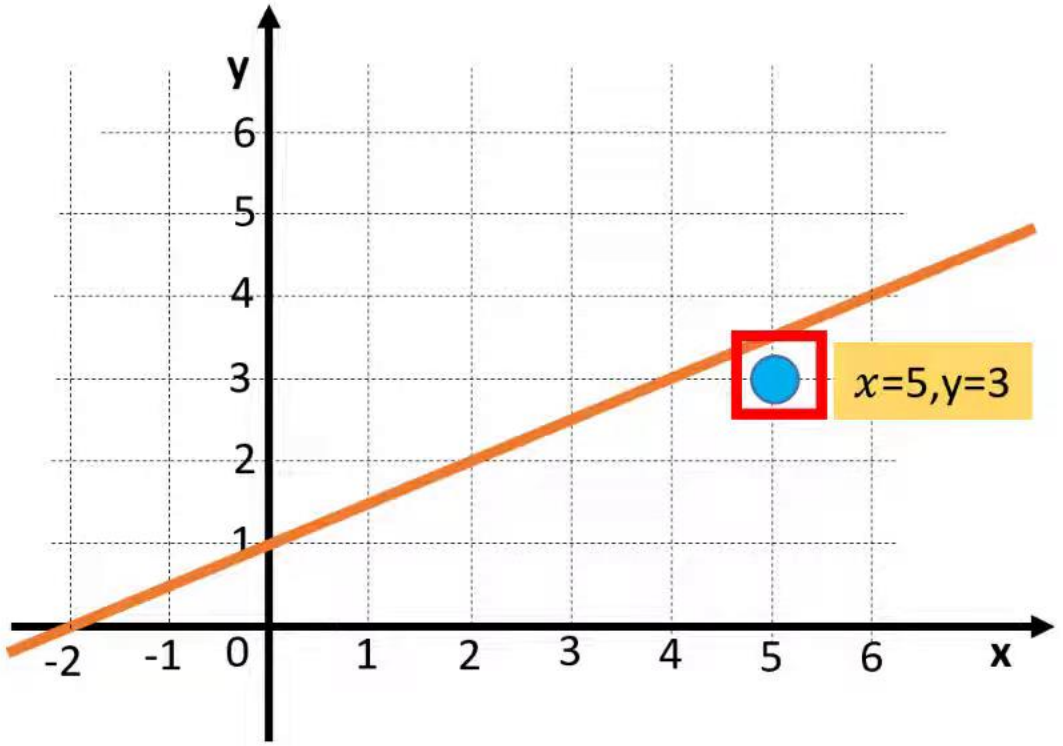
$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-2 \cdot 5 + 4 \cdot 3 - 4 = -2$$

In comparison, this data point results in a negative value, which tells us that the data point is below the line. This is handy when we later will classify data points based on the equation of the line.

The equation of a straight line



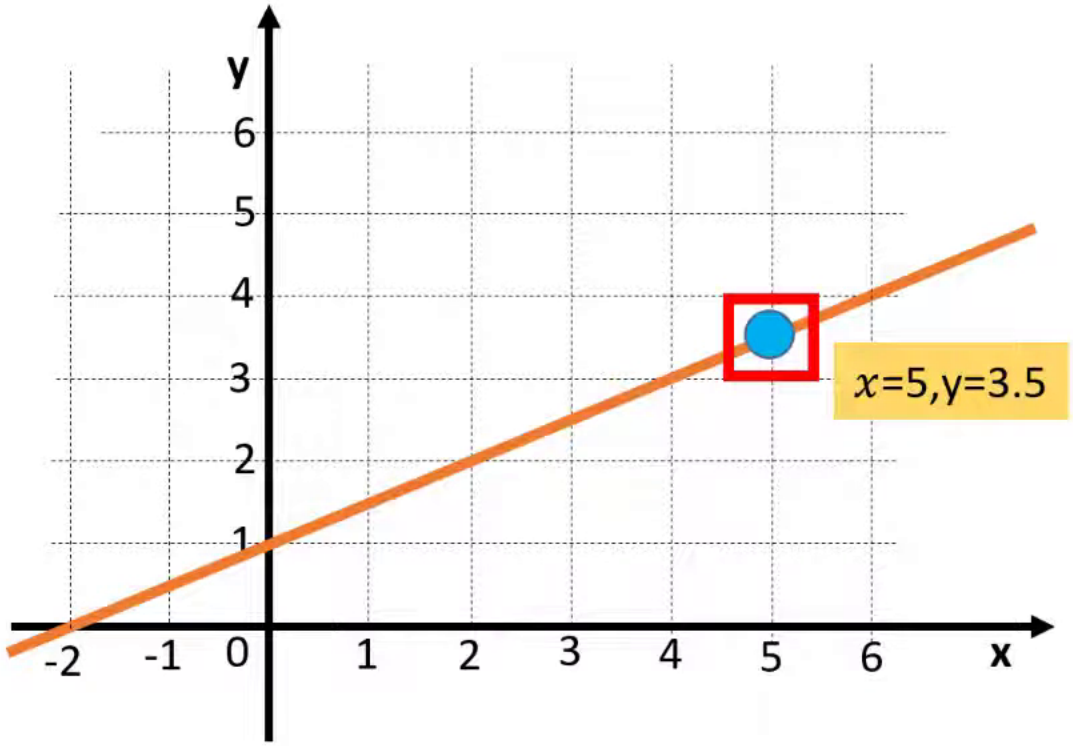
$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-2 \cdot 5 + 4 \cdot 3 - 4 = -2$$

In comparison, this data point results in a negative value, which tells us that the data point is below the line. This is handy when we later will classify data points based on the equation of the line.

The equation of a straight line



$$Ax + By + C = 0$$

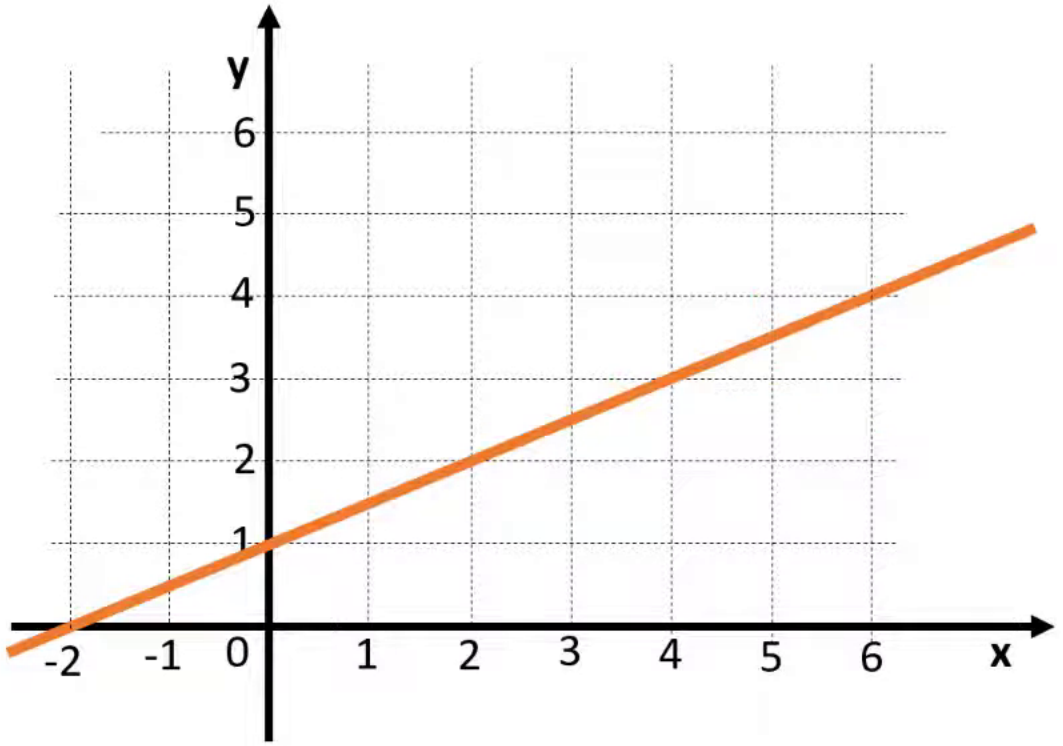
$$-2x + 4y - 4 = 0$$

$$-2 \cdot 5 + 4 \cdot 3.5 - 4 = 0$$

Note that any data point on the line will result in a value of zero.

The equation of a straight line

$$y = 0.5x + 1$$



$$Ax + By + C = 0$$

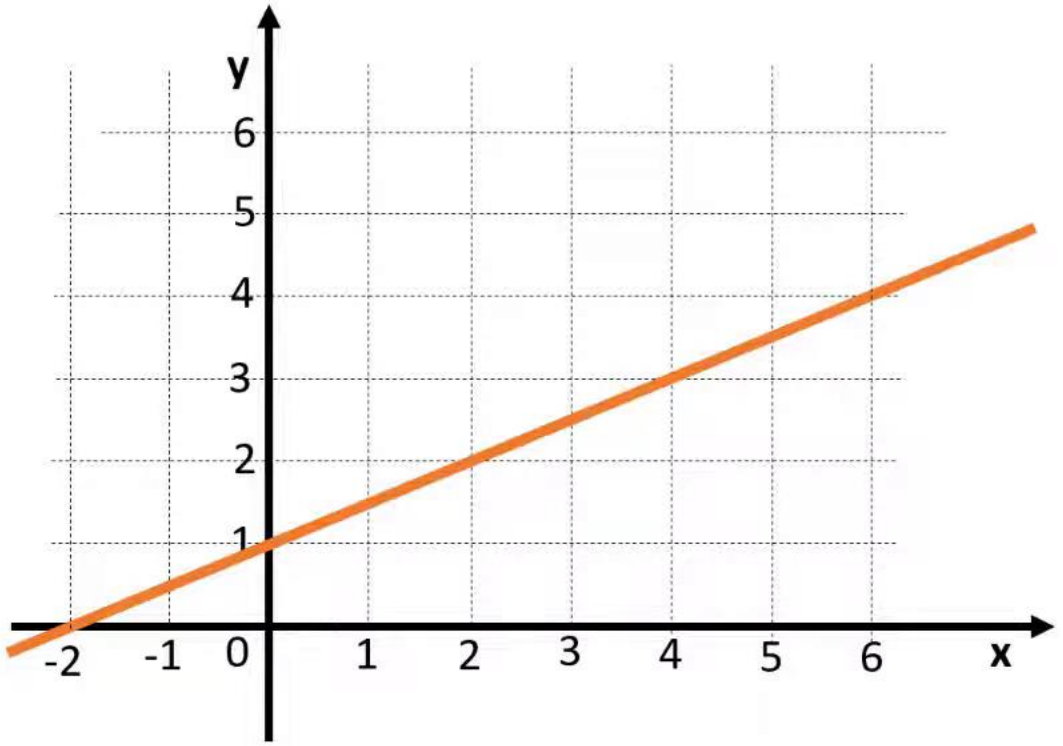
$$-2x + 4y - 4 = 0$$

$$-2x + 4y - 4 = 1$$

Let's see what happens if we add a one to the right-hand side of this equation.

The equation of a straight line

$y = 0.5x + 1$



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

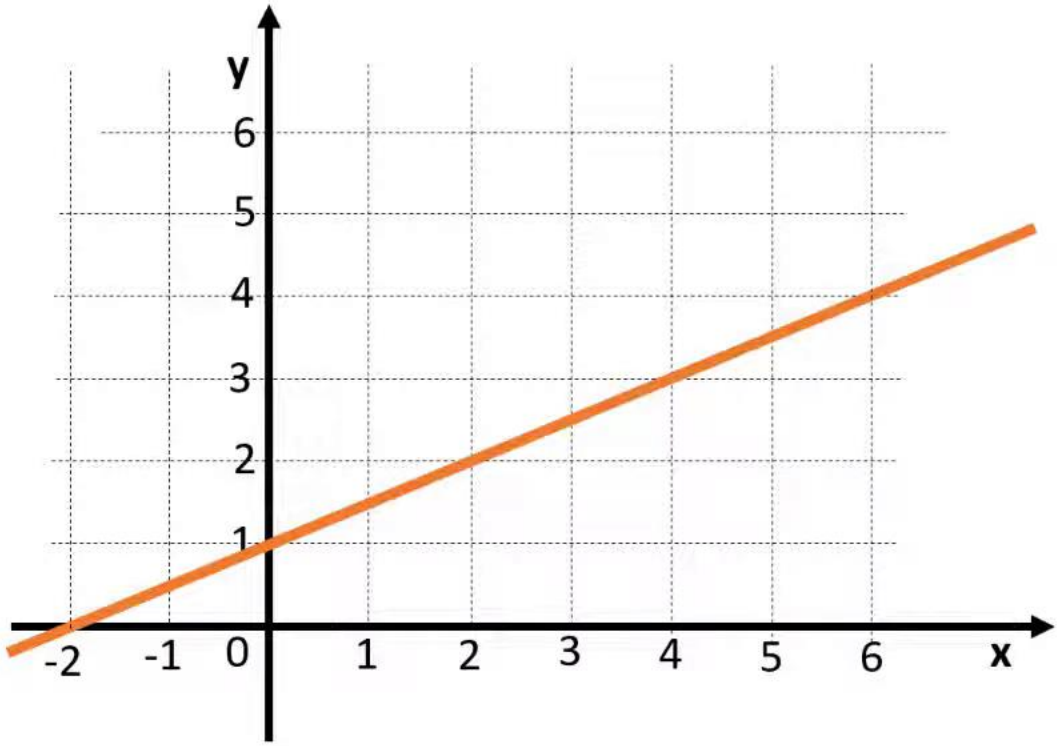
$$-2x + 4y - 4 = 1$$

$$4y = 2x + 5$$

$$y = 0.5x + 1.25$$

If we then solve for y,

The equation of a straight line



$y = 0.5x + 1$

$Ax + By + C = 0$

$-2x + 4y - 4 = 0$

$-2x + 4y - 4 = 1$

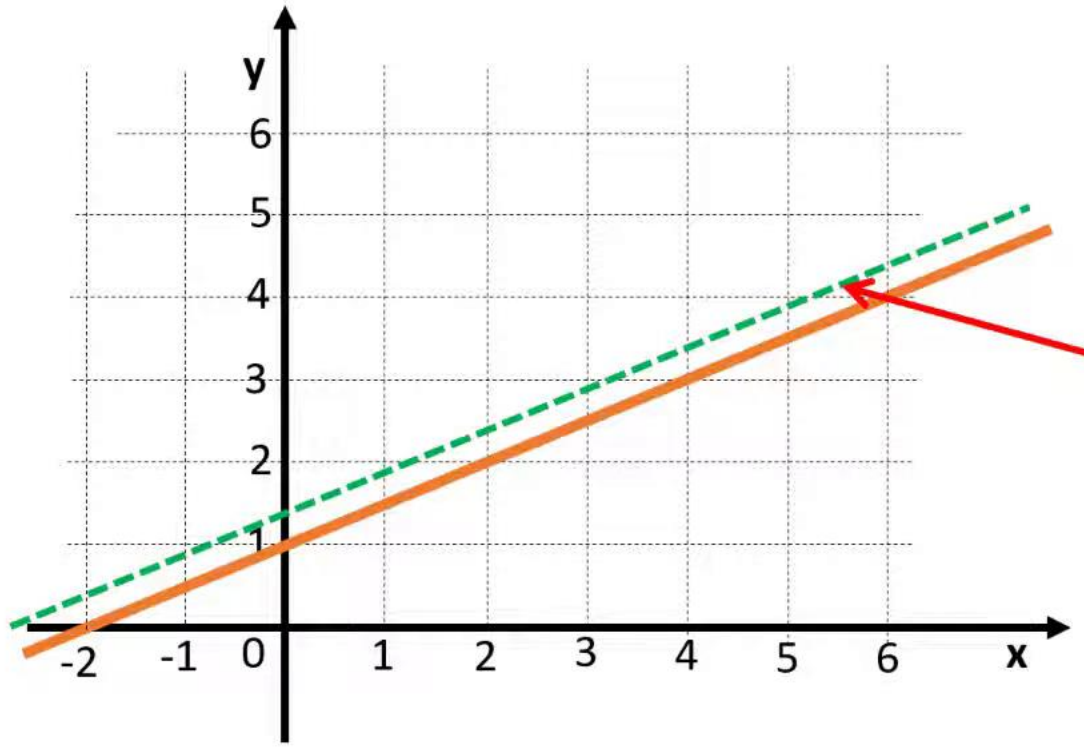
$4y = 2x + 5$

$y = 0.5x + 1.25$

we see that the slope is the same but that the intercept has increased from one to 1.25,

The equation of a straight line

$y = 0.5x + 1$



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$-2x + 4y - 4 = 1$$

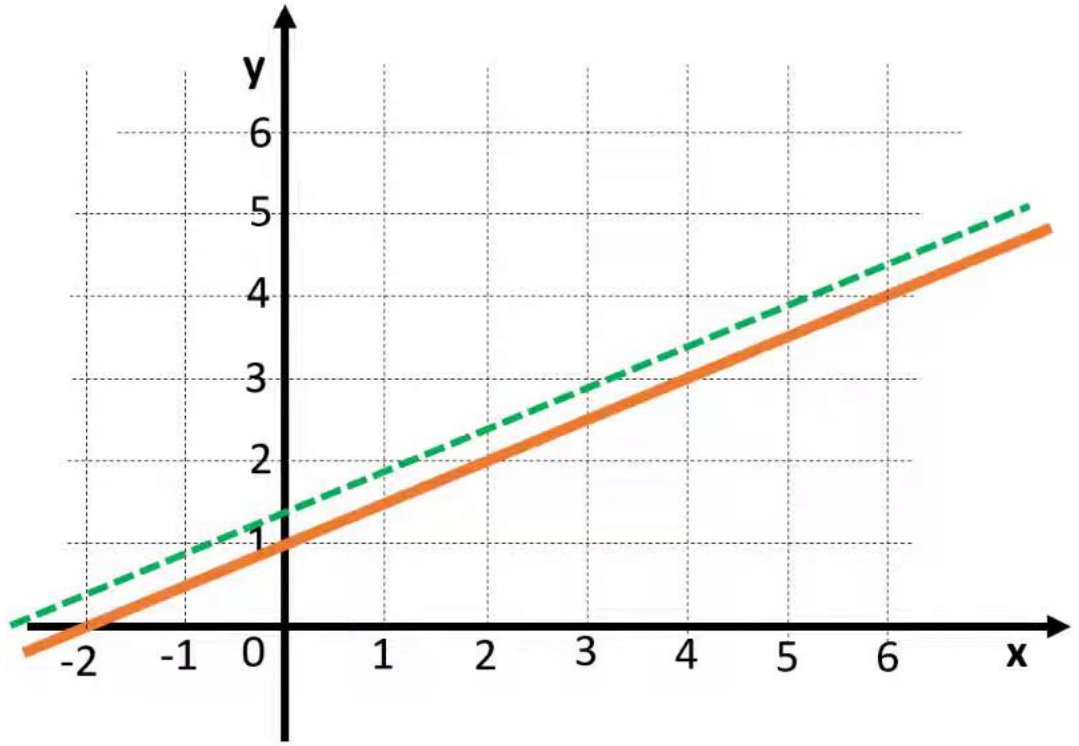
$$4y = 2x + 5$$

$$y = 0.5x + 1.25$$

which results in the following line.

The equation of a straight line

$y = 0.5x + 1$



$$Ax + By + C = 0$$

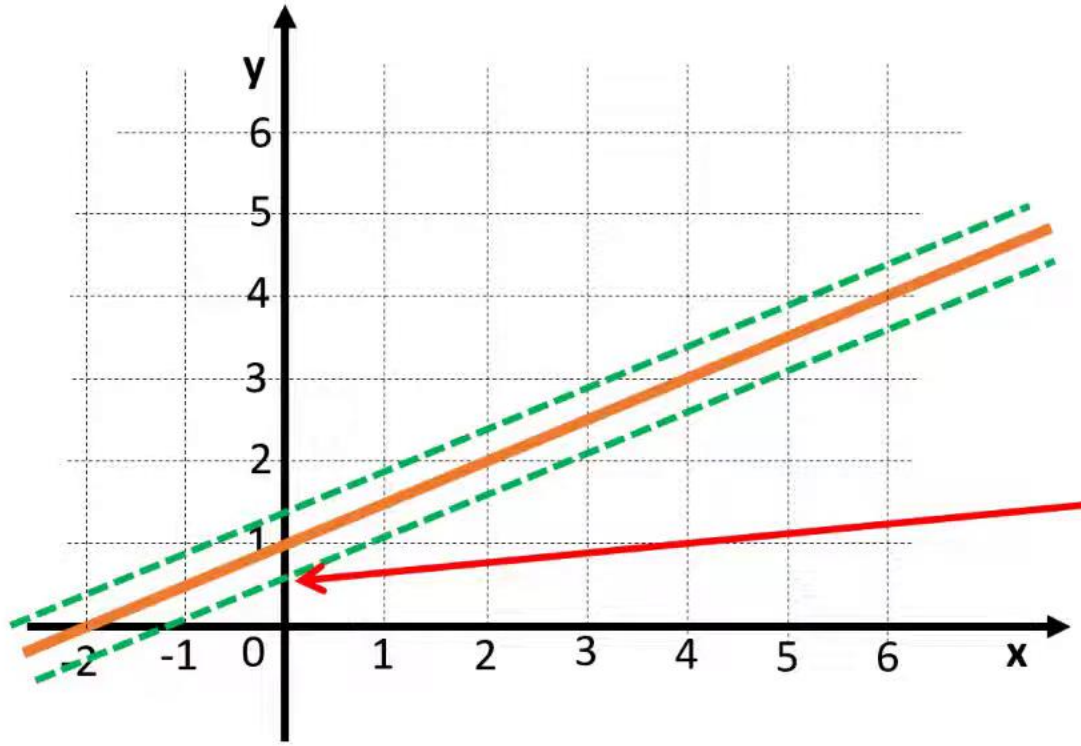
$$-2x + 4y - 4 = 0$$

$$-2x + 4y - 4 = -1$$

Similarly, if we add negative one here,

The equation of a straight line

$y = 0.5x + 1$



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

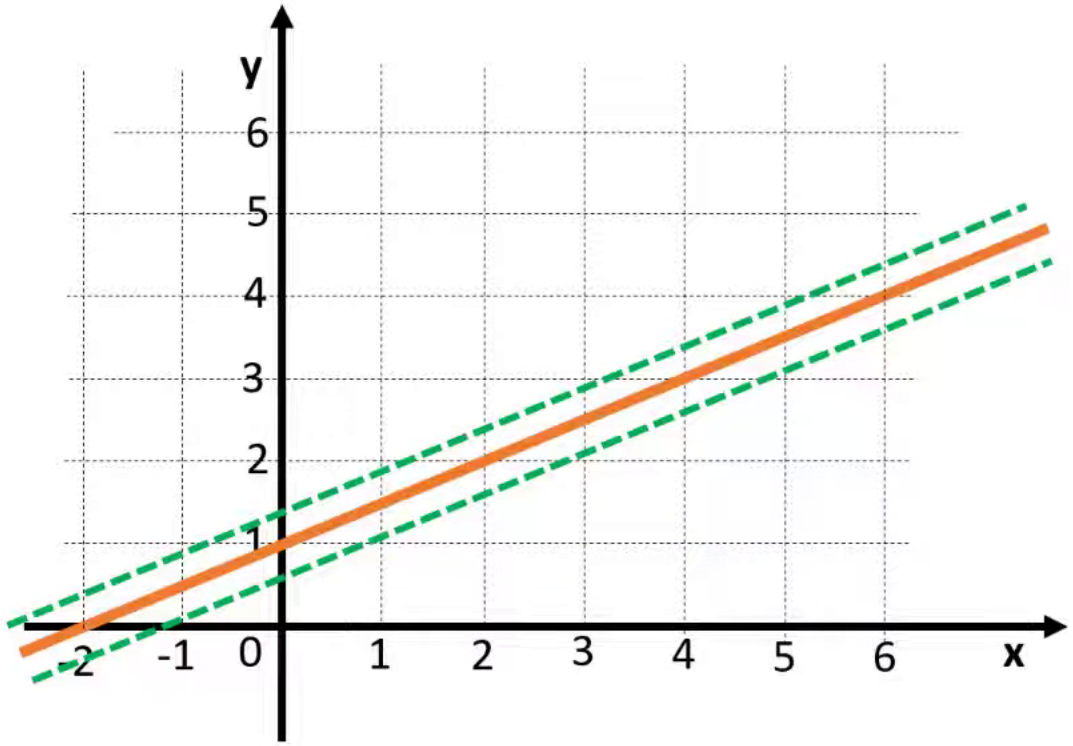
$$-2x + 4y - 4 = -1$$

$$4y = 2x + 3$$

$$y = 0.5x + 0.75$$

we will get this line that intercepts the y-axis at 0.75.

The equation of a straight line

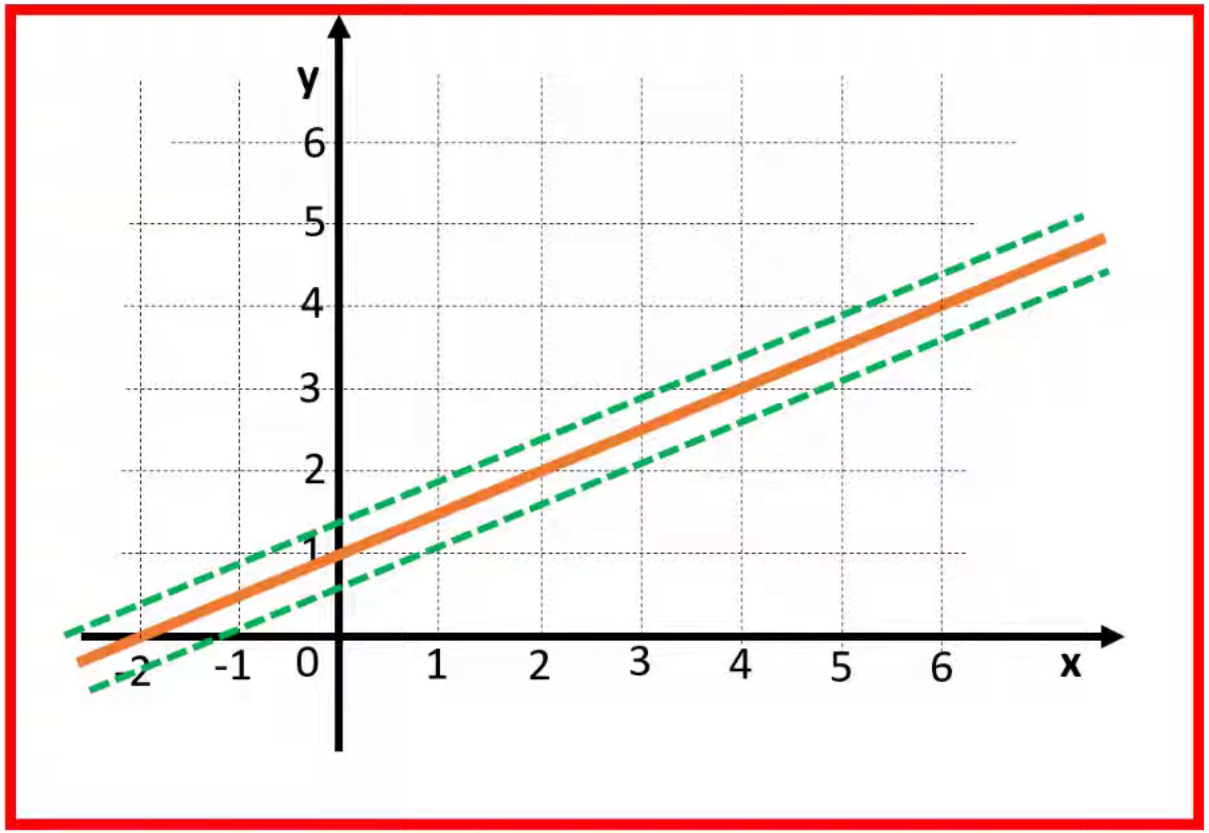


$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
$$-2x + 4y - 4 = -1$$

If we would multiply the terms of the left-hand side of these three equations with a factor smaller than 1,

The equation of a straight line

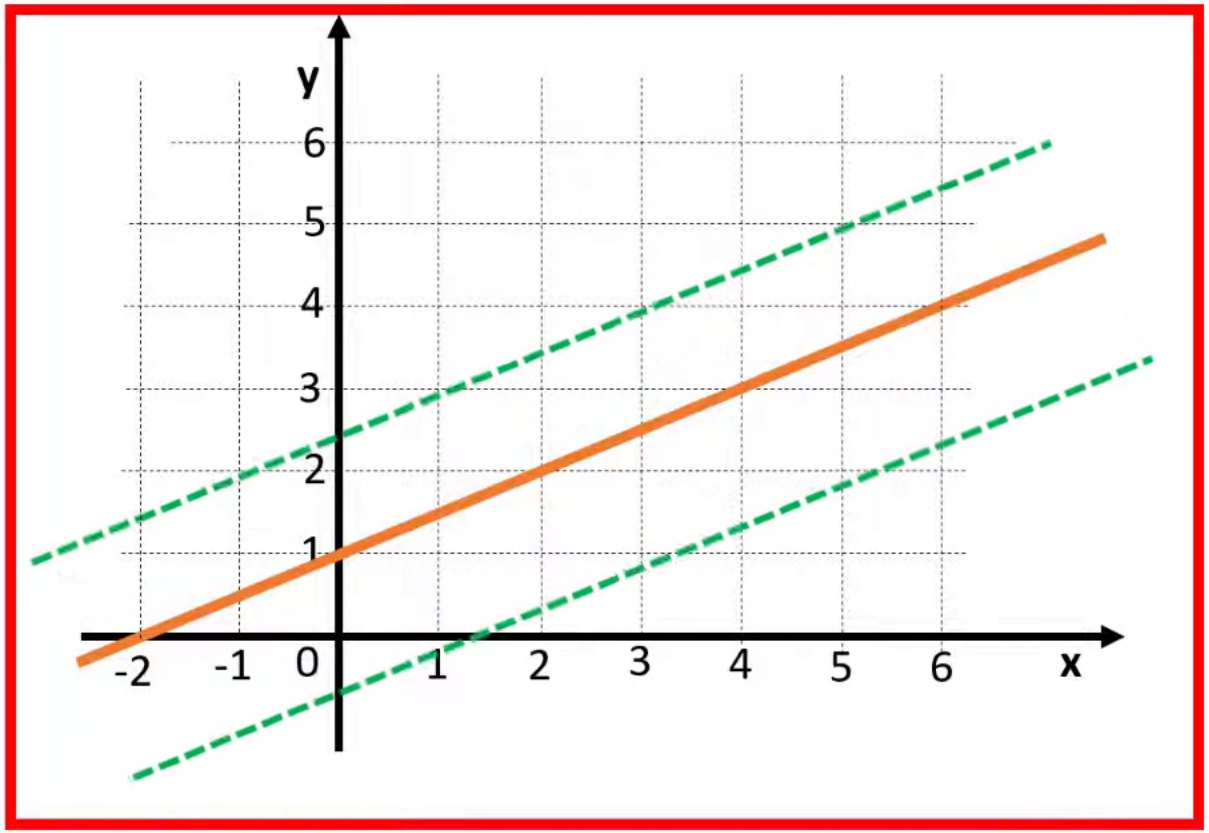


$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
$$-2x + 4y - 4 = -1$$

the green lines would move away from the original line, which stays in the same position.

The equation of a straight line

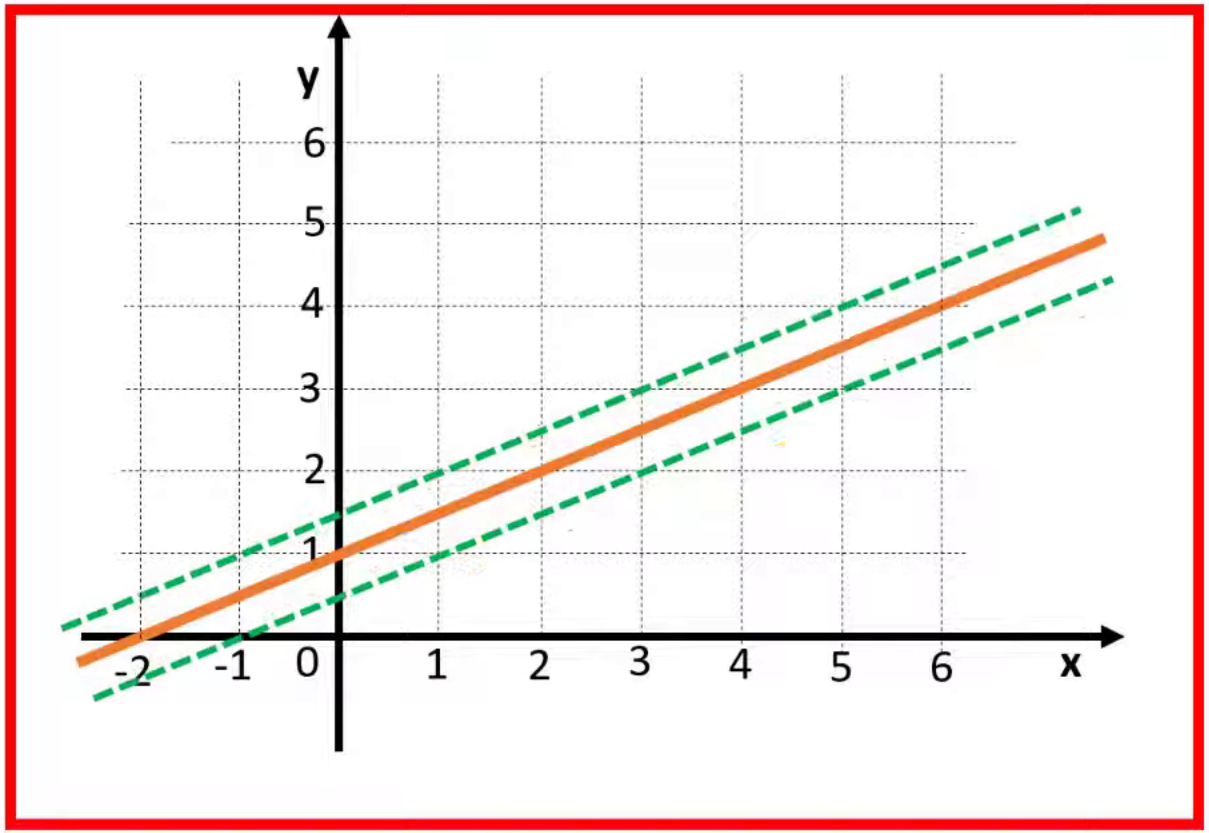


$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
$$-2x + 4y - 4 = -1$$

the green lines would move away from the original line, which stays in the same position.

The equation of a straight line

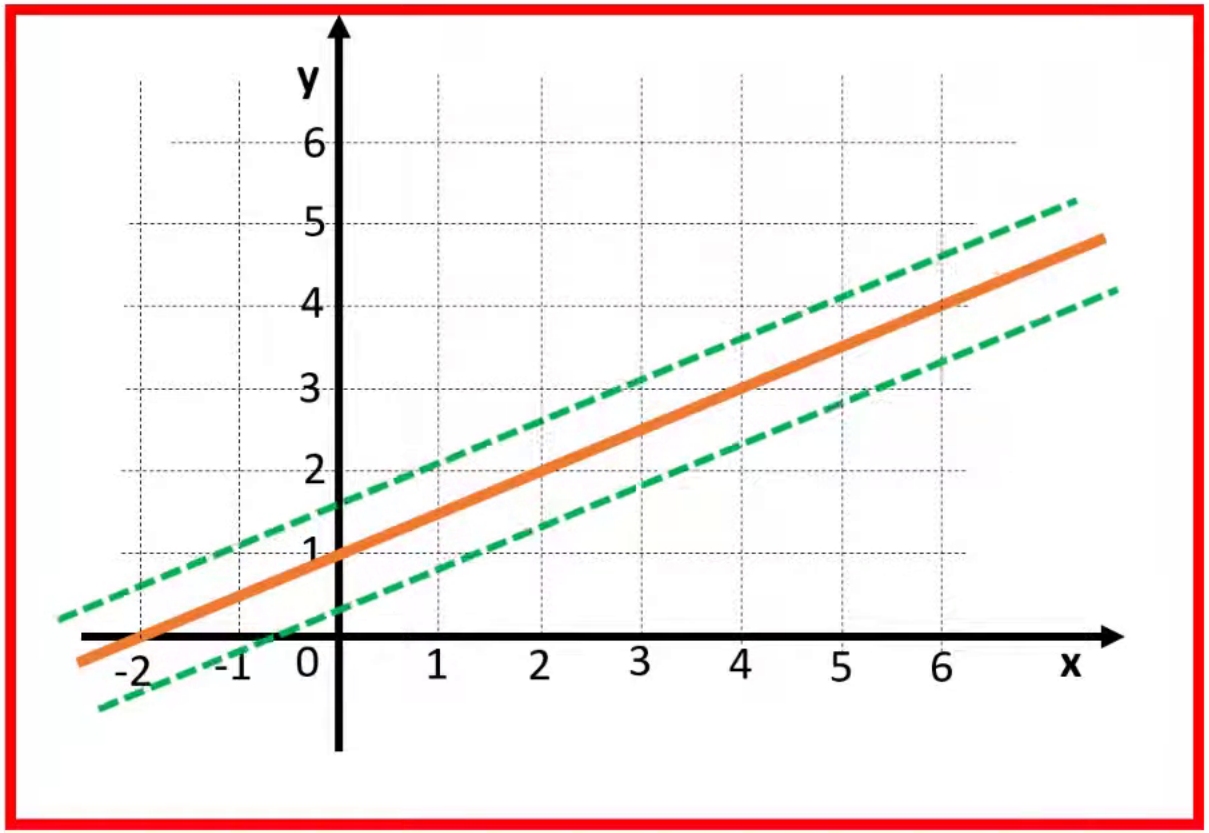


$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
$$-2x + 4y - 4 = -1$$

the green lines would move away from the original line, which stays in the same position.

The equation of a straight line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
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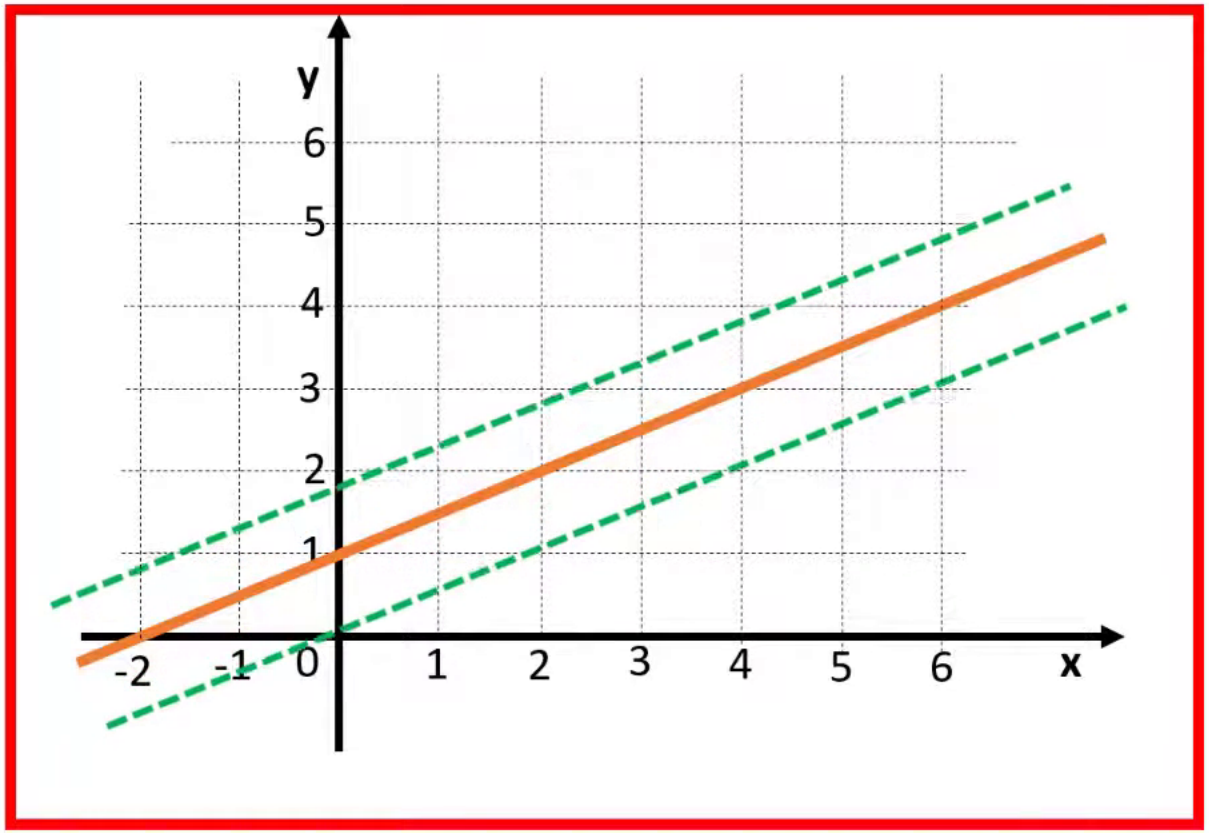
$$-2x + 4y - 4 = -1$$

the green lines would move away from the original line, which stays in the same position.

The equation of a straight line

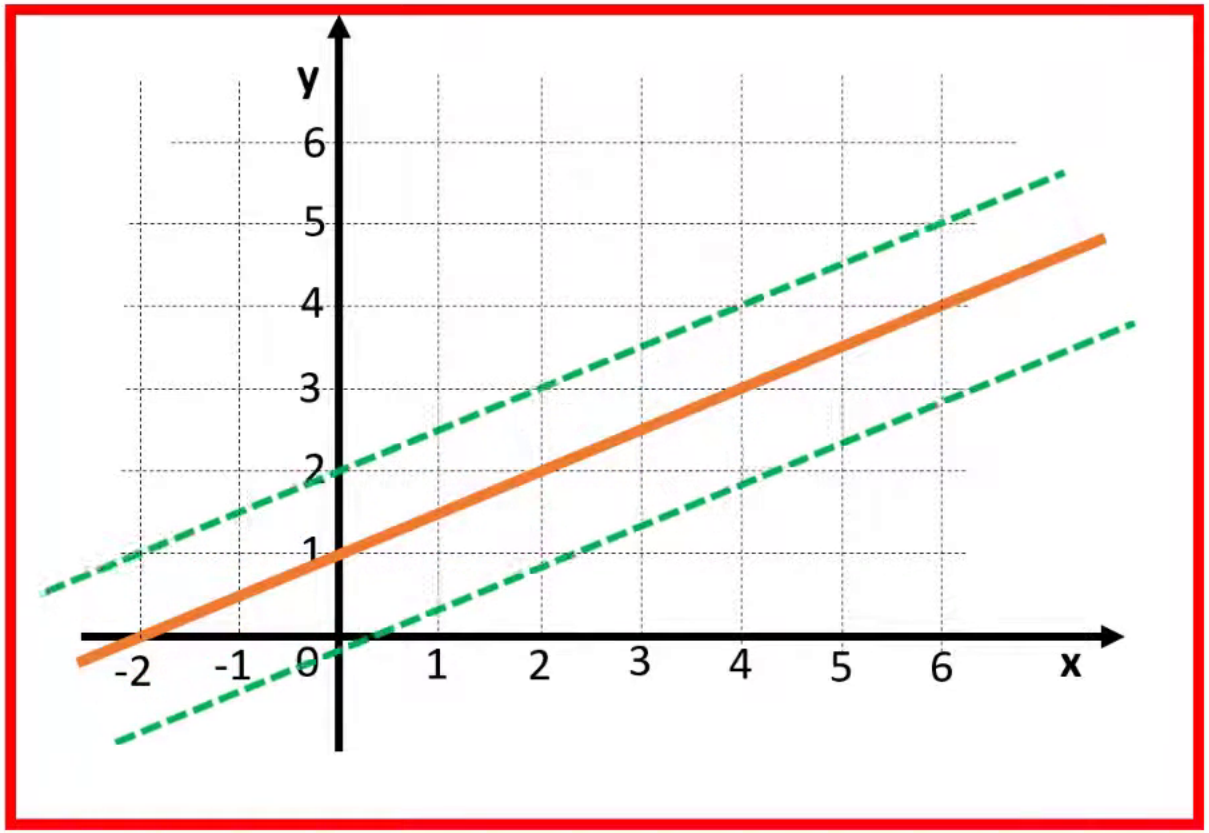


$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
$$-2x + 4y - 4 = -1$$

the green lines would move away from the original line, which stays in the same position.

The equation of a straight line

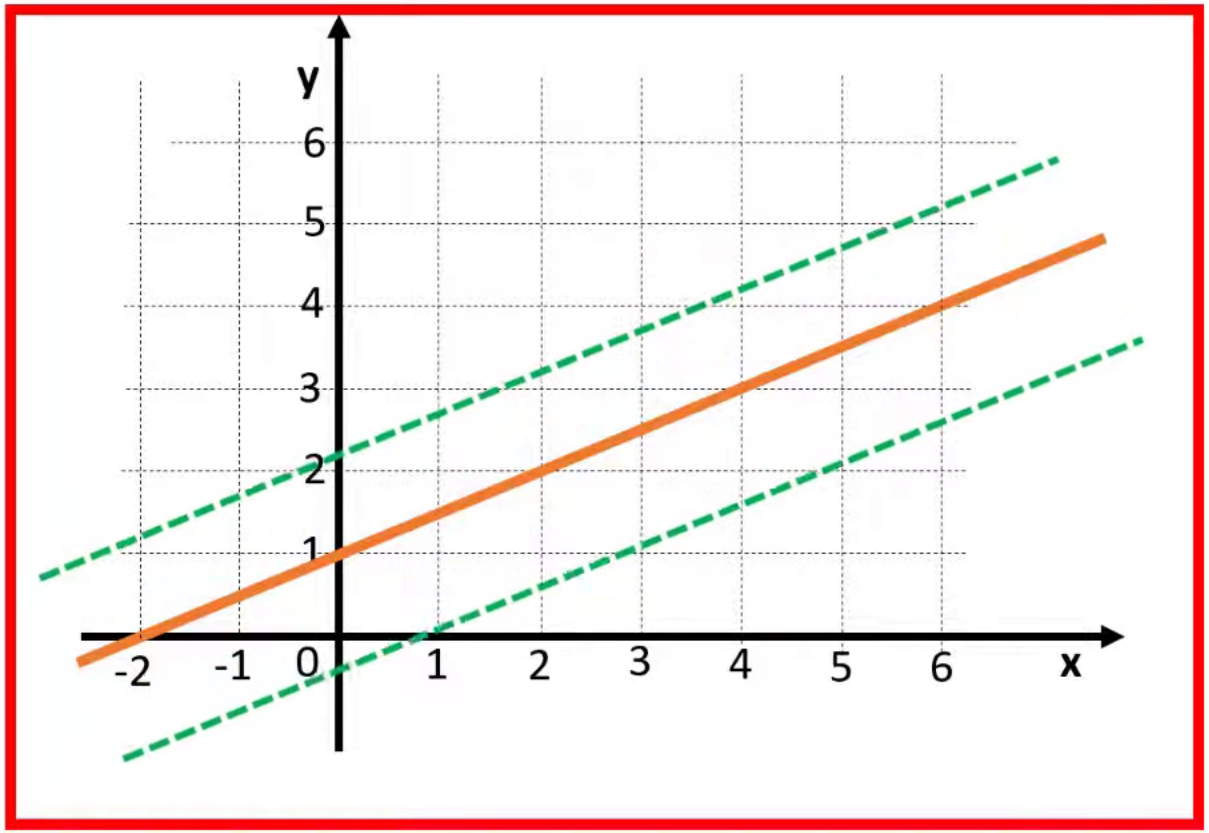


$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
$$-2x + 4y - 4 = -1$$

the green lines would move away from the original line, which stays in the same position.

The equation of a straight line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
$$-2x + 4y - 4 = -1$$

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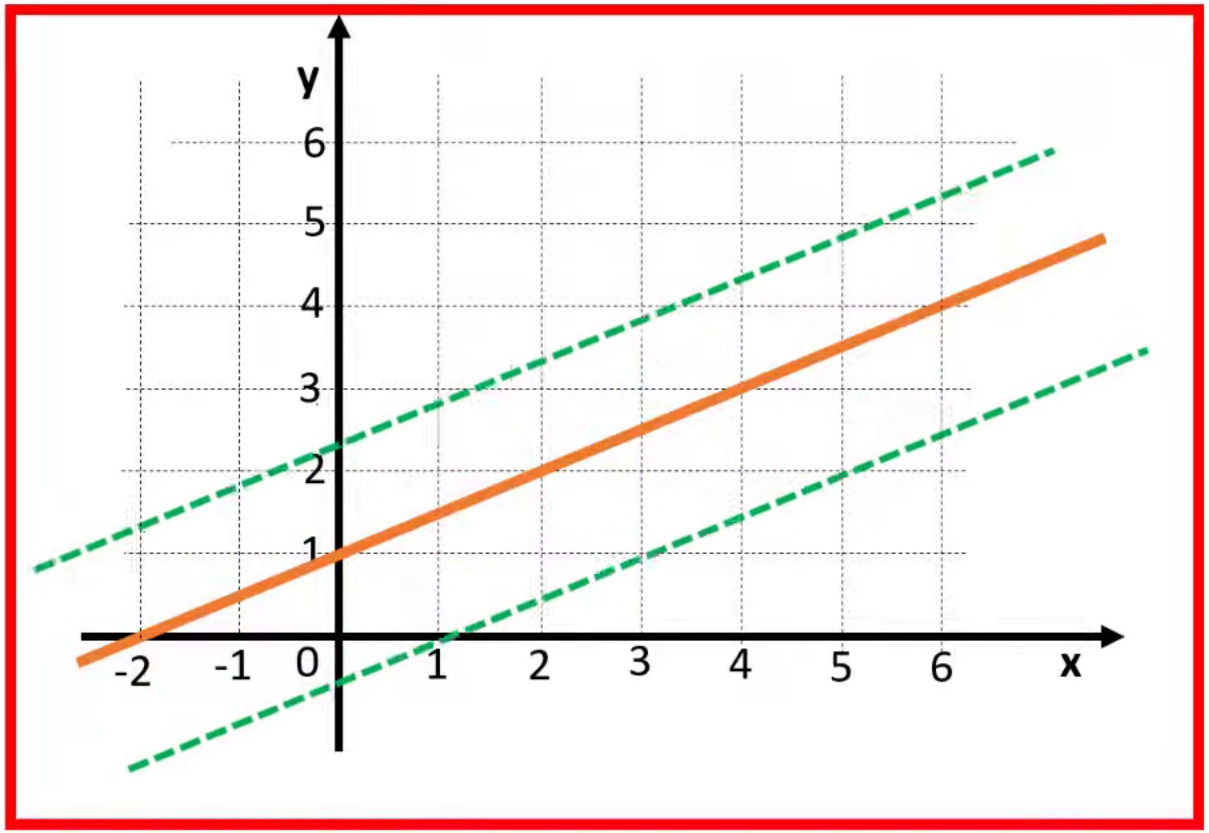
The equation of a straight line

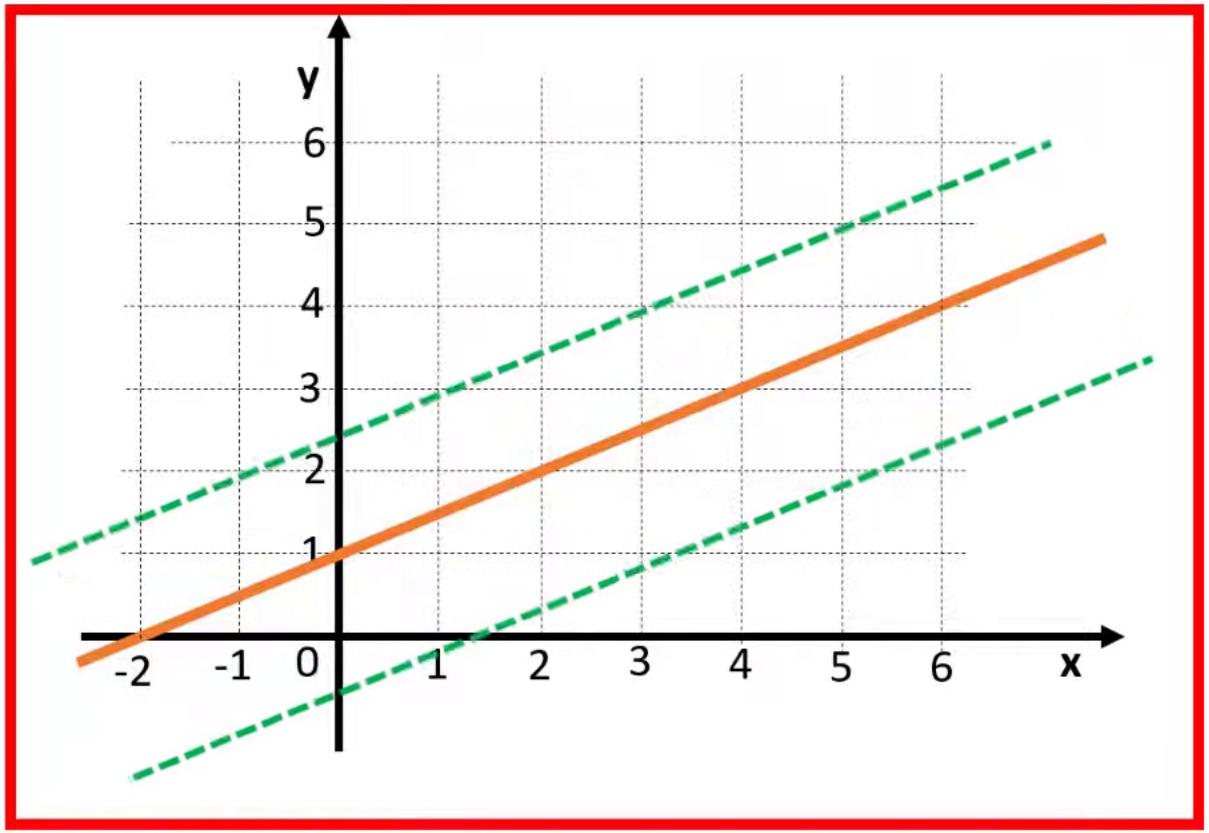


$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
$$-2x + 4y - 4 = -1$$

the green lines would move away from the original line, which stays in the same position.

The equation of a straight line



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$$-2x + 4y - 4 = 1$$
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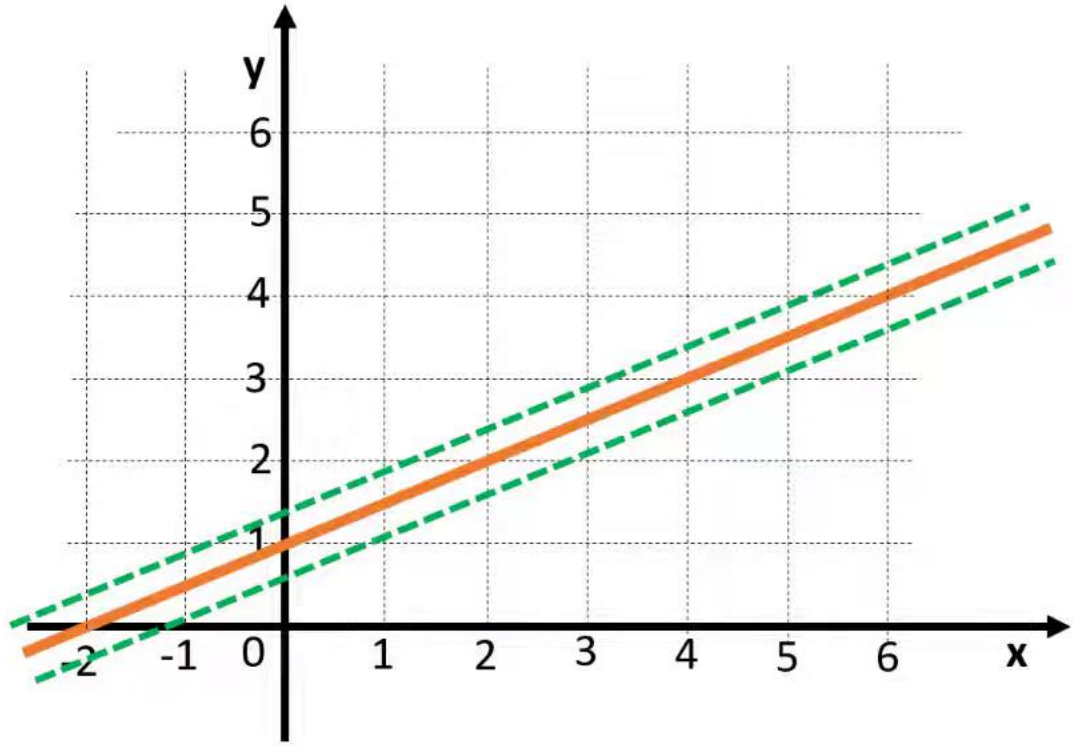
The equation of a straight line

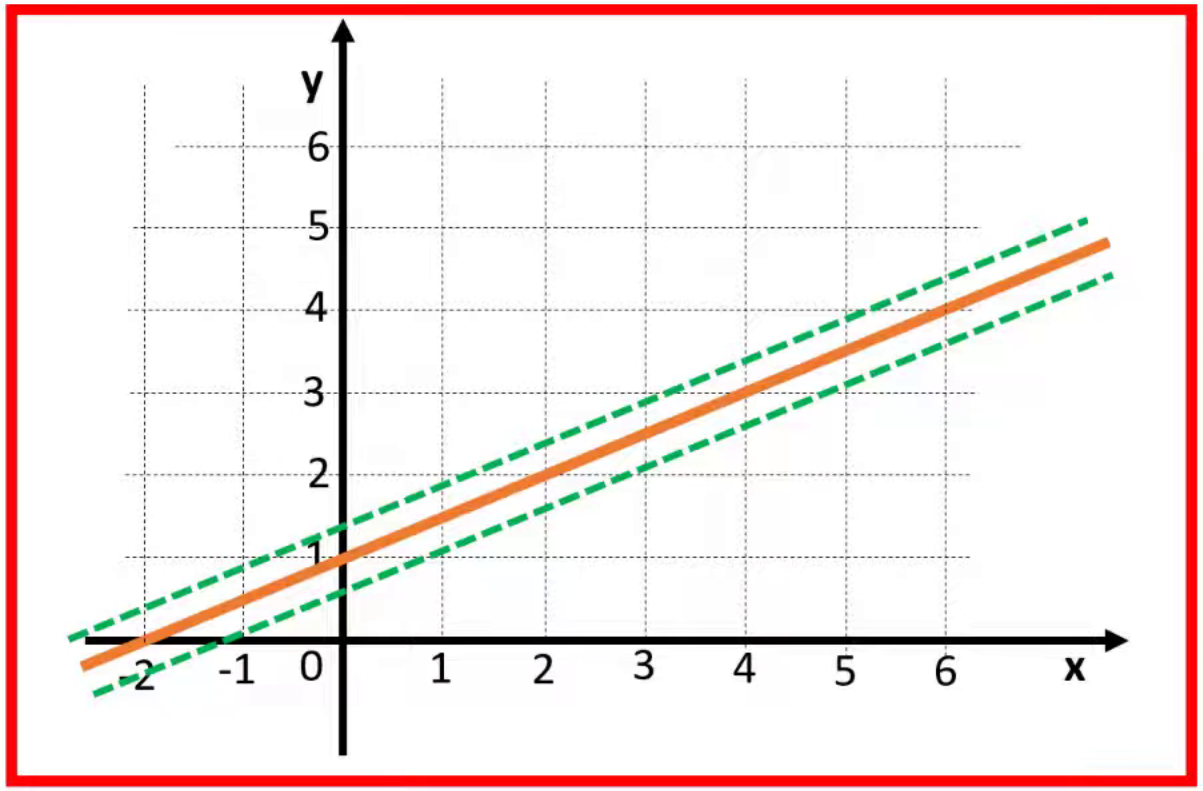


$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
$$-2x + 4y - 4 = -1$$

And if we would multiply with a factor greater than 1,

The equation of a straight line

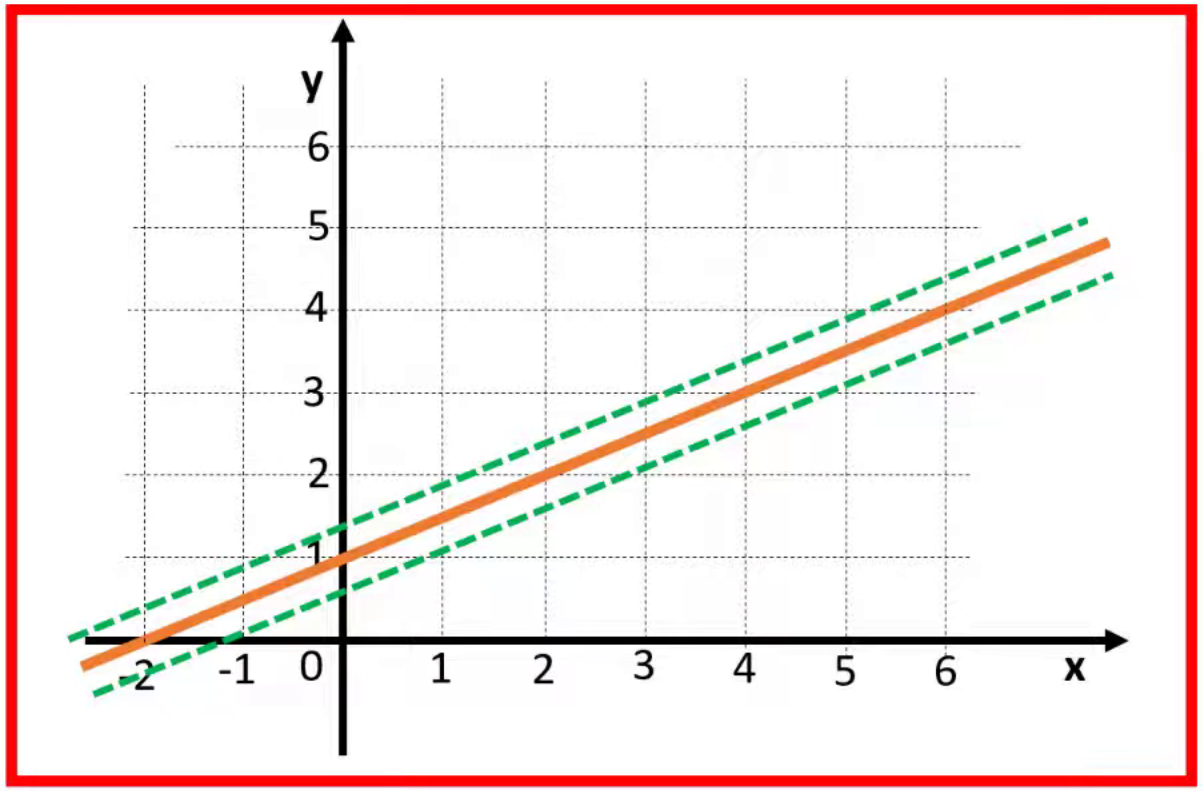


$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
$$-2x + 4y - 4 = -1$$

the green lines would move closer to the original line. This method is used on SVM to increase or decrease the so-called margin that we will discuss later on.

The equation of a straight line



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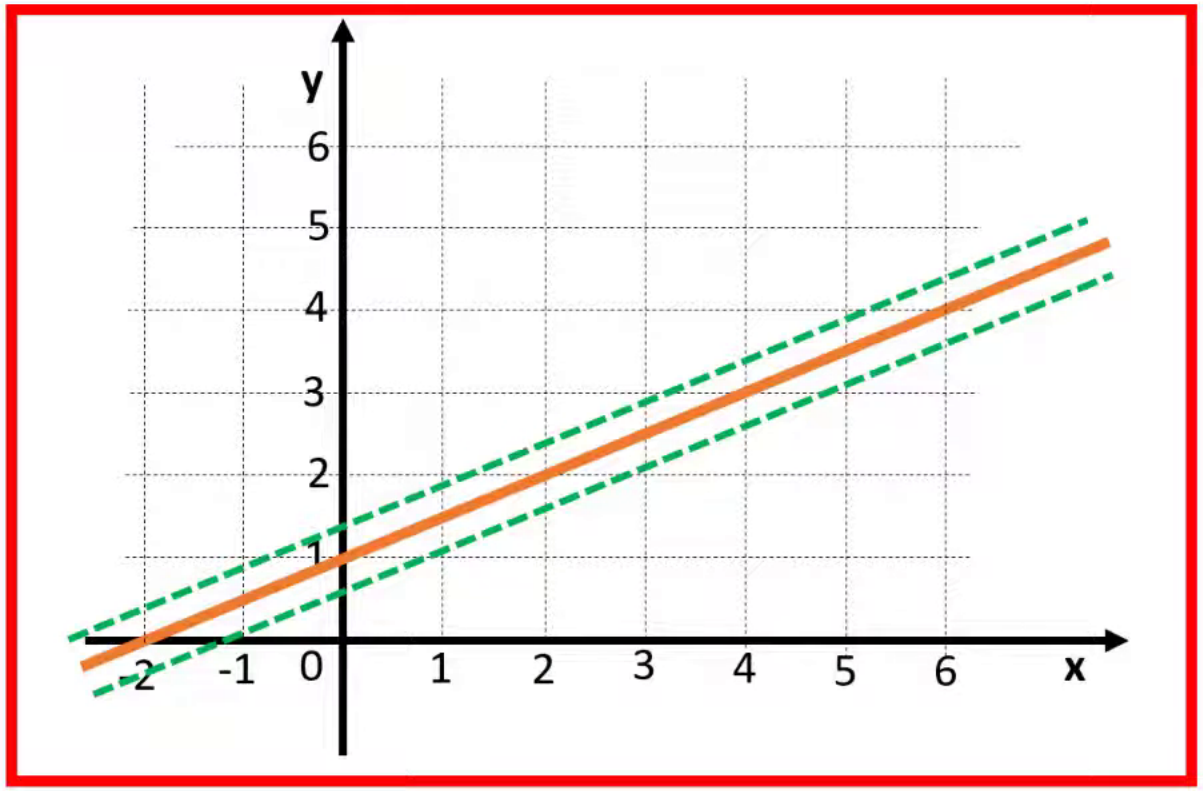

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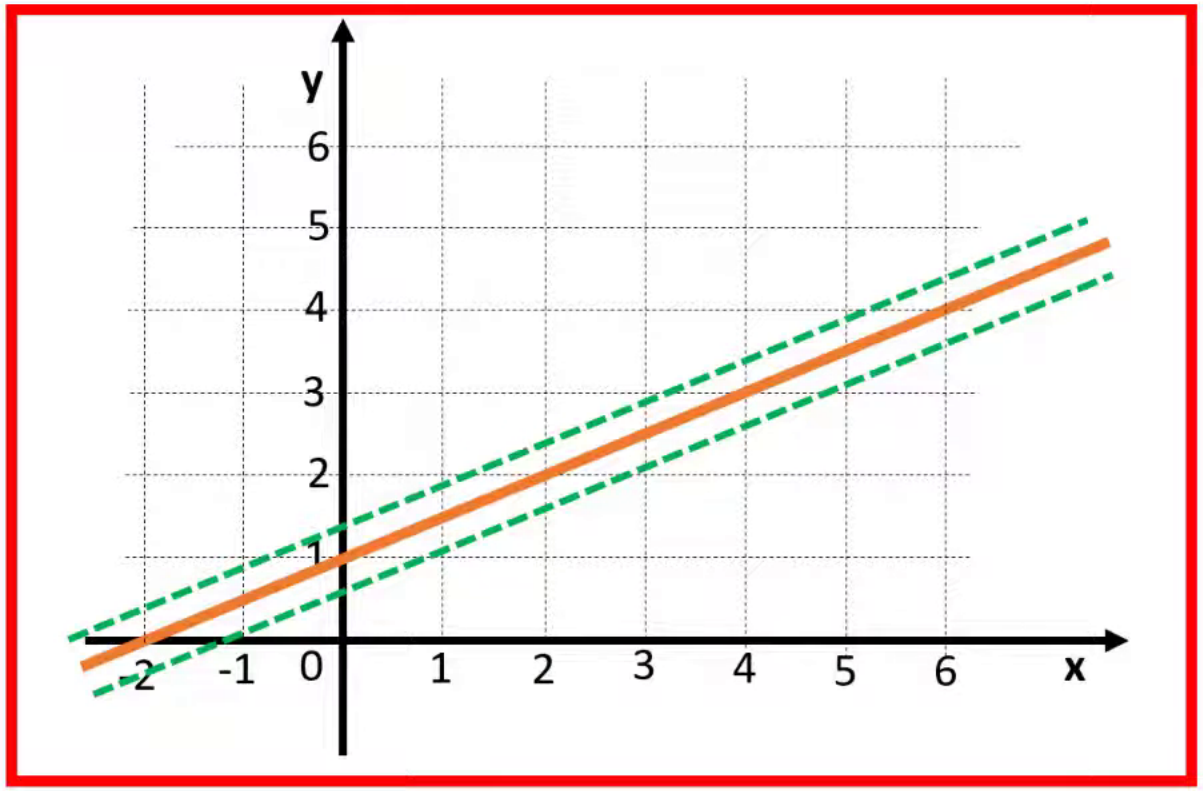

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$$-2x + 4y - 4 = 1$$
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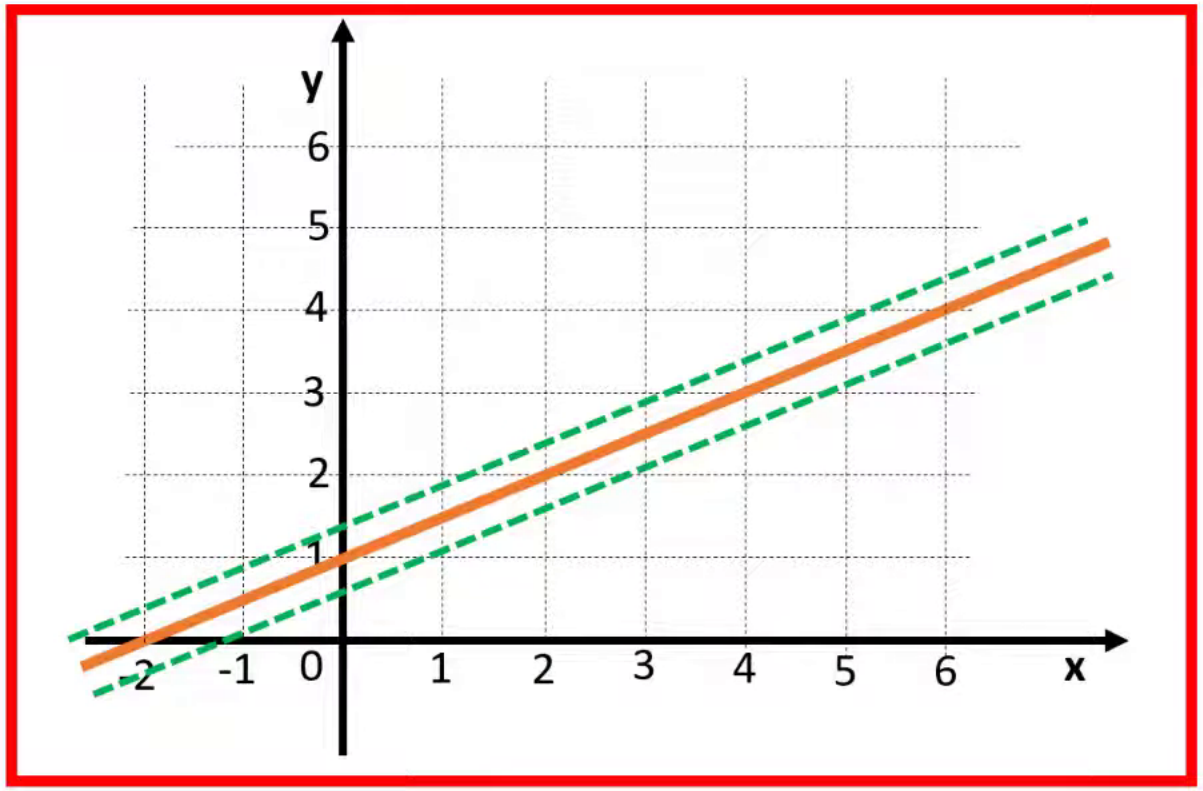

$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
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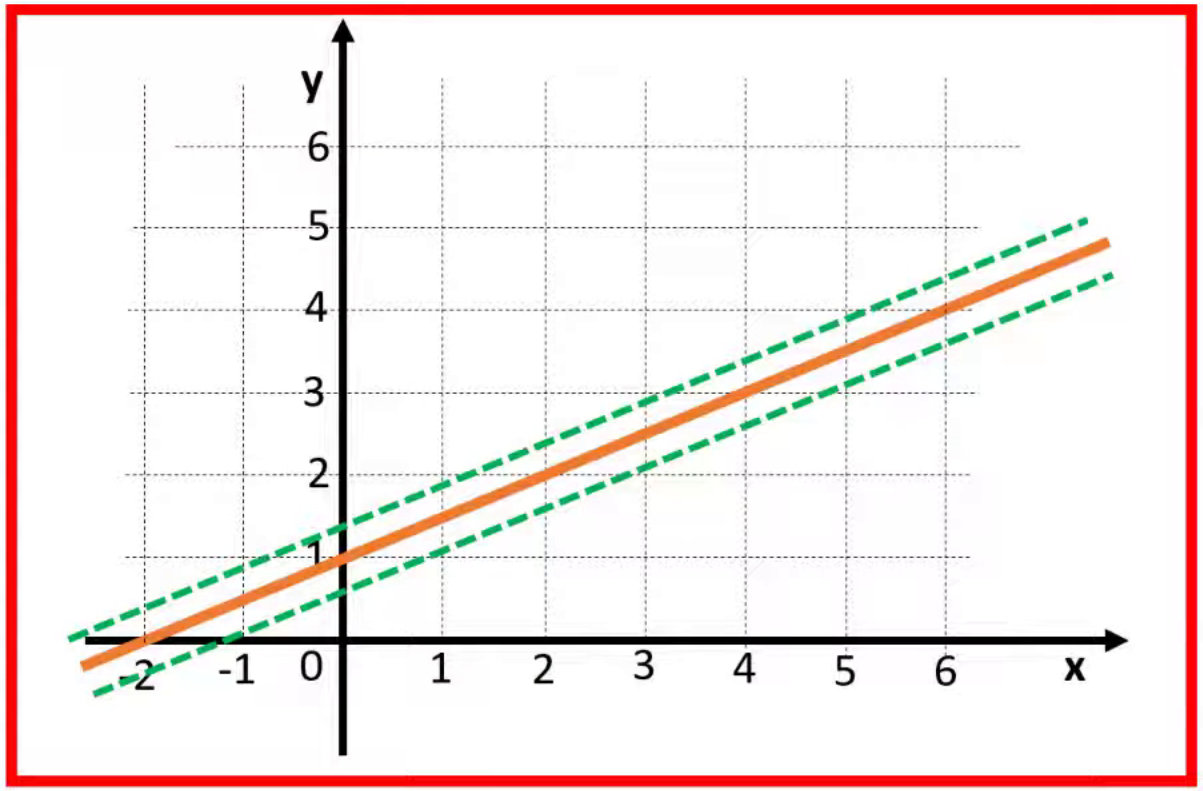

$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
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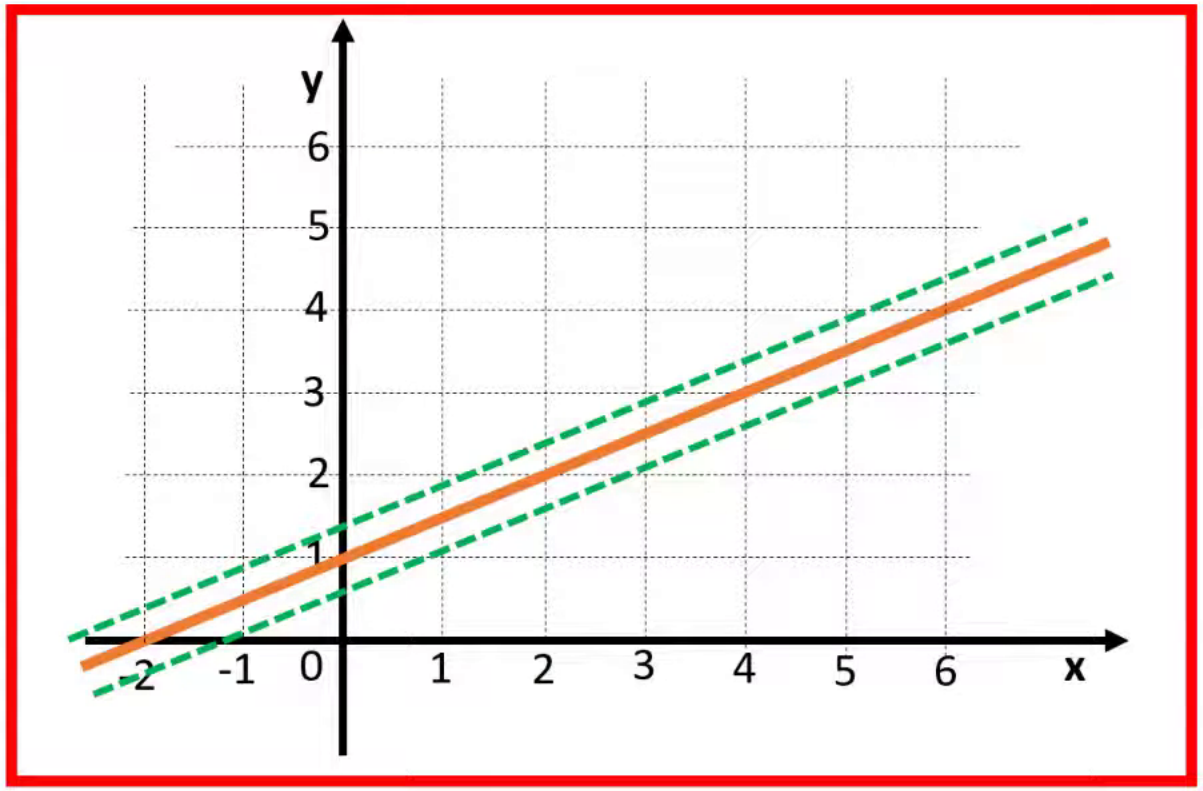

$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
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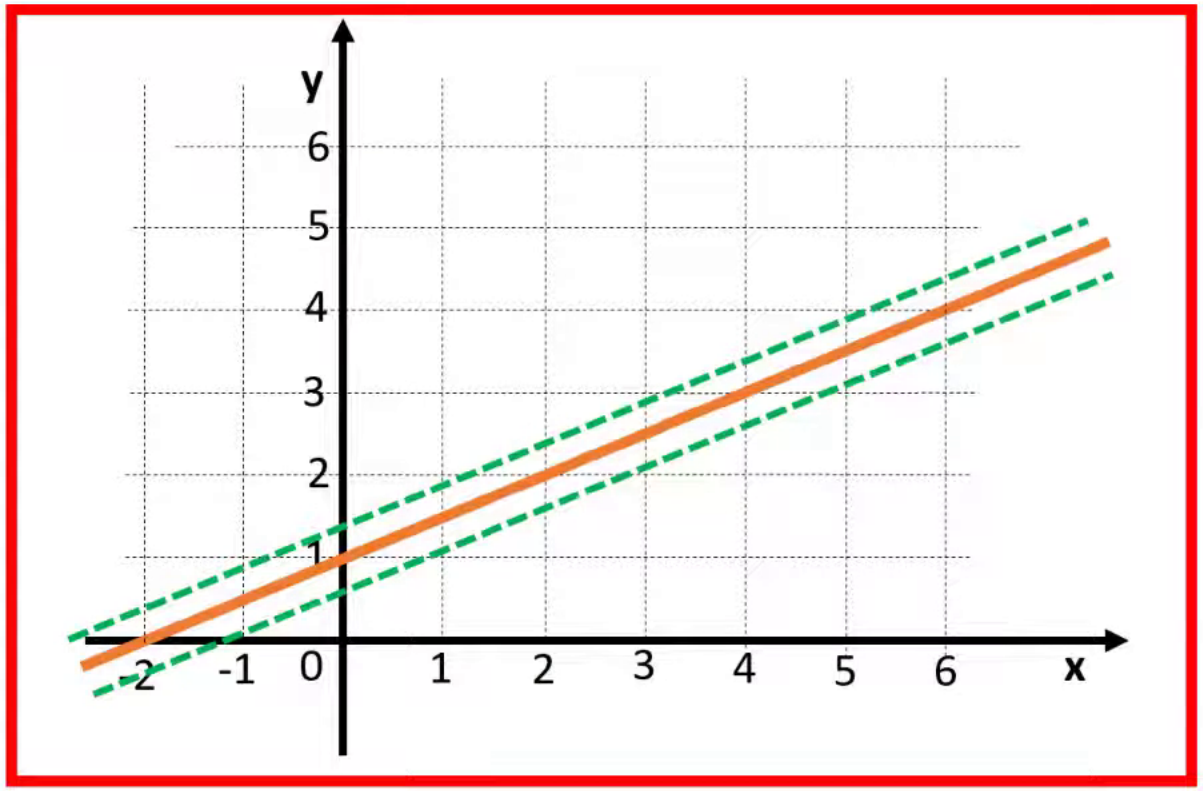

$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
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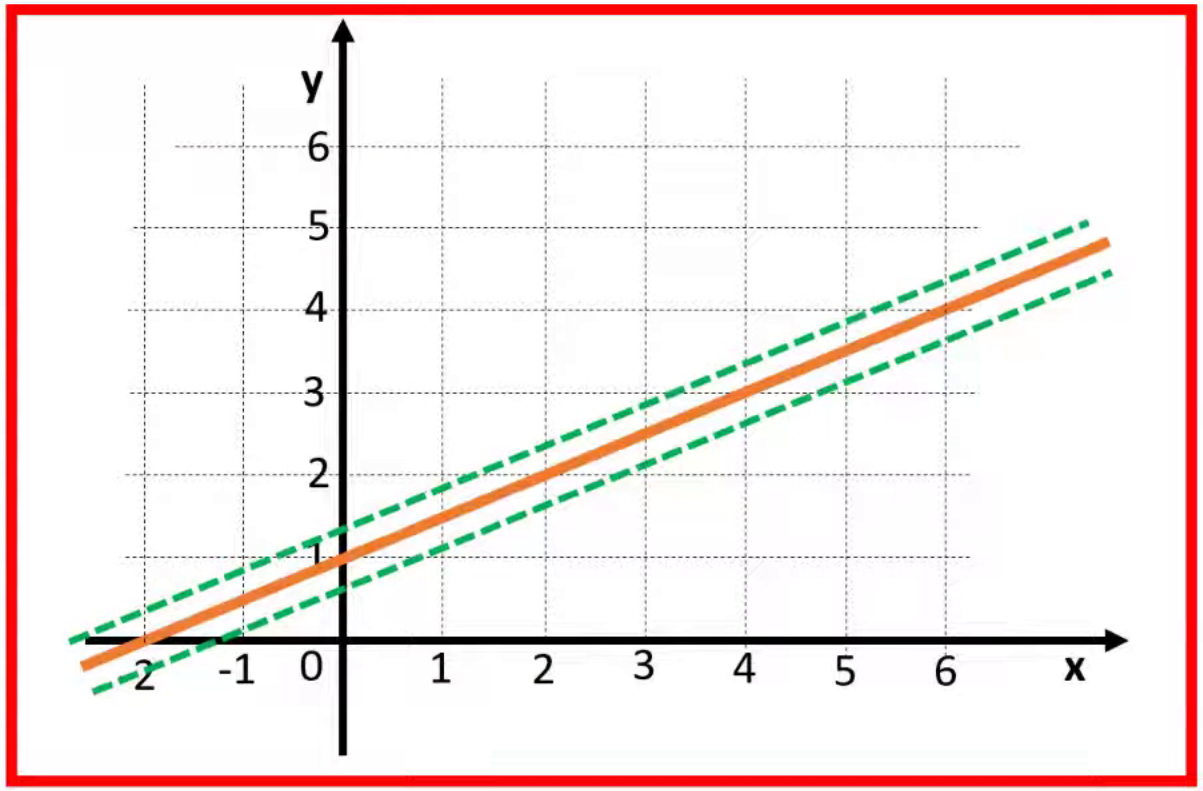
The equation of a straight line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
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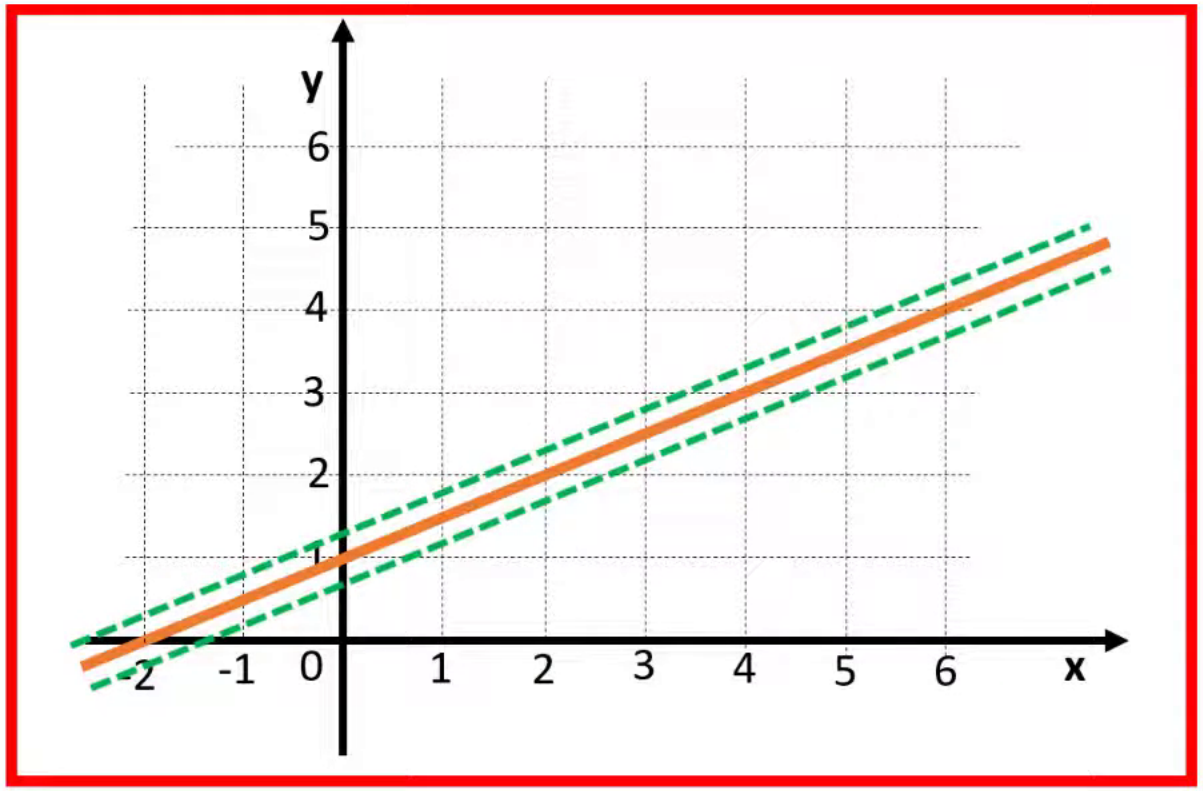
The equation of a straight line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
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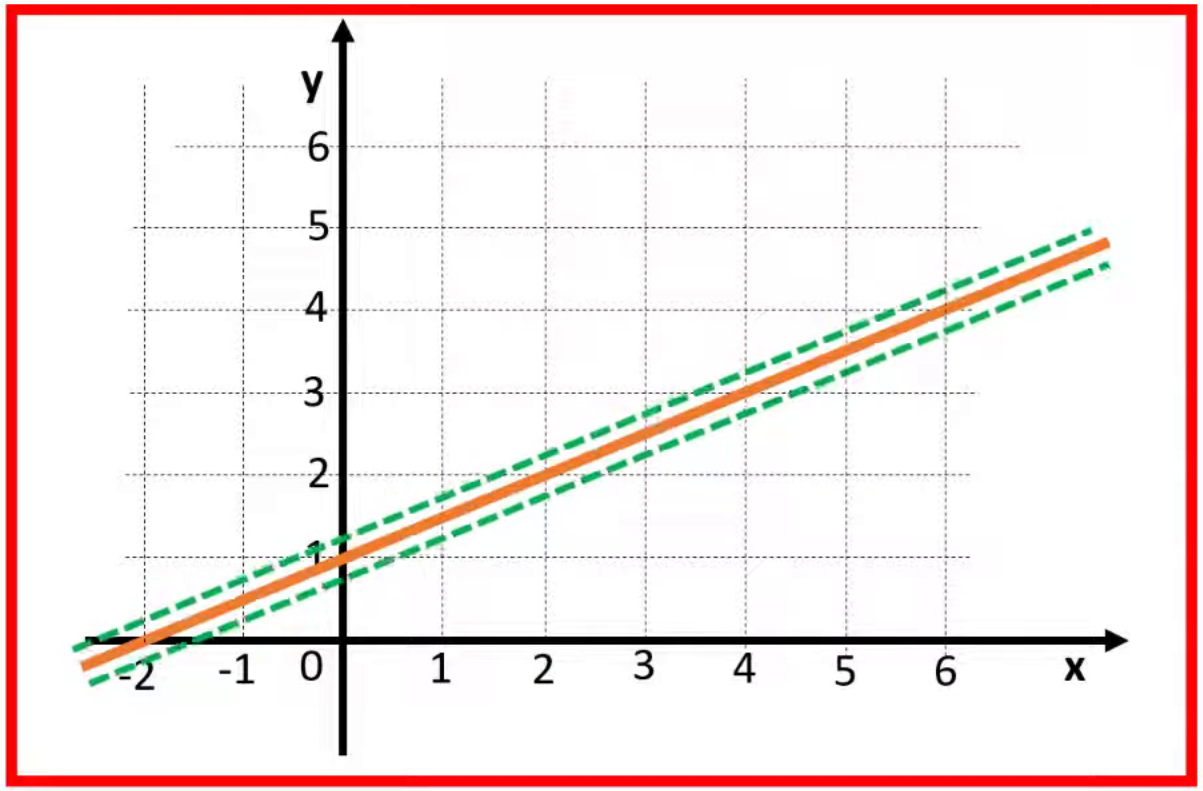
The equation of a straight line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
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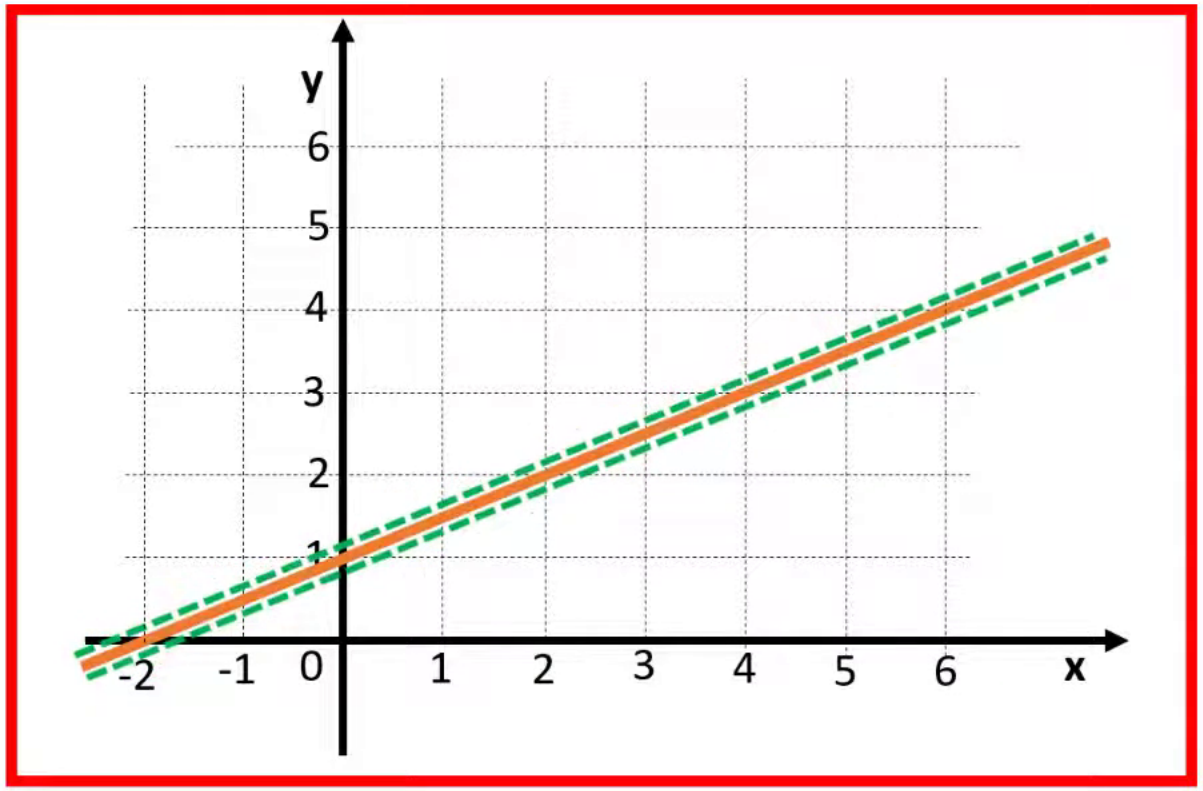

$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
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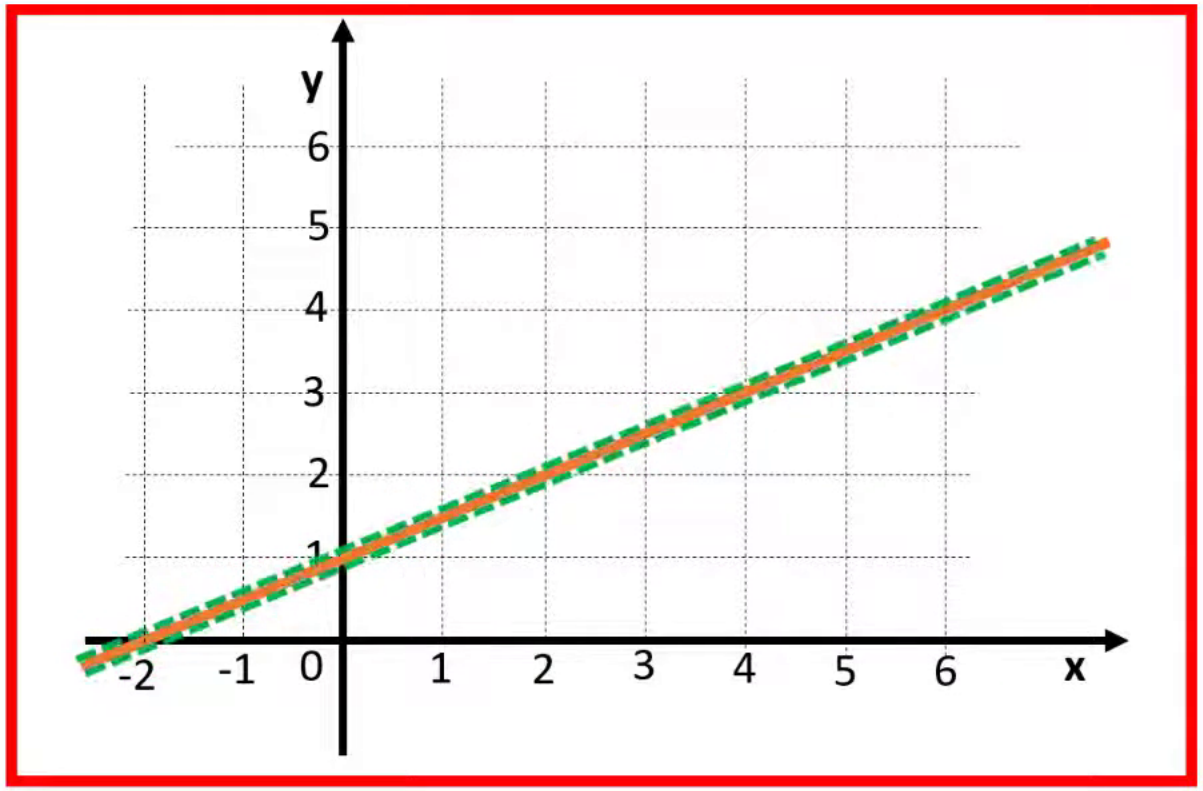

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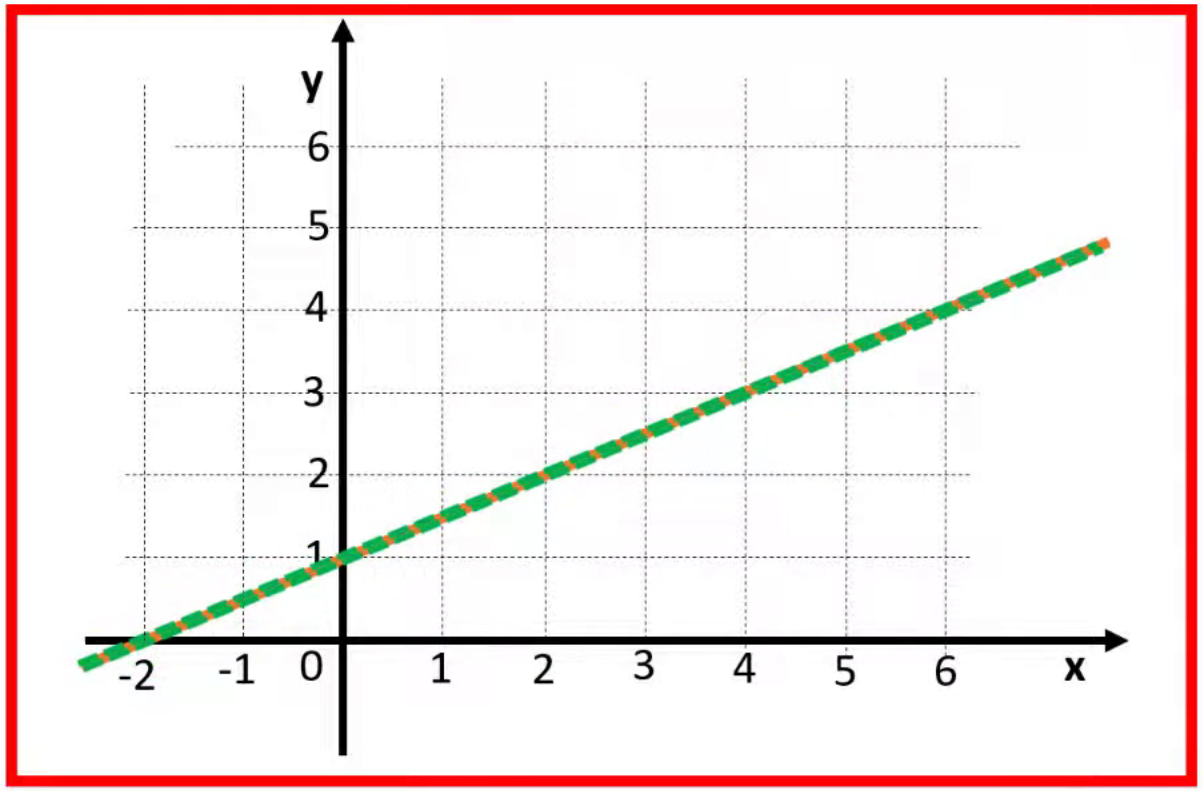

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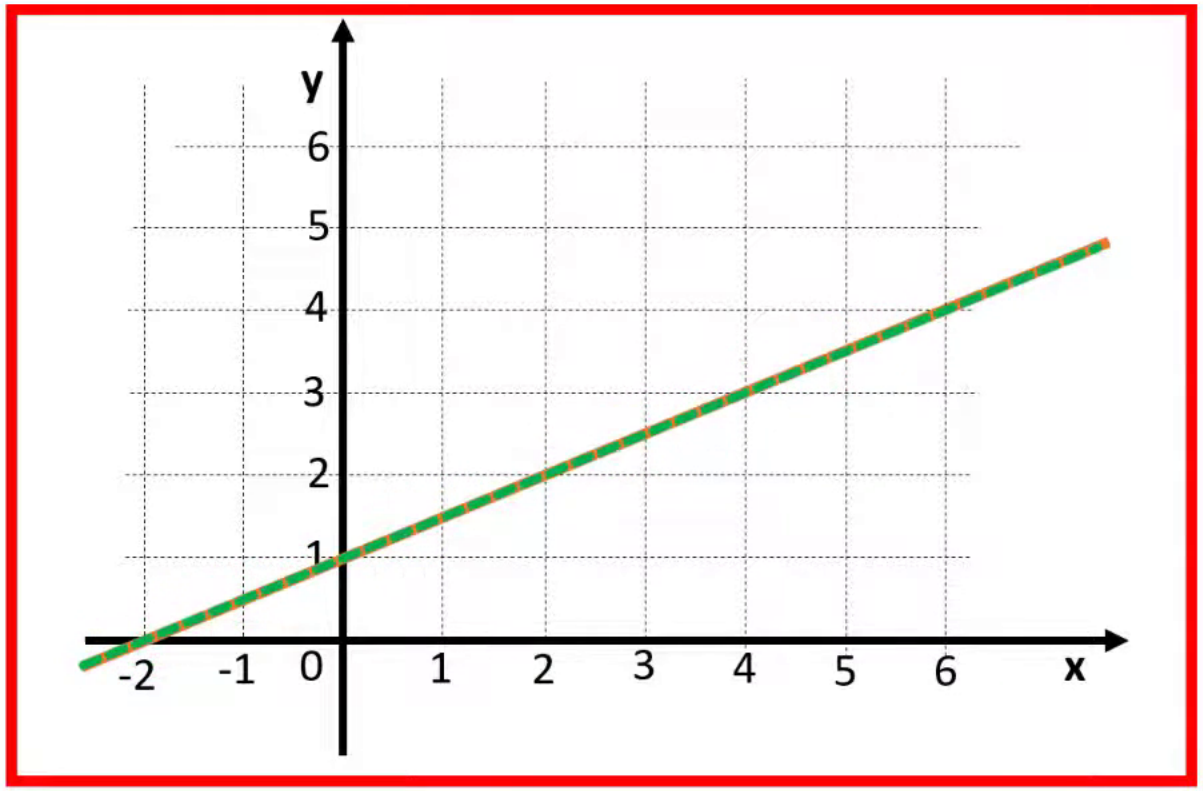

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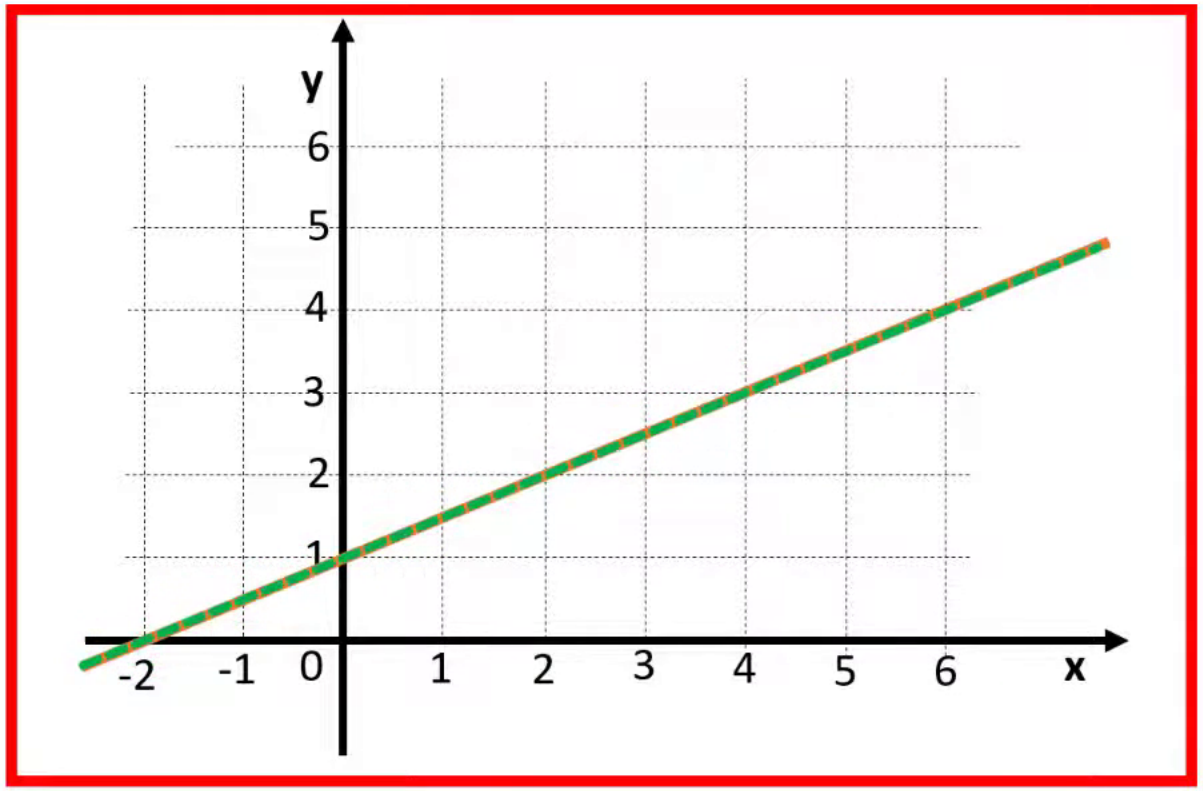

$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
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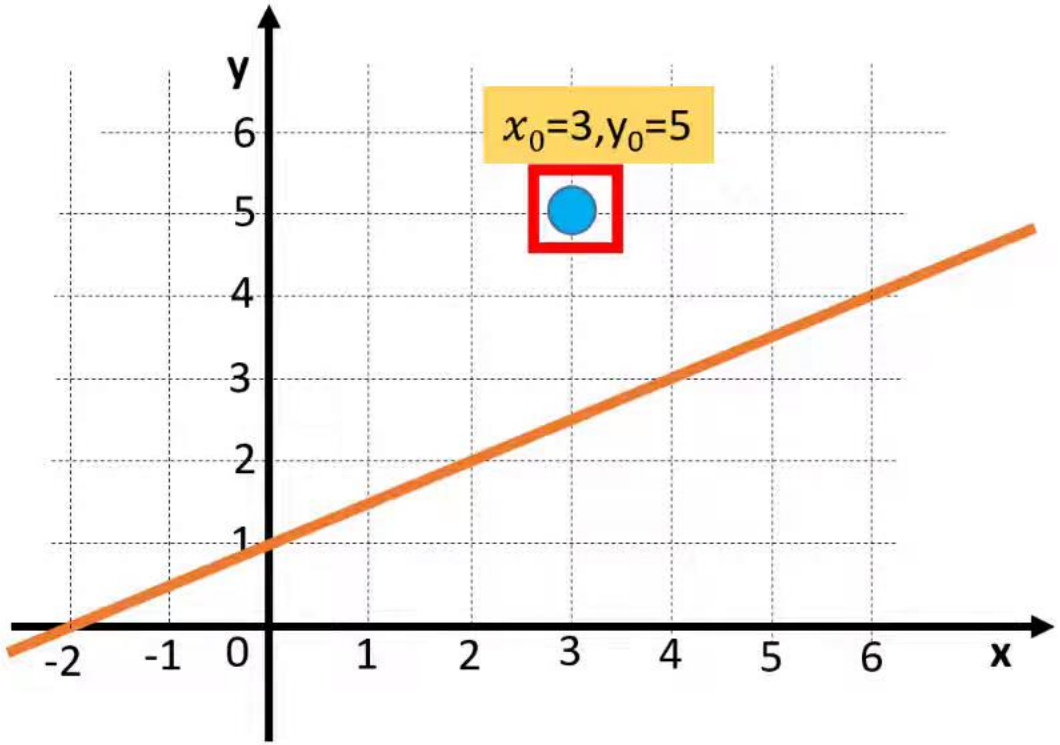
Distance between a point and a line

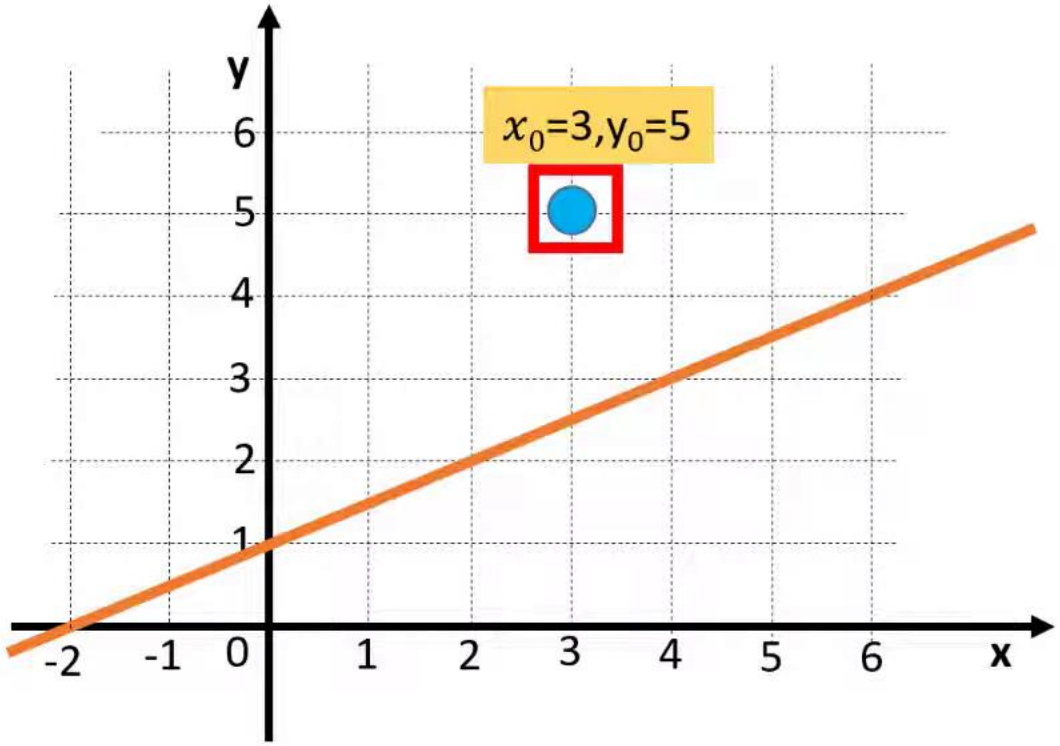


$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

Another thing we need to know before we look into the details about SVM is how to calculate the distance between a data point and a line.

Distance between a point and a line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

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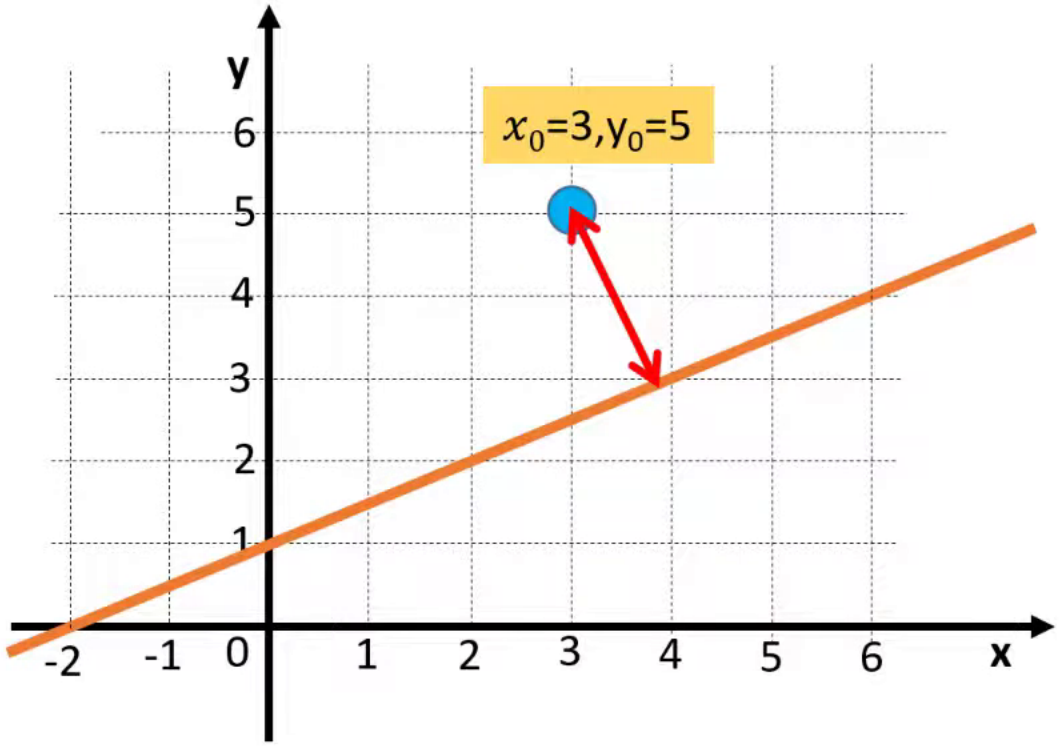


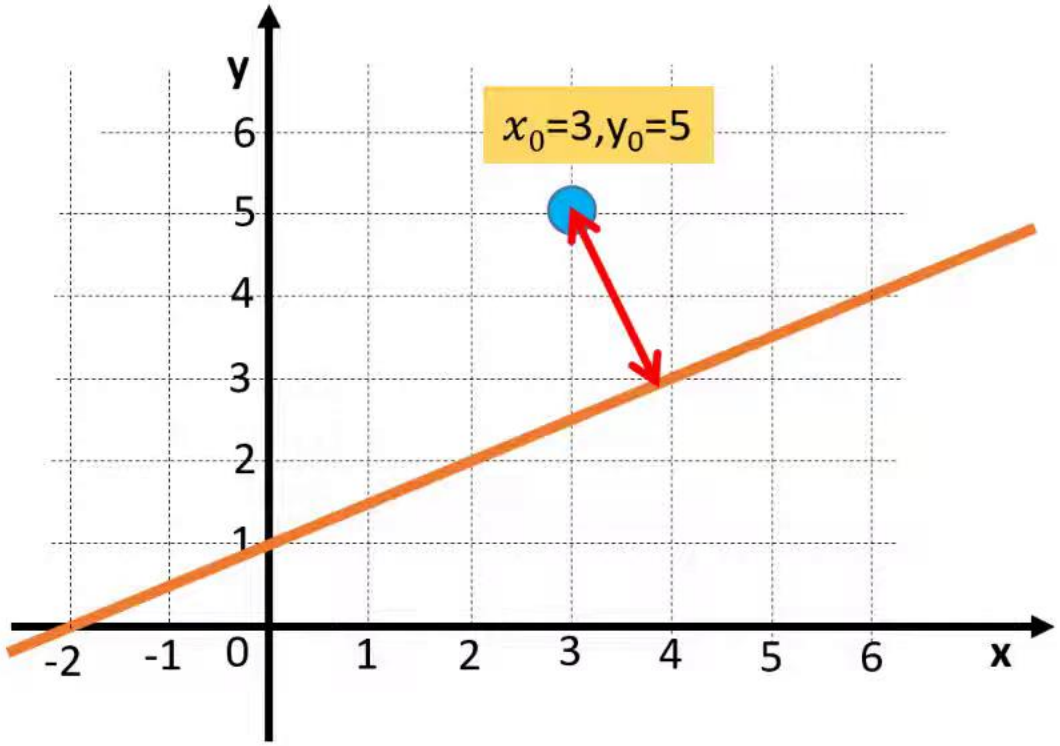
$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$d = \frac{|Ax_0 + By_o + C|}{\sqrt{A^2 + B^2}}$$

The shortest distance between a data point and a line can be calculated by the following formula.

Distance between a point and a line



$$Ax + By + C = 0$$

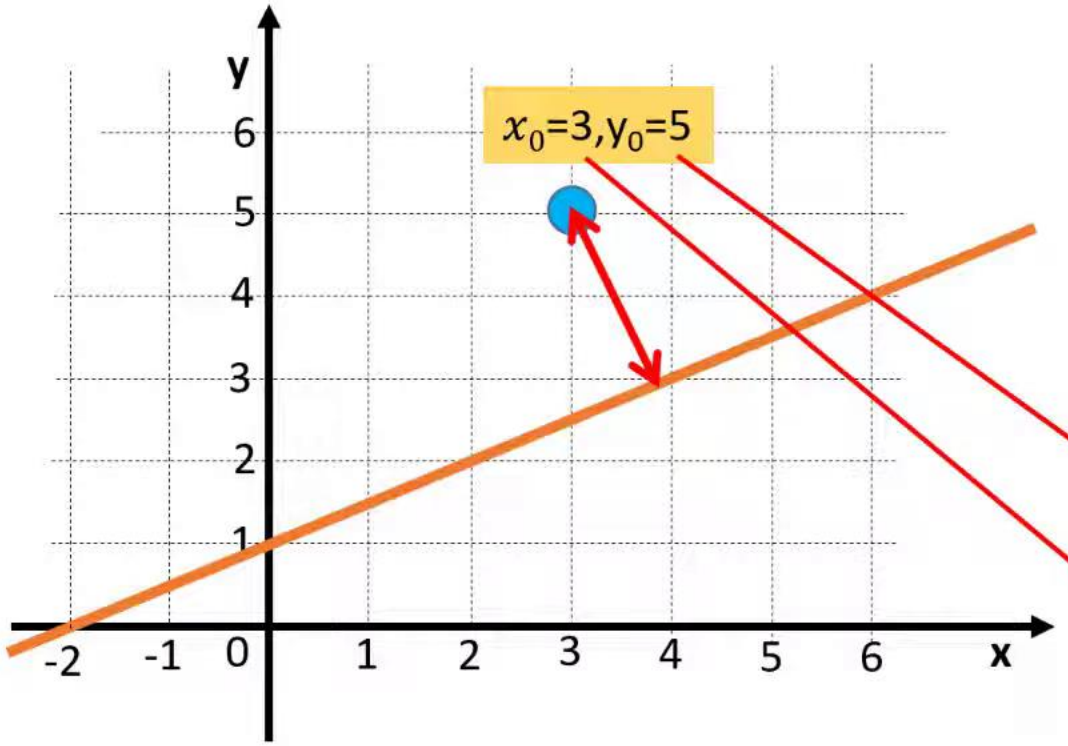
$$-2x + 4y - 4 = 0$$

$$d = \frac{|Ax_0 + By_o + C|}{\sqrt{A^2 + B^2}}$$

$$d = \frac{|(-2) \cdot 3 + 4 \cdot 5 + (-4)|}{\sqrt{(-2)^2 + 4^2}} = \frac{10}{\sqrt{20}} = 2.236$$

If we plug in the numbers from the equation of the line,

Distance between a point and a line



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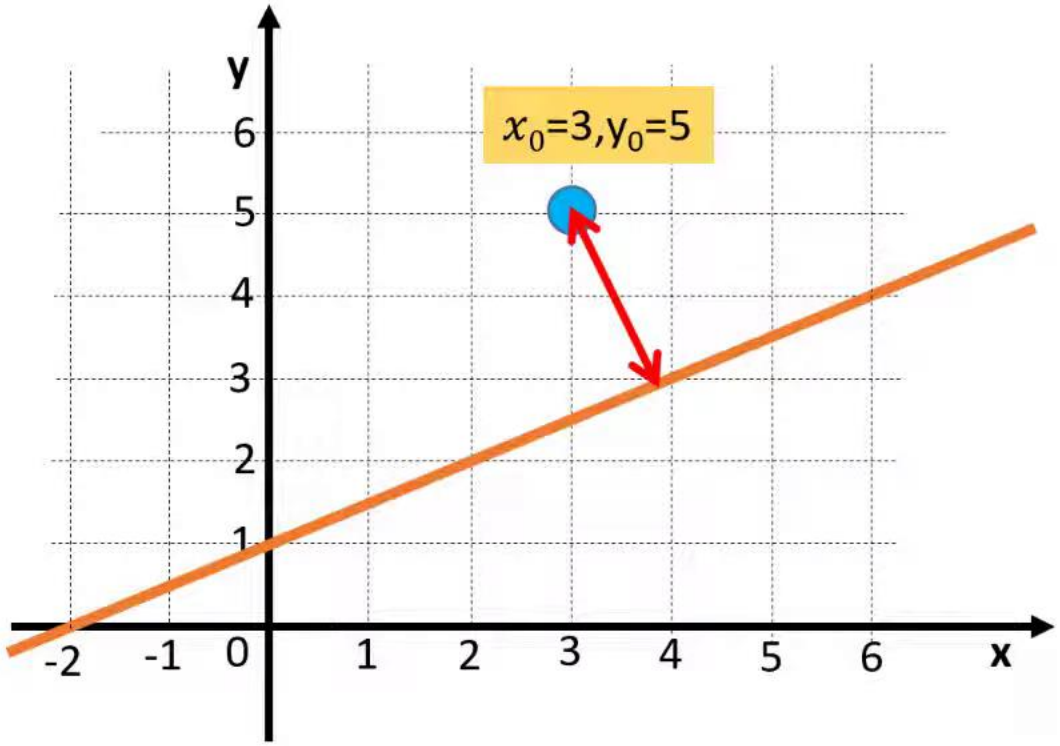
$$d = \frac{|Ax_0 + By_o + C|}{\sqrt{A^2 + B^2}}$$

$$d = \frac{|(-2) \cdot 3 + 4 \cdot 5 + (-4)|}{\sqrt{(-2)^2 + 4^2}} = \frac{10}{\sqrt{20}} = 2.236$$

and the x and y-coordinates of the data point,

Distance between a point and a line



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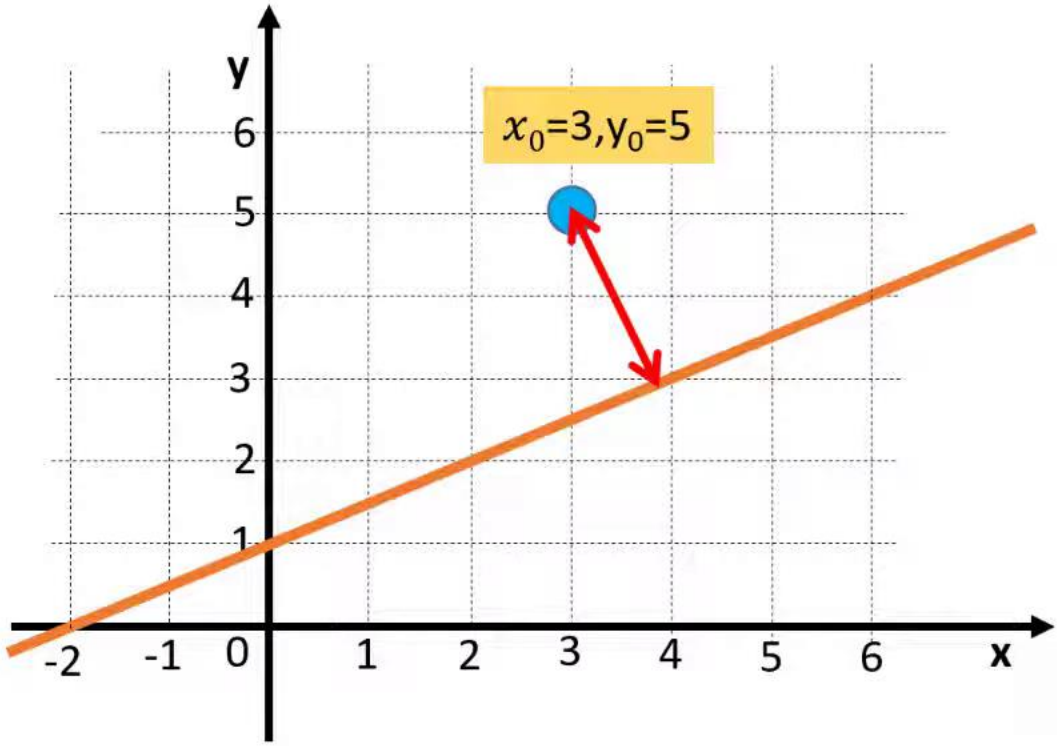
and do the math, we see that the distance is equal to about 2.24.

Distance between a point and a line



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 0$$

$$d = \frac{|Ax_0 + By_o + C|}{\sqrt{A^2 + B^2}}$$

$$d = \frac{|(-2) \cdot 3 + 4 \cdot 5 + (-4)|}{\sqrt{(-2)^2 + 4^2}} = \frac{10}{\sqrt{20}} = 2.236$$

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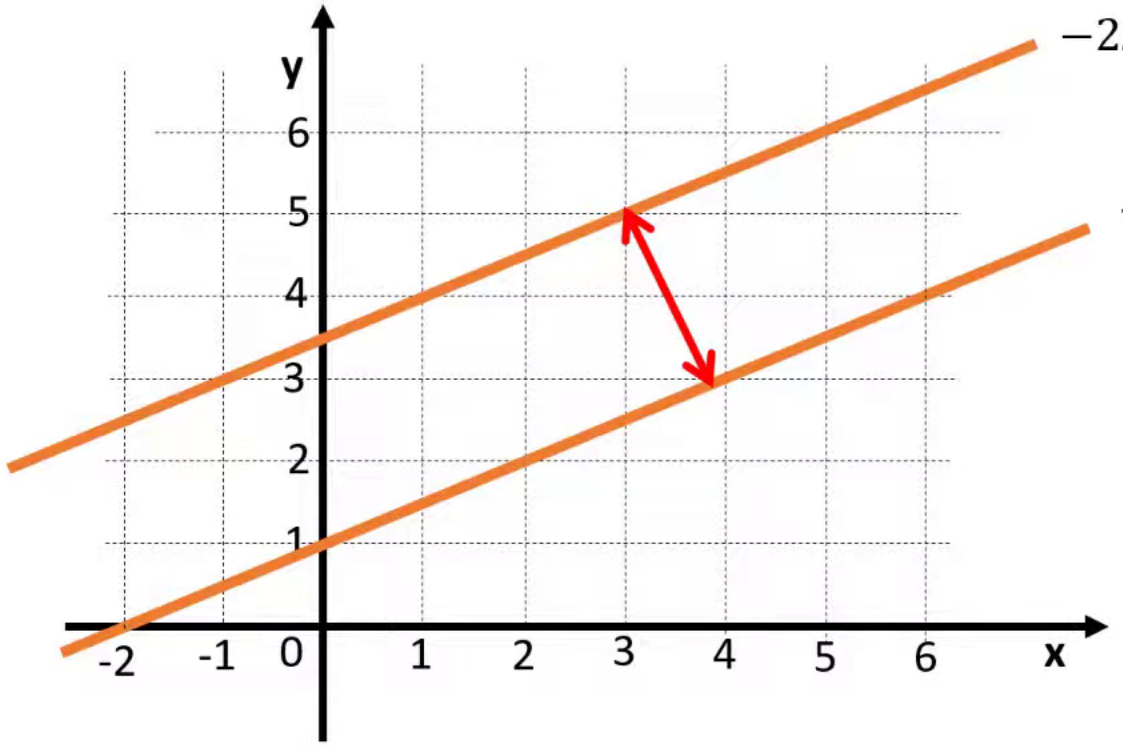
Distance between two parallel lines

$$Ax + By + C = 0$$



$$-2x + 4y - 14 = 0$$

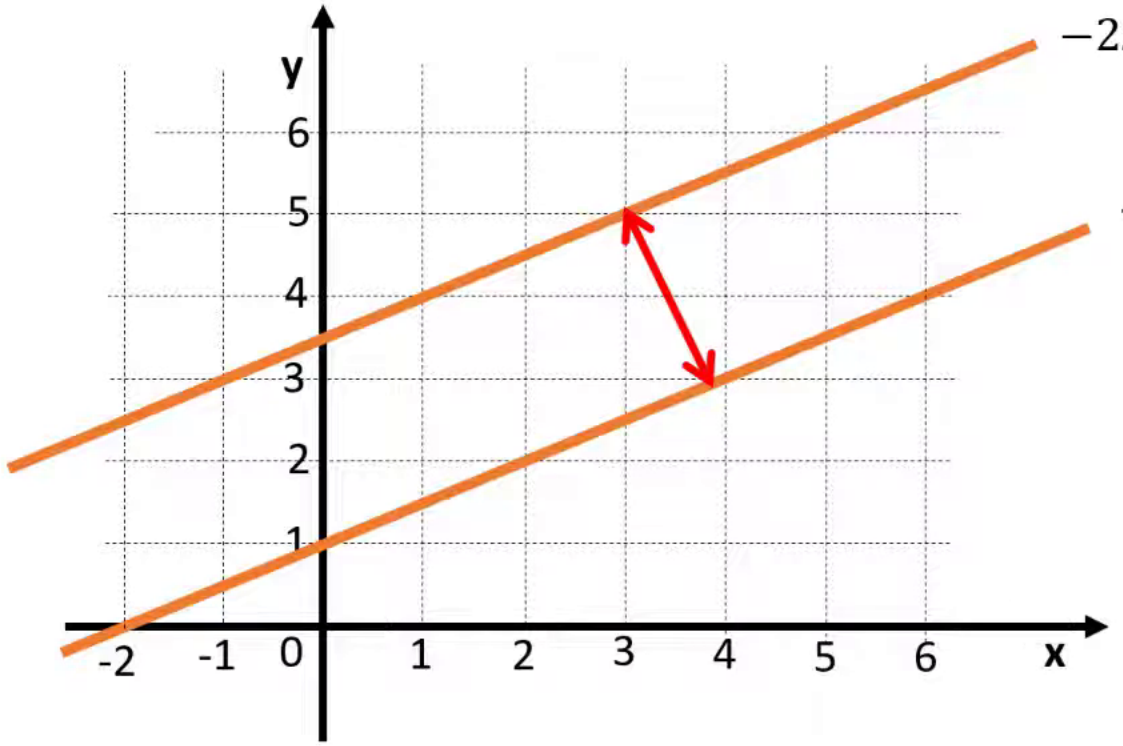
$$-2x + 4y - 4 = 0$$

$$d = \frac{|C_2 - C_1|}{\sqrt{A^2 + B^2}}$$

Similarly, the distance between two parallel lines can be calculated by the following formula.

Distance between two parallel lines

$$Ax + By + C = 0$$



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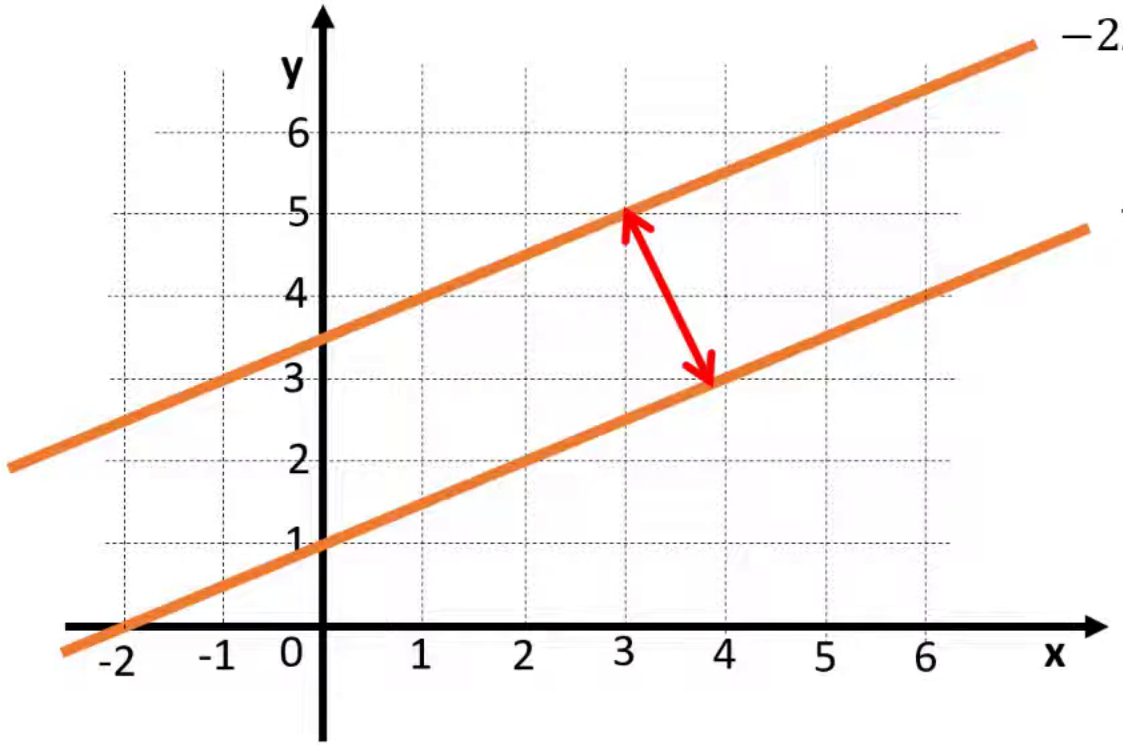


$$-2x + 4y - 14 = 0$$

$$-2x + 4y - 4 = 0$$

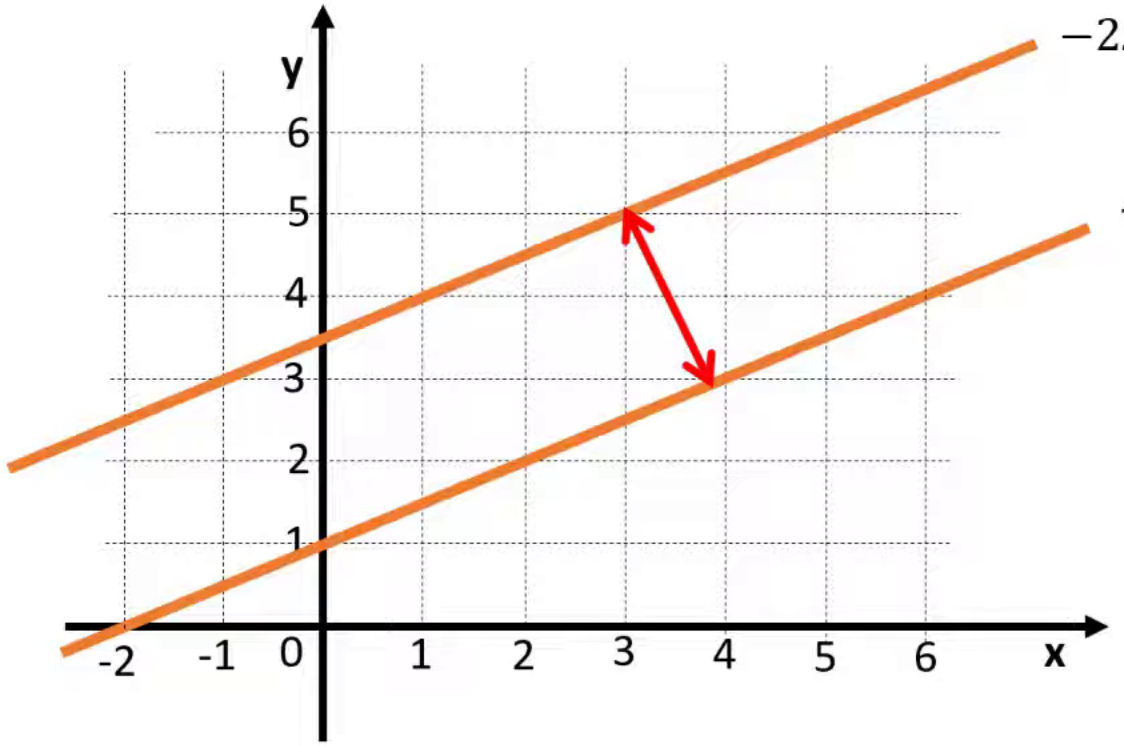
$$d = \frac{|C_2 - C_1|}{\sqrt{A^2 + B^2}}$$

$$d = \frac{|(-4) - (-14)|}{\sqrt{(-2)^2+4^2}} = \frac{10}{\sqrt{20}} = 2.236$$

If we plug in the coefficients of the equations of the two parallel lines,

Distance between two parallel lines

$$Ax + By + C = 0$$



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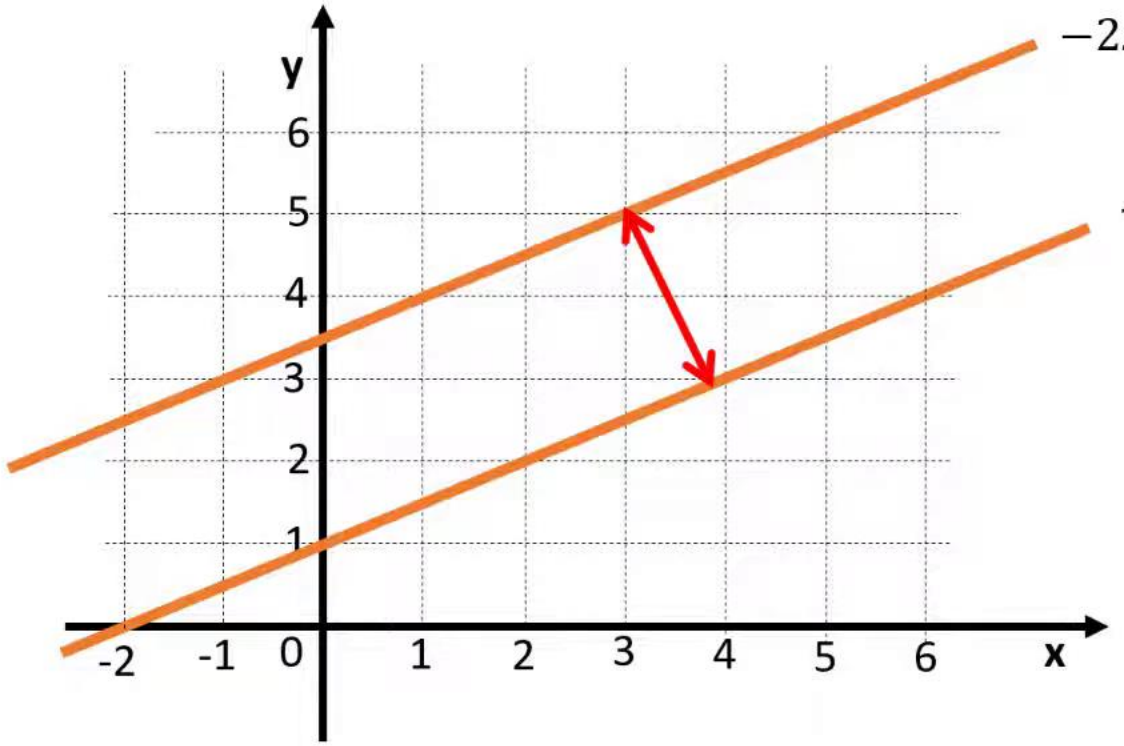

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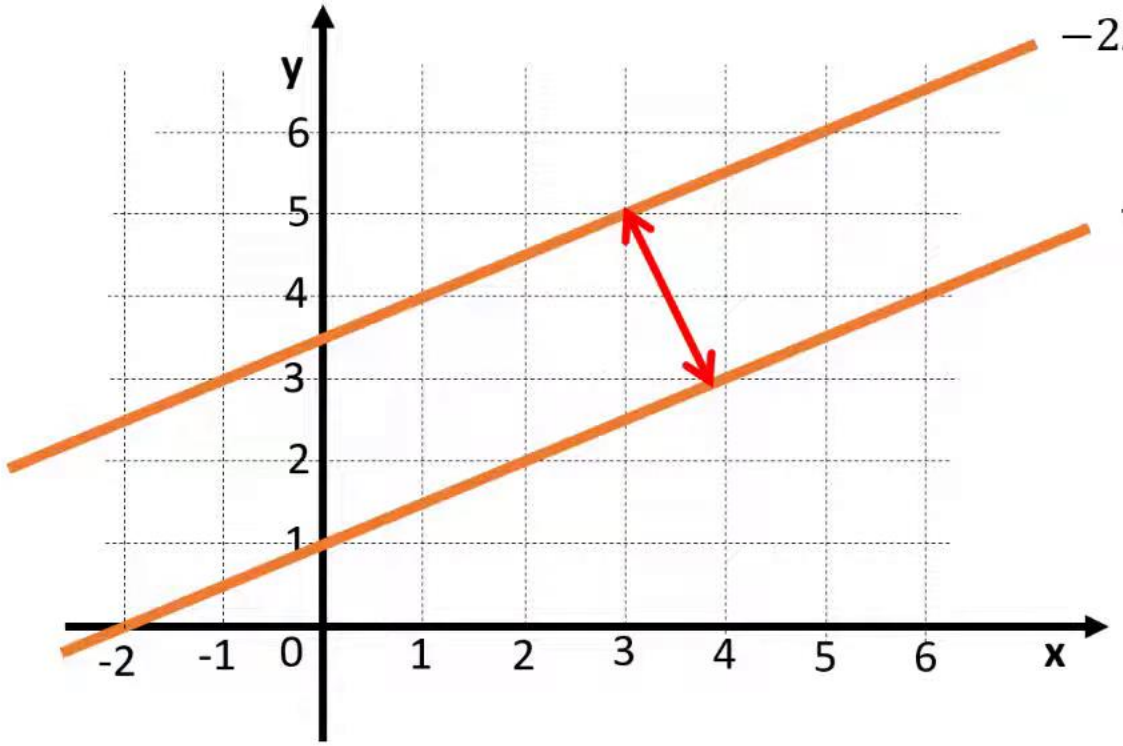
and the constants,

Distance between two parallel lines



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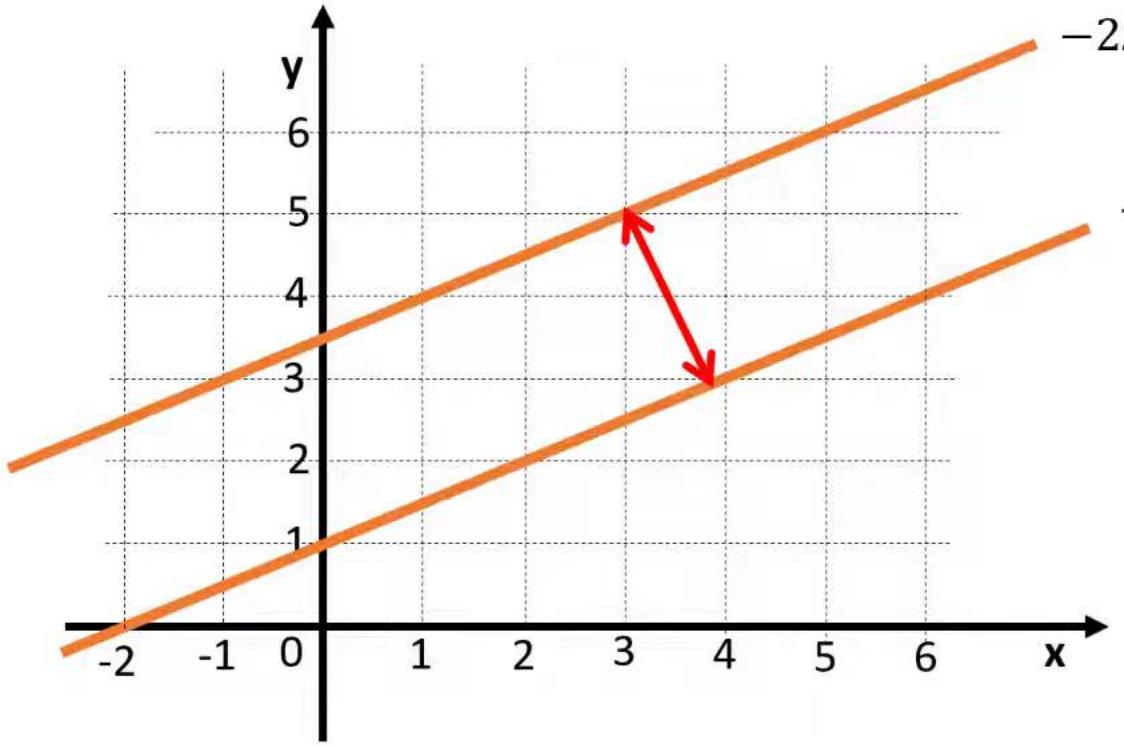

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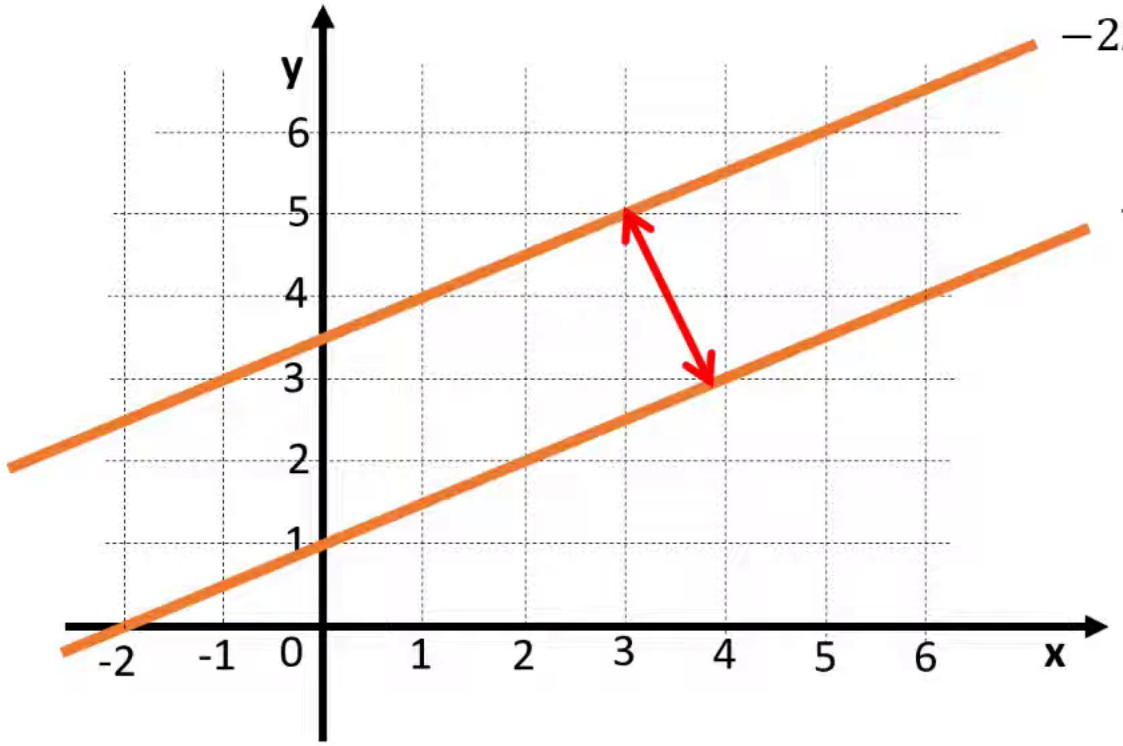
we see that the shortest distance between these two parallel lines is about 2.24.

Distance between two parallel lines

$$Ax + By + C = 0$$



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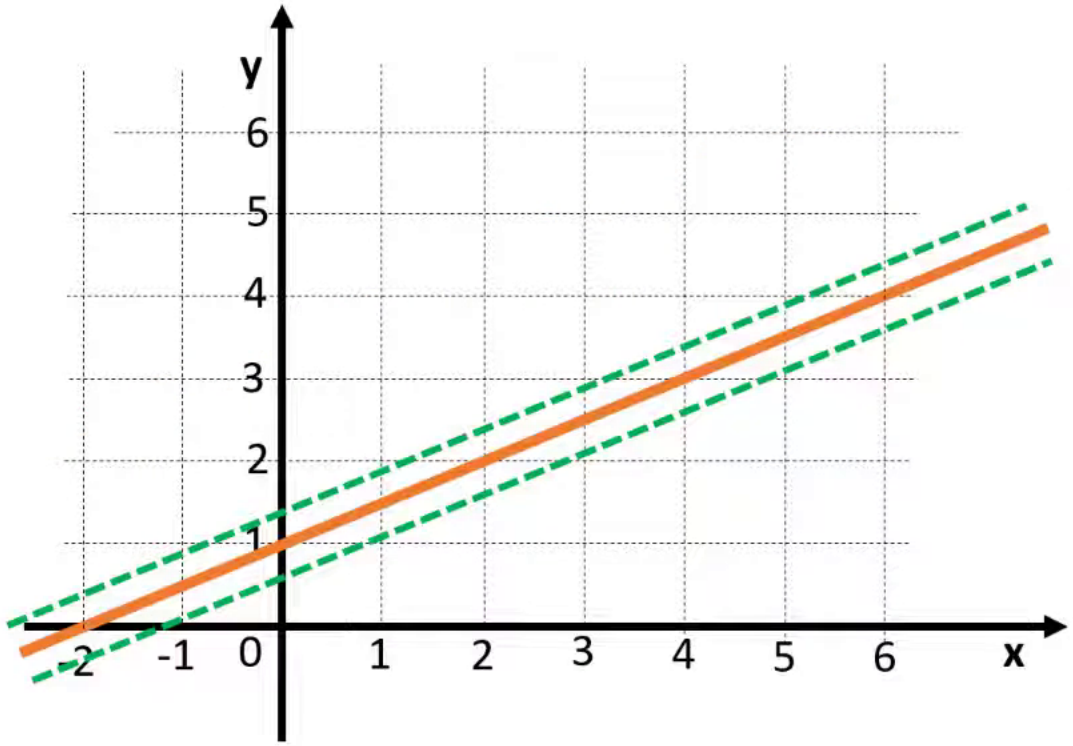
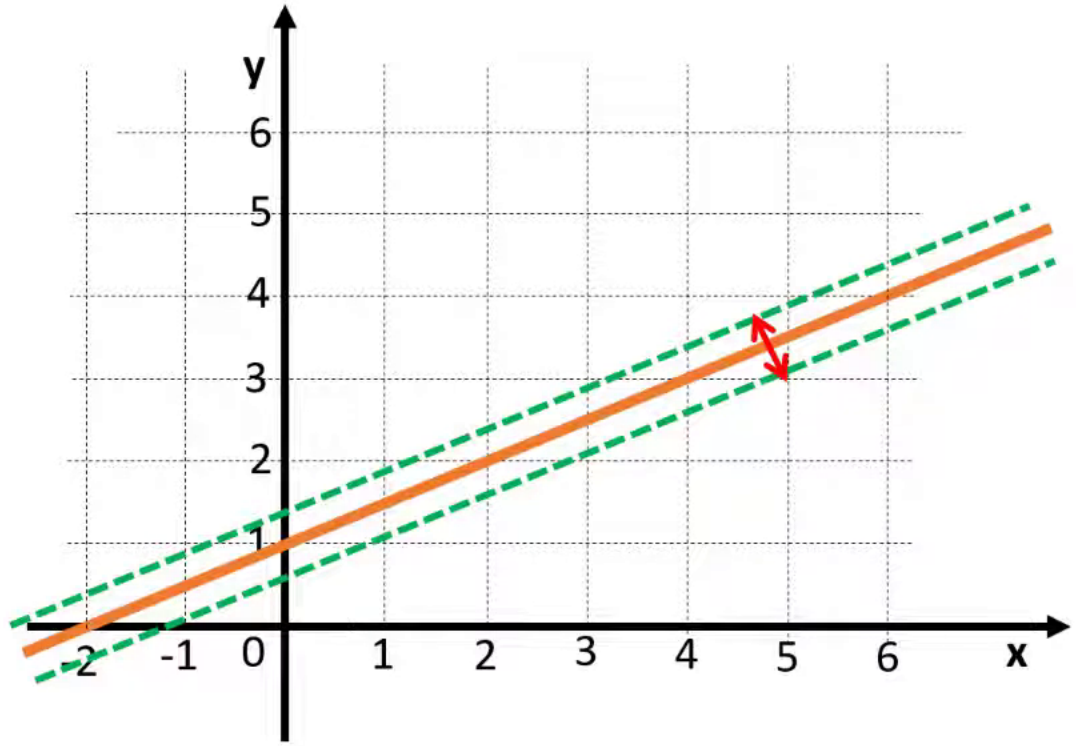


$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
$$-2x + 4y - 4 = -1$$

Remember that we previously added a one and a negative on the right-hand side to obtain the two green lines.

Distance between two parallel lines



$$Ax + By + C = 0$$

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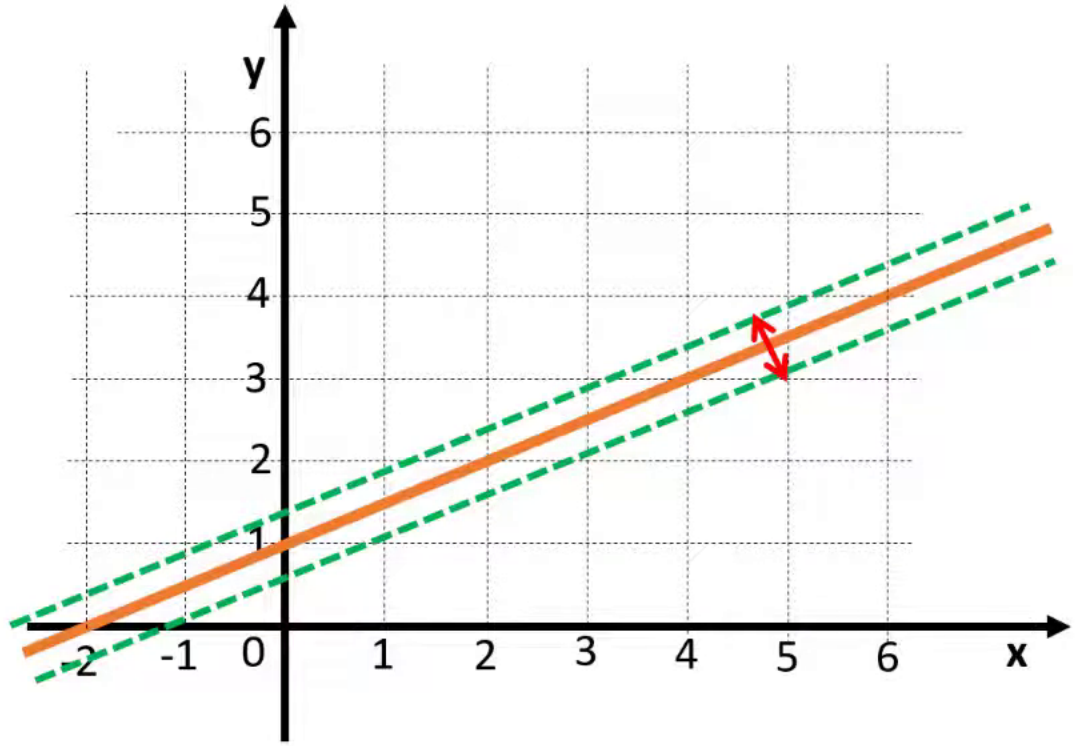
$$d = \frac{|C_2 - C_1|}{\sqrt{A^2 + B^2}}$$

$$d = \frac{|(-3) - (-5)|}{\sqrt{(-2)^2 + 4^2}} = \frac{2}{\sqrt{20}} = 0.447$$

The distance between these two green lines is calculated like this.

Distance between two parallel lines



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
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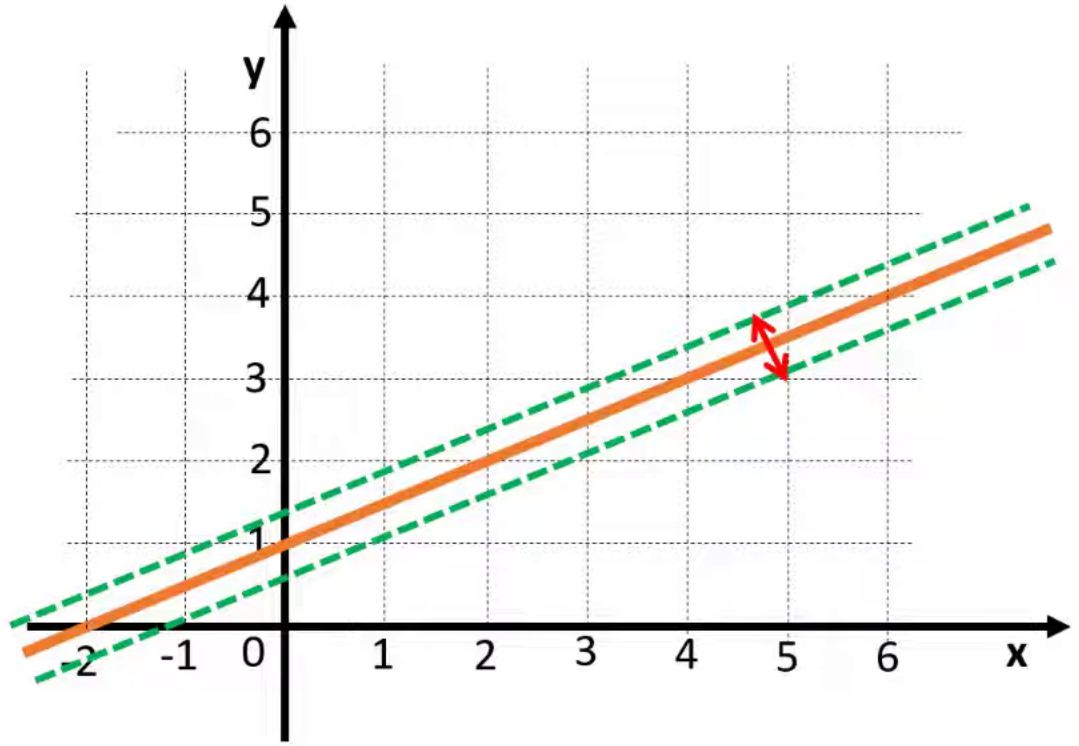
Note that we have a 2 in the numerator because the absolute difference between the constants is always two for such lines.

Distance between two parallel lines



$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
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$$-2x + 4y - 4 = -1$$

$$d = \frac{|C_2 - C_1|}{\sqrt{A^2 + B^2}}$$

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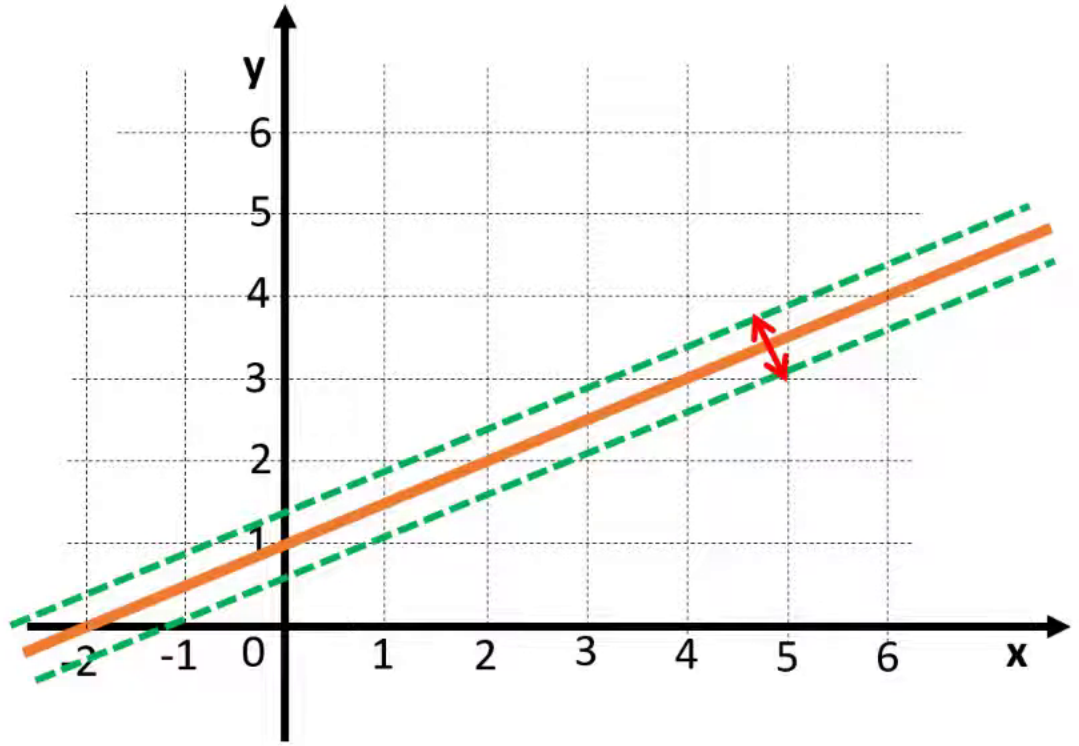
Distance between two parallel lines



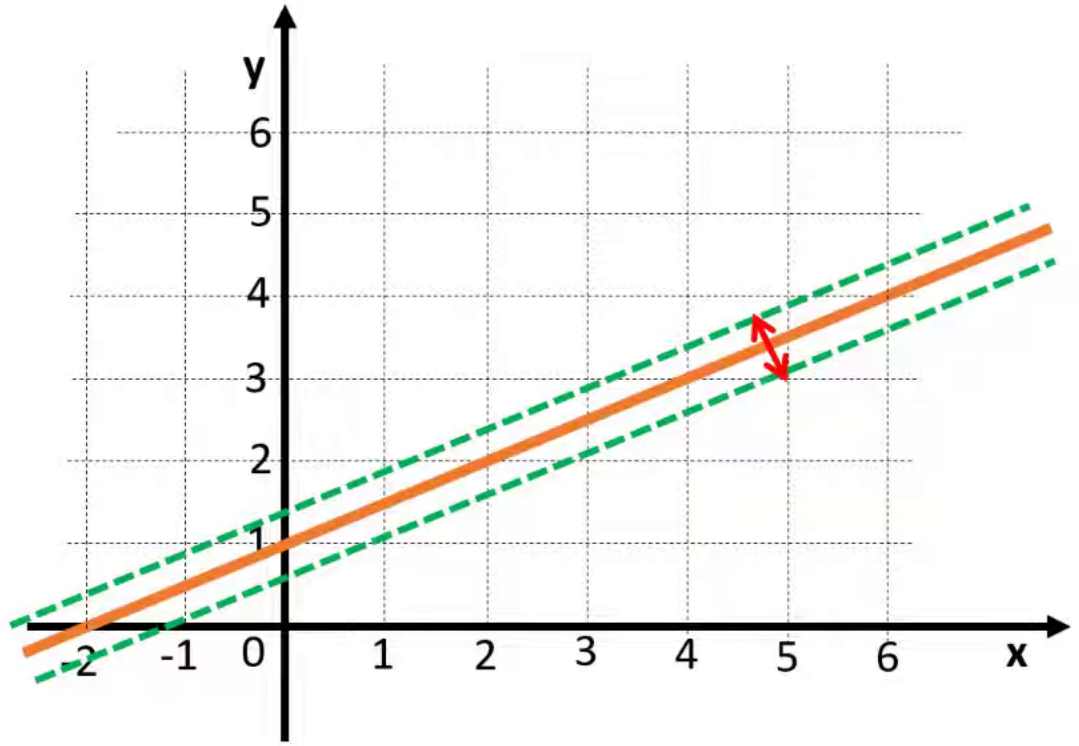
$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
$$-2x + 4y - 4 = -1$$

$$d = \frac{2}{\sqrt{A^2 + B^2}}$$

For such lines, we can simplify the equation to this,

Distance between two parallel lines



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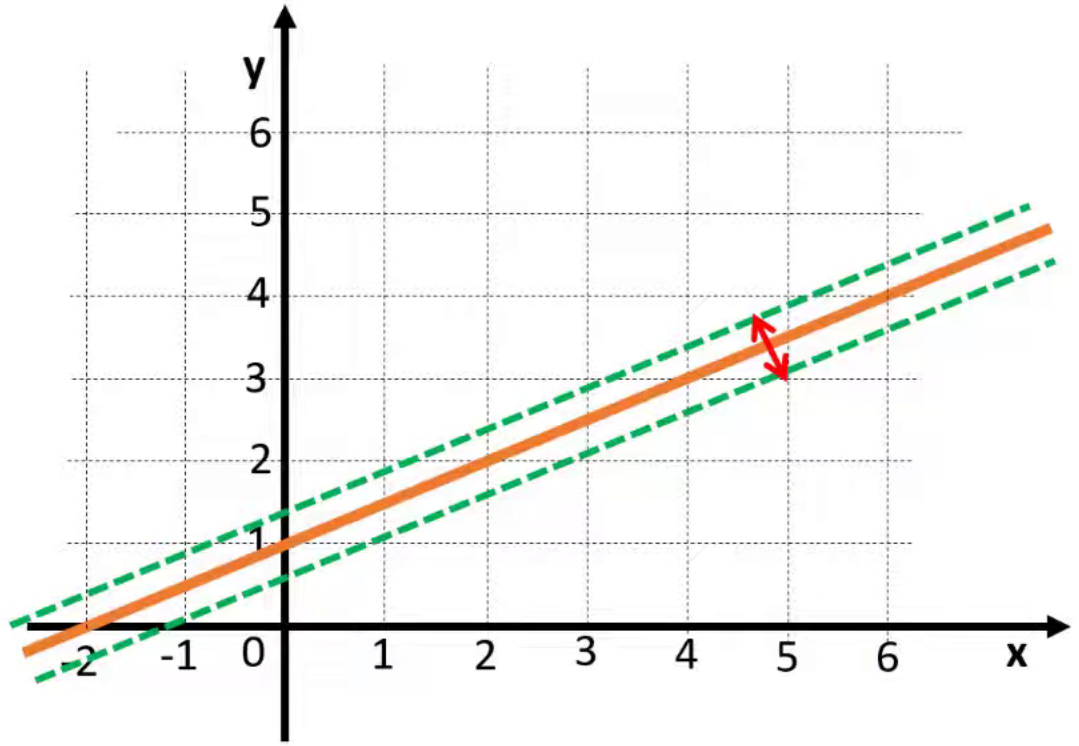


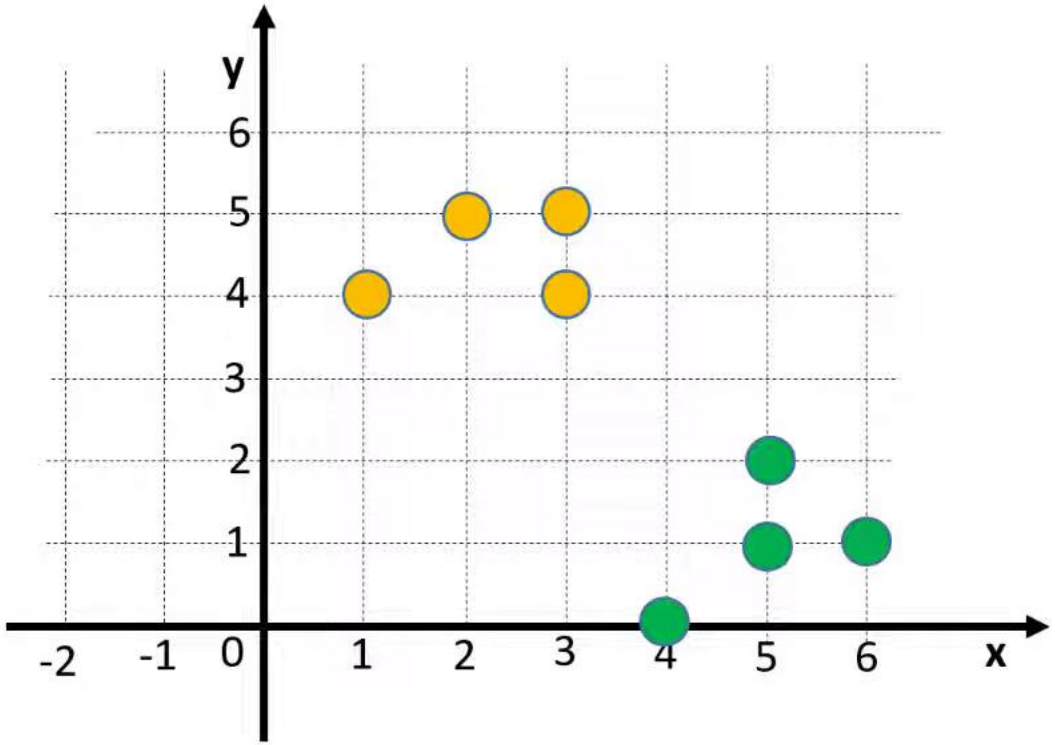
$$Ax + By + C = 0$$

$$-2x + 4y - 4 = 1$$
$$-2x + 4y - 4 = 0$$
$$-2x + 4y - 4 = -1$$

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \boxed{\frac{2}{\|w\|}}$$

or like this, which you will see later on.

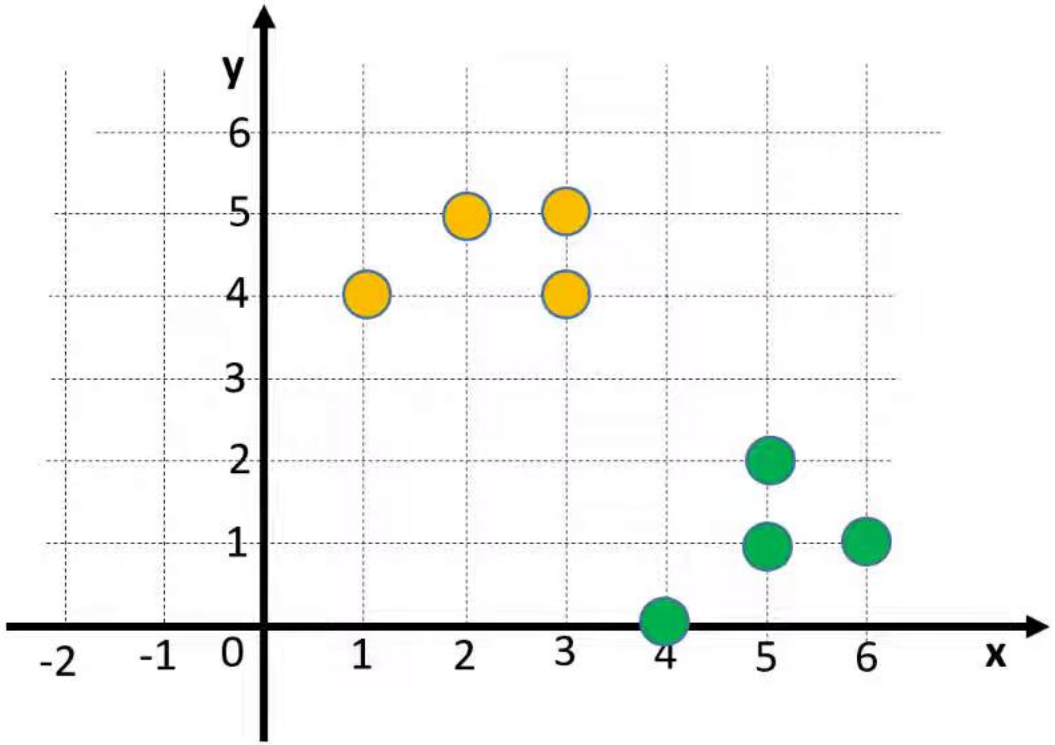
SVM



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

We will now have a look at a real data set, where we will use a SVM to generate a line that separates the two groups as good as possible.

SVM



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A	3	5
A	3	4
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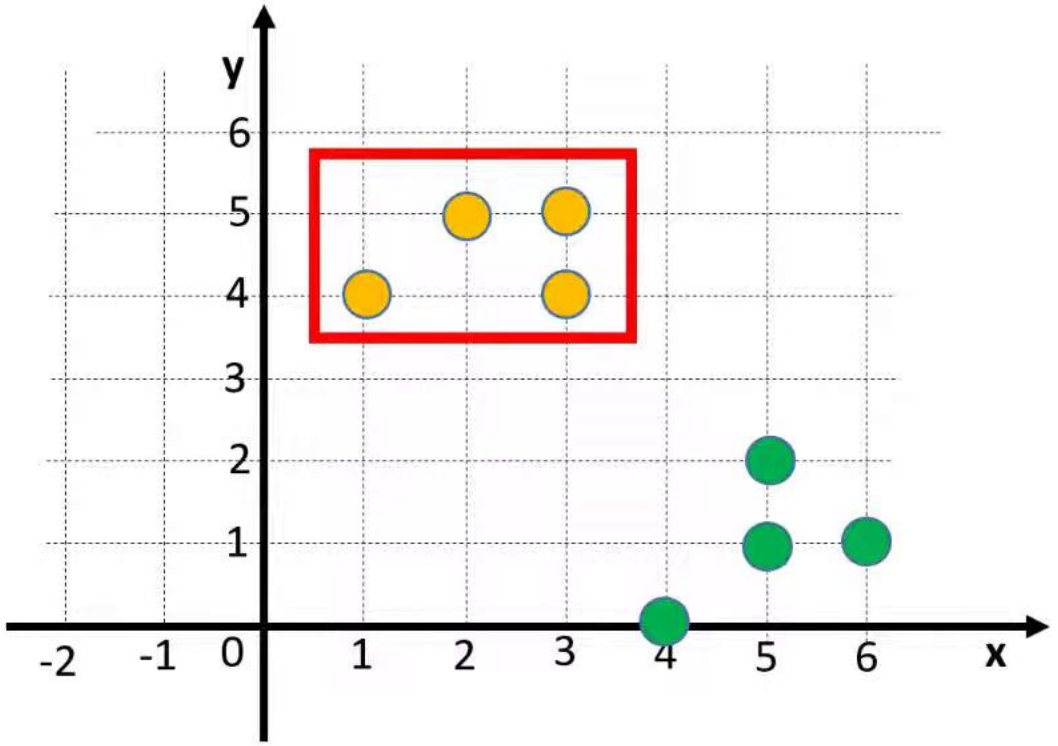
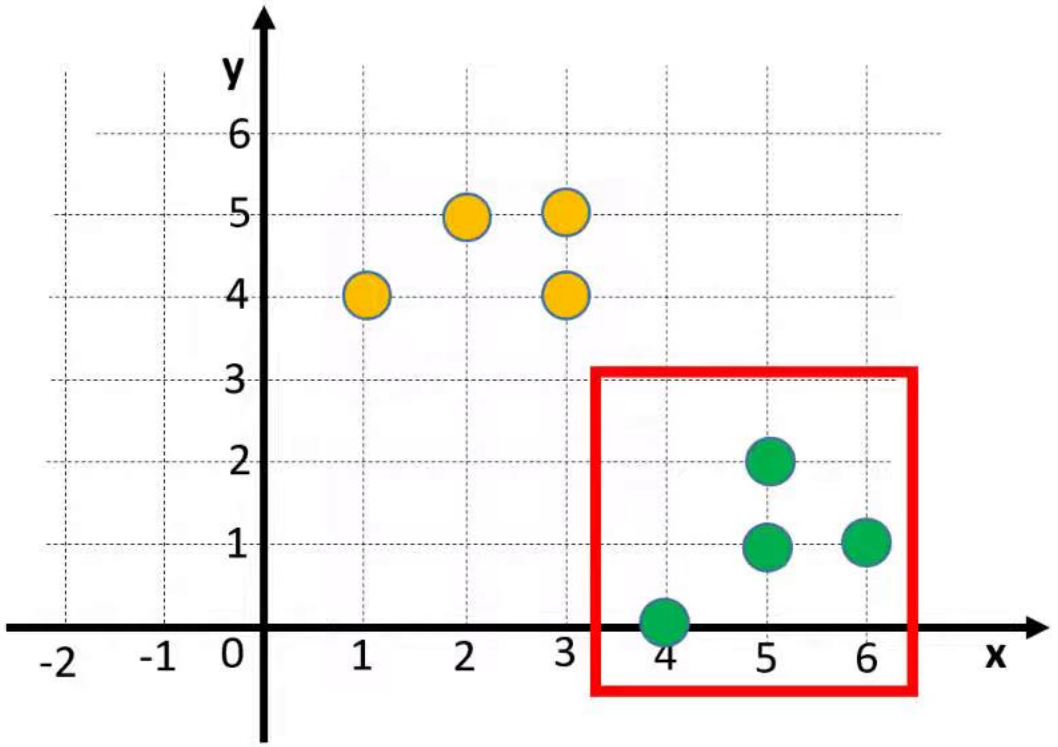


Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

The yellow points belong to group A, which, for example, could represent four individuals with a certain disease,

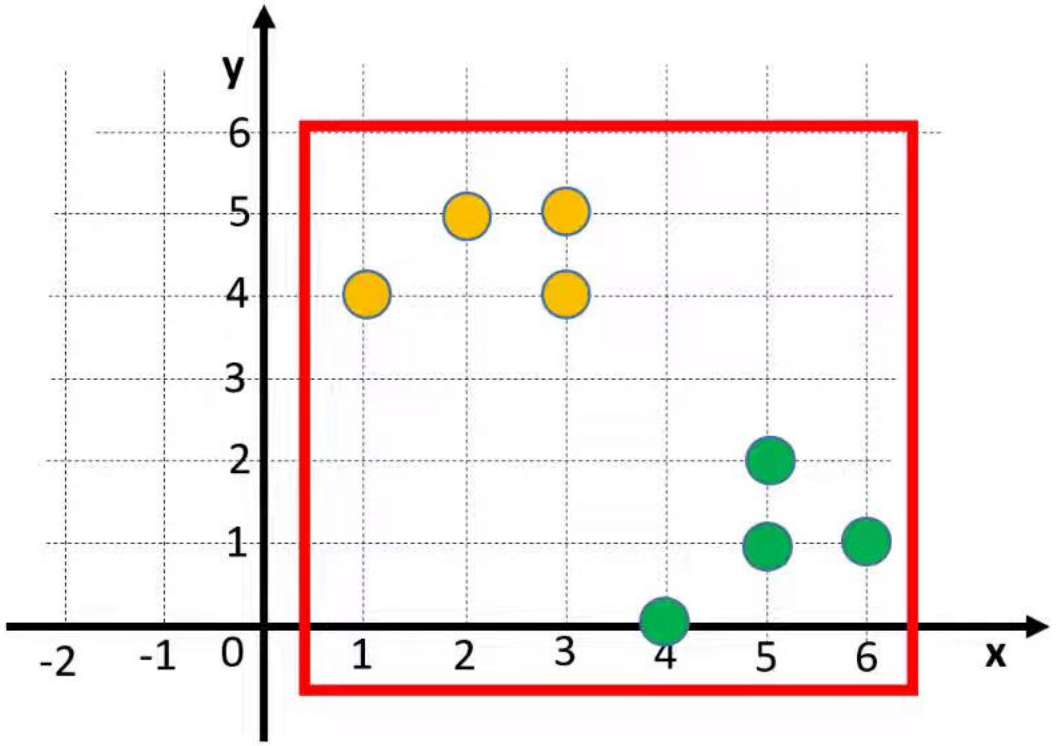
SVM



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

whereas these green points belong to group B, which could represent healthy individuals.

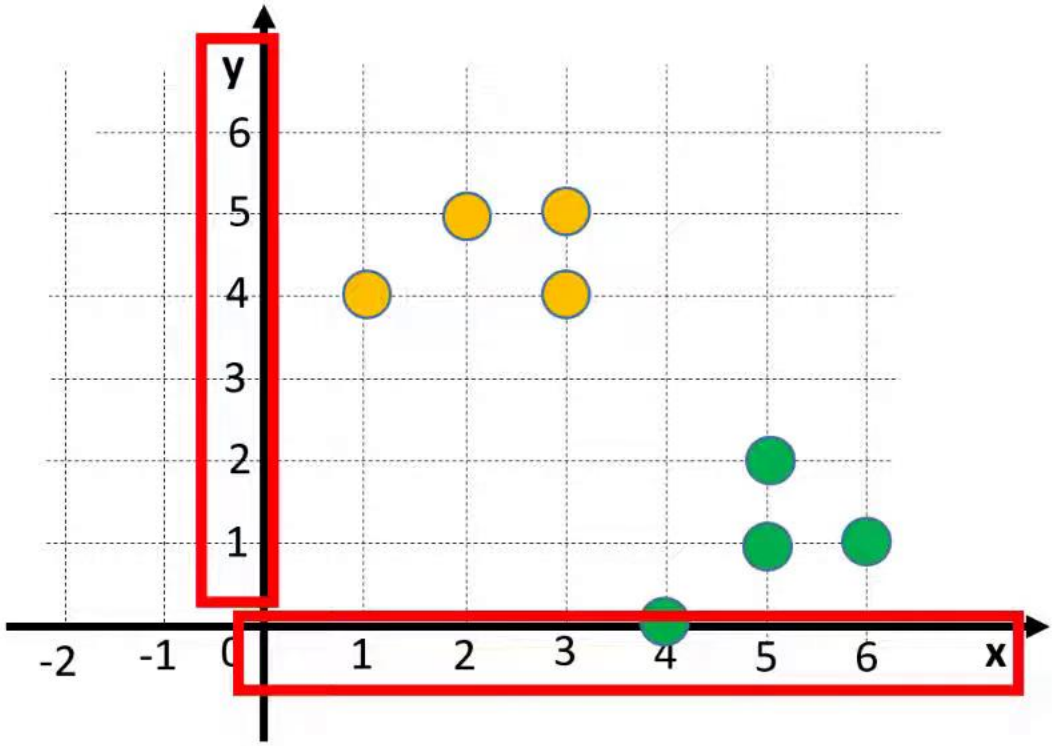
SVM



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

Since we know which group the data points belong to, we call this our training data because we will let the SVM train on this data. The training means that the SVM finds a line that separate the two groups in an optimal way.

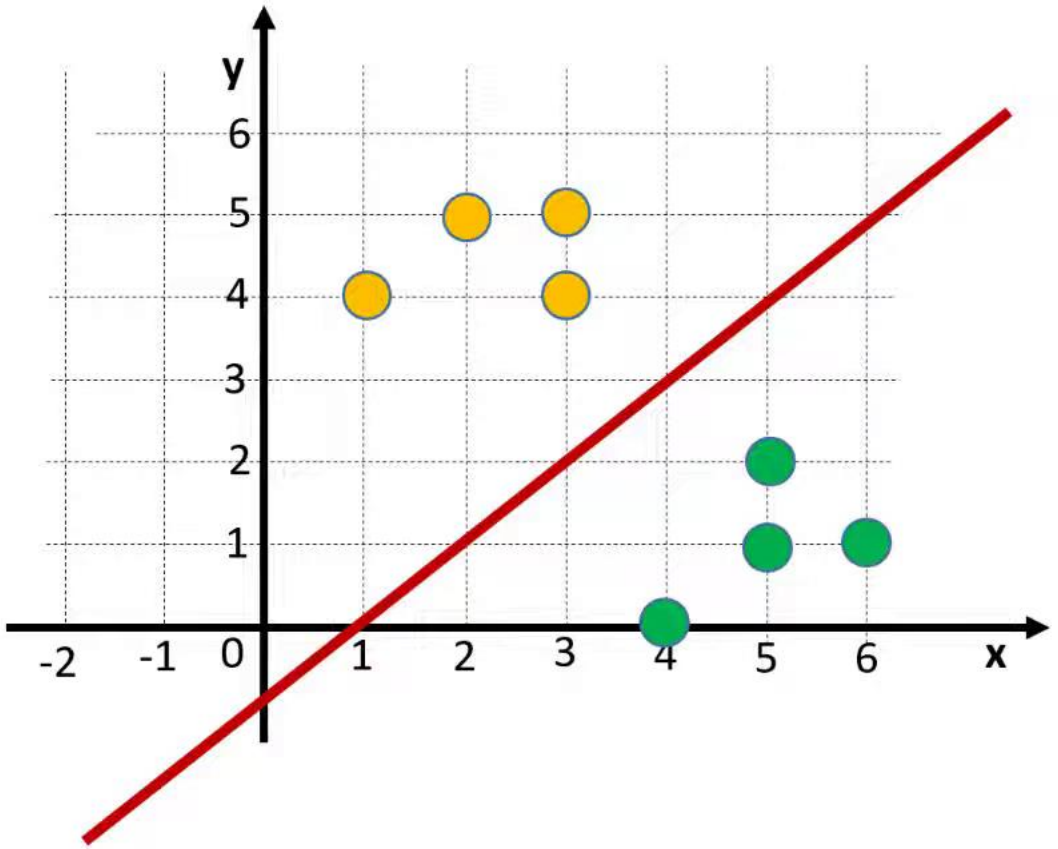
SVM



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

x and y are two variables, that could represent two measurements on the eight individuals, for example, blood pressure and cholesterol level.

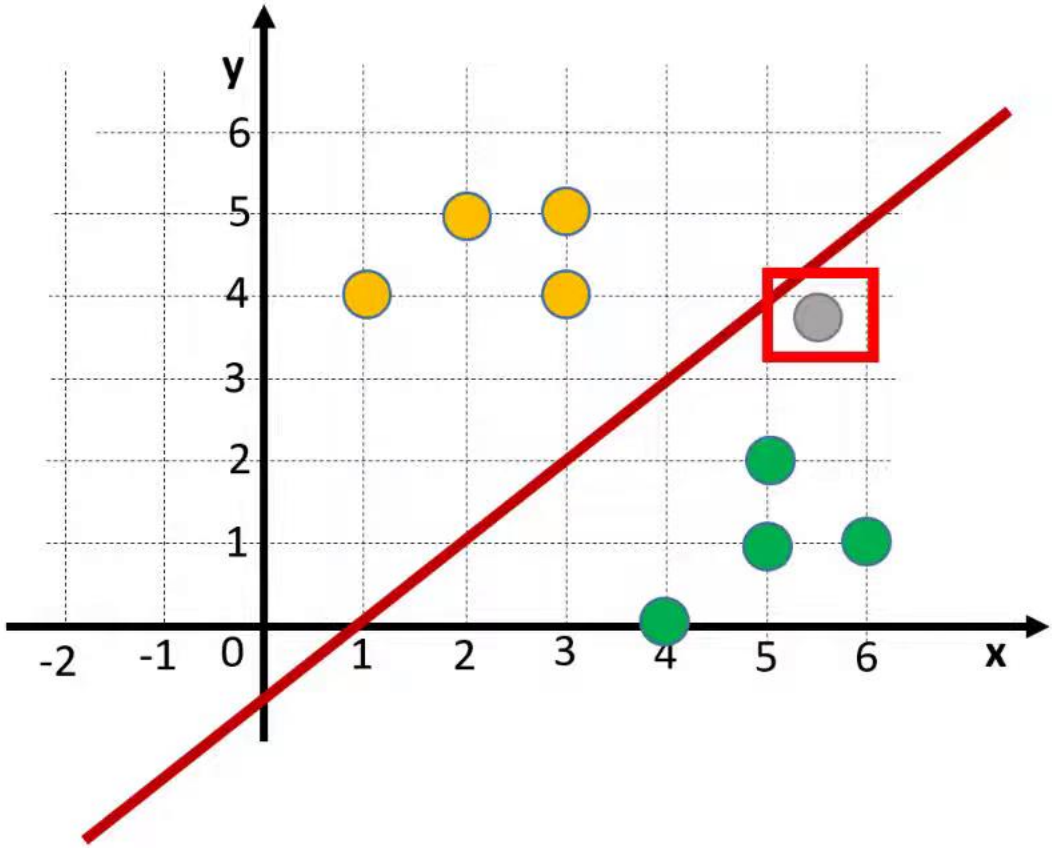
SVM



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

The following line is the best line to separate the two groups based on the training data.

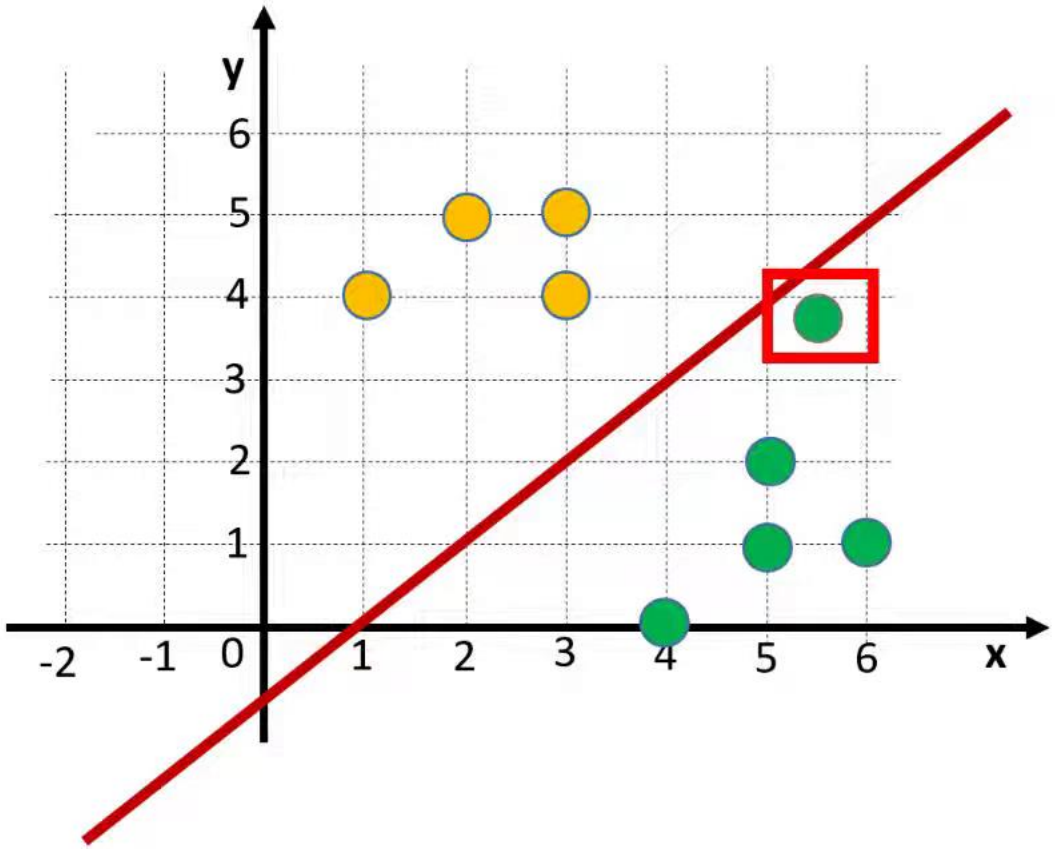
Predict



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

By using this line, we could predict if someone has a disease or not based on the value of the two variables.

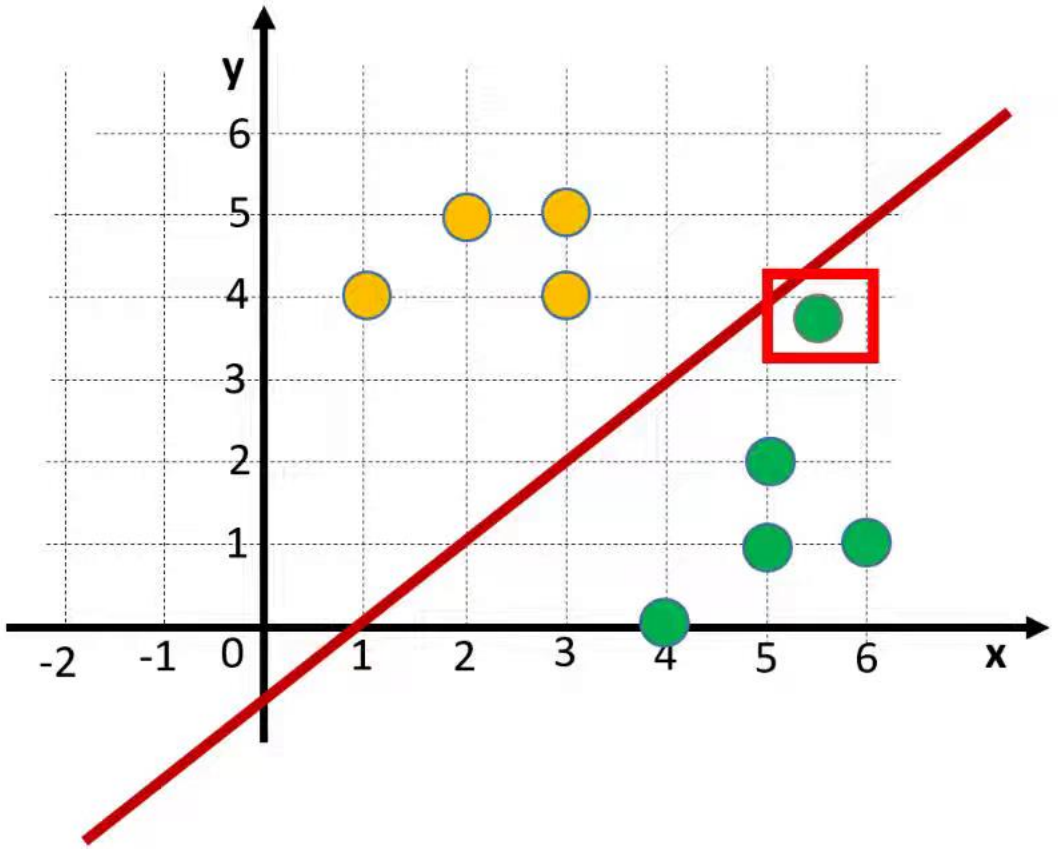
Predict



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

If the data point is below the line, we predict that it belongs to group B, the healthy group,

Predict



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
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B	5	1

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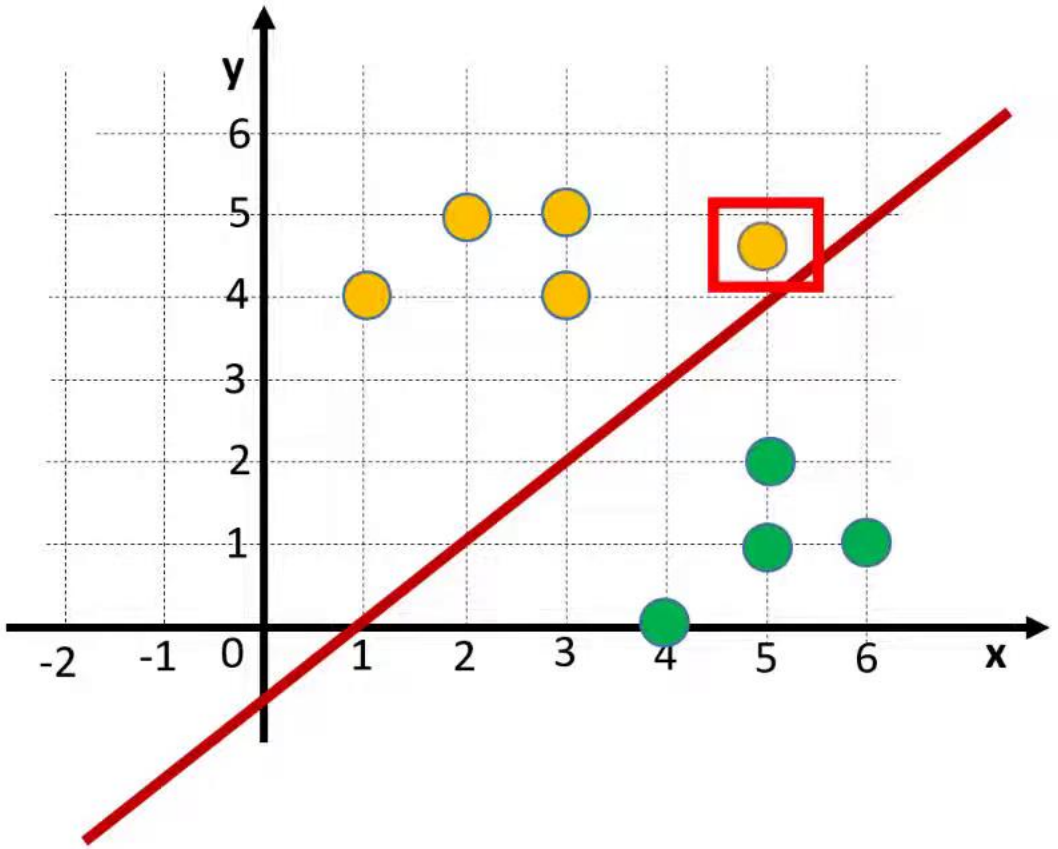
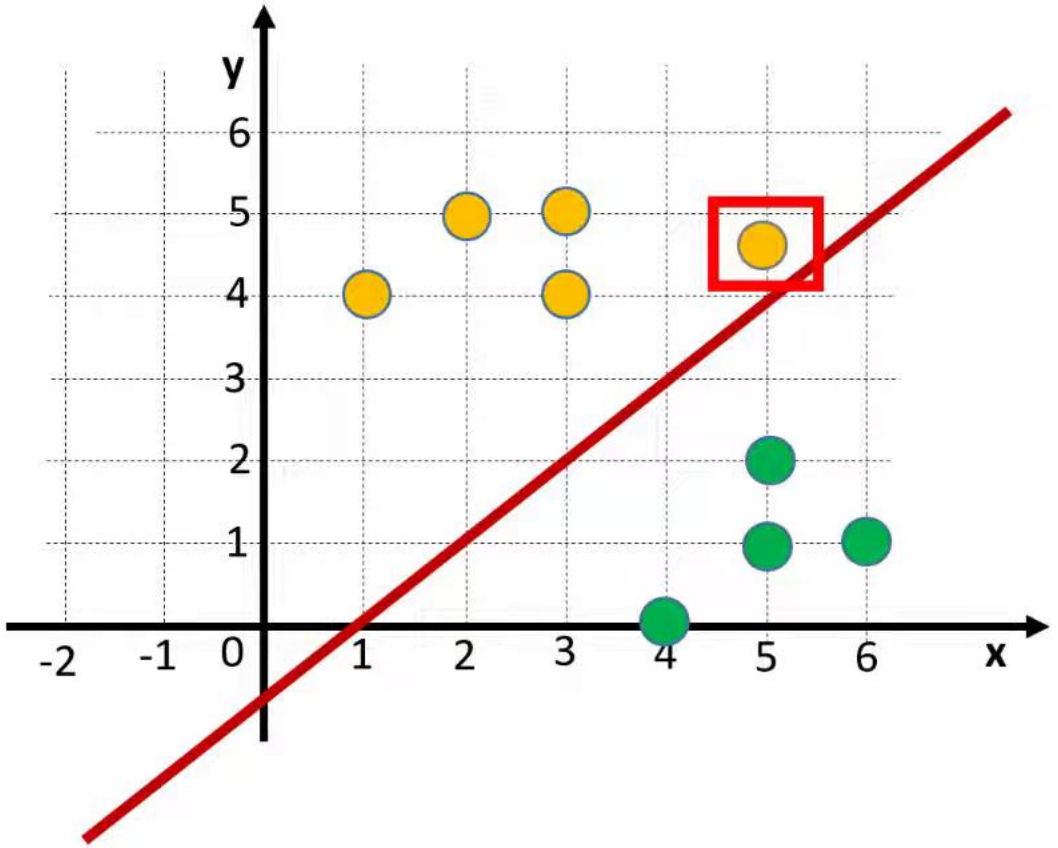
Predict



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

and if the data point is above the line, we predict that it belongs to group A, the disease group.

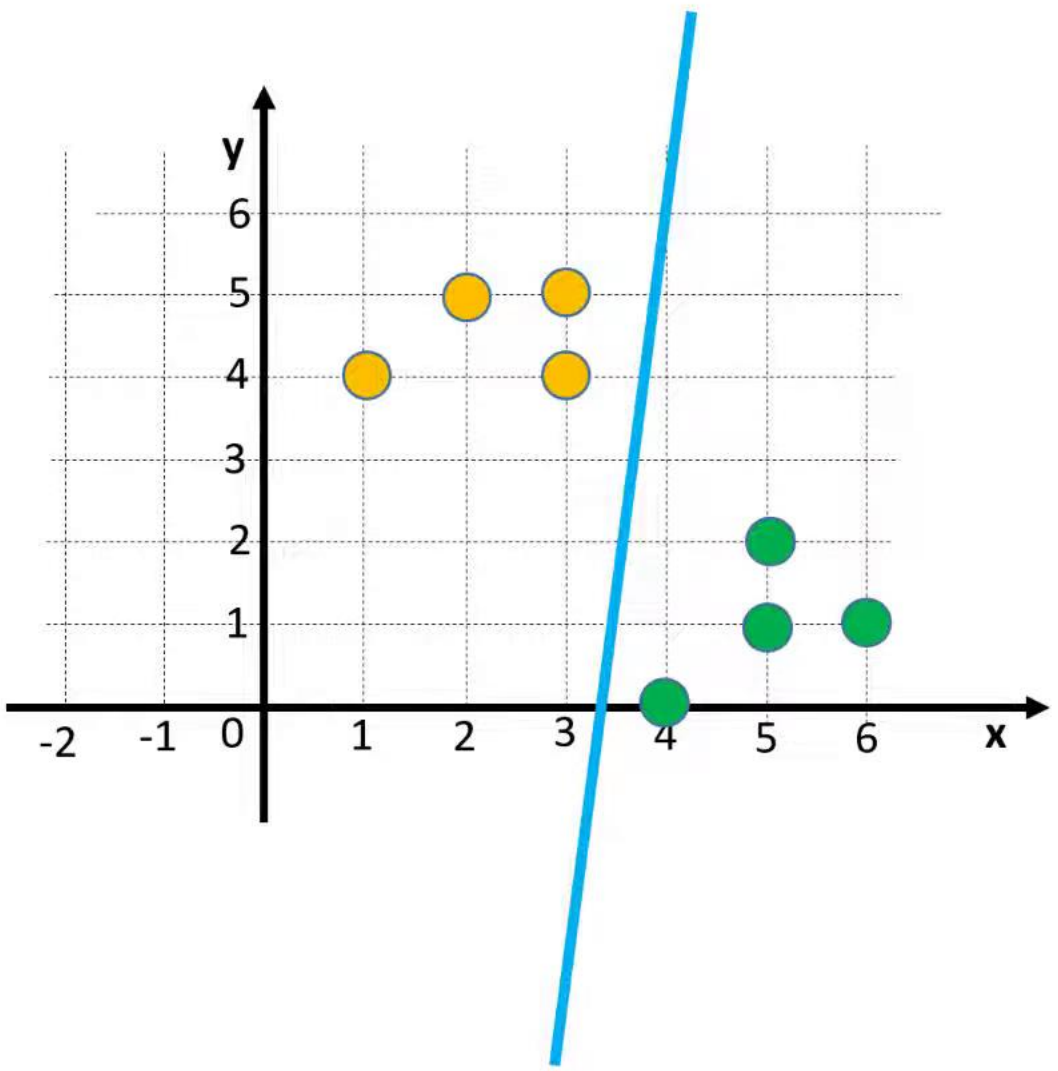
Predict



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

Based on the measured values of x and y, the blood pressure and cholesterol level, the SVM can help us to predict if someone has the disease or not.

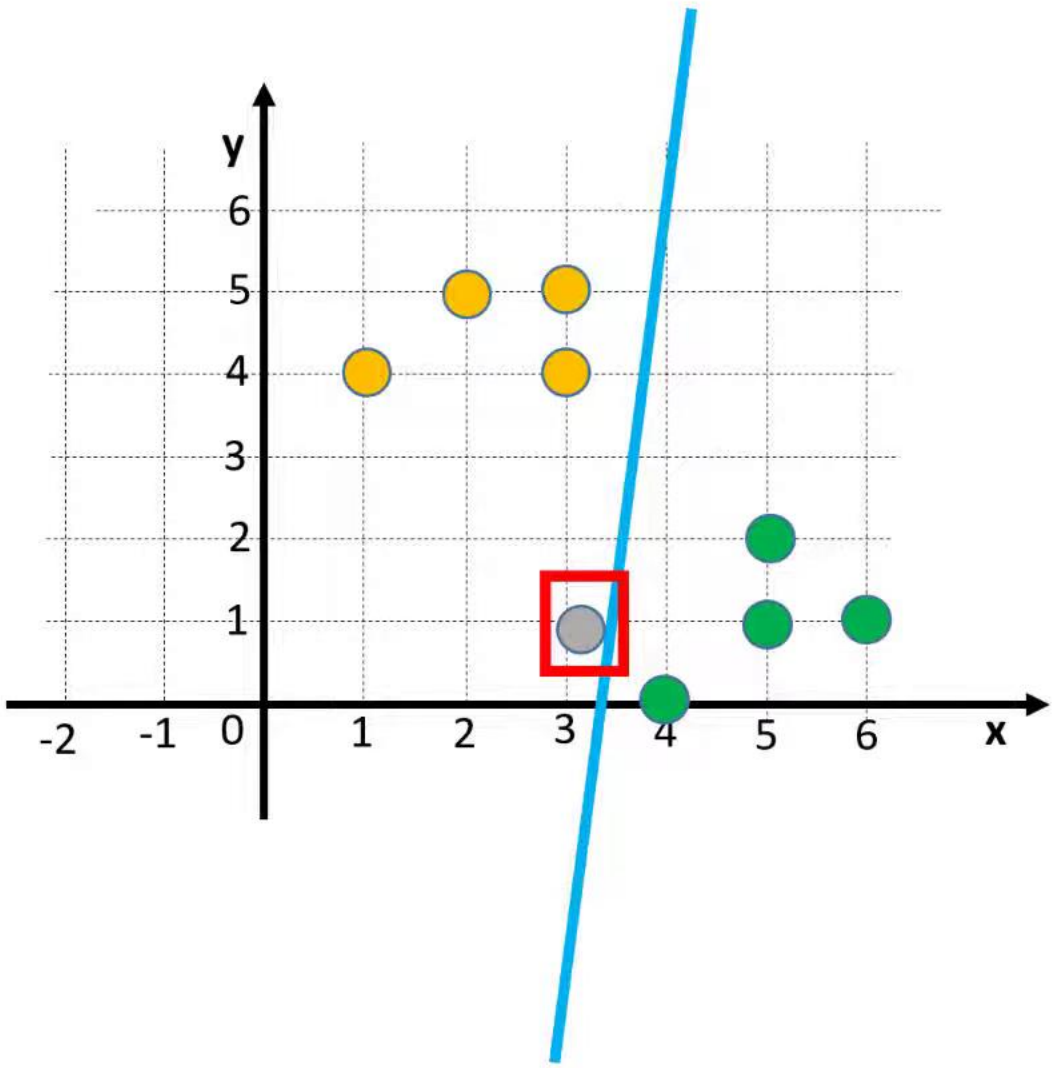
Predict



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

Although this blue line can also separate the two groups completely, it will not be good for predicting new data,

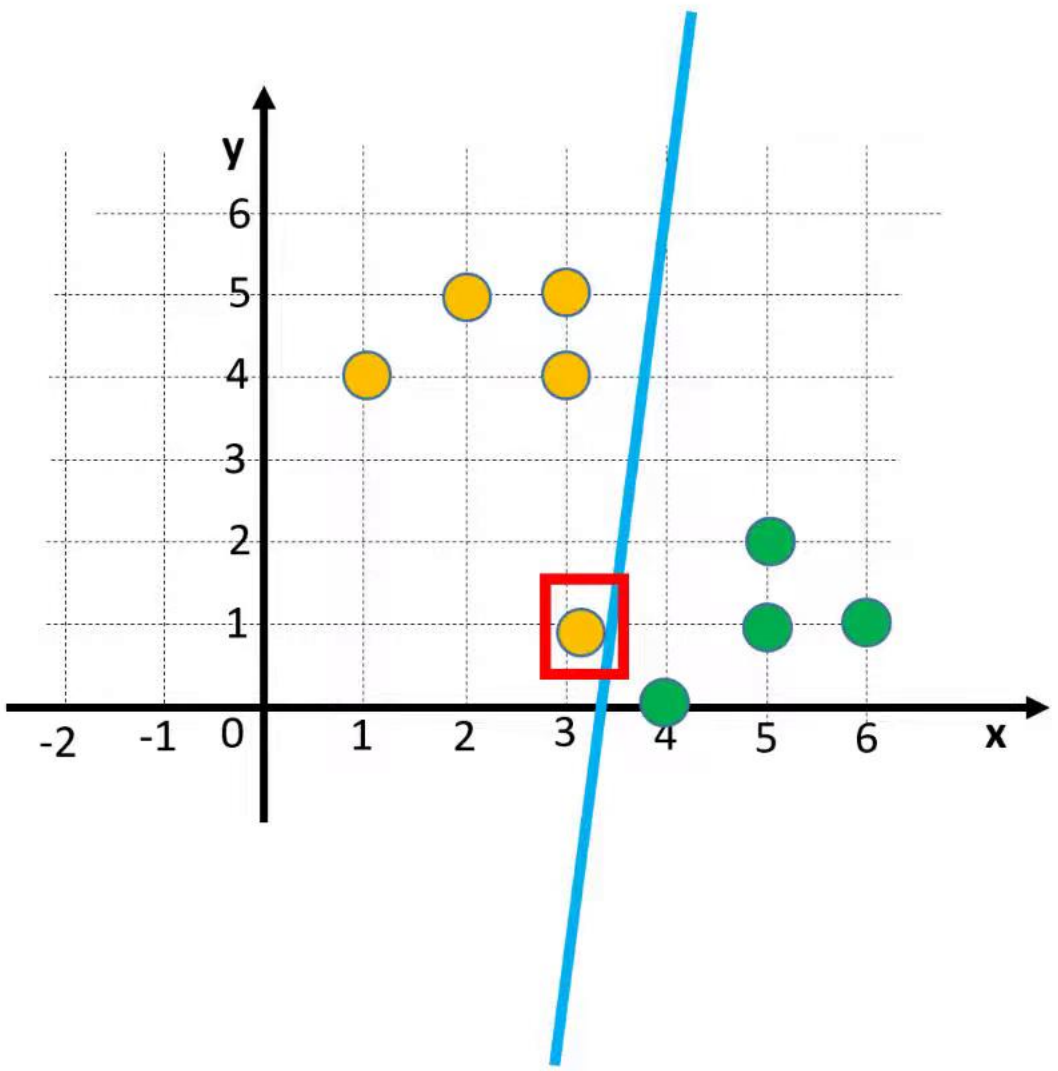
Predict



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

because if we would classify the following data point, according to this blue line,

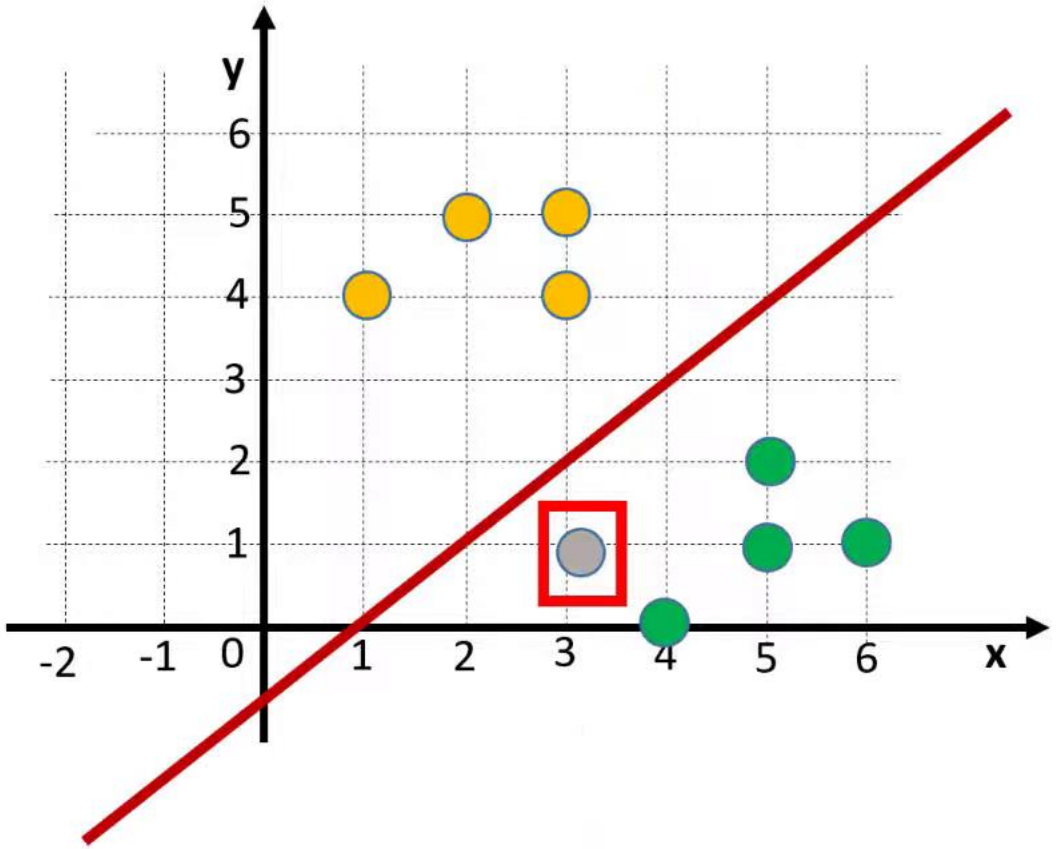
Predict



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

we would predict that it belongs to group A, which means that the person is predicted to have the disease. This does not make sense because the data point is much closer to the green points, which represent the healthy individuals.

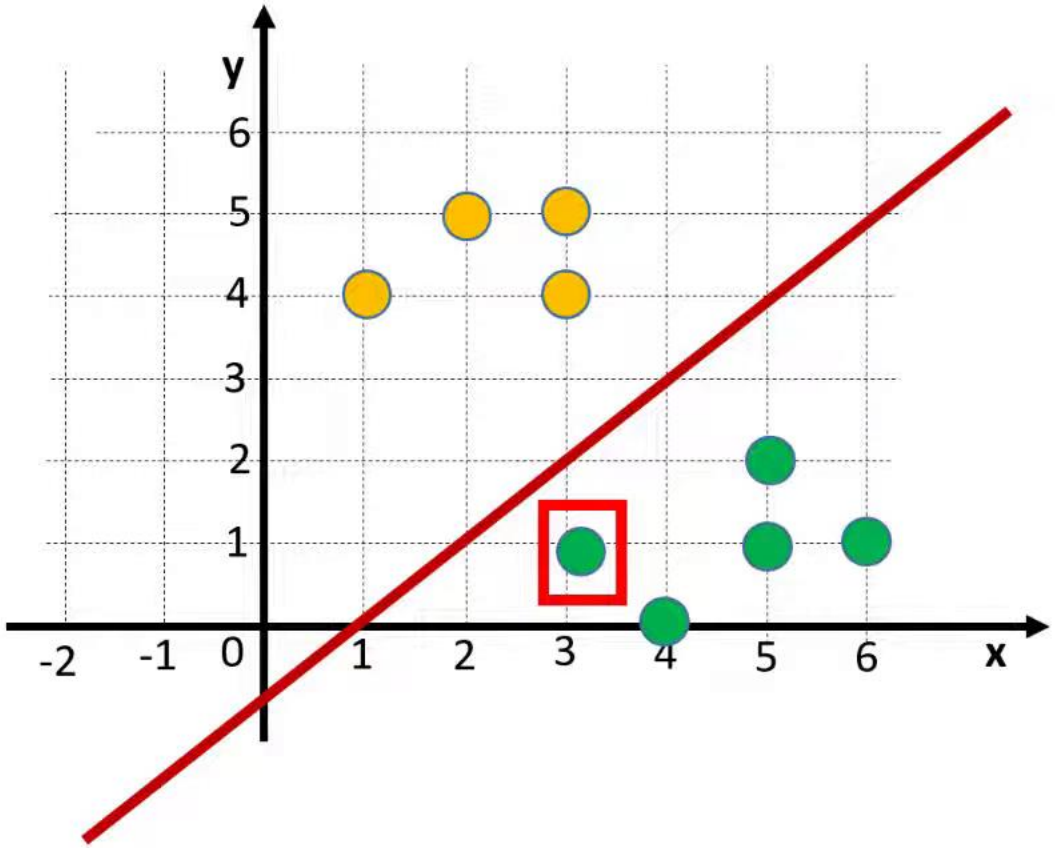
Predict



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

In comparison, by instead using the red line,

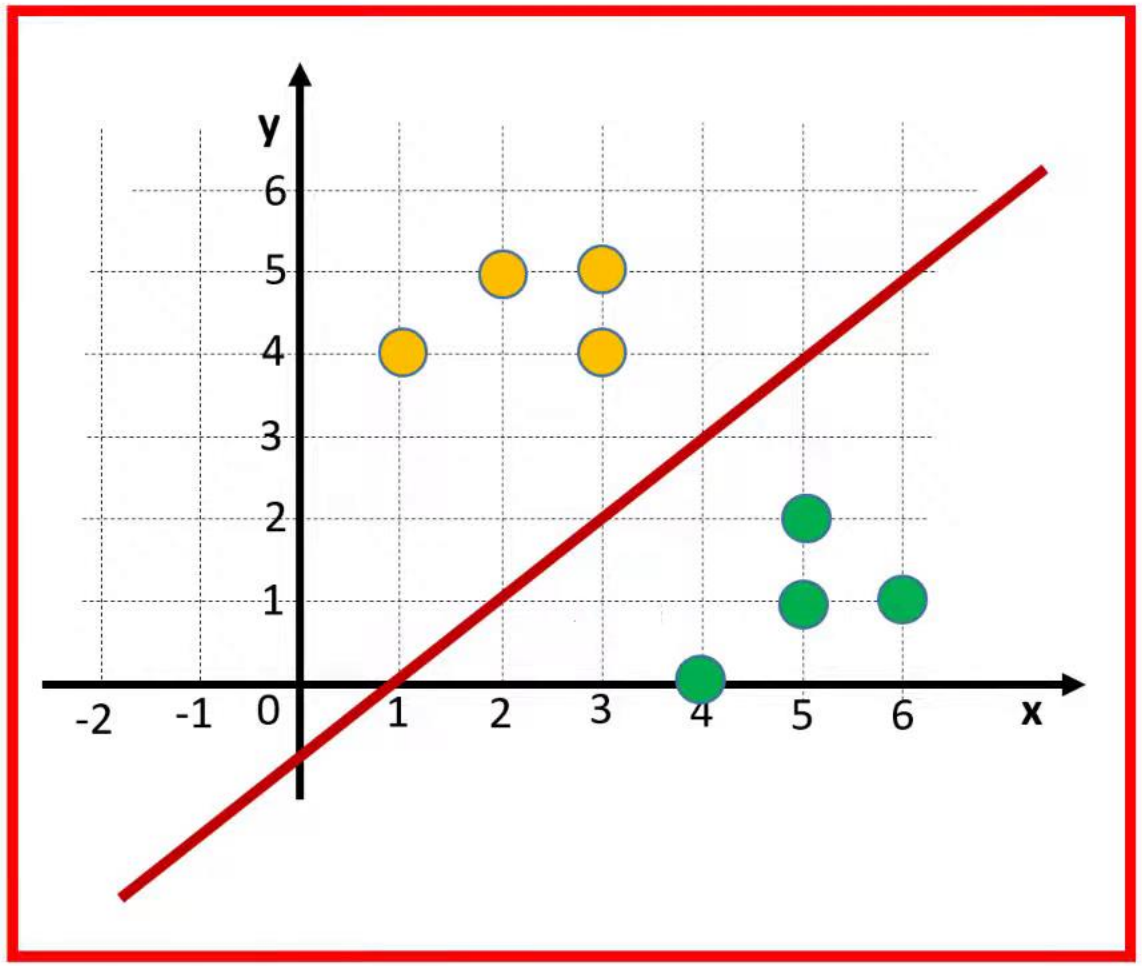
Predict



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

we would classify the data point as belonging to group B, the healthy group, which seems like a better prediction because the point is closer to the green data points.

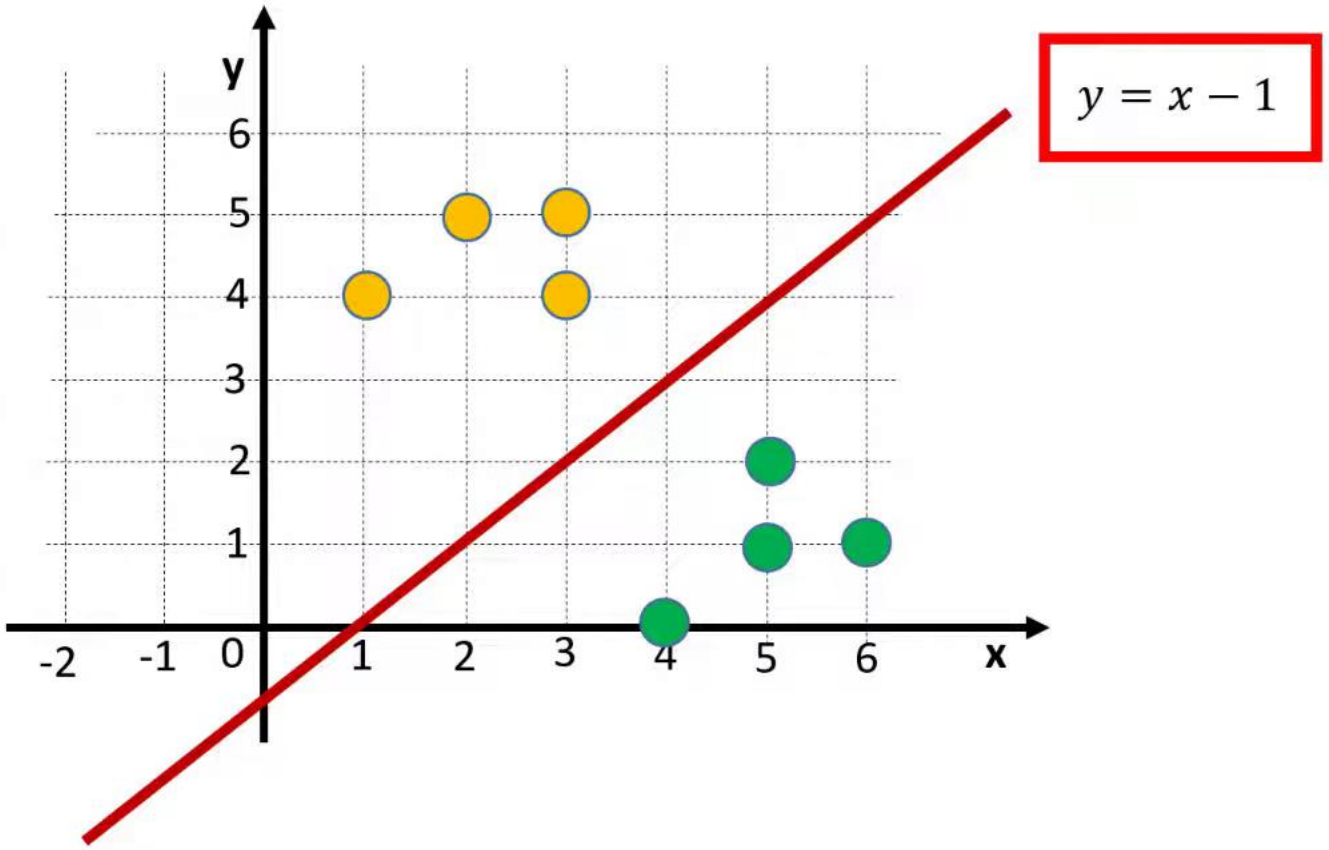
SVM – the math



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

Note that we here only have two dimensions, but SVM work just fine also for several dimensions. This means that we can use it to predict if someone has a disease, or not, based on many measured variables or features.

SVM – the math

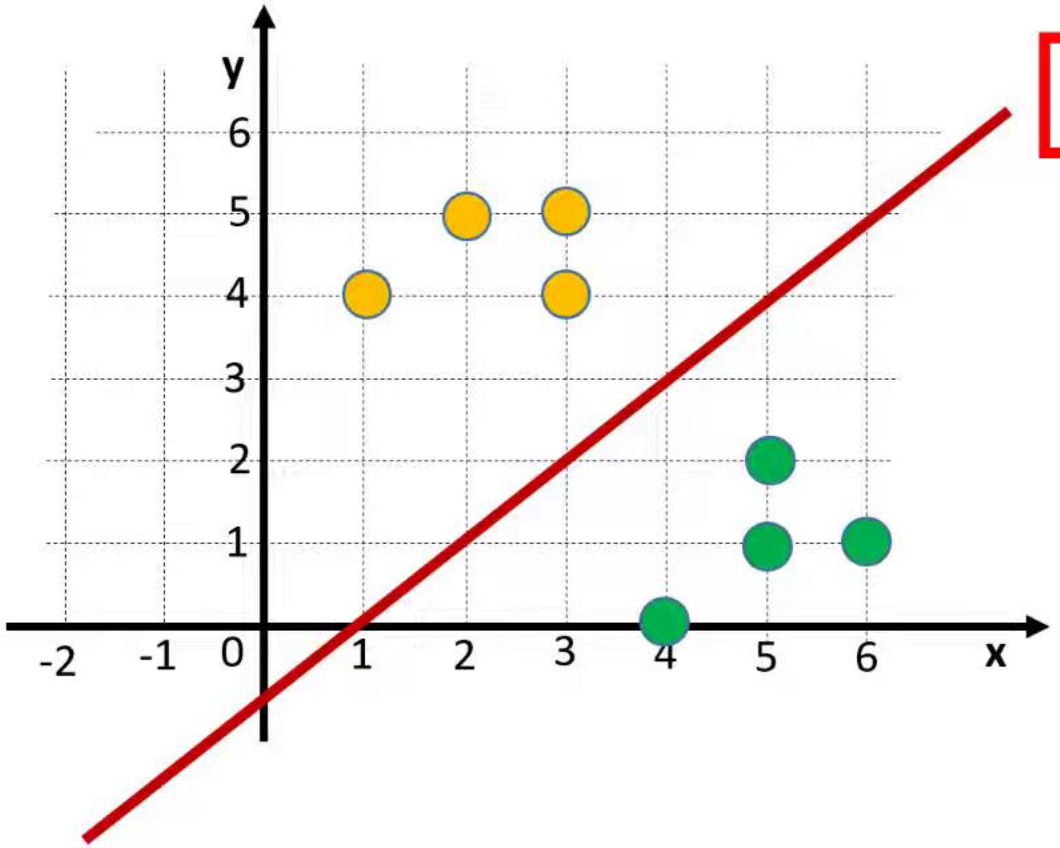


$$y = x - 1$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

This is the equation of the red line, which has an intercept of negative one and a slope of one.

SVM – the math

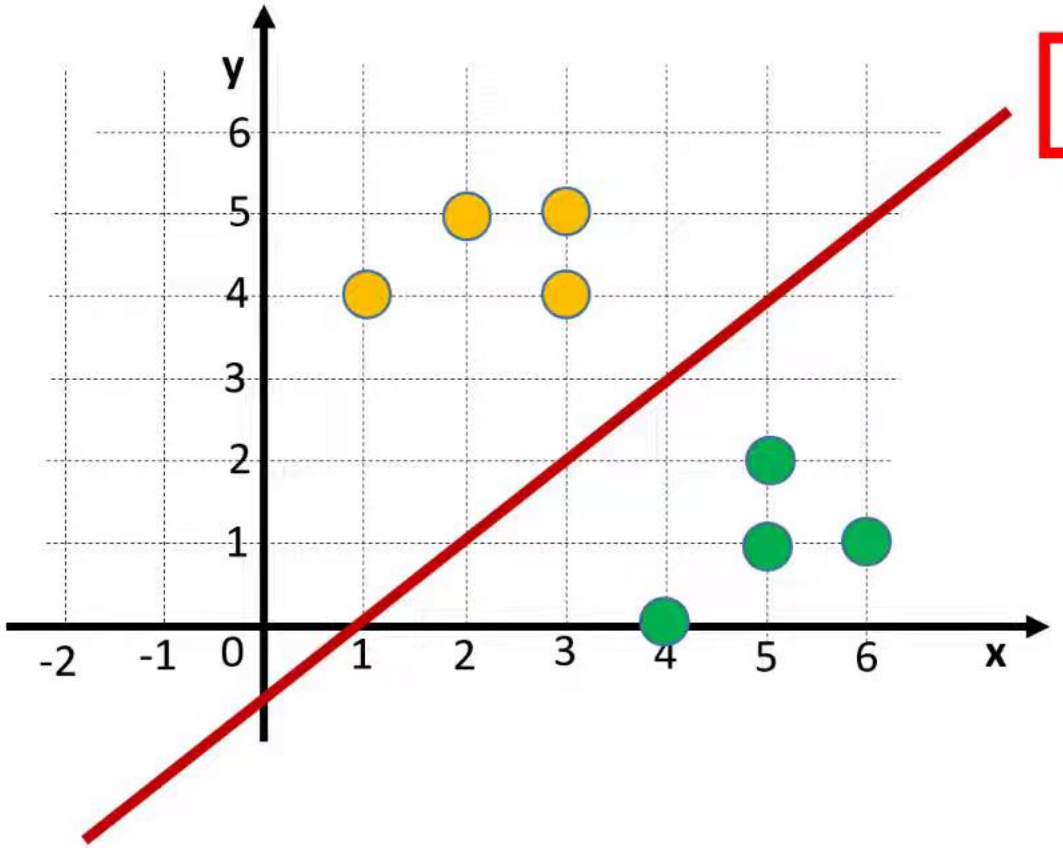


$$-x + y + 1 = 0$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

Let's reformulate the equation to the form we discussed earlier.

SVM – the math

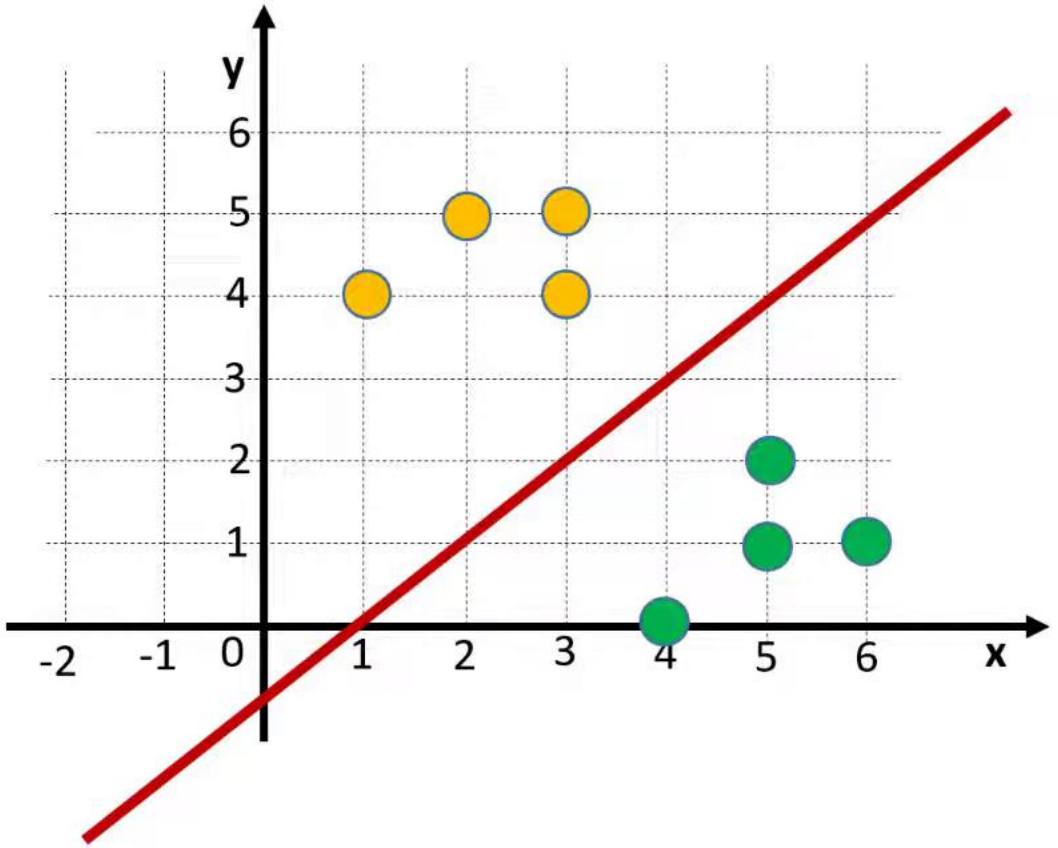


$$-4x + 4y + 4 = 0$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

We can, for example, multiply the terms by 4 because that does not change the position of the line.

SVM – the math

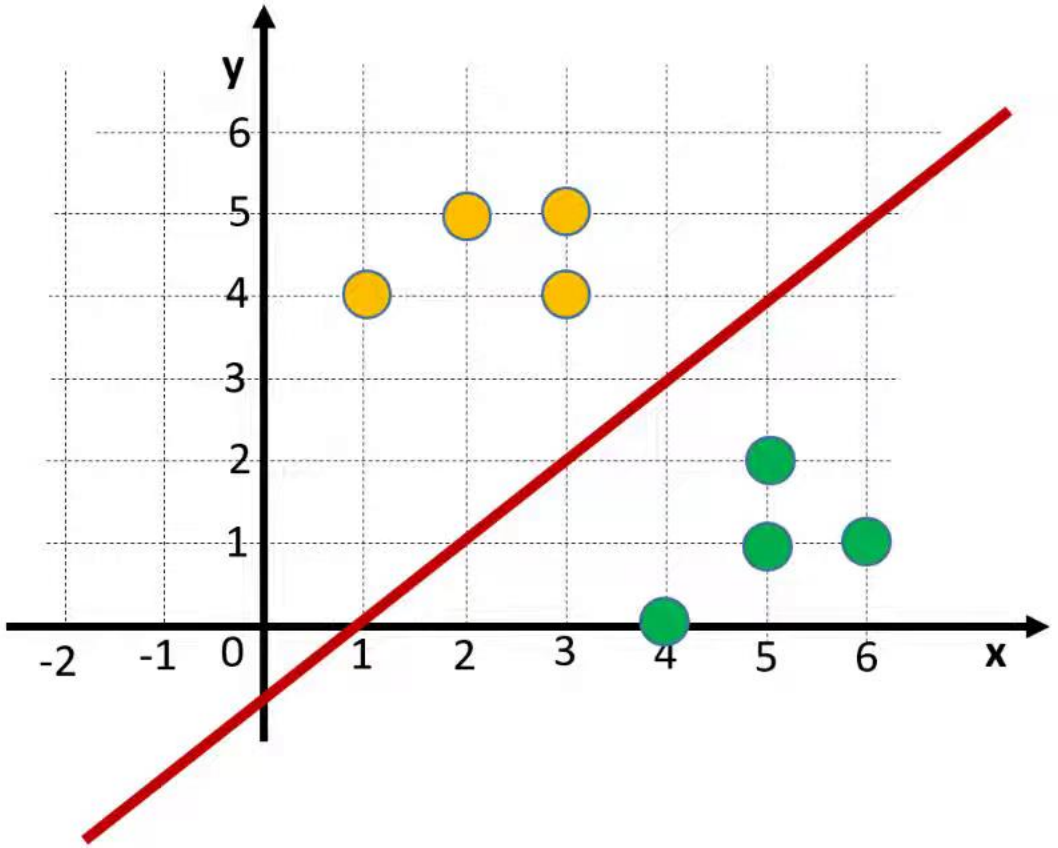


$$-4x + 4y + 4 = 0$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

Remember that if we solve this equation for y,

SVM – the math



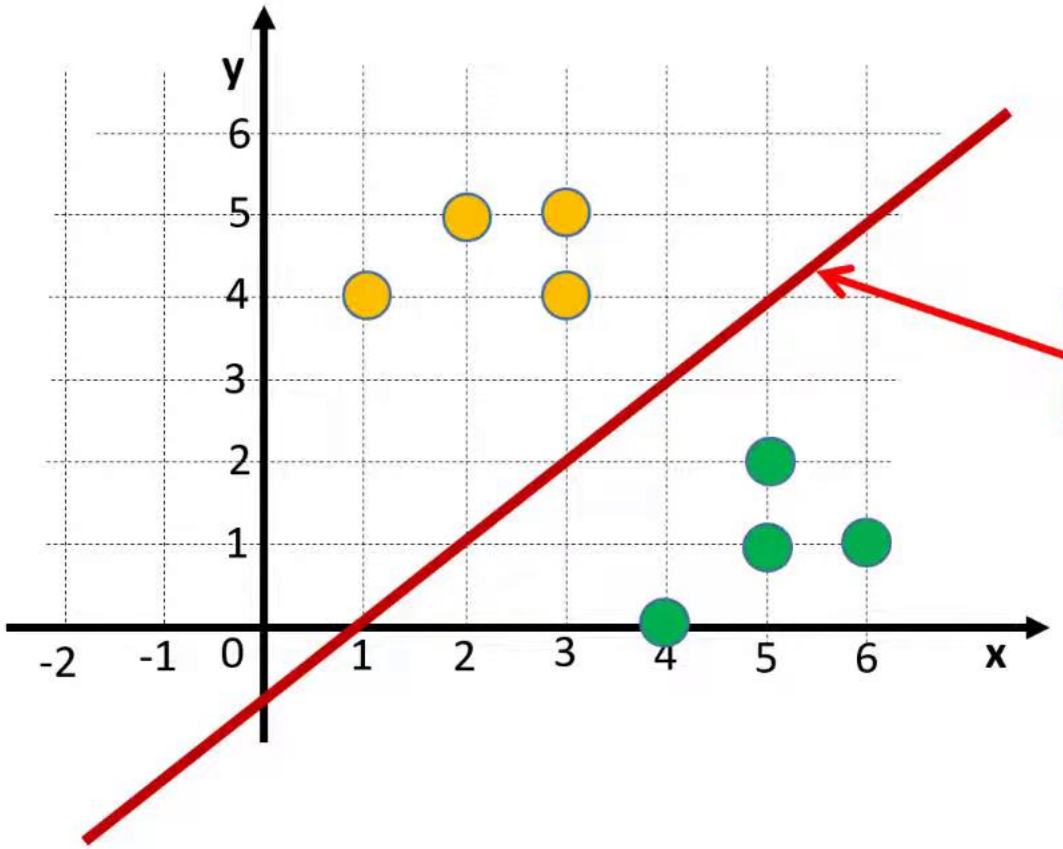
$$-4x + 4y + 4 = 0$$

$$y = x - 1$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

we will end up with the following equation that tells us that the line has a slope of one and an intercept of negative one.

SVM – the math



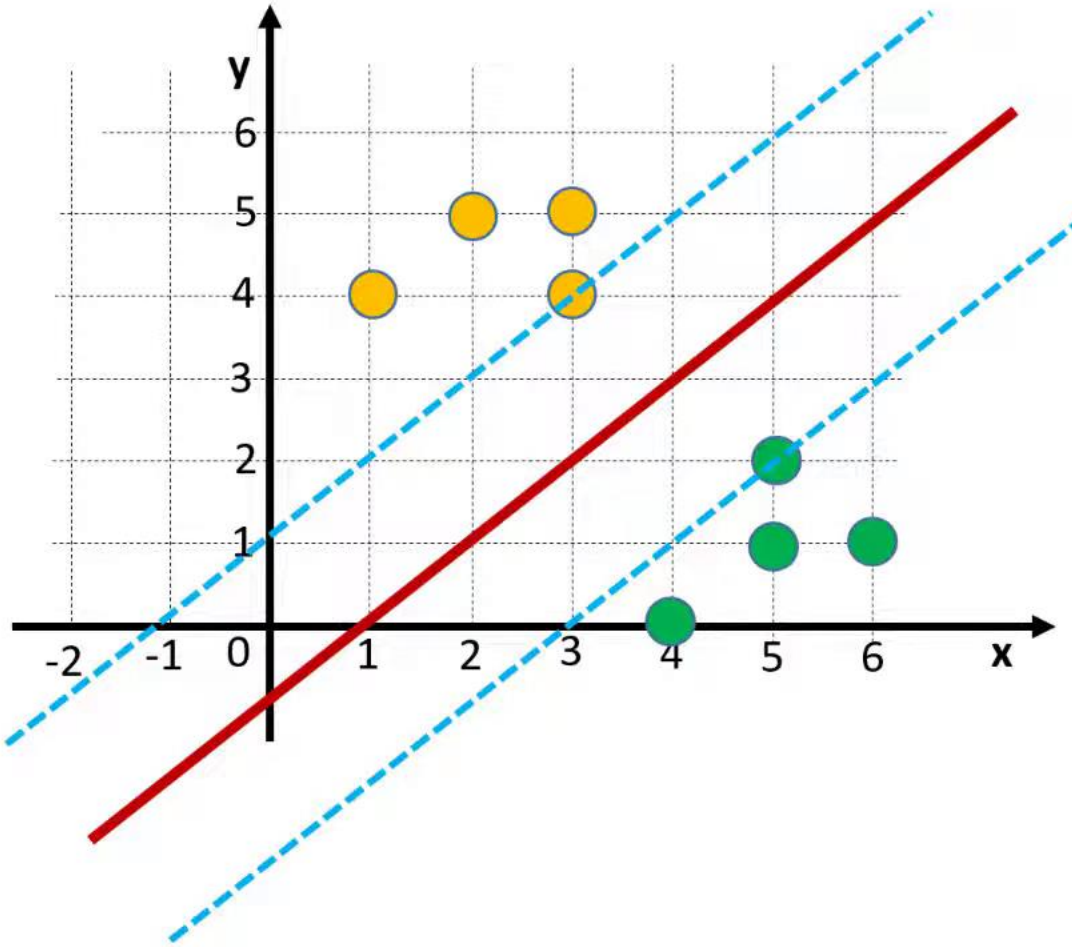
$$-4x + 4y + 4 = 0$$

Hyperplane

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

Note that this red line is called a hyperplane in SVM.

SVM – the math

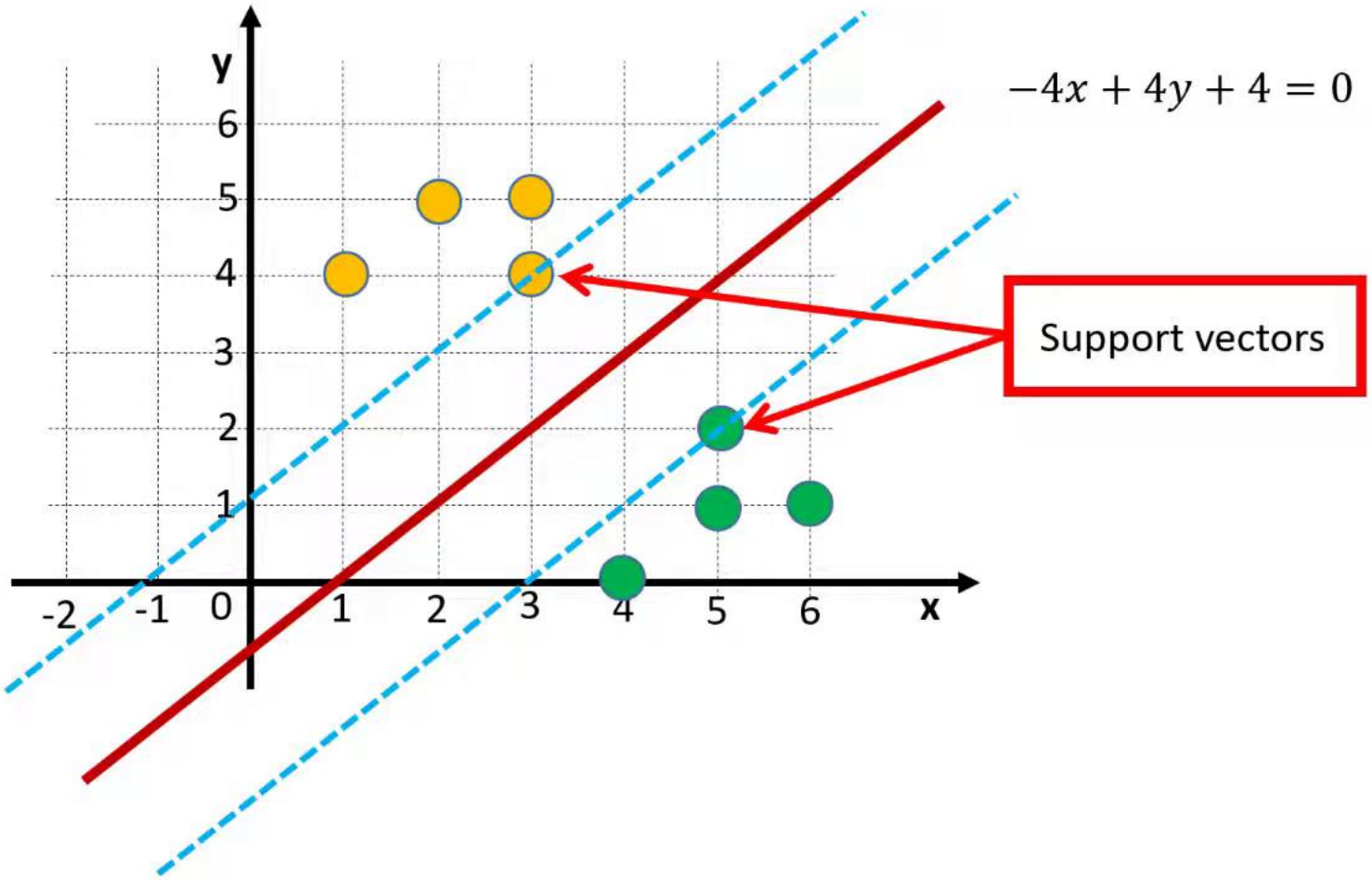


$$-4x + 4y + 4 = 0$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

We will now draw two parallel lines that intercept the two closest data points to the hyperplane.

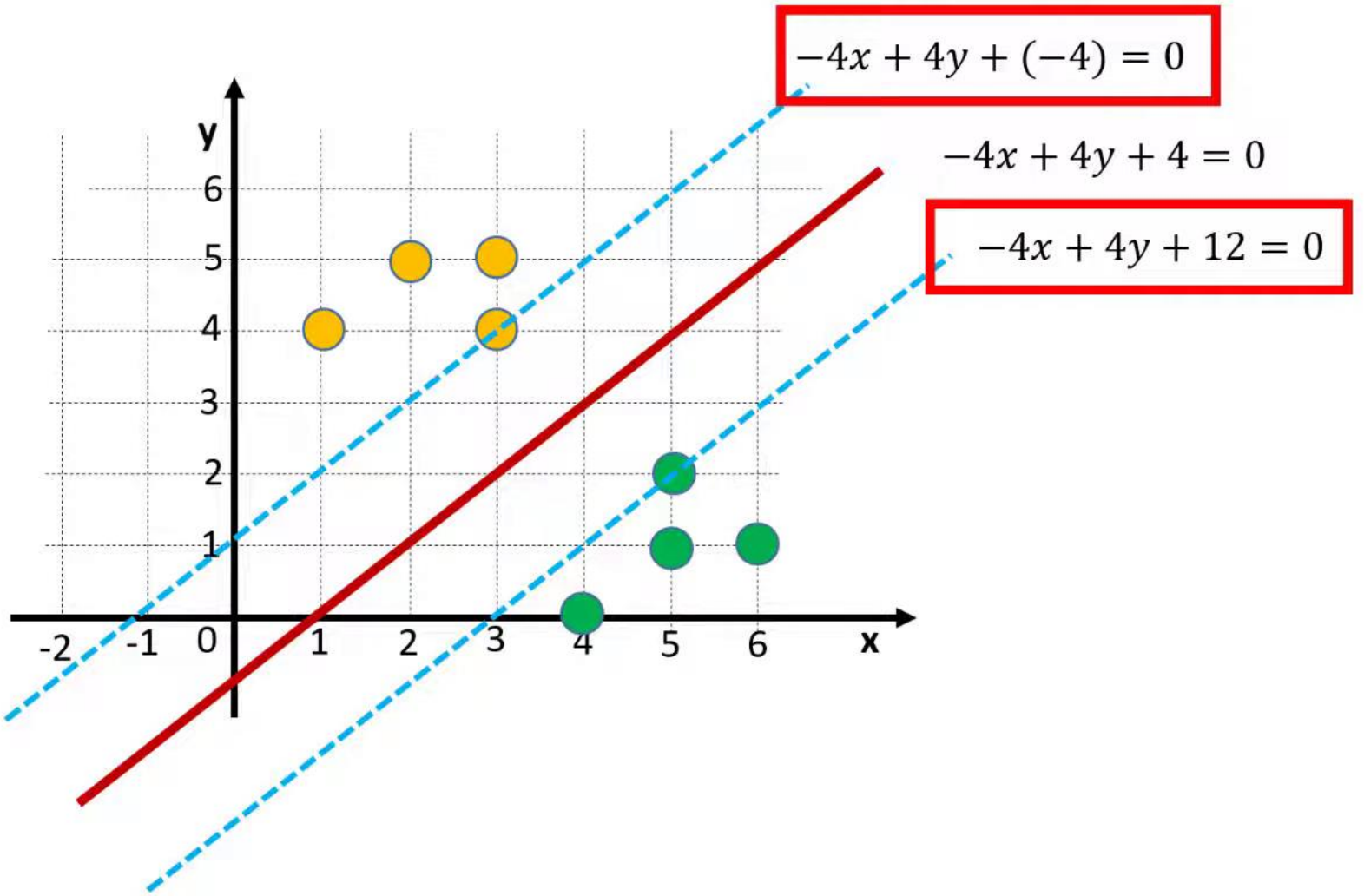
SVM – the math



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

Such data points are called support vectors.

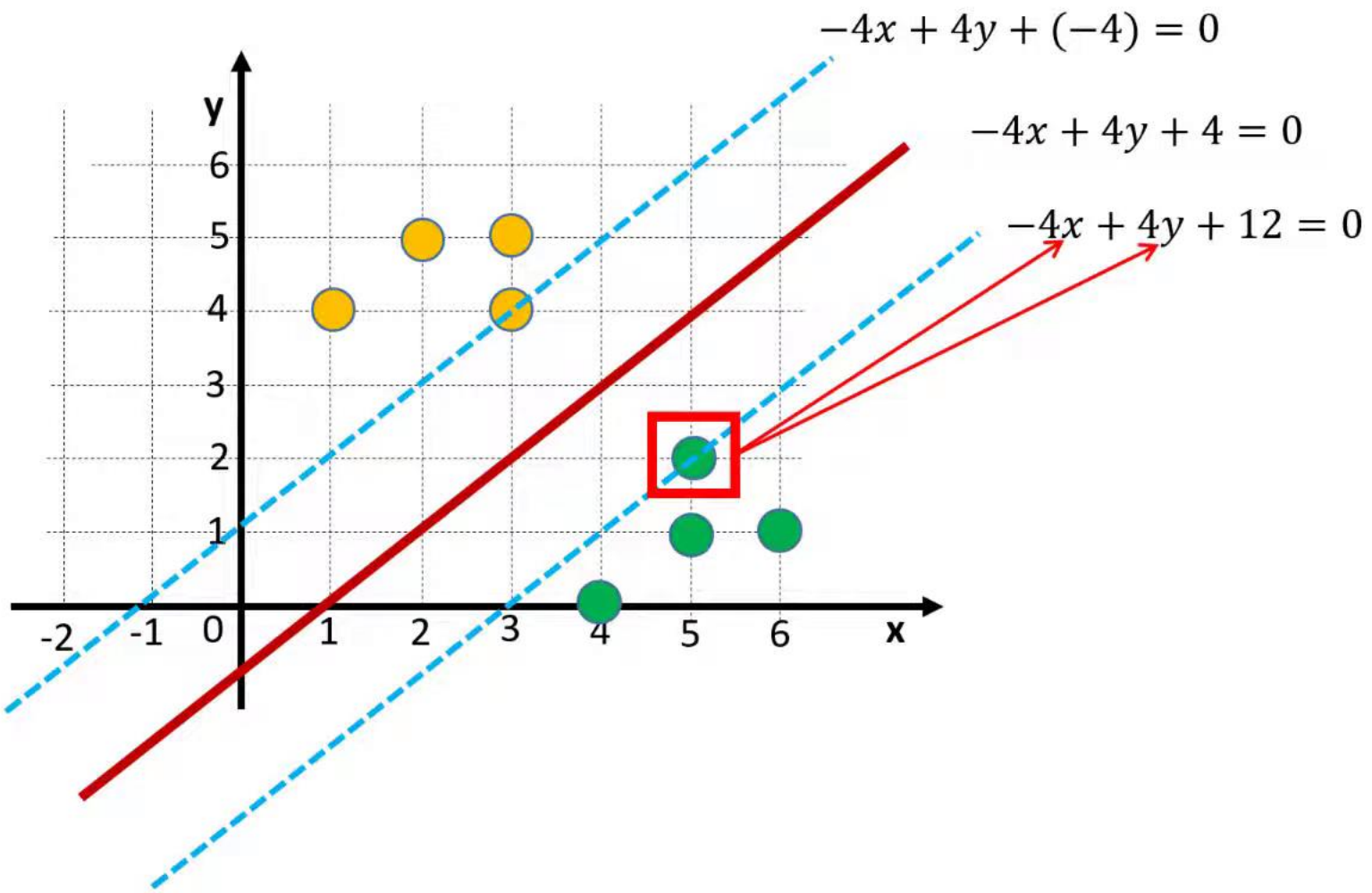
SVM – the math



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

The equations of these two blue lines look like this.

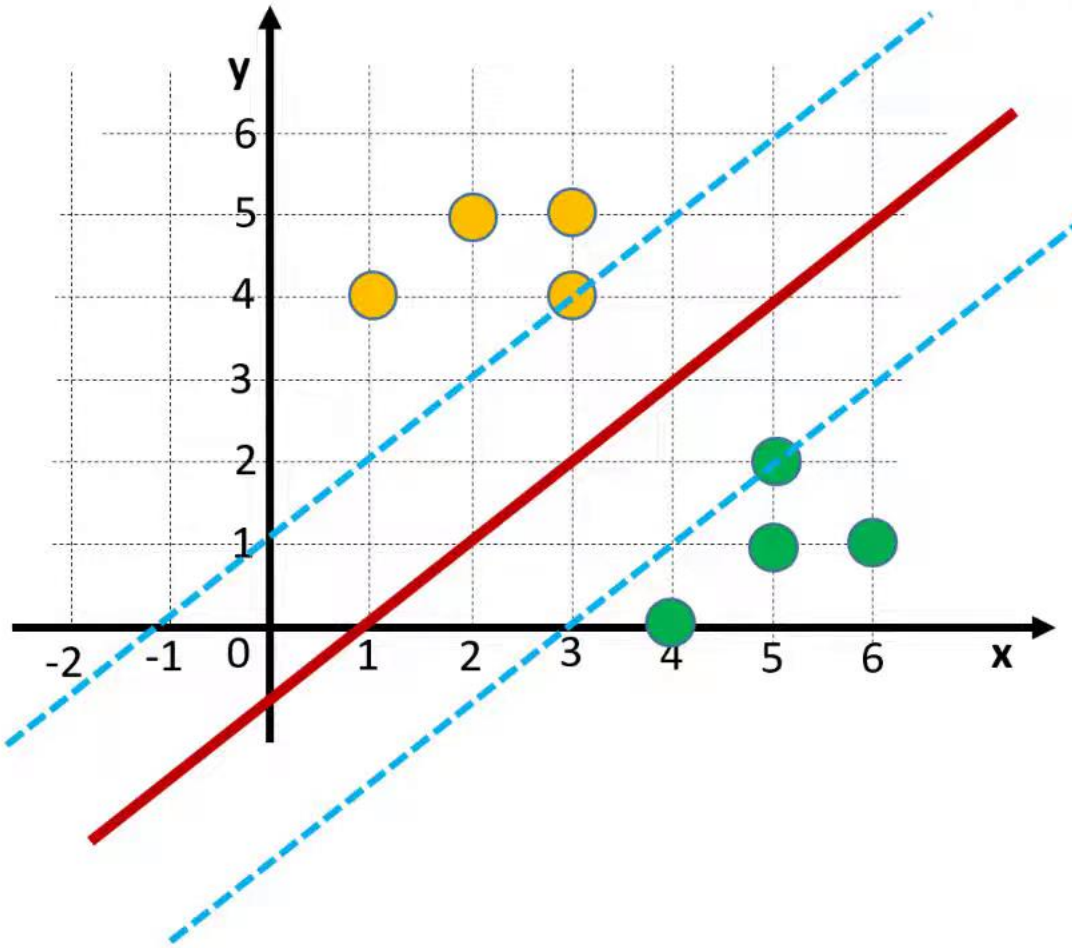
SVM – the math



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

For example, if we plug in the x and y-coordinates of a point on this blue line,

SVM – the math



$$-4x + 4y + (-4) = 0$$

$$-4x + 4y + 4 = 0$$

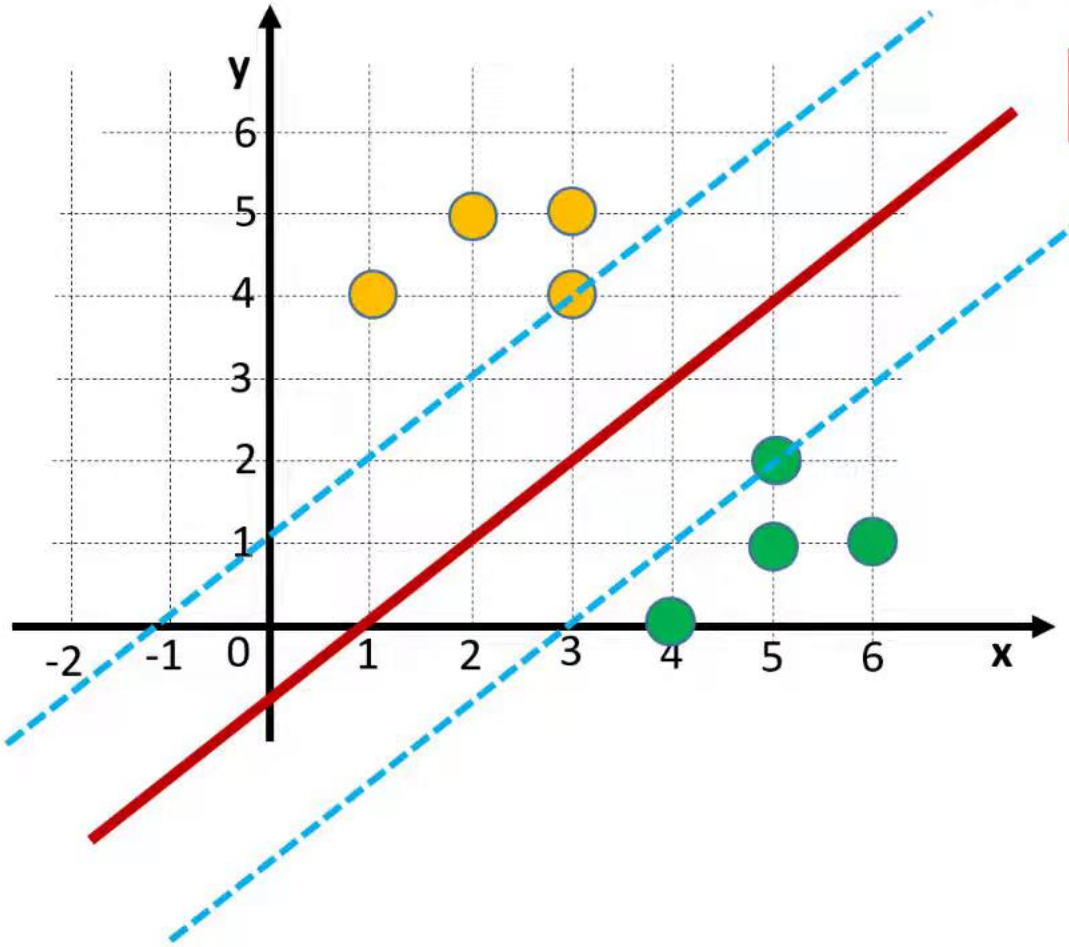
$$-4x + 4y + 12 = 0$$

$$-4 \cdot 5 + 4 \cdot 2 + 12 = 0$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

the left-hand side should be equal to zero, which is true in this case.

SVM – the math



$$-4x + 4y + (-4) = 0$$

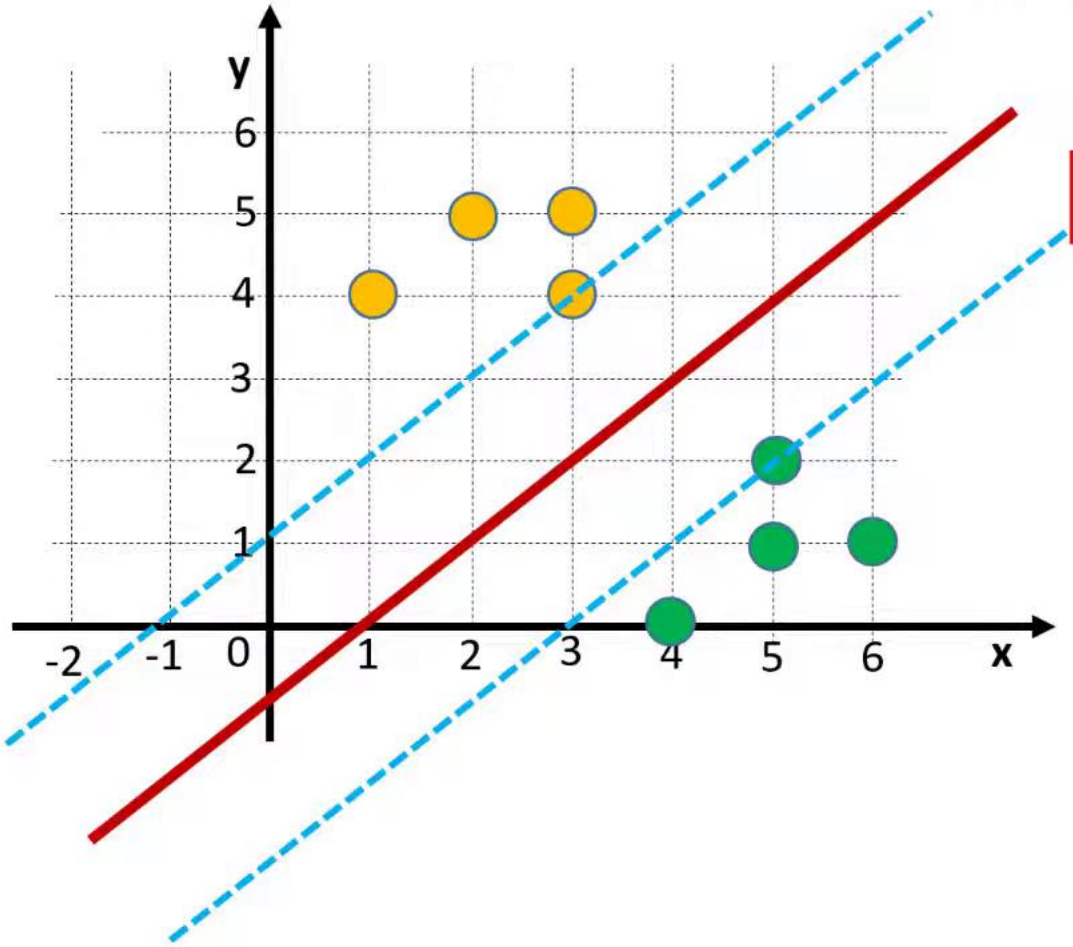
$$\boxed{-4x + 4y + 4 = 0}$$

$$-4x + 4y + 12 = 0$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

Can we somehow normalize this equation that represents the hyperplane,

SVM – the math



$$-4x + 4y + (-4) = 0$$

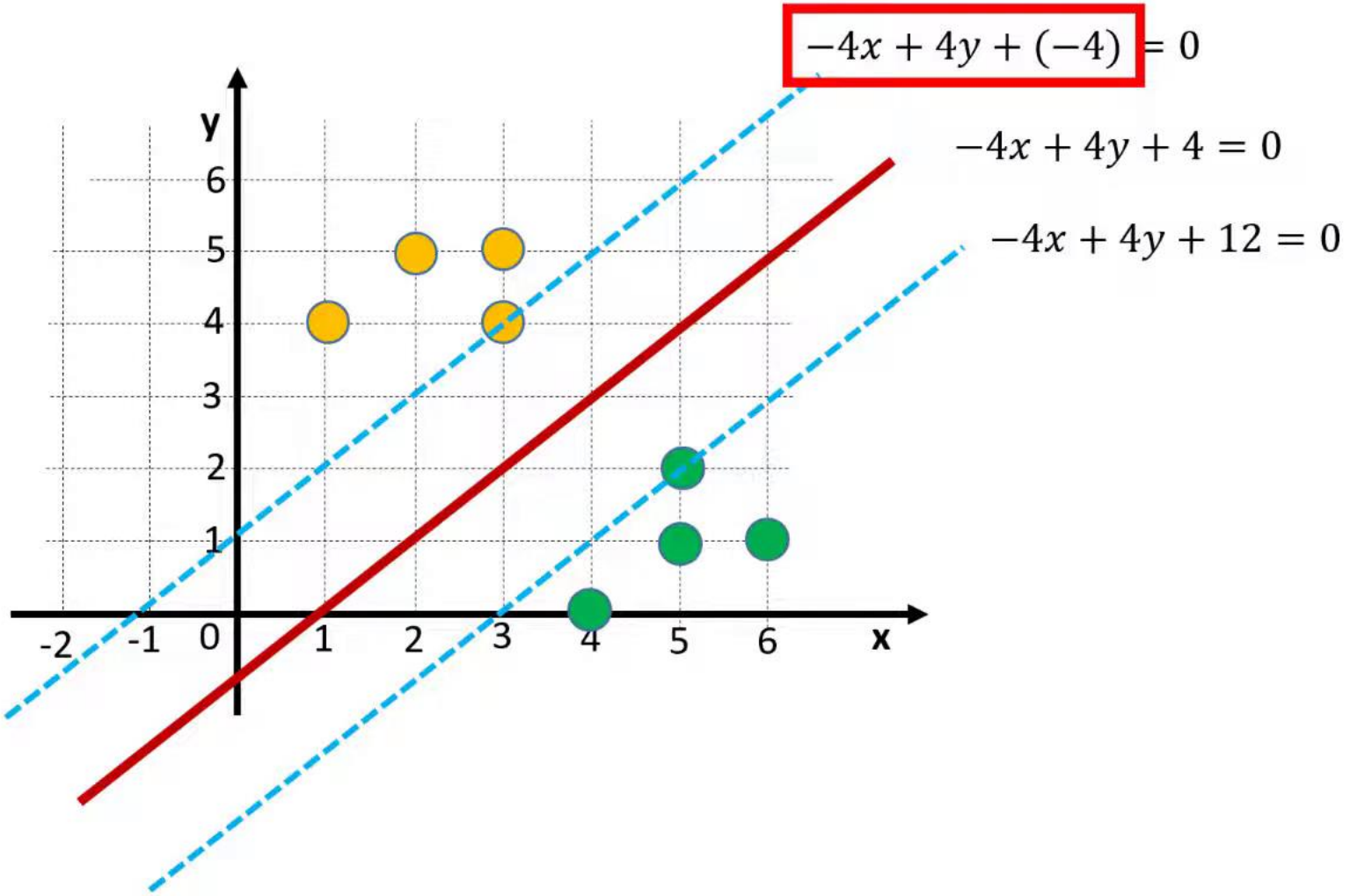
$$-4x + 4y + 4 = 0$$

$$-4x + 4y + 12 = 0$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

so that the left-hand side of this equation is equal to negative one,

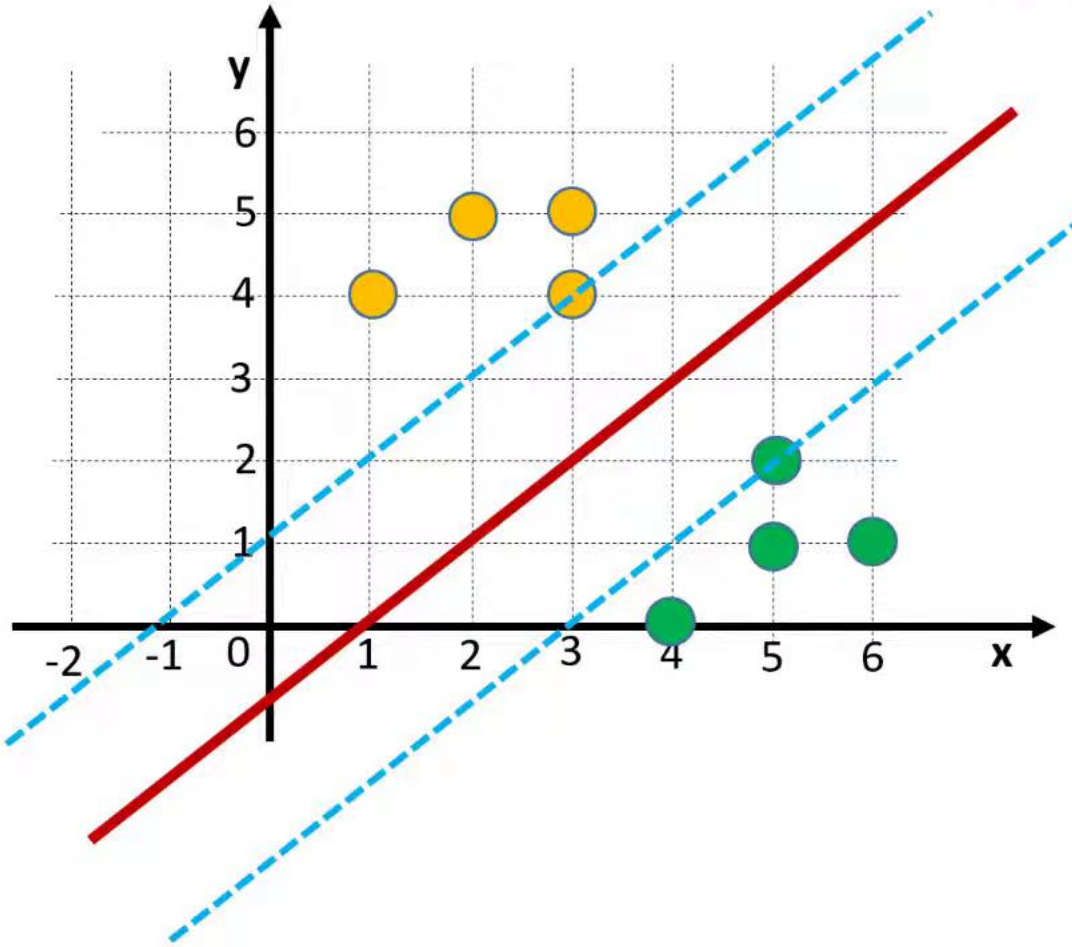
SVM – the math



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

and that the left-hand side of this equation is equal to positive one,

SVM – the math



$$-4x + 4y + \boxed{(-4)} = 0$$

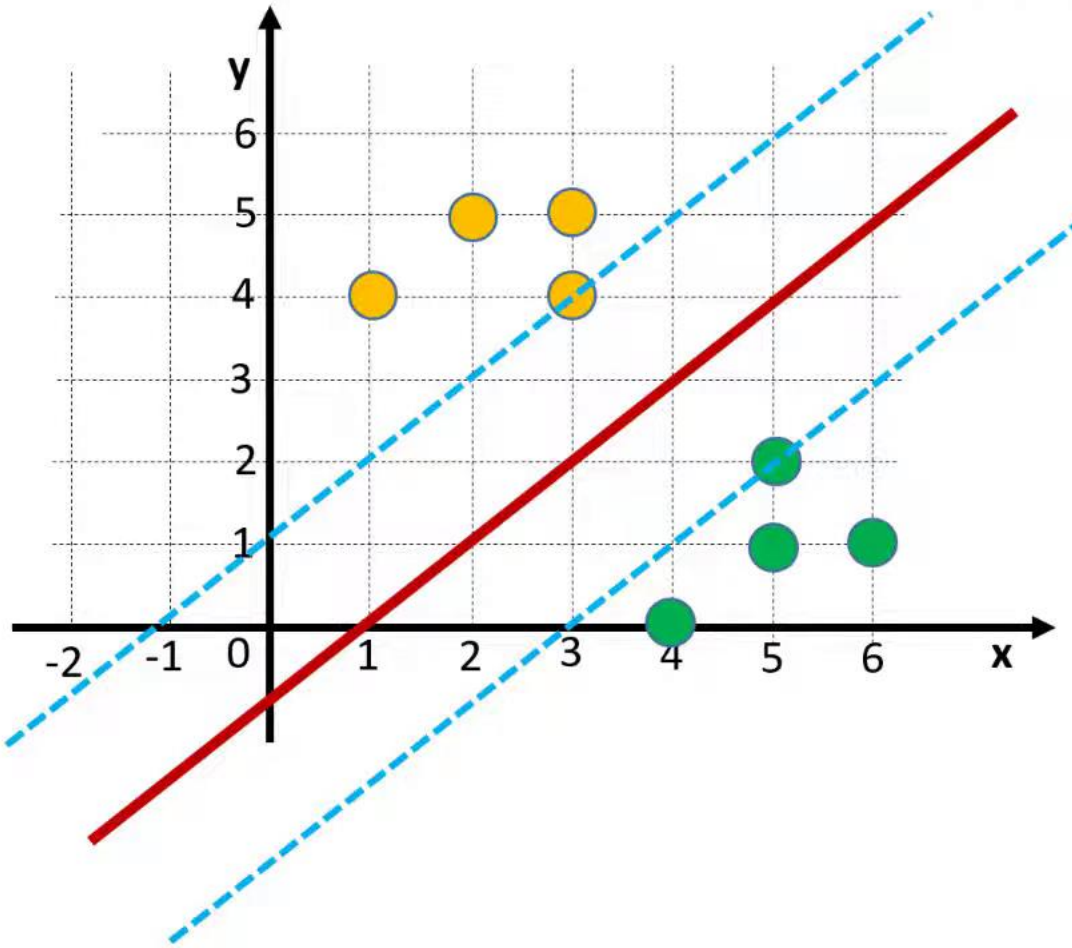
$$-4x + 4y \boxed{+ 4} = 0$$

$$-4x + 4y \boxed{+ 12} = 0$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

given that all three equations have the same constant term on the left-hand side as the equation of the hyperplane?

SVM – the math



$$-4x + 4y + (-4) = 0$$

$$-4x + 4y + 4 = 0$$

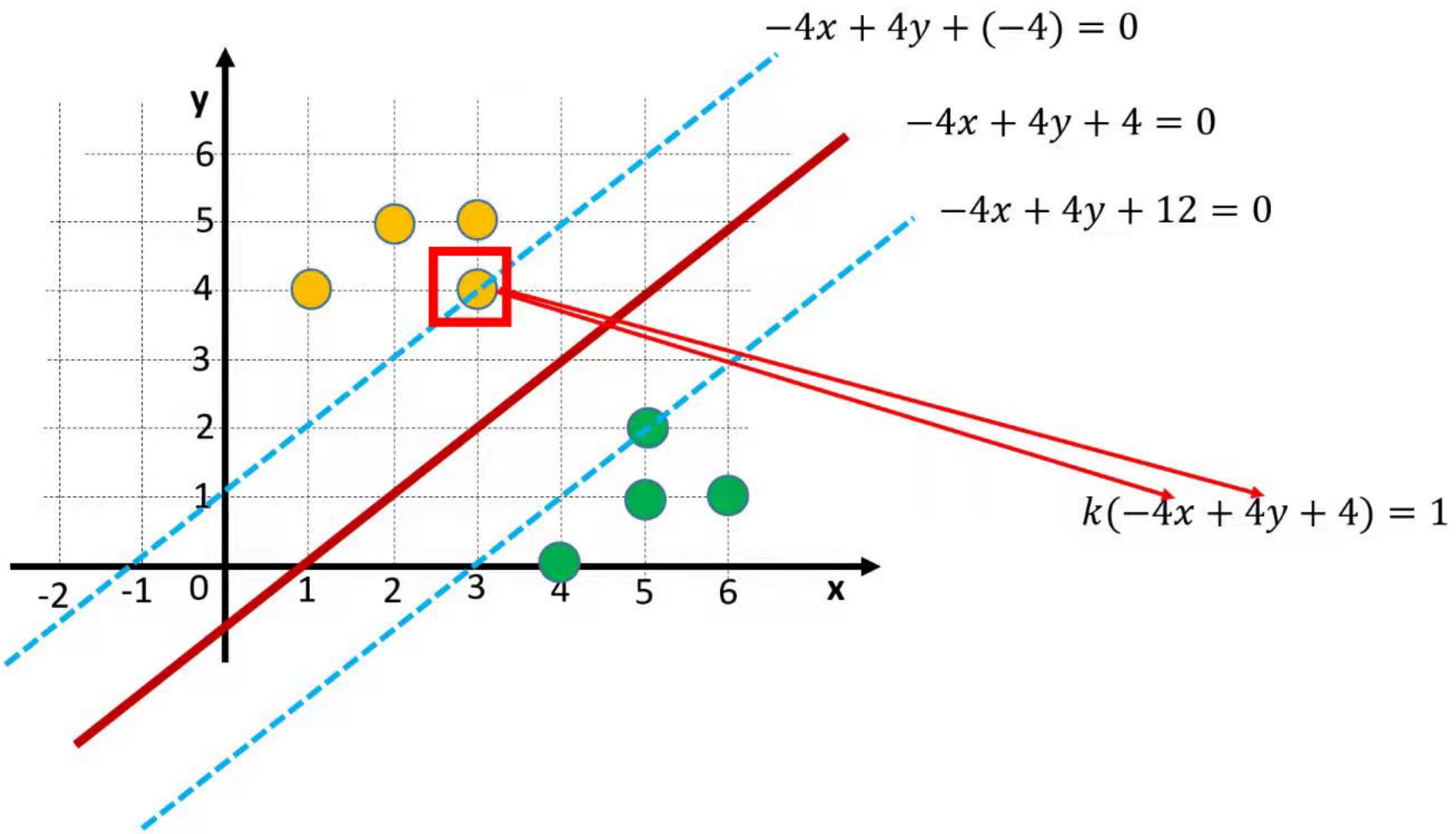
$$-4x + 4y + 12 = 0$$

$$k(-4x + 4y + 4) = 1$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

In other words, can we find a value of *k* so that the left-hand side is equal to one (or negative one)?

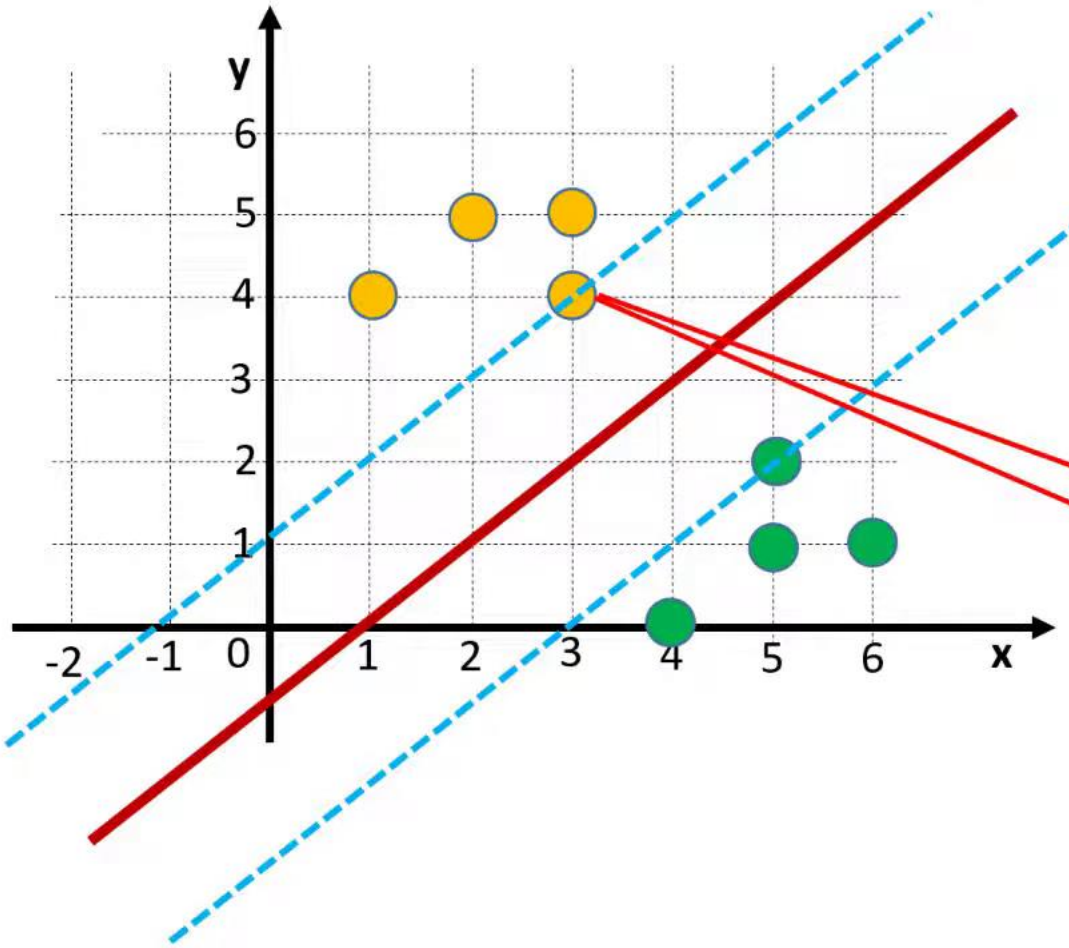
SVM – the math



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

Let's plug in the x and y-coordinates of this support vector,

SVM – the math



$$-4x + 4y + (-4) = 0$$

$$-4x + 4y + 4 = 0$$

$$-4x + 4y + 12 = 0$$

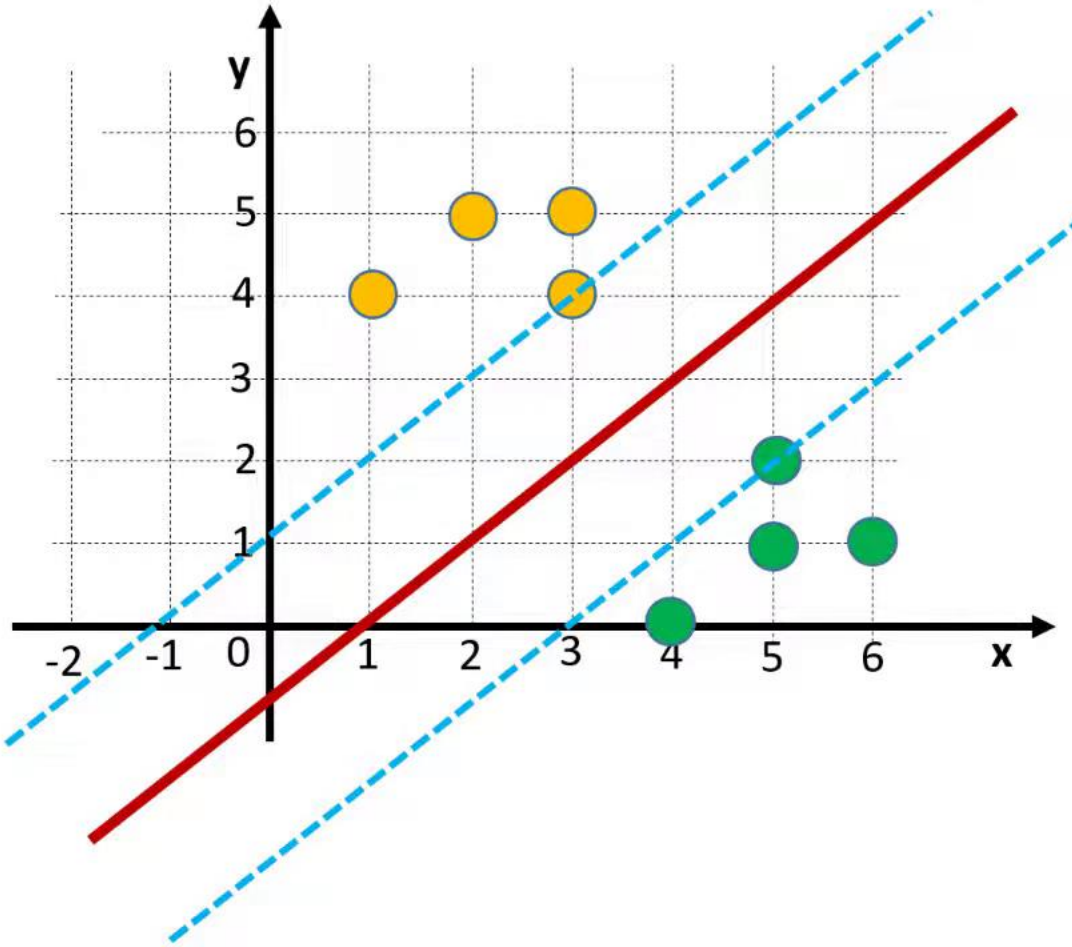
$$k(-4x + 4y + 4) = 1$$

$$k(-4 \cdot 3 + 4 \cdot 4 + 4) = 1$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

like this.

SVM – the math



$$-4x + 4y + (-4) = 0$$

$$-4x + 4y + 4 = 0$$

$$-4x + 4y + 12 = 0$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

$$k(-4x + 4y + 4) = 1$$

$$k(-4 \cdot 3 + 4 \cdot 4 + 4) = 1$$

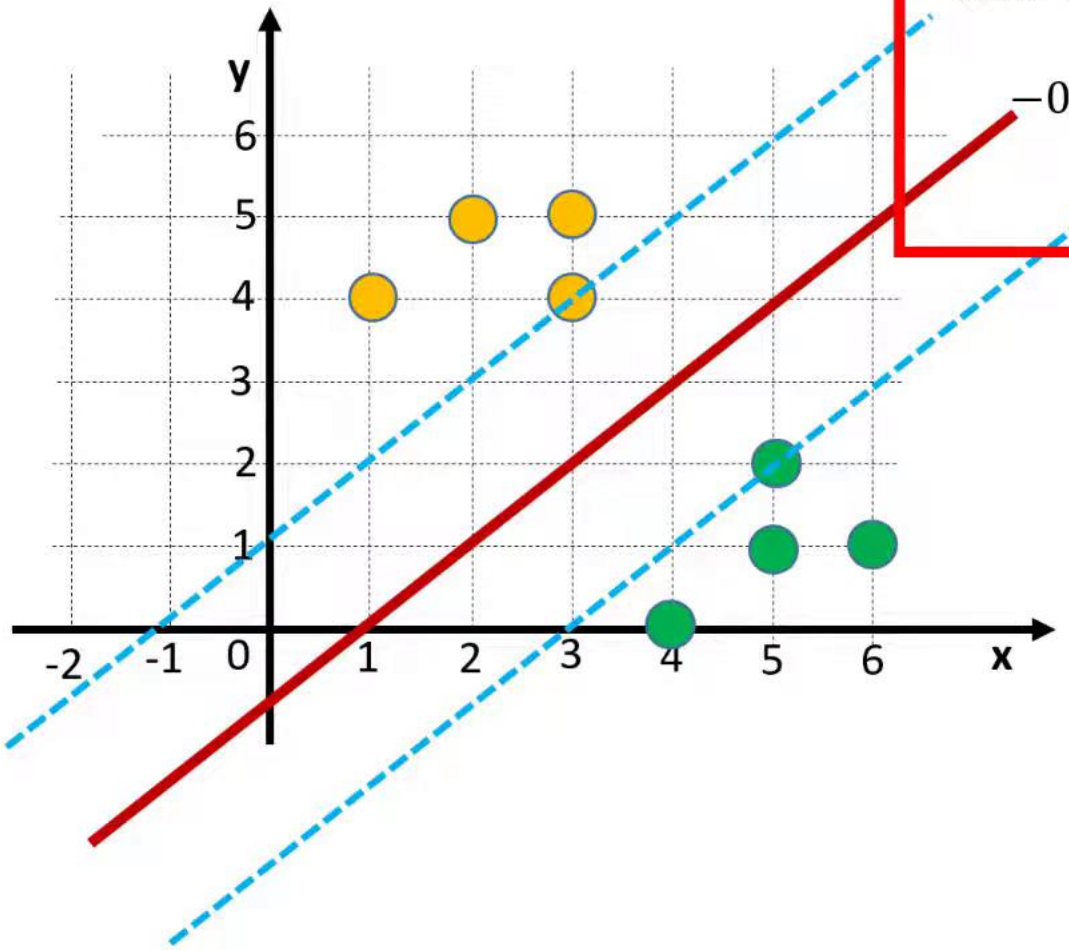
$$k(-12 + 16 + 4) = 1$$

$$k \cdot 8 = 1$$

$$\boxed{k = 1/8}$$

If we solve this equation, we see that we should multiply the terms on the left-hand side by one over eight, or simply divide the terms by eight.

SVM – the math



$$-0.5x + 0.5y + 0.5 = 1$$

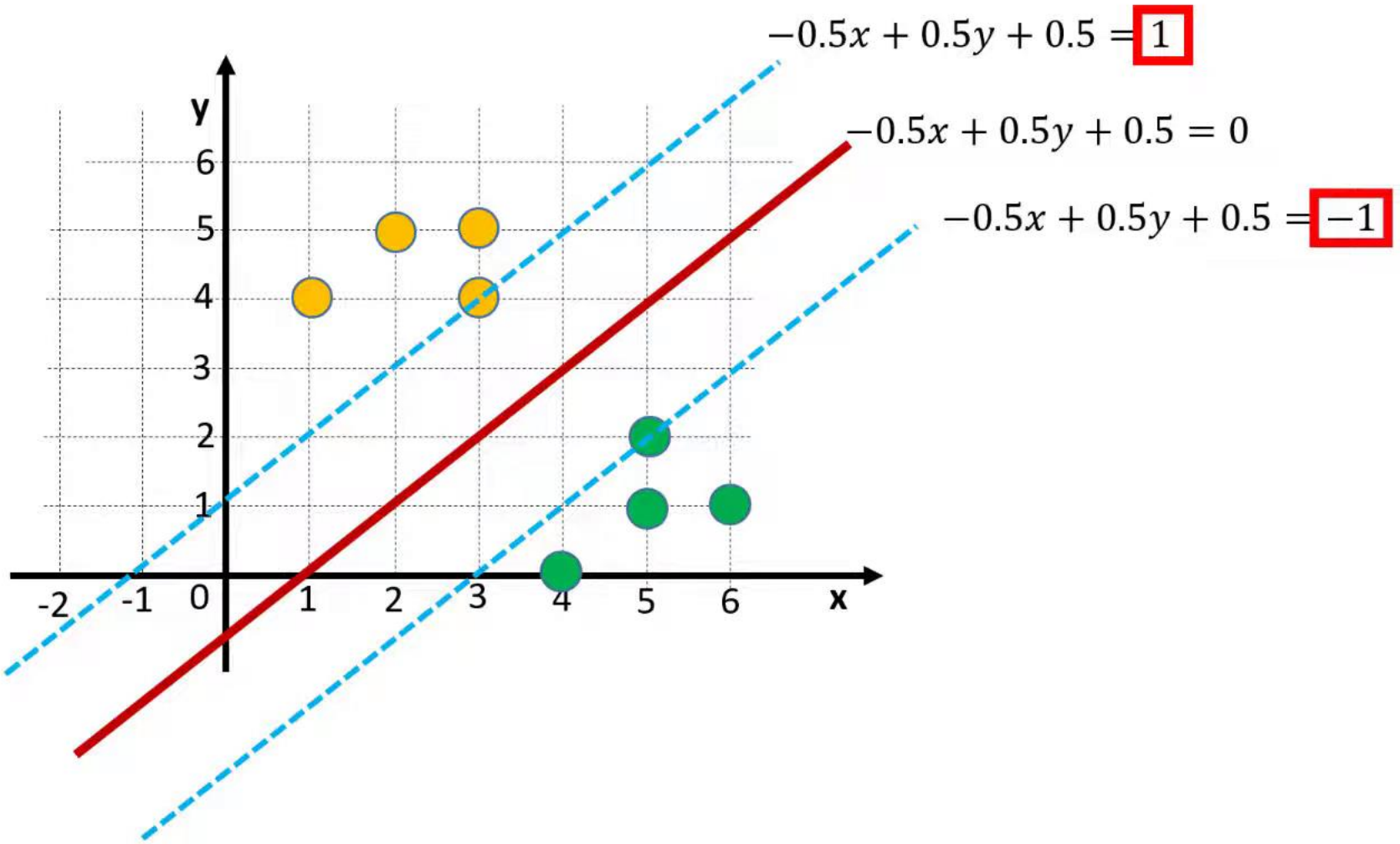
$$-0.5x + 0.5y + 0.5 = 0$$

$$-0.5x + 0.5y + 0.5 = -1$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

After dividing the terms by eight, the equations look like this, which is the standard form of the equations in SVM,

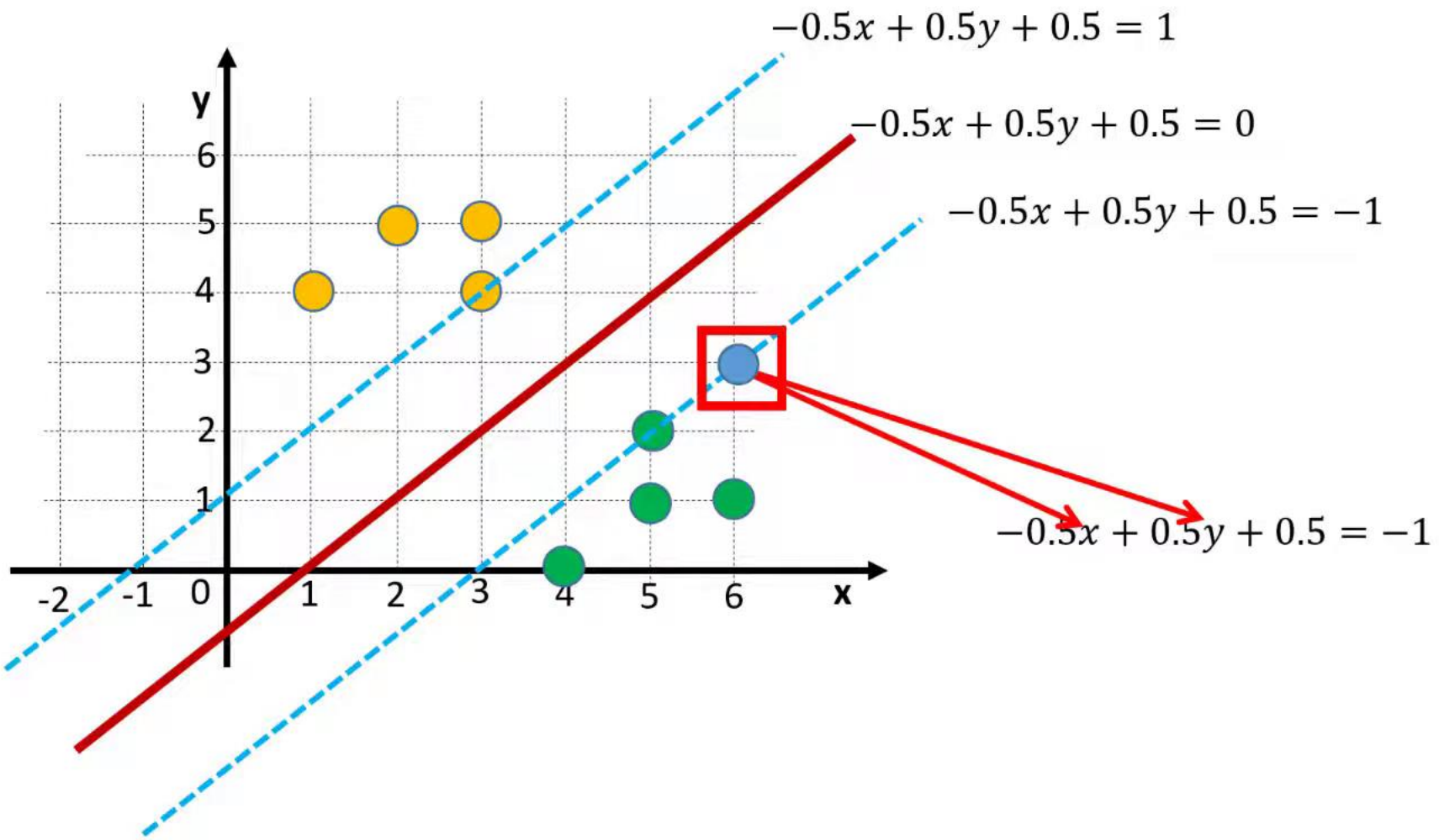
SVM – the math



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

where the right-hand side is equal to positive one and negative one of the two blue lines.

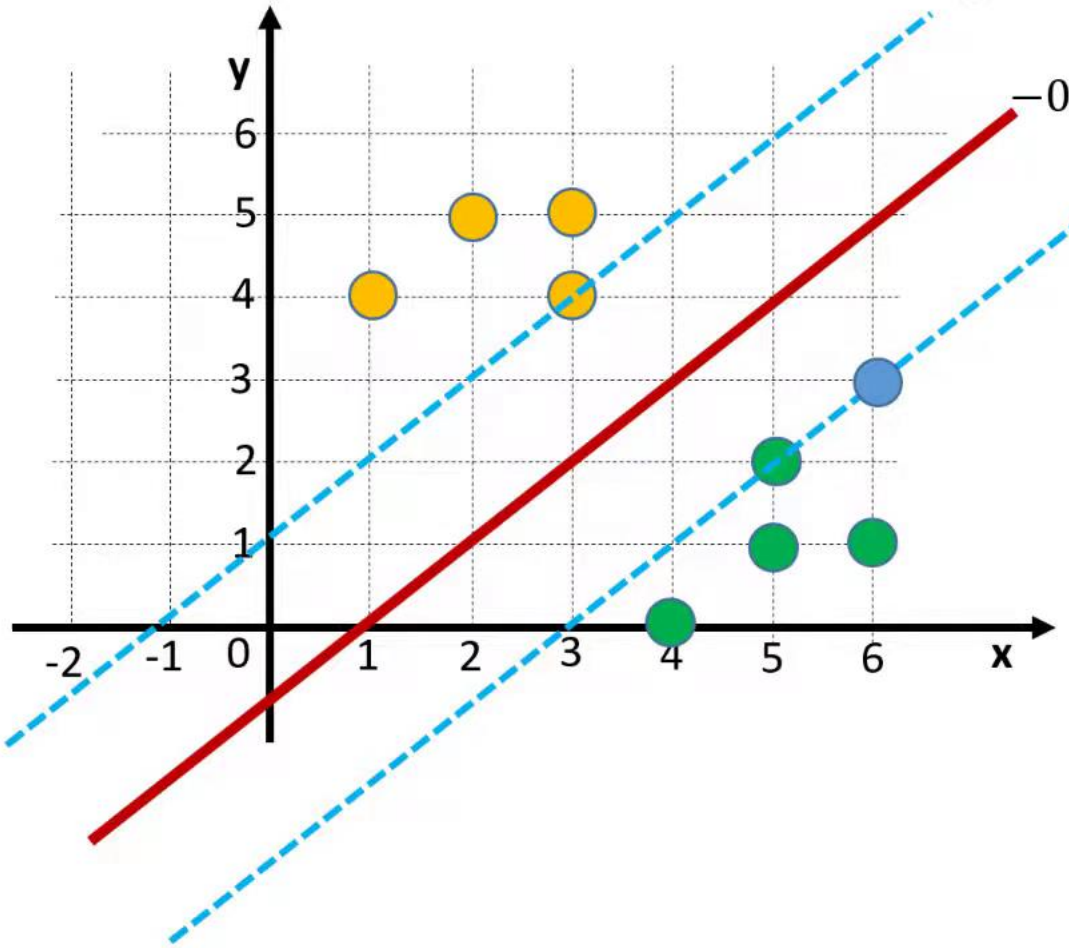
SVM – the math



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

For example, if we plug in the x and y-coordinates of an arbitrary point on this blue line,

SVM – the math



$$-0.5x + 0.5y + 0.5 = 1$$

$$-0.5x + 0.5y + 0.5 = 0$$

$$-0.5x + 0.5y + 0.5 = -1$$

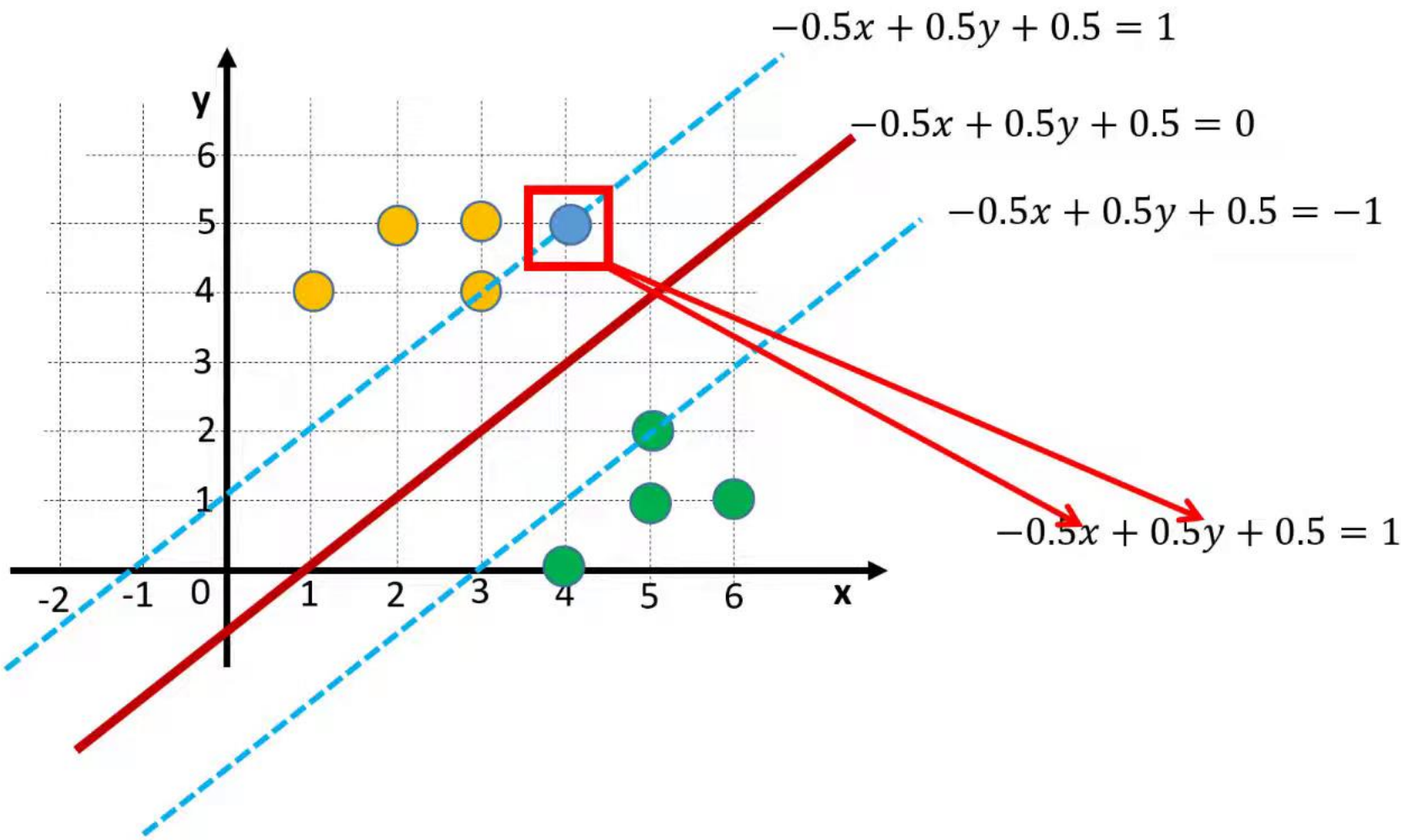
Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

$$-0.5x + 0.5y + 0.5 = -1$$

$$-0.5 \cdot 6 + 0.5 \cdot 3 + 0.5 = -1$$

we see that the left-hand side is now equal to negative one.

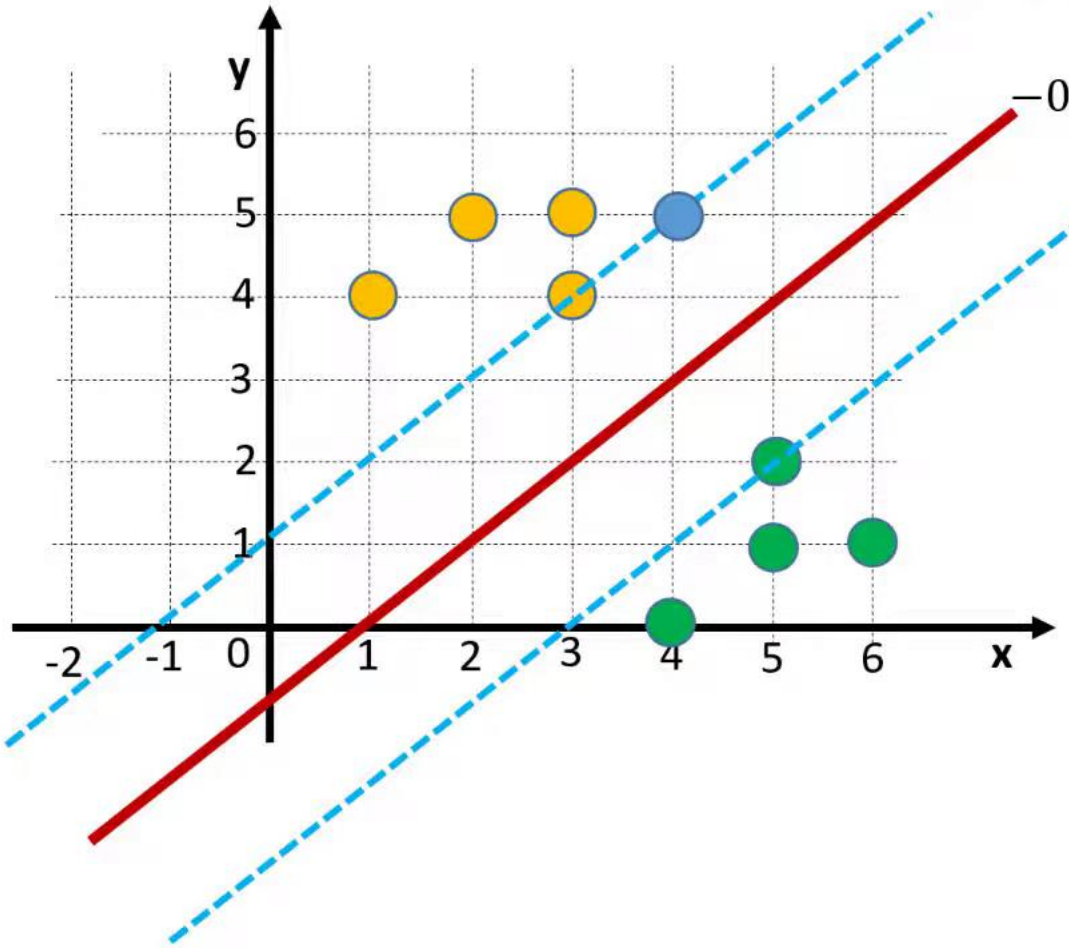
SVM – the math



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

And if we plug in the x and y-coordinates of an arbitrary point on the blue line above the hyperplane,

SVM – the math



$$-0.5x + 0.5y + 0.5 = 1$$

$$-0.5x + 0.5y + 0.5 = 0$$

$$-0.5x + 0.5y + 0.5 = -1$$

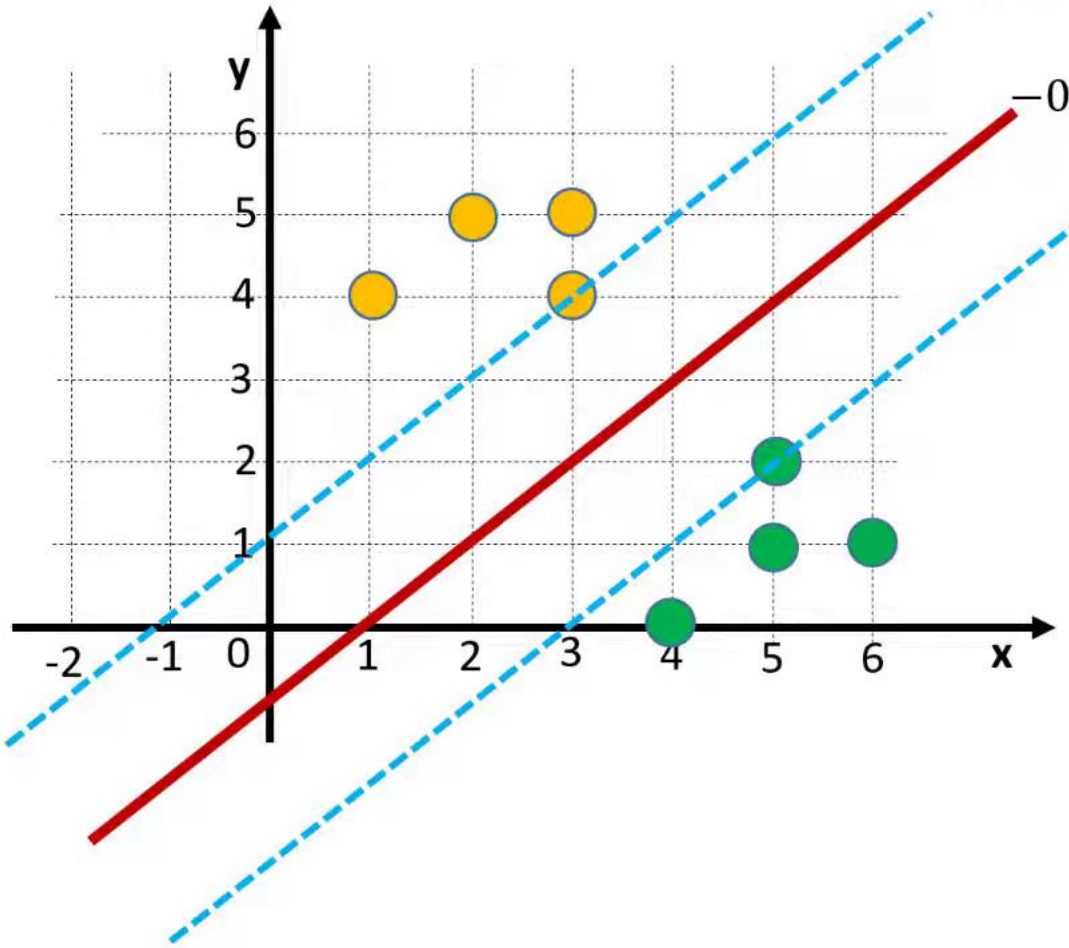
Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

$$-0.5x + 0.5y + 0.5 = 1$$

$$-0.5 \cdot 4 + 0.5 \cdot 5 + 0.5 = 1$$

we see that the left-hand side is now equal to positive one.

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

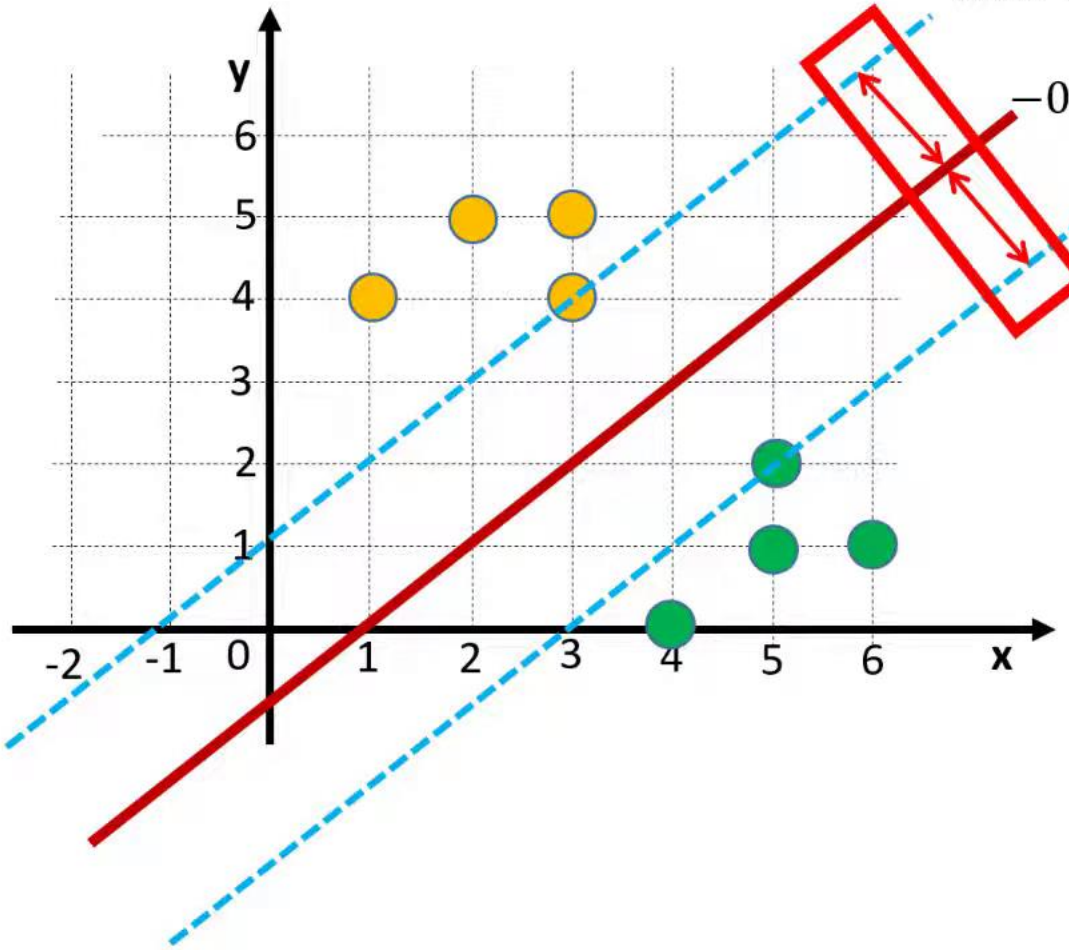
$-0.5x + 0.5y + b = -1$

$b = 0.5$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

Let's replace the constant 0.5 with b.

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

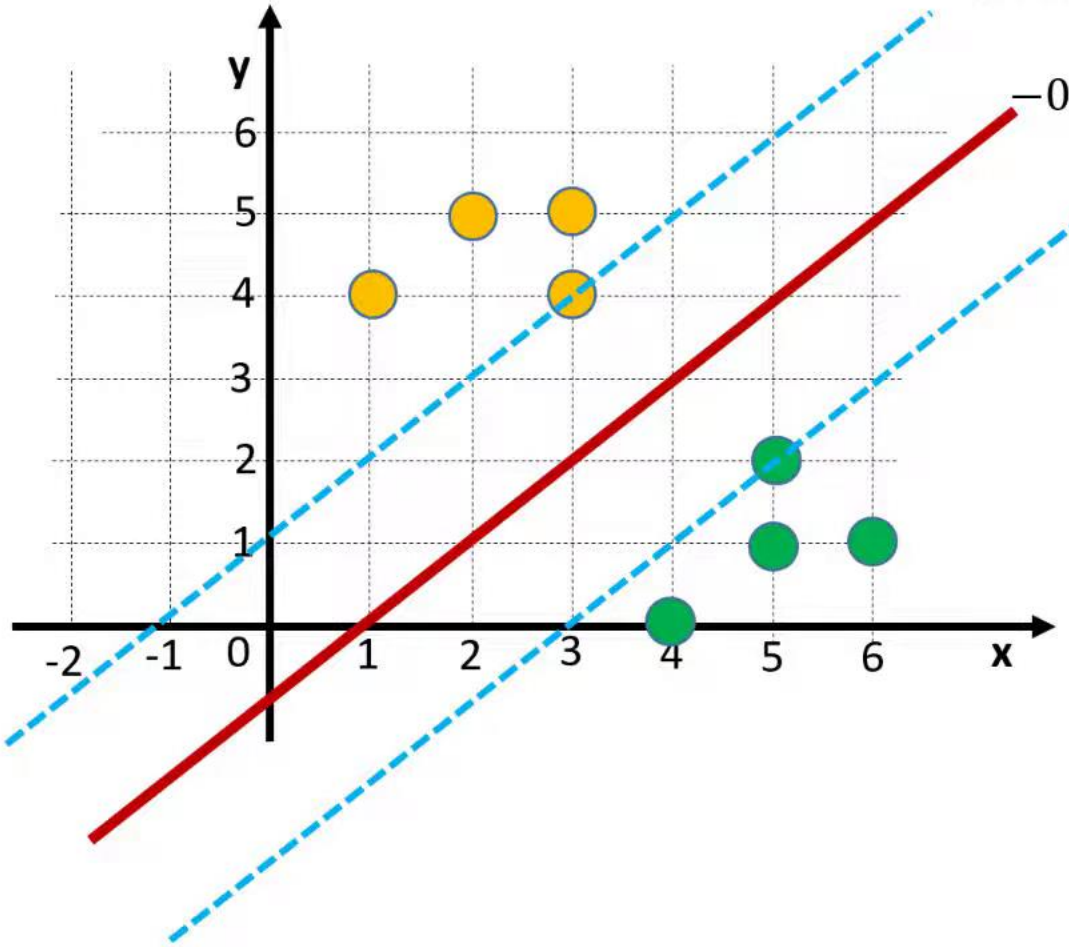
$-0.5x + 0.5y + b = -1$

$b = 0.5$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

Note that the two distances between the red line, our hyperplane, and the blue lines should always be equal so that the red line is in the center.

SVM – the math



$$-0.5x + 0.5y + b = 1$$

$$-0.5x + 0.5y + b = 0$$

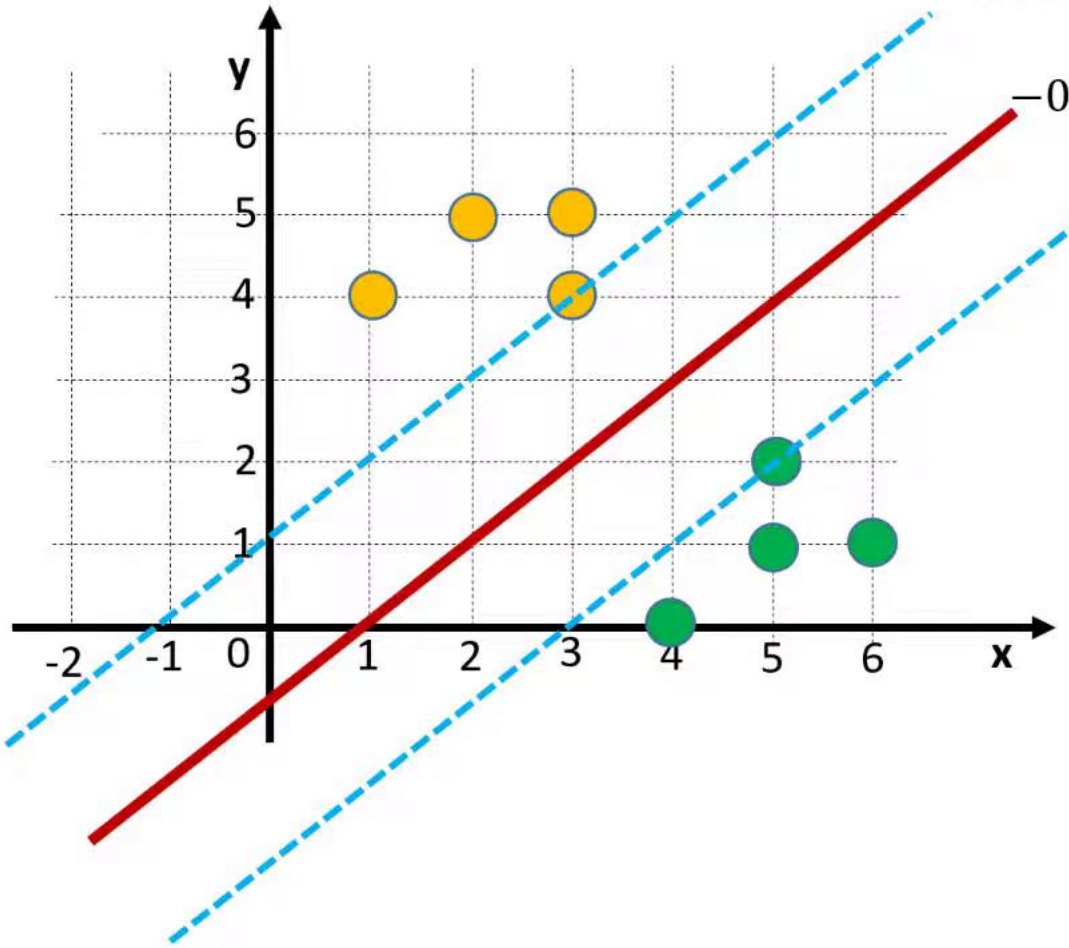
$$-0.5x + 0.5y + b = -1$$

$$b = 0.5$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

This explains why we use one and negative one as constants on the right-hand side.

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

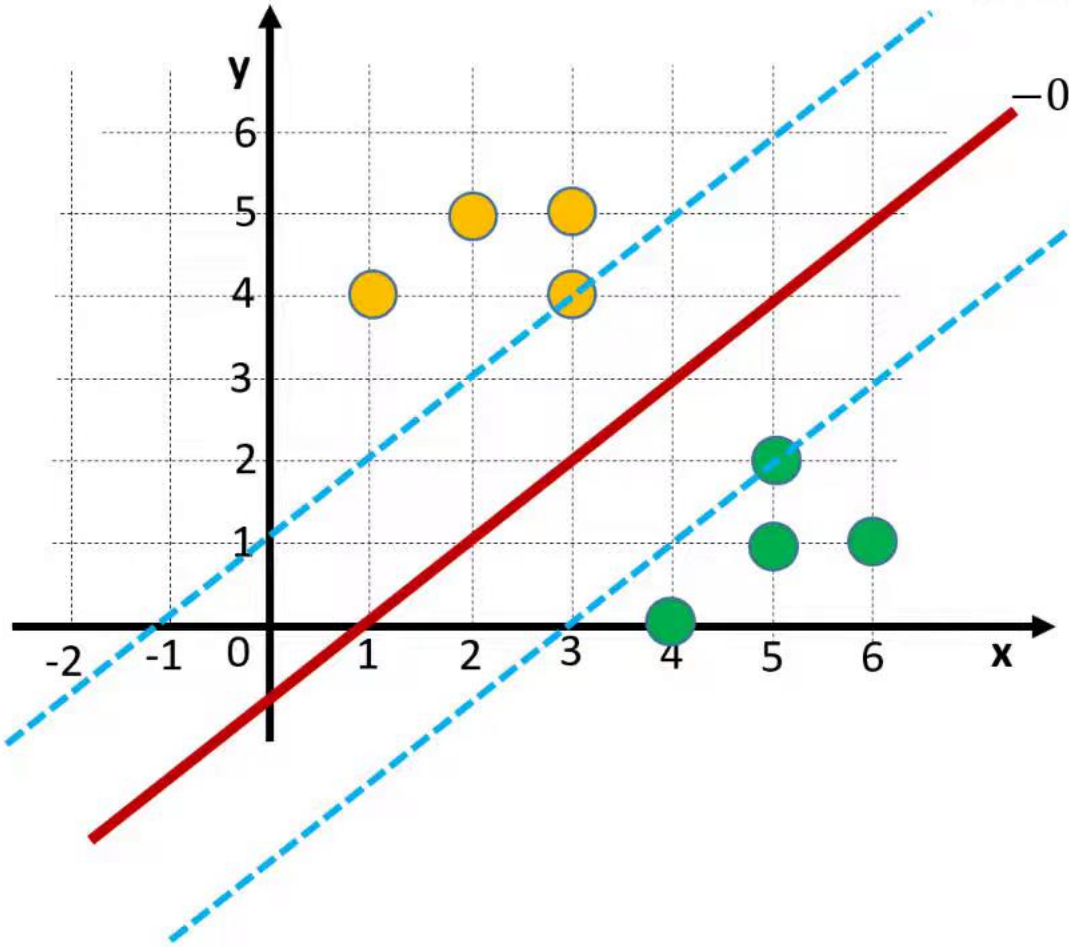
$b = 0.5$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

$$w^T x + b = 0$$

The equation of the hyperplane is usually expressed like this in SVM.

SVM – the math



$-0.5x + 0.5y + b = 1$

$b = 0.5$

$-0.5x + 0.5y + b = 0$

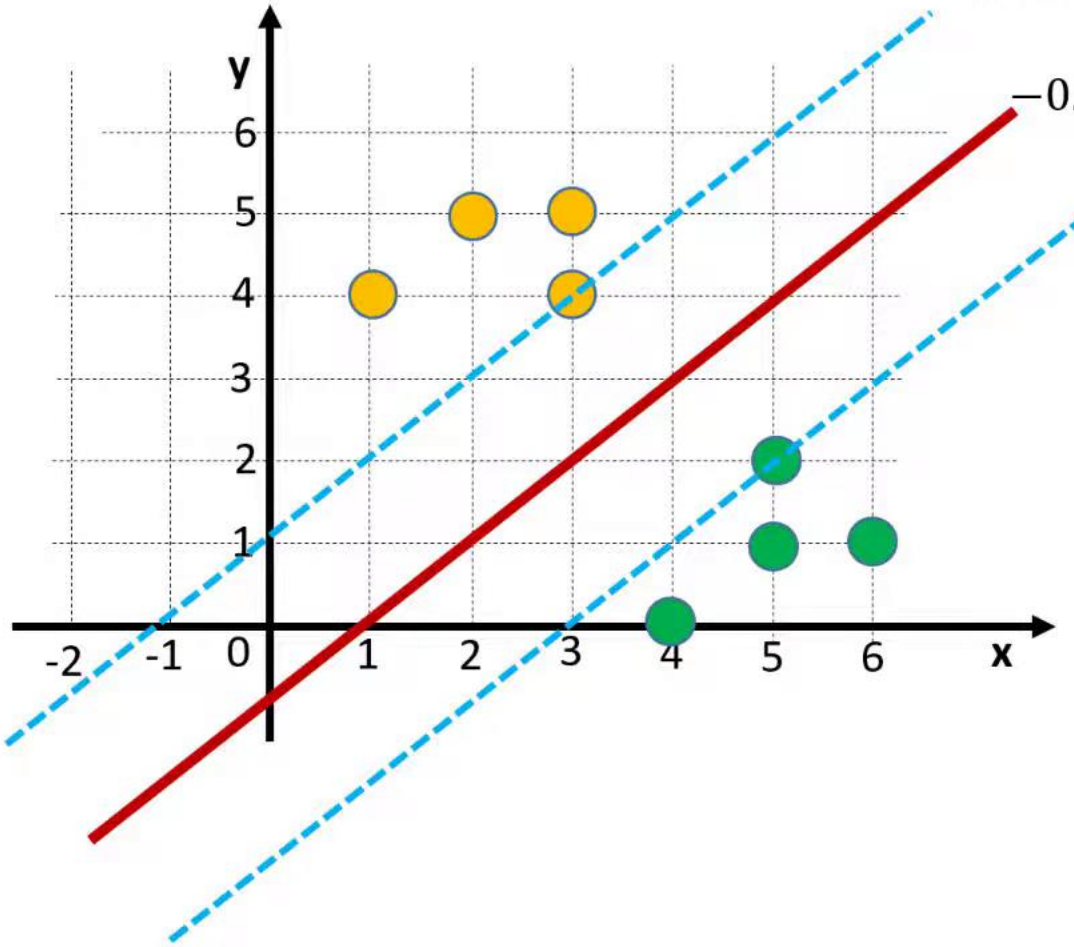
$-0.5x + 0.5y + b = -1$

$w^T x + b = 0$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

This represents a vector that holds our coefficients,

SVM – the math



$-0.5x + 0.5y + b = 1$

$b = 0.5$

$-0.5x + 0.5y + b = 0$

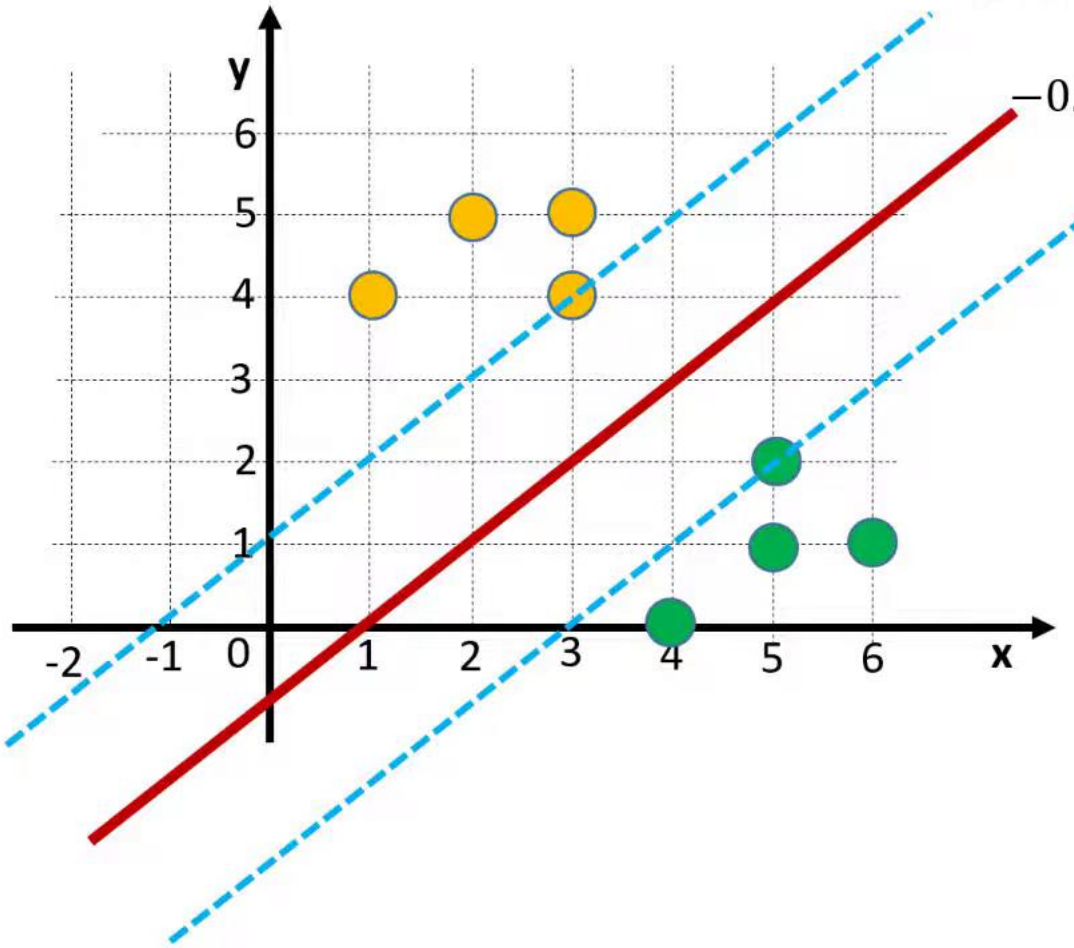
$-0.5x + 0.5y + b = -1$

$w^T x + b = 0$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

and this is a vector that includes the variables x and y in our example.

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

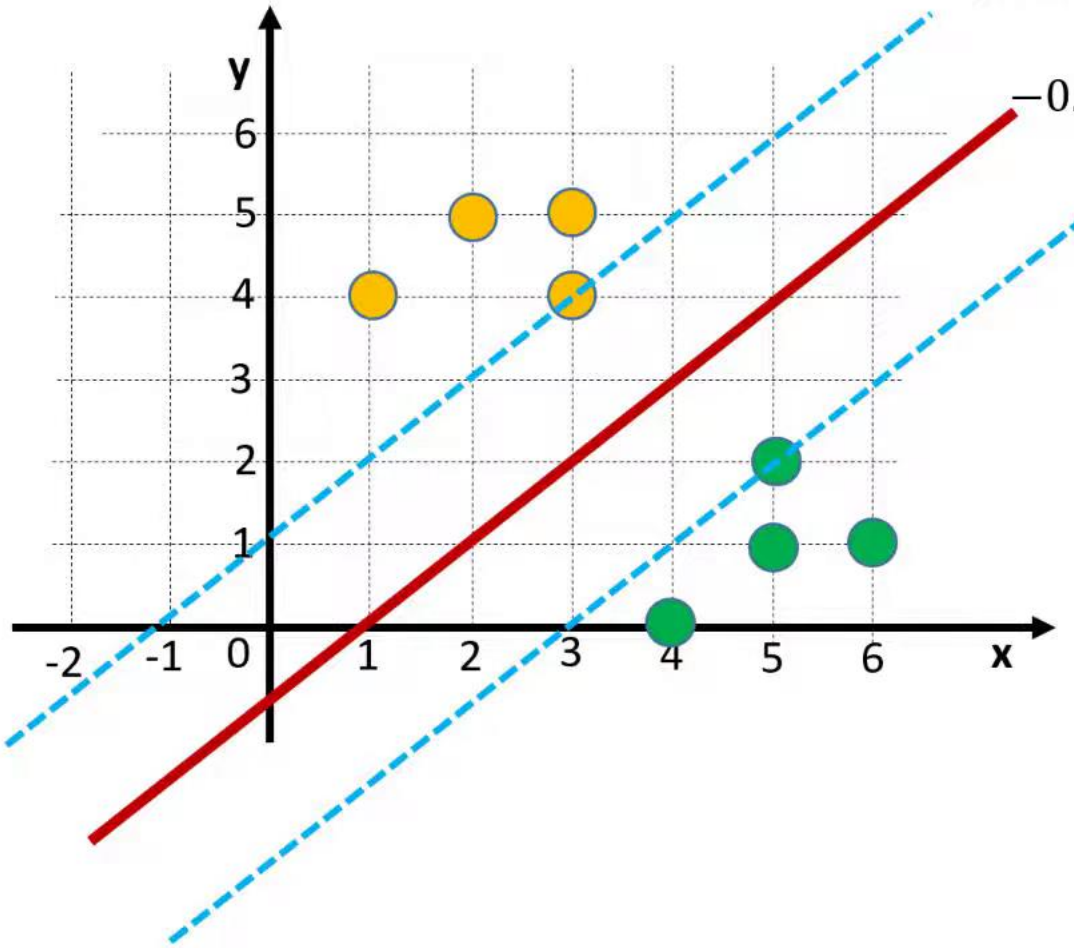
$b = 0.5$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

$$w^T x + b = 0$$

T represents that we should transpose vector w.

SVM – the math



$$-0.5x + 0.5y + b = 1$$

$$-0.5x + 0.5y + b = 0$$

$$-0.5x + 0.5y + b = -1$$

$$b = 0.5$$

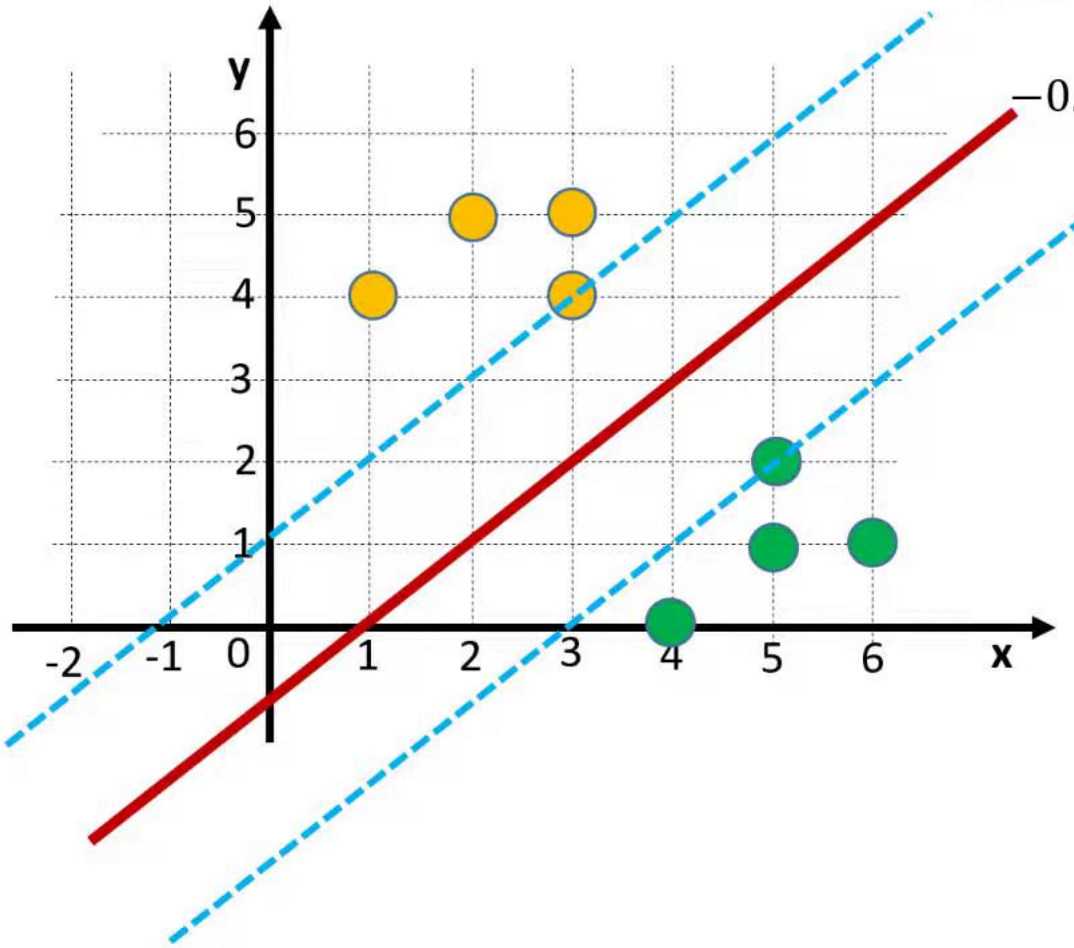
Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

$$w^T x + b = 0$$

$$\begin{bmatrix} -0.5 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} + b = 0$$

If we plug in the coefficients and x and y in this equation,

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

$b = 0.5$

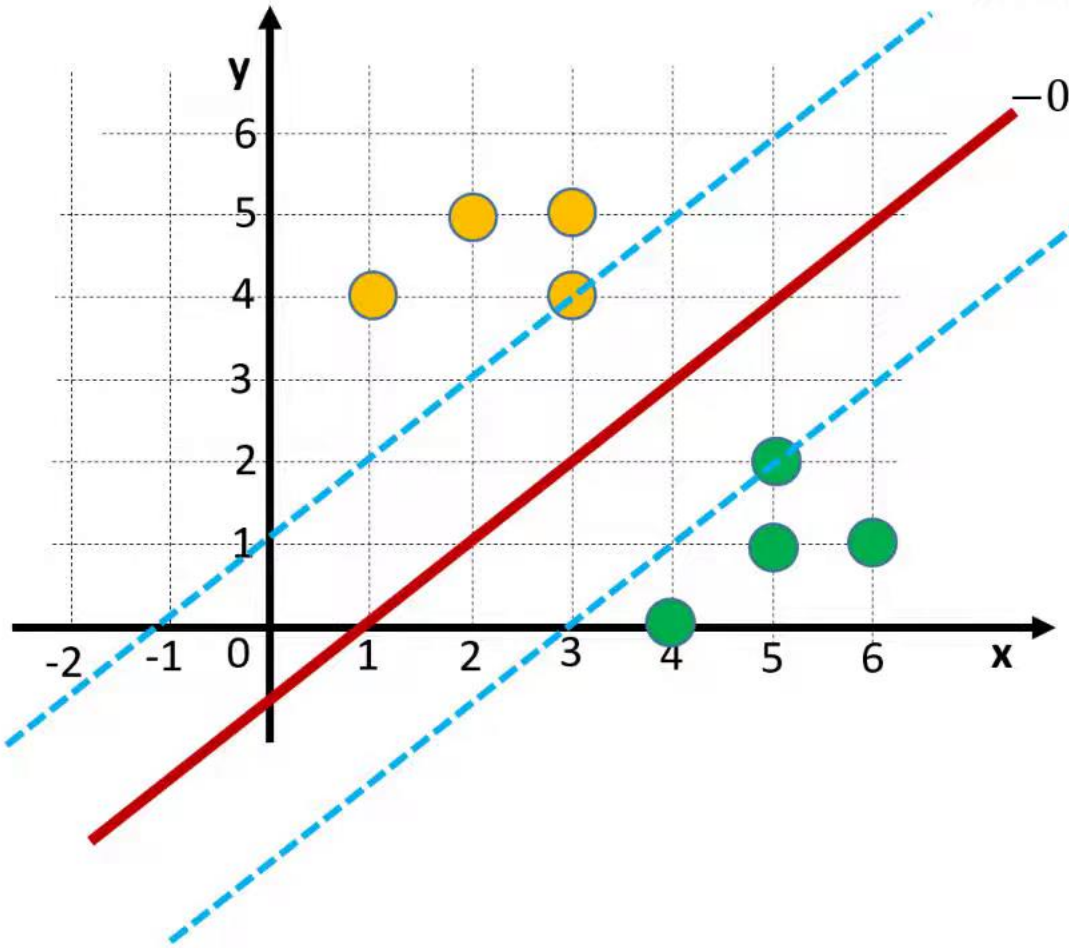
Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

$$w^T x + b = 0$$

$$\begin{bmatrix} -0.5 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} + b = 0$$

we see that if we multiply these two vectors,

SVM – the math



$-0.5x + 0.5y + b = 1$

$b = 0.5$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

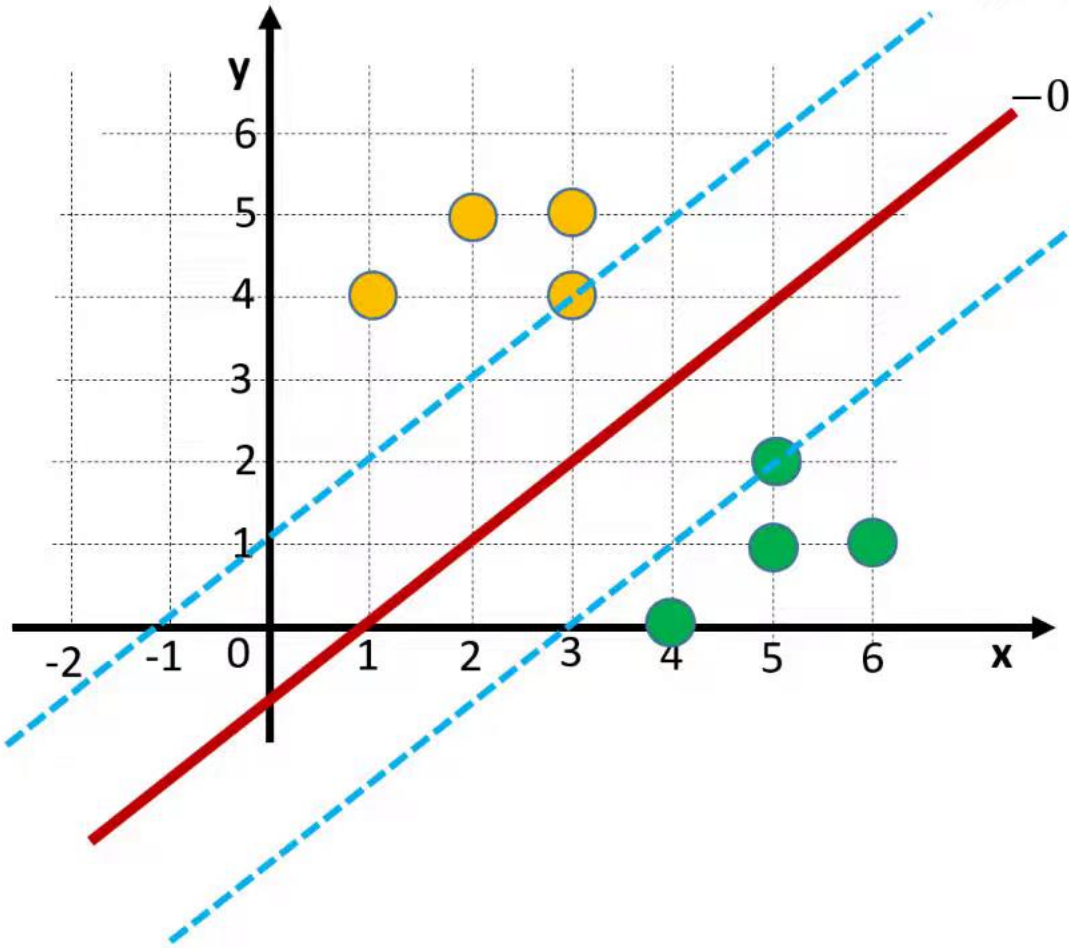
$$w^T x + b = 0$$

$$[-0.5 \quad 0.5] \cdot \begin{bmatrix} x \\ y \end{bmatrix} + b = 0$$

$$-0.5x + 0.5y + b = 0$$

we get the equation of our hyperplane.

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

$b = 0.5$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

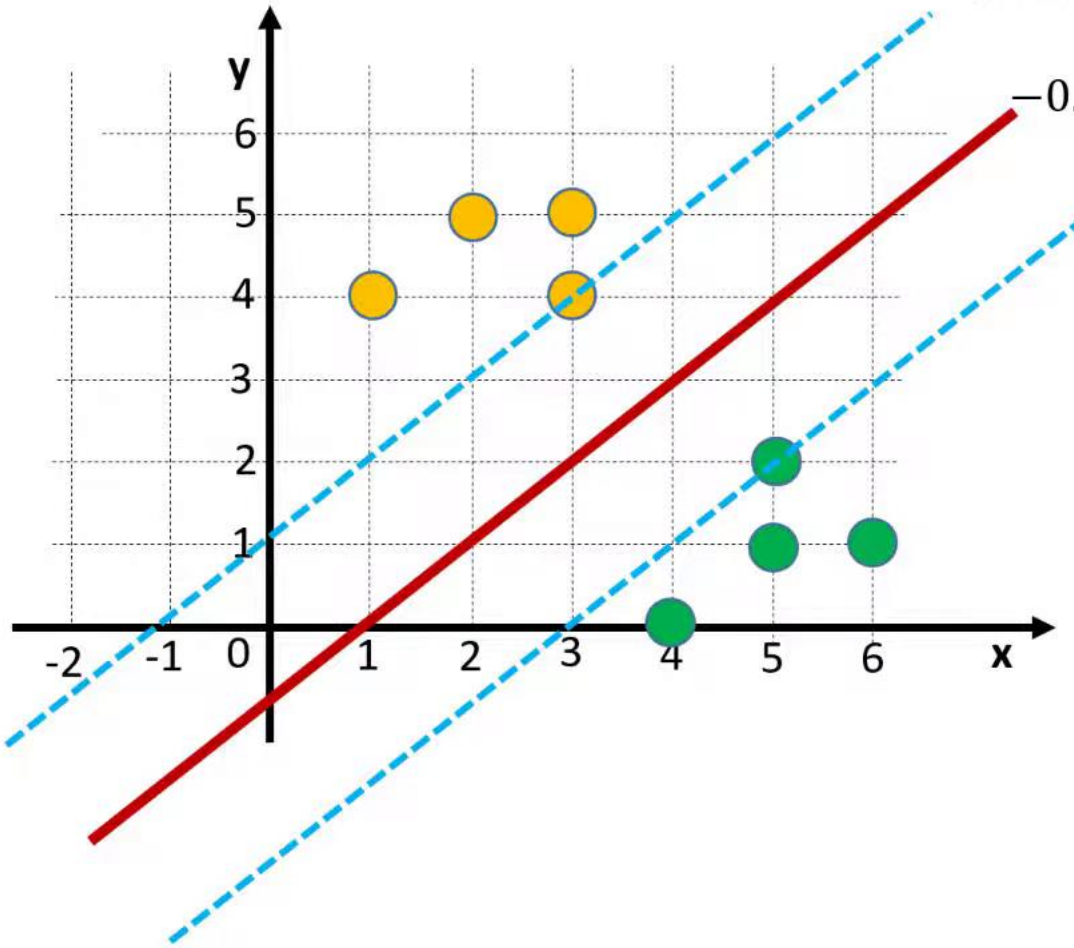
$$w^T x + b = 0$$

$$[-0.5 \quad 0.5] \cdot \begin{bmatrix} x \\ y \end{bmatrix} + b = 0$$

$$-0.5x + 0.5y + b = 0$$

Note that x in this equation is a vector of the variables x and y.

SVM – the math



$$-0.5x + 0.5y + b = 1$$

$$-0.5x + 0.5y + b = 0$$

$$-0.5x + 0.5y + b = -1$$

$$b = 0.5$$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

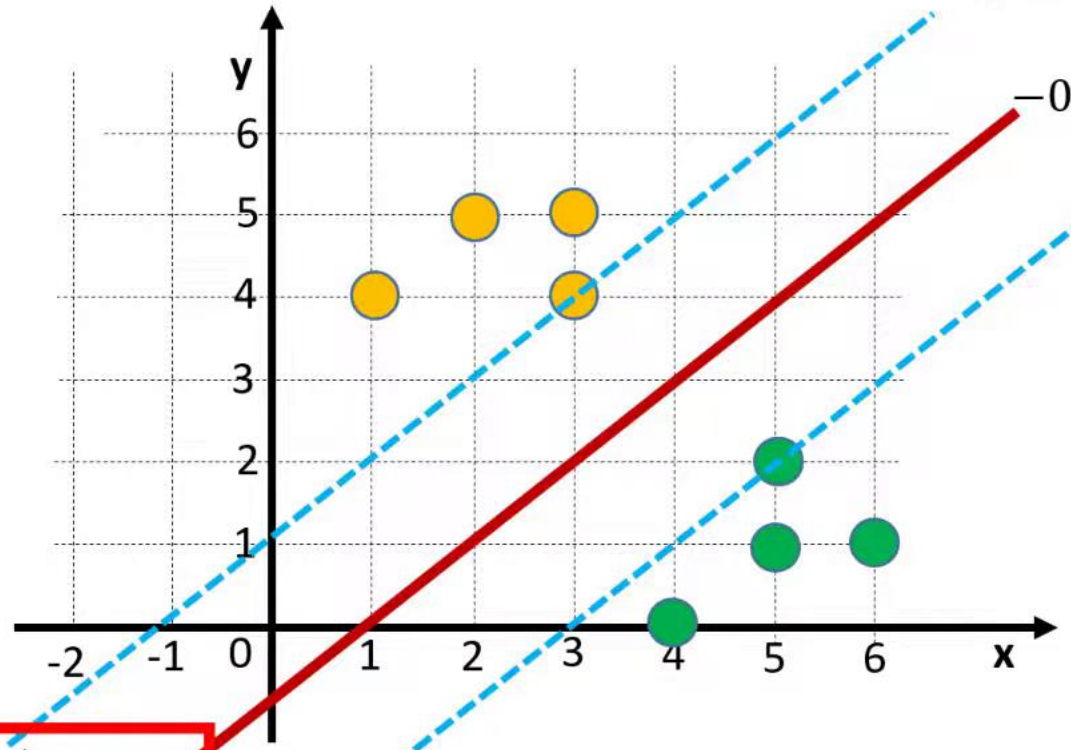
$$w^T x + b = 0$$

$$[-0.5 \quad 0.5] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + b = 0$$

$$-0.5x + 0.5y + b = 0$$

Usually, one defines the two dimensions as x1 and x2 in SVM instead of x and y but to keep things simple, I have here named the axes as x and y.

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

$b = 0.5$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

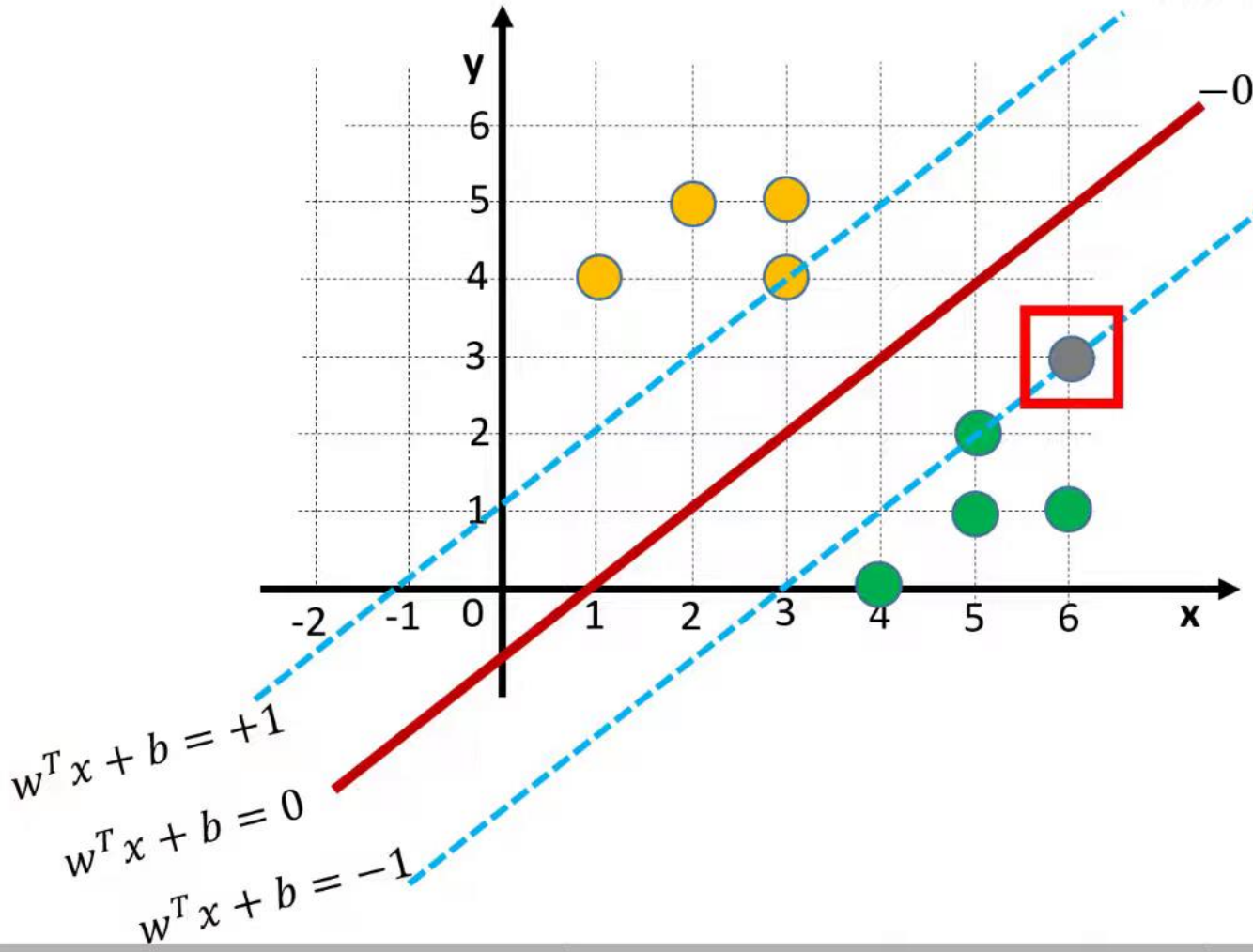
$w^T x + b = +1$

$w^T x + b = 0$

$w^T x + b = -1$

In the standard form, the equations of the three lines are defined like this, which is good to know because you will see these equations a lot when you read more about SVM.

SVM – the math



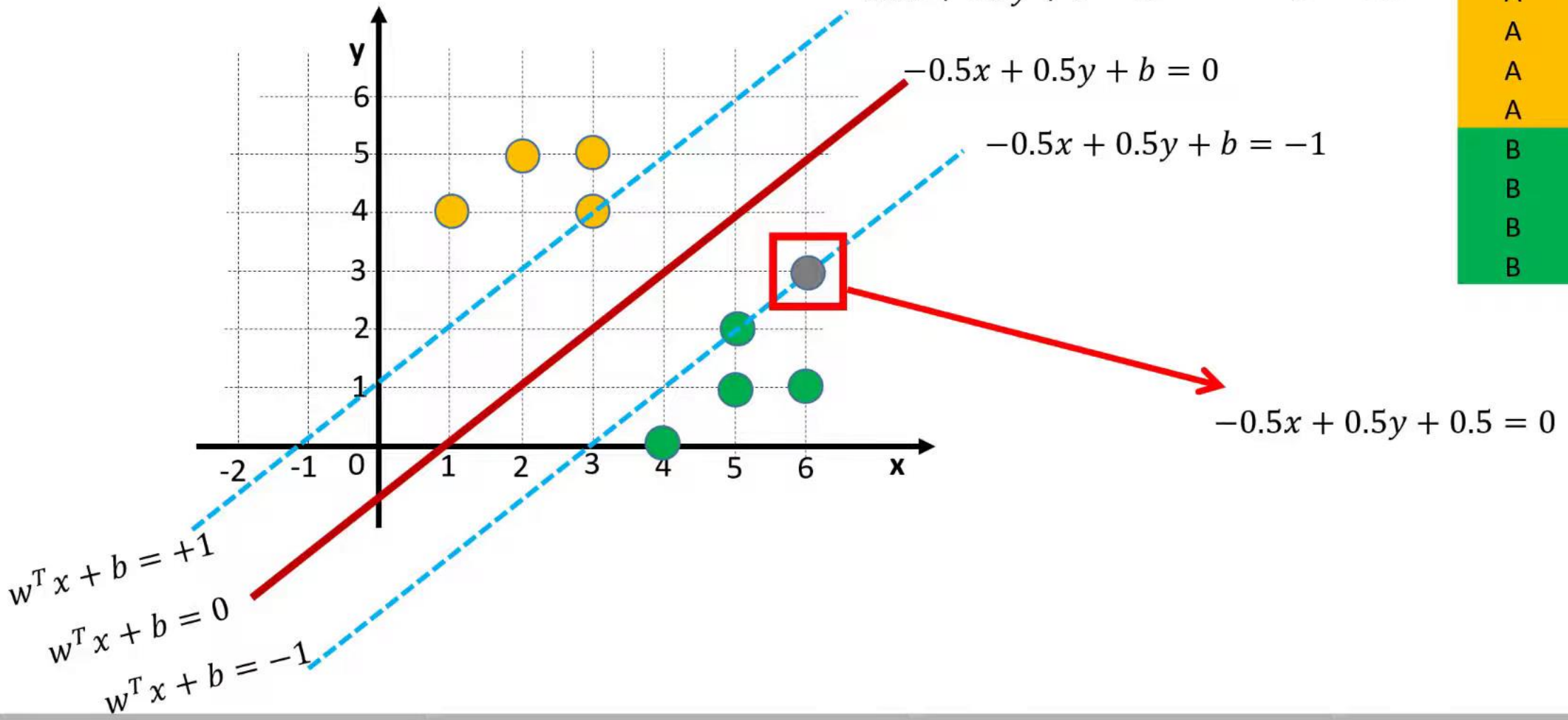
$b = 0.5$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

$-0.5x + 0.5y + 0.5 = 0$

Now, let's use our hyperplane to classify if a data point belongs to group A, the yellow points, or group B, the green points.

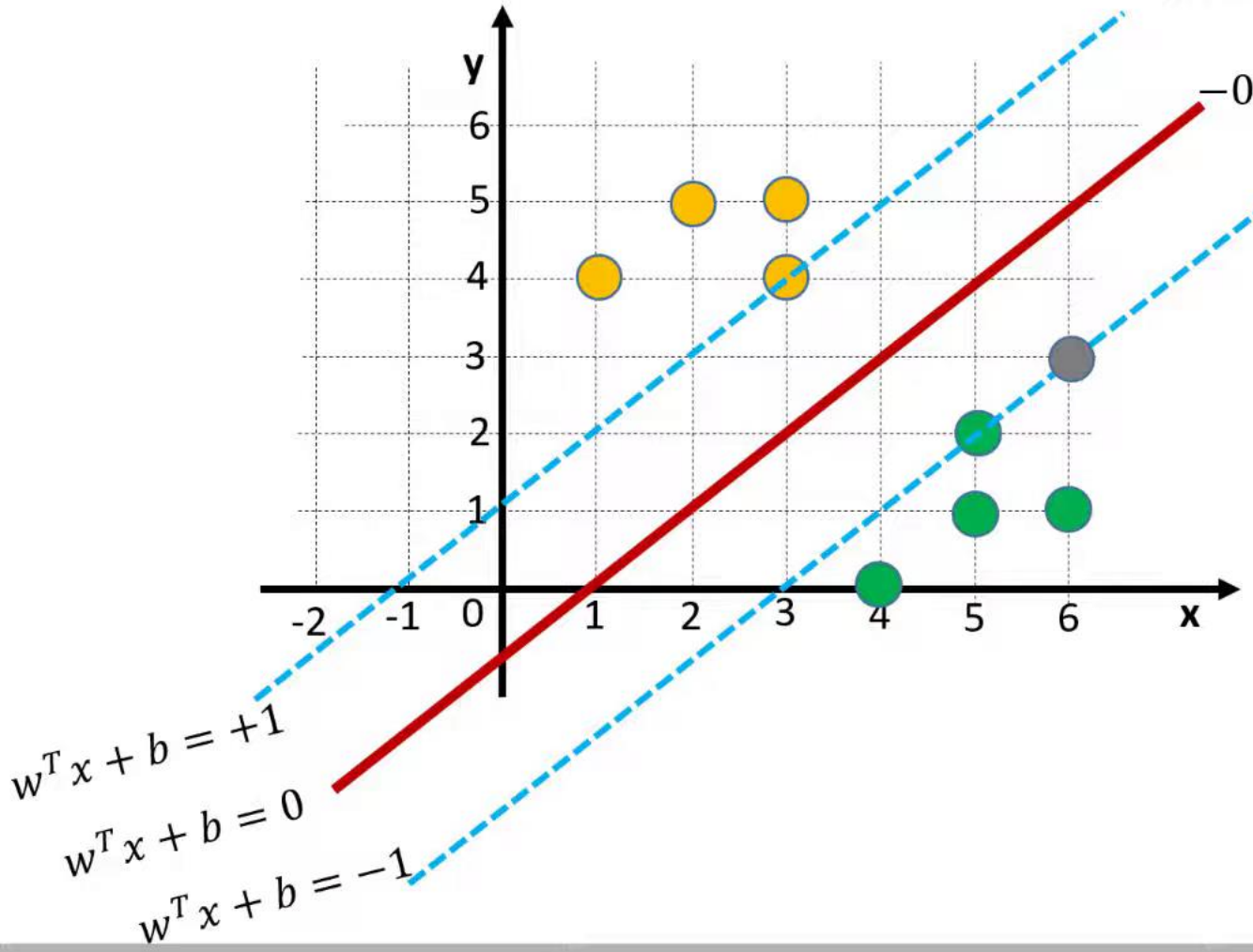
SVM – the math



Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

We plug in the x and y-coordinates of this data point of an unknown class in the equation of the hyperplane.

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

$b = 0.5$

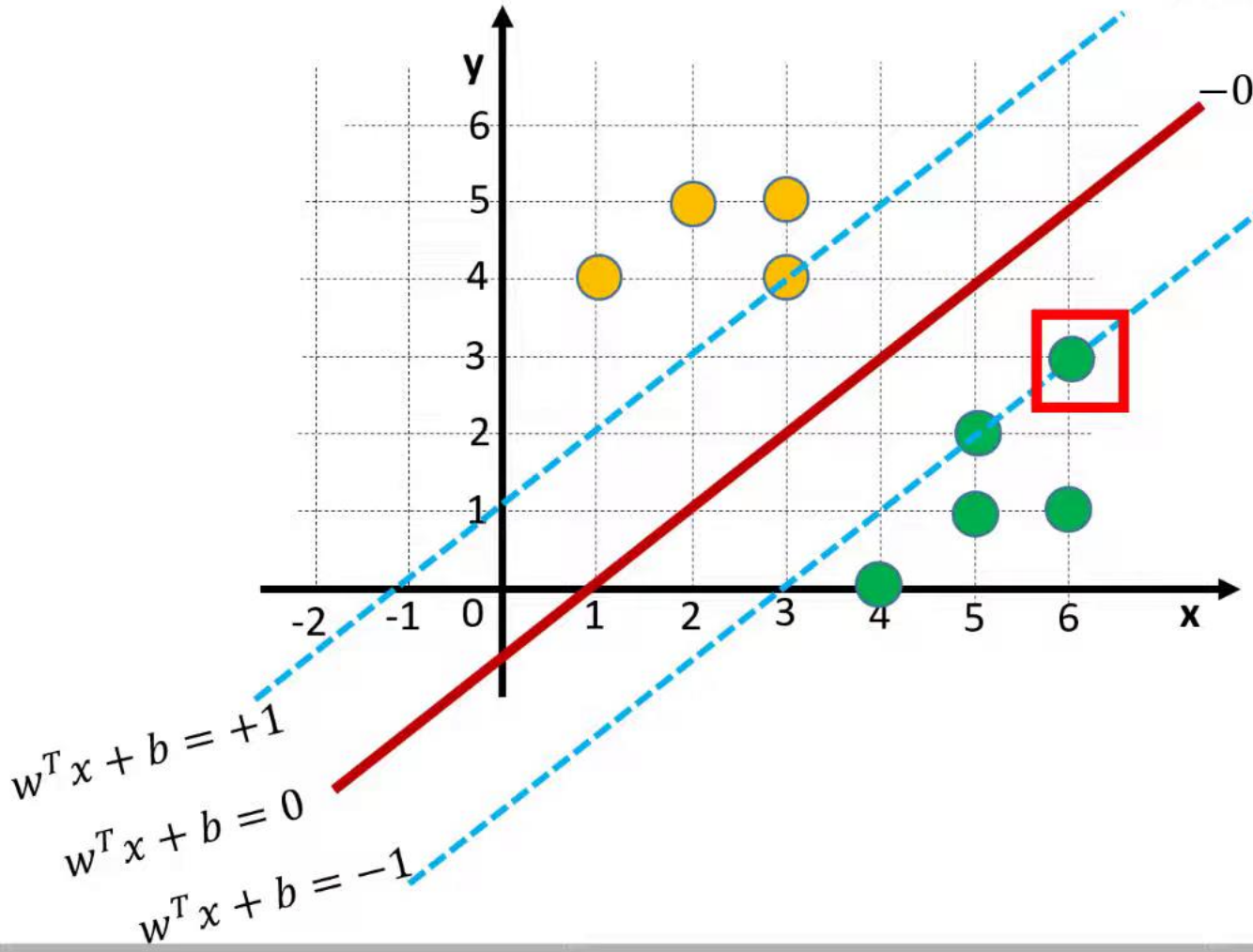
Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

$$-0.5x + 0.5y + 0.5 = 0$$

$$-0.5 \cdot 6 + 0.5 \cdot 3 + 0.5 = -1$$

Since this results in a negative value,

SVM – the math



$b = 0.5$

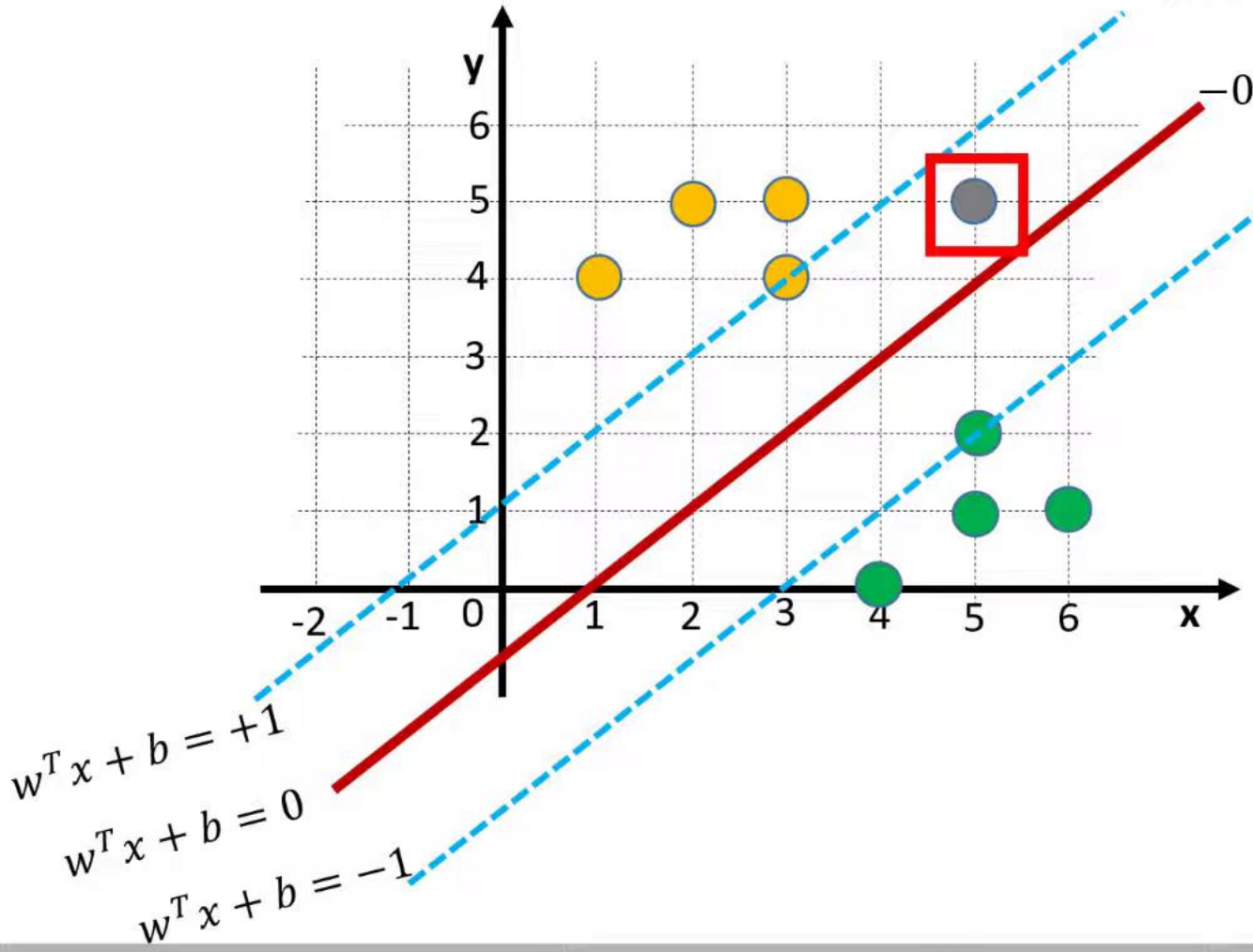
Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

$$-0.5x + 0.5y + 0.5 = 0$$

$$-0.5 \cdot 6 + 0.5 \cdot 3 + 0.5 = -1$$

we predict that the data point belongs to the negative group, which is group B in this case.

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

$b = 0.5$

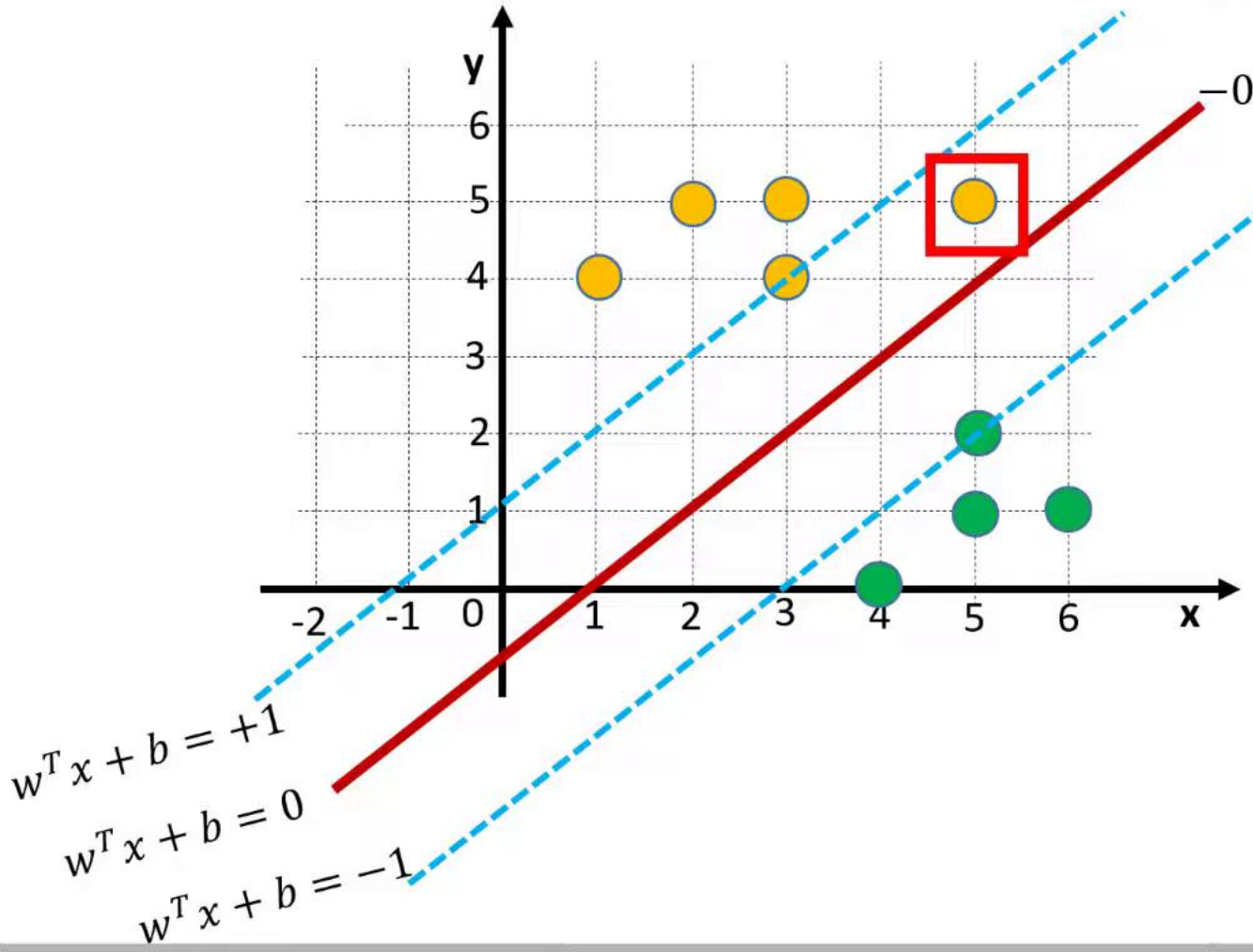
Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

$$-0.5x + 0.5y + 0.5 = 0$$

$$-0.5 \cdot 5 + 0.5 \cdot 5 + 0.5 = 0.5$$

In comparison, the following data point will result in a positive value,

SVM – the math



$b = 0.5$

Group	x	y
A	1	4
A	2	5
A	3	5
A	3	4
B	6	1
B	4	0
B	5	2
B	5	1

$$-0.5x + 0.5y + 0.5 = 0$$

$$-0.5 \cdot 5 + 0.5 \cdot 5 + 0.5 = 0.5$$

which means that we predict it to belong to the positive group, which is group A in this case.

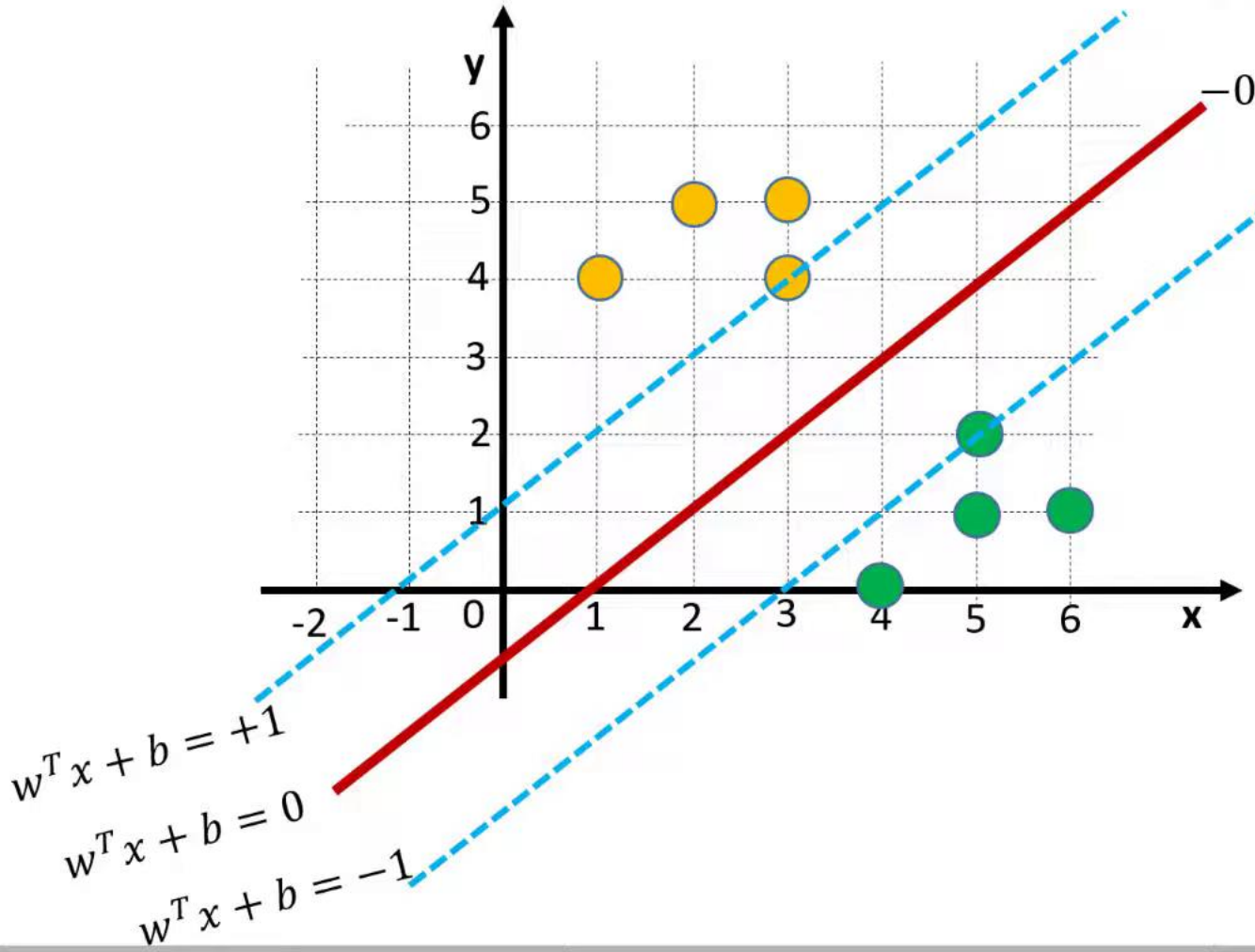
SVM – the math

$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

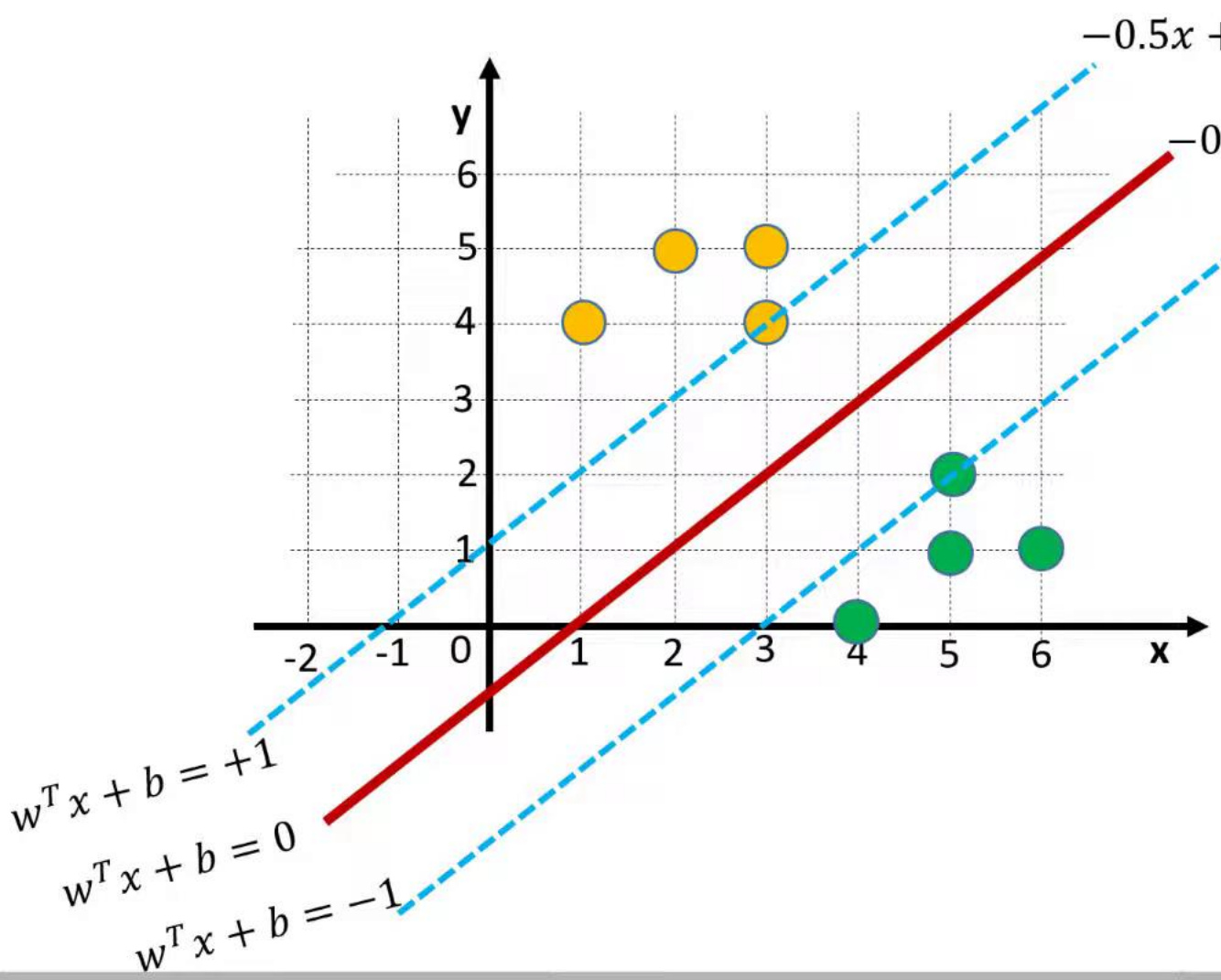
$b = 0.5$



Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

In SVM, one therefore divides the two groups into a positive and a negative group. In our example, group A, the yellow points, is the positive group whereas group B, the green points, is the negative group.

SVM – the math



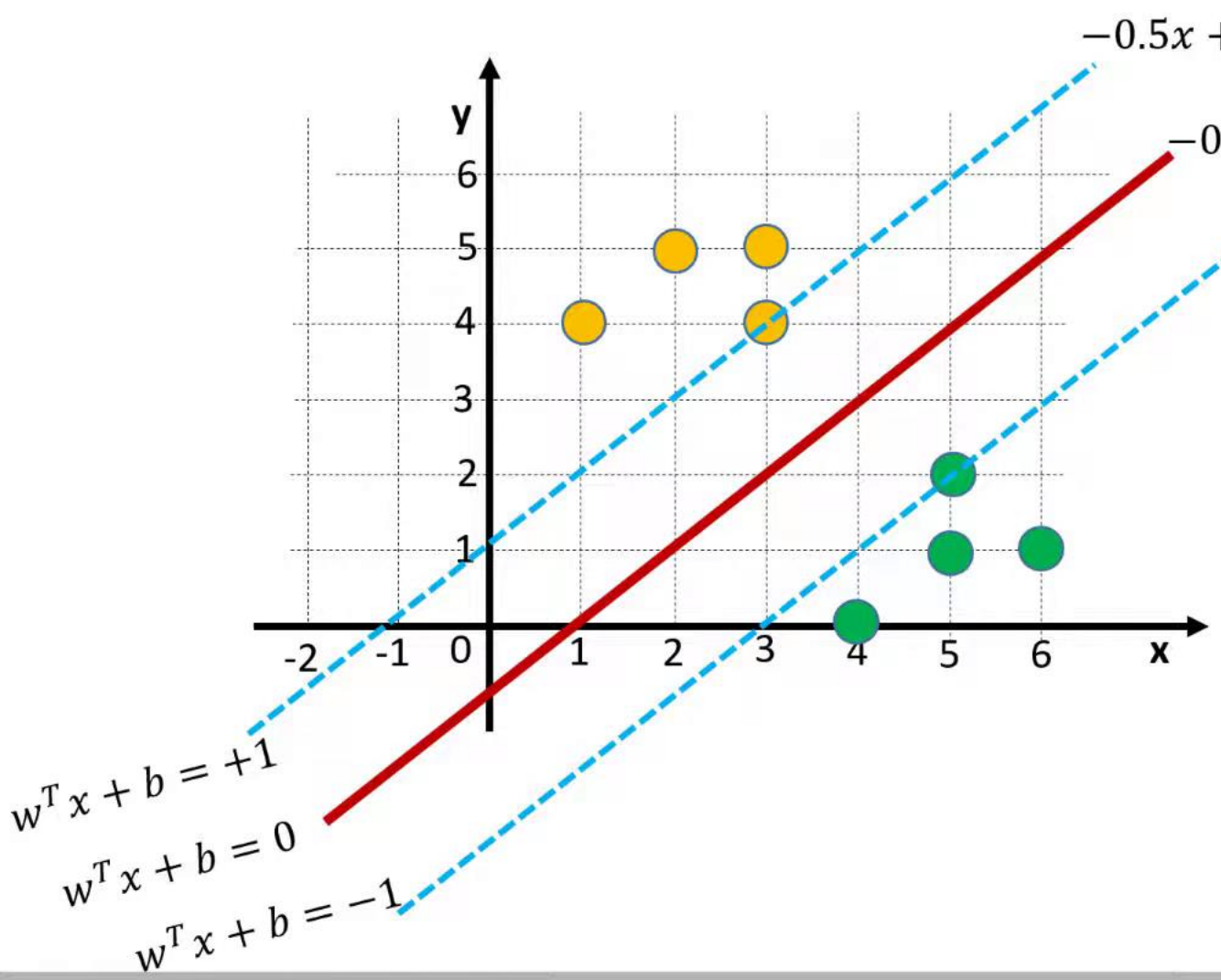
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y_i = [\,1, 1, 1, 1, -1, -1, -1, -1]$$

In support vector machines, Y-sub-i usually defines the class of the training data as one and negative one.

SVM – the math



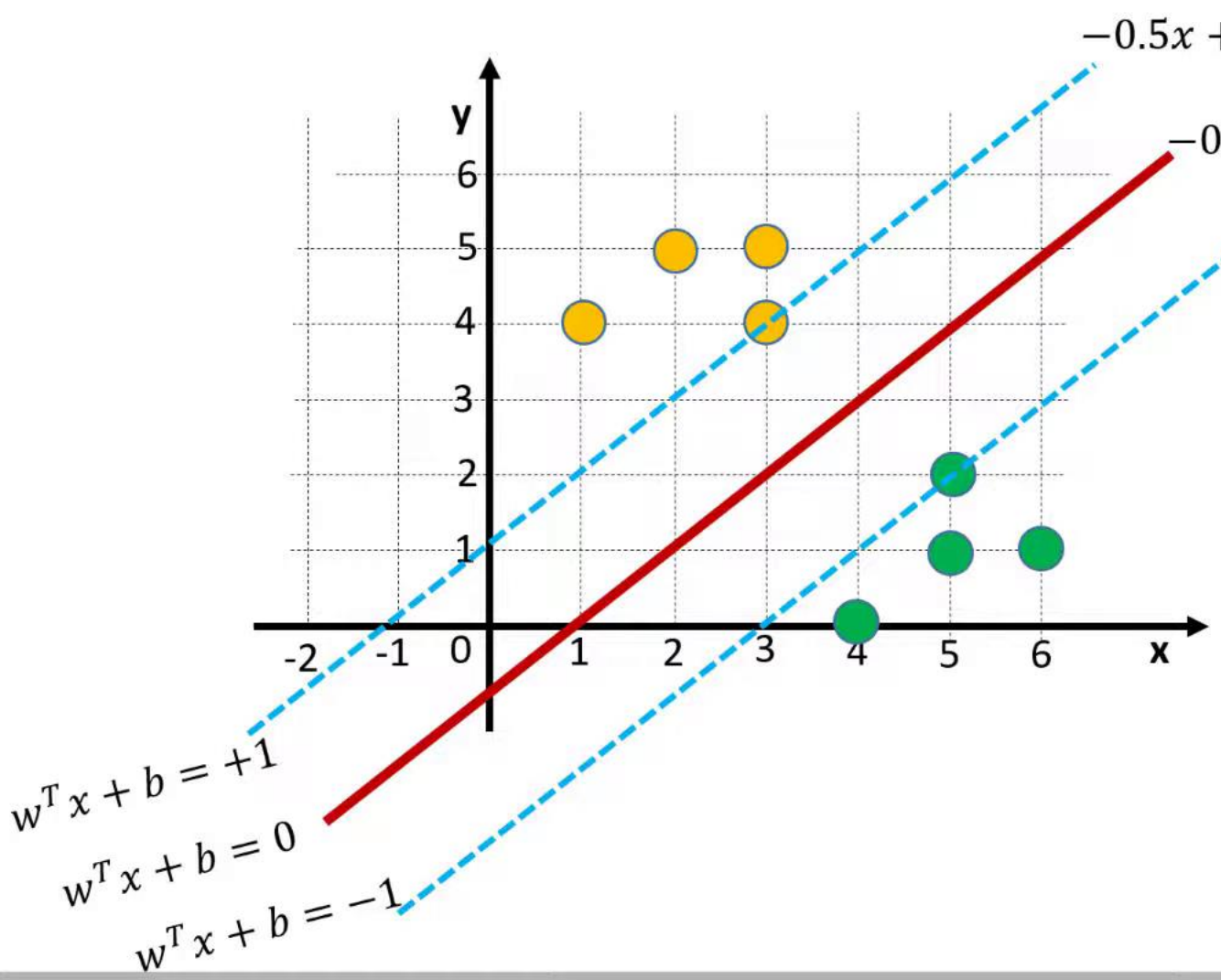
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$Y_i = [1, 1, 1, 1, -1, -1, -1, -1]$

For example, the first data point belongs to the positive group,

SVM – the math



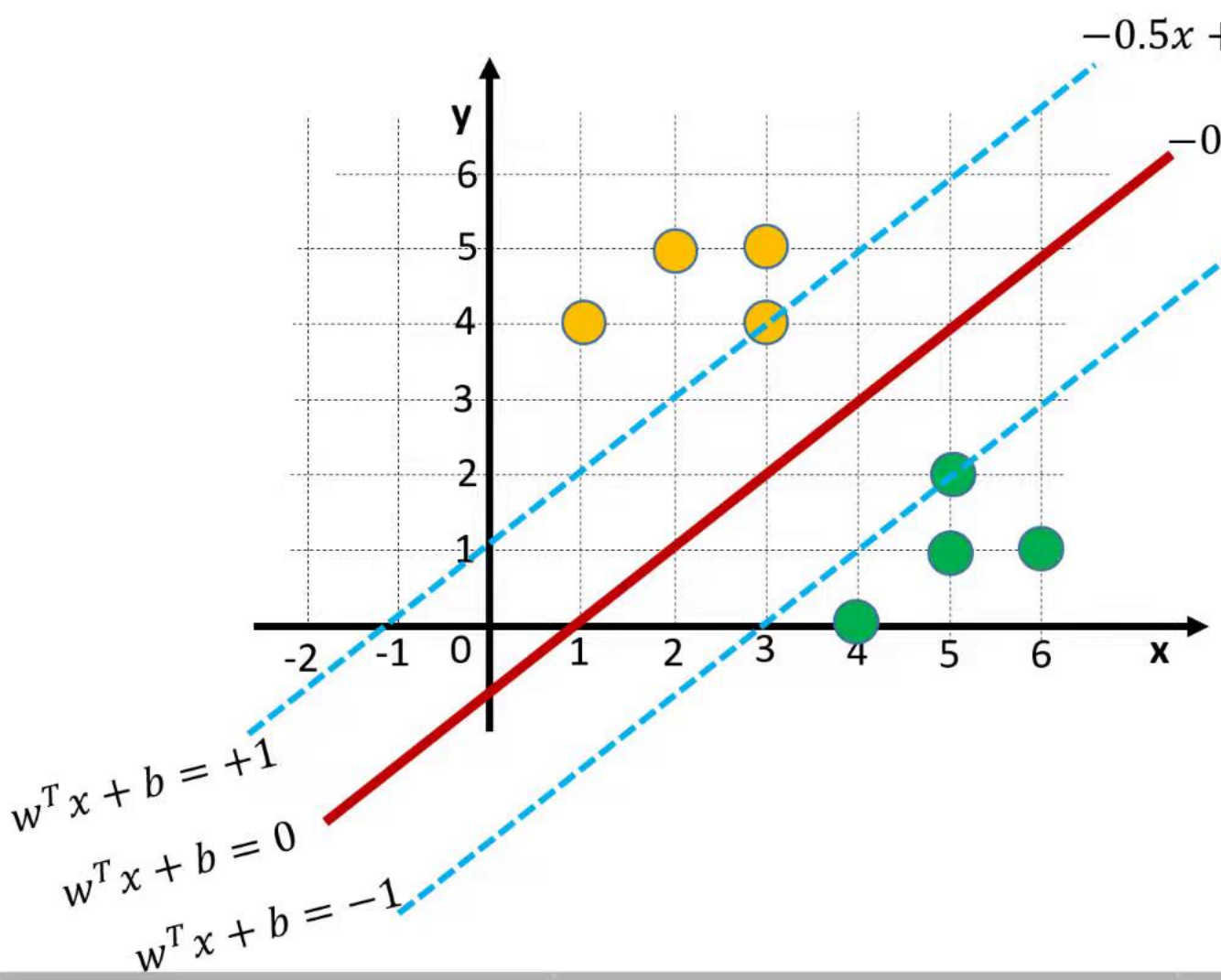
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$Y_i = [\,1, 1, 1, 1, -1, -1, -1, -1]$

whereas data point number five belongs to the negative group.

SVM – the math



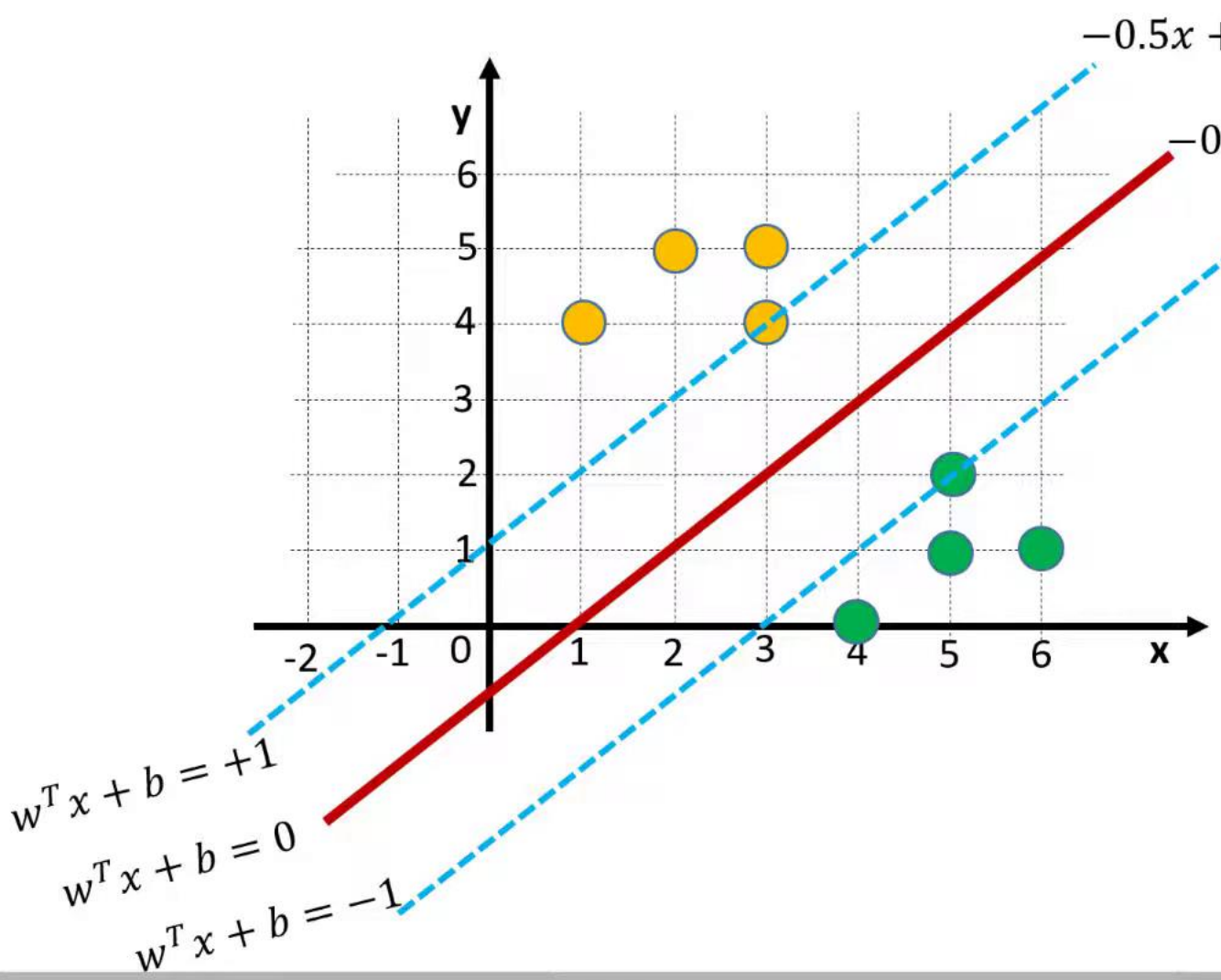
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y_i = [\,1, 1, 1, 1, -1, -1, -1, -1]$$

This is just another way to tell if a data point in our training data belongs to group A or B.

SVM – the math



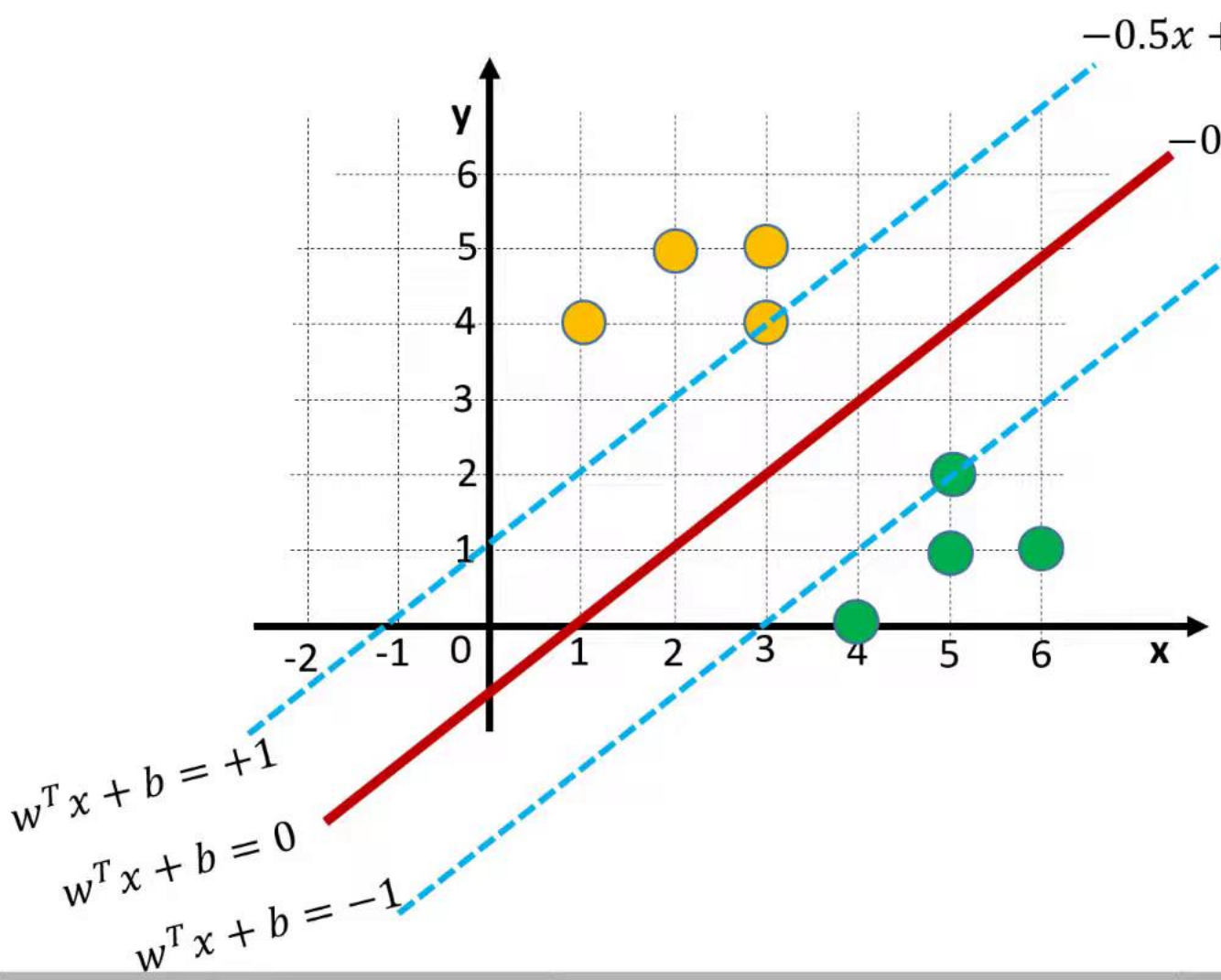
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y \begin{cases} +1 \text{ if } w^T x + b \geq 0 \\ -1 \text{ if } w^T x + b < 0 \end{cases}$$

The classification in SVM is usually defined like this.

SVM – the math



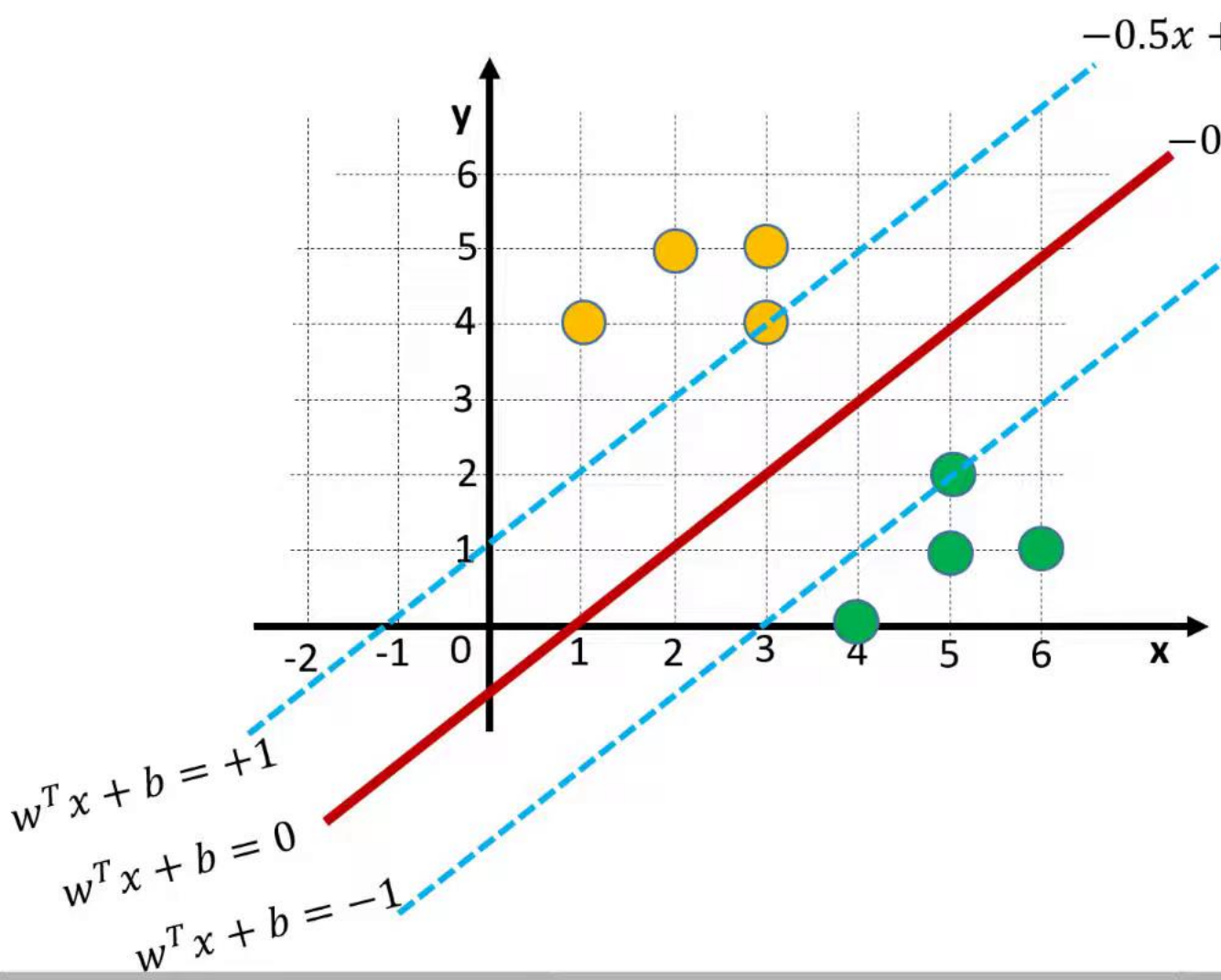
$b = 0.5$

Group (Y_i)	*x*	*y*
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y \begin{cases} +1 \text{ if } w^T x + b \geq 0 \\ -1 \text{ if } w^T x + b < 0 \end{cases}$$

We predict the class of a new observation,

SVM – the math



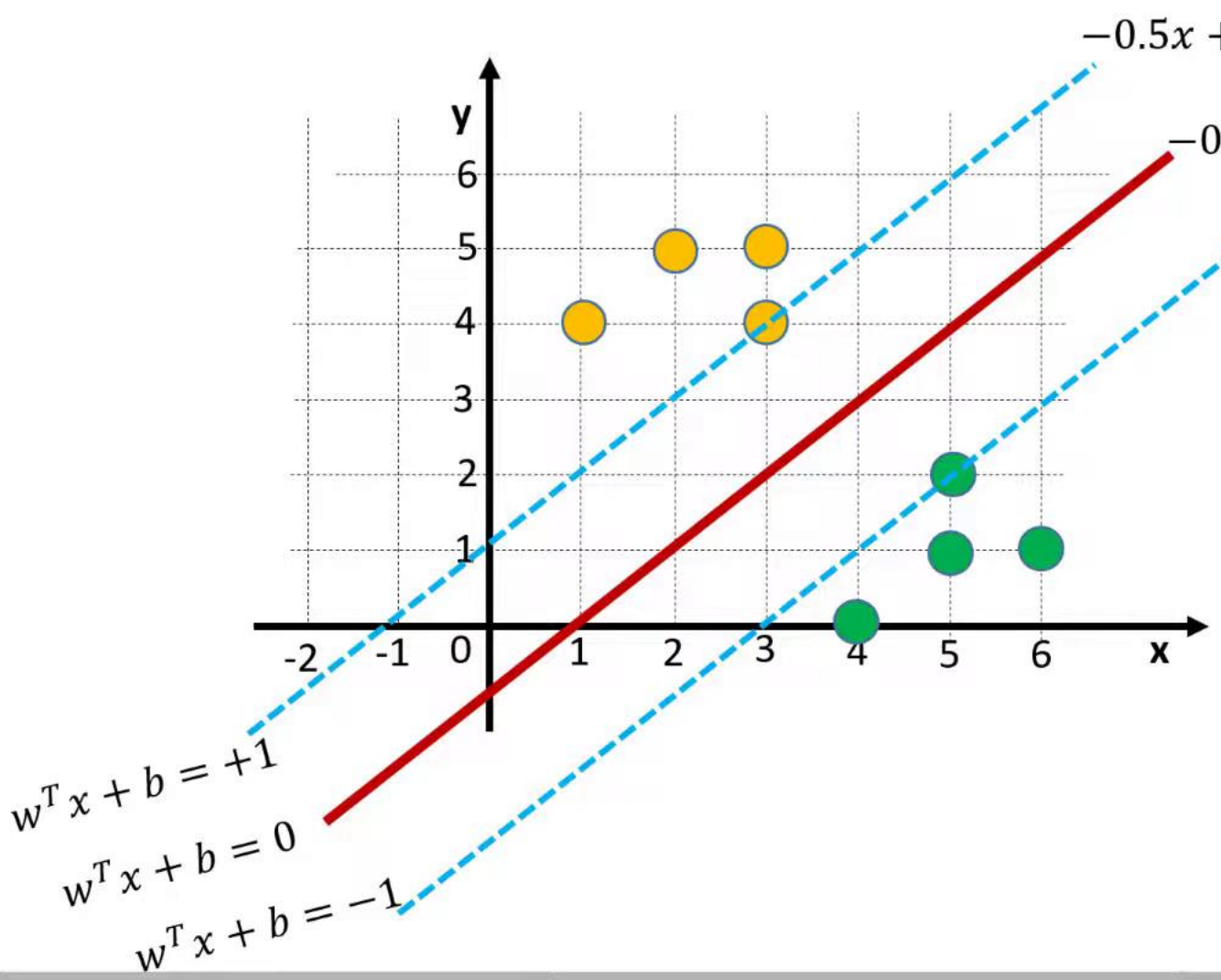
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y \begin{cases} +1 \text{ if } w^T x + b \geq 0 \\ -1 \text{ if } w^T x + b < 0 \end{cases}$$

to belong to the positive group,

SVM – the math



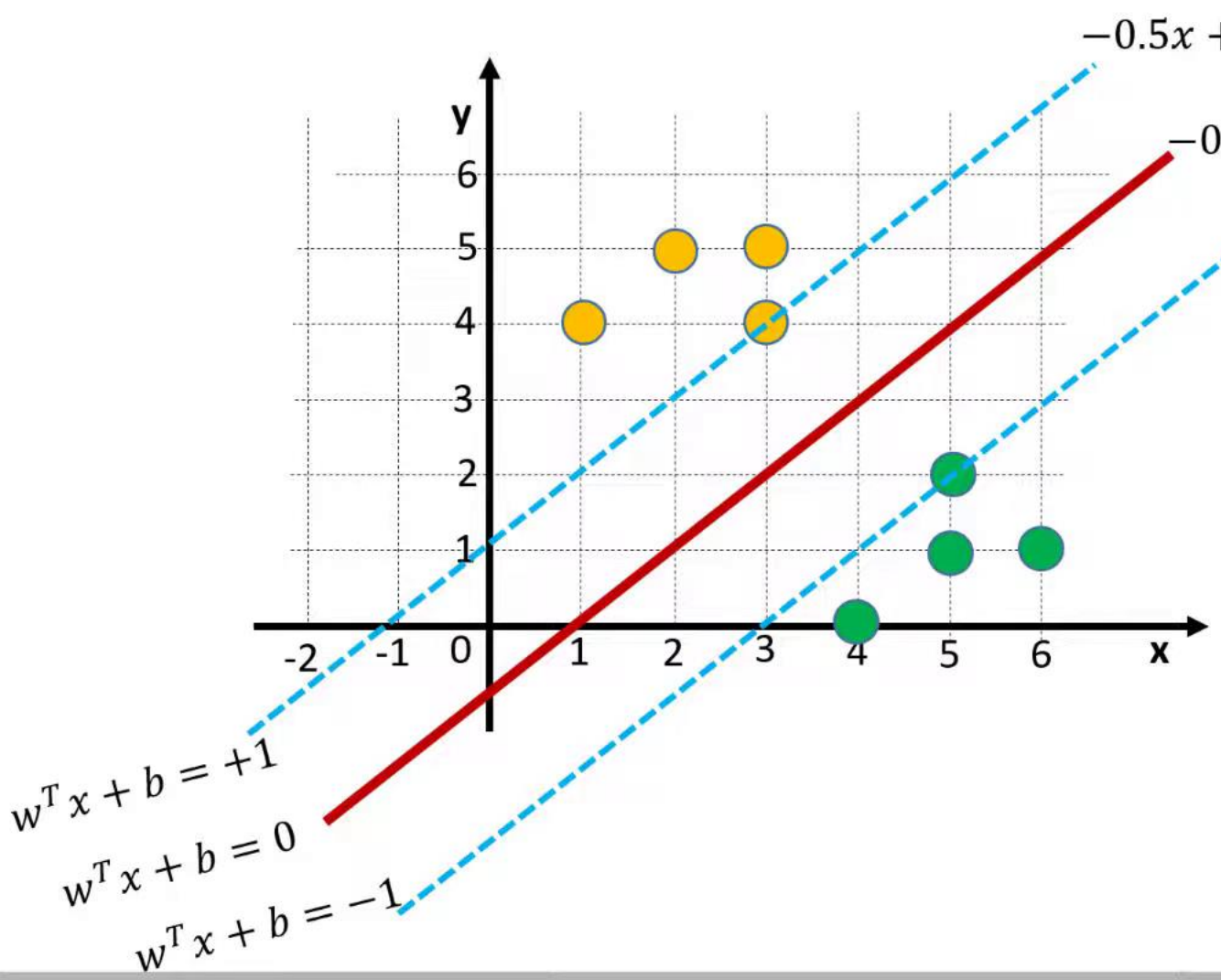
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y \begin{cases} +1 & \text{if } w^T x + b \geq 0 \\ -1 & \text{if } w^T x + b < 0 \end{cases}$$

if it is above, or on, the hyperplane in our example.

SVM – the math



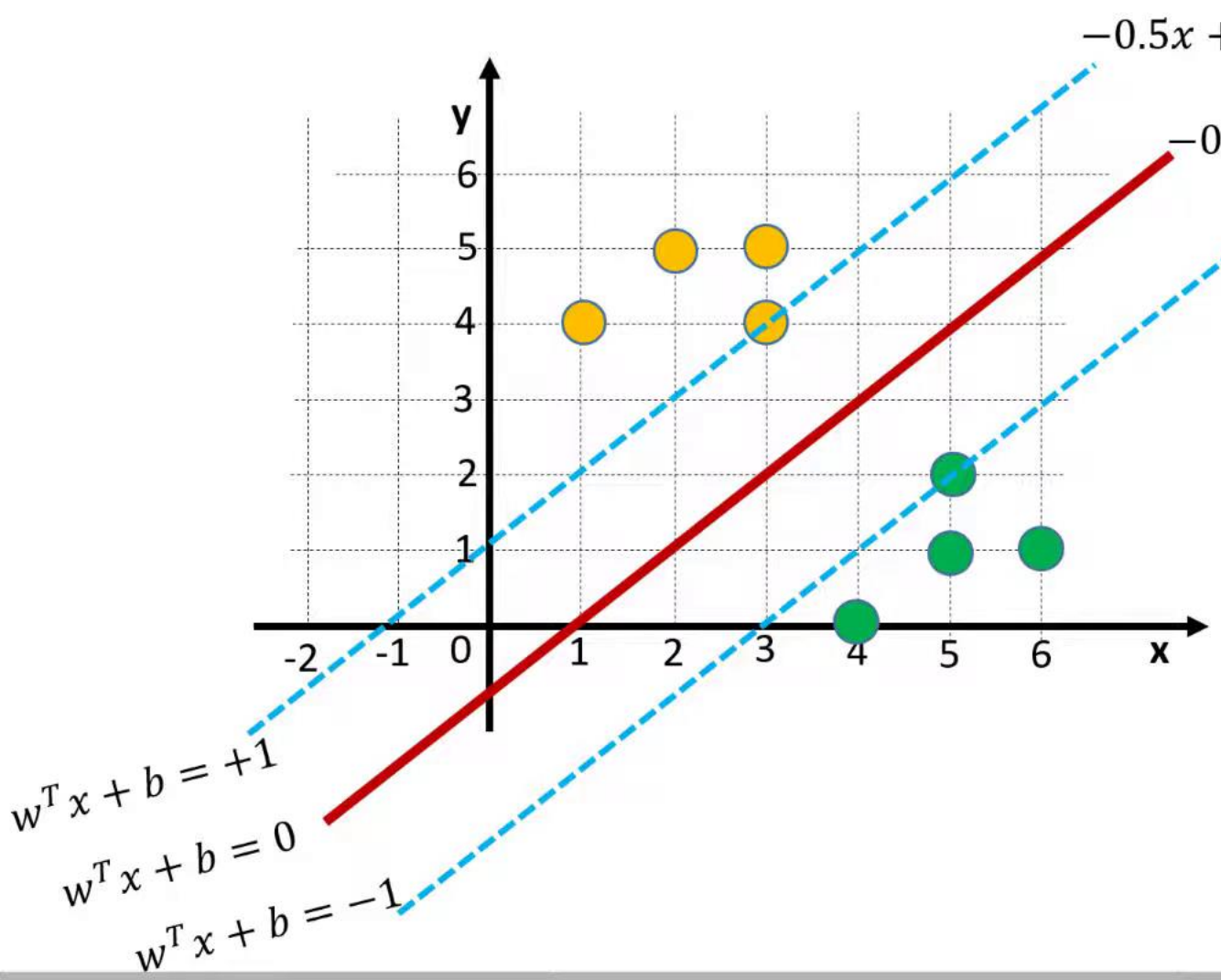
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y \begin{cases} +1 \text{ if } w^T x + b \geq 0 \\ -1 \text{ if } w^T x + b < 0 \end{cases}$$

If the data point is below the hyperplane, we predict that the new observation belongs to the negative group.

SVM – the math



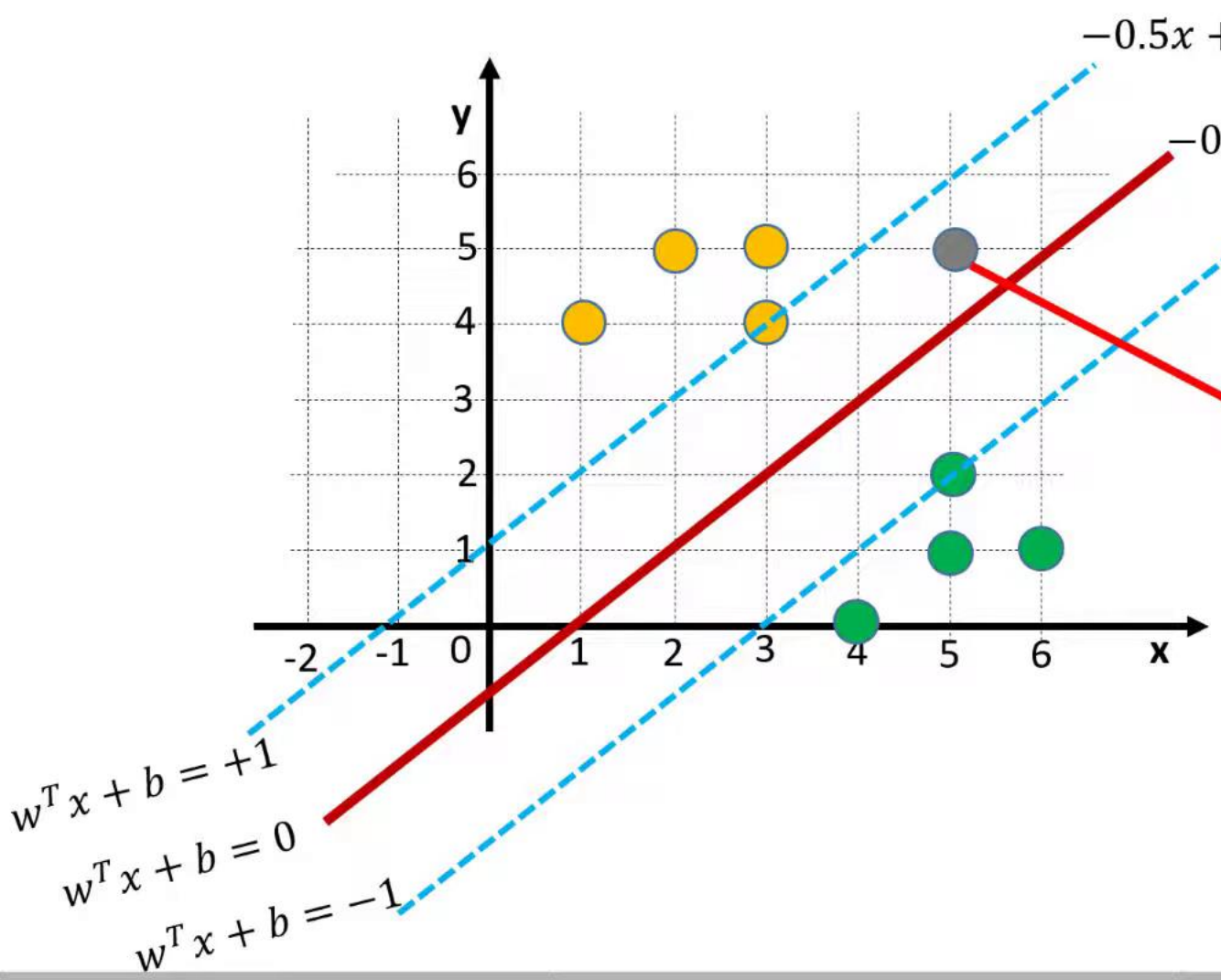
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y \begin{cases} +1 \text{ if } -0.5x + 0.5y + b \geq 0 \\ -1 \text{ if } -0.5x + 0.5y + b < 0 \end{cases}$$

If we plug in the equation of our hyperplane,

SVM – the math



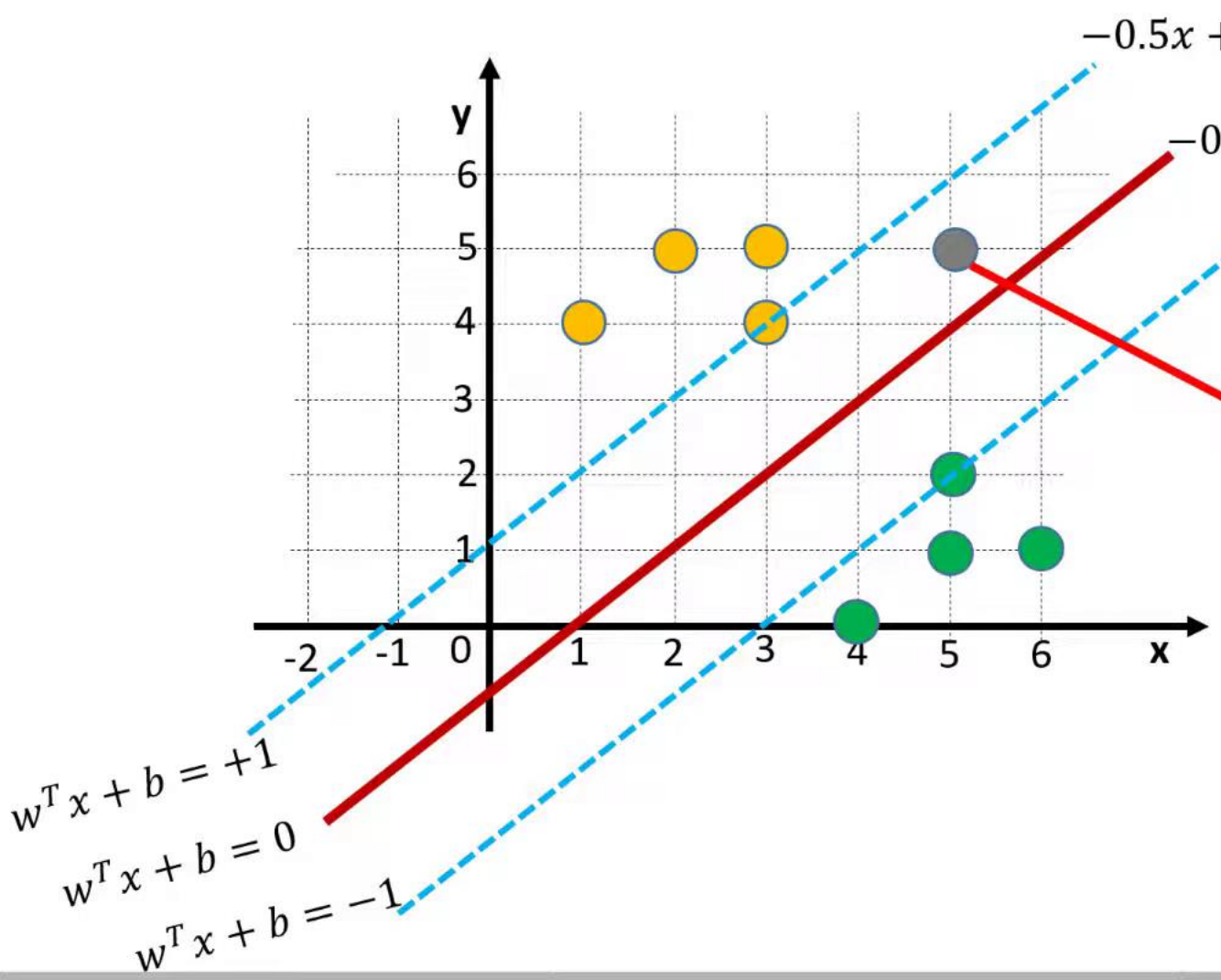
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y \begin{cases} +1 \text{ if } -0.5x + 0.5y + b \geq 0 \\ -1 \text{ if } -0.5x + 0.5y + b < 0 \end{cases}$$

and the x and y-coordinates of the following data point,

SVM – the math



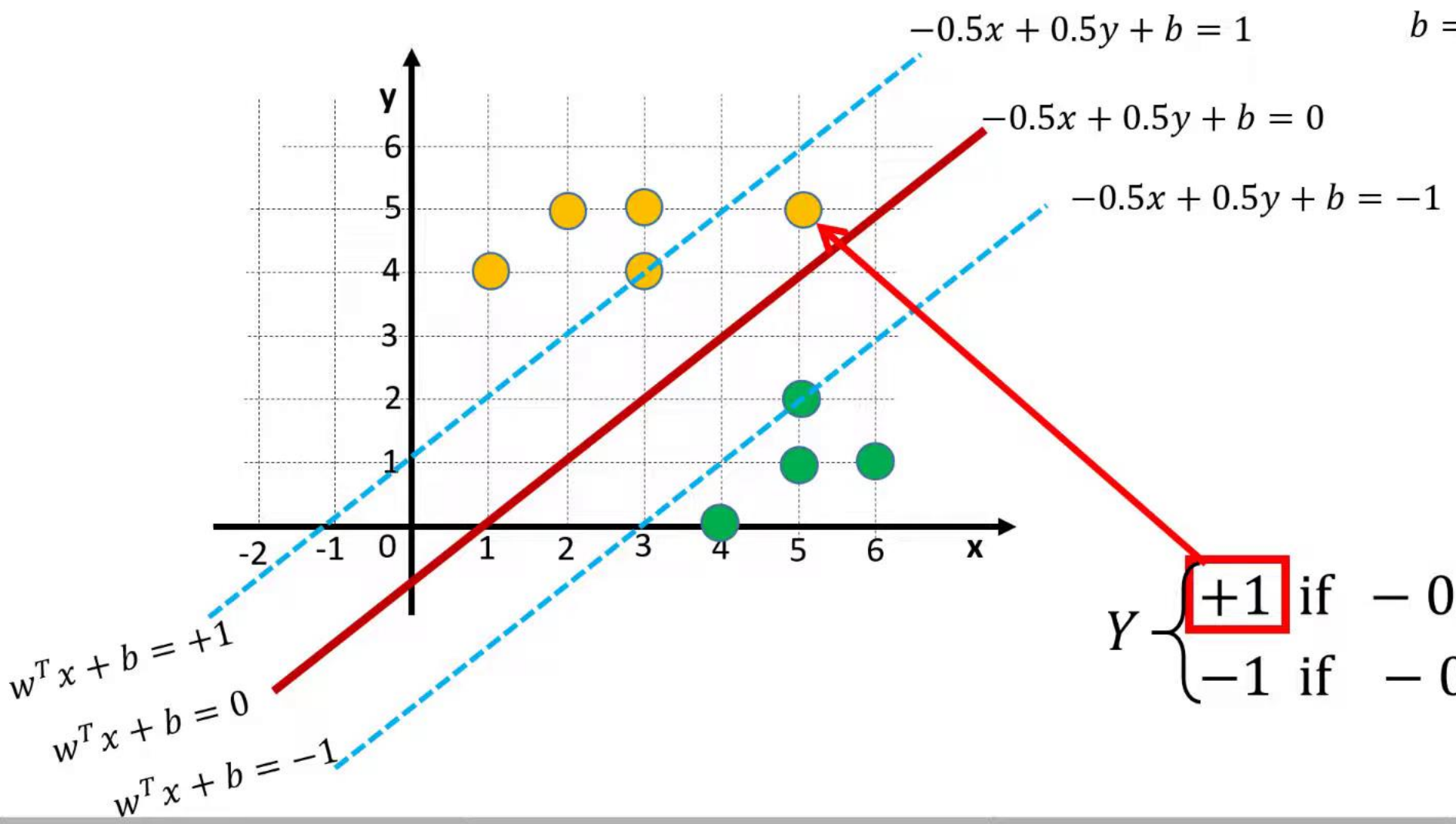
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y \begin{cases} +1 \text{ if } -0.5x + 0.5y + b \geq 0 \\ -1 \text{ if } -0.5x + 0.5y + b < 0 \end{cases}$$

we would see that the calculation would result in a value larger than zero,

SVM – the math



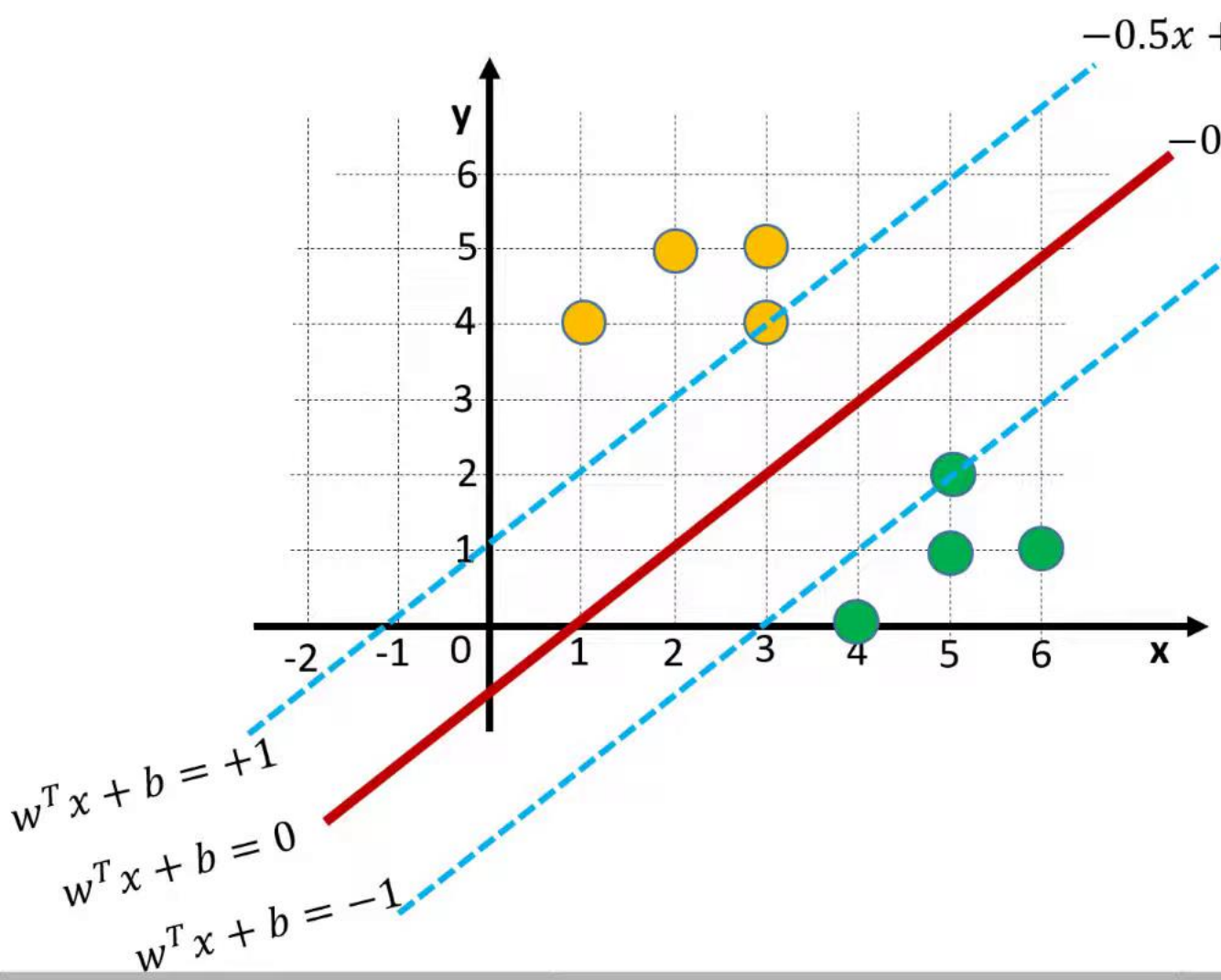
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y \begin{cases} +1 \text{ if } -0.5x + 0.5y + b \geq 0 \\ -1 \text{ if } -0.5x + 0.5y + b < 0 \end{cases}$$

which means that we would predict that the data point belongs to the positive group.

SVM – the math



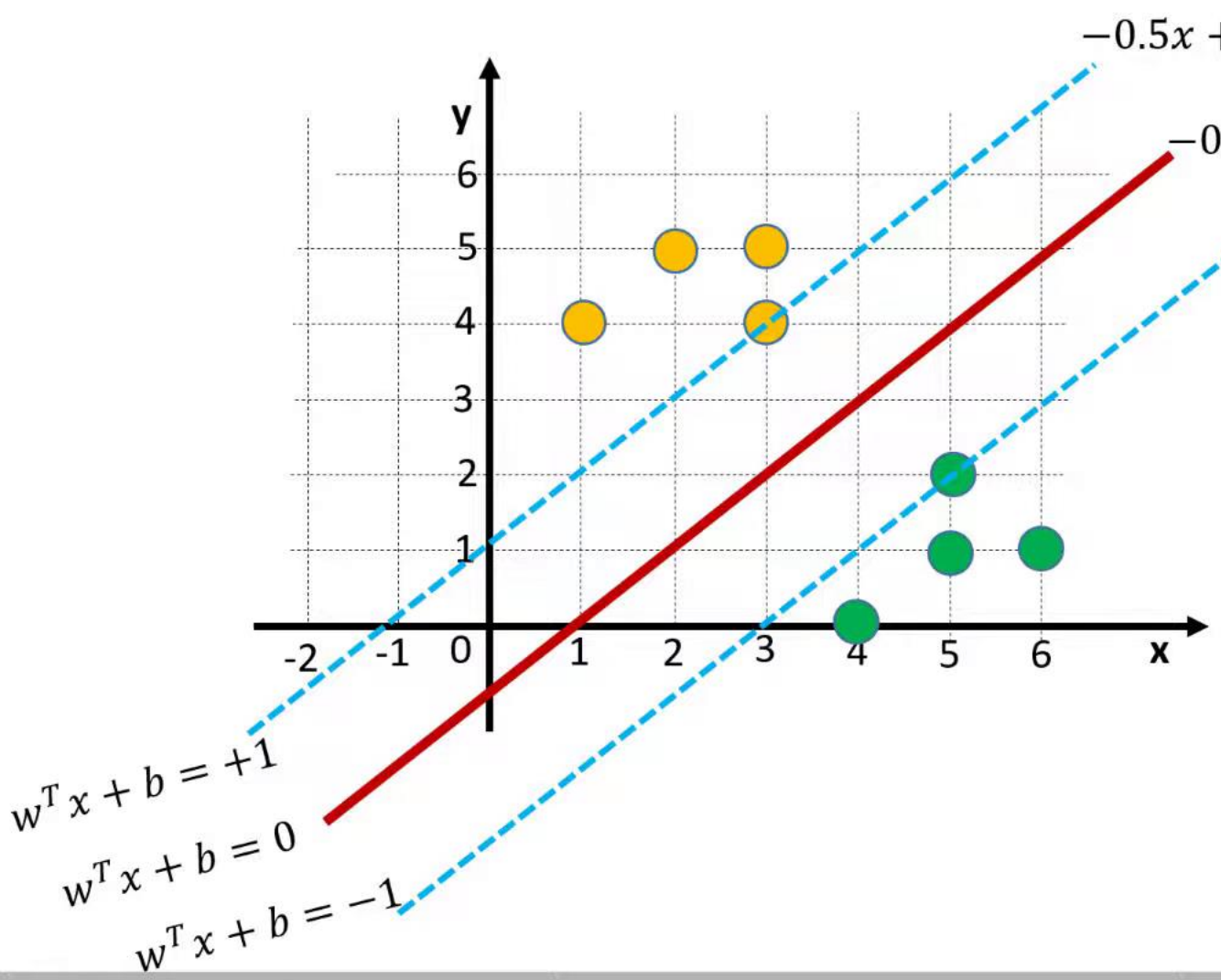
$b = 0.5$

Group (Y_i)	*x*	*y*
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{|C_2 - C_1|}{\sqrt{A^2 + B^2}}$$

Remember that we can calculate the distance between two parallel lines with this equation.

SVM – the math

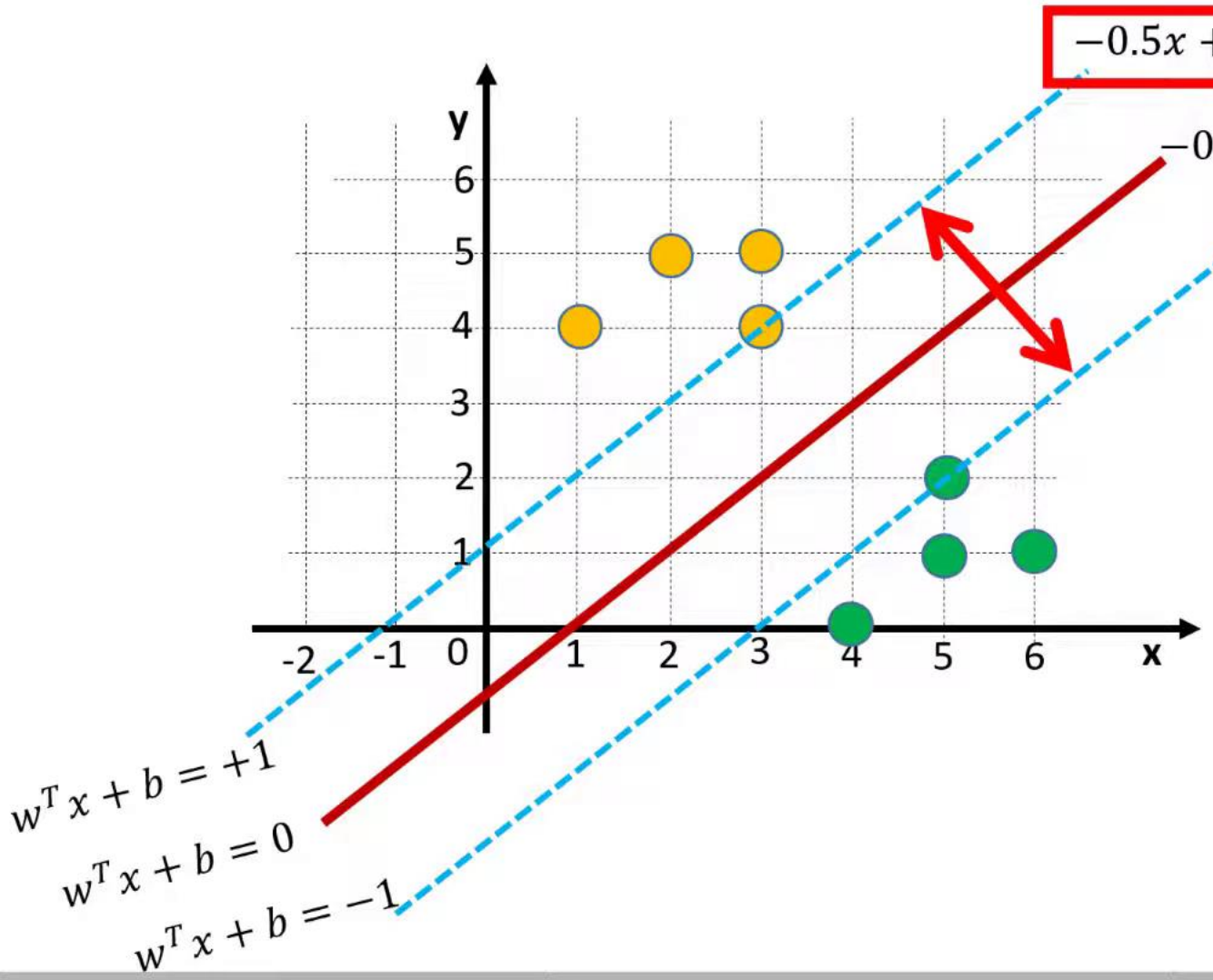


Group (Y_i)	*x*	*y*
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{|C_2 - C_1|}{\sqrt{A^2 + B^2}}$$

Remember that we can calculate the distance between two parallel lines with this equation.

SVM – the math



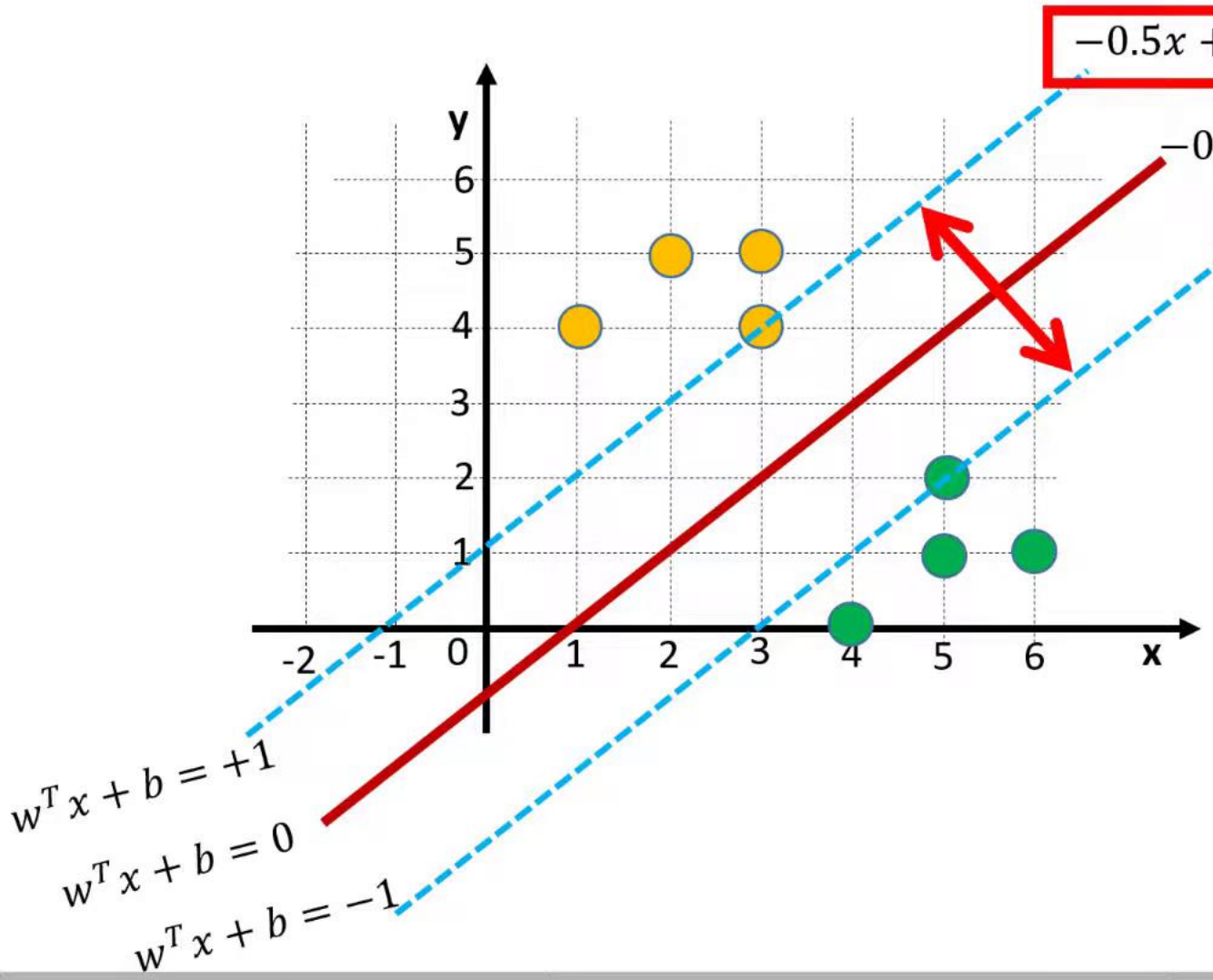
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{|C_2 - C_1|}{\sqrt{A^2 + B^2}}$$

And in the special case where we have normalized the equations like this,

SVM – the math



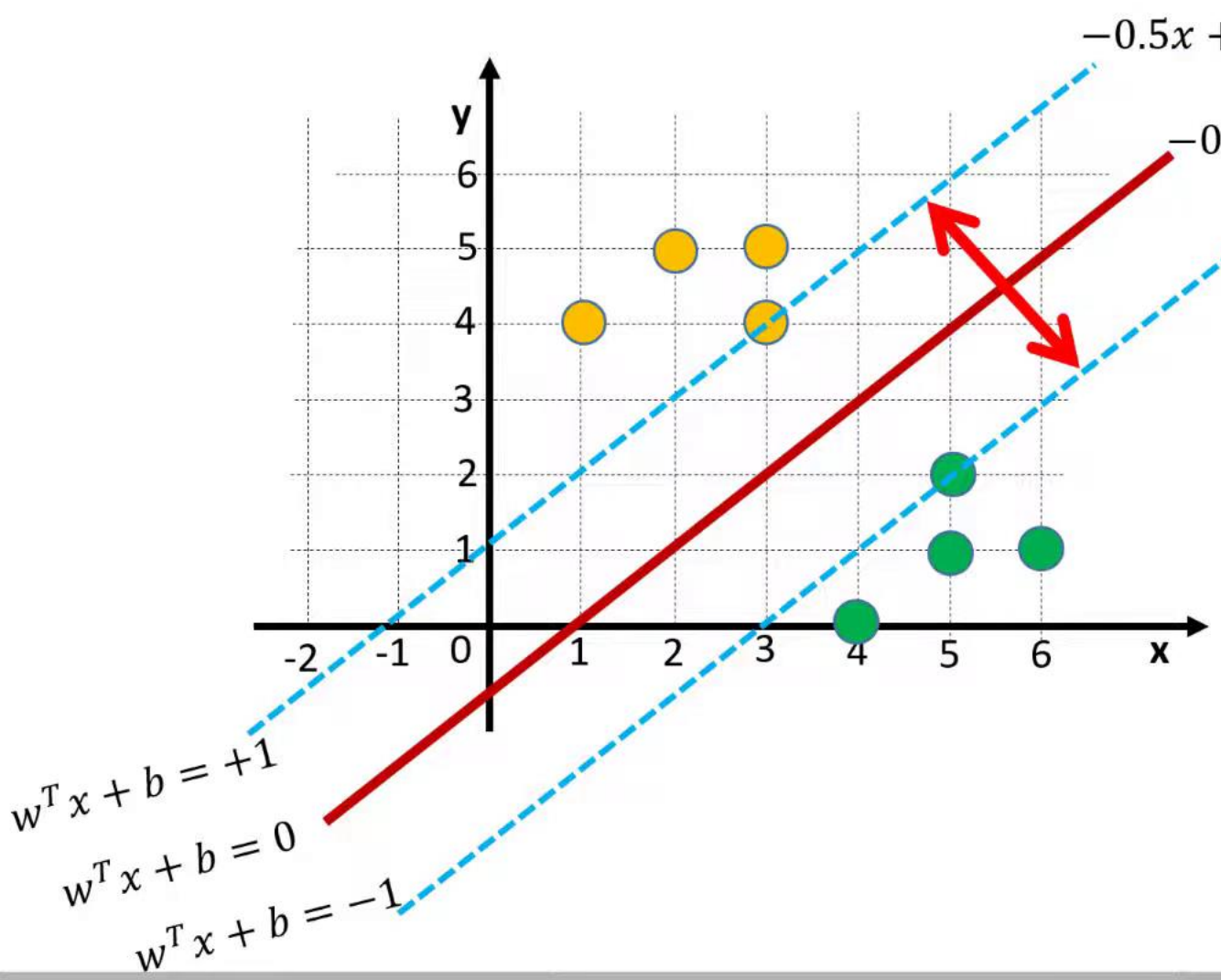
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{|C_2 - C_1|}{\sqrt{A^2 + B^2}}$$

And in the special case where we have normalized the equations like this,

SVM – the math



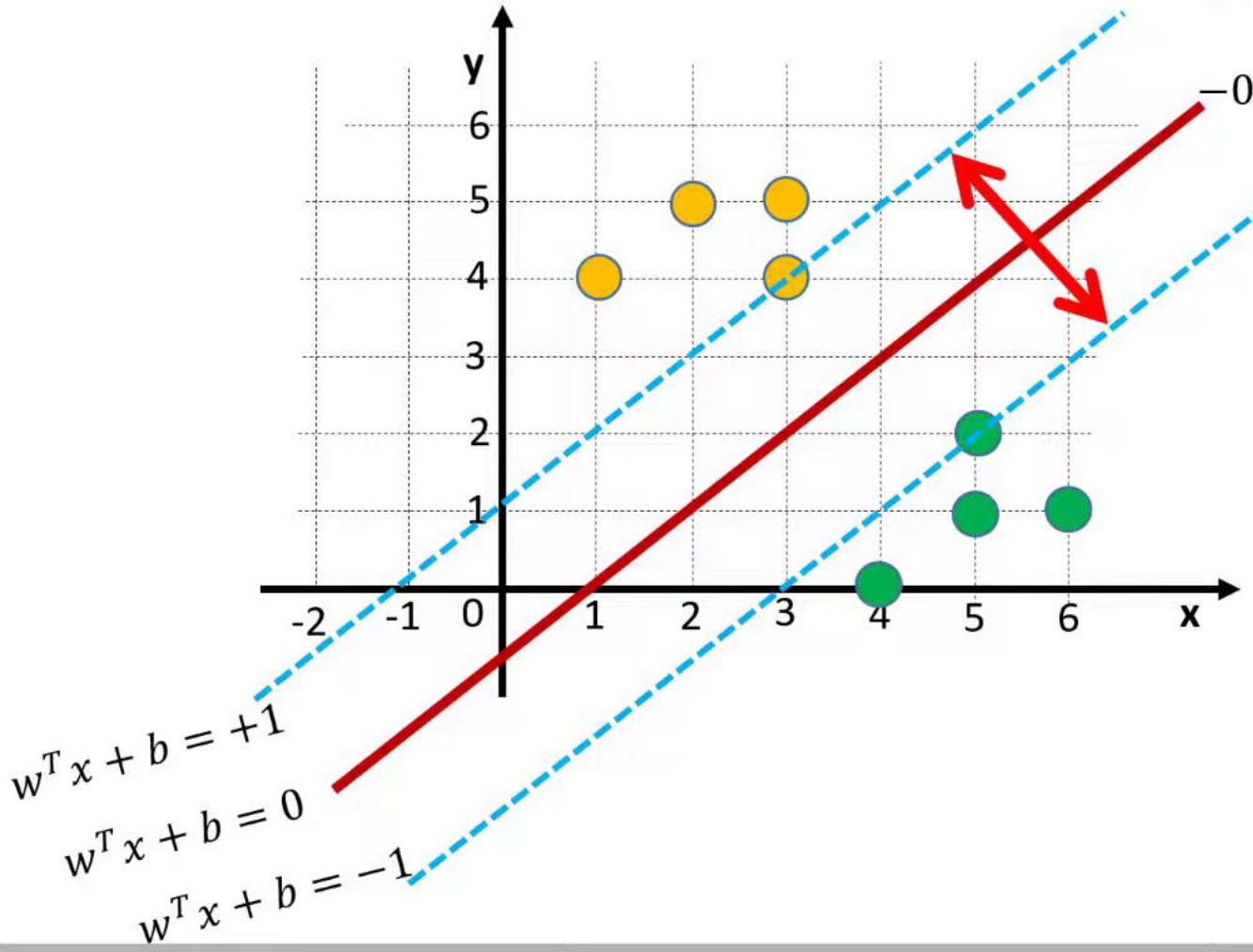
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}}$$

we can simplify the equation to this,

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

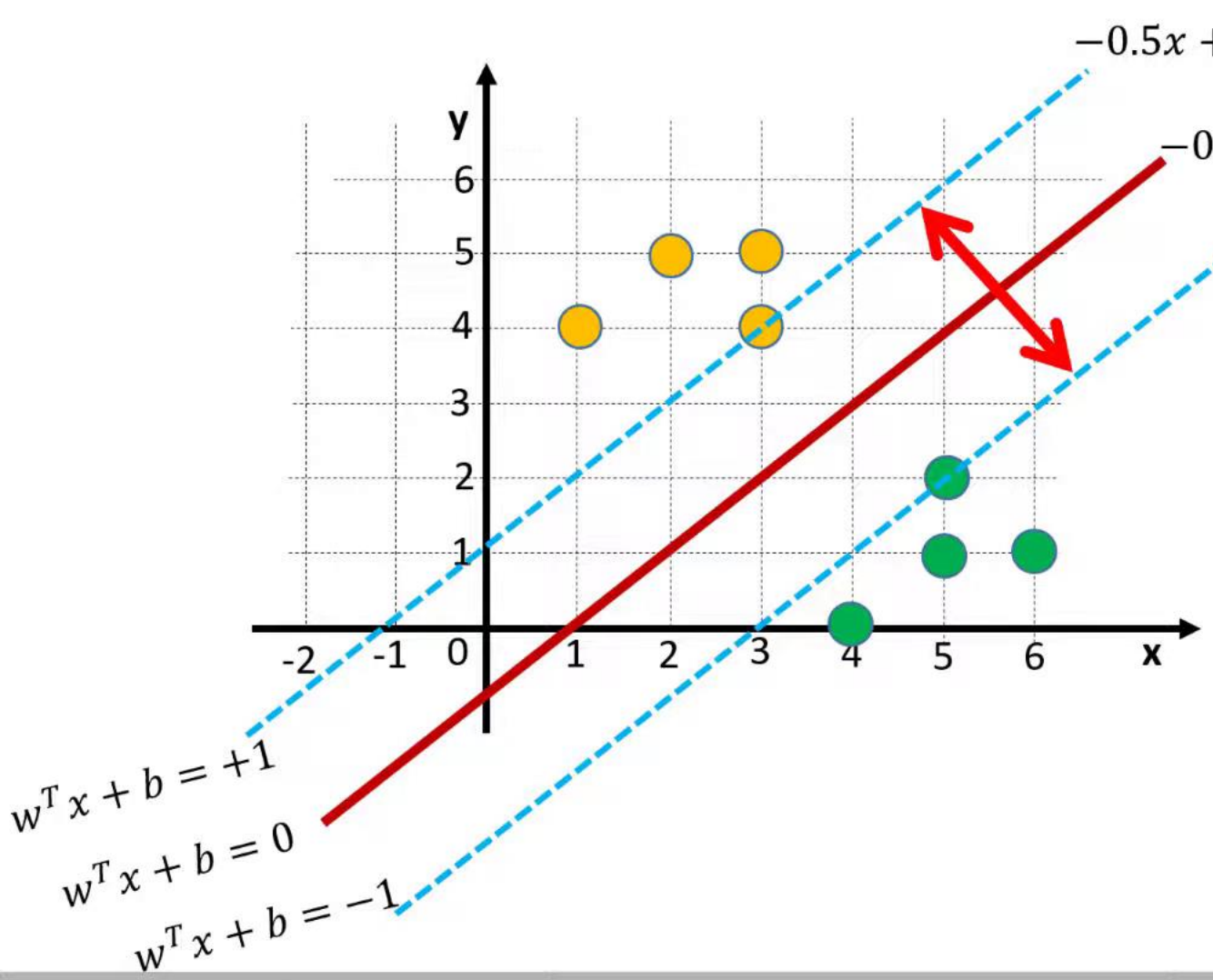
$b = 0.5$

Group (Y_i)	*x*	*y*
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}}$$

because the absolute difference between the constant terms is always two.

SVM – the math



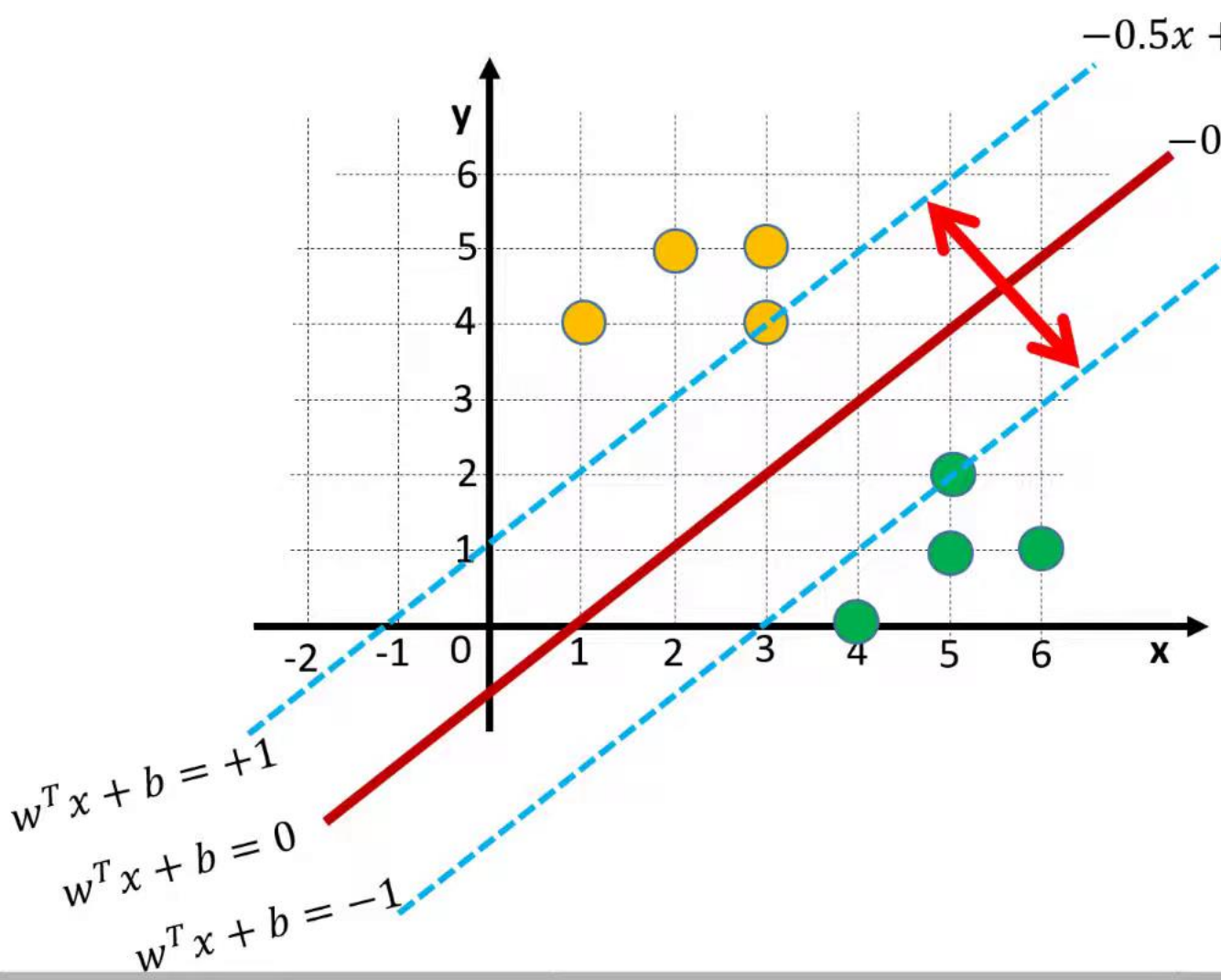
$b = 0.5$

Group (Y_i)	*x*	*y*
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}}$$

because the absolute difference between the constant terms is always two.

SVM – the math



$b = 0.5$

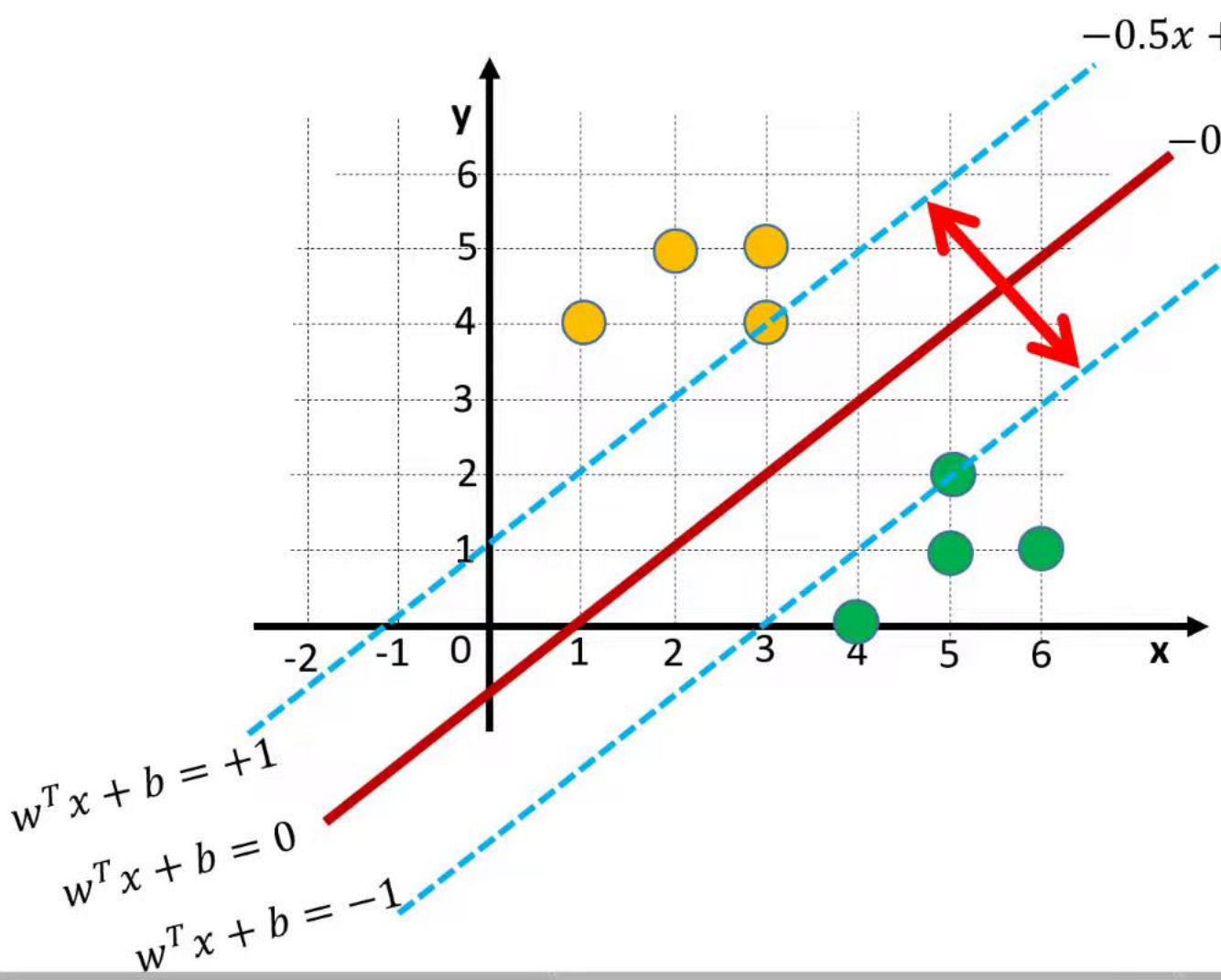
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}}$$

$$w^T x + b = 0$$

Remember that w is a vector of A and B in our example,

SVM – the math



$b = 0.5$

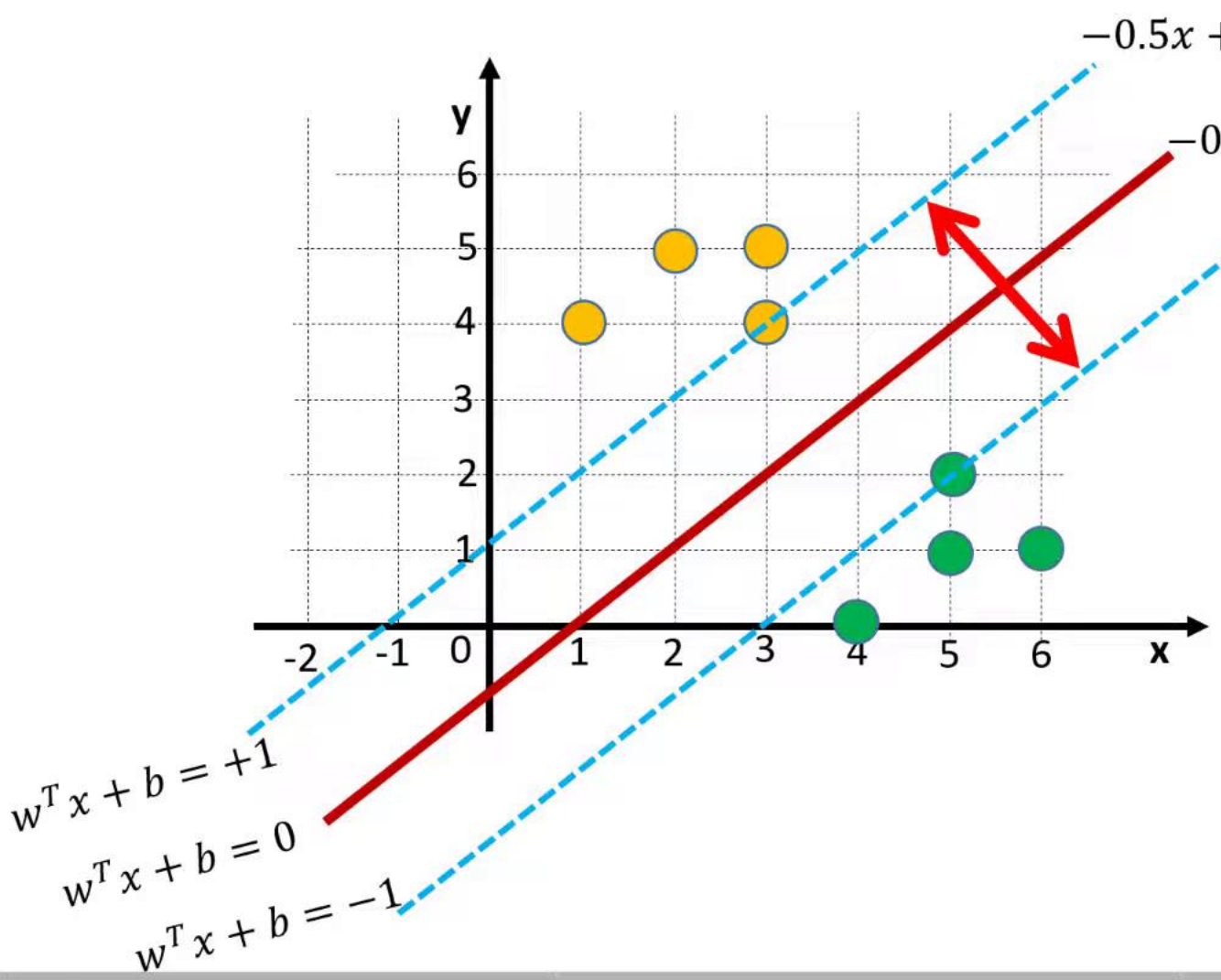
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$w^T x + b = 0$$

which explains why the distance between the two blue lines is usually expressed like this.

SVM – the math



$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$w^T x + b = 0$$

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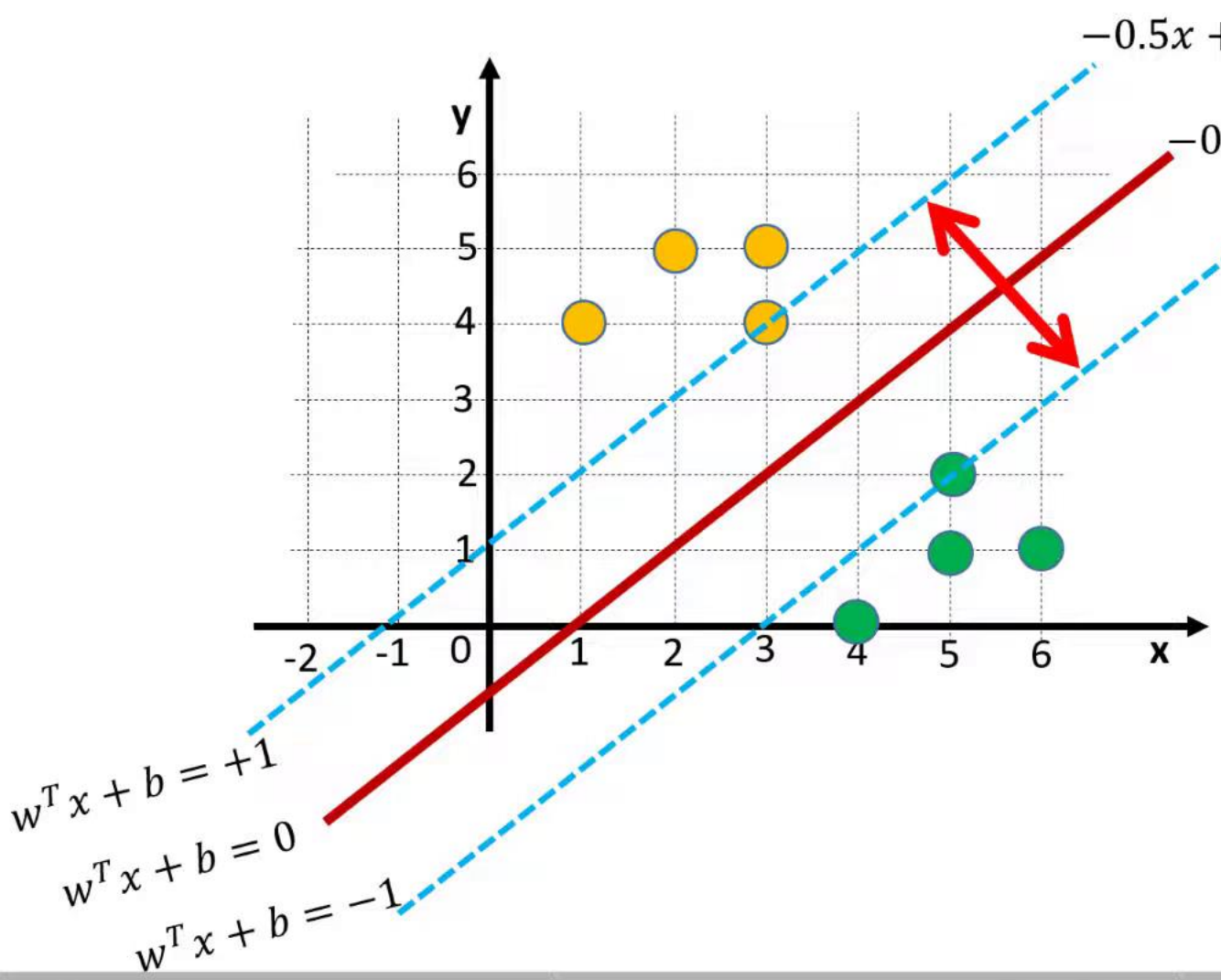
SVM – the math



$b = 0.5$

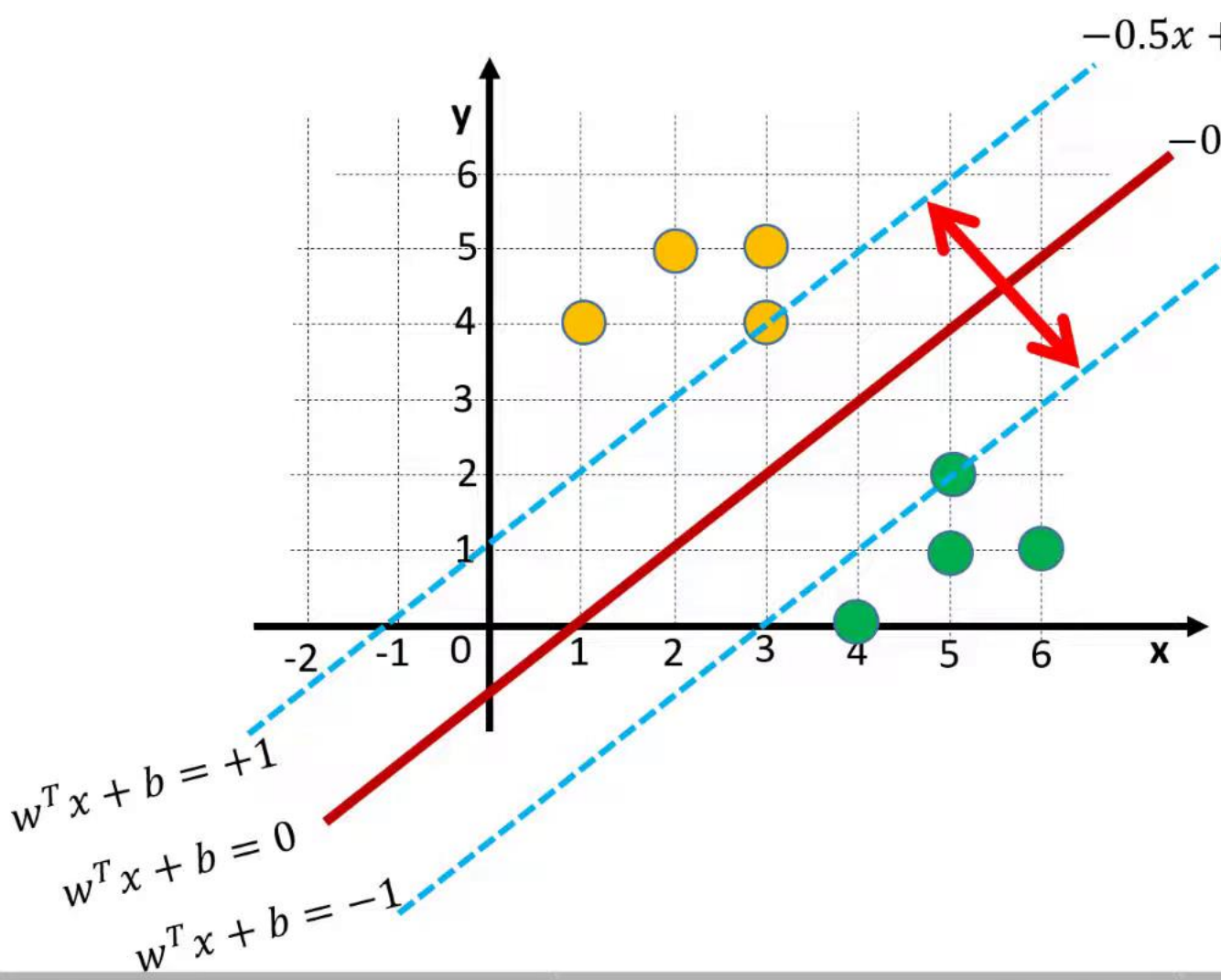
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$w^T x + b = 0$$

If there is a line that can separate the two groups completely, SVM tries to find the optimal values of A and B so that the distance between the two blue lines is maximized.

SVM – the math



$b = 0.5$

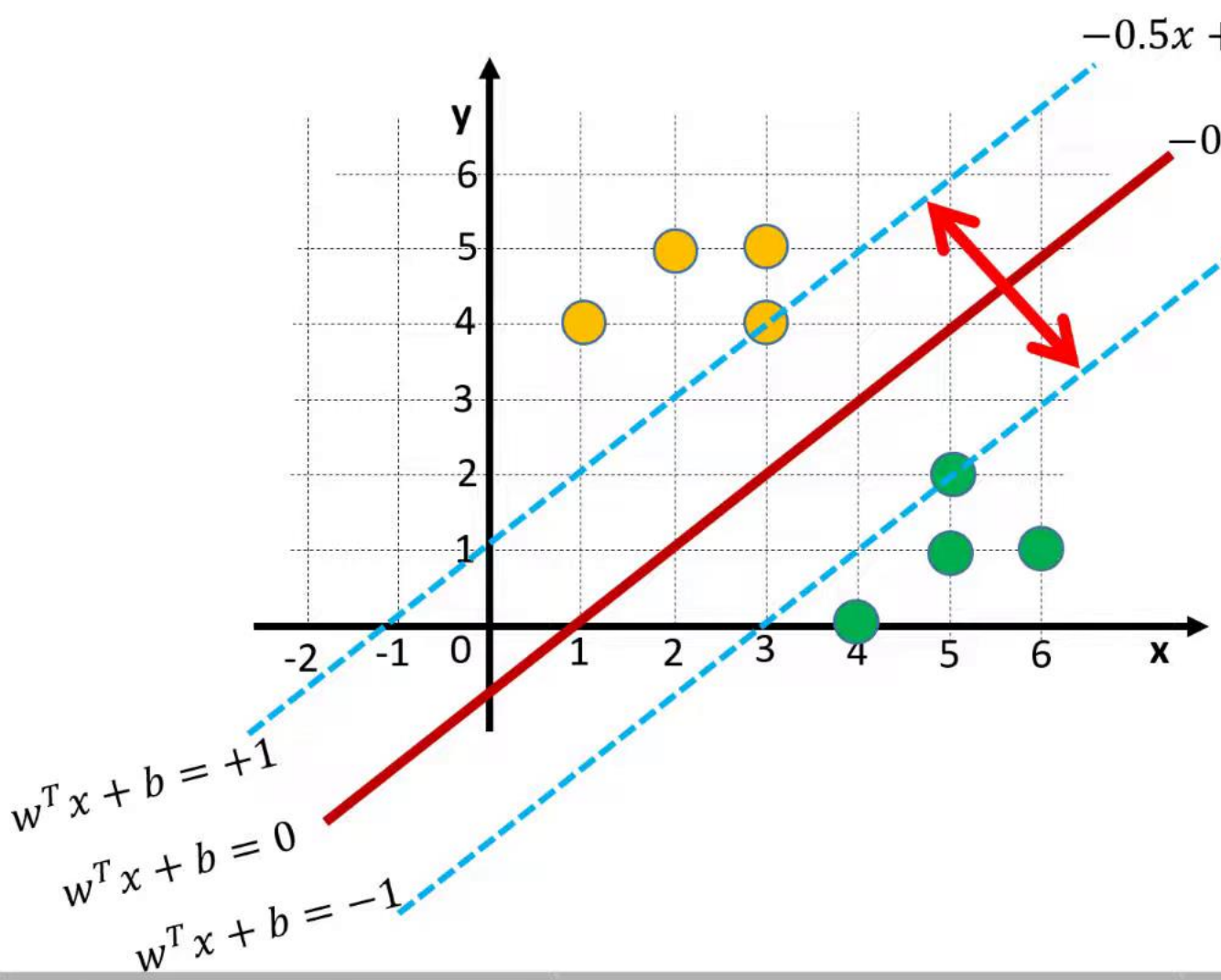
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$w^T x + b = 0$$

If there is a line that can separate the two groups completely, SVM tries to find the optimal values of A and B so that the distance between the two blue lines is maximized.

SVM – the math



$b = 0.5$

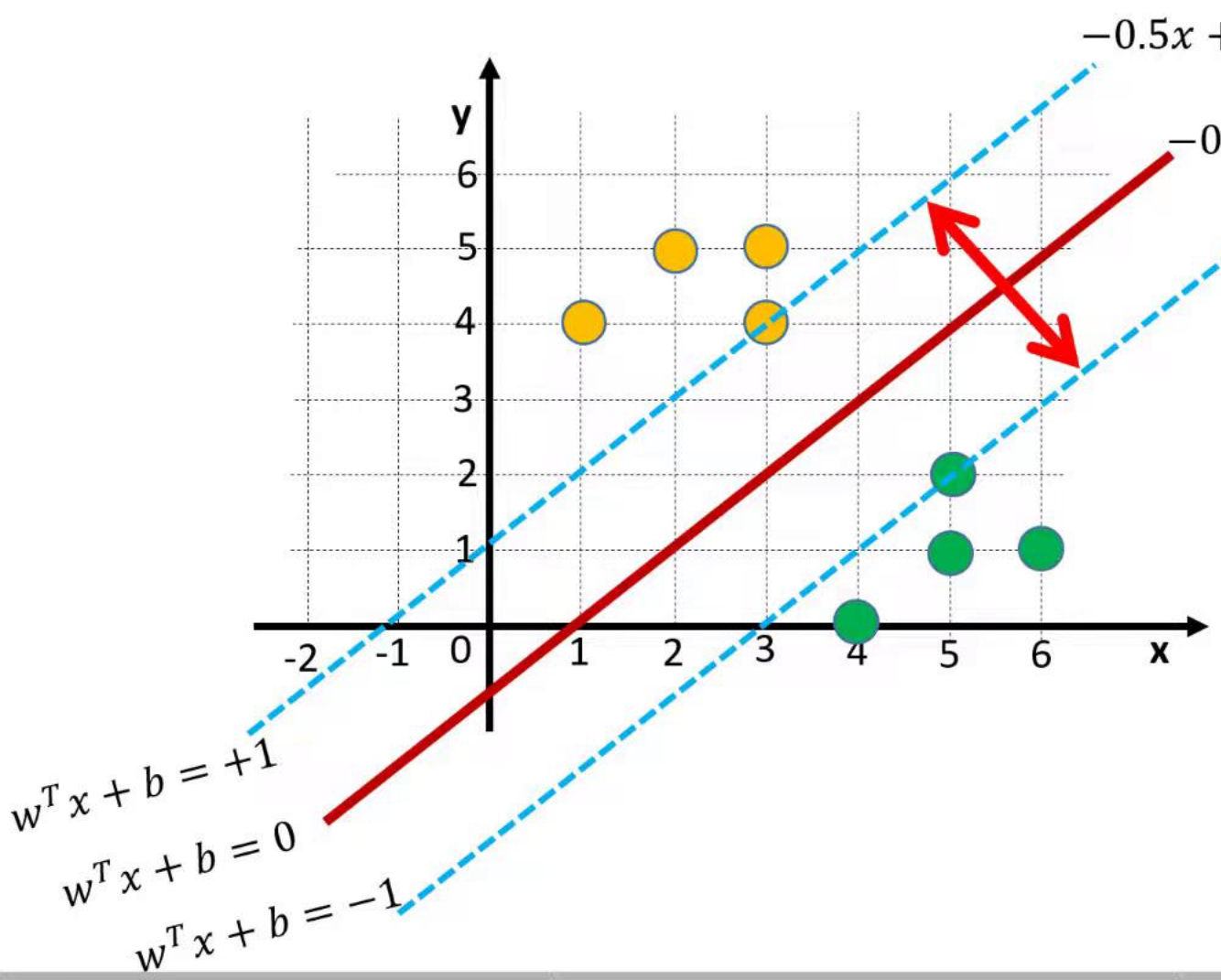
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$w^T x + b = 0$$

Or, we can also minimize the denominator because that will also maximize the distance between the two blue lines.

SVM – the math



$b = 0.5$

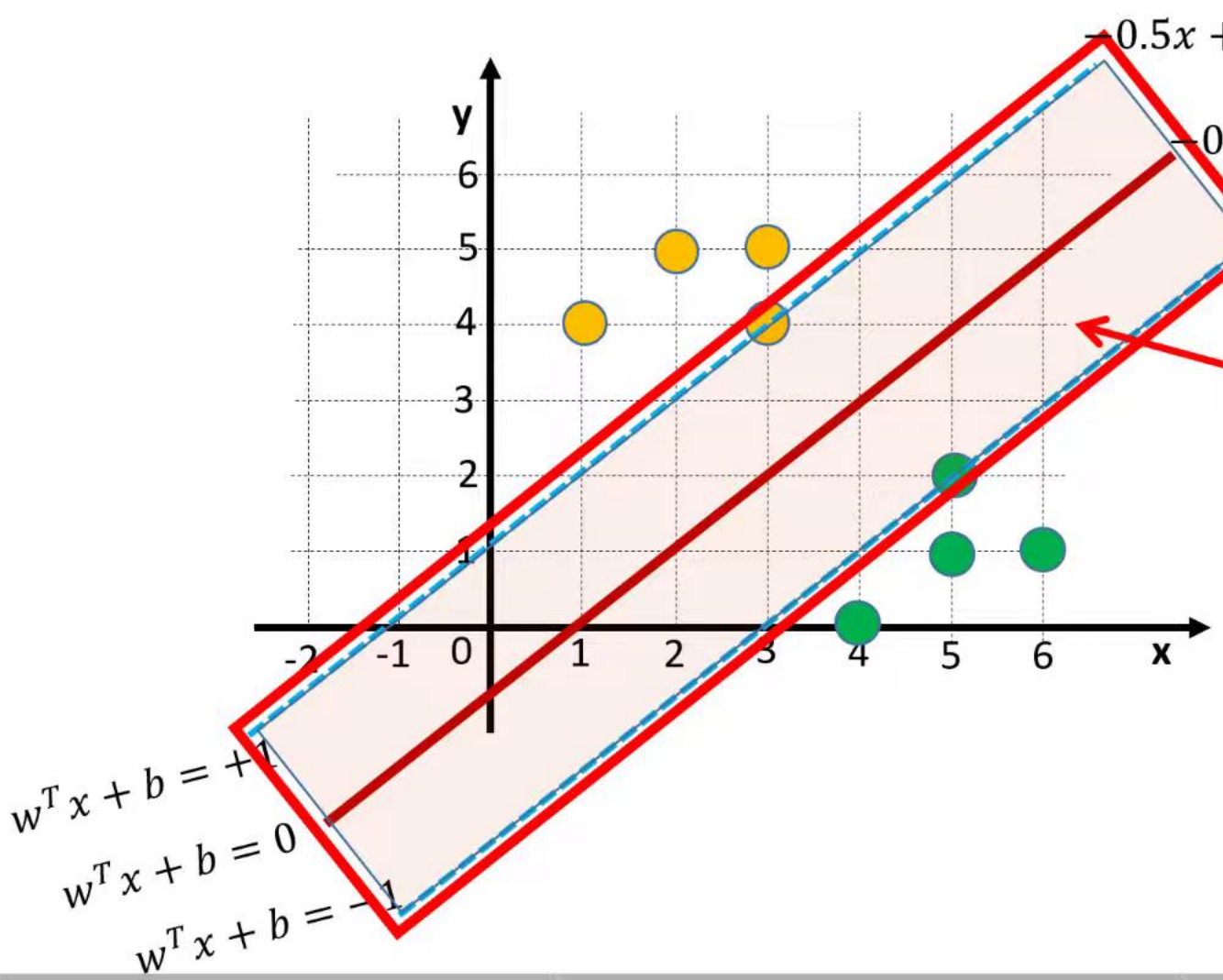
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$w^T x + b = 0$$

Or, we can also minimize the denominator because that will also maximize the distance between the two blue lines.

SVM – the math



$b = 0.5$

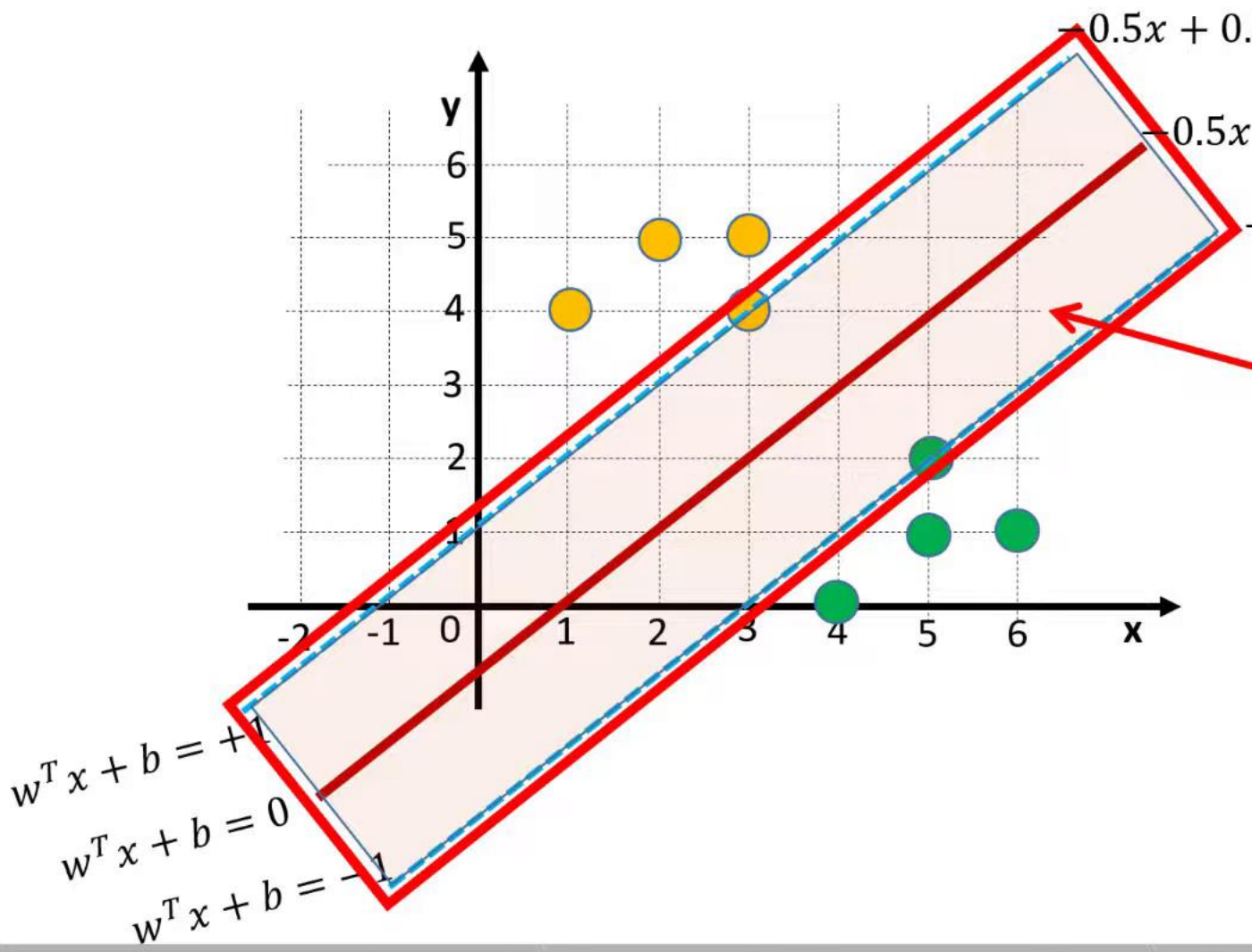
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

Margin

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

The area between the two blue lines is called the margin. The SVM therefore tries to find a hyperplane that maximizes the width of this margin.

SVM – the math



$b = 0.5$

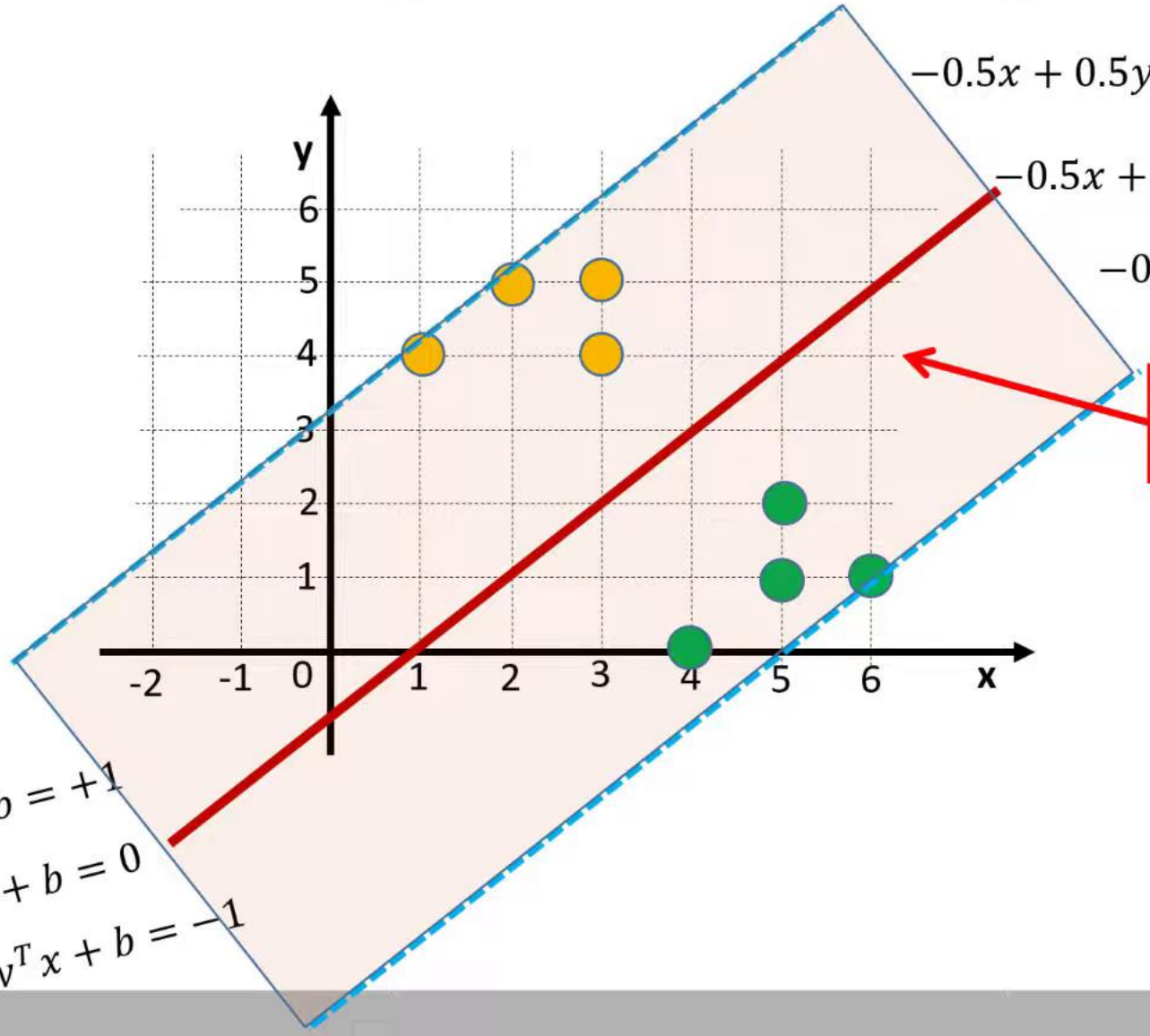
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

Margin

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

The area between the two blue lines is called the margin. The SVM therefore tries to find a hyperplane that maximizes the width of this margin.

SVM – the math



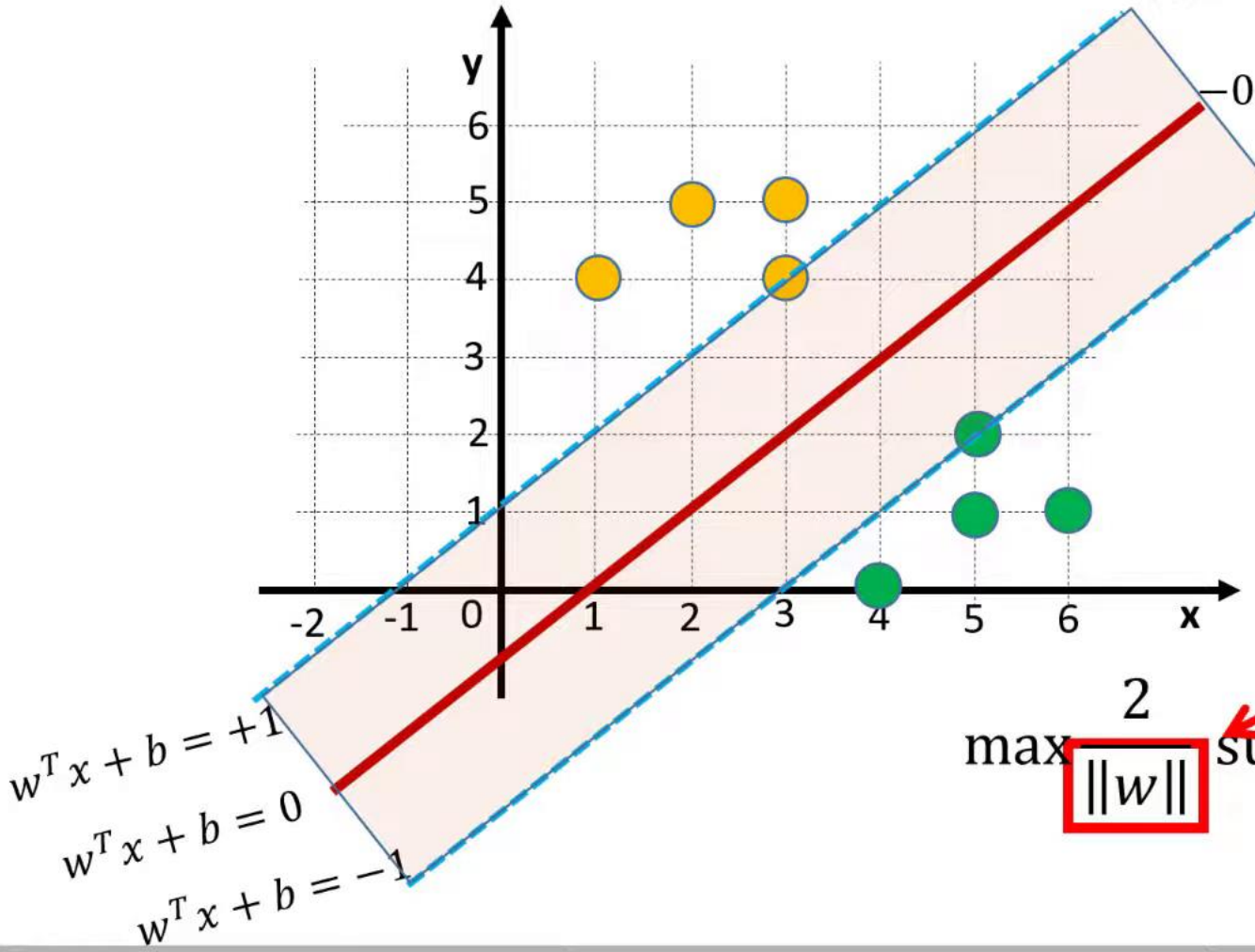
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

However, we need some constraints because the margin can be infinitely large.

SVM – the math



$b = 0.5$

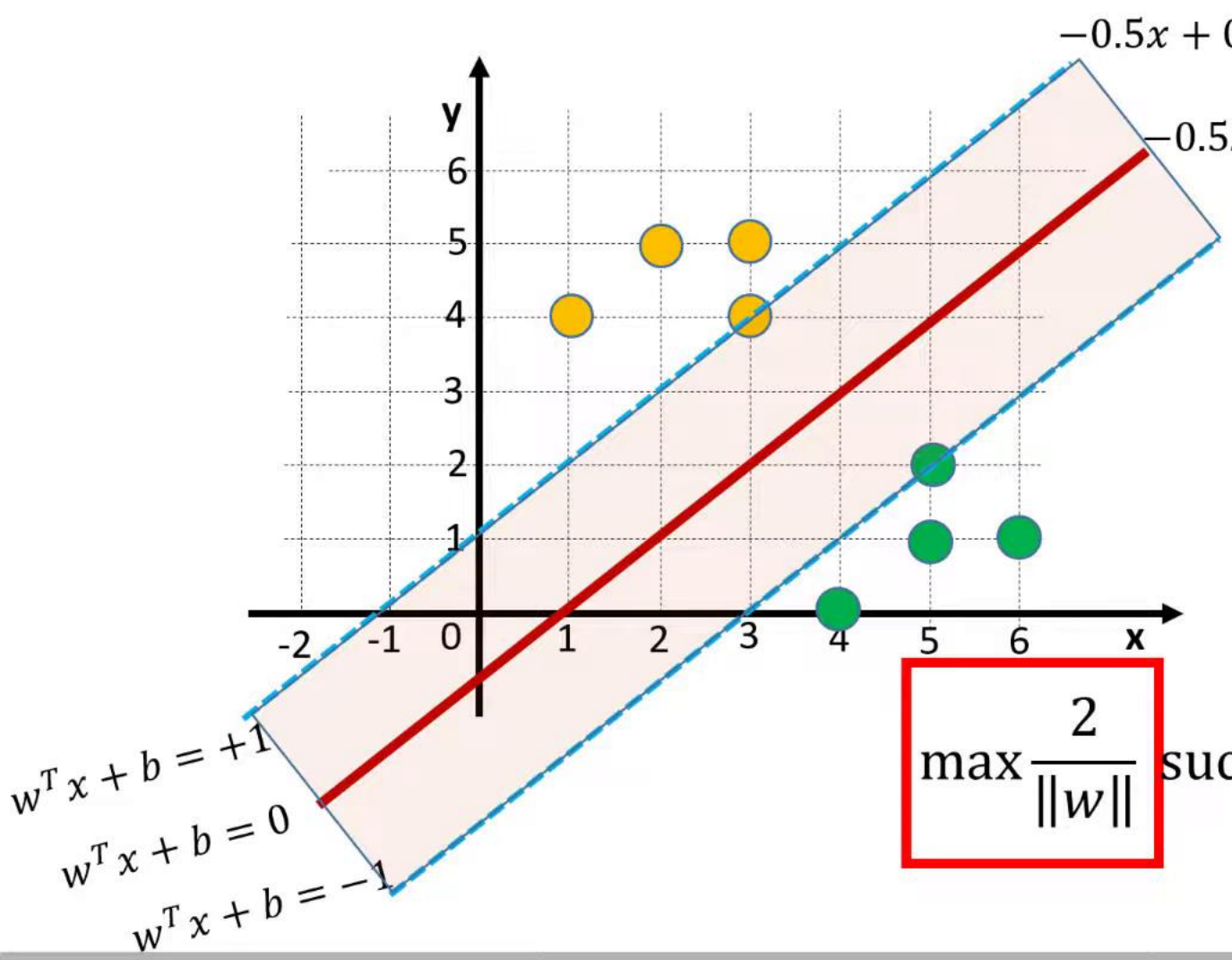
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{matrix} \geq 1 & \text{if } Y_i = +1 \\ \leq -1 & \text{if } Y_i = -1 \end{matrix}$$

Training the SVM can be defined as we try to find the optimal values of A and B of the hyperplane,

SVM – the math



$b = 0.5$

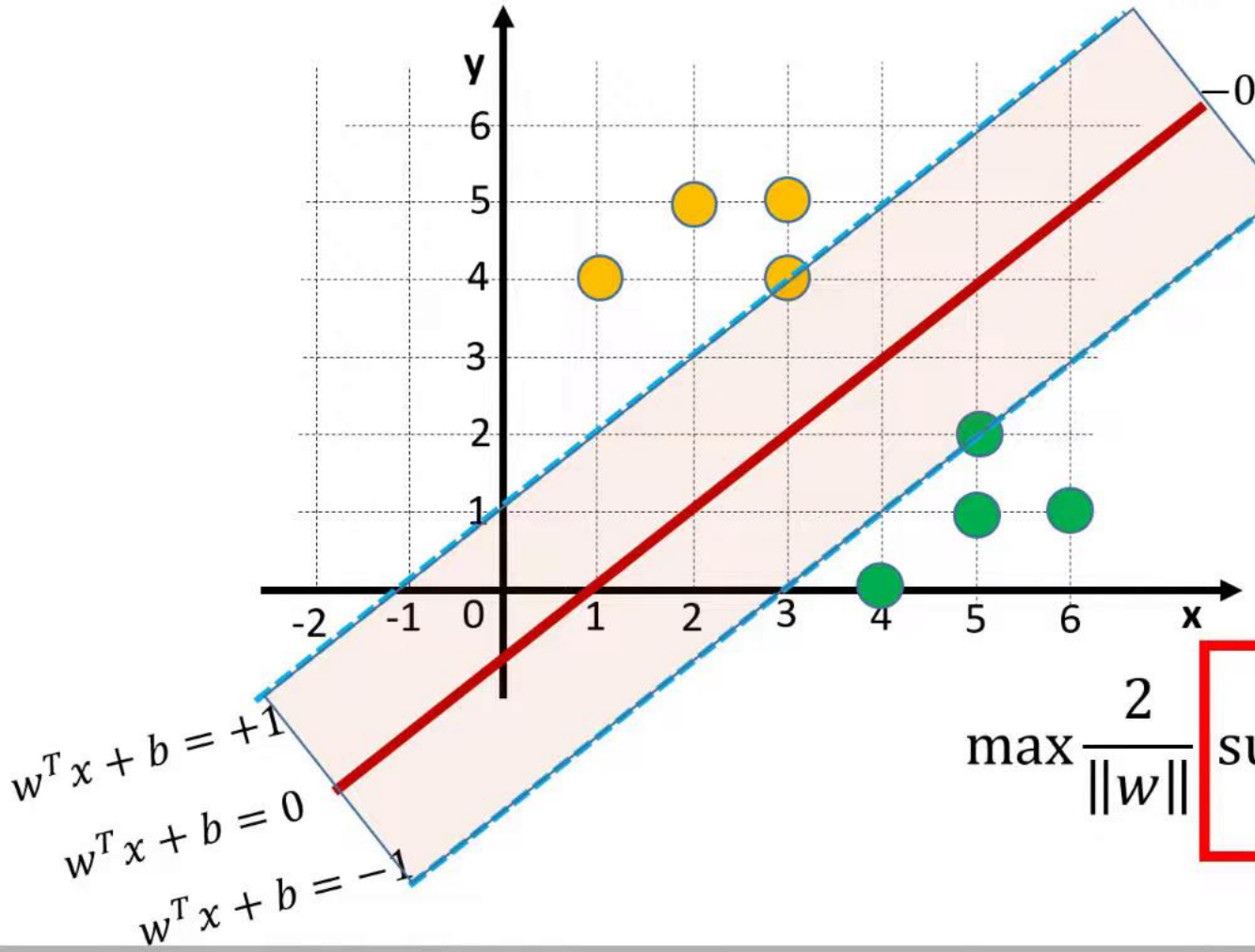
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{array}{l} \geq 1 \text{ if } Y_i = +1 \\ \leq -1 \text{ if } Y_i = -1 \end{array}$$

so that we maximize the margin,

SVM – the math



$b = 0.5$

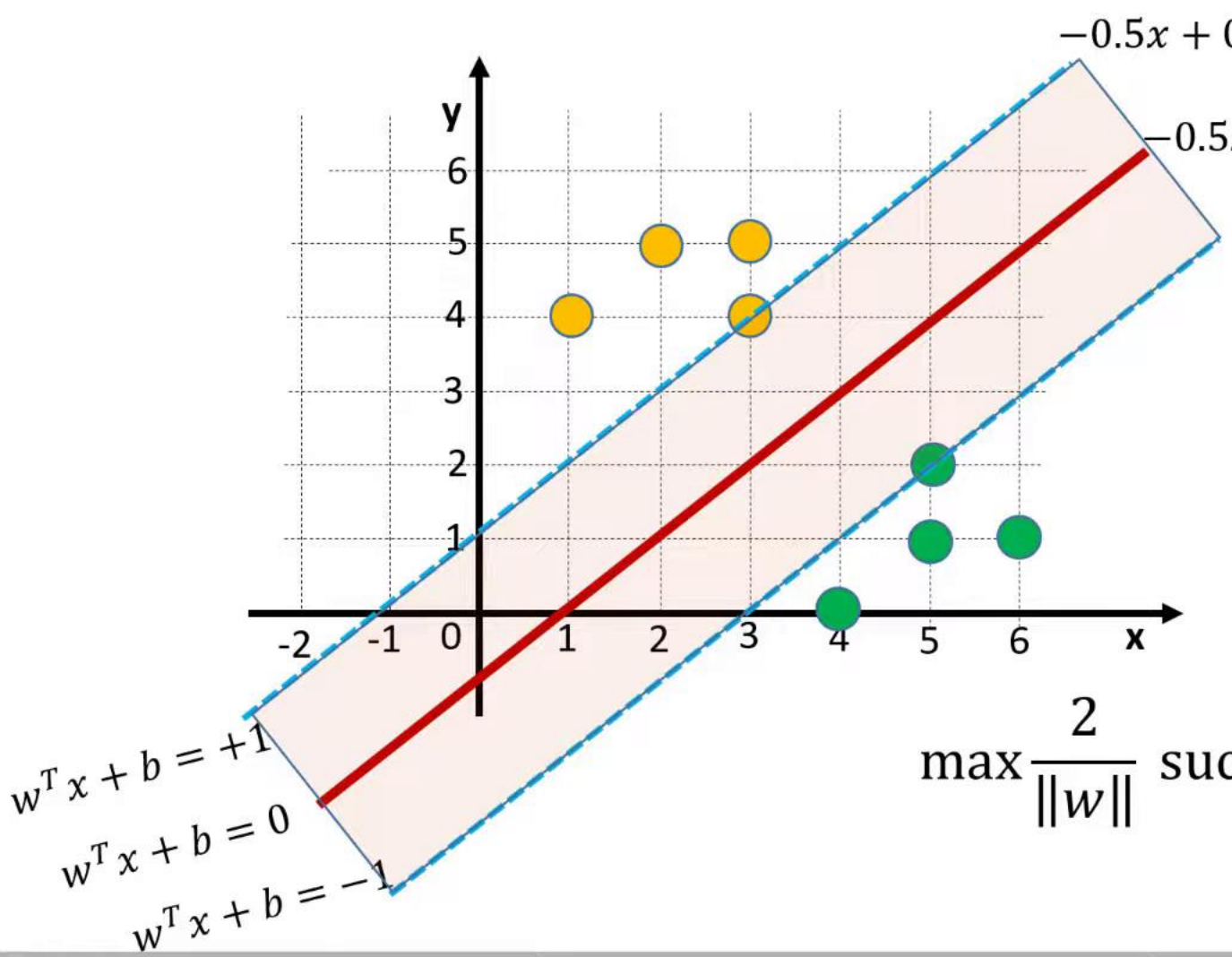
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$\max \frac{2}{\|w\|} \quad \text{such that } w^T x_i + b \quad \begin{matrix} \geq 1 & \text{if } Y_i = +1 \\ \leq -1 & \text{if } Y_i = -1 \end{matrix}$$

such that

SVM – the math



$b = 0.5$

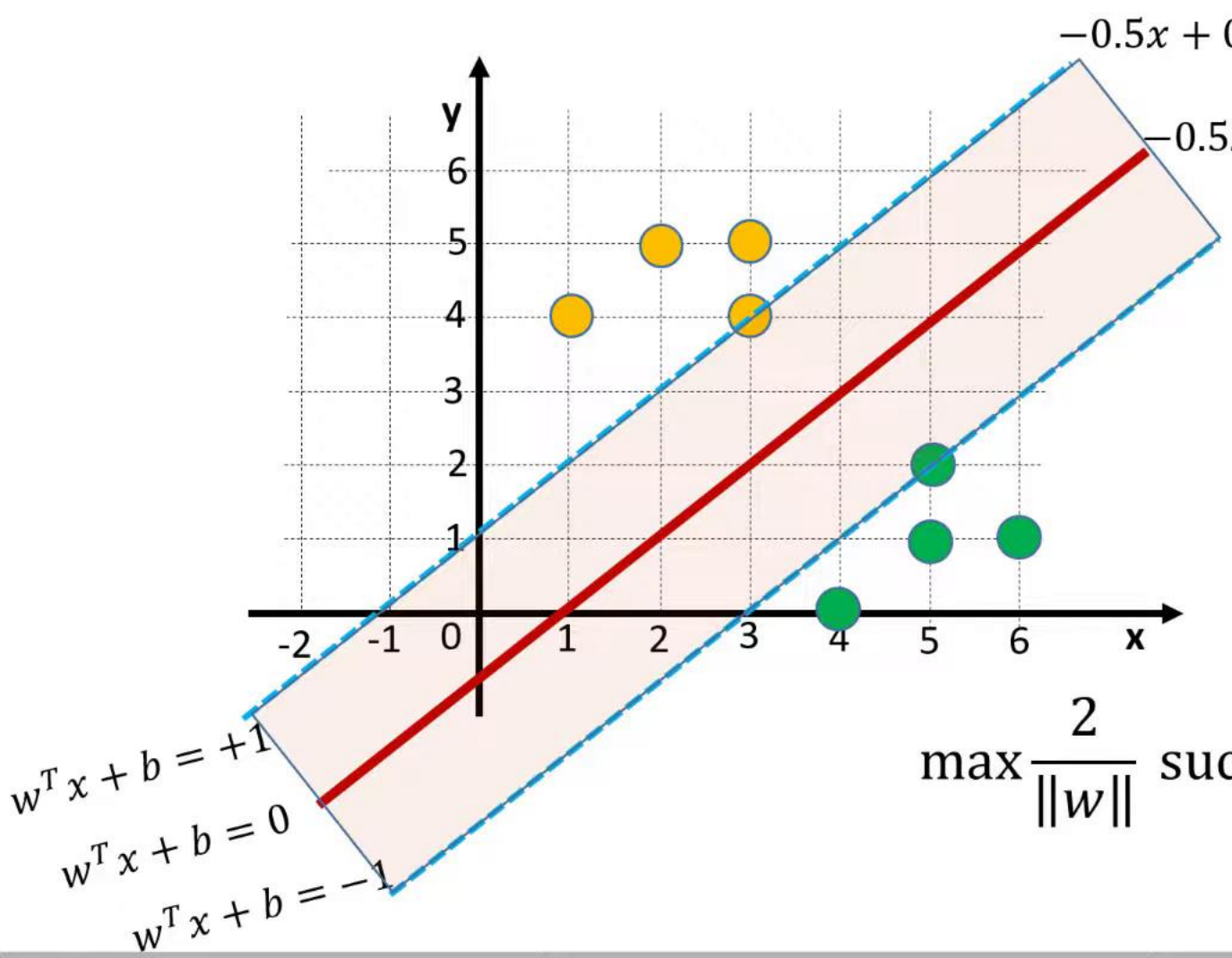
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{cases} \geq 1 & \text{if } Y_i = +1 \\ \leq -1 & \text{if } Y_i = -1 \end{cases}$$

results in a value, based on the equation of the hyperplane,

SVM – the math



$b = 0.5$

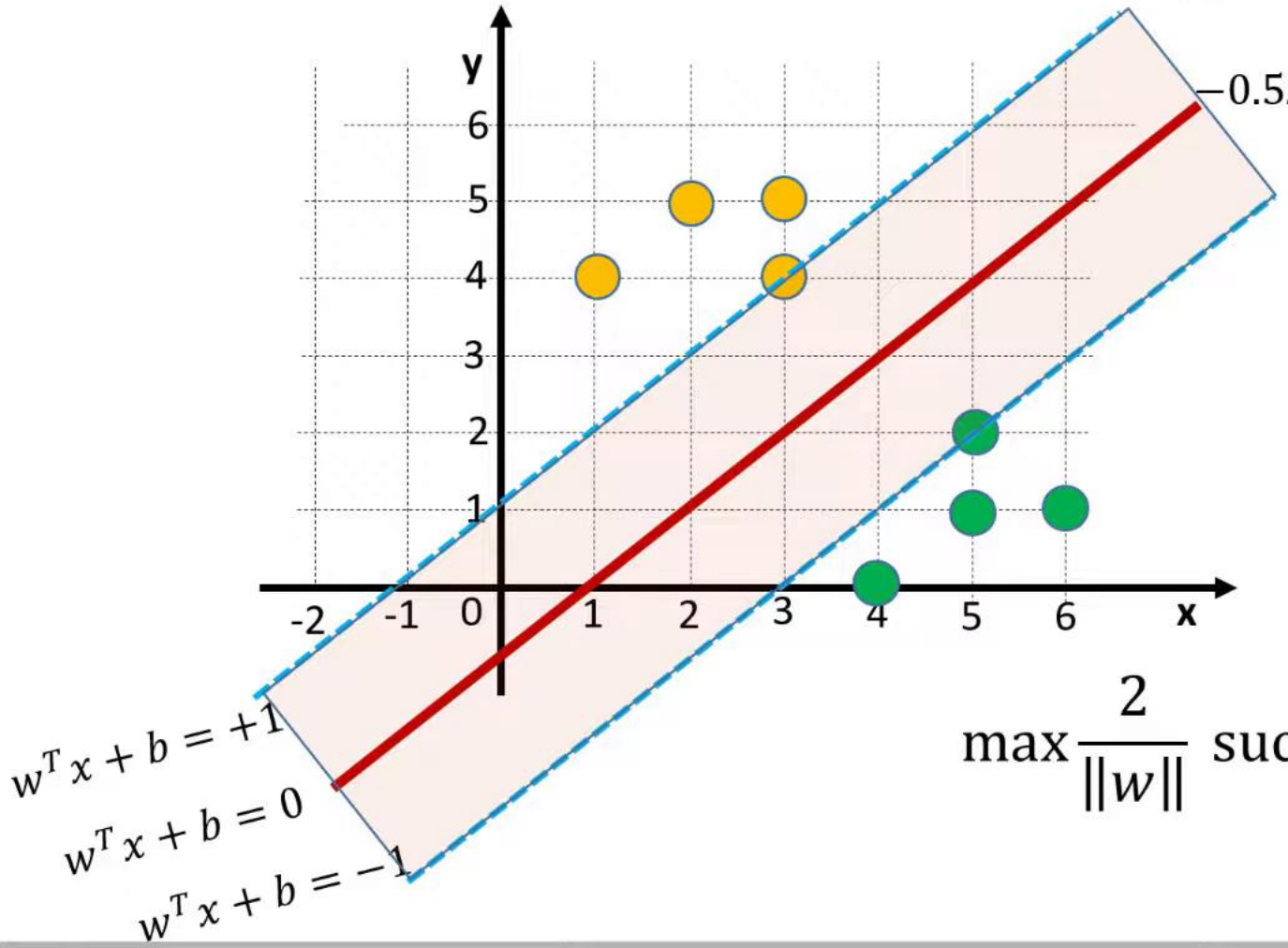
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{cases} \geq 1 & \text{if } Y_i = +1 \\ \leq -1 & \text{if } Y_i = -1 \end{cases}$$

that is equal to or greater than one,

SVM – the math



$b = 0.5$

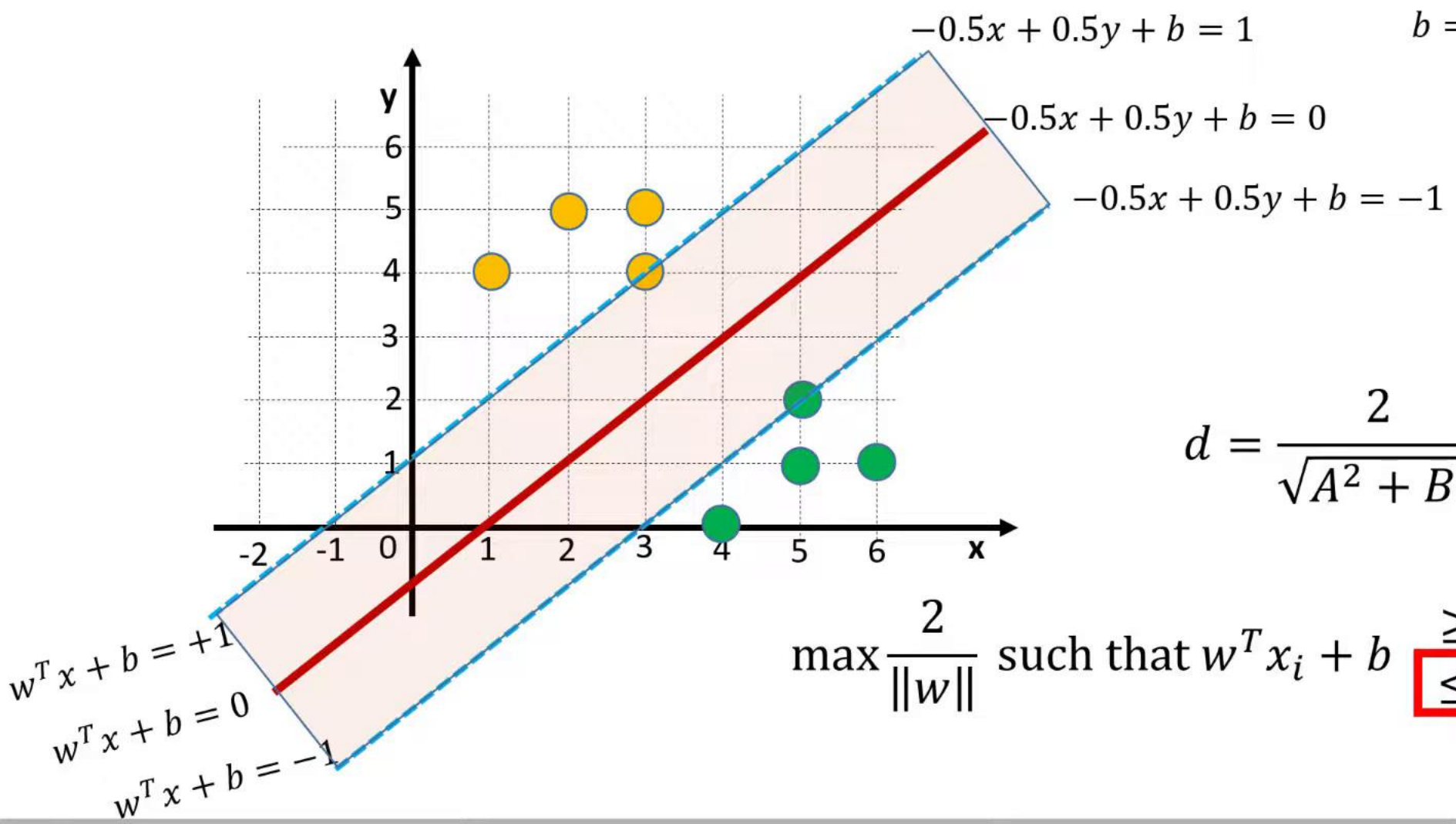
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{matrix} \geq 1 & \text{if } Y_i = +1 \\ \leq -1 & \text{if } Y_i = -1 \end{matrix}$$

and that the data points that belong to the negative group,

SVM – the math



$b = 0.5$

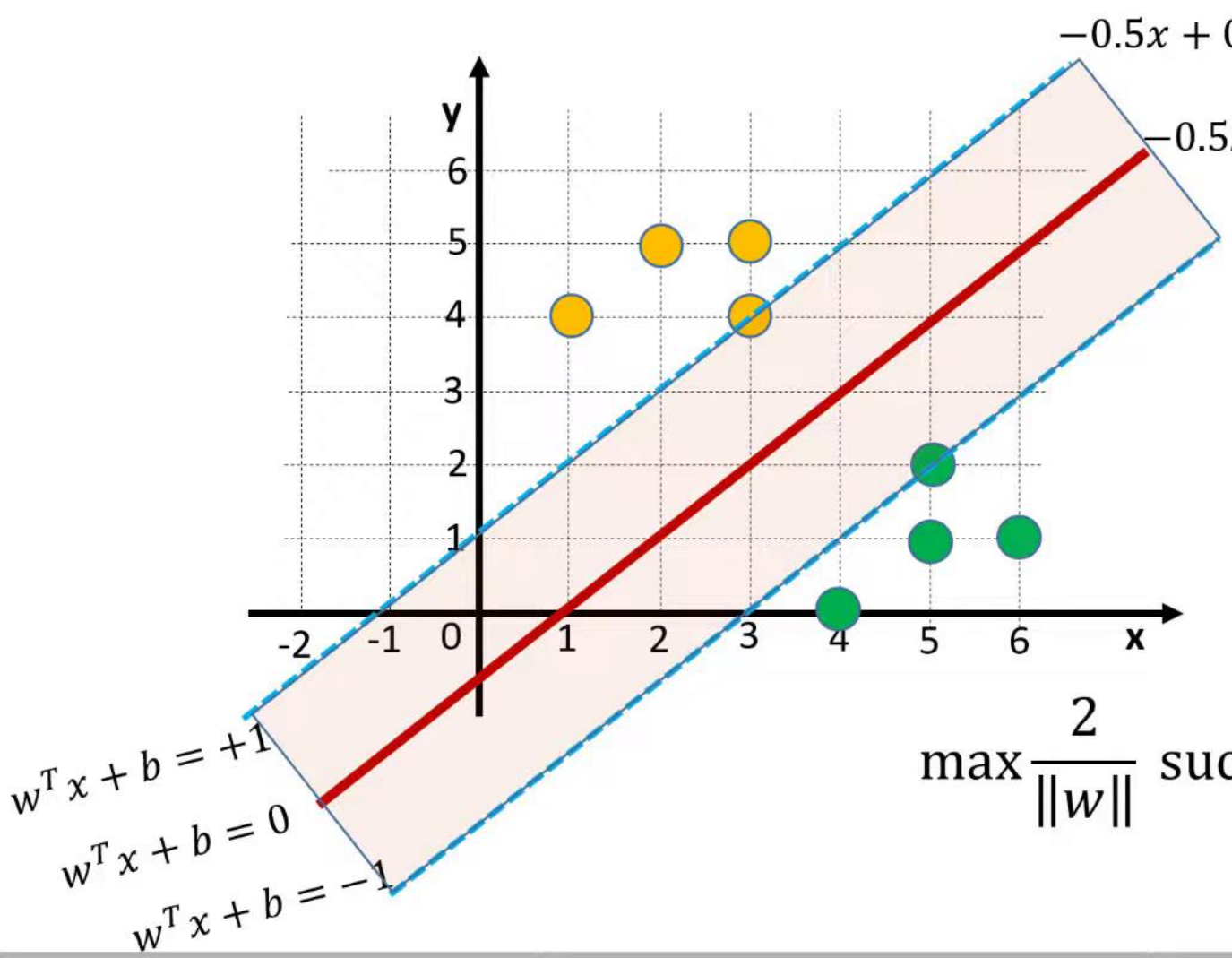
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{cases} \geq 1 & \text{if } Y_i = +1 \\ \leq -1 & \text{if } Y_i = -1 \end{cases}$$

results in a value that is equal to or less than negative one.

SVM – the math



$b = 0.5$

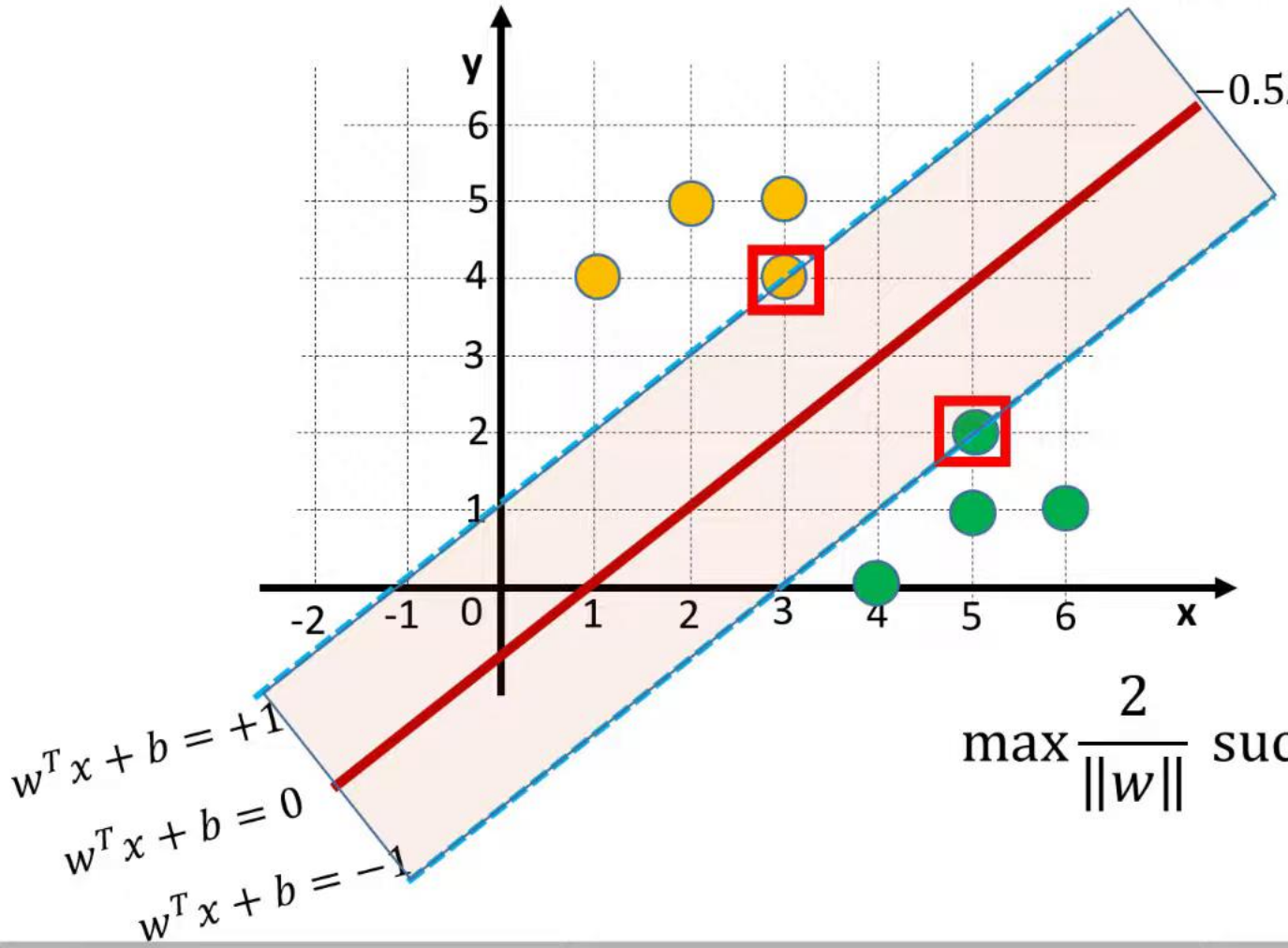
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{cases} \geq 1 & \text{if } Y_i = +1 \\ \leq -1 & \text{if } Y_i = -1 \end{cases}$$

results in a value that is equal to or less than negative one.

SVM – the math



$b = 0.5$

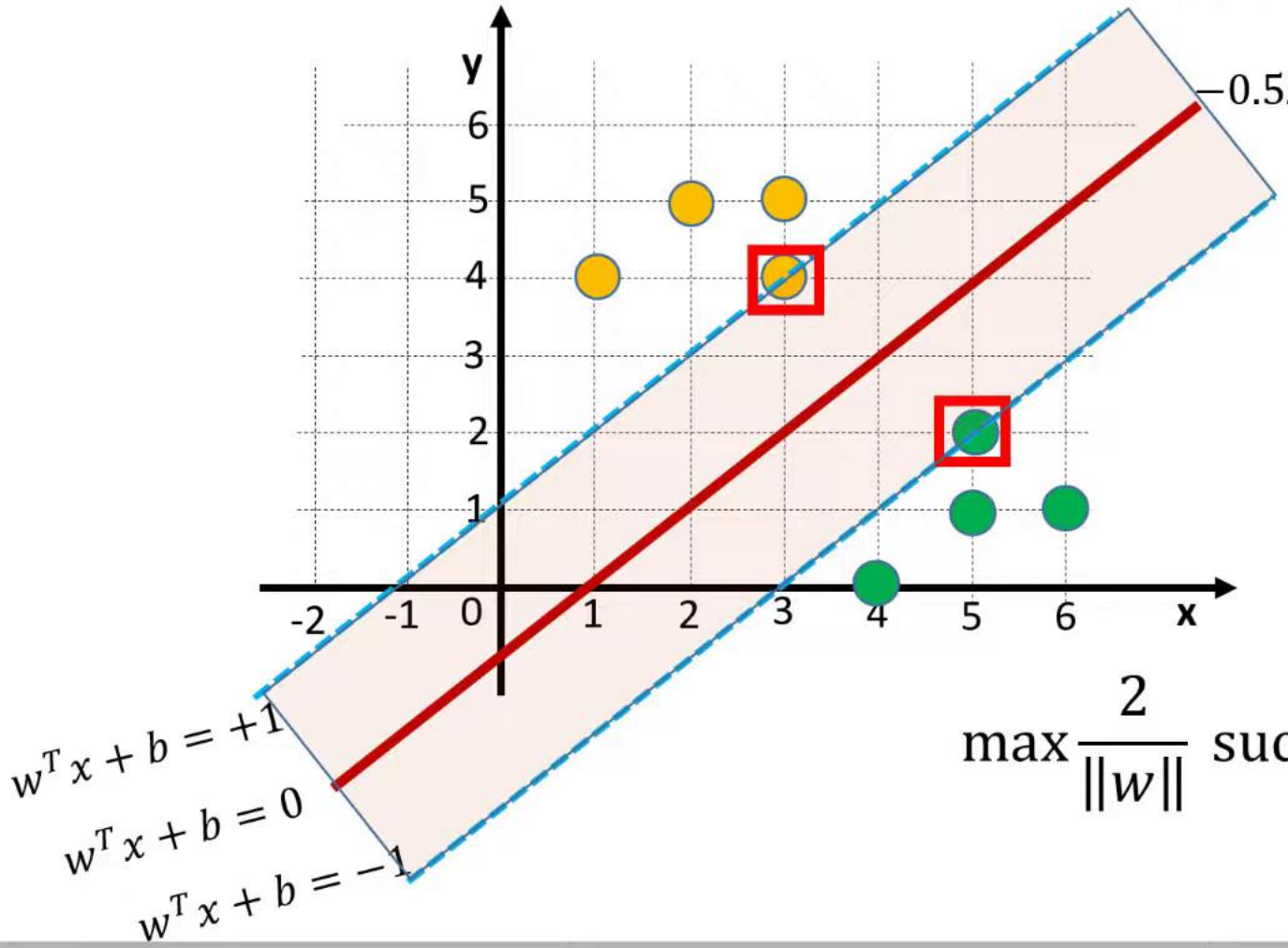
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{array}{l} \geq 1 \text{ if } Y_i = +1 \\ \leq -1 \text{ if } Y_i = -1 \end{array}$$

This means that the margin should not span beyond the data points closest to the hyperplane. Remember that these points are called support vectors.

SVM – the math



$b = 0.5$

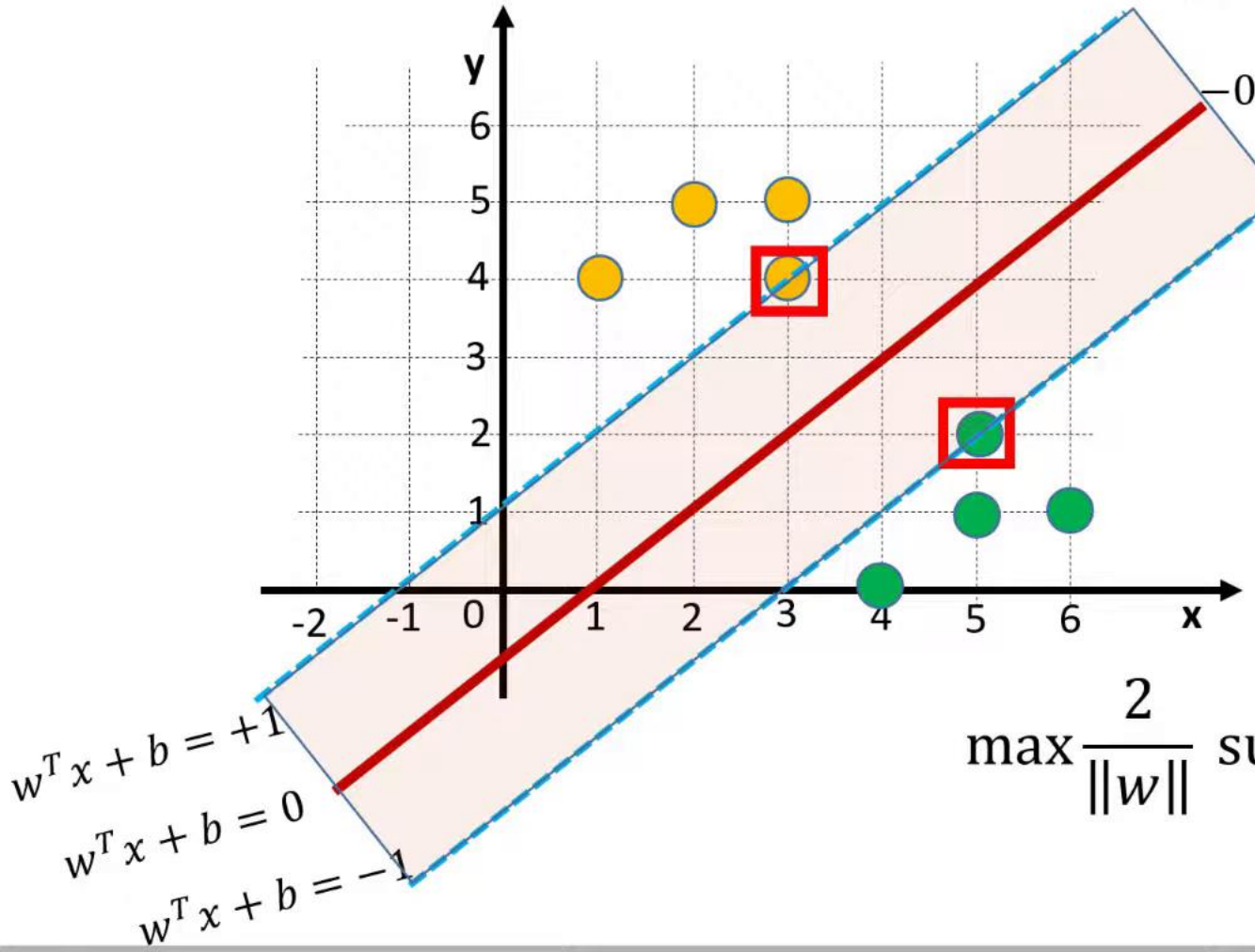
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{array}{l} \geq 1 \text{ if } Y_i = +1 \\ \leq -1 \text{ if } Y_i = -1 \end{array}$$

This means that the margin should not span beyond the data points closest to the hyperplane. Remember that these points are called support vectors.

SVM – the math



$b = 0.5$

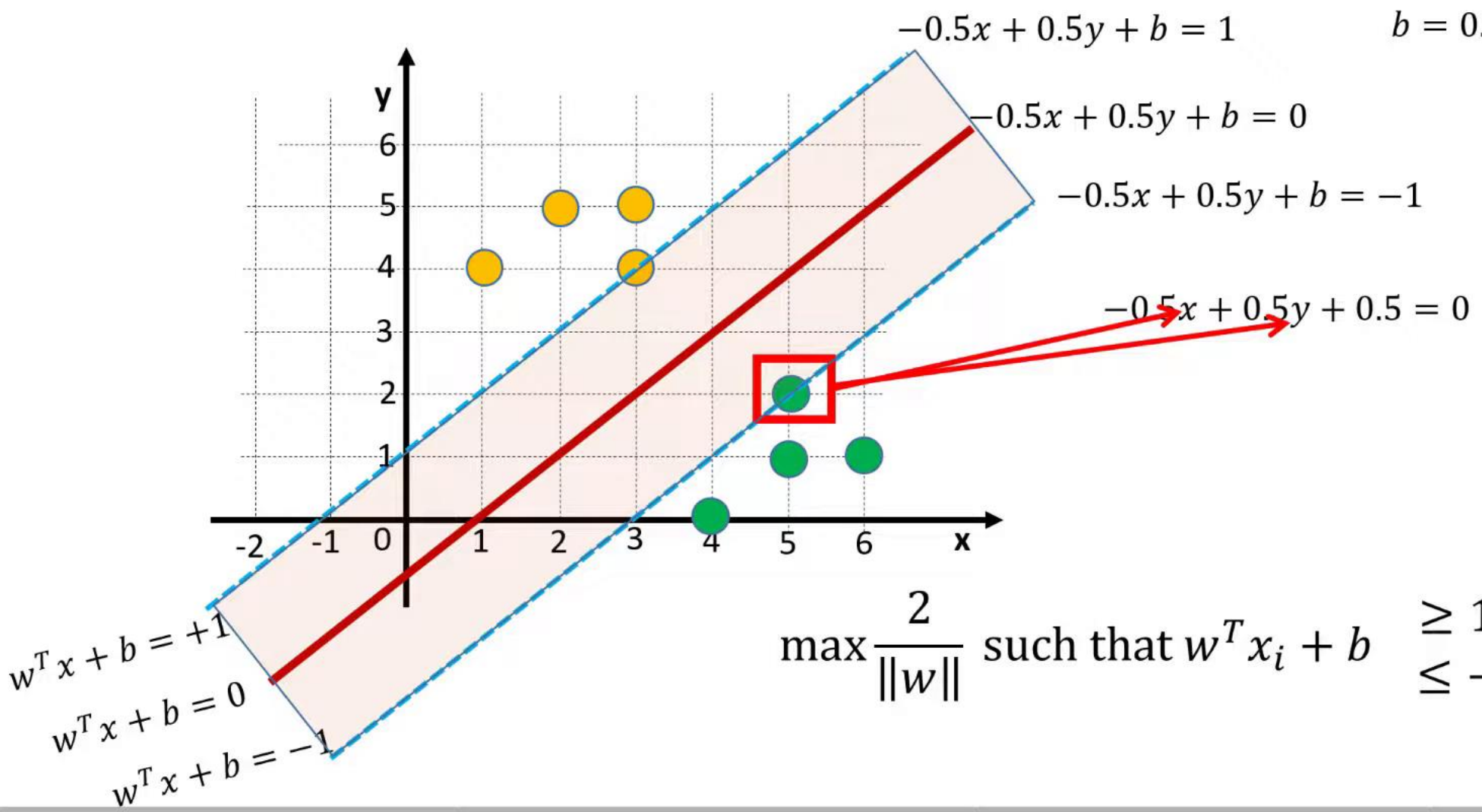
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \quad \begin{matrix} \geq 1 & \text{if } Y_i = +1 \\ \leq -1 & \text{if } Y_i = -1 \end{matrix}$$

This means that the margin should not span beyond the data points closest to the hyperplane. Remember that these points are called support vectors.

SVM – the math

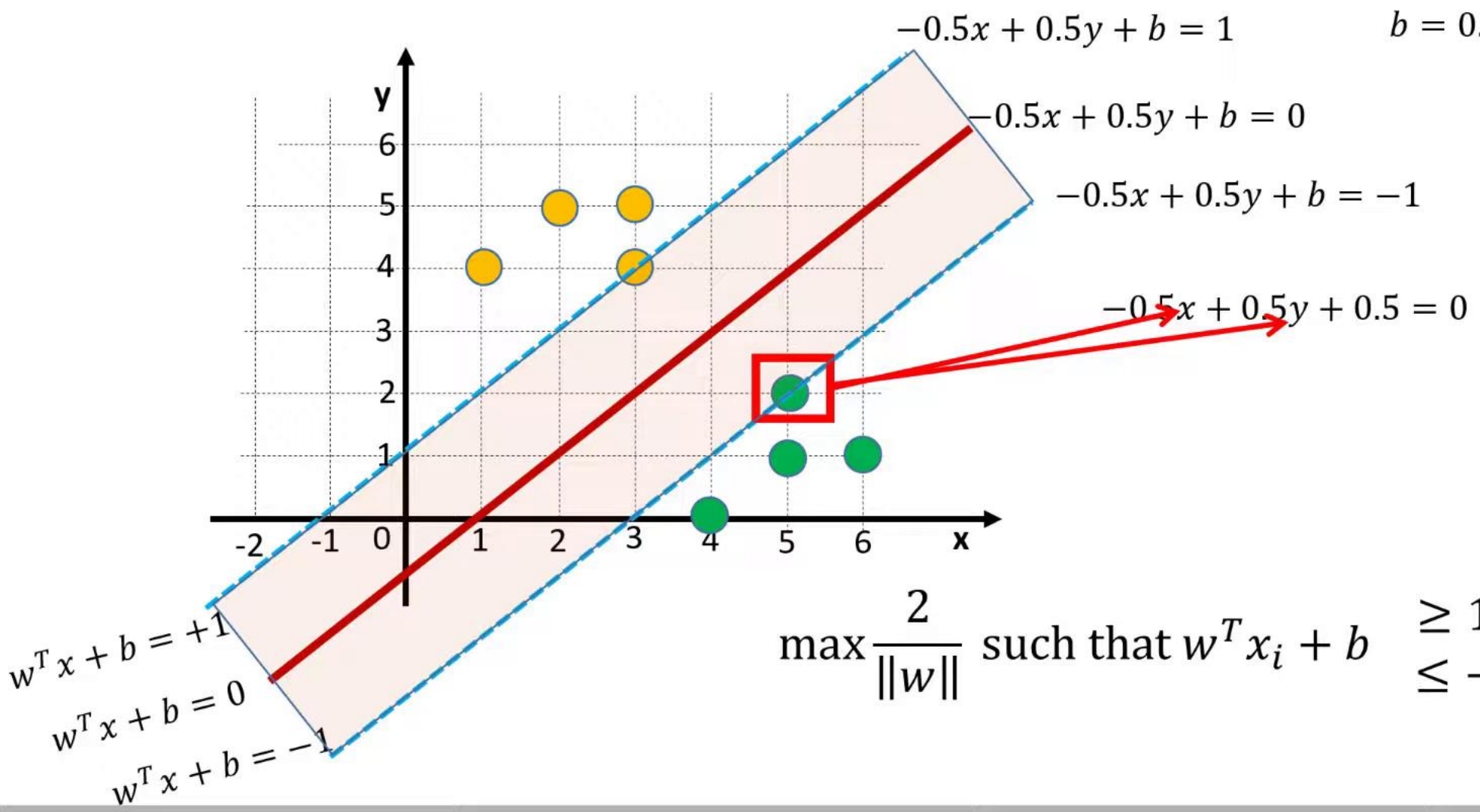


Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{array}{l} \geq 1 \text{ if } Y_i = +1 \\ \leq -1 \text{ if } Y_i = -1 \end{array}$$

Let's see what value we get if we plug in the x and y coordinates of this data point in the equation of our hyperplane.

SVM – the math

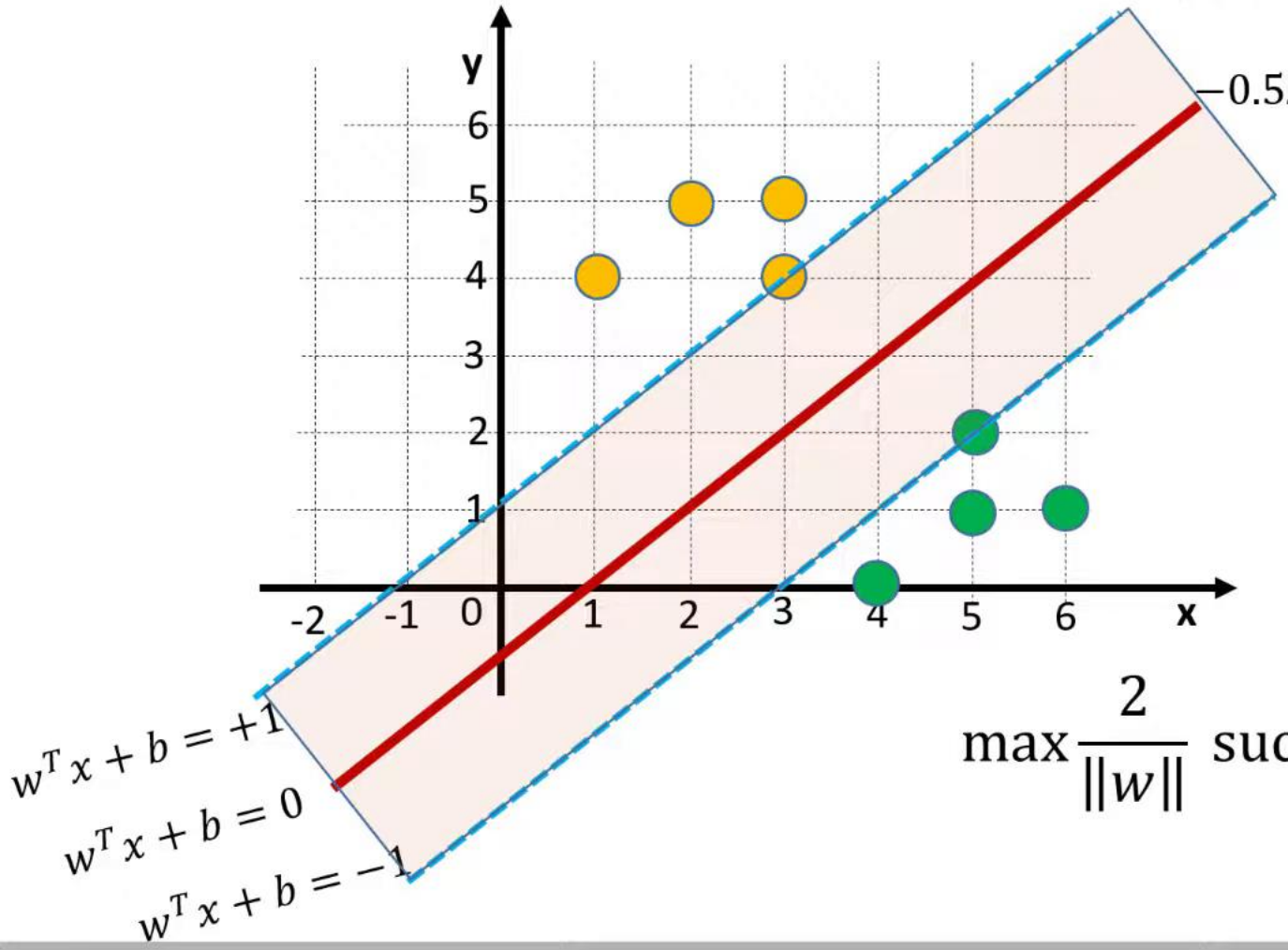


Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{array}{l} \geq 1 \text{ if } Y_i = +1 \\ \leq -1 \text{ if } Y_i = -1 \end{array}$$

Let's see what value we get if we plug in the x and y coordinates of this data point in the equation of our hyperplane.

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

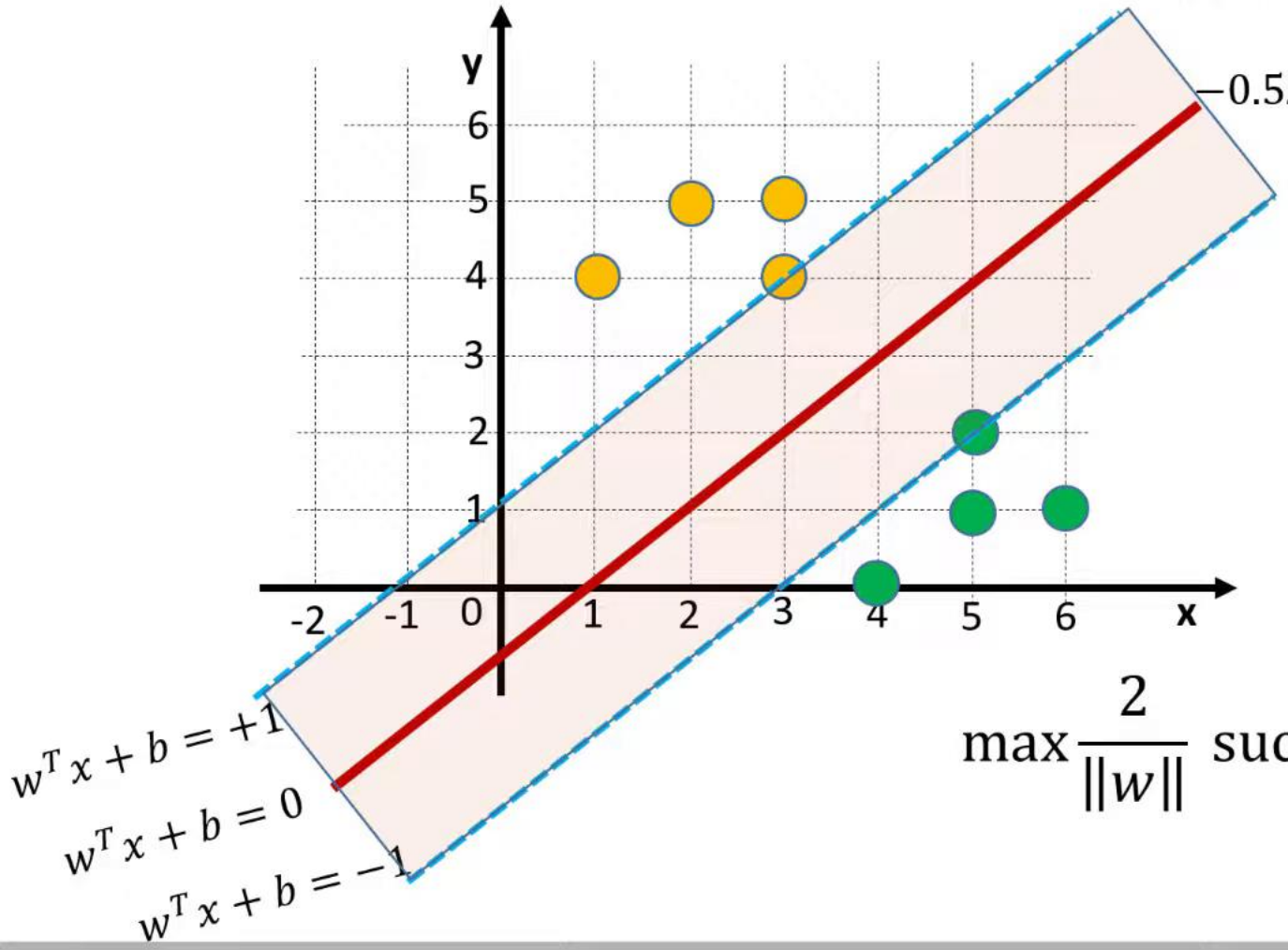
$-0.5x + 0.5y + 0.5 = 0$

$-0.5 \cdot 5 + 0.5 \cdot 2 + 0.5 = -1$

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{array}{l} \geq 1 \text{ if } Y_i = +1 \\ \leq -1 \text{ if } Y_i = -1 \end{array}$$

Since this point represents a support vector that is on the blue line below the hyperplane, it will result in a value of negative one.

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

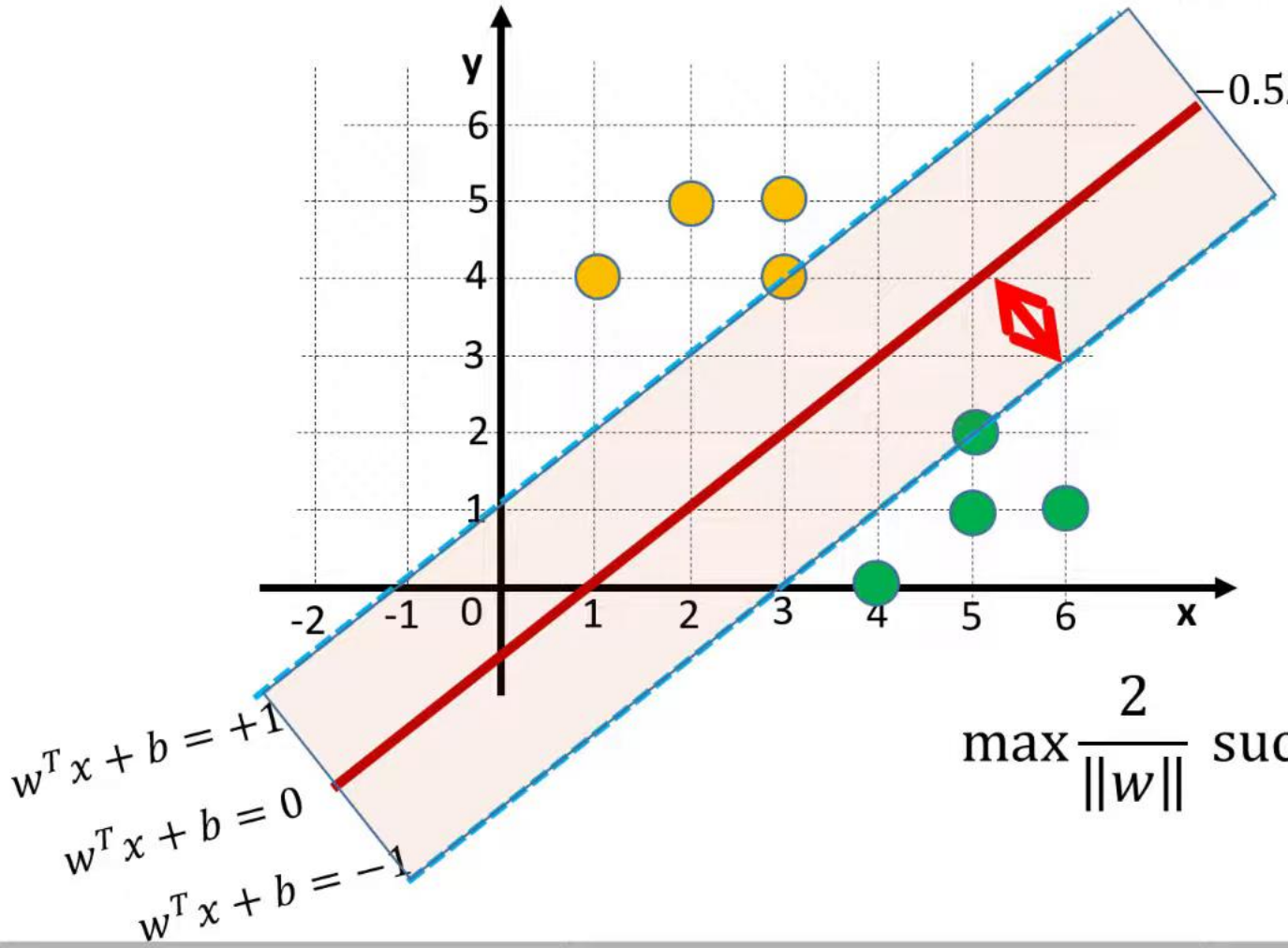
$-0.5x + 0.5y + 0.5 = 0$

$-0.5 \cdot 5 + 0.5 \cdot 2 + 0.5 = -1$

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{array}{l} \geq 1 \text{ if } Y_i = +1 \\ \leq -1 \text{ if } Y_i = -1 \end{array}$$

Since this point represents a support vector that is on the blue line below the hyperplane, it will result in a value of negative one.

SVM – the math



$-0.5x + 0.5y + b = 1$

$b = 0.5$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

$-0.5x + 0.5y + 0.5 = 0$

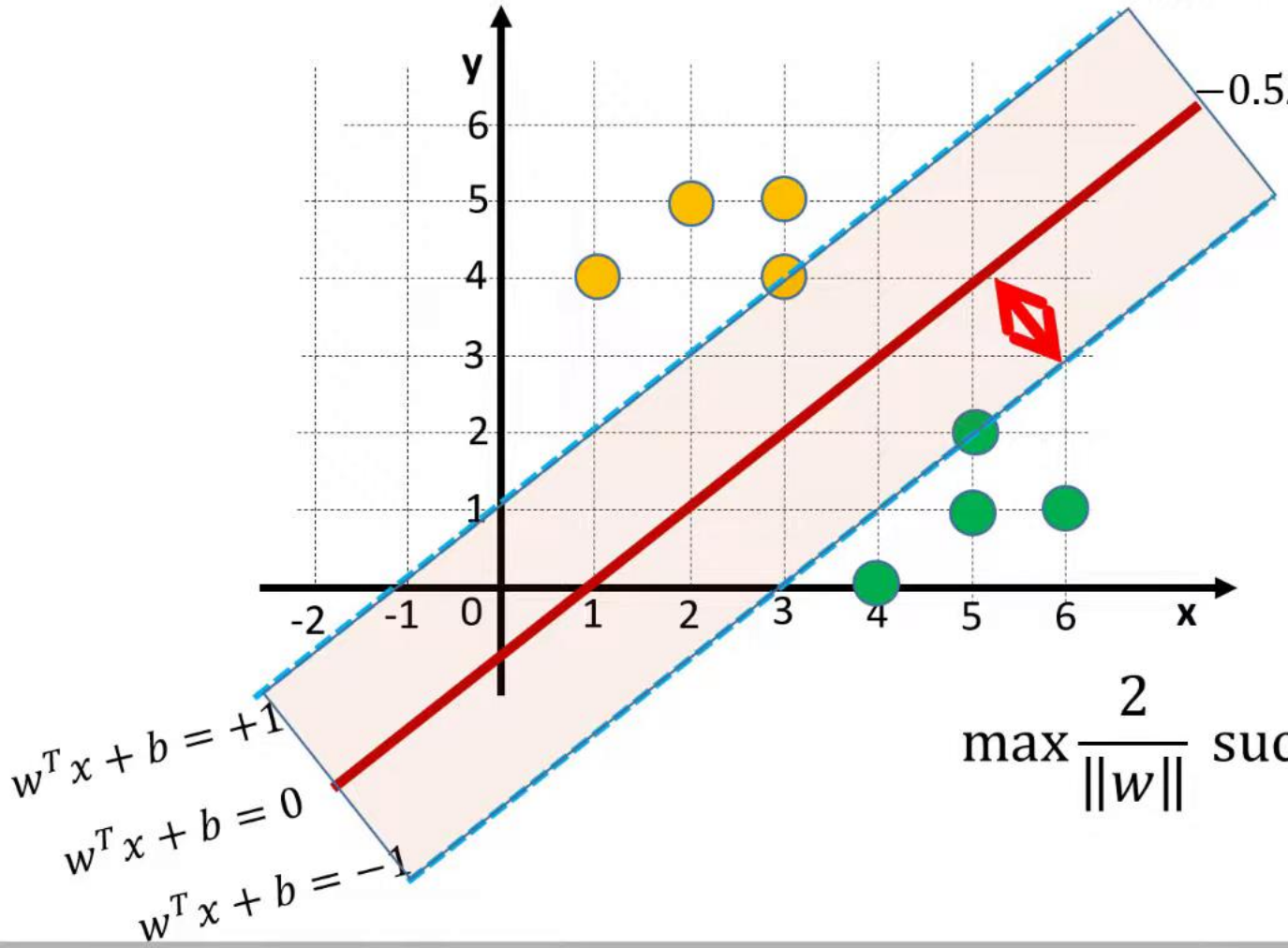
$-0.5 \cdot 5 + 0.5 \cdot 2 + 0.5 = -1$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{array}{l} \geq 1 \text{ if } Y_i = +1 \\ \leq -1 \text{ if } Y_i = -1 \end{array}$$

The width of the margin is therefore constrained so that it cannot span beyond the support vector if the hyperplane can separate the two groups completely.

SVM – the math



$-0.5x + 0.5y + b = 1$

$b = 0.5$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

$-0.5x + 0.5y + 0.5 = 0$

$-0.5 \cdot 5 + 0.5 \cdot 2 + 0.5 = -1$

Group (Y_i)	*x*	*y*
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{array}{l} \geq 1 \text{ if } Y_i = +1 \\ \leq -1 \text{ if } Y_i = -1 \end{array}$$

The width of the margin is therefore constrained so that it cannot span beyond the support vector if the hyperplane can separate the two groups completely.

SVM – the math

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$b = 0.5$

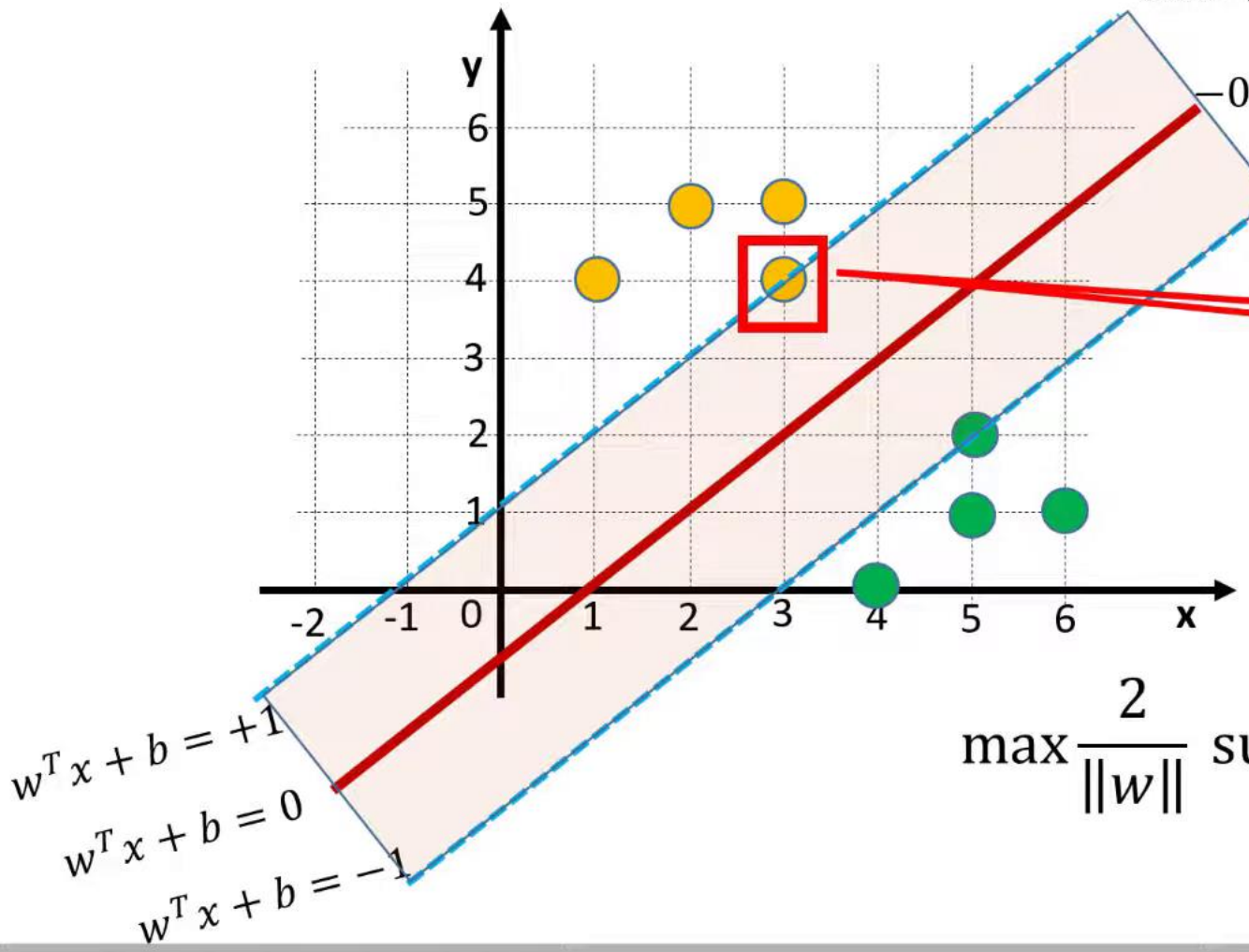
$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

$-0.5x + 0.5y + 0.5 = 0$

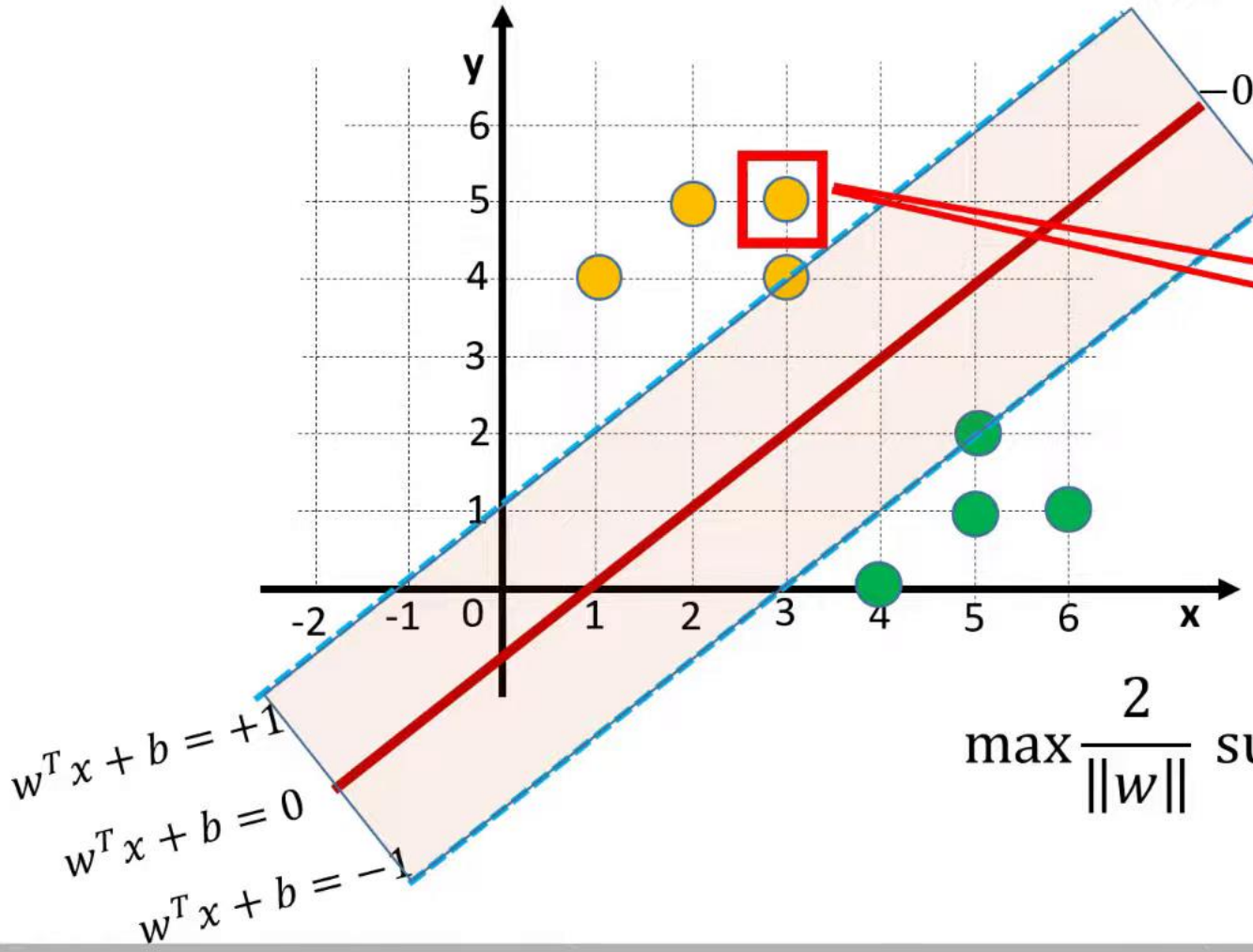
$-0.5 \cdot 3 + 0.5 \cdot 4 + 0.5 = 1$



$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{array}{l} \geq 1 \text{ if } Y_i = +1 \\ \leq -1 \text{ if } Y_i = -1 \end{array}$$

Similarly, the line that defines the boundary above our hyperplane cannot be further away because if we plug in the x and y coordinates of this support vector in the equation of the hyperplane, that results in a value that is equal to positive one.

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

$b = 0.5$

$-0.5x + 0.5y + 0.5 = 0$

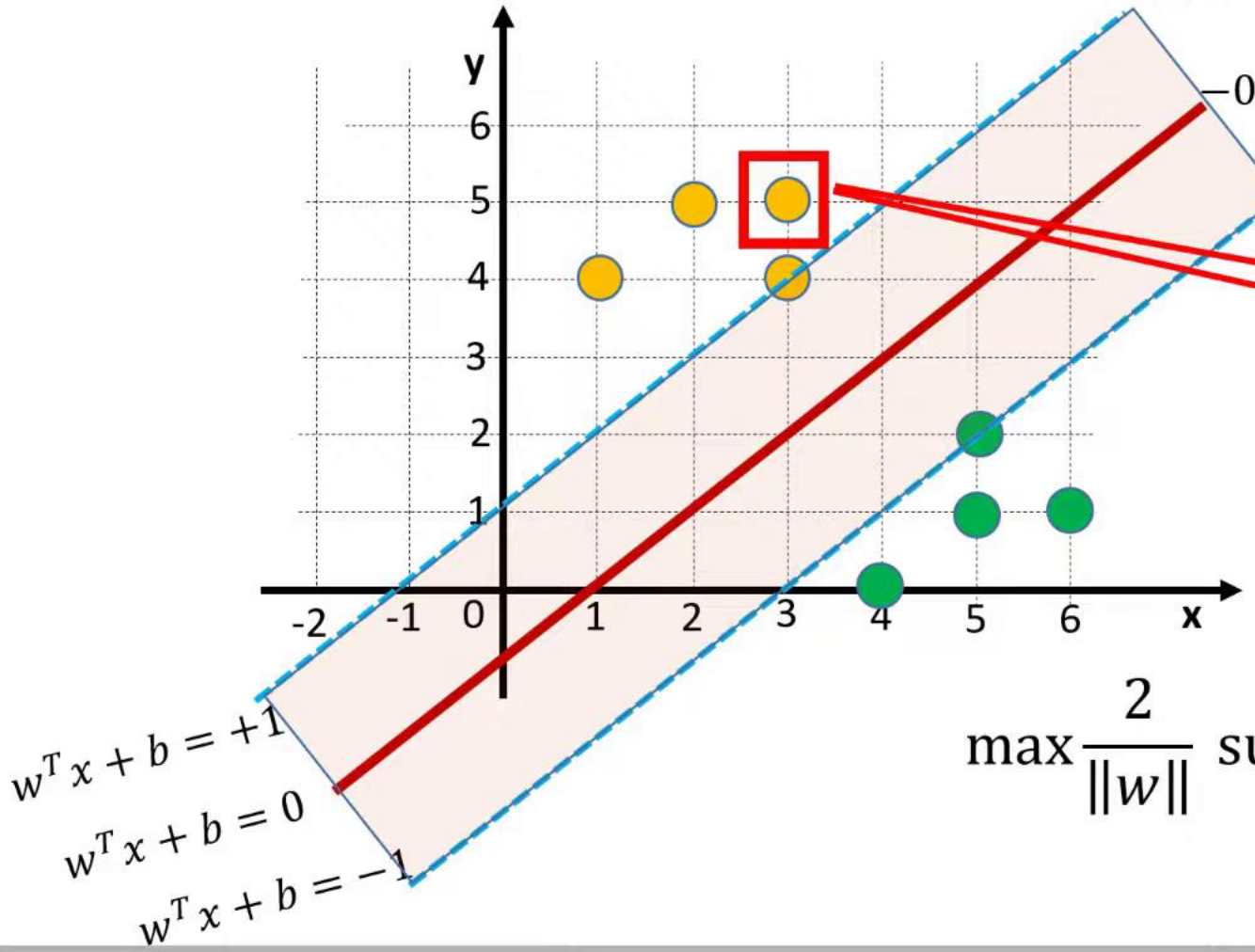
$-0.5 \cdot 3 + 0.5 \cdot 5 + 0.5 = 1.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{array}{l} \geq 1 \text{ if } Y_i = +1 \\ \leq -1 \text{ if } Y_i = -1 \end{array}$$

A data point that is further away will result in a value that is larger than positive one,

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

$-0.5x + 0.5y + 0.5 = 0$

$b = 0.5$

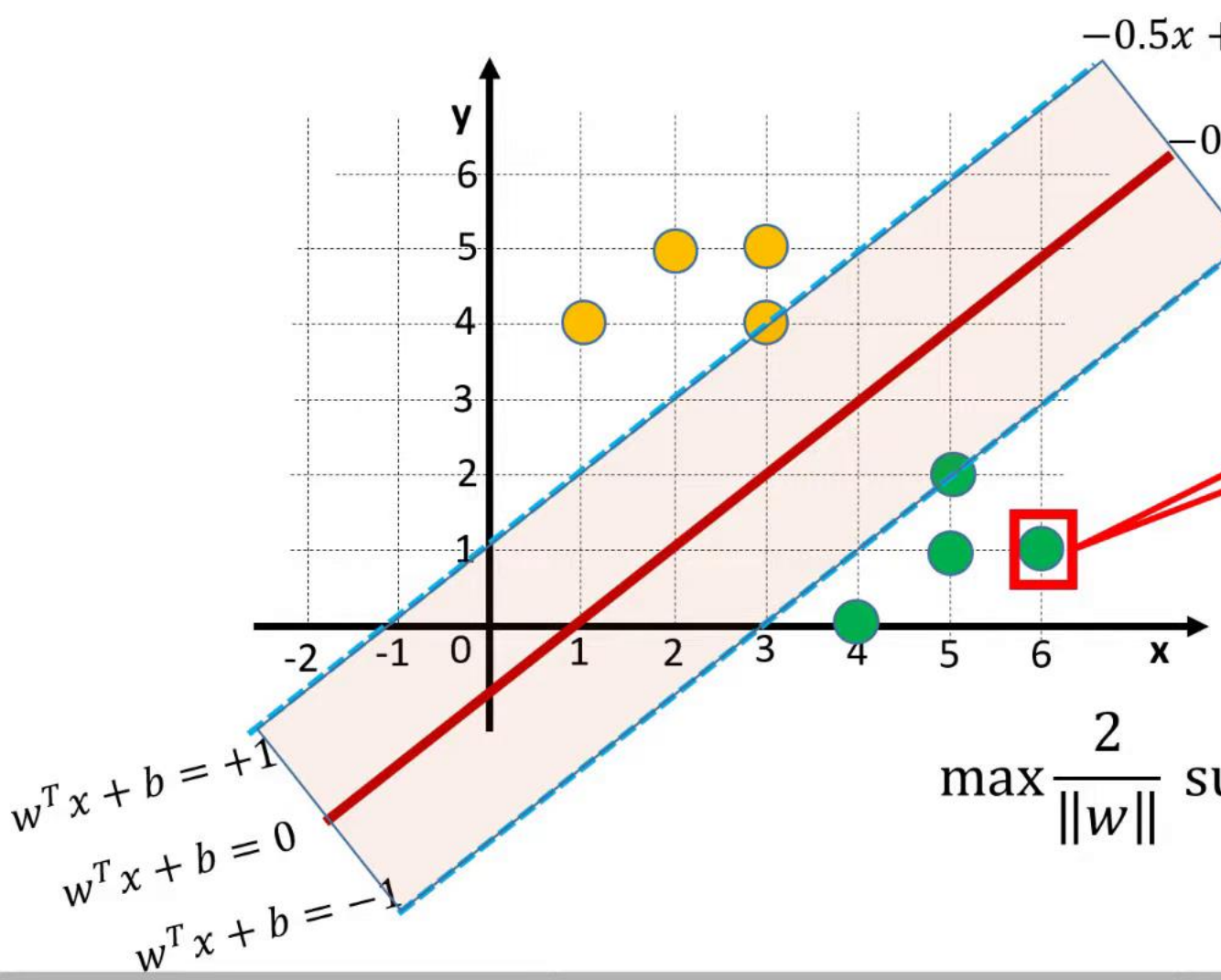
$-0.5 \cdot 3 + 0.5 \cdot 5 + 0.5 = 1.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{array}{l} \geq 1 \text{ if } Y_i = +1 \\ \leq -1 \text{ if } Y_i = -1 \end{array}$$

A data point that is further away will result in a value that is larger than positive one,

SVM – the math



$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

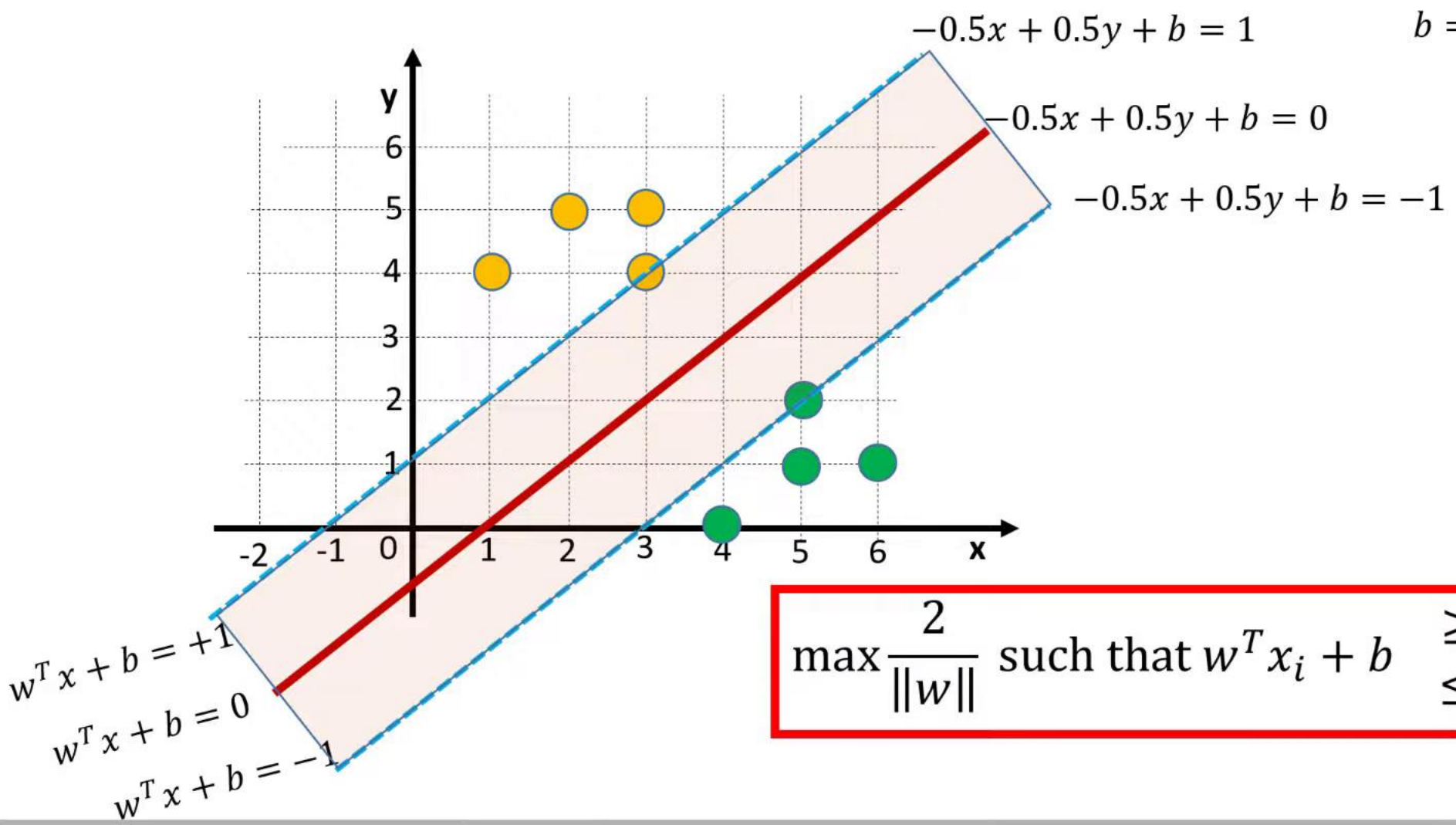
$-0.5x + 0.5y + 0.5 = 0$

$-0.5 \cdot 6 + 0.5 \cdot 1 + 0.5 = -2$

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{matrix} \geq 1 & \text{if } Y_i = +1 \\ \leq -1 & \text{if } Y_i = -1 \end{matrix}$$

or smaller than negative one.

SVM – the math



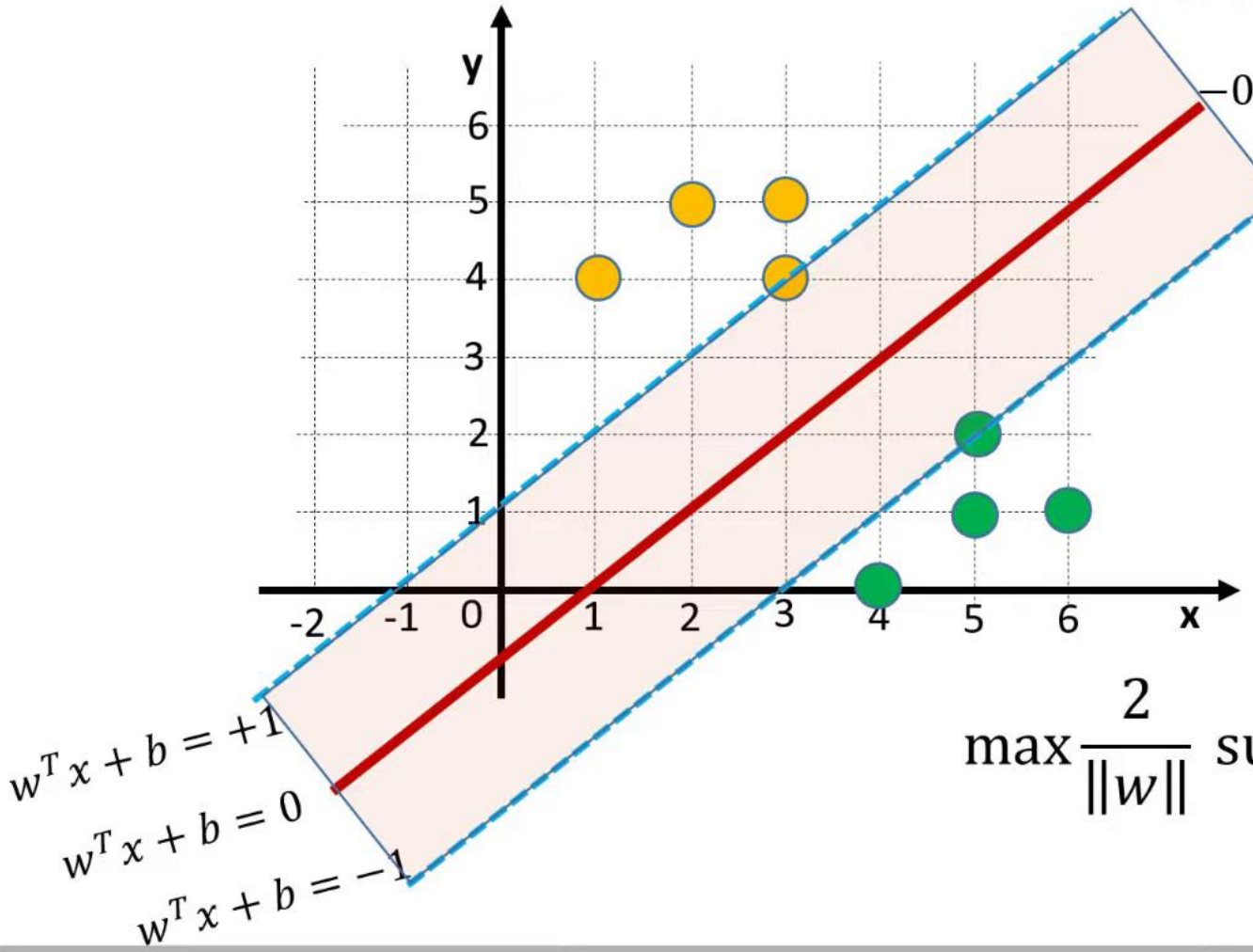
$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{array}{l} \geq 1 \text{ if } Y_i = +1 \\ \leq -1 \text{ if } Y_i = -1 \end{array}$$

We can simplify this equation a bit,

SVM – the math



$b = 0.5$

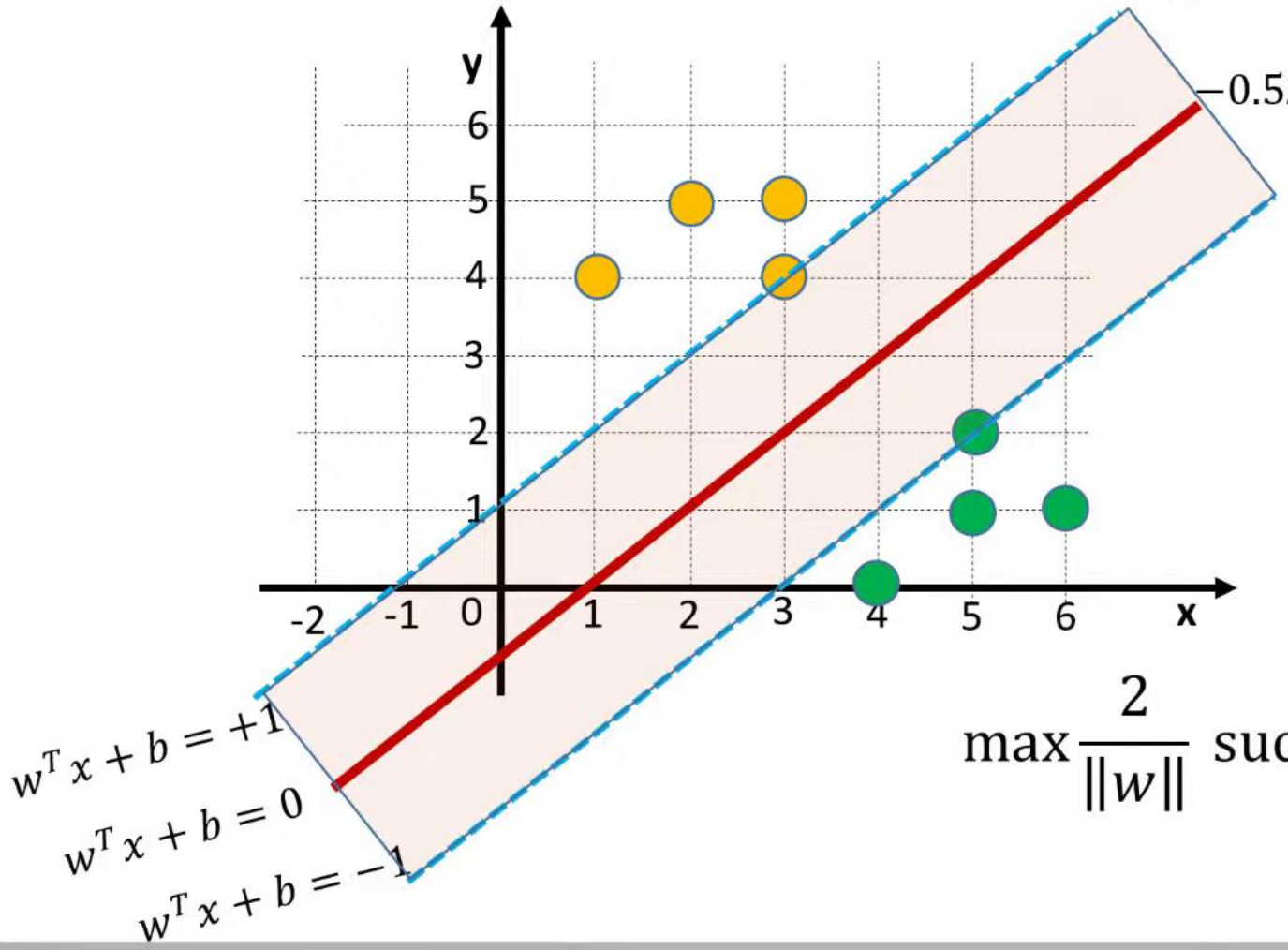
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y_i = [\,1, 1, 1, 1, -1, -1, -1, -1]$$

$$\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \begin{cases} \geq 1 & \text{if } Y_i = +1 \\ \leq -1 & \text{if } Y_i = -1 \end{cases}$$

if we multiply the equation by Y-sub-i,

SVM – the math



$b = 0.5$

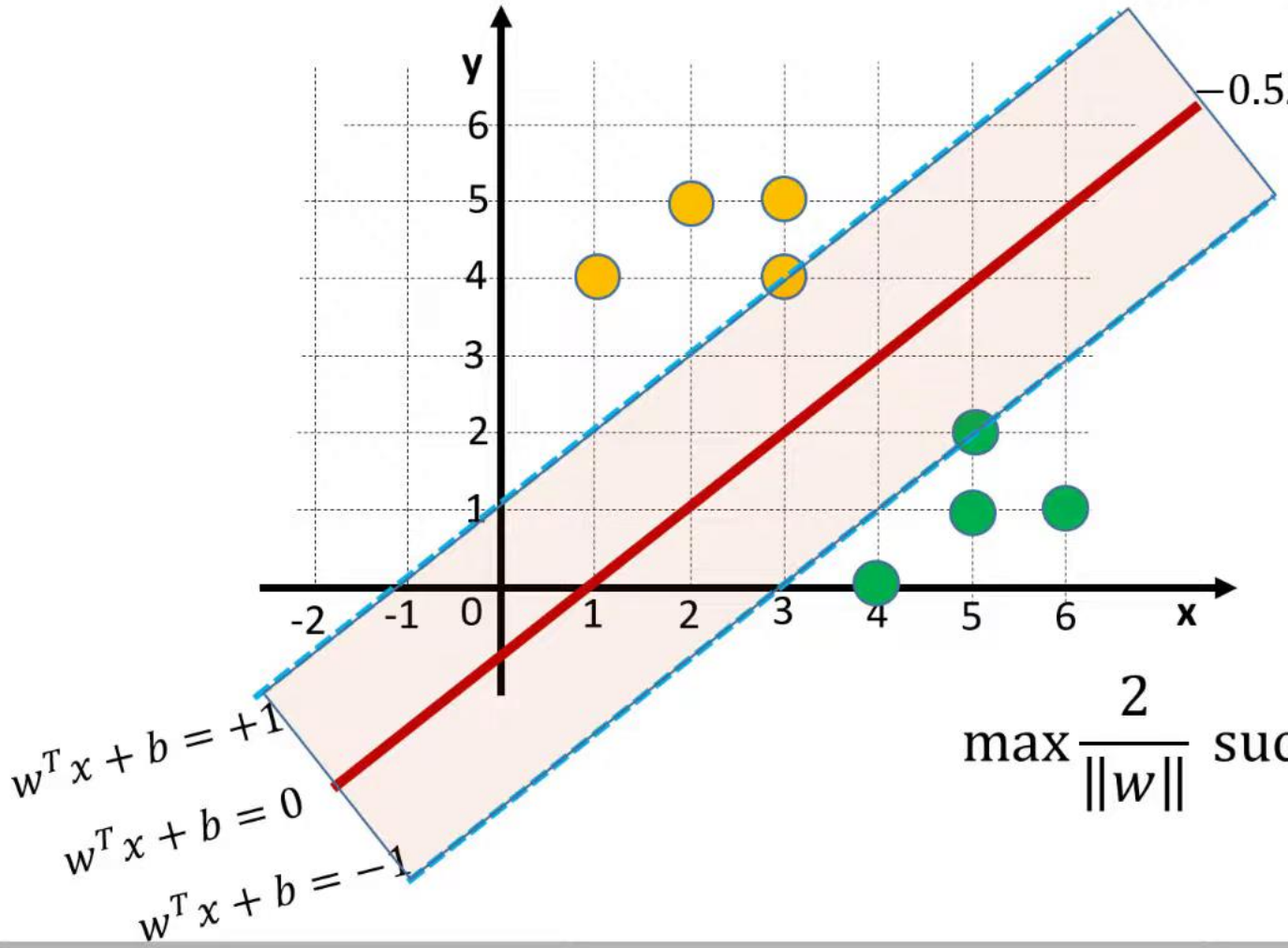
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y_i = [\,1, 1, 1, 1, -1, -1, -1, -1]$$

$$\max \frac{2}{\|w\|} \text{ such that } Y_i(w^T x_i + b) \geq 1$$

because Y-sub-i is equal to one if the data point belongs to the positive group, and equal to negative one if the data point belongs to the negative group.

SVM – the math



$b = 0.5$

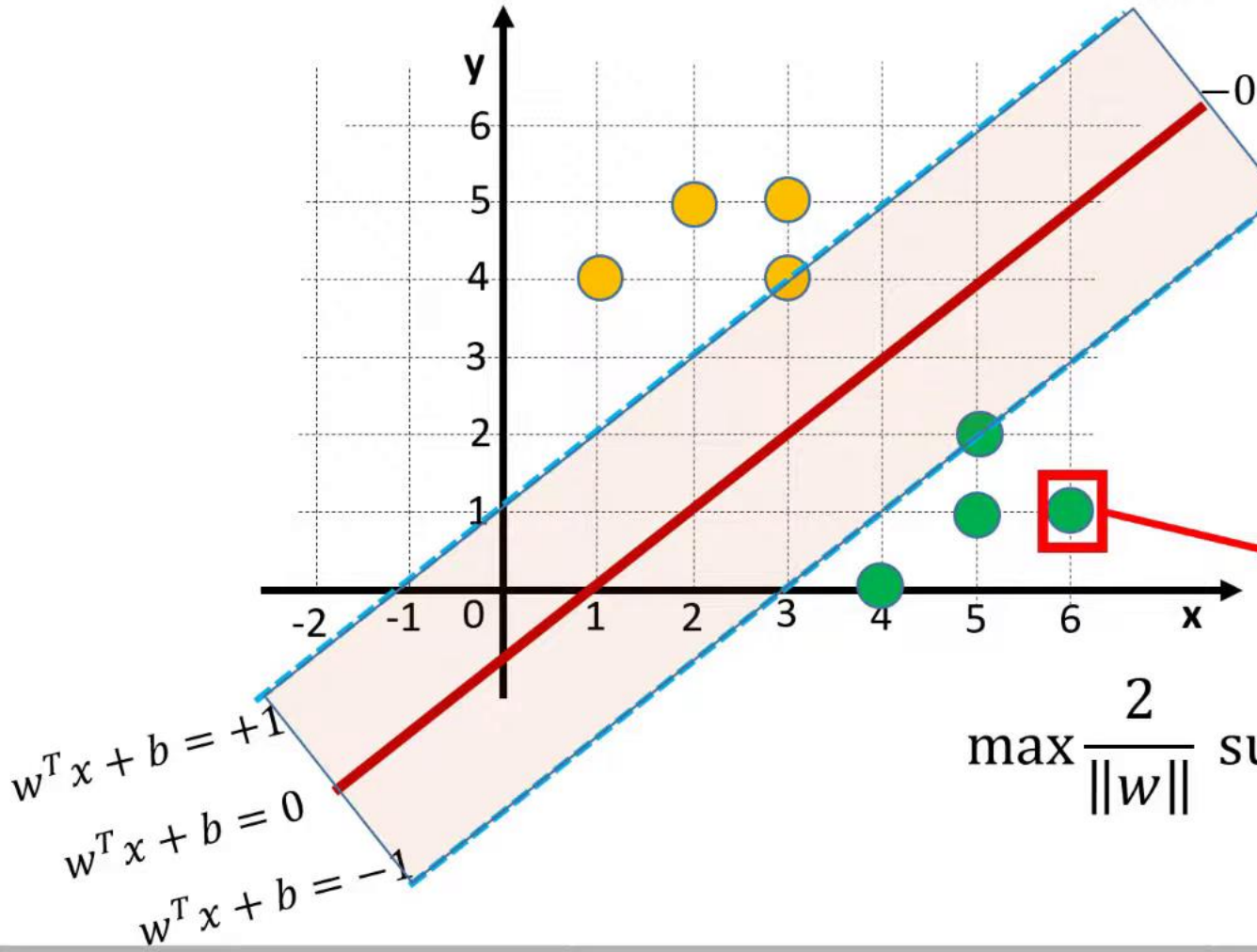
Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y_i = [\,1, 1, 1, 1, -1, -1, -1, -1]$$

$$\max \frac{2}{\|w\|} \text{ such that } Y_i(w^T x_i + b) \geq 1$$

because Y-sub-i is equal to one if the data point belongs to the positive group, and equal to negative one if the data point belongs to the negative group.

SVM – the math



$-0.5x + 0.5y + b = 1$

$-0.5x + 0.5y + b = 0$

$-0.5x + 0.5y + b = -1$

$b = 0.5$

Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

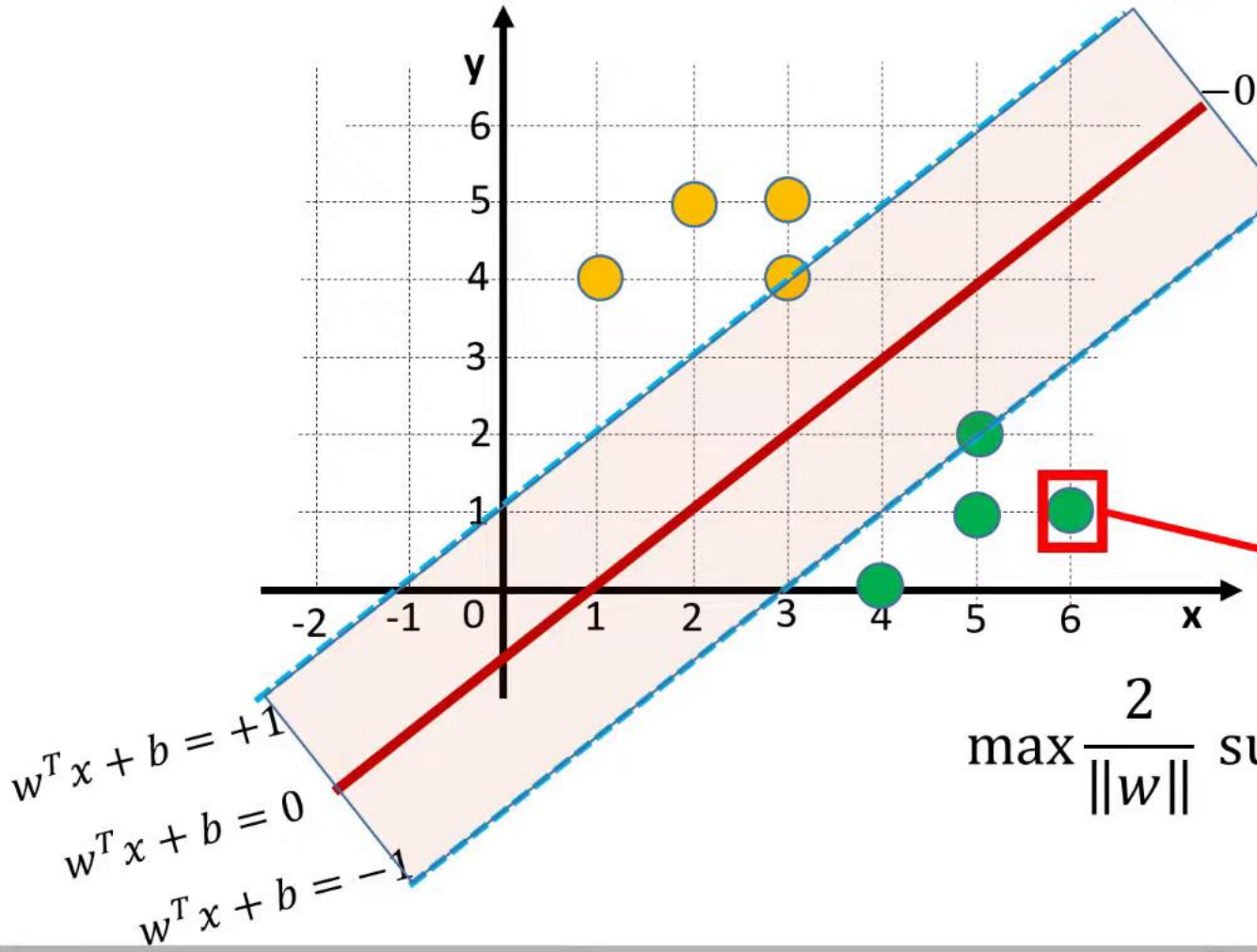
$Y_i = [\,1, 1, 1, 1, -1, -1, -1, -1]$

$(-1) \cdot (-0.5 \cdot 6 + 0.5 \cdot 1 + 0.5) = 2$

$$\max \frac{2}{\|w\|} \text{ such that } Y_i(w^T x_i + b) \geq 1$$

For example, this data point will now result in a positive value,

SVM – the math



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B (-1)	5	2
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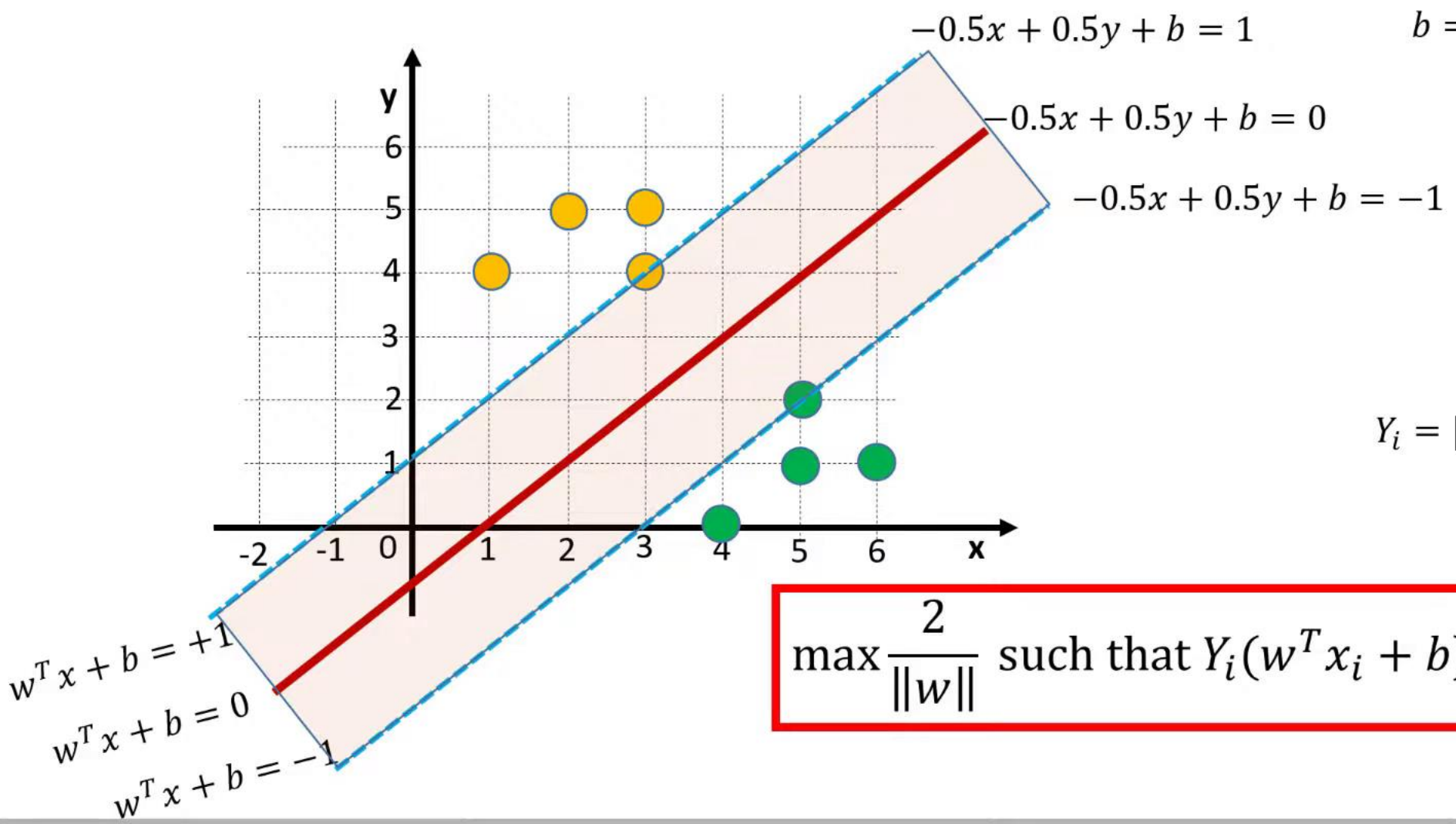

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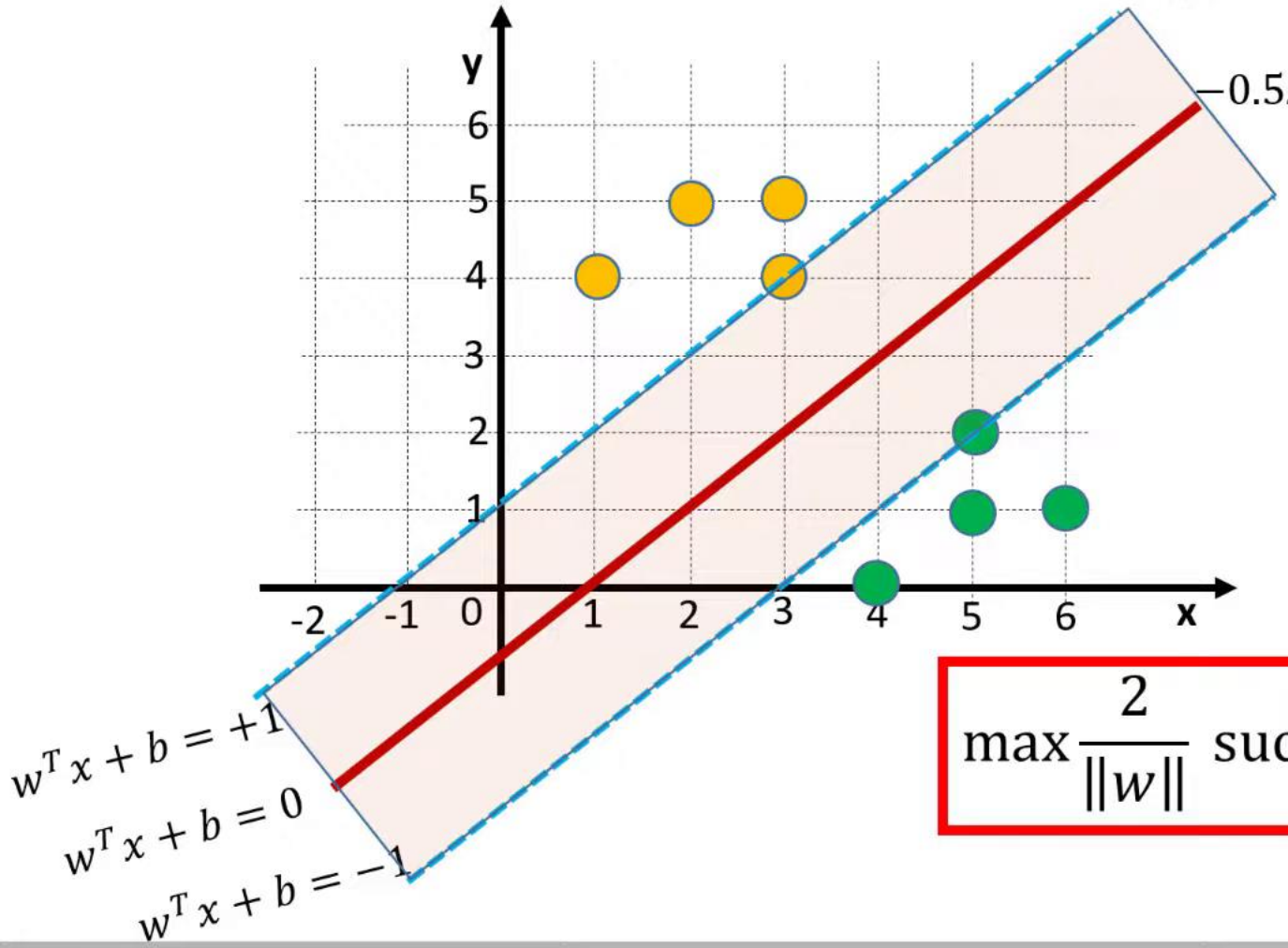
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

$$Y_i = [\,1, 1, 1, 1, -1, -1, -1, -1]$$

$$\max \frac{2}{\|w\|} \text{ such that } Y_i(w^T x_i + b) \geq 1$$

Note that this type of problem is applicable only when the data can be fully separated by the hyperplane.

SVM – the math



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$$Y_i = [\,1, 1, 1, 1, -1, -1, -1, -1]$$

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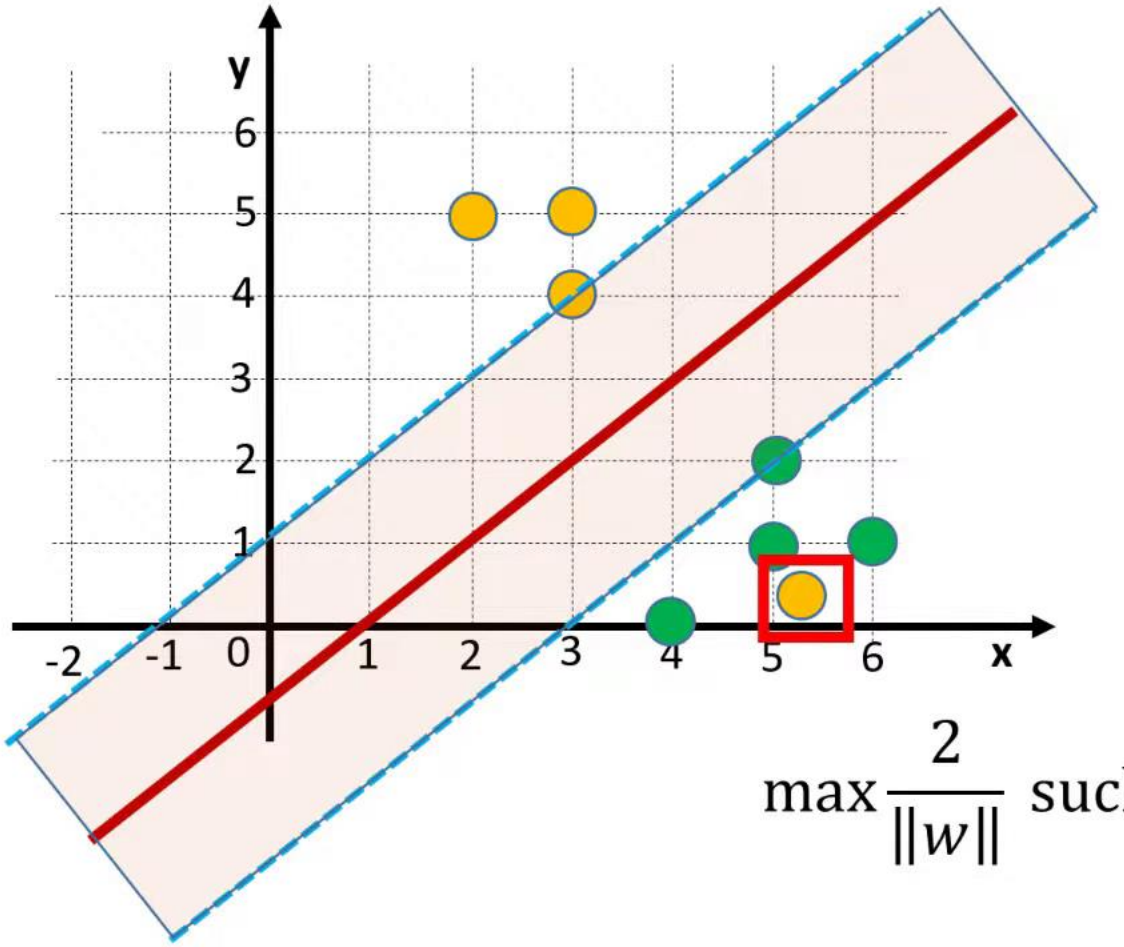
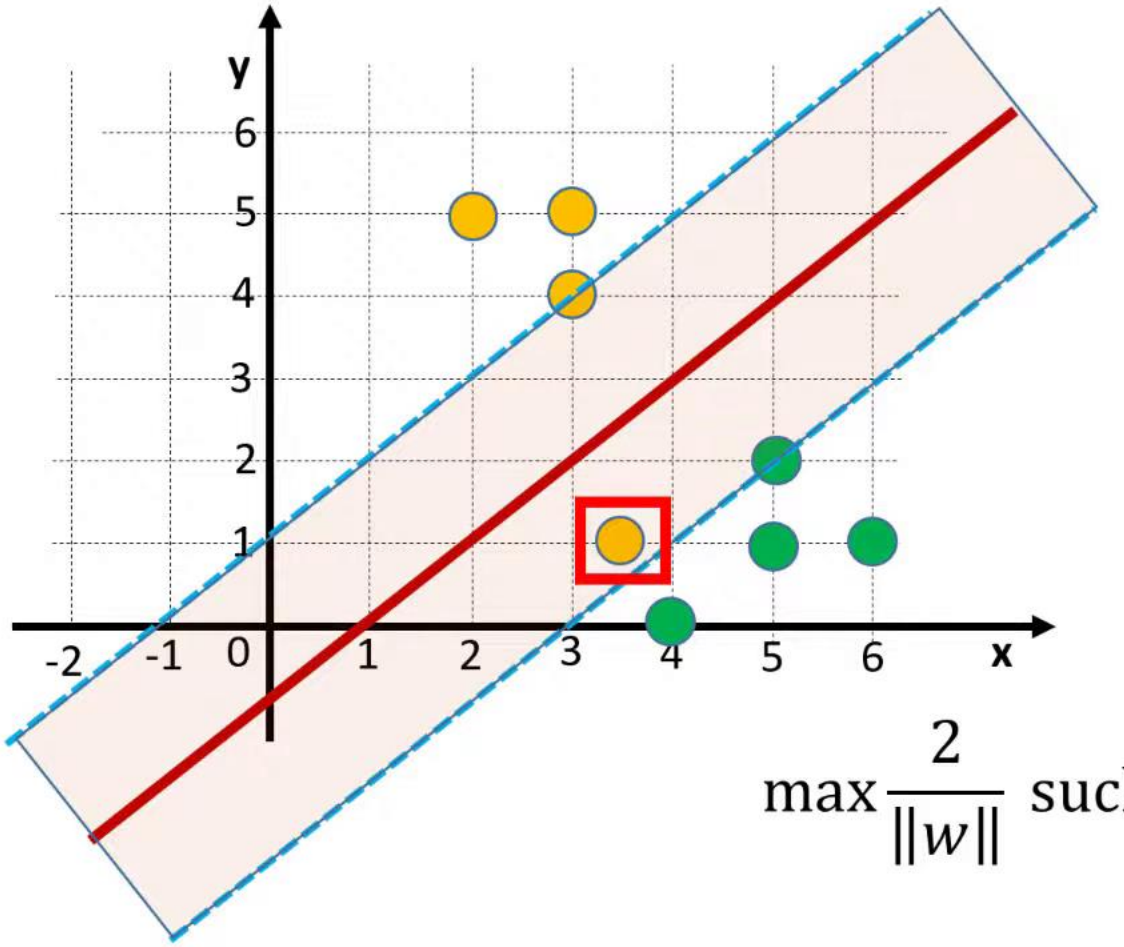
SVM – the math



$$\max \frac{2}{\|w\|} \text{ such that } Y_i(w^T x_i + b) \geq 1$$

What should we do if a yellow data point is down here?

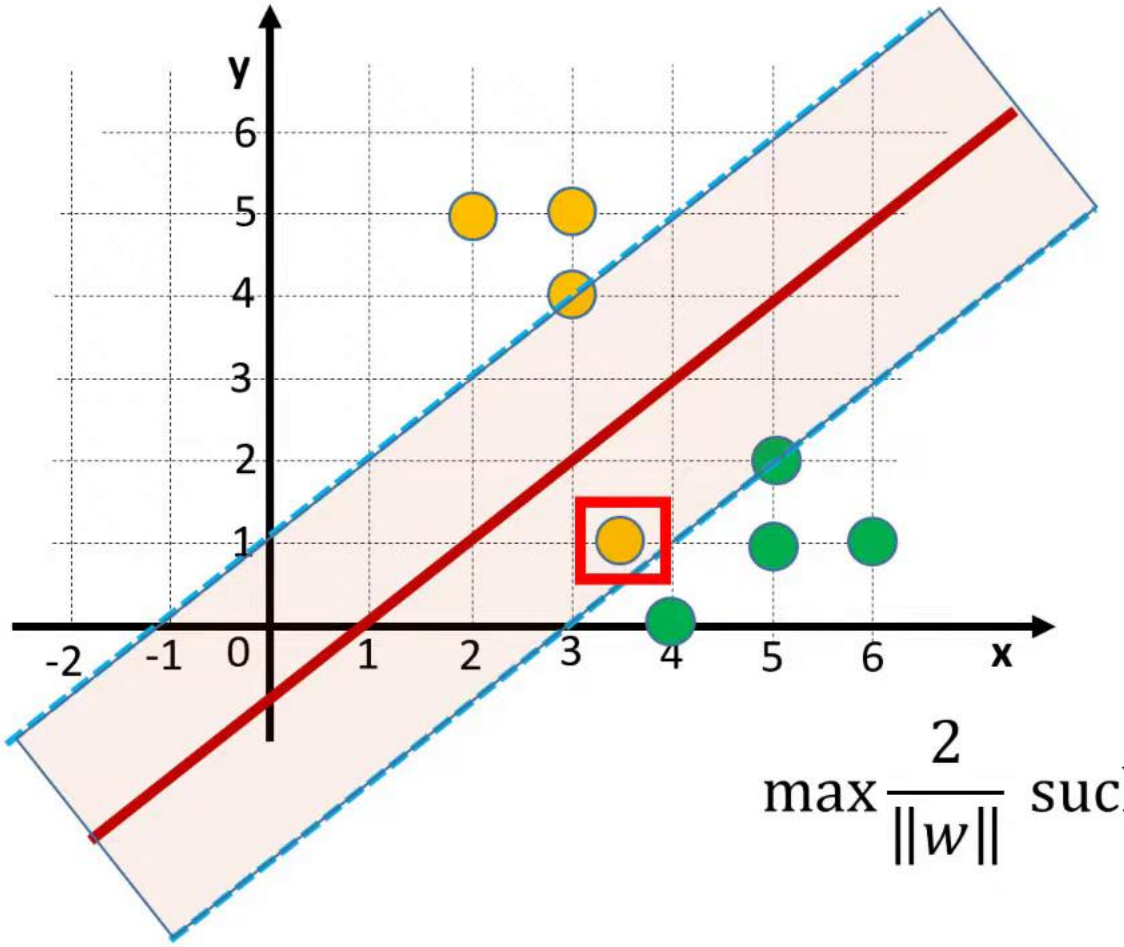
SVM – the math



$$\max \frac{2}{\|w\|} \text{ such that } Y_i(w^T x_i + b) \geq 1$$

Or here?

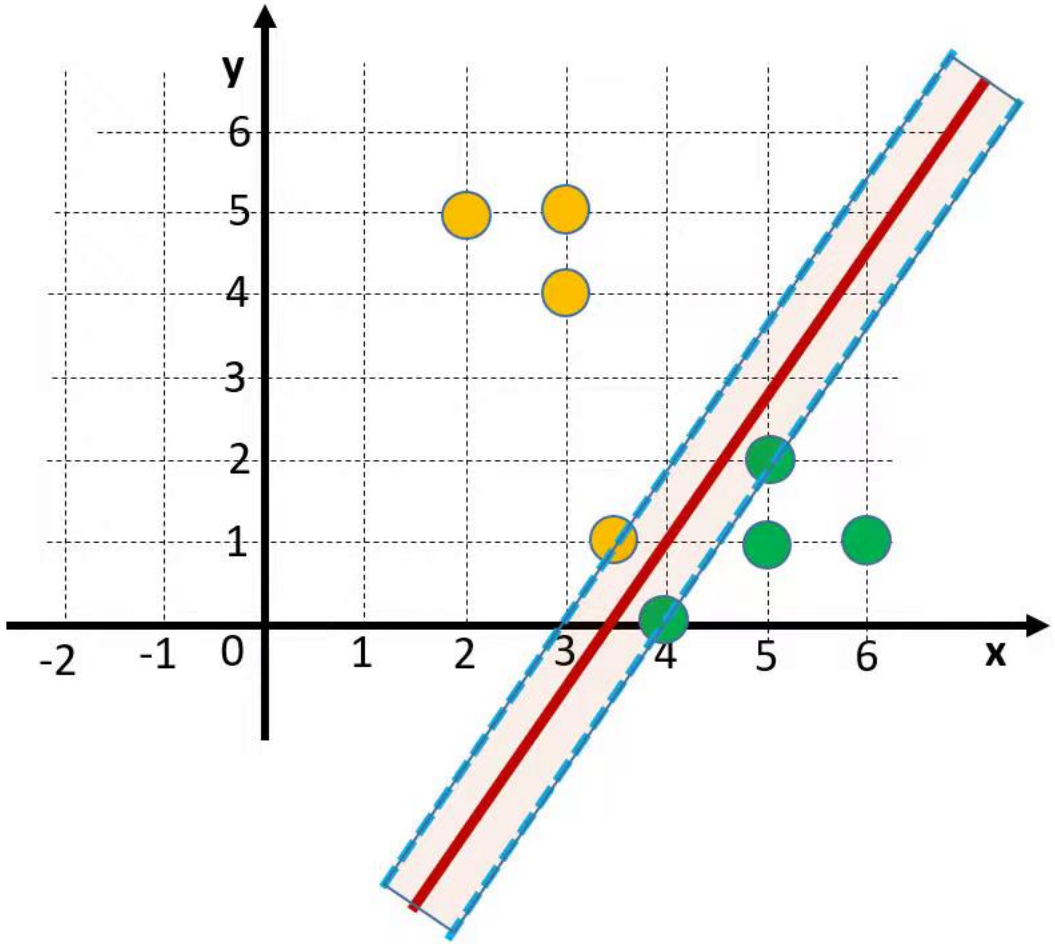
SVM – the math



$$\max \frac{2}{\|w\|} \text{ such that } Y_i(w^T x_i + b) \geq 1$$

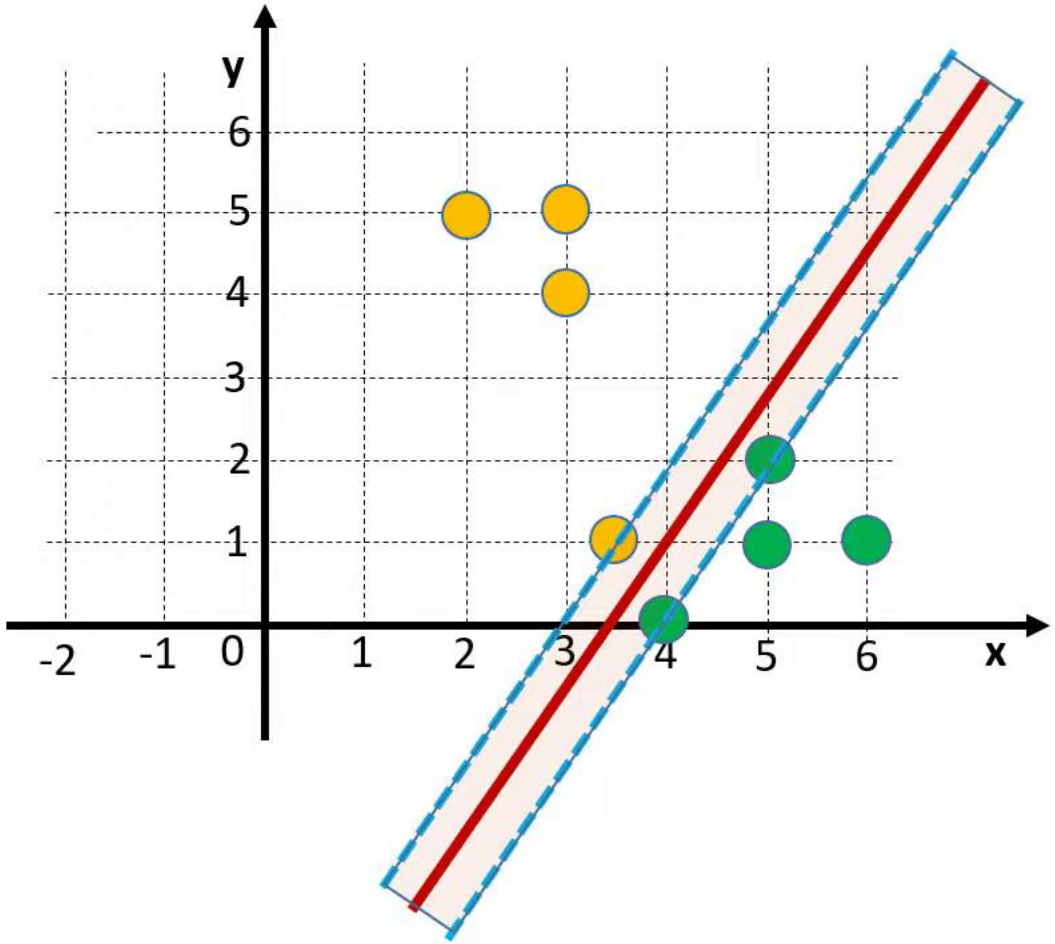
We can either accept that the data point will be incorrectly classified,

SVM – the math



or change the hyperplane so that we correctly classify all data points but with the cost of a much smaller margin.

SVM – the math



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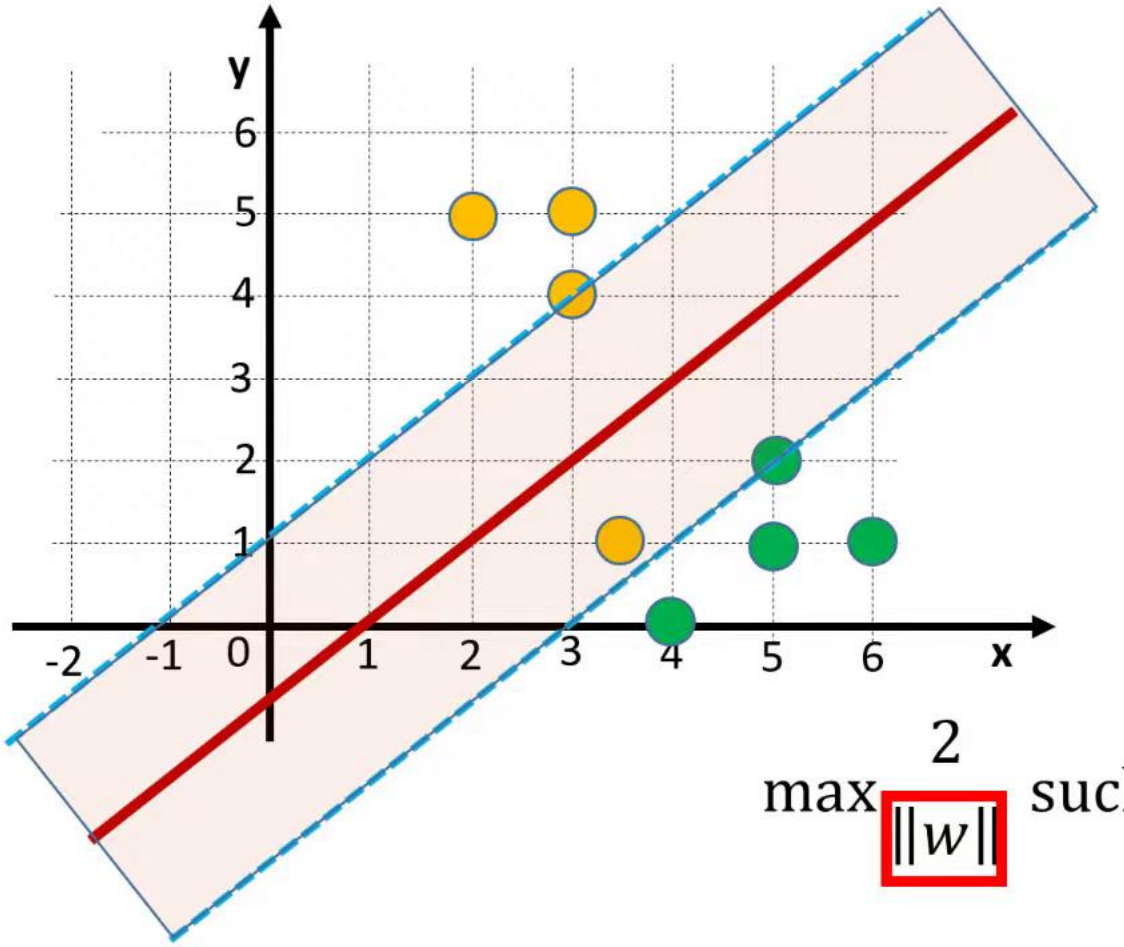
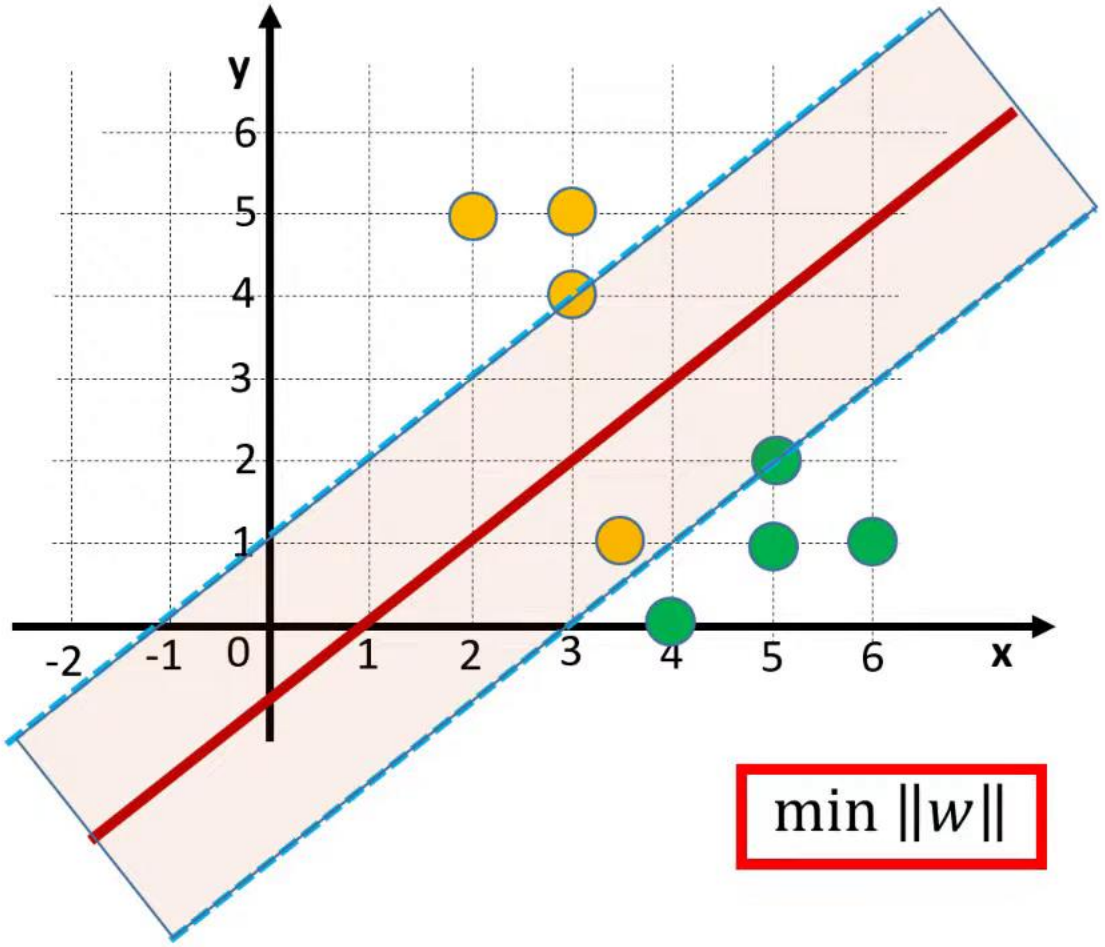
SVM – the math



$$\max \frac{2}{\|w\|} \text{ such that } Y_i(w^T x_i + b) \geq 1$$

Remember that, to maximize the margin, we need to minimize what is in the denominator.

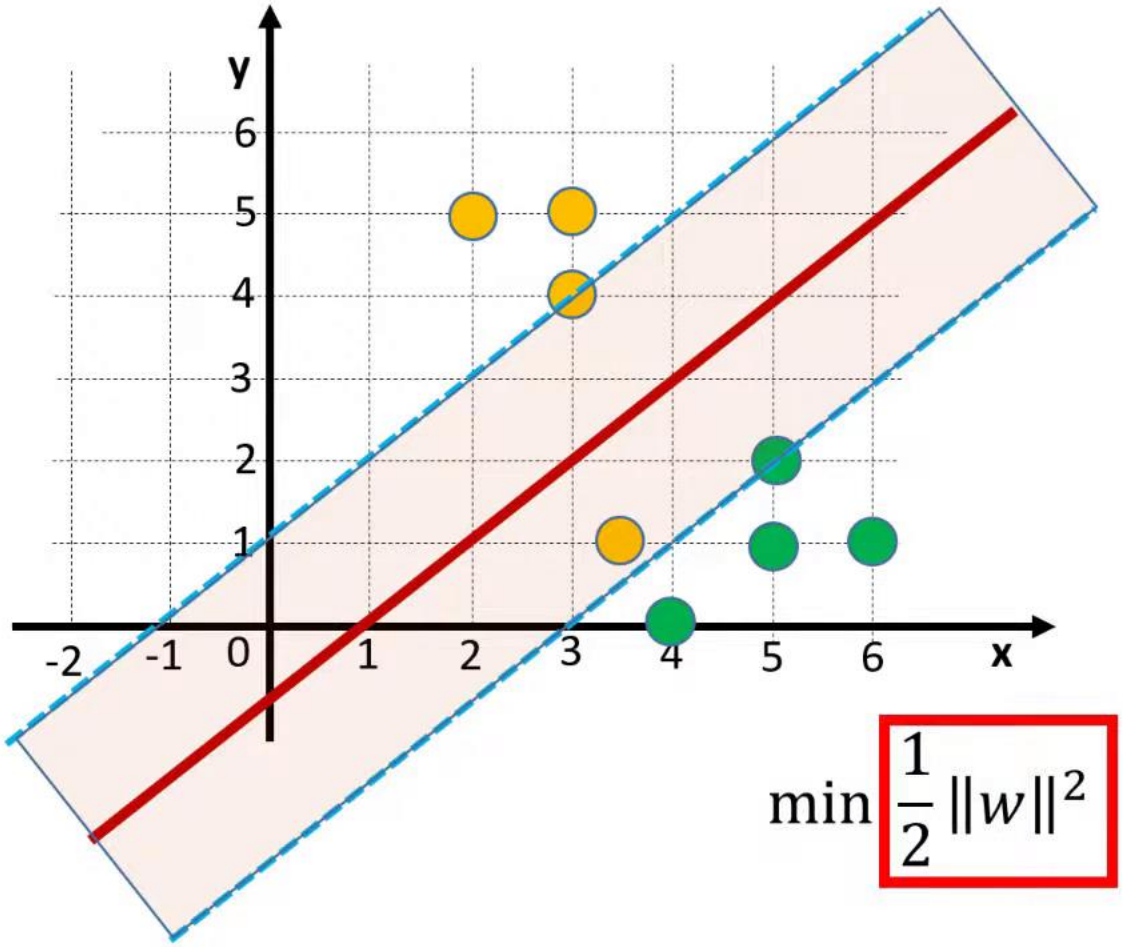
SVM – the math



$$\min \|w\|$$

We can see this as we minimize only the denominator.

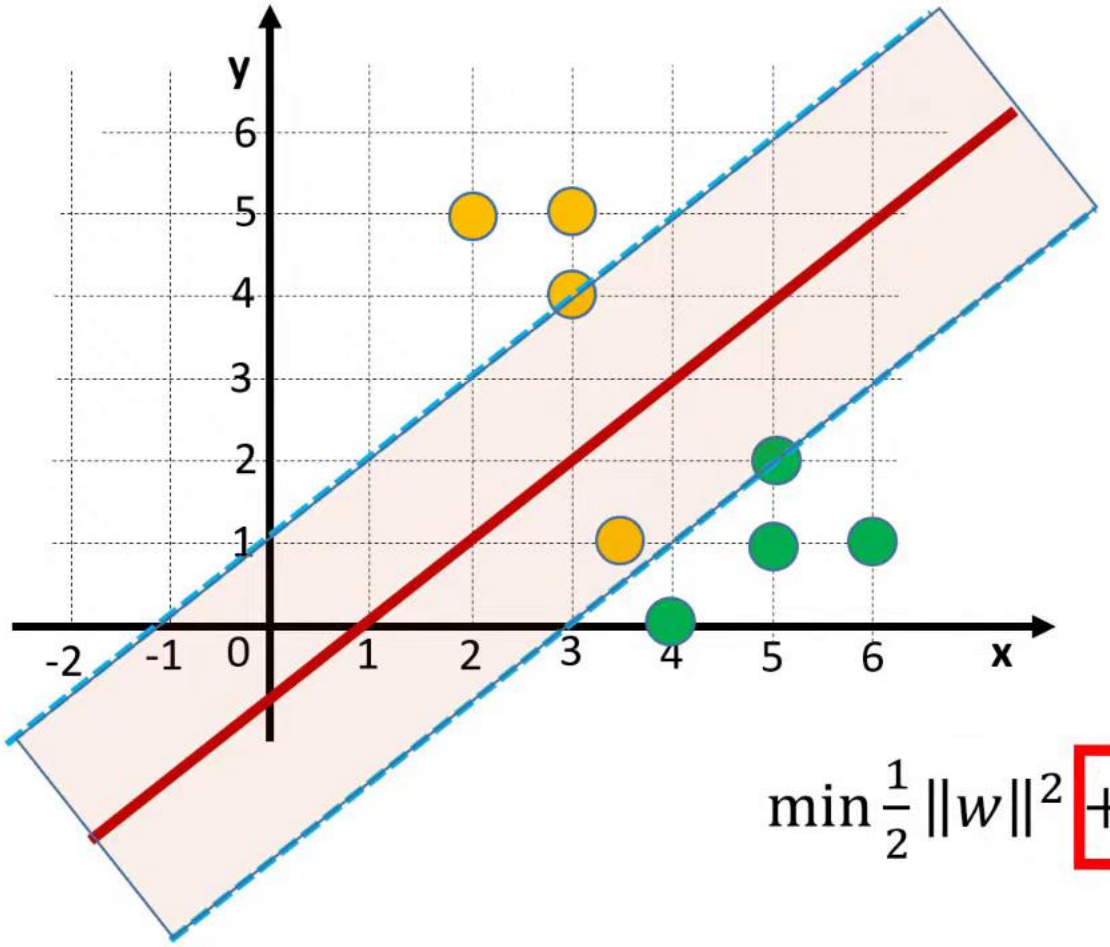
SVM – the math



$$\min \frac{1}{2}\|w\|^2$$

For computational reasons, we usually express the term like this.

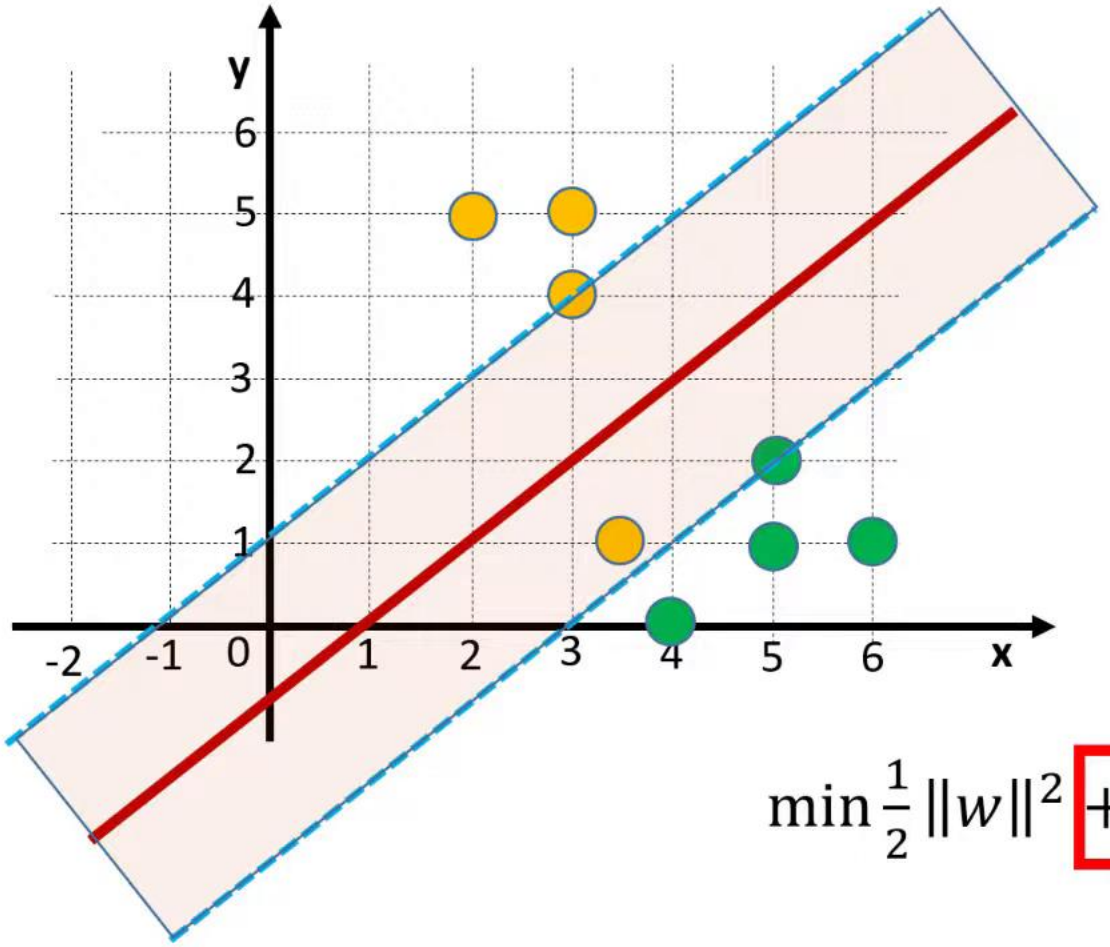
SVM – the math



$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

And to allow for misclassification, we can add this term,

SVM – the math



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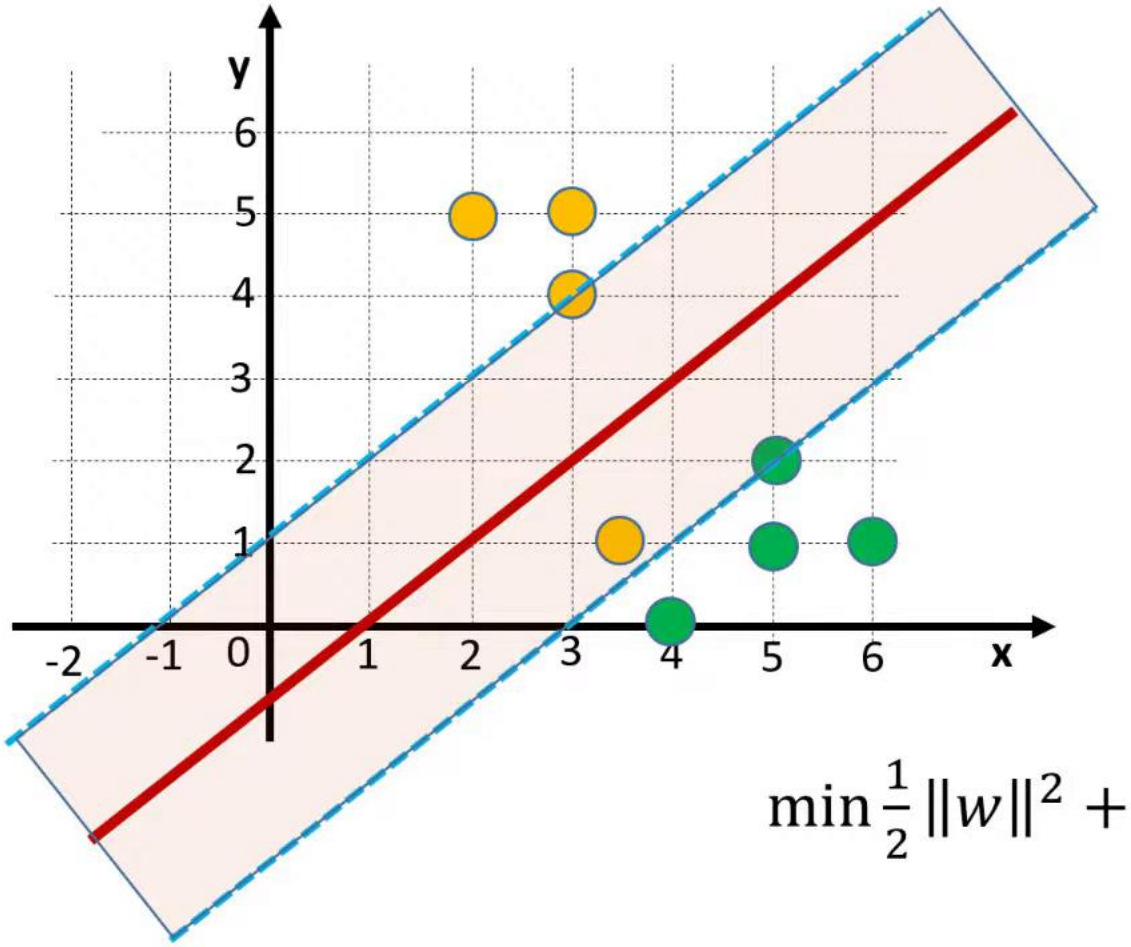
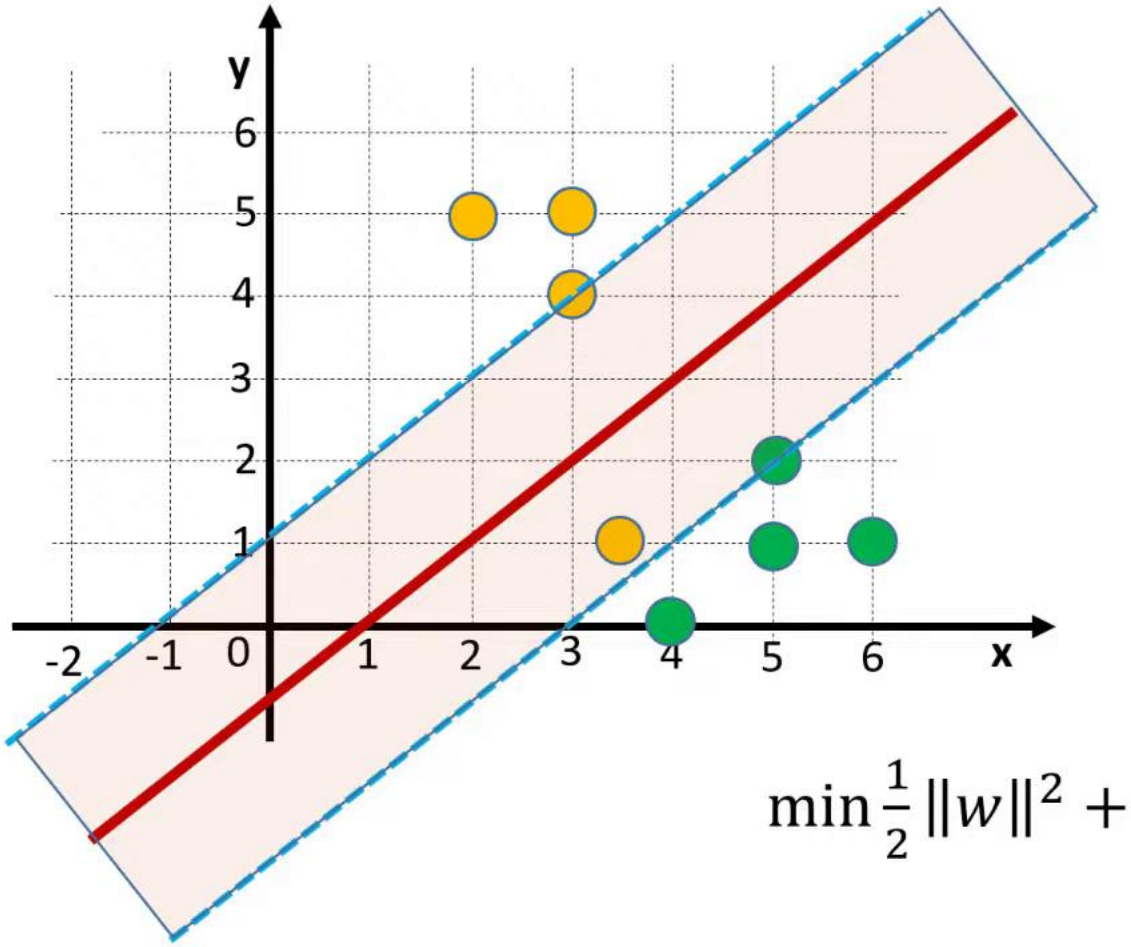
SVM – the math



$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

where epsilon is a distance measure of the data points from their corresponding blue line. This is called a slack variable in SVM.

SVM – the math



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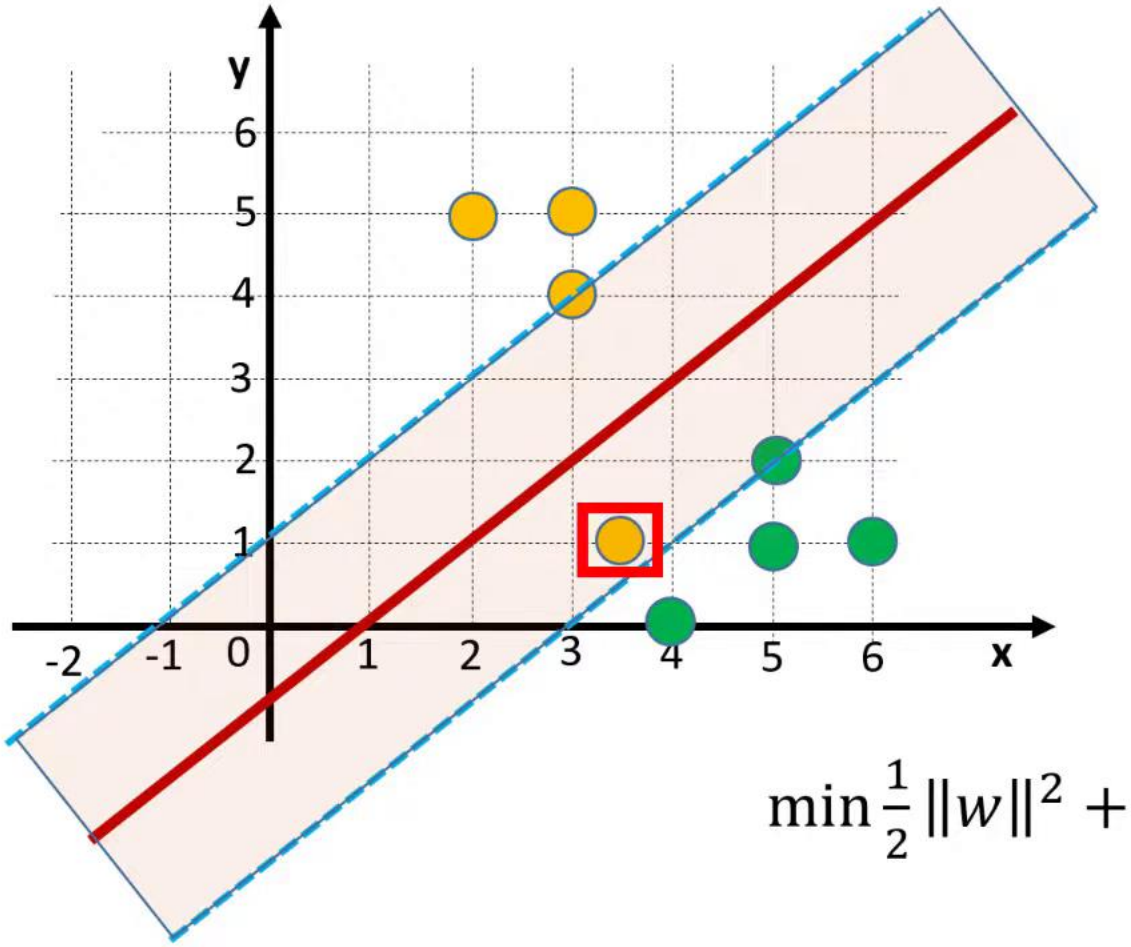
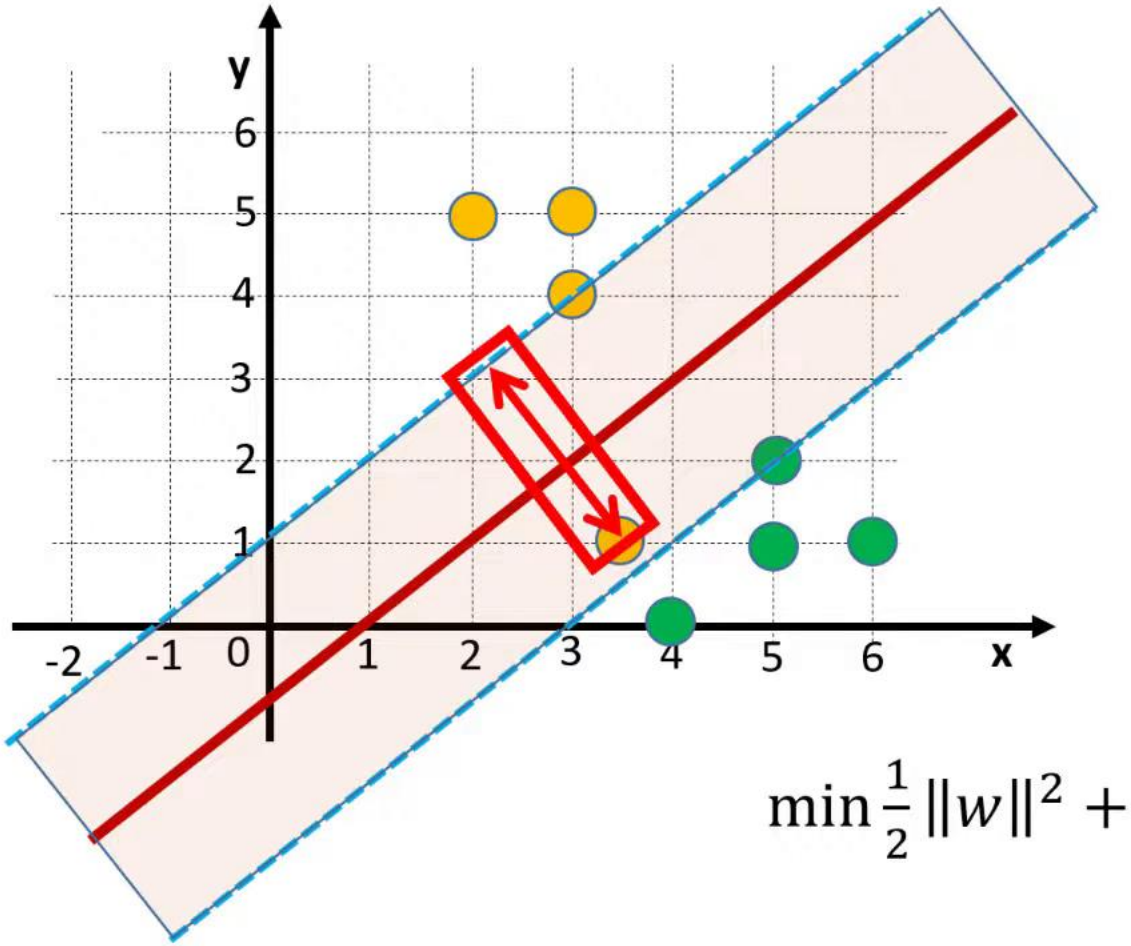
SVM – the math



$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

In this example, there is only one data point that is incorrectly classified because it is on the wrong side of the hyperplane.

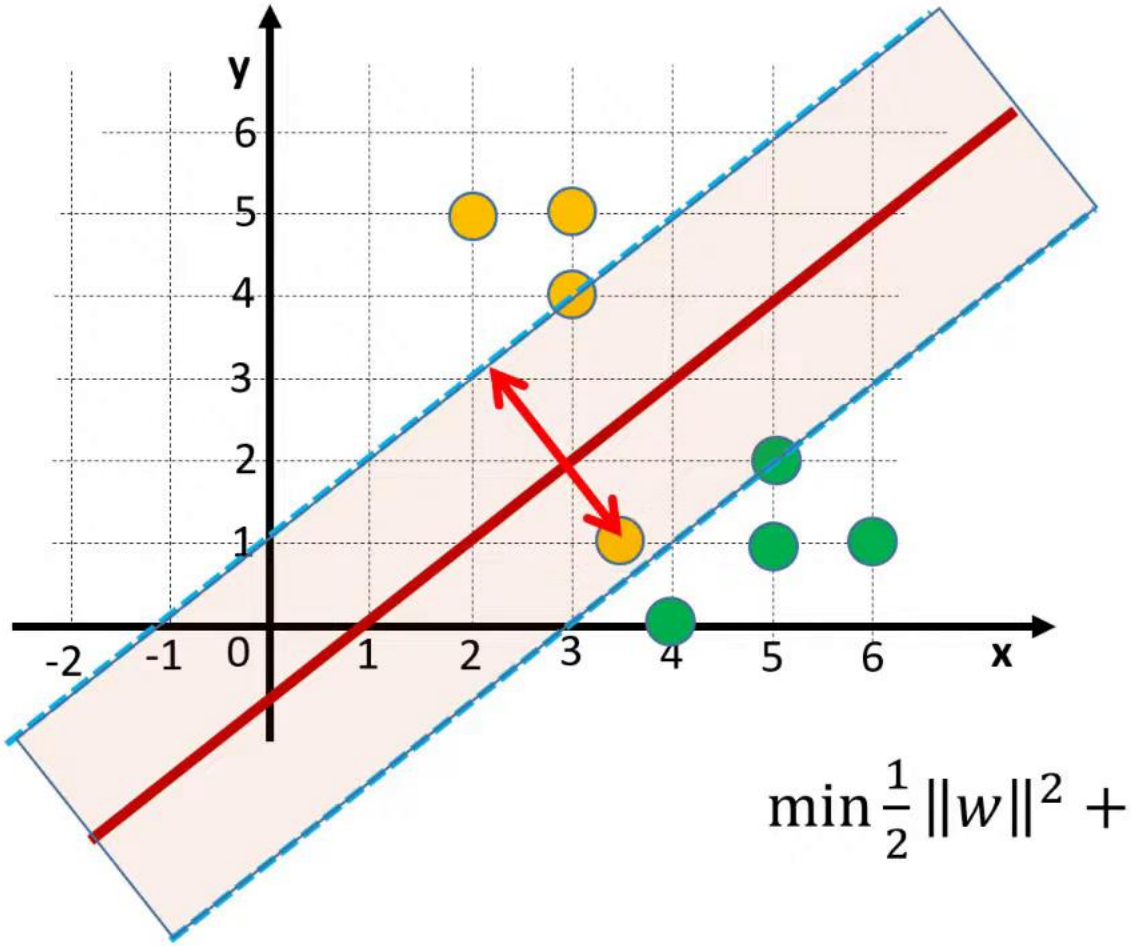
SVM – the math



$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

The distance between this data point and its corresponding blue line,

SVM – the math

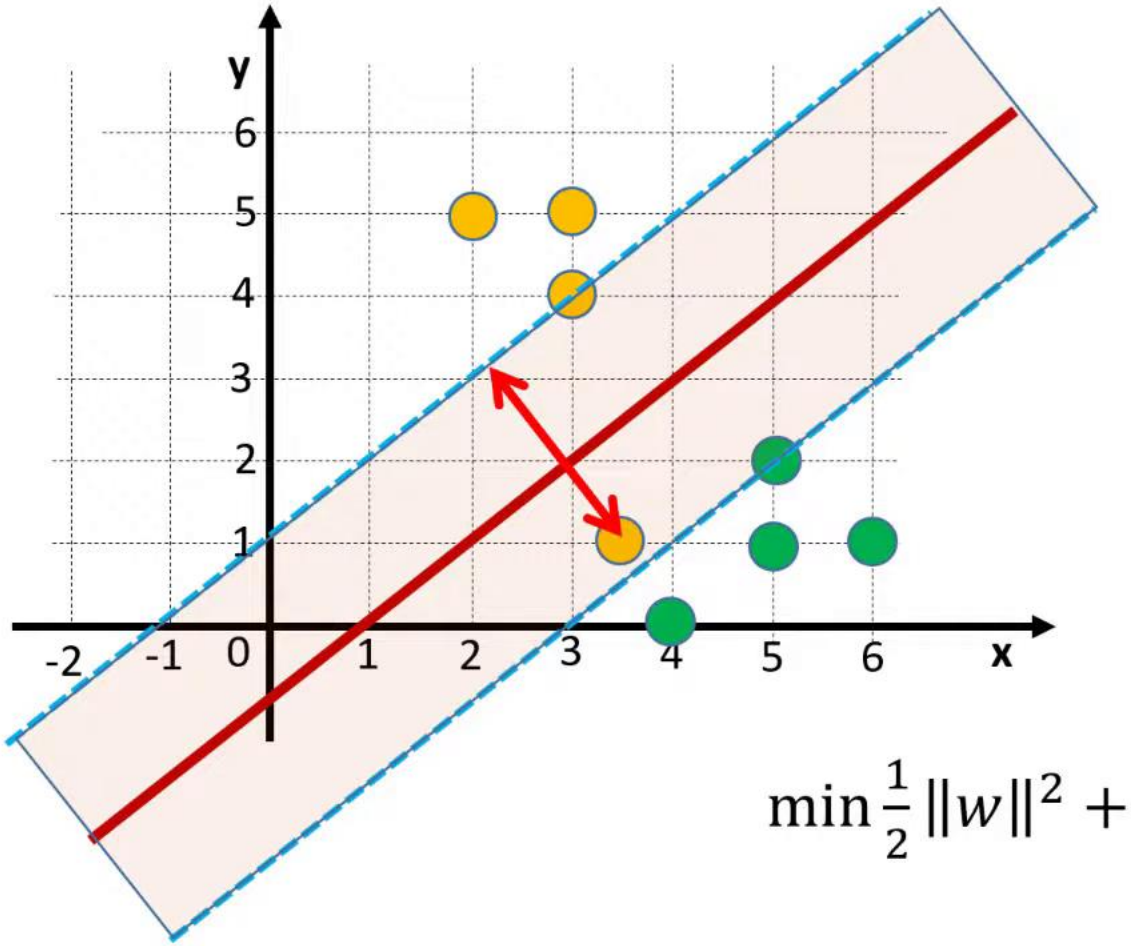


$$d = \frac{\varepsilon}{\|w\|}$$

$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

can be described by the following equation.

SVM – the math



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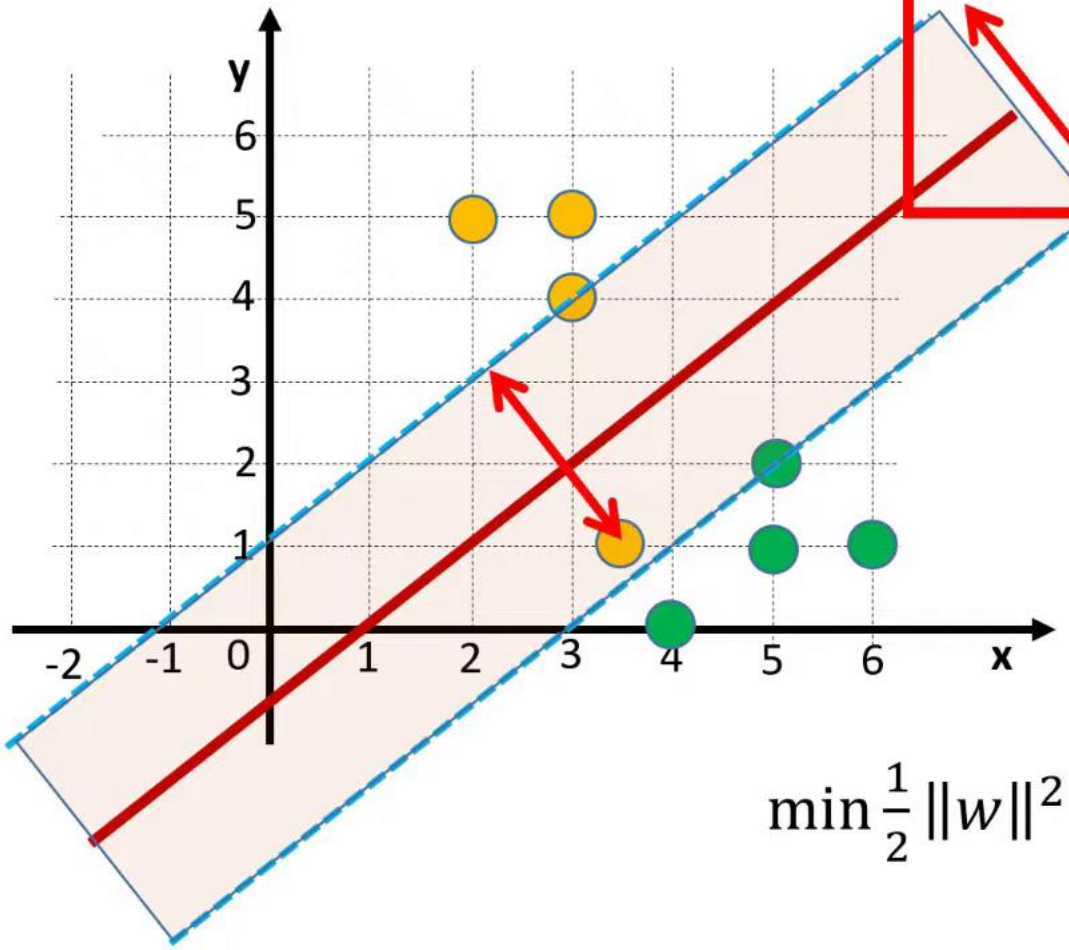
SVM – the math



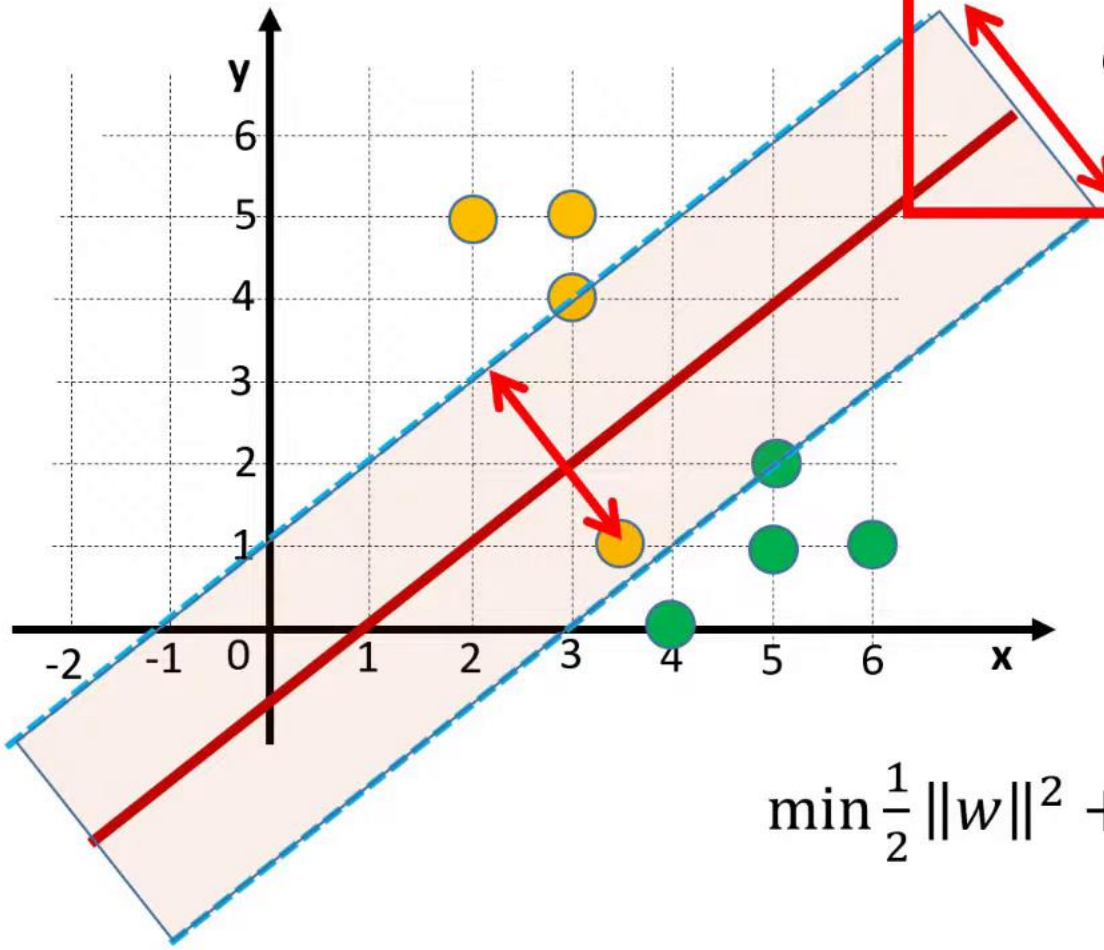
$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

$$d = \frac{\varepsilon}{\|w\|}$$

$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

Remember that the width of the margin can be calculated by the following formula,

SVM – the math



$$d = \frac{2}{\sqrt{A^2 + B^2}} = \frac{2}{\|w\|}$$

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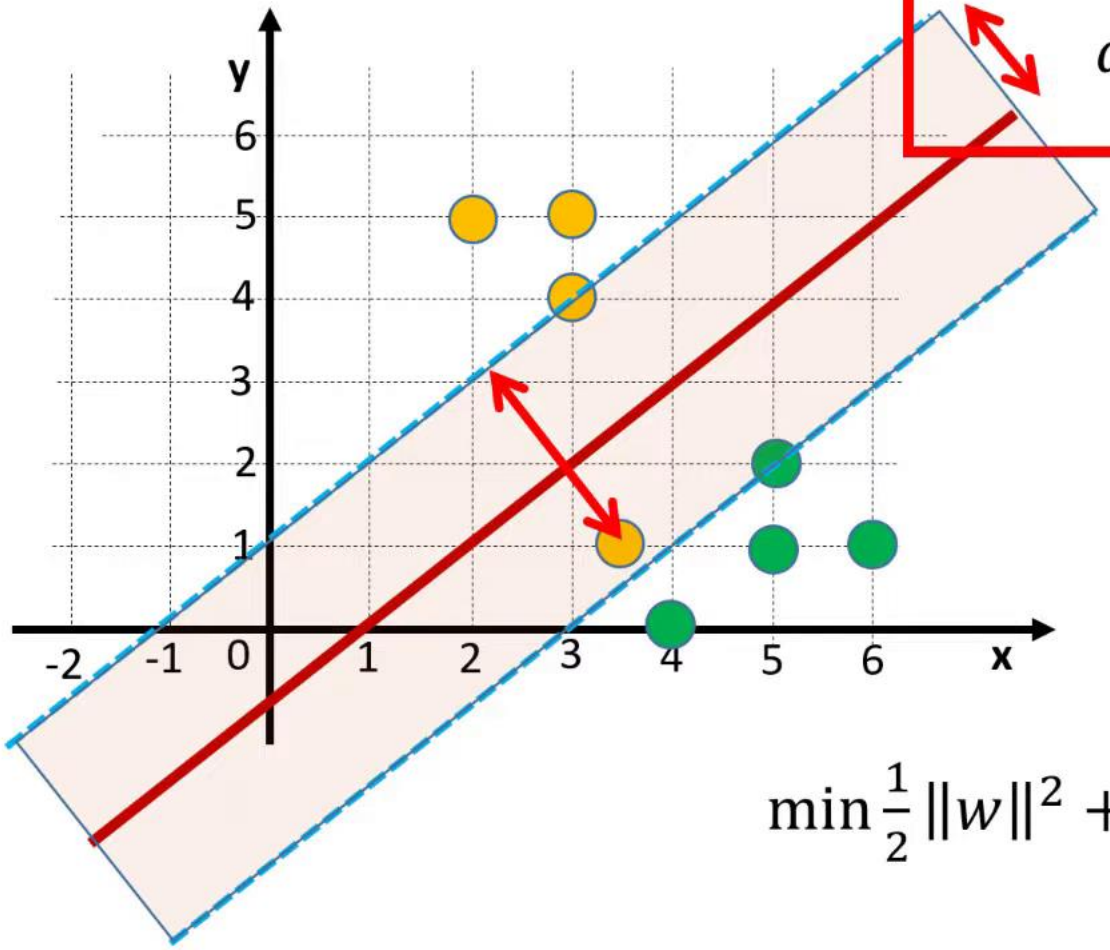
SVM – the math



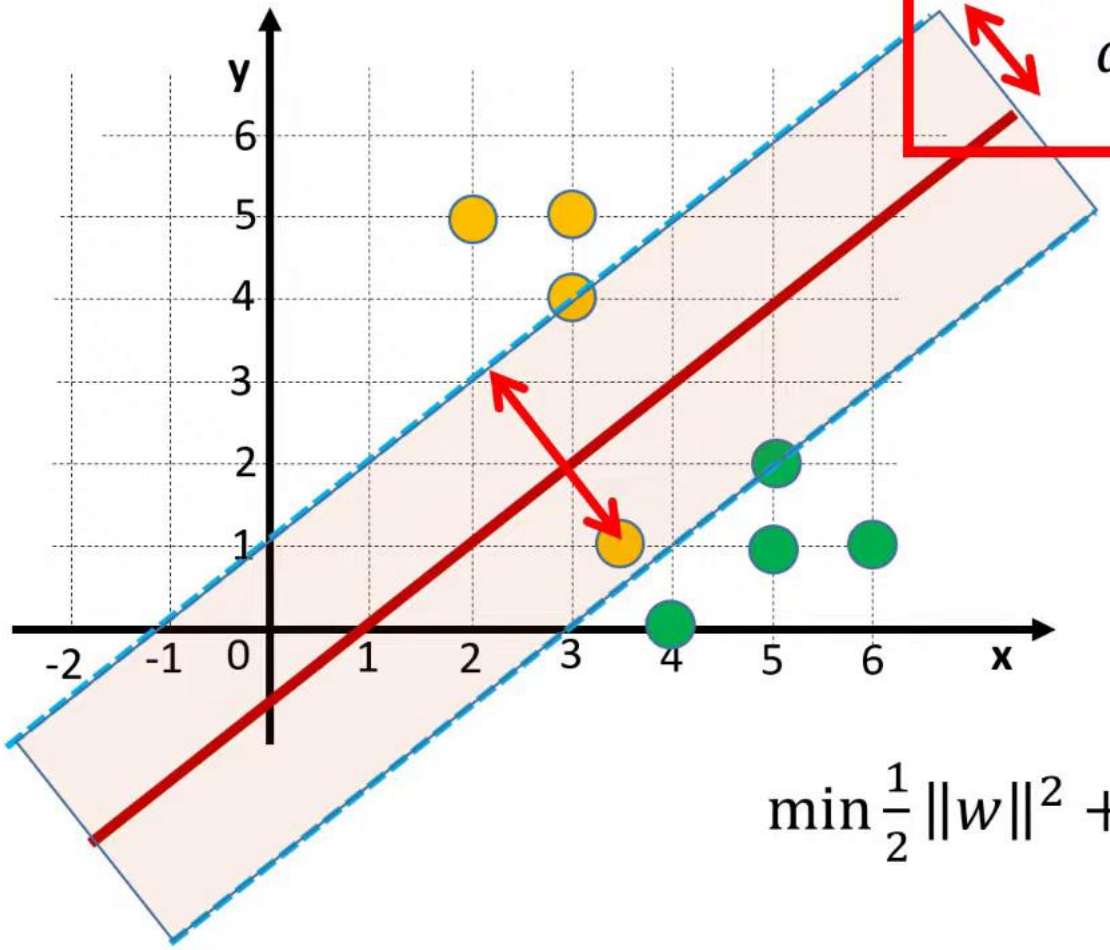
$$d = \frac{1}{\sqrt{A^2 + B^2}} = \frac{1}{\|w\|}$$

$$d = \frac{\varepsilon}{\|w\|}$$

$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

whereas half the width of the margin, which is the distance between one of the blue lines and the hyperplane, is calculated like this.

SVM – the math



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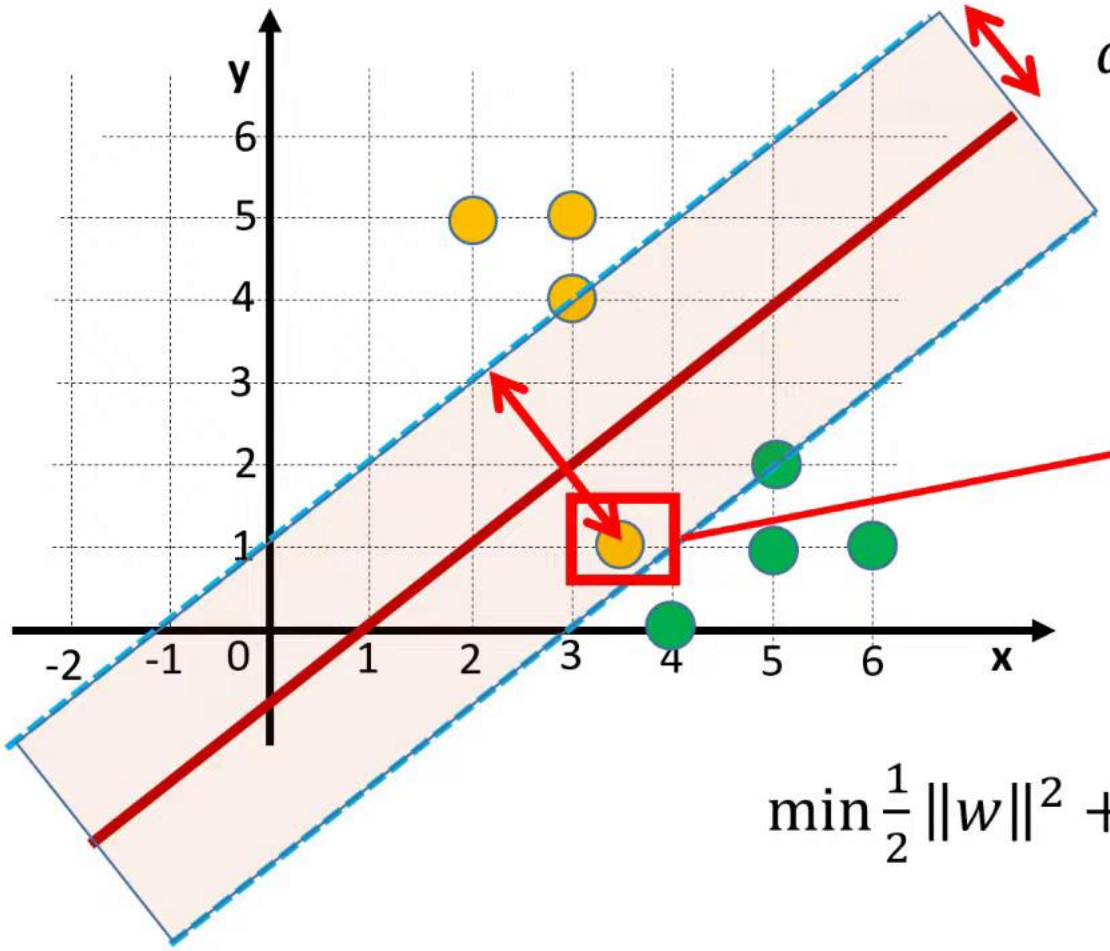
SVM – the math



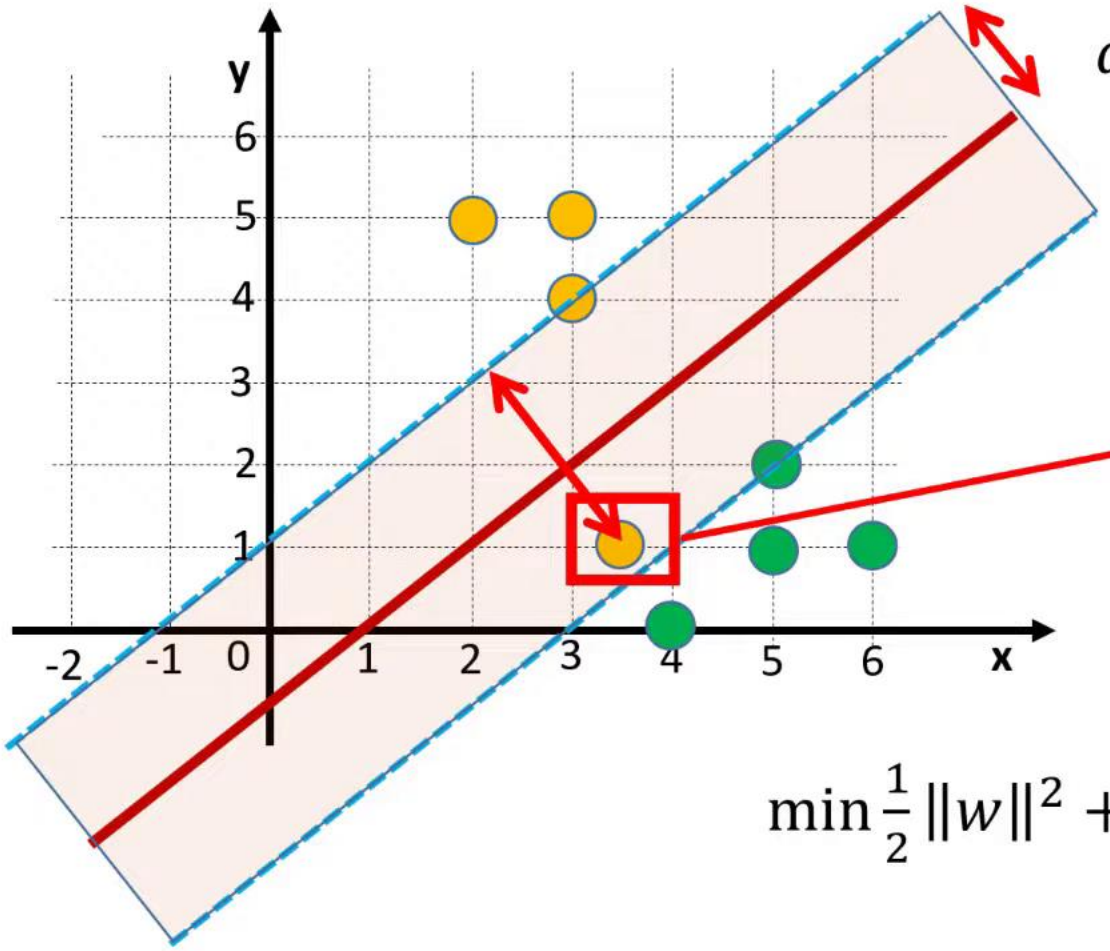
$$d = \frac{1}{\sqrt{A^2 + B^2}} = \frac{1}{\|w\|}$$

$$d = \frac{\varepsilon}{\|w\|}$$

$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

This means that epsilon will be greater than one for data points that are on the wrong side of the hyperplane,

SVM – the math



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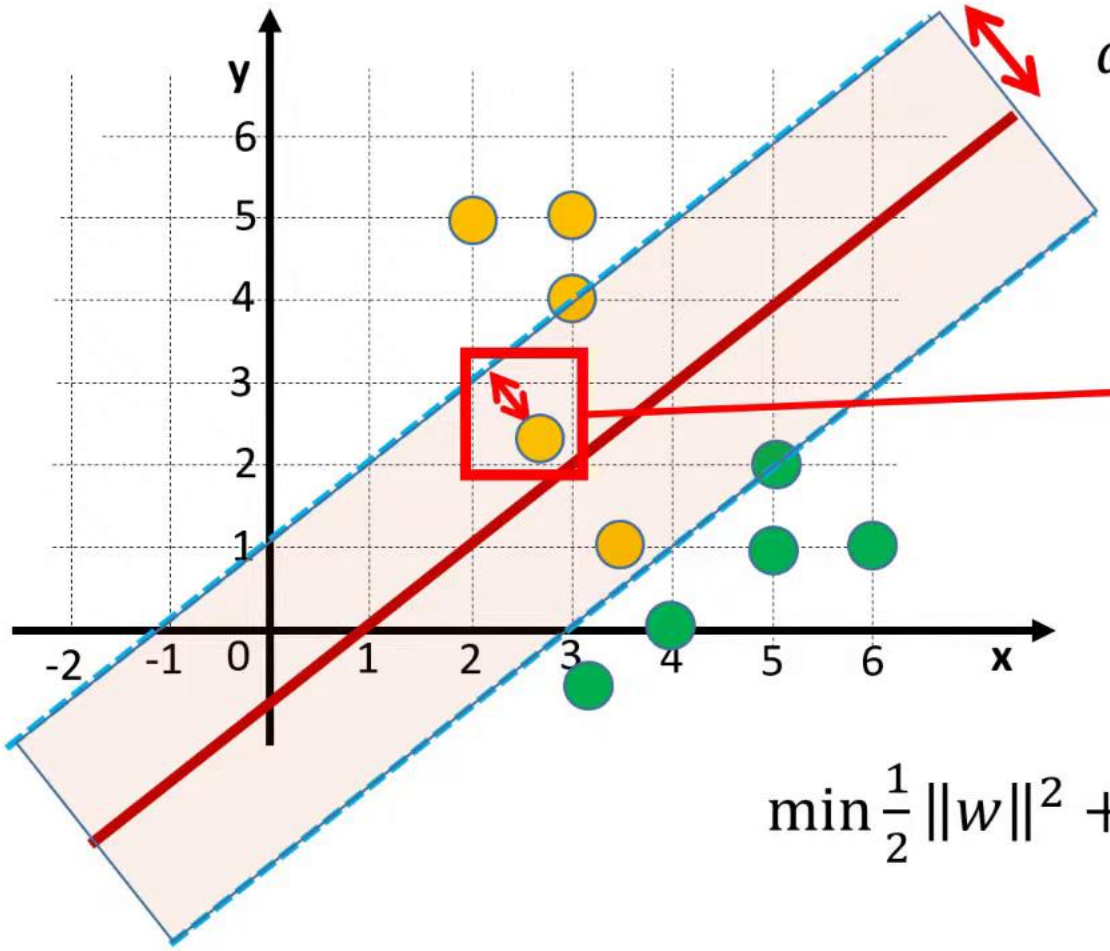
SVM – the math



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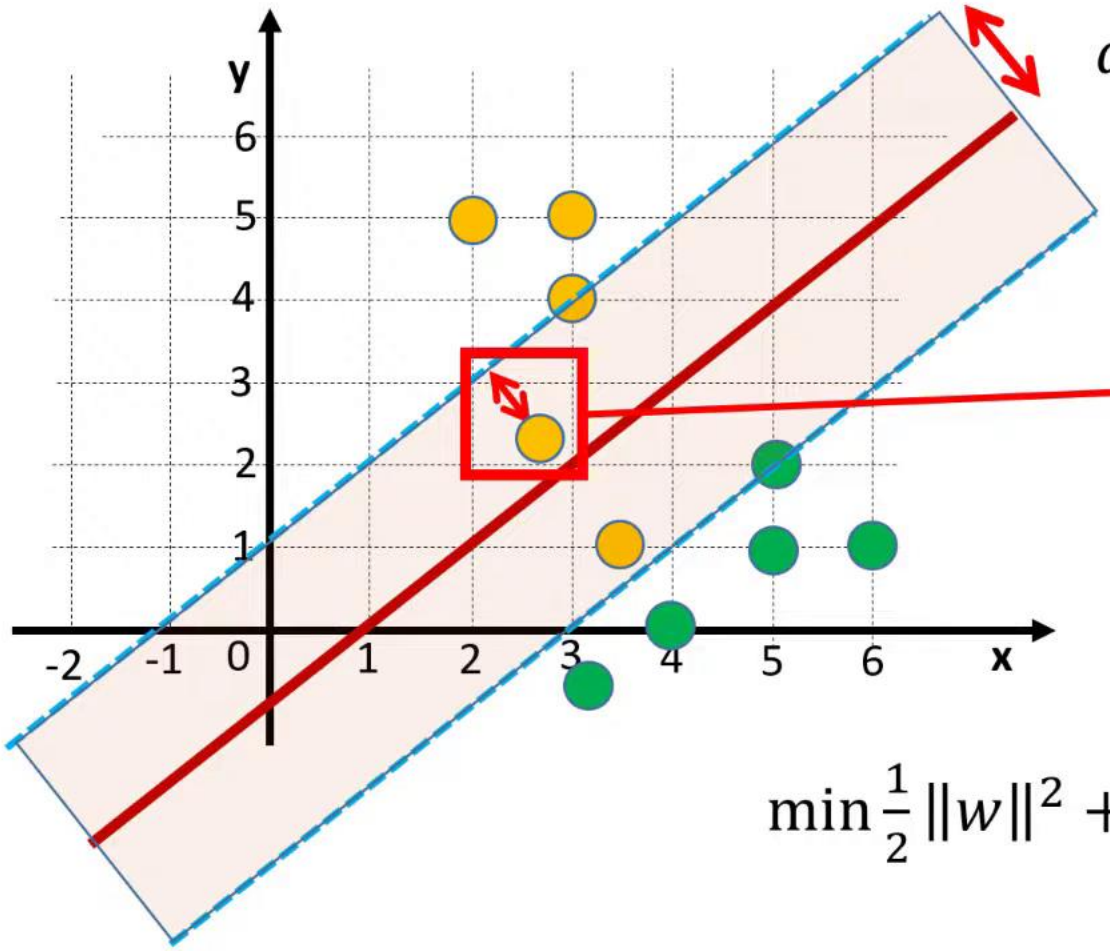
$$d = \frac{\varepsilon}{\|w\|}$$

$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

whereas a data point within the margin, but on the correct side of the hyperplane, will have an epsilon value that is between zero and one.

SVM – the math



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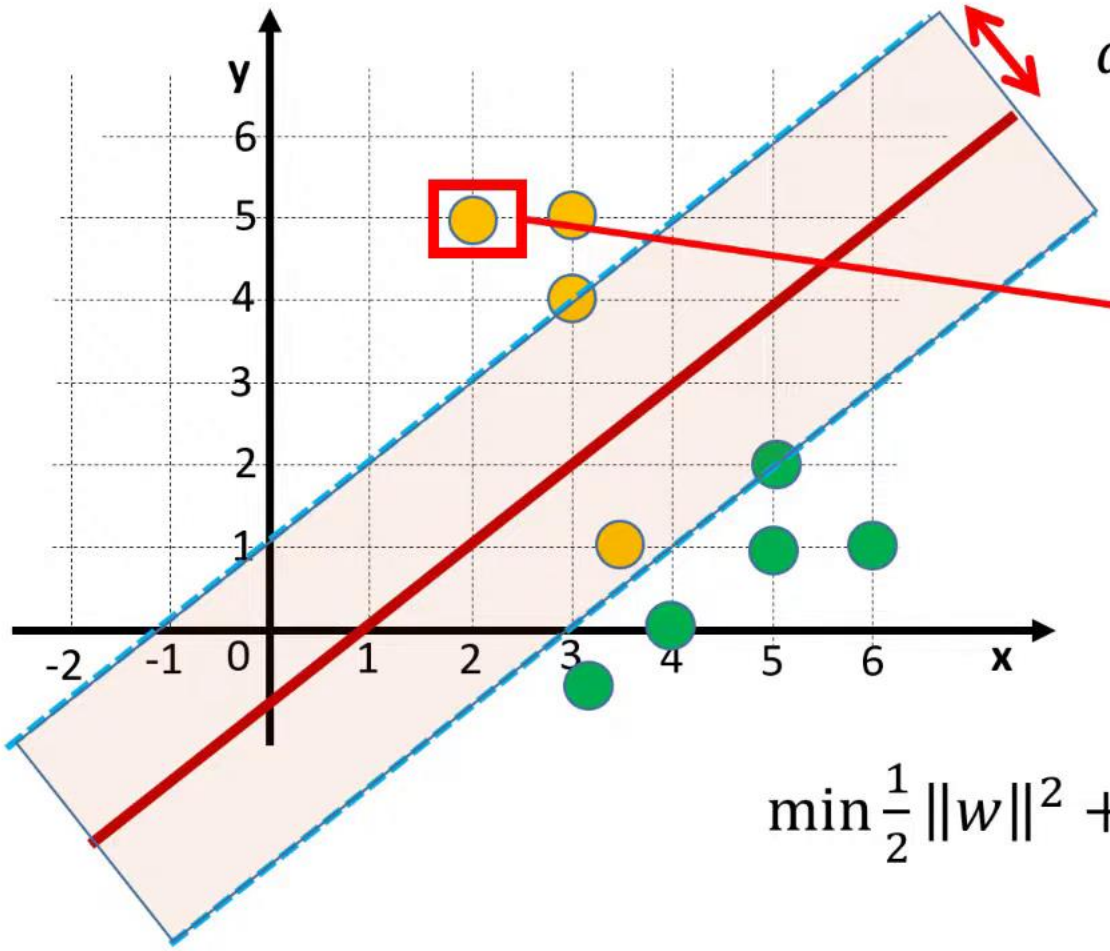
SVM – the math



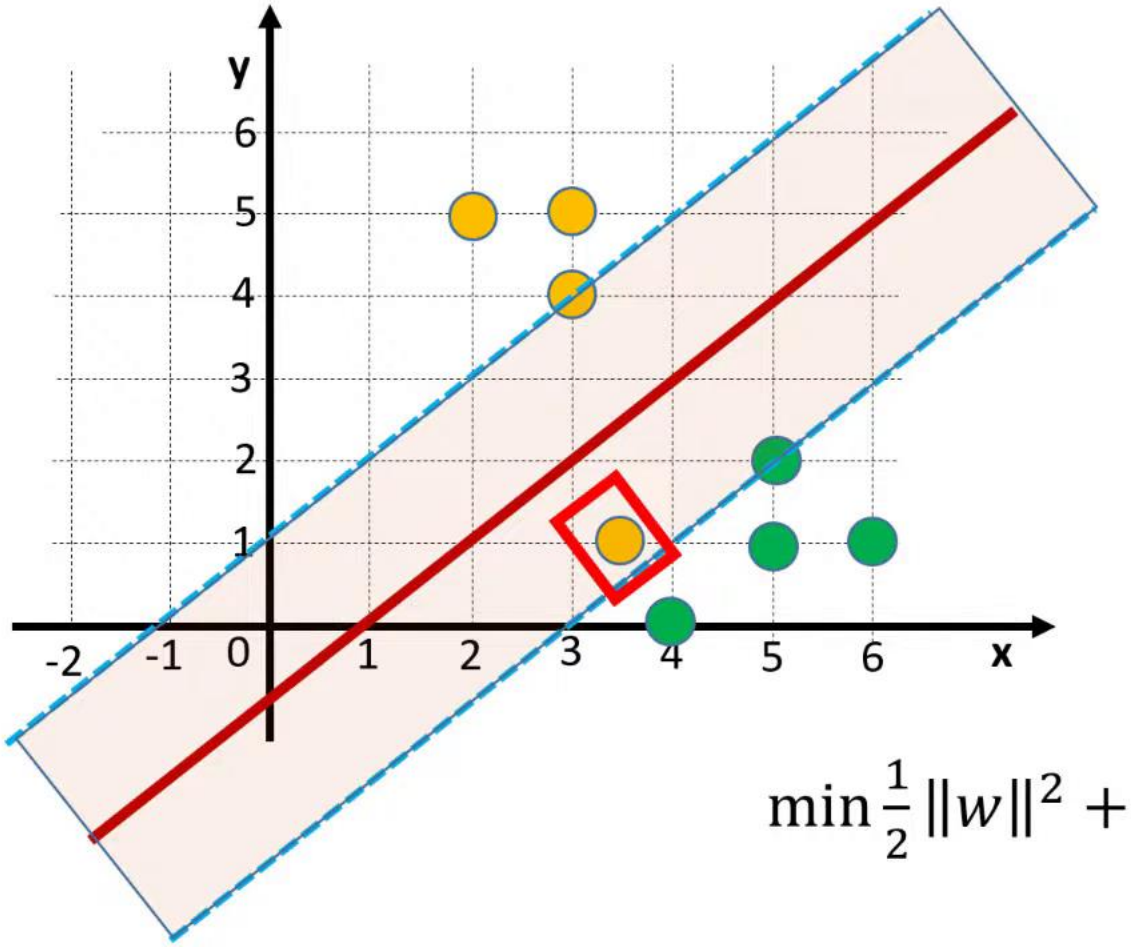
$$d = \frac{1}{\sqrt{A^2 + B^2}} = \frac{1}{\|w\|}$$

$$d = \frac{\varepsilon}{\|w\|}$$

$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

Epsilon is set to zero for all data points that are correctly classified and that are outside the margin.

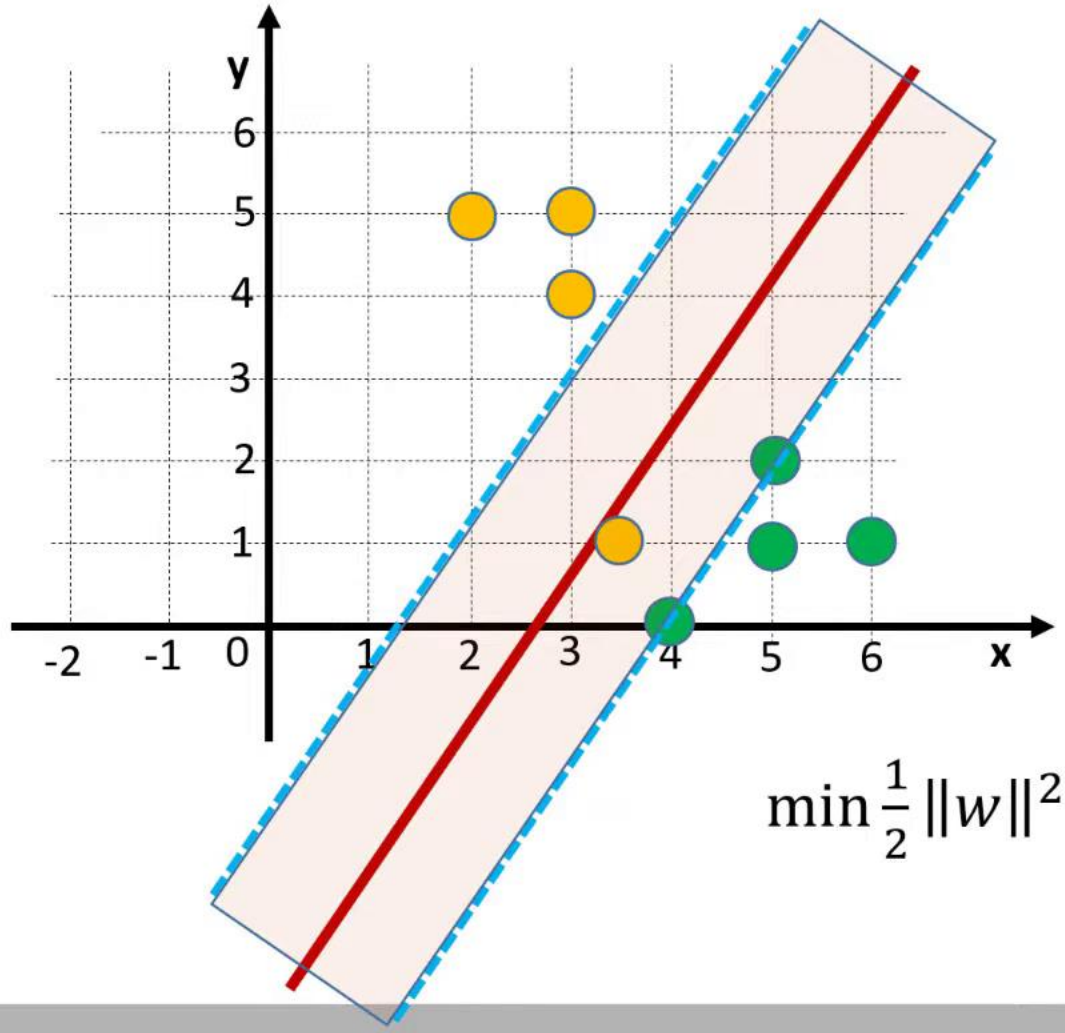
SVM – the math



$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

Note that this point is now also considered as a support vector because it will influence the orientation of the hyperplane,

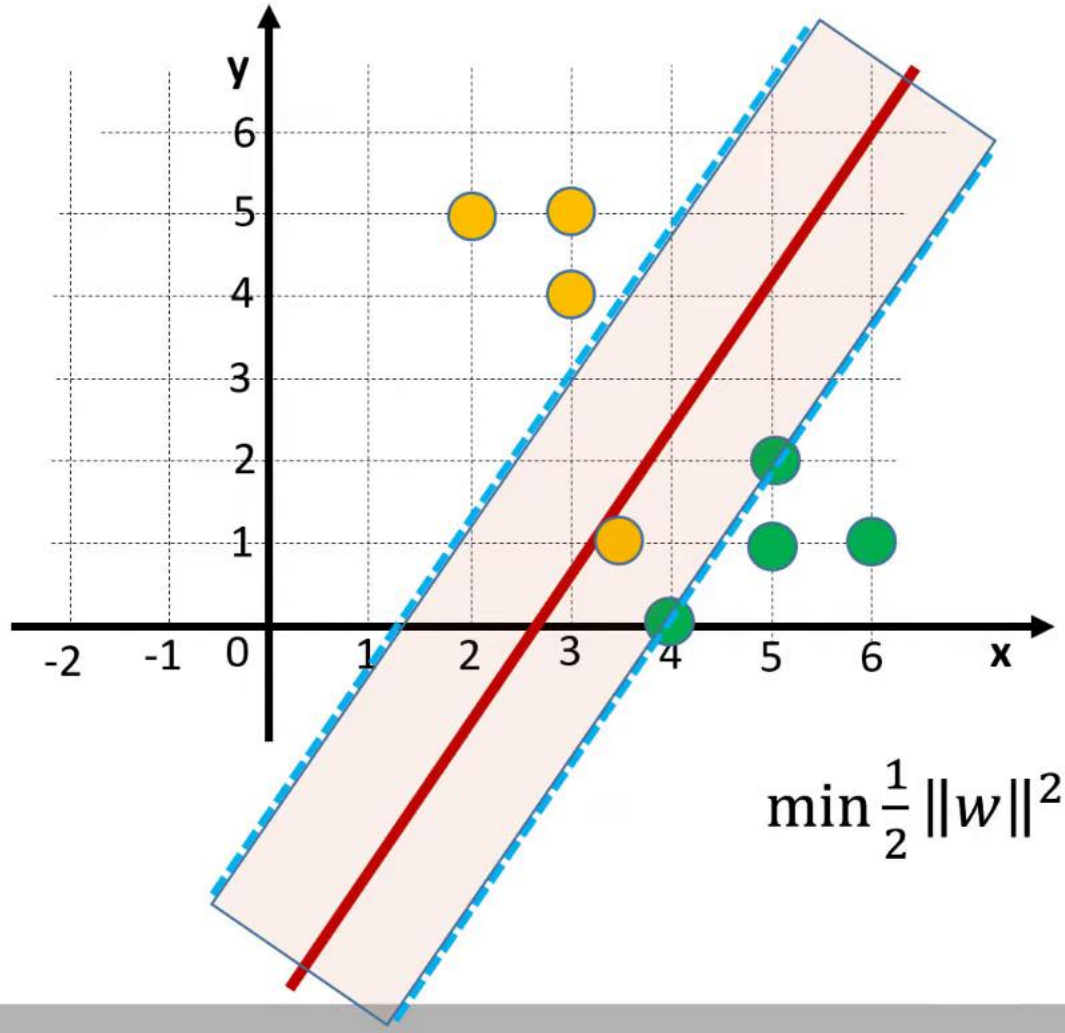
SVM – the math



$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

like this for a certain value of C.

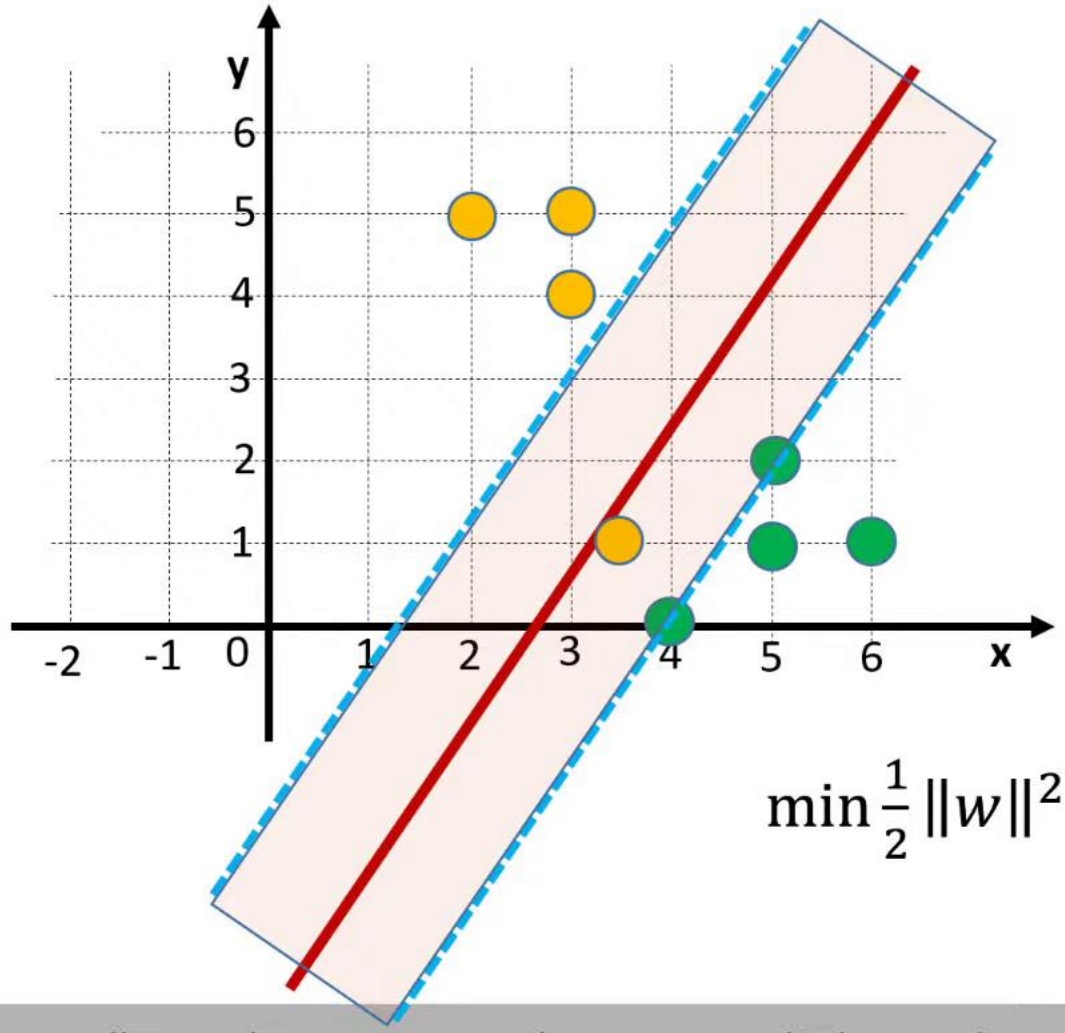
SVM – the math



$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

C is a tuning parameter that controls how much weight we should put on the data points that are incorrectly classified.

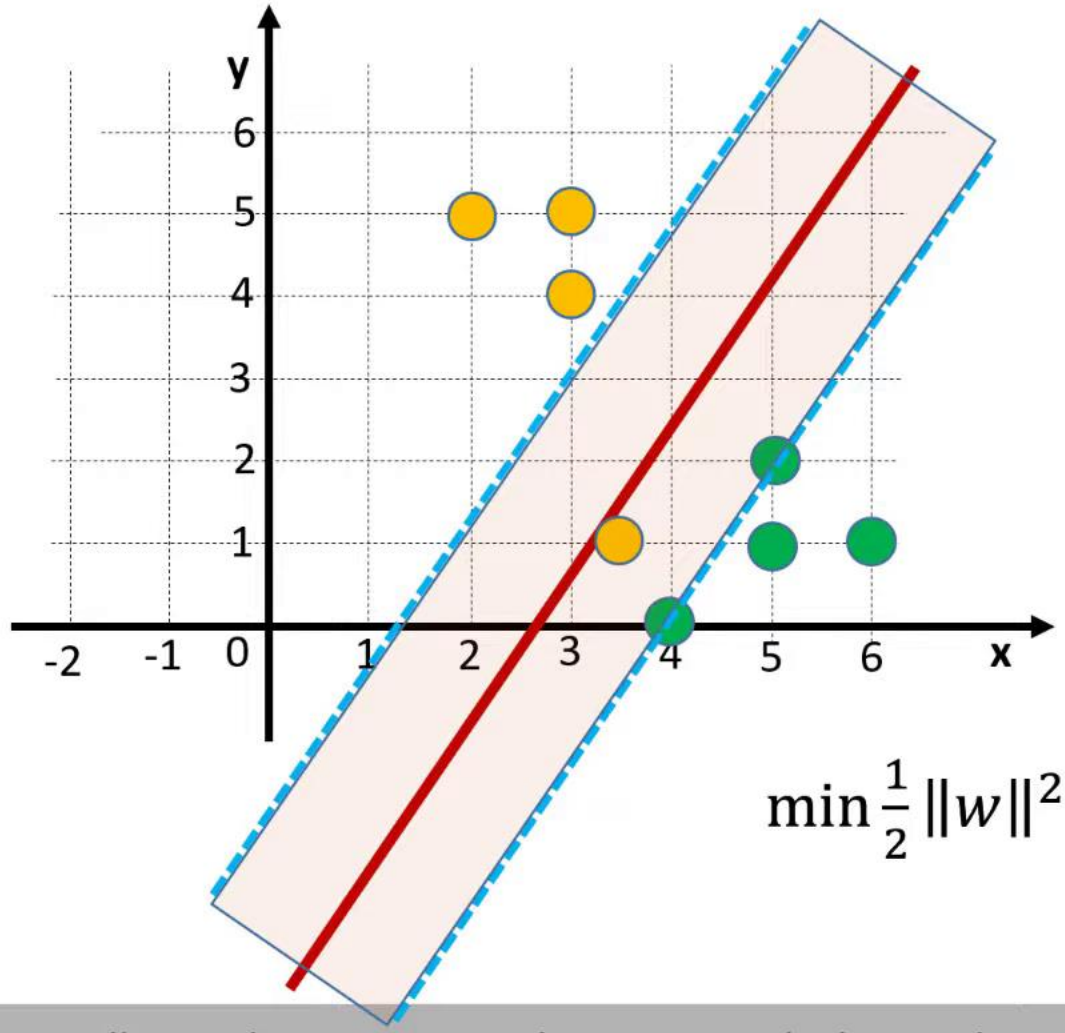
SVM – the math



$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

If C is small, we will get a large margin where we put little weight on data points that are misclassified. This will result in a so-called soft margin SVM.

SVM – the math



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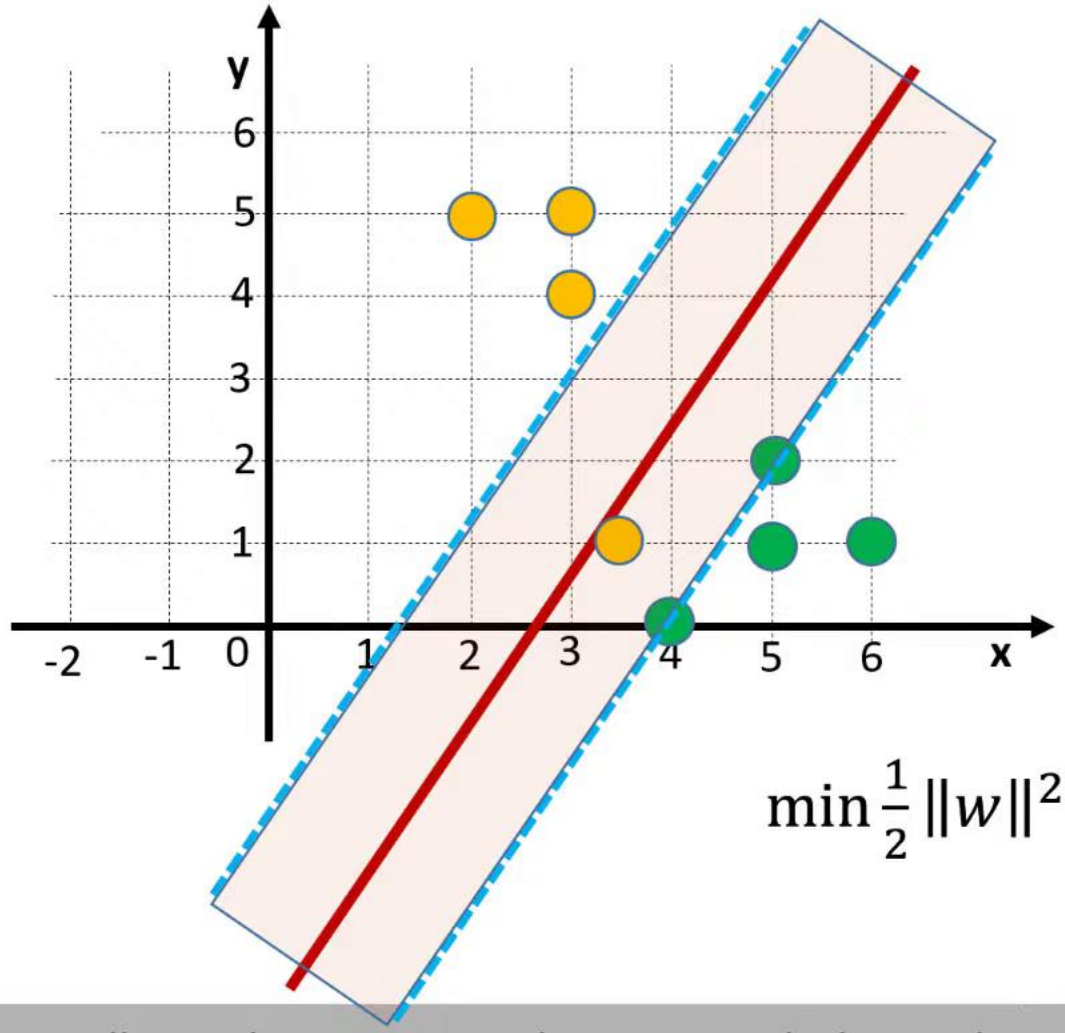
SVM – the math



$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

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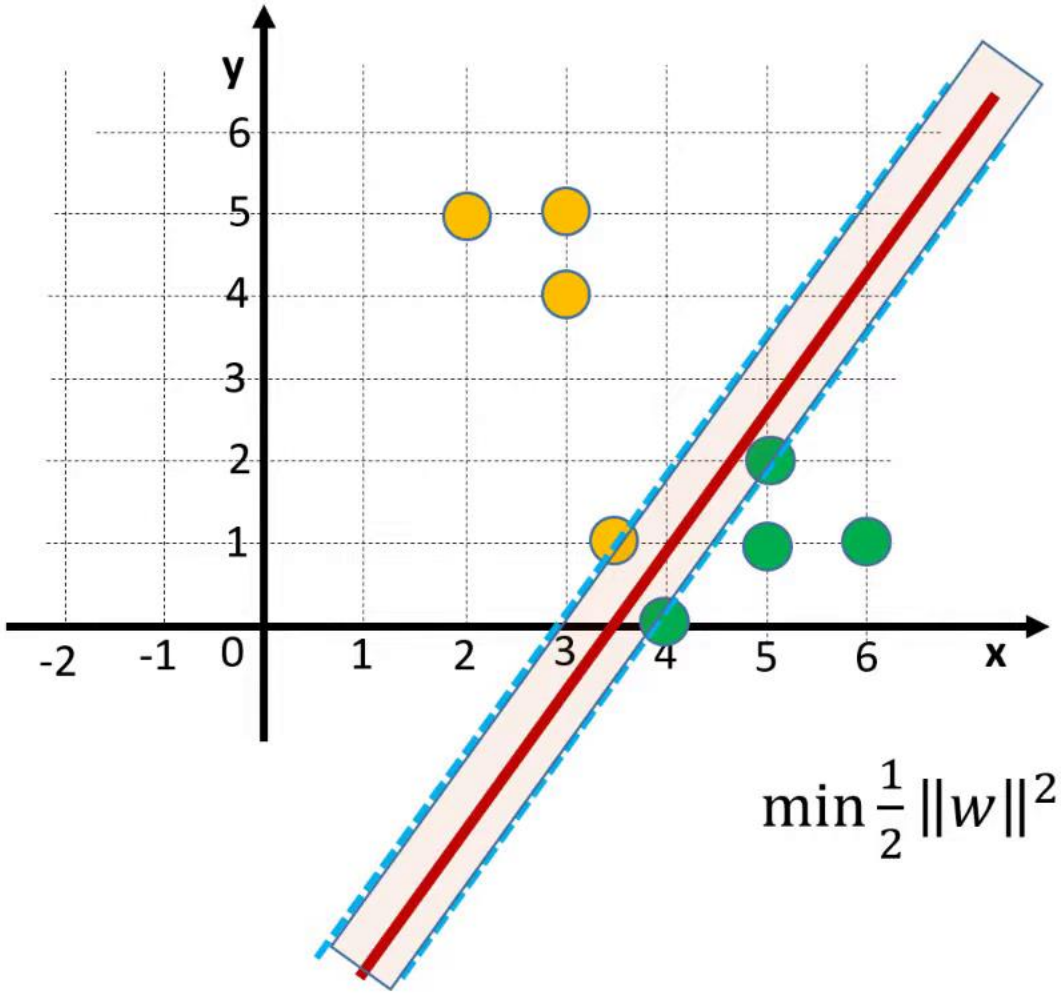
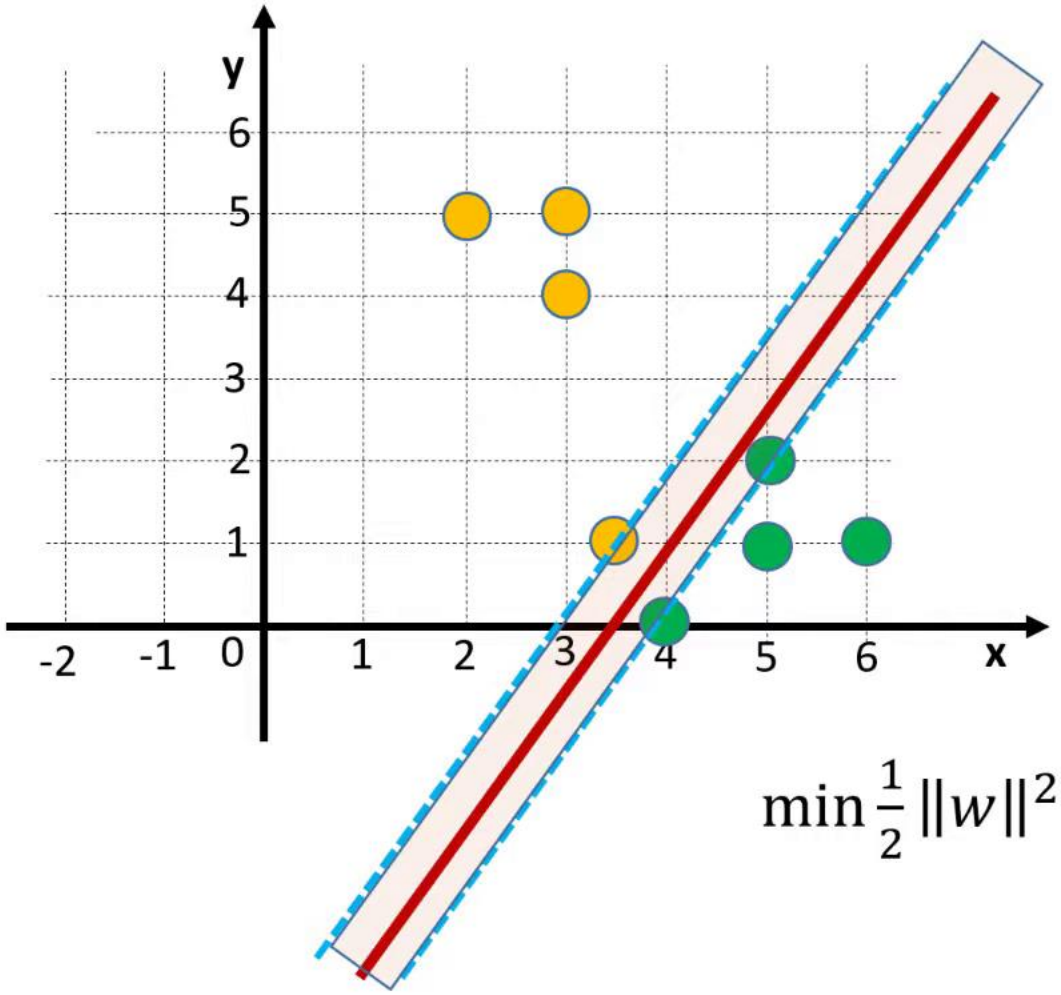
SVM – the math



$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

In contrast, if C is large, which puts more weight to avoid a misclassification, will result in a smaller margin. A large value of C will result in a SVM that is less tolerant to outliers that might cause overfitting.

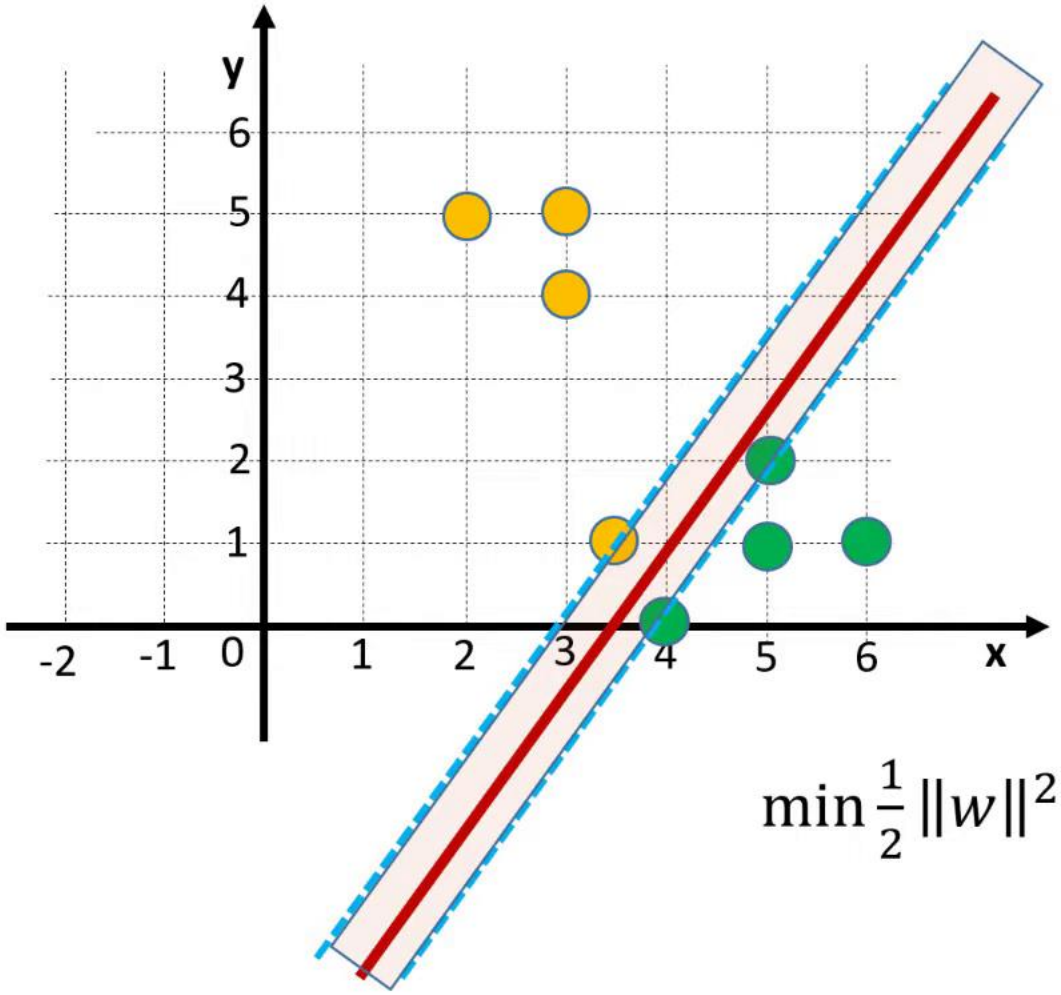
SVM – the math



$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

In contrast, if C is large, which puts more weight to avoid a misclassification, will result in a smaller margin. A large value of C will result in a SVM that is less tolerant to outliers that might cause overfitting.

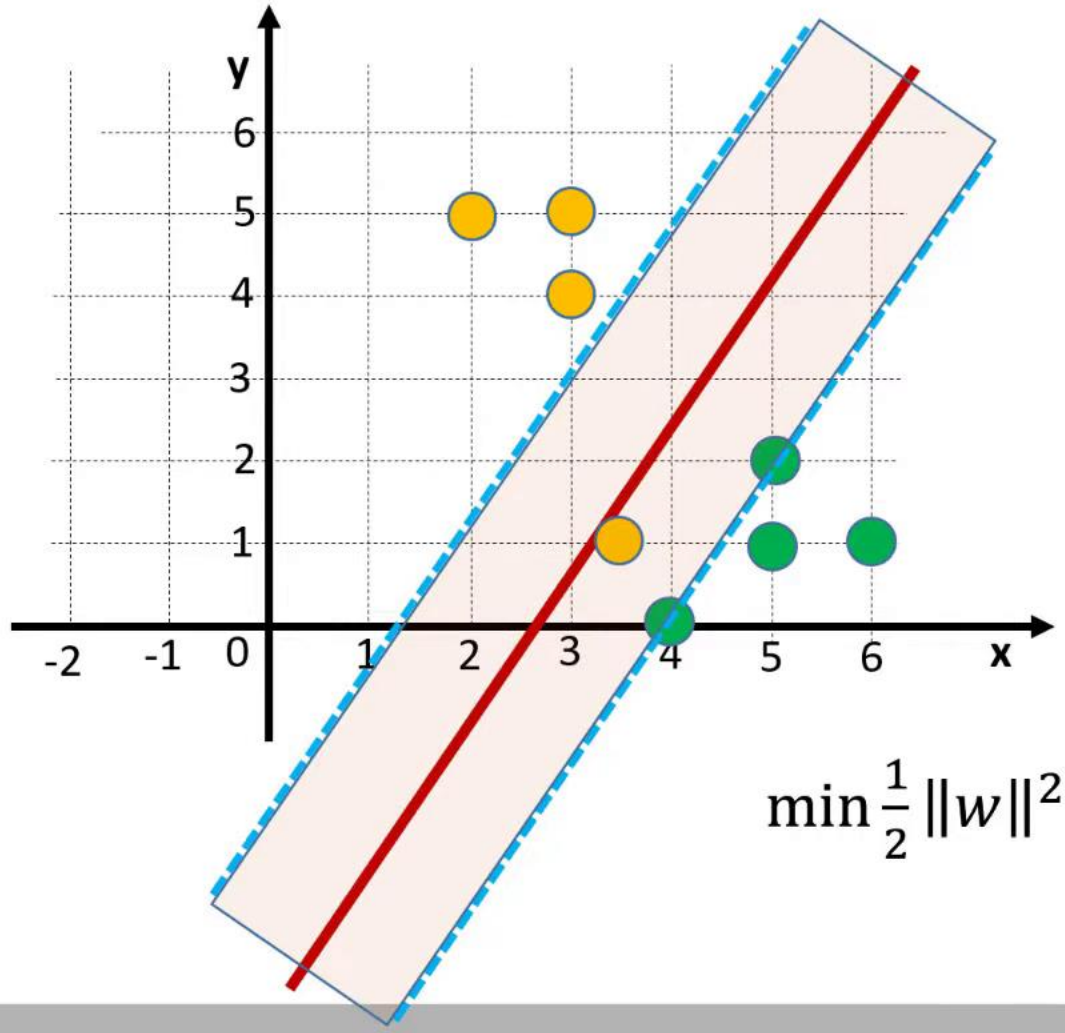
SVM – the math



$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

The value of C can be optimized by using, for example, cross-validation where we pick the value of C that results in the best performance according to the cross-validation.

SVM – the math

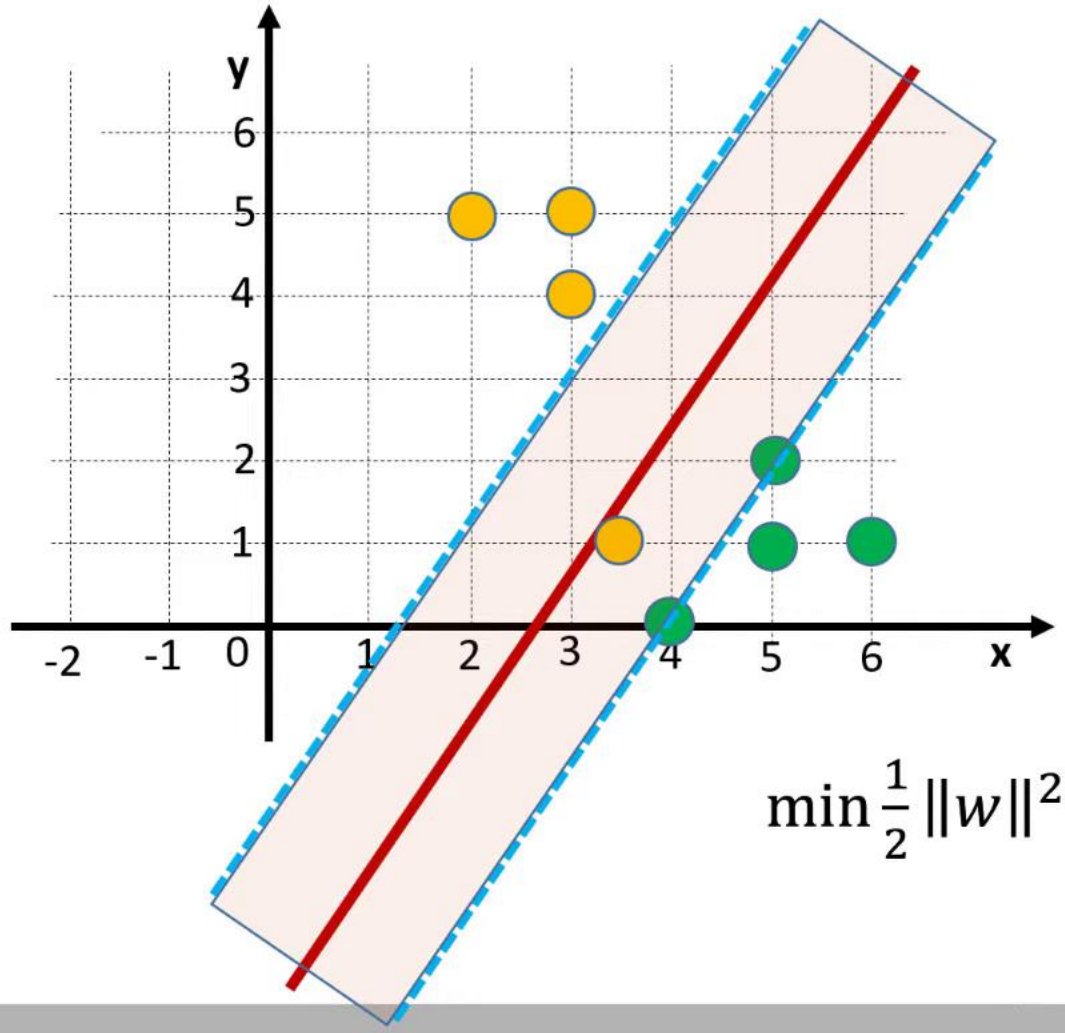


$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

$$\text{such that } Y_i(w^T x_i + b) \geq 1 - \varepsilon_i$$

Similar to before, optimizing the SVM is constrained to the following condition,

SVM – the math



$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i$$

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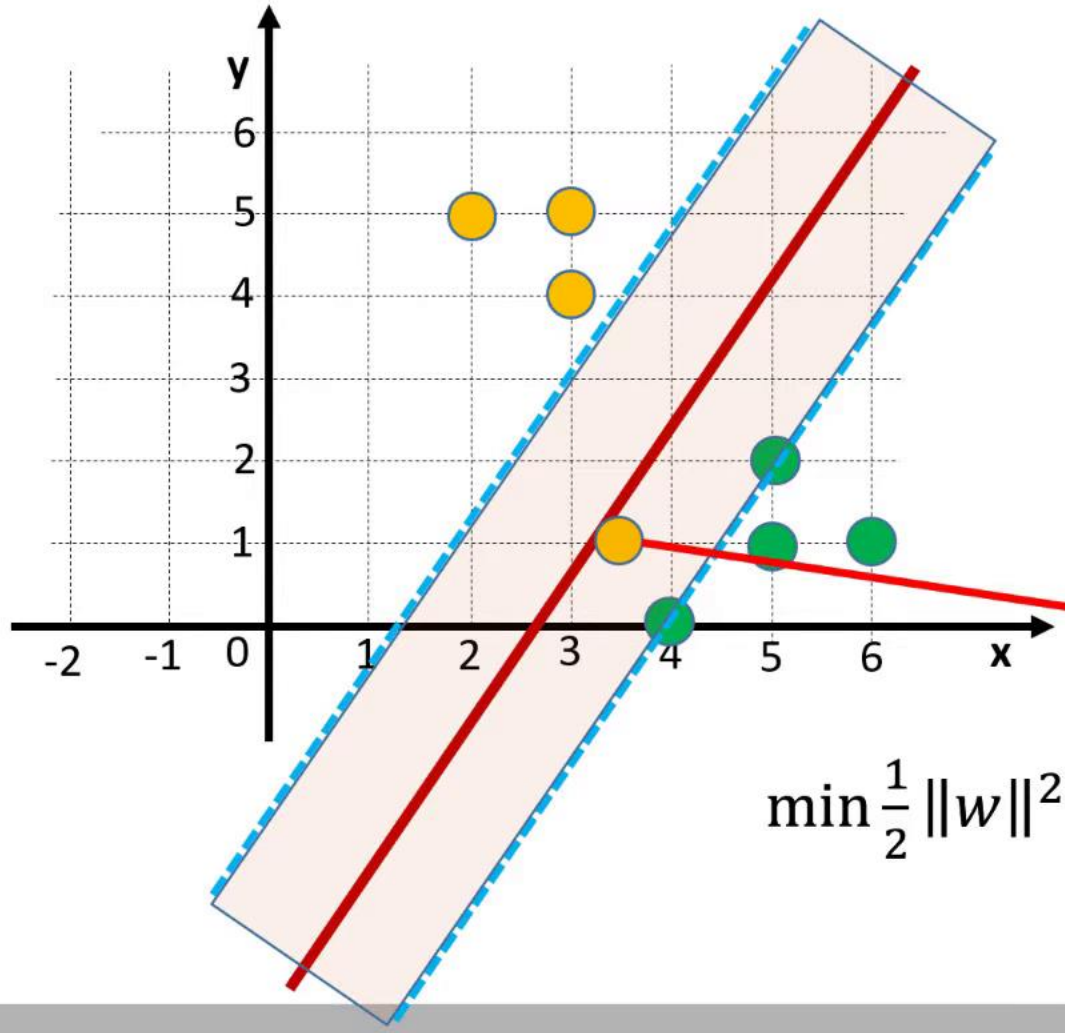
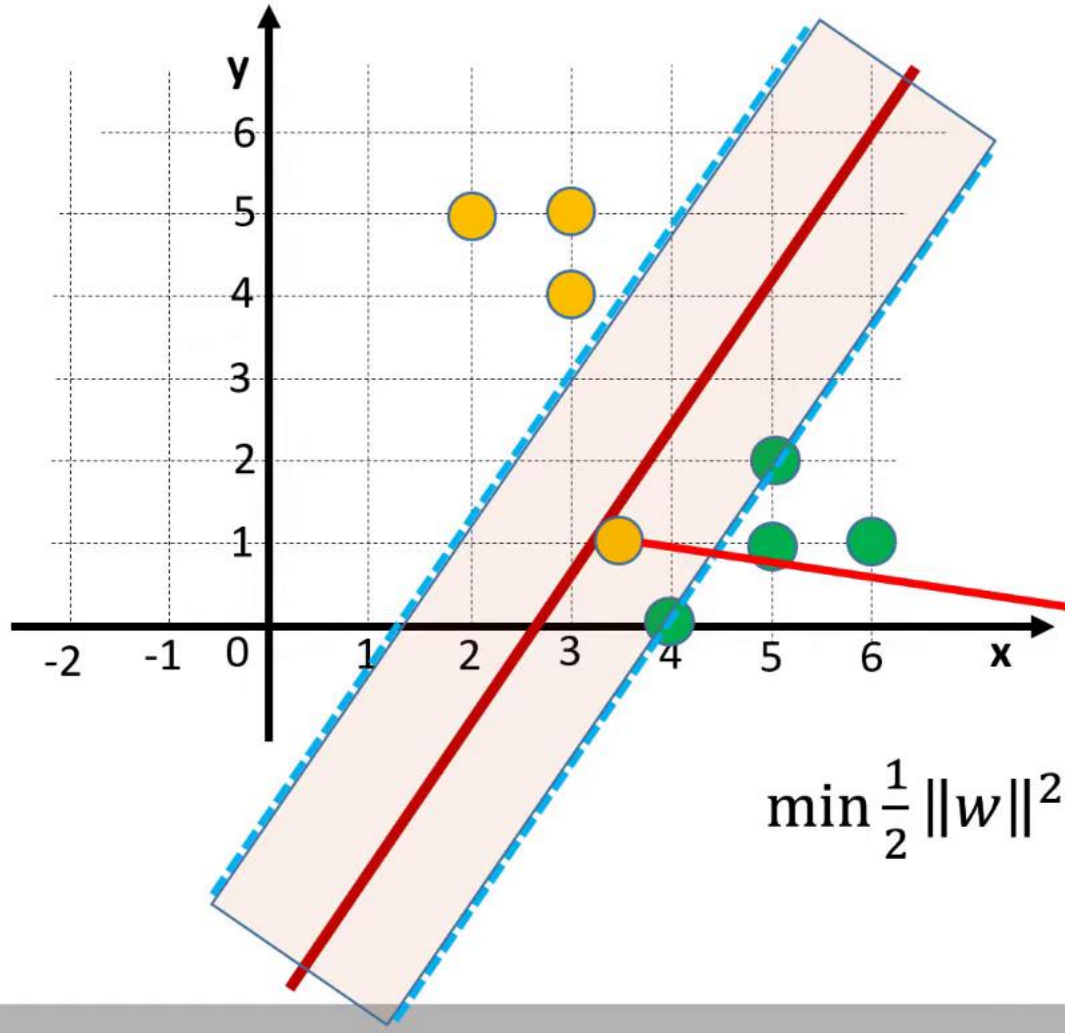


$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i \quad \text{such that } Y_i(w^T x_i + b) \geq 1 - \varepsilon_i$$

whereas incorrectly classified data points, and points within the margin, may generate a value that is smaller than one.

SVM – the math



$$\min \frac{1}{2}\|w\|^2 + C\sum_i^N \varepsilon_i \quad \text{such that } Y_i(w^T x_i + b) \geq 1 - \varepsilon_i$$

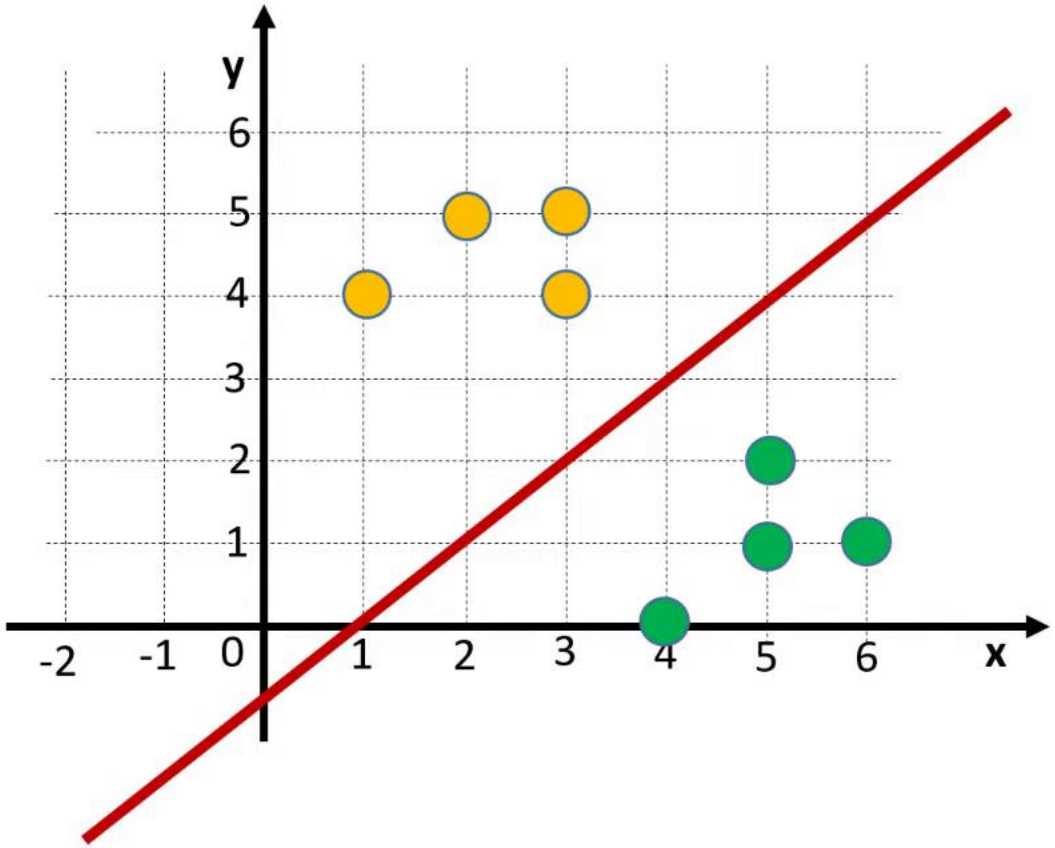
whereas incorrectly classified data points, and points within the margin, may generate a value that is smaller than one.

Evaluation

Evaluation

We will now have a quick look at how we can evaluate how good the SVM performs.

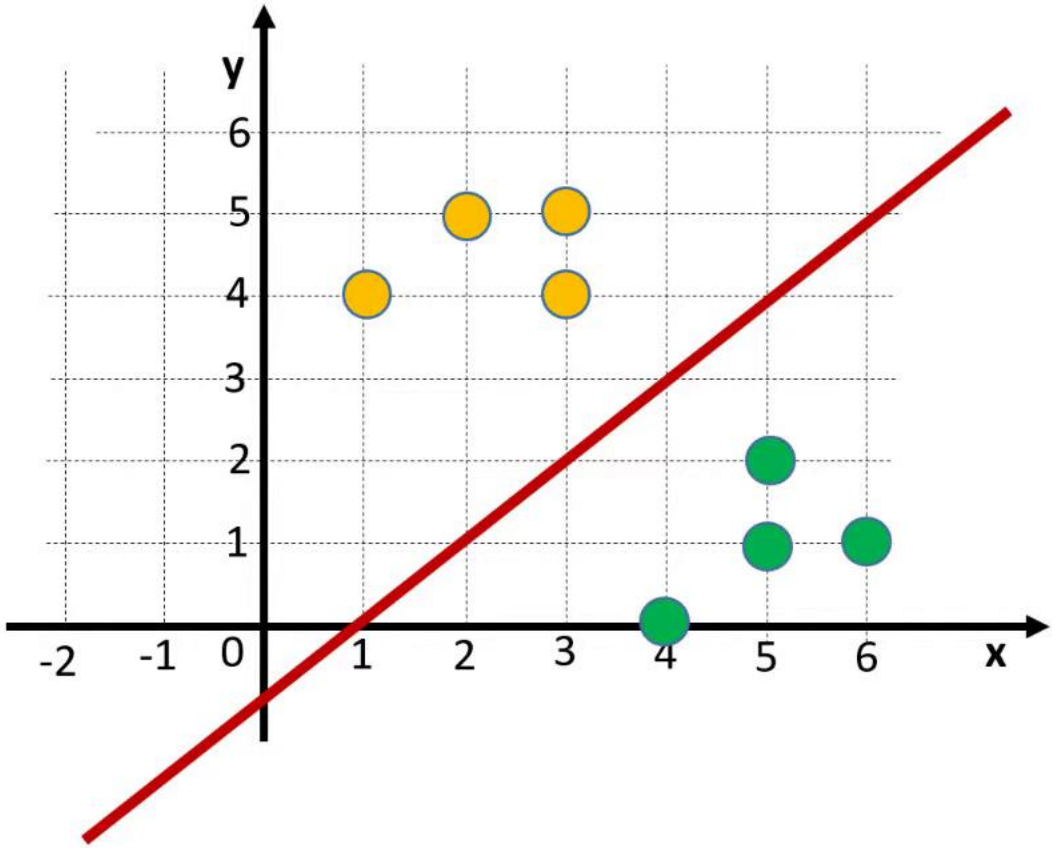
Evaluation



Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

Suppose that we have trained our SVM based on this training data. Note that you should use a lot more data points to train the model than shown in this simple example.

Evaluation



Group (Y_i)	x	y
A (+1)	1	4
A (+1)	2	5
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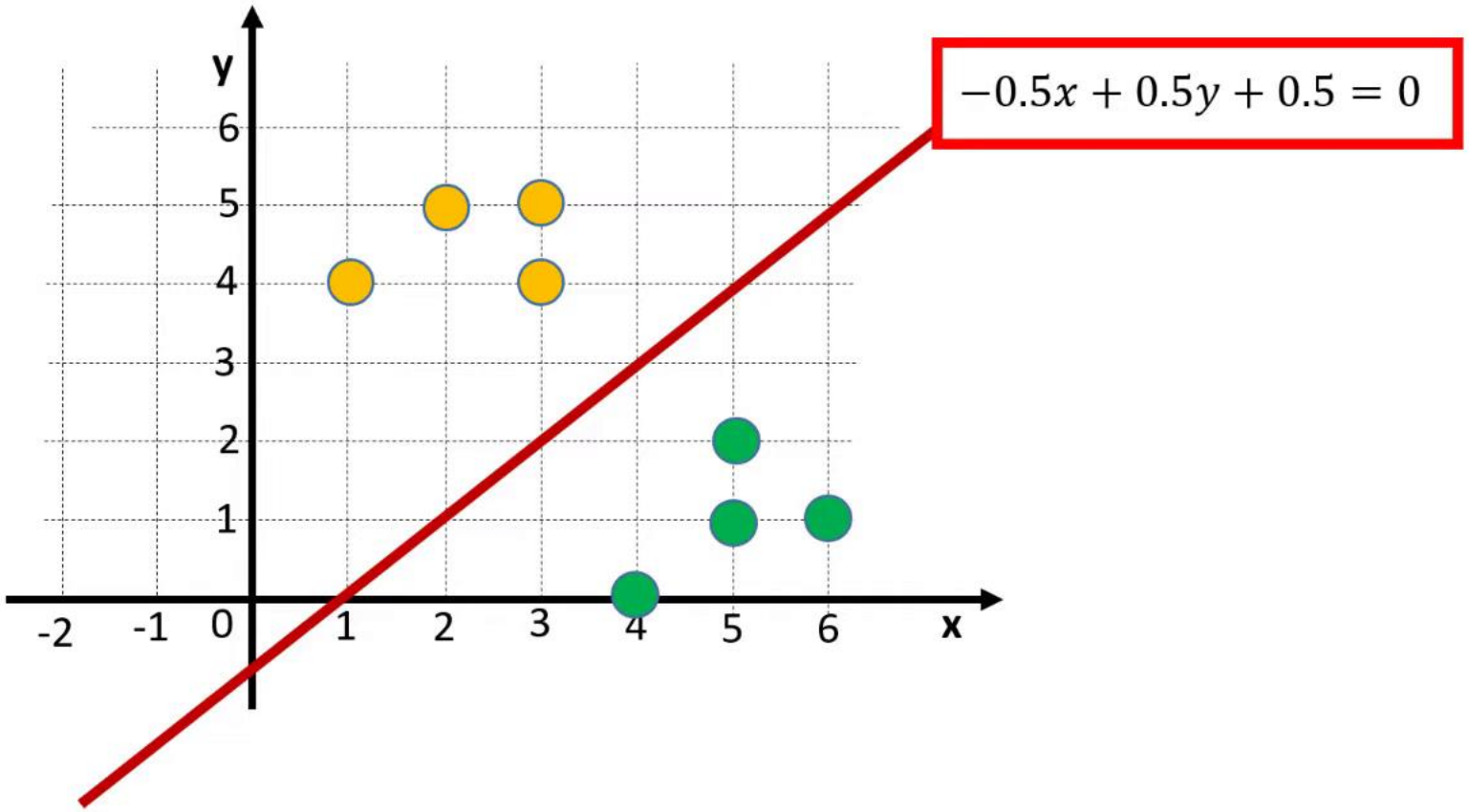
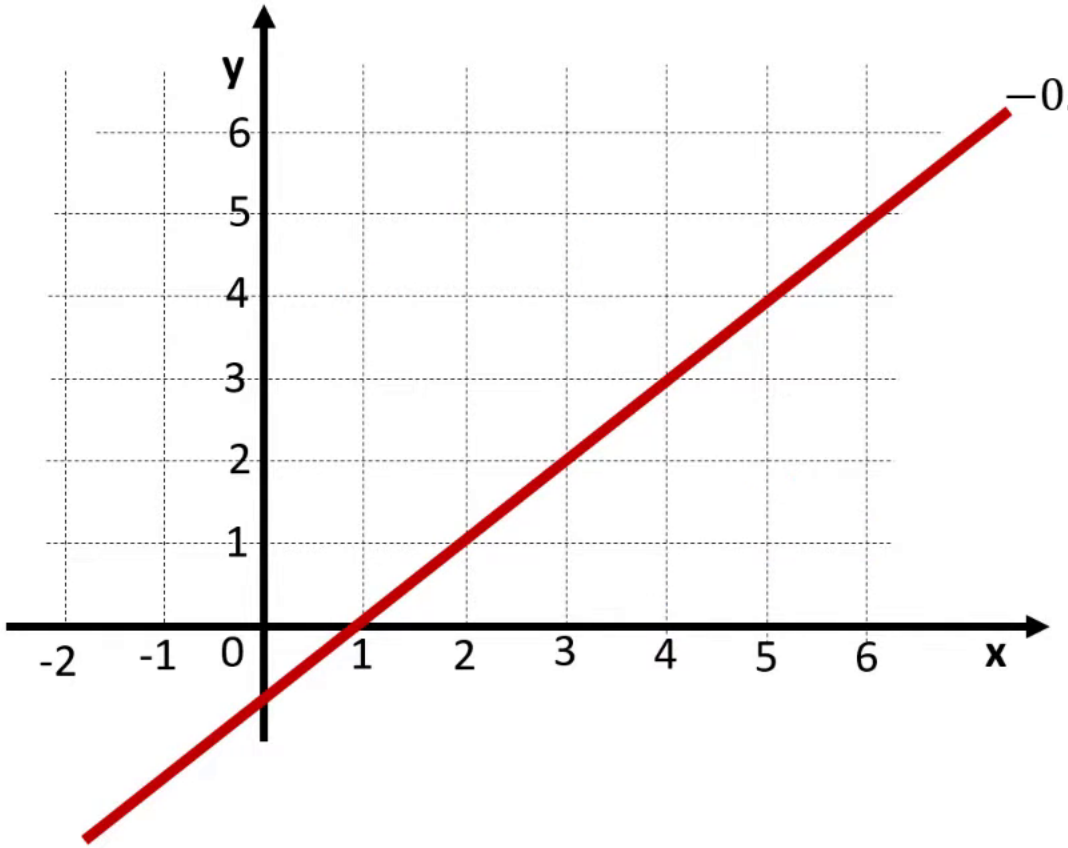


Group (Y_i)	*x*	*y*
A (+1)	1	4
A (+1)	2	5
A (+1)	3	5
A (+1)	3	4
B (-1)	6	1
B (-1)	4	0
B (-1)	5	2
B (-1)	5	1

Let's say that the SVM has optimized the coefficients to these values of our hyperplane according to the minimization function as we saw earlier.

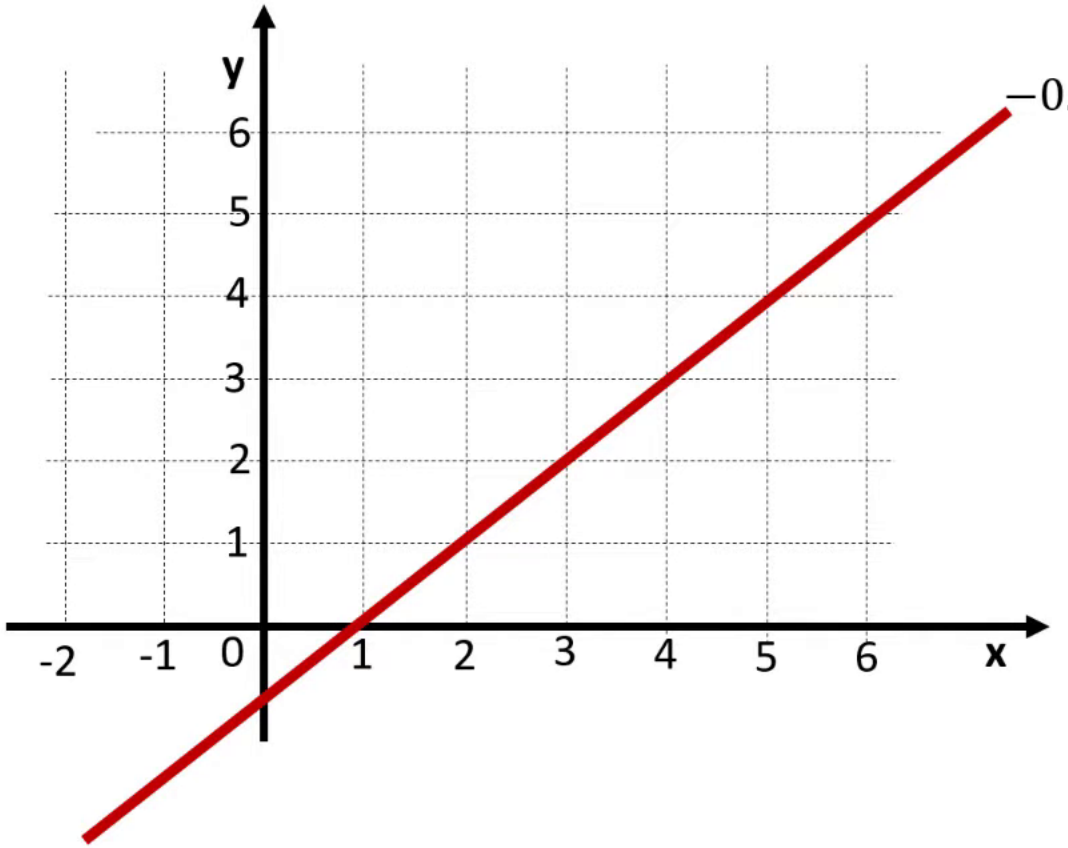
Evaluation



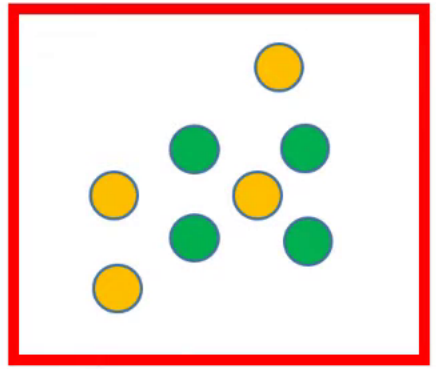
$$-0.5x + 0.5y + 0.5 = 0$$

We now remove the training data,

Evaluation

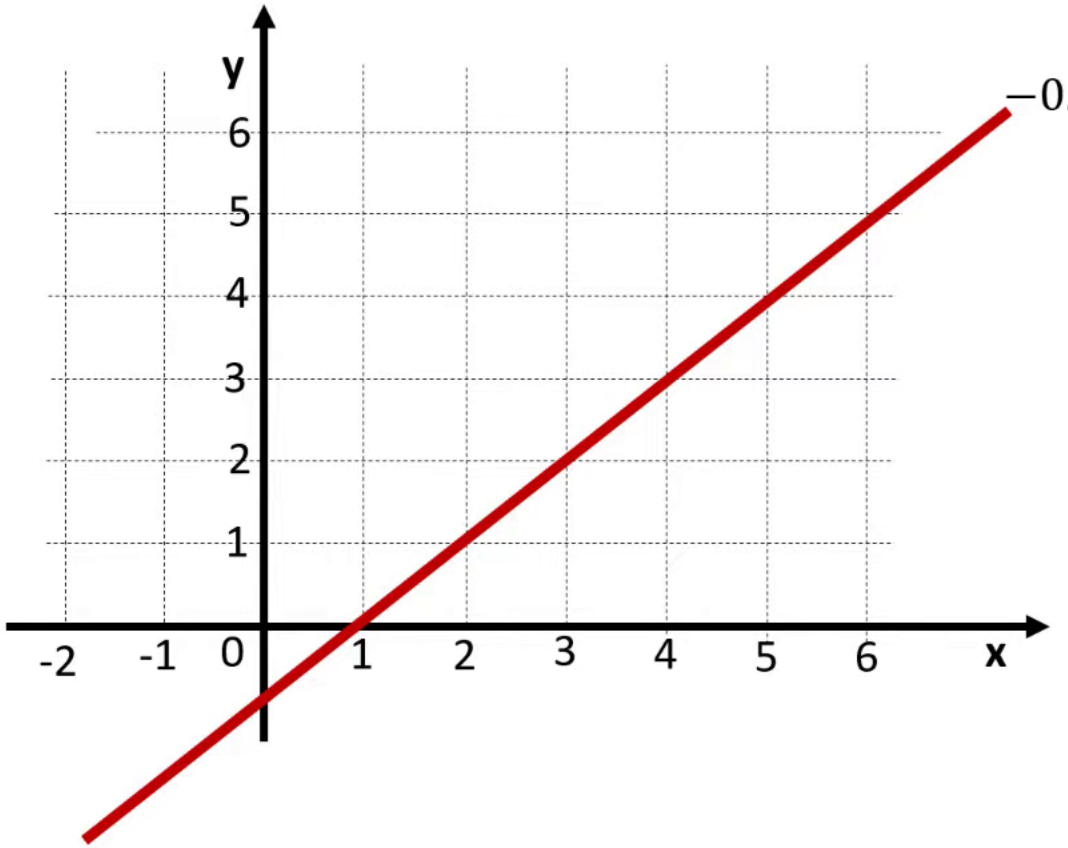


$$-0.5x + 0.5y + 0.5 = 0$$

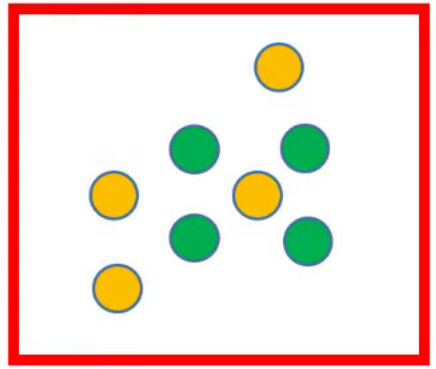


and use a new data set, a so-called test data set, where we also know the class of each observation. This test data set is similar to the training data set but which has not been used in training the SVM.

Evaluation

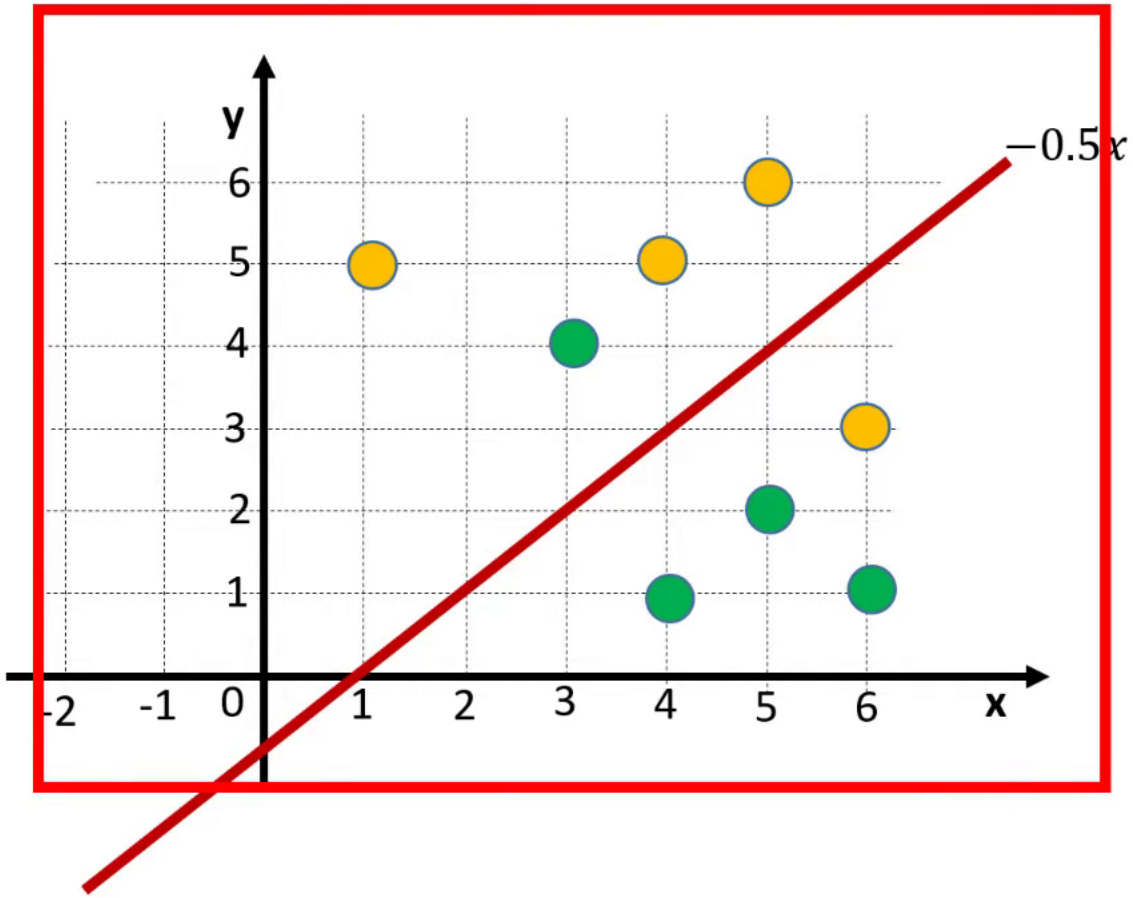


$$-0.5x + 0.5y + 0.5 = 0$$



and use a new data set, a so-called test data set, where we also know the class of each observation. This test data set is similar to the training data set but which has not been used in training the SVM.

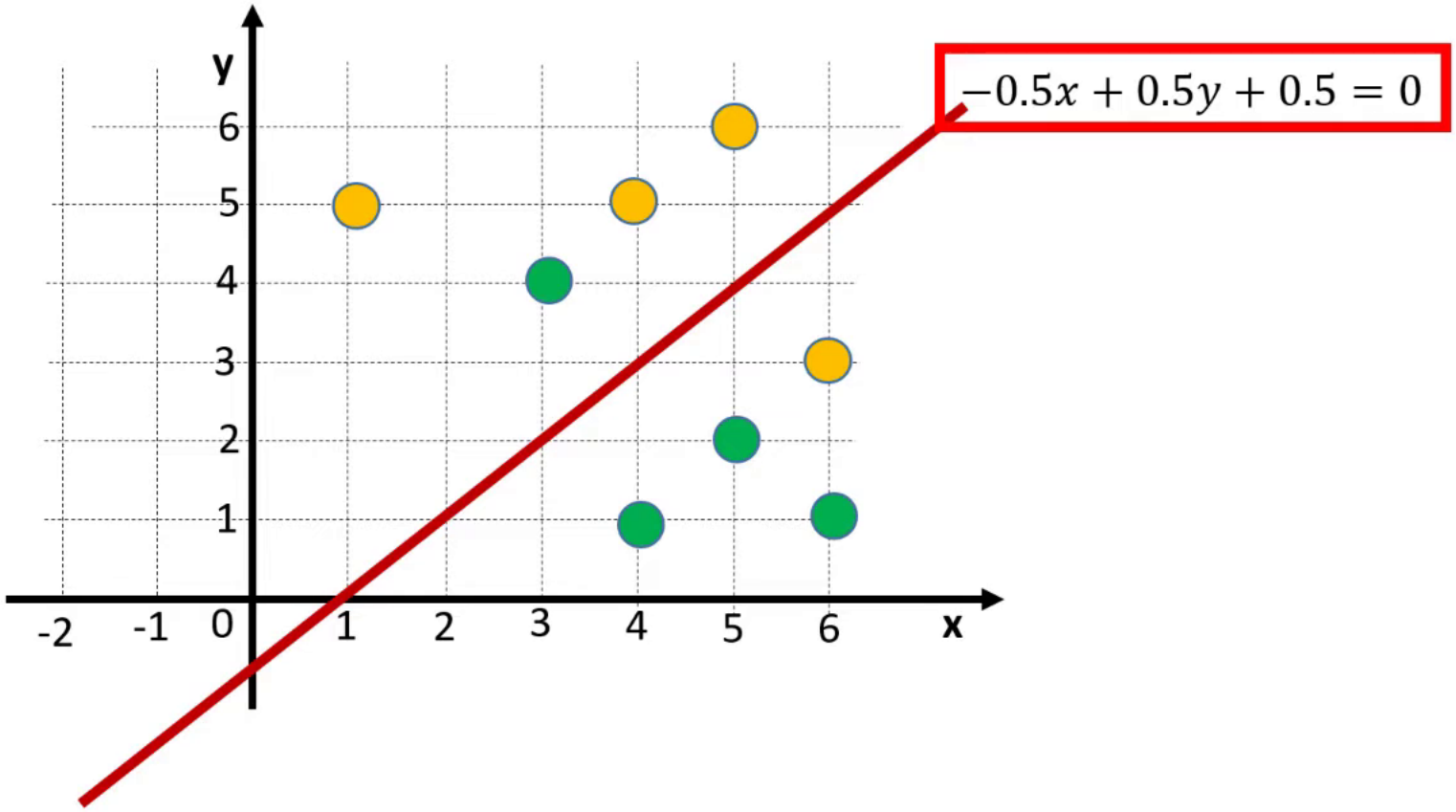
Evaluation



$$-0.5x + 0.5y + 0.5 = 0$$

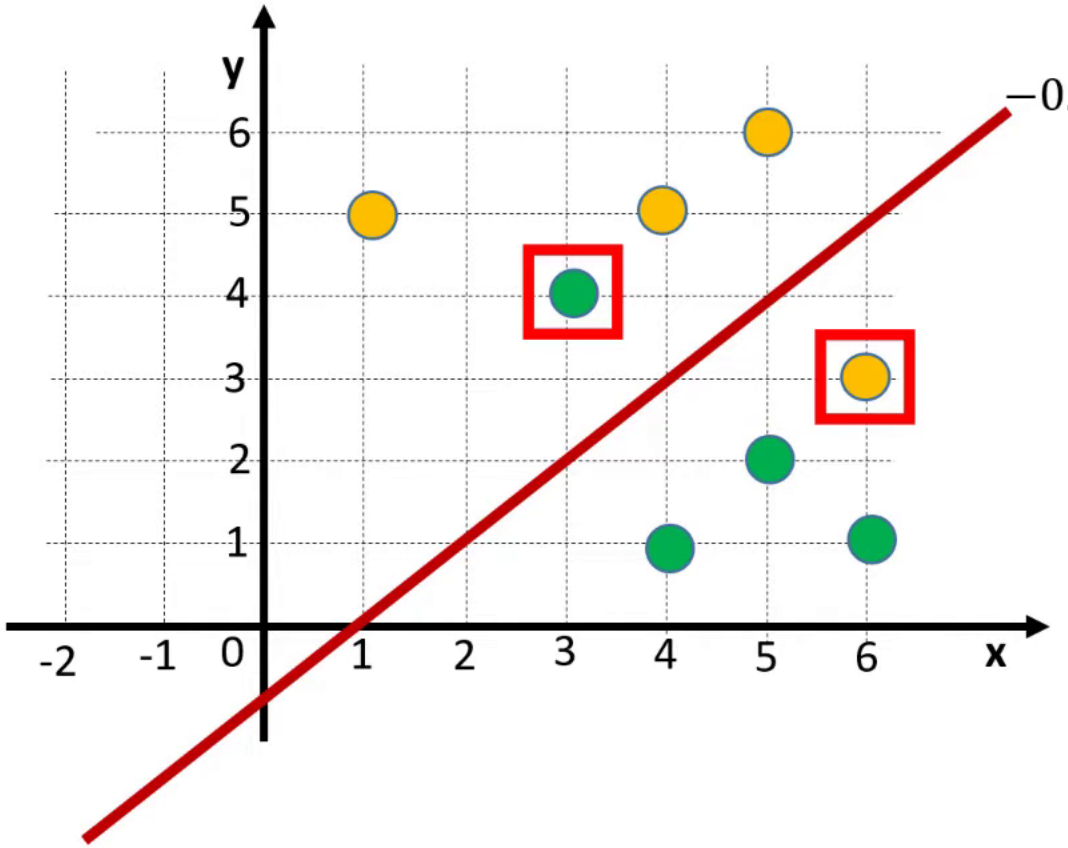
We then place the test data set in the plot to see if they are on the correct side of the line,

Evaluation



or simply plug in their coordinates in the equation of the hyperplane to see if they result in positive or negative values.

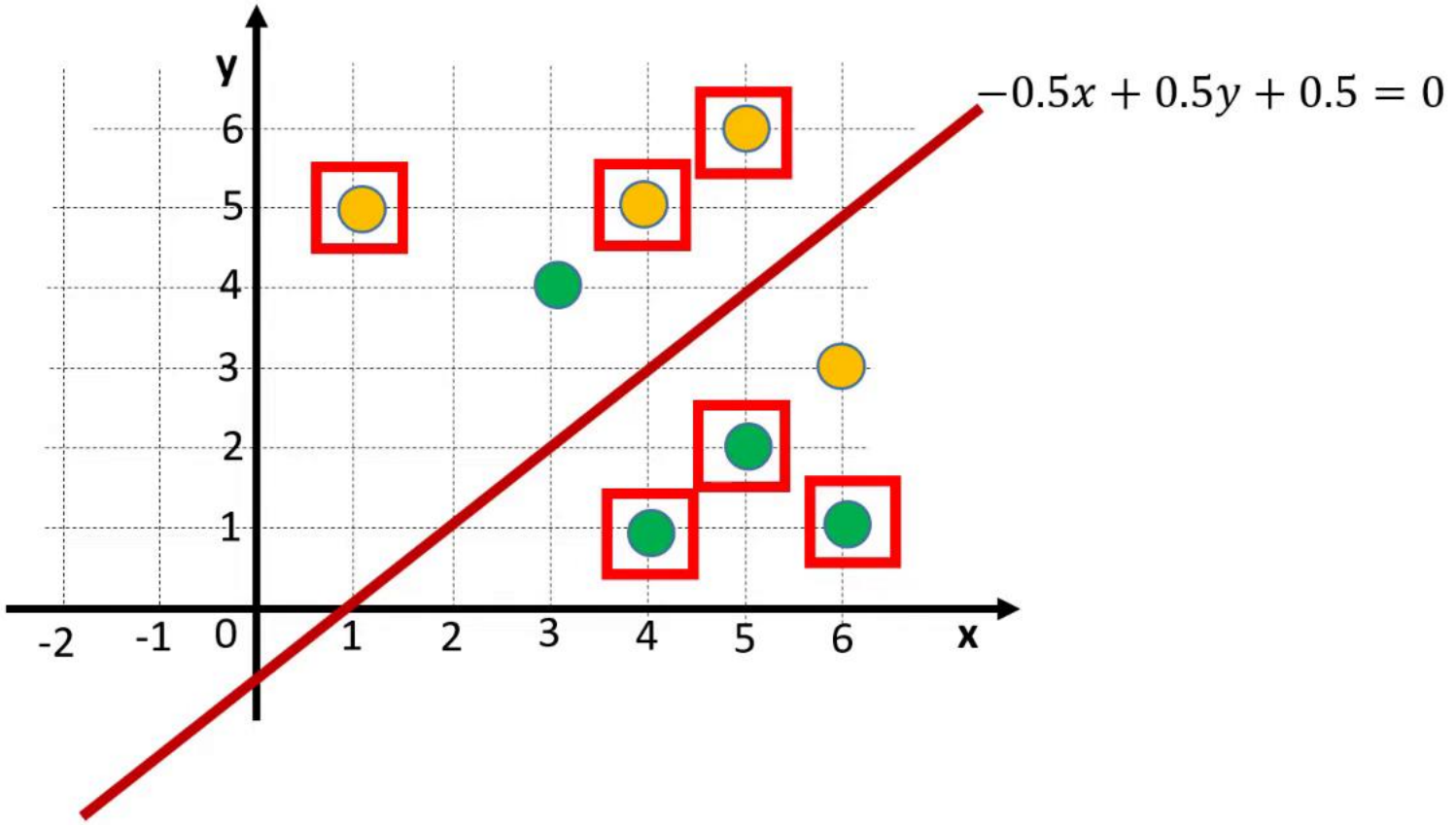
Evaluation



$$-0.5x + 0.5y + 0.5 = 0$$

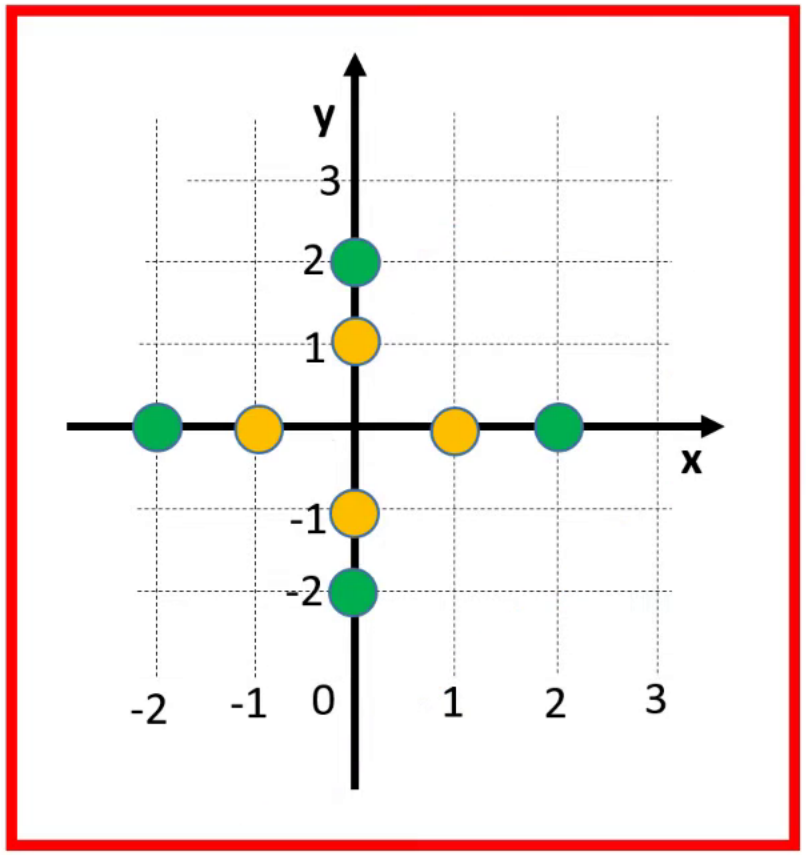
We see that we make two incorrect predictions, because these two data points are on the wrong side of the hyperplane,

Evaluation



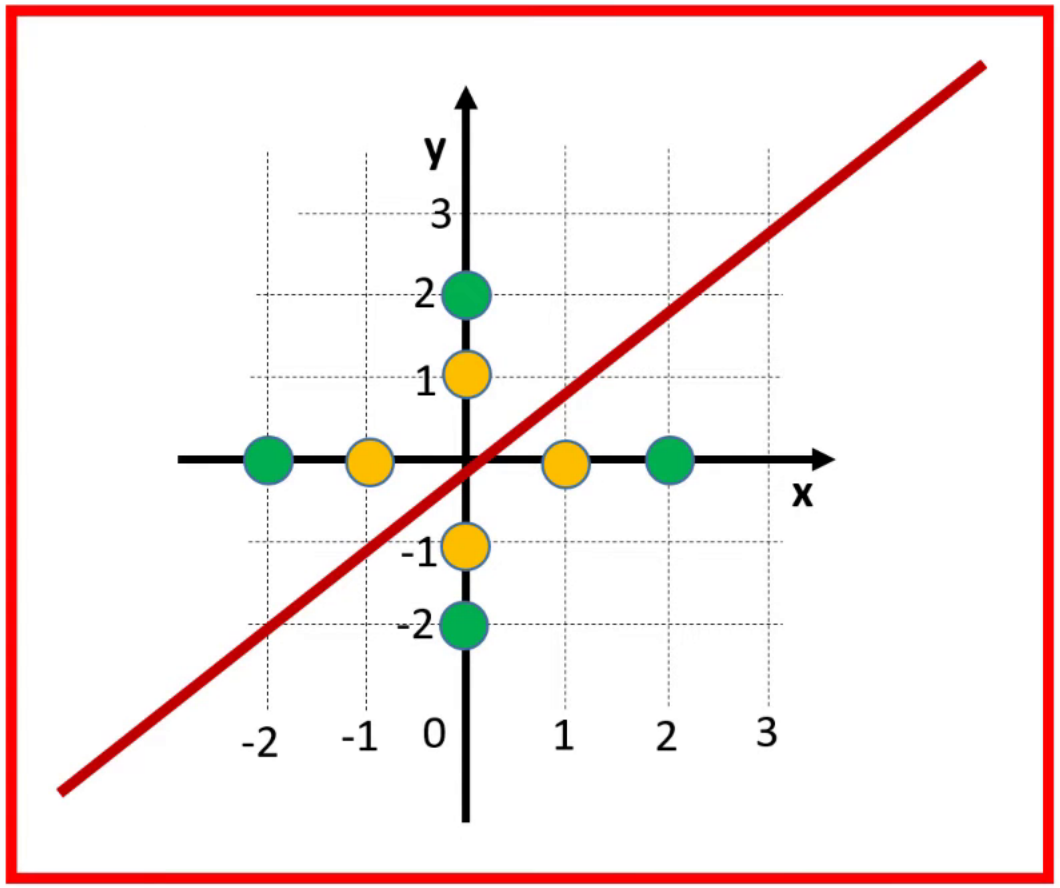
and six correct predictions. The accuracy of our model is therefore six over eight, or 75%. Note that using a test data set like this to evaluate the accuracy of the model is a general technique that is be applied in most supervised machine learning models.

Kernel Trick



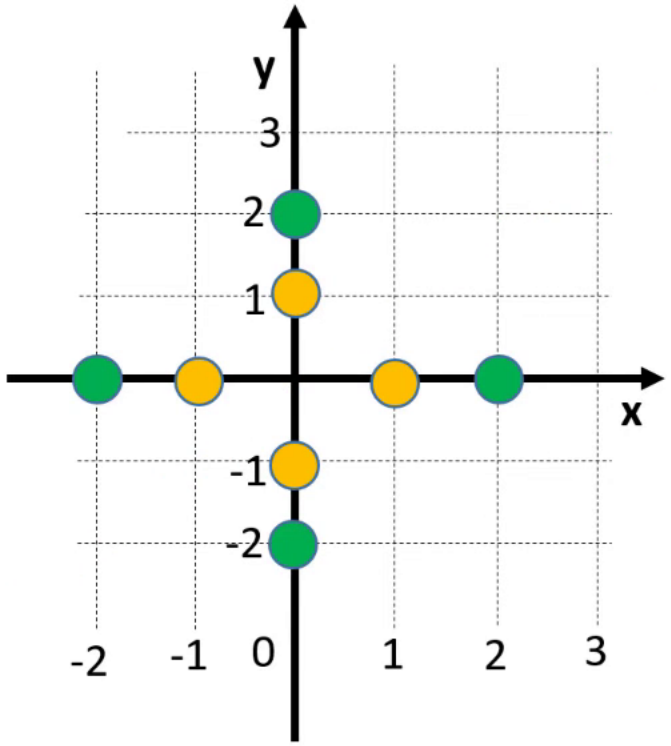
Finally, we will just briefly discuss the so-called Kernel Trick, which can be used when a line is not useful to separate the data.

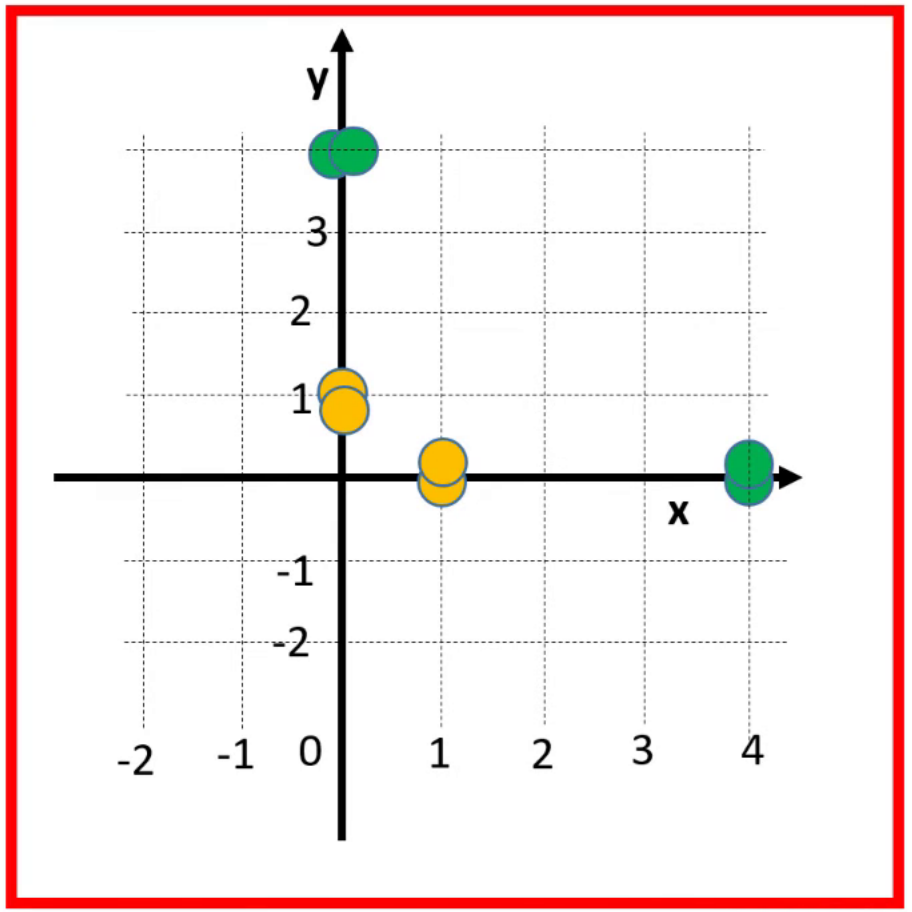
Kernel Trick



For this type of data, a straight line does not make a good job in separating the data points into the corrected classes.

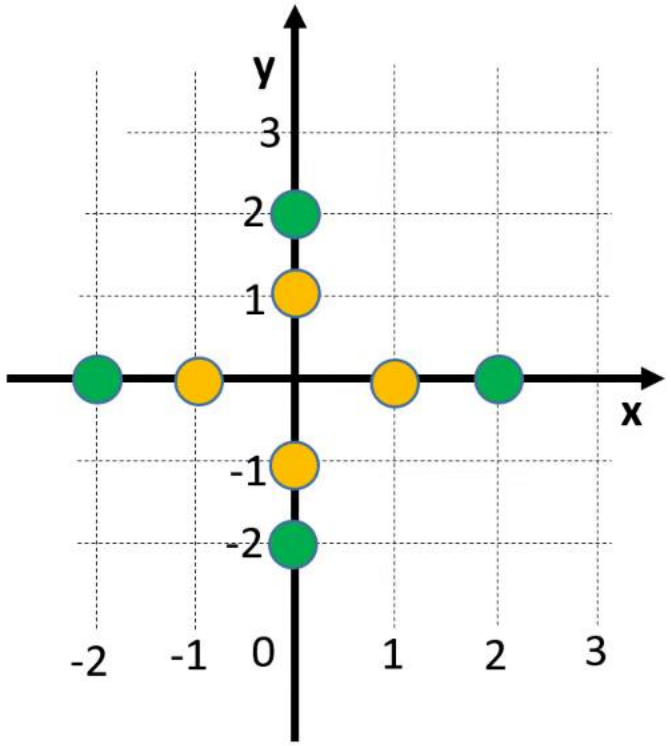
Kernel Trick

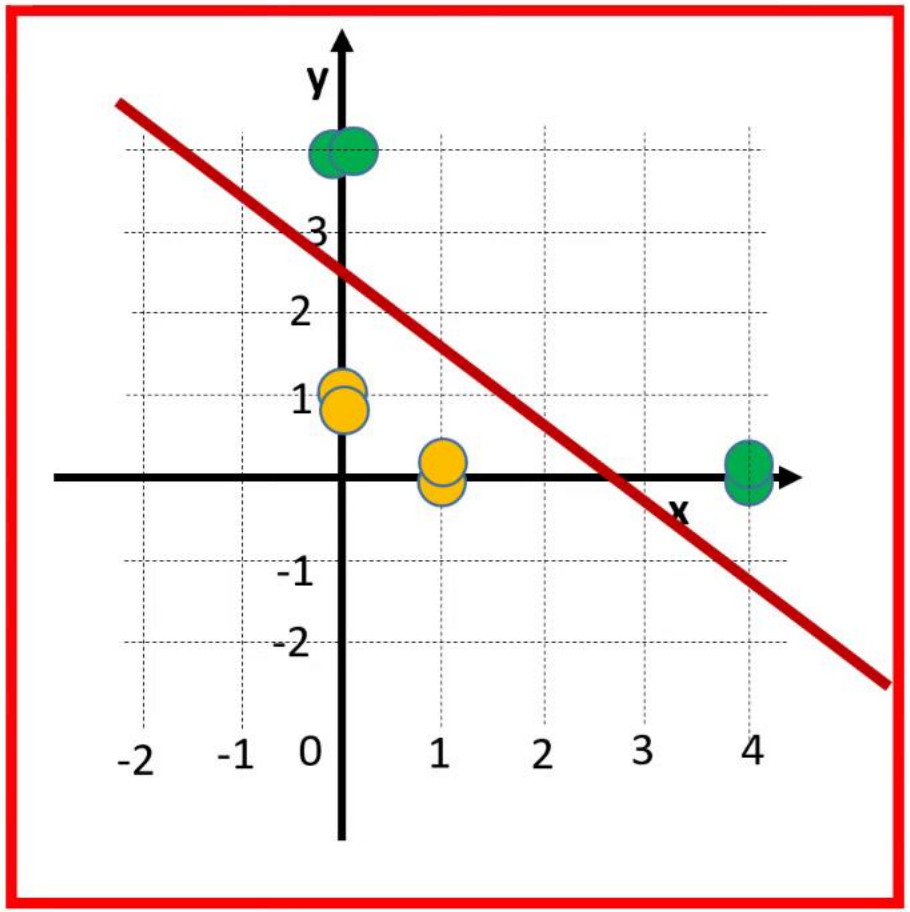




If we would, for example, square the coordinates of the data points,

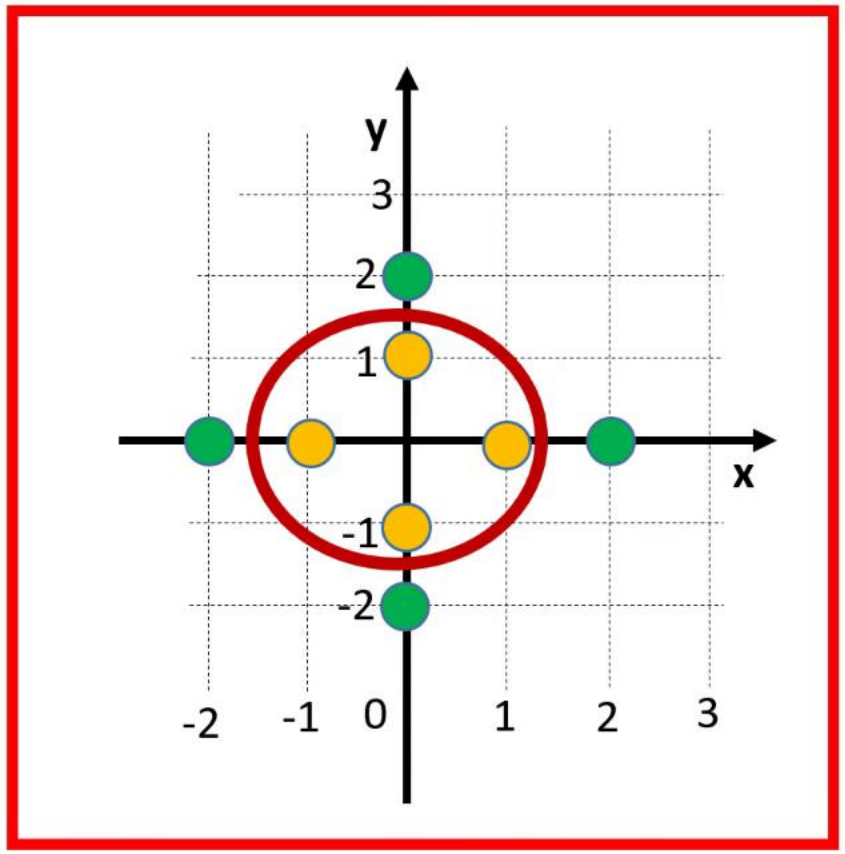
Kernel Trick

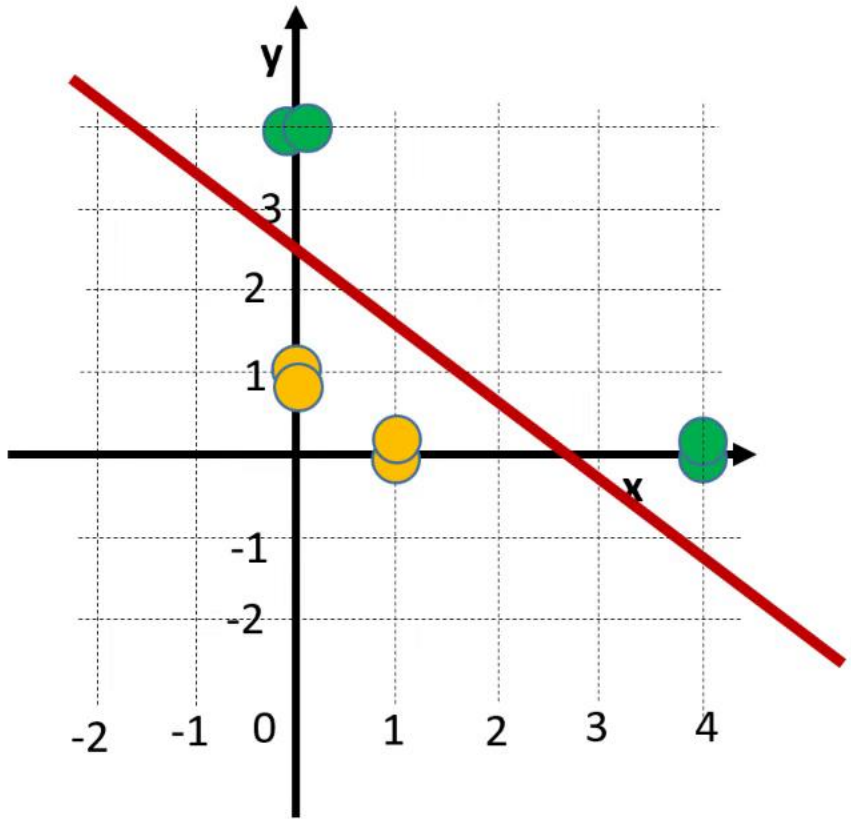




it would be possible to separate the two groups.

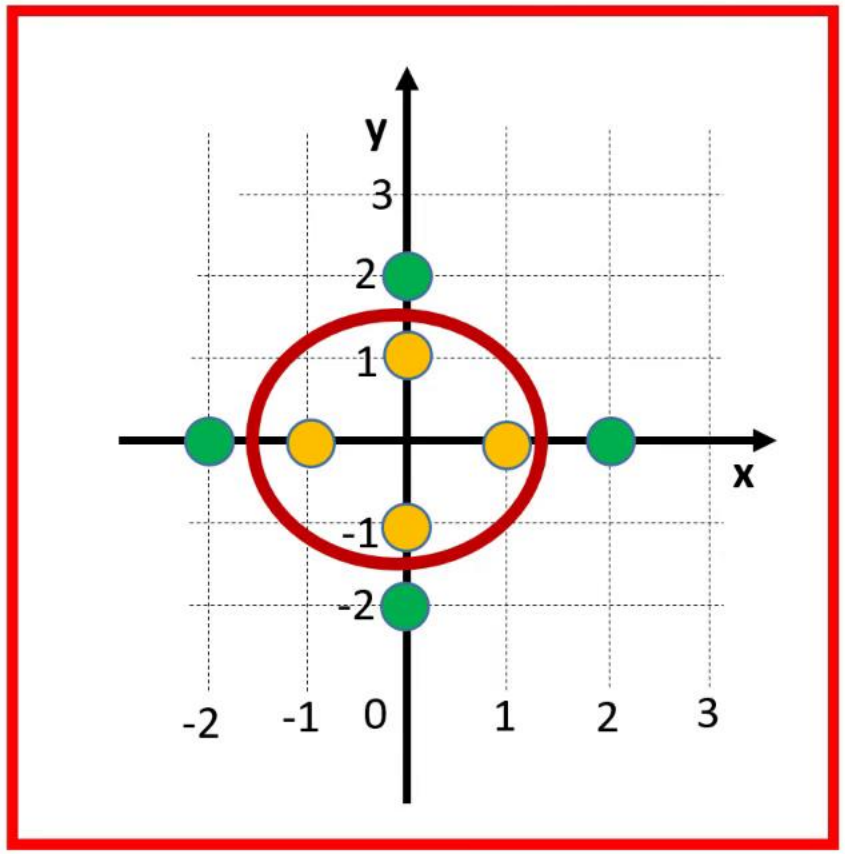
Kernel Trick

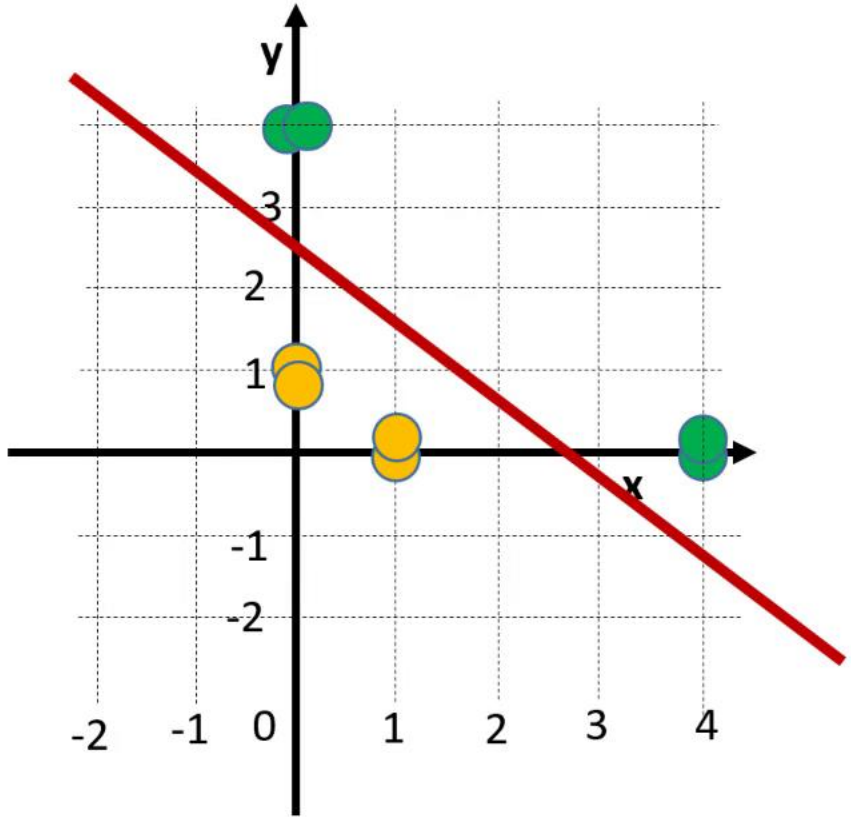




This can be seen as we use a circle for decision boundary, which is called the Kernel trick! Non-linear functions that can help us to separate this kind of data are called Kernels.

Kernel Trick





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